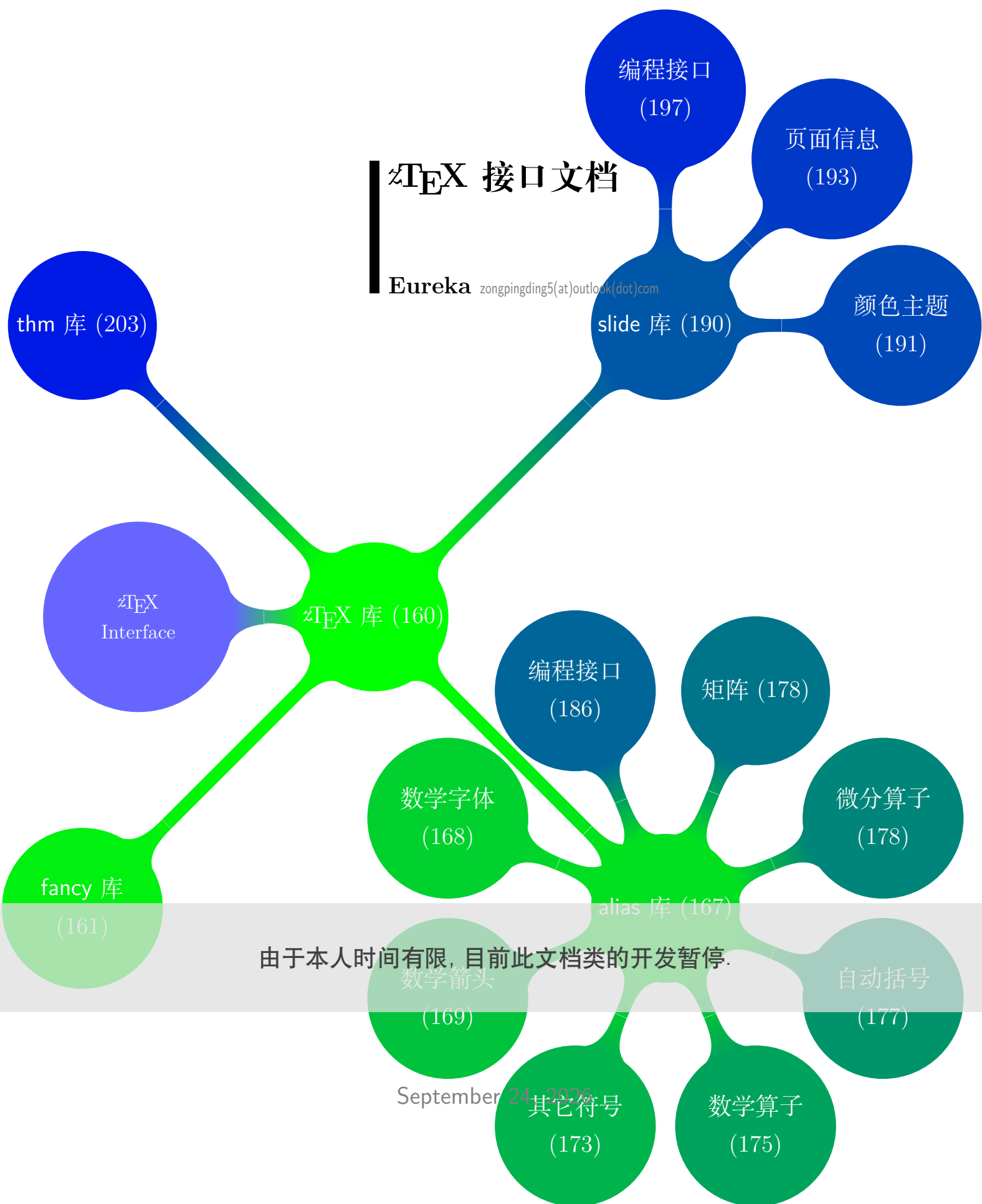




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1 基本介绍

$\LaTeX$ 文档类默认基于 `article` 文档类，但是你仍然可以在加载本文档类时选择加载其他的文档类，通过设置选项 `\class` 的值为 `article`, `book` 亦或者是 `ctexbook`. 通过更换默认的文档类, $\LaTeX$ 可以满足使用者的不同需求，目前本模板可以用于以下场景：

- 撰写书籍或者笔记
- 讨论班的 Slide 制作

$\LaTeX$ 的制作初衷: 让使用者可以方便进行书籍和笔记的撰写以及日常汇报 slide 的无缝切换. $\LaTeX$ 全部由 $\LaTeX 3$ 进行编写，采用 `\key-value` 的方式进行选项和命令的配置，对于作者来说: 方便后续的模板拓展和维护；对于用户来说: 使用键值对可以减轻用户记忆命令参数这一负担，方便用户使用模板内置命令. 如果用户熟悉 $\LaTeX$ ，那么花费不到 10min 的时间，用户便可以轻松使用本文档类完成如上任务，减少不必要的工作.

$\LaTeX$ 文档类会根据用户指定的选项自动处理和加载对应的宏包，所以 $\LaTeX$ 文档类在不同的导言区选项声明下加载的宏包和命令是不同的. 后文详细地介绍了不同导言区配置以及不同编译引擎下的宏包加载情况.

$\LaTeX$ 一直坚持“能自己实现就不依赖外部宏包”的原则. 比如，有些用户会用到 `lastpage` 宏包，它提供了一个名为 `LastPage` 的 label; $\LaTeX$ 也实现了类似功能，提供了“`ztex:lastpage`”这个 label (在页码正确的情况下，超链接跳转可能并不正确，这种情况下可以使用 `ztex@lastpage` 这一个 anchor). 为了在实现一些复杂“盒子”样式的同时，尽量保持较快的编译速度， $\LaTeX$ 引入了 `framedmulticol` 宏包. 有了它的辅助，用户在不依赖 `tikz` 或 `pstricks` 的前提下，也能实现比较复杂的盒子排版¹.

$\LaTeX$ 会加载一系列的基本宏包²，意味着无论用户的导言区如何配置，这部分宏包均会被加载. 具体的宏包加载情况如下：

<code>geometry</code>	<code>fancyhdr</code>	<code>graphicx</code>	<code>xcolor</code>
<code>amsmath</code>	<code>amsfonts</code>	<code>esint</code>	<code>etoolbox</code>
<code>framed</code>	<code>framedmulticol</code>	<code>cleveref/zref-clever</code>	

表 1: $\LaTeX$ 文档类基本宏包

$\LaTeX$ 默认只加载很少的一部分基础宏包，用户如果想要实现更加个性化的功能还请自行引入相关宏包；在默认情况下本模板即可呈现一个比较好的效果，不熟悉 $\LaTeX$ 的用户不用担心本模板配置选项过于复杂. 想要马上开始使用本模板？请参见“[节 \(2.3\)](#)”的最小写作示例.

¹用户可以参考 `longfbox` 宏包的文档，它能够很方便地制作一些精美的“盒子”，十分强大，而且编译速度很快. 因为它只依赖于 $\LaTeX 2\epsilon$ 自带的 `picture` 环境.

²加载 $\LaTeX$ 后，用户可以直接使用 `ctexpatch` 或 `xpatch` 宏包提供的命令.

2 安装使用

2.1 在线模板

为了让部分用户可以直接使用到 $\text{\texttt{zT}\TeX}$, 免去“繁杂”的环境配置. 我已将本模板部署在 $\text{\texttt{TeX}}\text{Page}$ 上, 地址为: [TeXPage \$\text{\texttt{zT}\TeX}\$ Project](https://github.com/zongpingding/zTeX_bundle), 直接打开此地址即可体验. Github 上的项目地址为:

https://github.com/zongpingding/zTeX_bundle

仓库中包含本手册以及 $\text{\texttt{zTi}\kern-0.05emZ}$ 宏集 (由于技术原因, $\text{\texttt{zTi}\kern-0.05emZ}$ 请在本地体验) 的源码, 用户手册以及部分的使用示例; 当前宏集的稳定版本于 2025 年 09 月发布, 最新的开发版请切换到“dev”分支; 本手册适用于当前最新的开发版.

2.2 本地安装

$\text{\texttt{zT}\TeX}$ 宏集目前还未上传 CTAN, 因为还没有开发完成. 本文档类使用的部分 $\text{\texttt{L}\TeX}\text{X3}$ 命令在老版本的 $\text{\texttt{T}\TeX}\text{Live}$ 下并不存在, 若用户的 $\text{\texttt{T}\TeX}\text{Live}$ 版本过低, 则无法正常使用本宏集. 目前 $\text{\texttt{zT}\TeX}$ 文档类在各平台的兼容情况如下:

Windows : $\text{\texttt{T}\TeX}\text{Live}$ 最低版本 2024

Linux : $\text{\texttt{T}\TeX}\text{Live}$ 最低版本 2024

MacOS : $\text{\texttt{MacT}\TeX}$ 还未测试

因 $\text{\texttt{zT}\TeX}$ 还未传入 CTAN(未来可能会考虑), 所以想要使用此文档类, 只有如下两种方法:

- 把此宏集 - $\text{\texttt{ztex}}$ 目录中的所有内容放入当前项目文件夹下;
- 在命令行运行命令: $\text{\texttt{kpsewhich-var-value=TEXMFHOME}}$, 在 Windows 上这个路径一般是: $\text{\texttt{C:/Users/<name>/texmf/}}$, 在 Linux 下一般是: $\text{\texttt{~/texmf/}}$; 具体路径以自己的实际情况为准. 在此路径下新建文件夹 $\text{\texttt{tex/latex/ztex}}$; 此文件夹对应的路径我们记为 $\langle \text{\texttt{zT}\TeX} \rangle$, 随后把 $\text{\texttt{ztex}}$ 目录中的所有内容放入 $\langle \text{\texttt{zT}\TeX} \rangle$ 下即可.

在本手册后续, 我们使用 $\langle \text{\texttt{zT}\TeX} \rangle$ 表示本宏集的根本目录.

2.3 快速开始

本节展示 $\LaTeX$ 的最小工作示例³：首先是中文写作示例， $\LaTeX$ 默认加载 `article` 文档类，如果用户希望使用 `book` 文档类，可以在加载文档类时指定文档类选项：`class = book`。

```
% !TeX program = XeLaTeX
\documentclass[lang=cn]{ztex}

\begin{document}
% some preface
% \tableofcontents

% writing your document here ...
\end{document}
```

例 1

其次是英文写作示例，更改基文档类 – `class = book`，更改语言选项 – `lang = en` (此为默认选项)，然后使用 `pdfLaTeX` 引擎编译此文档：

```
% !TeX program = pdfLaTeX
\documentclass[class=book]{ztex}

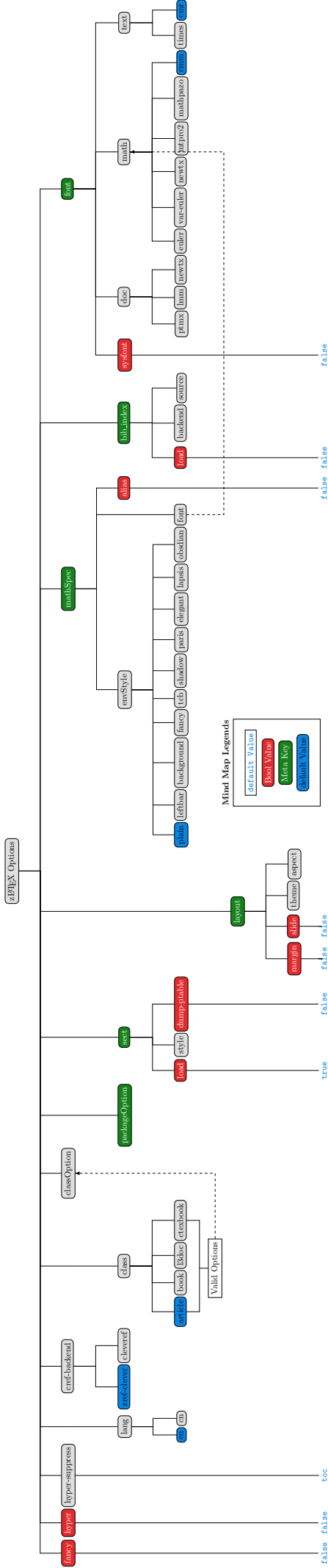
\title{Title}
\author{Author}
\date{\today}
\begin{document}
\maketitle
\frontmatter
% some preface
% \tableofcontents
% some claim etc.
\mainmatter

% writing your document here ...
\end{document}
```

例 2

在使用 `book` 文档类时，需要在特定的位置调用 `\frontmatter` 和 `\mainmatter` 两个命令，否则之后文档的页眉，页脚格式可能会不正确。有时甚至会破坏相关的超链接跳转。

³ 导言区的配置可能需要根据自己的实际情况加以调整，详细配置请参见后文



3 基本命令

在介绍后续命令的具体用法之前, 我们首先约定一套符号和标记规则. 这些约定适用于 $\text{\LaTeX}$ 所提供的一系列 $\text{\LaTeX} 2_{\epsilon}$ 与 $\text{\LaTeX} 3$ 命令, 它们能够帮助你更清晰、更高效地理解和使用这些命令:

- 名字后带有 $\star$ 号的命令, 可以在 x , e , f 型参数中被完全展开,
- 名字后带有 $\star$ 号的命令, 只能在 x , e 型参数中被完全展开, 无法在 f 型参数中被完全展开;

$\text{\LaTeX}$
$\text{\LaTeX}$
$\text{\LaTeX}$
$\text{\LaTeX}$
Updated: 2024-11-05

它们用于输出本宏集的标志 (logo), 命令名不区分大小写, 并且它们都提供了一个带星号 ($\star$) 的变体.

Fancy Style: $\text{\LaTeX}$ VS $\text{\LaTeX}$; $\text{\LaTeX}$ VS $\text{\LaTeX}$; $\text{\LaTeX}$ VS $\text{\LaTeX}$; $\text{\LaTeX}$ VS $\text{\LaTeX}$. Plain Style: $\text{\LaTeX}$ VS $\text{\LaTeX}$; $\text{\LaTeX}$ VS $\text{\LaTeX}$; $\text{\LaTeX}$ VS $\text{\LaTeX}$; $\text{\LaTeX}$ VS $\text{\LaTeX}$. hologo Package: $\text{\LaTeX}$; $\text{\LaTeX}$; $\text{\LaTeX}$; $\text{\LaTeX}$. ----- Fancy Style: $\text{\LaTeX}$ VS $\text{\LaTeX}$; $\text{\LaTeX}$ VS $\text{\LaTeX}$; $\text{\LaTeX}$ VS $\text{\LaTeX}$. Plain Style: $\text{\LaTeX}$ VS $\text{\LaTeX}$; $\text{\LaTeX}$ VS $\text{\LaTeX}$; $\text{\LaTeX}$ VS $\text{\LaTeX}$. hologo Package: $\text{\LaTeX}$; $\text{\LaTeX}$; $\text{\LaTeX}$; $\text{\LaTeX}$.	例 3
--	----------------

$\text{\LaTeX}$
Updated: 2025-04-25

$\text{\LaTeX}$

该命令用于打印 $\text{\LaTeX}$ 传入当前文档类的所有选项, 可以在调试模板时使用.

$\text{\LaTeX}$	例 4
cn , 11pt	

\ztexset Updated: 2025-04-25	\ztexset{<key-value>} 此命令用于配置 $\text{\LaTeX}$ 选项, 部分的配置仅可以在加载文档类时指定, 这部分键的使用说明请参照后续: 节 (4) – 文档类选项.
\ztexloadmod \ztexloadlib Updated: 2025-04-25	\ztexloadmod{<module name>} \ztexloadlib{<library name>} $\text{\LaTeX}$ 由一系列的模块 (module) 和库 (library) 组成, 用户需要使用这两个命令加载 $\text{\LaTeX}$ 的模块和库; 所有模块默认都会被加载, 而库 (library) 默认则不会自动加载, 需由用户手动指定. 在本小节的后续, 我将介绍一些分散于 <code>ztex.cls</code>、<code>graphics</code> 模块、<code>counter</code> 模块以及 <code>item</code> 模块中的命令. 由于这些命令较为零散, 且缺乏系统性, 我们将其集中在此做统一说明, 以便用户查阅.
\zpw \zph New: 2024-12-05	此二命令表示当前纸张的宽和高, 命令原型为 <code>\paperwidth</code> 和 <code>\paperheight</code>.
\c_ztex_quad_dim	此命令表示当前文档中一个空格的宽度.
\uncompressPDF New: 2025-09-20	这个命令来自 <code>primitive</code> 库, 可以取消 PDF 压缩 (一定程度上提高文档编译速度), 只能在导言区使用. 针对 <code>pdf$\text{\TeX}$</code>, <code>X$\text{\TeX}$</code> 以及 <code>Lua$\text{\TeX}$</code> 均有适配, 用户无需再介入处理.
\ztextitle \ztexauthor \ztexdate Updated: 2025-04-25	此三个命令用于分别保存导言区 <code>\@title</code>, <code>\@author</code>, <code>\@date</code> 三个变量的值, 用户可以在正文部分使用此三个变量. 一个基本的使用样例如下:

`\ztextitle\par`

`\ztexauthor\par`

`\ztexdate`

$\text{\LaTeX}$ 接口文档

Eureka `zongpingding5(at)outlook(dot)com`

September 24, 2026

例 5

4 文档类选项

$\LaTeX$ 的文档类选项可以在加载文档类时指定, 也可以后续通过 `\ztexset` 命令设置. $\LaTeX$ 中的 $\langle \text{key-value} \rangle$ 被划分为两个层级: 第一层中的 $\langle \text{layout} \rangle$, $\langle \text{mathSpec} \rangle$, $\langle \text{sect} \rangle$, $\langle \text{packageOption} \rangle$, $\langle \text{classOption} \rangle$, $\langle \text{font} \rangle$, $\langle \text{bib_index} \rangle$ 具有自己的独立子键, 我们称它们为元键 (meta key); 其余的键则比较简单, 可以直接指定. `ztex.cls` 中的键值关系请参见节首图示.

总体而言, $\LaTeX$ 的文档类选项相对较为复杂. 对于刚接触该文档类的用户而言, 无需掌握所有配置选项; 在默认设置下, $\LaTeX$ 即可生成视觉效果良好的文档.

接下来, 我们将详细介绍 $\LaTeX$ 中各个 $\langle \text{key} \rangle$ 的设置方式及其具体含义. 在进入正题之前, 我们先约定一组符号和格式规则, 以便更好地理解后续内容:

- 名字后带有 ☆ 号的选项, 只能作为宏包/文档类选项, 需要在引入宏包/文档类的时候指定;
- 名字后带有 ★ 号的选项, 只能通过 $\LaTeX$ 宏集提供的用户接口 `\ztexset` 来设定;
- 名字后不带有特殊符号的选项, 既可以作为宏包/文档类选项, 也可以通过 `\ztexset` 来设定.

 ztex/lang ☆

Updated: 2024-11-05

lang = $\langle \text{en|cn} \rangle$ 初始值: **en**

LaTeX 目前仅对中英文做了适配, 对于法语有部分的支持. 根据不同的文档类语言设置, LaTeX 会加载不同的 (和语言相关的) 宏包; 在不同的 $\langle \text{lang} \rangle$ 设置下, 语言类宏包的详细加载情况如下:

- lang = en: inputenc(若使用pdfTeX), fontenc, babel, microtype;
- lang = cn: fontspec, ctex;

NOTE: 目前 ztex 文档类已移除如下配置

```
\sys_if_engine_pdfTeX:T
{ \RequirePackage[utf8]{inputenc} }
\RequirePackage[english]{babel}
\ztex_hook_preamble_last:n
{
  \RequirePackage{csquotes}
  \RequirePackage{microtype}
}
```

例 6

 ztex/hyper ☆

ztex/hyper-suppress ☆

Updated: 2025-07-07

hyper = $\langle \text{true|false} \rangle$ 初始值: **false**

hyper-suppress = $\langle \text{clist} \rangle$ 初始值: **toc**

是否开启文档内部的超链接以及 PDF 书签, 默认为 **false**. 建议在最后的成稿中启用此选项, 在草稿阶段置为 **false** 可以加快文档的编译速度; $\langle \text{hyper-suppress} \rangle$ 用于禁用 hyperref 的 Patch(es), 默认禁用对目录的 Patch; $\langle \text{hyper-suppress} \rangle$ 的可选值有: “footnote, amsmath@tag, counter, mathenv, caption, longtable, bib, thm”.

 ztex/fancy ☆

Updated: 2024-11-05

fancy = $\langle \text{true|false} \rangle$ 初始值: **false**

此选项用于控制文档的外观, 包括章节样式, 定理类环境样式, 默认为 **false**.

ztex/sect/load	☆	load = <code><true false></code> 初始值: <code>true</code>
ztex/sect/style	☆	style = <code><ltx fancy></code> 初始值: <code>ltx</code>
ztex/sect/dump-ptable	☆	dump-ptable = <code><false true></code> 初始值: <code>false</code>

Updated: 2025-09-05

因 $\LaTeX$ 的 `sect` 模块重写了章节命令和目录相关的接口, 同时也提供了此选项用于禁用这些更改; 当 “load = false” 时, 便可成功禁用 (定理目录之类的目录命令也将会失效); `<style>` 用于指定章节命令的样式, $\LaTeX$ 仅提供了 “ltx” 一种样式 (其余样式需要用户自定义), $\LaTeX$ 默认使用 “ltx” 样式; `<dump-ptable>` 用于控制是否生成相应的 “*.p<table>” 文件, 此选项主要用于调试, ‘*.p<table>’ 内数据的存储结构请参见: 节 (6.8.5); 当 `dump-ptable = true` 时生成相应文件, 比如 “*.ptoc, *.plot, ...”

NOTE: `sect` 模块和 $\LaTeX$ 中的部分模块或库目前处于耦合状态, 请慎用 “sect/load=false” 设置.

ztex/class	☆	class = <code><article bool ctexbook></code> 初始值: <code>article</code>
------------	---	--

Updated: 2024-11-05

此选项用于指定加载的基文档类, 默认为 `article`. 加载不同的文档类, 用户可以使用不同的命令: 比如 `ctexbook` 提供了 `\ctexset` 命令进行章节样式设置.

NOTE: 命令 `\ctexset` 或 `ctex` 宏包中 `heading` 部分的设置可能与 `sect` 模块相冲突, 请谨慎加载.

ztex/classOption	☆	classOption= <code><key-value></code> 初始值: <code>oneside, 12pt</code>
------------------	---	---

Updated: 2024-11-05

此选项接受一个逗号分隔的列表, 用于传递基文档类选项, 针对默认的 `article` 文档类, 此项为 `oneside, 12pt`.

ztex/packageOption	☆	packageOption= <code><key-value></code>
--------------------	---	---

Updated: 2024-11-20

此选项接受一个键值对, 用于向目标宏包传递选项, 一个基本的使用样例如下:

```
\documentclass[
  packageOption={
    fontspec=quiet,
    ctex={scheme=plain, punct=quanjiao},
  },
]{ztex}
```

例 7

ztex/font/sysfont	sysfont = $\langle \text{true} \textbf{false} \rangle$初始值: false
ztex/font/doc	doc = $\langle \text{lmm} \text{ptmx} \text{newtx} \rangle$初始值: cm
ztex/font/math	math = $\langle \text{euler} \text{var-euler} \text{newtx} \text{mtpro2} \text{mathpazo} \rangle$初始值: cmm
ztex/font/text	text = $\langle \text{times} \rangle$初始值: cm
Updated: 2024-12-06	
此选项主要用于文档的字体配置, 用户可以通过此键来分别定义文档中的正文或数学字体. 注意: 其中的子键 $\langle \text{sysfont} \rangle$ 默认为 <code>false</code> , 在启用此选项后, $\mathcal{T}_{\text{E}}\text{X}$ 会自动加载 <code>fontspec</code> 宏包, 此时需更换引擎为 $\text{X}_{\mathcal{T}}\text{E}\text{X}$ 或者 $\text{LuaT}_{\text{E}}\text{X}$.	
ztex/layout/margin ☆	margin = $\langle \text{true} \textbf{false} \rangle$初始值: false
ztex/layout/slide ☆	slide = $\langle \text{true} \textbf{false} \rangle$初始值: false
ztex/layout/aspect ☆	aspect = $\langle \text{浮点数} \mid \text{浮点数} \rangle$初始值: 12 9
ztex/layout/theme ☆	theme = $\langle \text{主题名} \rangle$初始值: AnnArborDefault
Updated: 2024-11-05	
设置文档布局, 如果设置 $\langle \text{slide} \rangle = \text{true}$, 那么此时 $\mathcal{T}_{\text{E}}\text{X}$ 会自动加载 <code>slide</code> 库, 最终的文档将变为一个演示文档. 注意: $\langle \text{aspect} \rangle$ 选项仅在 $\langle \text{slide} \rangle = \text{true}$ 时有效.	
ztex/bib_index/load	load = $\langle \text{true} \textbf{false} \rangle$初始值: false
ztex/bib_index/source	source = $\langle \text{字符串} \rangle$初始值: ref.bib
ztex/bib_index/backend	backend = $\langle \text{biber} \text{bibtex} \rangle$初始值: biber
Updated: 2024-12-05	
此选项用于控制索引与参考文献的生成; $\langle \text{load} \rangle$ 用于指定是否加载 <code>biblatex</code> 宏包, 默认为 <code>false</code> ; $\langle \text{source} \rangle$ 用于指定参考文献源文件, 默认为: <code>ref.bib</code> ; $\langle \text{backend} \rangle$ 用于指定处理参考文献的后端, 默认为 <code>biber</code> .	
ztex/mathSpec/alias	alias = $\langle \text{true} \textbf{false} \rangle$初始值: false
ztex/mathSpec/envStyle	envStyle = $\langle \text{主题名} \rangle$初始值: plain
ztex/mathSpec/font	font = $\langle \text{euler} \text{var-euler} \text{newtx} \text{mtpro2} \text{mathpazo} \rangle$初始值: cmm
Updated: 2024-11-05	
此键用于配置数学排版相关的选项. 其中, $\langle \text{alias} \rangle$ 默认为 <code>false</code> ; 当设为 <code>true</code> 时, $\mathcal{T}_{\text{E}}\text{X}$ 将自动加载 <code>alias</code> 库. 该库提供了一系列与数学符号, 微分算子, 矩阵相关的简写命令, 例如: 使用 <code>\ZZ</code> 代替 <code>\mathbb{Z}</code> , <code>\mat</code> 用于快速输入矩阵, ... 最后, $\langle \text{envStyle} \rangle$ 用于指定数学环境的样式, 默认值为 <code>plain</code> .	
出于编译速度的考虑, 虽然 $\mathcal{T}_{\text{E}}\text{X}$ 预定义了一系列定理环境样式, 但它们并不会默认加载. 其中部分样式被移入了 <code>thm</code> 库中, 用户按需加载即可. $\mathcal{T}_{\text{E}}\text{X}$ 中预定义的定理类环境样式包括以下几种:	

thm module 定义样式:

- plain
- background
- leftbar
- fancy

thm library 定义样式:

- shadow
- paris
- tcb
- elegant
- obsidian
- lapsis

$\langle font \rangle$ 用于指定数学公式字体,预定义的字体有: `newtx`, `euler`, `var-euler`, `mtpro2`, `mathpazo`, `ptmx`. 其中 `mtpro2` 为付费字体,需用户自行安装.

5 状态检测

因 $\text{\LaTeX}$ 的选项配置比较庞大，其中涉及到诸多的宏包和命令的加载，在文档编译时，我们可能需要对文档的各种状态进行检测；于是， $\text{\LaTeX}$ 提供了一系列的命令用于检测文档中各个变量以及库的加载情况。

$\text{\textbackslash ztexhyperTF}$	*	$\text{\textbackslash ztexhyperTF}\{\langle true code \rangle\}\{\langle false code \rangle\}$
$\text{\textbackslash ztexfancyTF}$	*	此命令用于检测当前文档中是否开启了超链接功能，如果开启了，那么执行 $\langle true code \rangle$ ，否则执行 $\langle false code \rangle$ ；其余命令的使用方法同理；各个检测命令的基本使用样例如下：
$\text{\textbackslash ztexmarginTF}$	*	
$\text{\textbackslash ztexslideTF}$	*	
$\text{\textbackslash ztexsysfontTF}$	*	
$\text{\textbackslash ztexaliasTF}$	*	
$\text{\textbackslash ztexbibindTF}$	*	
$\text{\textbackslash ztethmllibTF}$	*	

New: 2025-01-15

```

\ztexhyperTF{Hyperref enable.}{Hyperref does NOT enable.}\par例 8
\ztexfancyTF{Fancy lib is loaded.}{Fancy lib is NOT loaded.}\par
\ztexmarginTF{Margin does set.}{Margin does NOT set.}\par
\ztexslideTF{Slide lib is loaded.}{Slide is NOT loaded.}\par
\ztexsysfontTF{System Font config is loaded.}{System Font
config is NOT loaded.}\par
\ztexaliasTF{Math alias is loaded.}{Math alias is NOT loaded.}\par
\ztexbibindTF{Bib index enable.}{Bib index does NOT enable.}\par
\ztethmllibTF{Thm lib is loaded.}{Thm lib is NOT loaded.}

```

Hyperref enable.
Fancy lib is NOT loaded.
Margin does NOT set.
Slide is NOT loaded.
System Font config is NOT loaded.
Math alias is loaded.
Bib index does NOT enable.
Thm lib is loaded.

6 zT_EX 模块

本节对应的所有 module 默认自动加载, 除此之外, 用户还可以通过命令 `\ztexloadmod` 调用自己编写的 module. 目前已有的 module 列表如下:

- `ztex.module.box.tex`
- `ztex.module.item.tex`
- `ztex.module.cmd.tex`
- `ztex.module.page.tex`
- `ztex.module.color.tex`
- `ztex.module.ref.tex`
- `ztex.module.counter.tex`
- `ztex.module.sclist.tex`
- `ztex.module.font.tex`
- `ztex.module.thm.tex`
- `ztex.module.graphics.tex`
- `ztex.module.sect.tex`

用户也可以编写你自己的 module, 不妨假设其名称为 `<moduleA>`; 将此文件命名为 `ztex.module.<moduleA>.tex`, 然后将其放入路径 `<zTEX>/module/` 下, 最后使用 `\ztexloadmod{<moduleA>}` 即可加载此 module. `<moduleA>` 中程序的基本框架如下:

```
\ProvidesExplFile{ztex.module.<moduleA>.tex}
    {2025/04/26}
    {1.0.0}
    {<discription>}

\newcommand\YourCmd{<definition>}
```

例 9

6.1 font 模块

本模块主要用于配置 XeTeX 的字体, 尽管 fontspec 和 unicode-math 已经在很大程度上简化了字体的配置, 但是对于一些用户来说, 仍然会感到困惑. 本模块的目的就是为了简化字体的配置, 让普通的 LaTeX 用户也能够方便的配置字体, 用上自己喜欢的字体.

6.1.1 字体机制

一个很经典的问题: 当调用一个新字体时, 我到底是使用 font name(字体名) 还是 file name(文件名)? fontspec 宏包中记录着此问题的详细解答:

- 当通过 font name(字体名) 调用系统字体时: 诸如 ~/Library/Fonts(MacOS), C:\Windows\Fonts(Windows) 这样的默认搜索路径 (search path), 其下的字体可以直接使用 XeTeX 或 LuaTeX 通过字体名调用. 需要注意的是: 任何系统中, TEXMF 下的字体都可以通过 LuaTeX 直接调用; 对于 XeTeX, Windows 或 Linux 的 TEXMF 路径下的字体能通过字体名直接调用. 通过字体名调用字体有一个好处: fontspec 能 (如果对应的字体文件存在) 自动完成斜体, 加粗等 font face 配置.
- 当通过 file name(文件名) 调用字体时: 此时在 /usr/local/texlive/2025/texmf-dist/fonts/opentype/public 下的字体仅可以通过文件名的形式让 XeTeX 调用, 然而 LuaTeX 则没有这样的限制. 且对于在**默认搜索路径或当前路径**下的字体文件, 在调用时不用指明路径; 此时请尽量给出完整的字体名, 如 lmroman10-regular.otf. (其实也可以仅给出 lmroman10-regular, 但是此时请给出 Path 这个键 – 无论是否赋值, 这样 fontspec 会自动去查找字体文件而非字体名.)

本节中所有命令参数中的 $\langle font \rangle$ 既可以是字体名 (font name), 也可以是字体文件名 (file name), 用户需要根据自己的实际情况选择适合自己的方式.

NOTE: 请尊重字体版权, 不要随意发布和传播商用字体!!!

怎么查看 font name ? T_EXLive 提供了 `otfinfo` 这一命令行工具, 比如我们想要查看 Latin Modern Roman 字体, 其对应的命令为: `otfinfo -i `kpsewhich lmroman10-regular.otf``. 命令的运行结果如下 (Linux 下):

```
> otfinfo -i `kpsewhich lmroman10-regular.otf`
Family:                LM Roman 10
Subfamily:             Regular
Full name:             LMRoman10-Regular
PostScript name:       LMRoman10-Regular
Preferred family:      Latin Modern Roman
Preferred subfamily:   10 Regular
Mac font menu name:    LM Roman 10 Regular
Version:               Version 2.004;PS 2.004;hotconv 1.0.49;makeotf.lib2.0.14853
Unique ID:             2.004;UKWN;LMRoman10-Regular
Trademark:             Please refer to the Copyright section for the font trademark attribution notices.
Copyright:             Copyright 2003, 2009 B. Jackowski and J. M. Nowacki (on behalf of TeX users groups). This work is released under the GUST Font License -- see http://tug.org/fonts/licenses/GUST-FONT-LICENSE.txt for details.
Vendor ID:             UKWN
Permissions:           Unknown (12)
```

X_YTeX 通常使用 `fontconfig` 库查找和调用字体, 因此, 可以用 `fc-list` 命令显示可用的字体. 一个基本的查找示例如下:

```
> fc-list | grep adobe
/usr/share/fonts/adobe-source-code-pro/SourceCodePro-BlackIt.otf:
Source Code Pro,Source Code Pro Black:style=Black Italic,Italic
/usr/share/fonts/adobe-source-code-pro/SourceCodeVF-Upright.otf:
SourceCodeVF:style=Semibold
/usr/share/fonts/adobe-source-code-pro/SourceCodePro-LightIt.otf:
Source Code Pro,Source Code Pro Light:style=Light Italic,Italic
/usr/share/fonts/adobe-source-code-pro/SourceCodeVF-Upright.otf:
SourceCodeVF:style=Medium
/usr/share/fonts/adobe-source-code-pro/SourceCodeVF-Italic.otf:
SourceCodeVF:style=Medium Italic
/usr/share/fonts/adobe-source-code-pro/SourceCodePro-Bold.otf:
Source Code Pro:style=Bold
```

6.1.2 默认字体族

<code>\rmdefault</code>	<code>\rmdefault</code> 初始值: <code>rm</code>
<code>\sfdefault</code>	<code>\sfdefault</code> 初始值: <code>sf</code>
<code>\ttdefault</code>	<code>\ttdefault</code> 初始值: <code>tt</code>

New: 2025-04-26

这三个命令保存了西文字体的默认字体族. 更改这三个默认字体族即可改变文档中的西文字体, 一个基本的使用示例如下 (将文档更改为类 Times 字体风格):

```
\renewcommand{\rmdefault}{ptm}
\renewcommand{\sfdefault}{phv}
\renewcommand{\ttdefault}{pcr}
```

例 10

<code>\CJKrmdefault</code>	<code>\CJKrmdefault</code> 初始值: <code>rm</code>
<code>\CJKsfdefault</code>	<code>\CJKsfdefault</code> 初始值: <code>sf</code>
<code>\CJKttdefault</code>	<code>\CJKttdefault</code> 初始值: <code>tt</code>

New: 2025-04-26

这三个命令和上述西文字体中的三个变量类似, 但其保存了 CJK 字体三个默认字体族的名称.

<code>\familydefault</code>	前者保存了 <code>\textnormal</code> , <code>\normalfont</code> 中西文字体所使用的字体族, 后者保存了对应的 CJK 字体的默认字体族.
<code>\CJKfamilydefault</code>	

New: 2025-04-26

<code>\setmainfont</code>	<code>\setmainfont{}[]</code>
<code>\setsansfont</code>	<code>\setsansfont{}[]</code>
<code>\setmonofont</code>	<code>\setmonofont{}[]</code>

New: 2025-04-26

这三个命令来自 `fontspec` 宏包, 用于设置西文字体的默认字体族 (`\setmainfont` 用于设置正文罗马族的西文字体).

<code>\setCJKmainfont</code>	<code>\setCJKmainfont{}[]</code>
<code>\setCJKsansfont</code>	<code>\setCJKsansfont{}[]</code>
<code>\setCJKmonofont</code>	<code>\setCJKmonofont{}[]</code> 或
	<code>\setCJKmainfont[]{}</code>
	<code>\setCJKsansfont[]{}</code>
	<code>\setCJKmonofont[]{}</code>

New: 2025-04-26

这三个命令来自 `xeCJK` 宏包, 用于设置 CJK 字体的默认字体族 (`\setCJKmainfont` 用于设置正文罗马族的 CJK 字体).

6.1.3 新建字体族

<code>\newfontfamily</code>	<code>\newfontfamily<cmd>{}[]</code>
<code>\setfontfamily</code>	<code>\setfontfamily<cmd>{}[]</code>
<code>\renewfontfamily</code>	<code>\renewfontfamily<cmd>{}[]</code>
<code>\providefontfamily</code>	<code>\providefontfamily<cmd>{}[]</code>

New: 2025-04-26

这系列命令来自 `fontspec` 宏包, `\newfontfamily` 会检查字体族是否存在, 如果不存在则创建一个新的字体族, 如果存在则抛出错误; `\setfontfamily` 无论字体族存在与否, 都会创建一个新的字体族, 如果存在则覆盖原字体族; `\renewfontfamily` 会检查字体族是否存在, 如果存在则覆盖原字体族, 如果不存在则抛出错误; `\providefontfamily` 会检查字体族是否存在, 如果存在则不做任何操作, 如果不存在则创建一个新的字体族.

<code>\newCJKfontfamily</code>	<code>\newCJKfontfamily[<family>]<cmd>{}[]</code>
<code>\setCJKfamilyfont</code>	<code>\setCJKfamilyfont{<family>}{}[]</code>

New: 2025-04-26

这两个命令来自 `xeCJK` 宏包, 用于创建一个新的 CJK 字体族, 作用和上述的 `\newfontfamily` 和 `\setfontfamily` 类似. 事实上, `\newCJKfontfamily` 是 `\setCJKfamilyfont` 和 `\CJKfamily` 的合并, 例如, 下面的两种写法等价:

```
\newCJKfontfamily[song]\songti{SimSun}
\setCJKfamilyfont{song}{SimSun}
\newcommand*\songti{\CJKfamily{song}}
```

例 11

<code>xeCJK/options/AutoFakeBold</code>	<code>AutoFakeBold = {<true false 浮点数>}</code>	初始值: <code>true</code>
<code>xeCJK/options/AutoFakeSlant</code>	<code>AutoFakeSlant = {<true false 浮点数>}</code>	初始值: <code>true</code>

New: 2025-04-26

局部启用或禁用当前字体族的伪粗和伪斜属性, 如果没有在局部给出这些选项, 将使用全局设定. **注意:** 当把 `<AutoFakeBold>` 和 `<AutoFakeSlant>` 设置为浮点数时, 此时将启用伪粗和伪斜; 此种方式和后续的 `<EmboldenFactor>` 和 `<SlantFactor>` 来设置伪粗和伪斜属性是等价的; 如果伪粗和伪斜二者均启用了, 那么后续的粗斜体也将启用此伪属性; 在西文字体的设置下, 以下两种设置等价:

```
\fontspec[AutoFakeBold=1.5]{Charis SIL}
\fontspec[BoldFeatures={FakeBold=1.5}]{Charis SIL}
```

例 12

xeCJK/options/EmboldenFactor	EmboldenFactor = {\浮点数 4}\dots\dots\dots初始值: 4
xeCJK/options/SlantFactor	SlantFactor = {\浮点数 0.167}\dots\dots\dots初始值: 0.167

New: 2025-04-26

全局设置当前字体族的伪粗和伪斜属性, 如果没有在局部给出这些选项, 将使用全局设定. 伪斜因子取值范围为: [-0.99, 0.99].

6.1.4 切换字体

<code>\newfontface</code>	<code>\newfontface{<cmd>}{}[]</code>
New: 2025-04-26	此命令来自 fontspec 宏包, 用于给西文字体创建单一 font face 的字体族, 仅在某一个 font face 对应的指令 (比如仅在 \textit) 下有效果 (此时 \textbf\textit 等组合命令只能得到其中一个轴上的效果).
<code>\fontspec</code>	<code>\fontspec{}[]</code>
<code>\CJKfontspec</code>	<code>\CJKfontspec{}[]</code> 或 <code>\CJKfontspec[]{}</code>
New: 2025-04-26	此二命令, 前者来自 fontspec 宏包, 用于临时切换字体. 后者来自 XeCJK 宏包, 作用和前者类似. 此二命令多用于测试, 普通用户不应该在正文中使用

6.1.5 ℒ_{TEX} 接口

<code>\resetfont</code>	<code>\resetfont[cm lm]</code>
<code>New: 2025-07-14</code>	此命令用于切换回默认的 Computer Modern 或 Latin Modern 字体 (X _Ǝ TEX/LuaTEX 下的默认西文字体), 默认切换到 Computer Modern 字体.
<code>\zfontfamilynew</code>	<code>\zfontfamilynew[⟨lang⟩]{⟨cmd⟩}{⟨key-value⟩}</code>
<code>New: 2025-04-26</code>	当 <code>⟨sysfont⟩=true</code> 时可用 (此时需更换 X _Ǝ TEX 或 LuaTEX 引擎). 此命令用于创建一个新的字体族, 其整合了西文字体族和中日韩字体族设置的接口; 如果对应的字体族已存在, 则它会被覆盖掉 . <code>⟨lang⟩</code> 用于指定生成的字体族对应的语言, 默认为 <code>en/CJK</code> (同时设置中英文, 用户需确保当前字体兼具中英文对应的 glyph), 另有可选值 <code>en, CJK</code> . <code>⟨cmd⟩</code> 用于调用新建立的字体族. <code>⟨key-value⟩</code> 用于指定新字体族的一系列属性, 目前支持的属性有请参见后续说明. 注意: 由此命令生成的字体族无法由 <code>AutoFakeBold</code> , <code>AutoFakeSlant</code> 等选项来设置伪粗和伪斜属性, 因为此命令生成的字体族中已经默认设置了 <code>BoldFont</code> , <code>ItalicFont</code> , <code>SlantedFont</code> 等为原始的 Regular 字体.
<code>ztex/fontcfg/new/name</code>	<code>name = ⟨字体名 文件名⟩..... 初始值: 无</code>
<code>ztex/fontcfg/new/path</code>	<code>path = ⟨字体路径⟩..... 初始值: 默认路径</code> <code>⟨name⟩</code>(必要参数): 用于指定字体的字体名或文件名, 如 Times New Roman 或 times.ttf. 字体设置时和 fontspec 中提供的命令相同, 也支持缩写; 可以使用 <code>*</code> 表示当前字体文件名, 即 <code>⟨name⟩</code> 的值. 用户可以通过命令 <code>fc-list</code> 来查看当前可供 X_ƎTEX 或 LuaTEX 调用的字体, 用法参见本节导言. <code>⟨path⟩</code>: 字体文件的路径, 默认为当前文档目录以及 X_ƎTEX 或 LuaTEX 的默认搜索目录.

ztex/fontcfg/new/feat/ext	ext	= <字体格式>.....	初始值: 无
ztex/fontcfg/new/feat/up	up	= <字体名 文件名>.....	初始值: *
ztex/fontcfg/new/feat/bd	bd	= <字体名 文件名>.....	初始值: *
ztex/fontcfg/new/feat/it	it	= <字体名 文件名>.....	初始值: *
ztex/fontcfg/new/feat/sc	sc	= <字体名 文件名>.....	初始值: *
ztex/fontcfg/new/feat/sl	sl	= <字体名 文件名>.....	初始值: *
ztex/fontcfg/new/feat/bdit	bdit	= <字体名 文件名>.....	初始值: *
ztex/fontcfg/new/feat/bdsl	bdsl	= <字体名 文件名>.....	初始值: *
ztex/fontcfg/new/feat/other	other	= <其他设置>.....	初始值: 空

`<feat>` 用于设置字体的一系列属性, 其中包含的子键有: `<up>`, `<bd>`, `<it>`, `<sl>`, `<sc>`, `<bdit>`, `<bdsl>`, `<other>`, 分别表示 `upright`, `bold`, `italic`, `slant`, `bold italic`, `boldslant` 7 种字体特性 + 额外的字体特性. `<ext>` 用于指定字体文件的后缀 (字体格式), 当 `<name>` 中已经含有了后缀时, 此时 `<ext>` 可以省略也可以再次给出. 更多的字体特性设置可以通过 `<other>` 键进行设置, 详情请参见 `fontspec` 和 `XeCJK` 的宏包文档. **注意:** 字体名和文件名不可在同一个字体声明命令的过程中混用; 当 `<name>` 为字体名时, 请不要设置 `<ext>` 的值, 这会导致无法找到字体.

ztex/./feat/Extension	Extension	= <字体格式>.....	初始值: 无
ztex/./feat/UprightFont	UprightFont	= <字体名 文件名>.....	初始值: *
ztex/./feat/BoldFont	BoldFont	= <字体名 文件名>.....	初始值: *
ztex/./feat/ItalicFont	ItalicFont	= <字体名 文件名>.....	初始值: *
ztex/./feat/SmallCapsFont	SmallCapsFont	= <字体名 文件名>.....	初始值: *
ztex/./feat/SlantedFont	SlantedFont	= <字体名 文件名>.....	初始值: *
ztex/./feat/BoldItalicFont	BoldItalicFont	= <字体名 文件名>.....	初始值: *
ztex/./feat/BoldSlantedFont	BoldSlantedFont	= <字体名 文件名>.....	初始值: *

`<feat>` 中含有字体的一系列属性, `fontspec` 宏包中的原始接接口.

关于 `\zfontnew` 命令的一个简单使用样例如下:

```
%% \zfontset{sysfont}
%% begin preamble
\zfontfamilynew\YaHei{
  name = msyh.ttc,
  path = ./support/Fonts/,
  feat = { ext=.ttc, bd=*bd }
}
\zfontfamilynew[en]\Arial{
  name = arial.ttf,
```

例 13

```
path = ./support/Fonts/,
feat = {Extension=.ttf, ItalicFont=*i}
}
\zfontfamilynew[en]\SourceCodePro{
  name = Source Code Pro,
  feat = { bd=Source Code Pro Bold }
}
%% end preamble
{\YaHei 你好世界 Hello world,\bfseries 你好世界 Hello world.}\par
{\Arial Hello world,\itshape Hello world.}\par
{Hello world,\SourceCodePro Hello world,\bfseries Hello world.}
-----
你好世界 Hello world, 你好世界 Hello world.
Hello world,Hello world.
Hello world,Hello world,Hello world.
```

注意事项:

- 在 fontspec 中, $\langle \text{BoldFont} \rangle$ 和 $\langle \text{ItalicFont} \rangle$ 也是必要参数, 但 ℒ_TE_X 已经帮用户默认配置了这两个选项, 默认为当前 UprightFont 对应的字体.
- 不能在声明一个字体族时混用 font name 和 file name, 否则 fontspec 会因字体无法找到而报错.

<code>\zfontset</code>	<code>\zfontset{⟨key-value⟩}</code>
<code>New: 2024-04-26</code>	此命令用于统一设置整个文档中的西文, 中文以及数学字体.
<code>ztex/font/sysfont</code>	<code>sysfont = ⟨true false⟩</code>初始值: <code>false</code> 此选项用于控制 ℒ _T E _X 是否启用系统字体配置, 默认为 <code>false</code> , 即默认不启用. 当设置 $\langle \text{sysfont} \rangle = \text{true}$ 时, 此时需使用 X _Y ℒ _T E _X 或 Lua _T E _X 引擎编译文档.
<code>ztex/font/doc/lmm</code> <code>ztex/font/doc/newtx</code> <code>ztex/font/doc/ptmx</code>	<code>lmm</code>不可设置值 <code>newtx</code>不可设置值 <code>ptmx</code>不可设置值 这三个选项会同时设置整个文档中的正文字体和数学字体, 目前仅在 pdf _T E _X 下可用. 注意: 如果在设置了此选项的同时也设置了后续的 $\langle \text{text} \rangle$ 或 $\langle \text{math} \rangle$ 选项, 那么此时后续的字体配置会覆盖前面的配置. <code>newtxtext</code> 字体宏包目前并不推荐使用, $\langle \text{newtx} \rangle$ 选项仅作为一个备用设置.

ztex/font/text/cmr
ztex/font/text/times

cmr 不可设置值
times 不可设置值
⟨*cmr*⟩ 即为文档在 pdf_YT_EX 下的默认字体, ⟨*times*⟩ 用于设置文档中的正文字体为 Times 风格.

ztex/font/math/euler
ztex/font/math/newtx
ztex/font/math/mtpro2
ztex/font/math/mathpazo

euler 不可设置值
newtx 不可设置值
mtpro2 不可设置值
mathpazo 不可设置值
⟨*euler*⟩ 用于设置文档中的数学字体为 Euler 风格, 使用 euler 宏包; ⟨*newtx*⟩ 用于设置文档中的数学字体为 NewTx 风格, 使用 newtxmath 宏包; ⟨*mtpro2*⟩ 用于设置文档中的数学字体为 MTPro2 风格, 使用 mtpro2 宏包; ⟨*mathpazo*⟩ 用于设置文档中的数学字体为 Palatino 风格, 使用的宏包为 mathpazo.

\zfontfamilyset

Updated: 2025-09-18

\zfontfamilyset[⟨*lang*⟩]{⟨*key-value*⟩}

此命令用于设置整个文档的字体族, 其整合了西文字体族和中日韩字体族设置的接口. ⟨*lang*⟩ 为一个逗号分割列表, 默认为 “en”, 可选值有: “en, CJK”. 注意: 目前此命令还未进行完整的测试, 可能存在一些潜在的问题. 此命令只能在导言区使用, 基本使用方法如下:

```

\zfontfamilyset
{
  main={Times New Roman}{},
  mono={Latin Modern Mono}{SlantedFont=Latin Modern Mono
Slanted},
  sans={Arial}{},
}
\zfontfamilyset[en, CJK]
{
  % overwrite the above 'sans' setting:
  sans={STSong}{BoldFont=FangSong},
}

```

例 14

6.1.6 杂项

<code>\removeCJKecglue</code>	命令 <code>\removeCJKecglue</code> 用于取消原始的 CJKecglue, 命令 <code>\restoreCJKecglue</code>
<code>\restoreCJKecglue</code>	用于恢复原始的 CJKecglue 设置.
New: 2025-09-03	
<code>\cinzel</code>	<code>\cinzel</code>
Updated: 2025-04-25	本命令用于临时切换 Cinzel 字体 (此时需使用 XeTeX 或 LuaTeX 引擎), 本字体在 $\langle fancy \rangle = \text{true}$ 时, 会自动应用于 chapter 页的字体.
<code>\blacktriangleright</code>	本命令 (符号) 来自 AMSa 字体, $\langle slot \rangle = "49$. 主要用于在 $\langle slide \rangle = \text{true}$ 时对此符号进行 Patch.
Updated: 2024-12-05	

Lorem ipsum dolor sit amet, consectetuer adipiscing elit.	Lorem ipsum dolor sit amet, consectetuer adipiscing elit.
--	--

\ztexlinkclass New: 2025-09-20	\ztexlinkclass{<class>}{<title>}{<link text>} \ztexlinkclass*{<class>}{<title>}{<link material>} 此命令会给 <link text> 或 <link material> 创建超链接, 跳转到由 <class> 和 <title> 确定的章节标题处. \ztexlinkclass* 中的 <link material> 可以包含 \verb 等命令. 备注: 此命令依赖于目录文件, 当目录不存在时, 此命令无效.
\hyper@linkstart New: 2024-12-05	\hyper@linkstart{<context>}{<destination name>} 此命令用于开启一个超链接域, 此域中的内容可以是任意的文本或其它图片对象. 此命令需结合后续的 \hyper@linkend 命令使用, 此二命令结合使用时基本和上述的 \hyper@link 命令基本等效.
\hyper@linkend New: 2024-12-05	用于结束由 \hyper@linkstart 开启的域.
\hyper@linkfile New: 2024-12-05	\hyper@linkfile{<link text>}{<filename>}{<destname>} 此命令用于创建一个超链接, 点击<link text> 即可跳转到对应的 <filename> 文件中的 <destname> 处.
\MakeLinkTarget \MakeLinkTarget* New: 2024-12-05	\MakeLinkTarget[<prefix>]{<counter>} \MakeLinkTarget*{<target>} 此二命令用于在用户层面创建超链接跳转目标, 其中 <prefix> 和 <counter> 可以作为命令 \hyper@link 的参数使用. <counter> 可以为 chapter, section, subsection 等. 针对 \MakeLinkTarget*, 其中 <target> 可以为任意的 Unicode 文本 (但为了兼容性考虑, 请尽量使用 ASCII 字符).
\zsetHcnt New: 2025-05-15	\zsetHcnt{<counter>}{<content>} 此命令用于设置 \theH<counter> 的值为 <content>, 其在制作一些附录相关的内容时是十分有用的.
\NextLinkTarget New: 2024-12-05	\NextLinkTarget{<target>} 此命令设置下一个由 \MakeLinkTarget 或 \refstepcounter 创建的 target. 此命令的作用和 \hypersetup 中的 next-anchor 类似.

`\SetLinkTargetFilter`

New: 2024-12-05

`\SetLinkTargetFilter{<filter>}`

此命令用于给当前文档中所有的 Link Target 添加一个前缀，此命令在合并多个不同的 PDF 时是十分有用的。

`\LinkTargetOn`

`\LinkTargetOn``\LinkTargetOff``\LinkTargetOff`

New: 2024-12-05

此命令常在一个局部中用于取消由 `\MakeLinkTarget` 或 `\refstepcounter` 创建的 Target. 在使用 `\LinkTargetOff` 后，你仍然可以在一个局部里重新启用超链接然后创建对应的 Target, 示例如下：

```

\LinkTargetOff % suppress anchor in internal refstepcounter 例 16
...
\refstepcounter{...}
...
{\LinkTargetOn\MakeLinkTarget*{mytarget}} % create manual  ✓
anchor for future reference
...
\LinkTargetOn

```

6.2.2 标签与引用

\cref New: 2025-04-21	\cref{<labels>} \cref[<options>]{<labels>} $\LaTeX$ 基于 <code>cleveref</code> 和 <code>zref-clever</code> 宏包提供“聪明引用”命令 <code>\cref</code>. 为统一命令, $\LaTeX$ (仅) 将 <code>zref-clever</code> 中的 <code>\zcref</code> 重定义为 <code>\cref</code>, 方便用户的使用. 注意: 尽管二者名称相同但各命令的需要的参数格式是不同的, 其余命令同理, 详情请参考对应的手册. 用户可以通过本文档类的 <code><cref-backend></code> 选项进行后端的设置, 默认后端为 <code>zref-clever</code> 一个简单的设置样例如下: \documentclass [cref-backend=zref-clever] {ztex} 例 17 NOTE: 目前 <code>cleveref</code> 宏包的维护情况不太明朗, 且和 <code>ctex</code> 宏包中的部分设置冲突. 于是, $\LaTeX$ 预设了二者的配置.
\ztexpageall * New: 2025-08-23	\ztexpageall * 此命令返回当前文档的总页数, 与 <code>\pageref{ztex:lastpage}</code> (或 <code>slide</code> 库中的 <code>\pageref{zslide:lastpage}</code>) 命令不同. 使用该命令时一定要检查其返回值 (使用 <code>\tl_if_empty:NTF</code> 命令), 如果返回值为 <code>\c_empty_tl</code>, 则表示该结果不可信; 反之, 则表示该结果正确. \ExplSyntaxOn \tl_if_empty:NTF \ztexpageall {Wrong}{Correct:\ztexpageall} \ExplSyntaxOff 例 18 Correct:442
\RecordProperties New: 2025-09-20	\RecordProperties{<label>}{<clist>} 此命令来自 <code>expl3</code> 的 <code>ltproperties</code> 模块. 它将 <code><clist></code> 中的一系列属性写入“.aux”文件, 它们的标签为 <code><label></code>. <code><clist></code> 中可以包含的值有: <code>abspage</code>, <code>page</code>, <code>pagenum</code>, <code>label</code>, <code>title</code>, <code>target</code>, <code>pagetarget</code>, <code>counter</code>.
\RefProperty * New: 2025-09-20	\RefProperty{<label>} 此命令来自 <code>ltproperties</code> 宏包, 用于获取 <code><label></code> 对应的属性 (值).
\zrefsavepos New: 2025-09-20	\zrefsavepos{<label>} 此命令会保存当前的位置信息, 留待之后使用.

`\zrefgetpos` ★`\zrefgetpos{⟨label⟩}{⟨xpos|ypos⟩}`

`New: 2025-09-20`

此命令用于取得之前由 `\zrefsavepos` 保存的位置信息, 返回浮点数, 其单位为“pt”.

6.2.3 杂项

ztex:titlepage

ztex:lastpage

Updated: 2025-04-25

`\pageref{ztex:titlepage}`

`\pageref{ztex:lastpage}`

引用当前文档的最后一页，可以在制作页眉页脚格式时使用。但对应的超链接跳转也许并不正确，此时应使用 `ztex@lastpage` 这一 anchor。一个基本的使用样例如下：

`\pageref{ztex:titlepage}--\pageref{ztex:lastpage}`

例 19

1-440

ztex@titlepage

ztex@lastpage

Updated: 2025-04-25

`\hyper@link{<context>}{ztex@titlepage}{<link text>}`

`\hyper@link{<context>}{ztex@lastpage}{<link text>}`

上述两 Targets 由命令 `\hyper@anchor` 设置，分别应用于引用当前文档的第一页和最后一页，在 zT_EX 中，标题页的页码为 1。

注意：普通用户不应该直接使用这两个 Targets，此二 Targets 主要提供给模板的开发者，用户应使用位于首页和尾页的 `ztex:titlepage` 和 `ztex:lastpage` 两 label。

6.3.2 页眉页脚

用户可以使用 `ltmarks` 模块提供的 `\NewMarkClass`, `\InsertMark`, `\TopMark`, `\FirstMark`, `\LastMark`, `\IfMarksEqualTF` 等命令; x₂TeX 目前并没有增加新的 `<region>`, 可用的 `<region>` 有 `page`, `previous-page`, `column` 等; 关于 new mark mechanism 的详细使用说明请参考: [ltmarks-doc.pdf](#).

NOTE:

1. 本节的命令由 `sect` 模块提供, 但因为它们与页面设置相关, 故而放置于此;
2. mark 相关接口的实现使用了 `ltmarks` 的内部变量 `\g__mark_classes_seq`.

```
ztex-1st
ztex-left
ztex-right
ztex-right-nonempty
ztex-4th
```

New: 2025-08-21

此为 x₂TeX 文档类提供的一系列的 mark class, x₂TeX 并没有使用内置 `2e-left`, `2e-right`, `2e-right-nonempty` 这几个 class. `ztex-1st` 记录了 part 层级的 marks; `ztex-4th` 在不同的文档类中所记录的标题层级不同, 比如在 `article` 文档类中, 其记录了 `subsubsection` 相关的 marks.

```
\leftmark
\rightmark
```

New: 2025-08-31

x₂TeX 重定义了原始的 `\leftmark` 和 `\rightmark` 命令, 但其意义仍然与原始命令相同. 更推荐用户使用 `\ztexleftmark` 和 `\ztexrightmark` 命令.

```
\ztexleftmark
\ztexrightmark
\robustleftmark
\robustrightmark
```

New: 2025-08-21

在这一系列命令中保存了一组 mark 值. 与原始 L^AT_EX 2_ε 提供的 `\leftmark` 和 `\rightmark` 相比, x₂TeX 的实现更加健壮且实用. `\robustleftmark` 同 `\ztexleftmark`, `\robustrightmark` 同 `\ztexrightmark`.

```
\markright
\markboth
```

New: 2025-08-21

x₂TeX 重定义了原始的 `\markright` 和 `\markboth` 命令, 用户应该尽量避免使用这两个命令.

注意: 因 x₂TeX 没有使用原始的 mark class, 当用户需要手动添加 mark 时, 请尽量不要使用 `\markboth`, `\markright` 命令, 部分情况下可能会导致 x₂TeX 中的部分 mark class 无法同步更新.

<code>\chaptermark</code>	<code>\chaptermark{⟨mark⟩}</code>
<code>\sectionmark</code>	<code>\sectionmark{⟨mark⟩}</code>
<code>\subsectionmark</code>	<code>\subsectionmark{⟨mark⟩}</code>

New: 2025-08-21

用户除了可以使用原始的 `\markright` 和 `\markboth` 命令外, 还可以使用 zT_EX 提供的这两个命令. 它们的用法和作用与 L^AT_EX 2_ε 中一致, 用户还可参见命令: `\zsectmarkinsert`.

<code>\zsect_marker_form:n</code>	<code>\zsect_marker_form:n {⟨format⟩}</code>
-----------------------------------	--

<code>\zsect_marker_form:(o e)</code>

New: 2025-09-24

此命令用于定义页眉 (页脚) 中的 mark 样式, 使用说明请参见后续 `\zsectmarkform` 命令.

<code>\zsectmarkform</code>	<code>\zsectmarkform{⟨format⟩}</code>
-----------------------------	---------------------------------------

New: 2025-09-24

此命令用于定义页眉 (页脚) 中的 mark 样式, 只能在导言区使用. `⟨format⟩` 中的可以使用的宏包括: `\zsecclass`, `\zsecnum`, `\zsecname`, `\zsectocnum` 和 `\zsectocHnum`, 它们的具体含义请参见后续: 节 (6.8.3). 一些简单的使用案例如下:

```

\ExplSyntaxOn
\zsectmarkform
{
  \bfseries
  {\fboxsepOpt\fbox{\zsecnum}}\enskip
  % Method I:
  \exp_args:Ne \ztextlink
    {\zsecclass.\zsectocHnum}
    {\zsecname}
  % Method II: relies on toc file
  \exp_args:Noo \ztextlinkclass{\zsecclass}
    {\zsecname}{\zsecname}
}
\ExplSyntaxOff

```

例 21

`\zsectmarkinsert`

New: 2025-08-21

Updated: 2025-09-24

`\zsectmarkinsert{<class>}{<content>}`

此命令用于更新 mark 列表, 和 L^AT_EX 2_ε 中的 `\markright`, `\markboth` 作用类似. `<class>` 表示标题的类型, 可以选择 `chapter`, `section`, `subsection` 等; `<content>` 为自定义的 mark 内容.

注意: 在 `article` 中, `\zsectmarkinsert{section}{content}` 的作用相当于原始的 `\markboth{content}{}{}` 命令; 但是在 `book` 文档类中, 其作用相当于 `\markright{content}` 命令.



6.3.3 页面水印

`\zpagenmask` `\zpagenmask[⟨key-value⟩]{⟨item⟩}`

`\zpagenmask*`

Updated: 2025-04-25

命令 `\zpagenmask` 用于给当前页面添加水印, `\zpagenmask*` 用于给当前页面及其之后的所有页面添加水印. `⟨item⟩` 可以为一段文字, 也可以为一系列的图片 (需要使用 `\includegraphics` 进行导入).

NOTE: 使用 X_YTeX 时, tikz 中与透明度相关的设置在 `layer=background` 时可能会失效.

`ztex/page/mask/layer`
`ztex/page/mask/label`
`ztex/page/mask/anchor`
`ztex/page/mask/position`

`layer` = `⟨foreground|background⟩` 初始值: `background`
`label` = `{⟨标签⟩}` 初始值: `DEFAULT`
`anchor` = `⟨X⟩⟨Y⟩` 初始值: `c`
`position` = `(⟨dim1⟩, ⟨dim2⟩)` 初始值: `(.5\zpw, .5\zph)`

其中 `⟨position⟩` 以页面的左下角为原点, 向上向右为正方向. `⟨anchor⟩` 中 `XY` 两个字符 (也可以只填入单个字符 `c`): 一个表示水平位置 – `X`, 另一个表示垂直位置 – `Y`. 其中水平位置包括: 左 (`l`)、中 (`c`)、右 (`r`)、内侧 (`i`)、外侧 (`o`); 垂直位置包括: 顶部 (`t`)、中部 (`m`)、底部 (`b`).

注意: `transparent` 宏包仅能在 pdfTeX 或 LuaTeX 引擎下正常工作. 下面是一个简单的示例, 用于给当前页面添加水印:

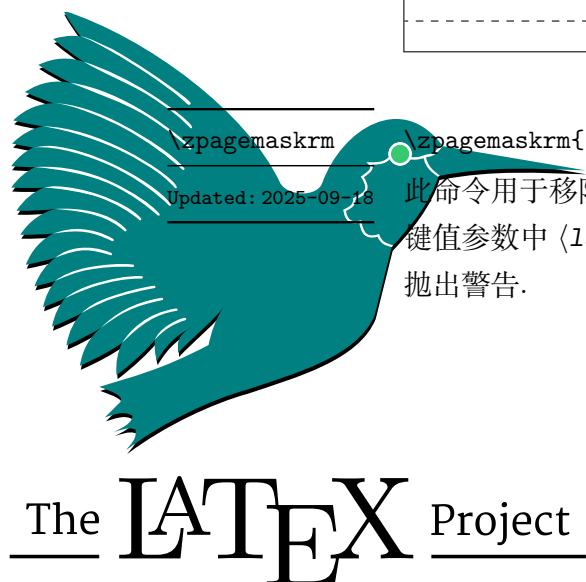
```
% \usepackage{tikzlings}
\zpagenmask[anchor=bl, position={⟨0pt, 0pt⟩}]{
  % \transparent{.5} % available in 'luatex'
  \includegraphics{./support/pics/latex-logo.pdf}
}
\zpagenmask[anchor=tr, position={⟨\zpw, \zph⟩}]{
  \begin{tikzpicture}[scale=2]
    \marmot
  \end{tikzpicture}
}
```

例 22

`\zpagenmaskrm` `\zpagenmaskrm{⟨label⟩}`

Updated: 2025-09-18

此命令用于移除由 `\zpagenmask` 命令添加的页面水印, `⟨label⟩` 即为 `\zpagenmask` 键值参数中 `⟨label⟩` 对应的 (标签). 如果 `⟨label⟩` 对应的水印并不存在, x_YTeX 会抛出警告.



`\zpagenaskrule`

New: 2025-09-18

`\zpagenaskrule{<label1>}{<relation>}{<label2>}`

此命令用于调整不同水印之间的遮盖关系. `<label1>` 和 `<label2>` 为两水印的标签, 它们必须属于同一个 `<layer>`; `<relation>` 可以为 “>(after), <(before), voids” 等: “<” 表示 `<label1>` 对应的水印代码置于 `<label2>` 对应的水印代码之前, 后者相反.

备注: 如果 `<label1>` 或 `<label2>` 中的任何一个不存在, 则此命令不会执行任何操作; 如果希望 `<label1>` 在 `<label2>` 图层上方, 将 `<relation>` 设为 “>” 即可.

`\ztex_page_annotate:nnnnn`

Updated: 2025-04-25

`\ztex_page_annotate:nnnnn {<foreground|background>}{<position>}{<anchor>}{<object>}{<hook range>}`

此命令为 `\zpagenask` 的底层命令, 用户可以依据此命令创建更加具有针对性的水印命令.

6.3.4 杂项

<code>\maketitle</code>
Updated: 2025-04-25

`\maketitle`
`\maketitle*`
`\maketitle[⟨dim⟩]`

LaTeX 对原始的 `\maketitle` 进行了重定义，以适应不同的文档类和页面布局。`\maketitle*` 为 LaTeX 中的 `\maketitle` 的原始定义。`\maketitle[⟨dim⟩]` 会忽略所有的文档类选项或者是页面布局，在新的页面布局中插入 LaTeX 中 `\maketitle` 的原始定义，`⟨dim⟩` 表示新的页面布局的 `margin` 的宽度，默认为空，可以接受一个合法的长度。

<code>\frontmatter</code>
<code>\mainmatter</code>
<code>\appmatter</code>
<code>\backmatter</code>
Updated: 2025-04-25

此系列命令用于分割文档，当加载的 `⟨class⟩` 为 `book` 或 `ctexbook` 时，这系列命令会自动处理页眉页脚，计数器和超链接等相关设置。备注：这些命令中均含有 `\pagestyle` 命令，若用户希望更改页面样式，可以重定义它们。

6.4 color 模块

本模块主要用于文档色彩定制，在本模块中定义了一系列的颜色主题，这系列主题可以应用于文章中的各个元素，包括但不限于章节标题, 定理环境, 超链接跳转, (子) 目录样式.

在颜色指定上, ℒ_TX 实现了一套自己的颜色指定方式 – 指定颜色时可以不必要提前定义. ℒ_TX 将文档中的元素分为如下的 3 类:

- 章节标题类: fancychap;
- 超链接类: link, cite, url;
- 数学环境类: axiom, definition, theorem, lemma, corollary, proposition, remark, proof, exercise, example, solution, problem.

ℒ_TX 的 (部分) 默认配色如下表所示:⁴

Struct	link	url	cite		fancychap		
Color							
MathEnv	axiom	definition	theorem	lemma	corollary	proposition	remark
Color							

表 2: ℒ_TX 文档类默认配色

⁴其余颜色设置请参见后续 \zcolorset 命令的说明.

`\zcolorset`

Updated: 2025-04-25

`\zcolorset{\langle key-value \rangle}`

此命令可以用于设置文档中各种元素的色彩, 此命令仅可在导言区使用. 当 $\langle hyper \rangle = \text{true}$ 时, 可以设置超链接相关的颜色. **备注:** 在指定特定键的色彩时: 一方面可以为普通的预定义色彩名, 如 `red`, `orange` 等; 另一方面, 也可以是 XeTeX 新定义的色彩格式 (后续称此为 XeTeX 色彩格式). 一个具体的设置样例如下:

```
\zcolorset{
  fancychap = red,
  link = {HTML}{d9d9d9},
  theorem = {RGB}{136, 63, 214}
}
```

例 23

`ztex/color/fancychap`

`fancychap = \langle color spec \rangle` 初始值: `ztex@color@fancychap`其中 $\langle color spec \rangle$ 为一个合法的 XeTeX 色彩格式.

`ztex/color/link`

`link = \langle color spec \rangle` 初始值: `purple``ztex/color/cite``cite = \langle color spec \rangle` ..... 初始值: `blue`

`ztex/color/url`

`url = \langle color spec \rangle` 初始值: `ztex@color@royalred`其中 $\langle color spec \rangle$ 为一个合法的 XeTeX 色彩格式.

`ztex/color/axiom`

`axiom = \langle color spec \rangle` 初始值: `ztex@color@axiom``ztex/color/definition``definition = \langle color spec \rangle` ..... 初始值: `ztex@color@definition``ztex/color/theorem``theorem = \langle color spec \rangle` ..... 初始值: `ztex@color@theorem``ztex/color/lemma``lemma = \langle color spec \rangle` ..... 初始值: `ztex@color@lemma``ztex/color/corollary``corollary = \langle color spec \rangle` ..... 初始值: `ztex@color@corollary``ztex/color/proposition``proposition = \langle color spec \rangle` ..... 初始值: `ztex@color@proposition`

`ztex/color/remark`

`remark = \langle color spec \rangle` 初始值: `ztex@color@remark`

其中 $\langle color spec \rangle$ 为一个合法的 XeTeX 色彩格式. 定理类环境的色彩保存于变量 `ztex@color@⟨name⟩` 中, 其中 $\langle name \rangle$ 为对应环境的名称. 不推荐用户使用命令 `\definecolor`, `\colorlet` 直接对这类色彩变量进行重定义, XeTeX 鼓励用户通过 `\zcolorset` 命令进行色彩的重定义.

注意: 后续的 `\zthmcolorset` 仅能用于数学类环境的色彩自定义, 所以如果出现 $\langle link \rangle$, $\langle chapter \rangle$ 等键, 那么此时 XeTeX 会抛出错误; 此时推荐使用 `\zcolorset` 命令进行色彩设置.

ztex/color/proof	proof	=	⟨color spec⟩		初始值: ztex@color@proof
ztex/color/exercise	exercise	=	⟨color spec⟩		初始值: ztex@color@exercise
ztex/color/example	example	=	⟨color spec⟩		初始值: ztex@color@example
ztex/color/solution	solution	=	⟨color spec⟩		初始值: ztex@color@solution
ztex/color/problem	problem	=	⟨color spec⟩		初始值: ztex@color@problem

其中 ⟨color spec⟩ 为一个合法的 x_Y 色彩格式. x_Y 对证明类环境的颜色处理与定理类环境相同, 这里不再说明.

`\ztex_color_set:n` `\ztex_color_set:n {⟨color spec⟩}`

Updated: 2025-04-25

此命令可以自动解析 ⟨color spec⟩, 并以此创建或定义对应的色彩. ⟨color spec⟩ 可以为普通的预定义色彩名, 如 red, orange 等. 亦或者是 HTML, RGB, CMYK 等色彩模型, 但此时的格式略有不同. 此命令仅能在 `\keys_define:nn` 中使用, 新定义的色彩名为: `ztex@color@l_keys_key_str`. 下面是关于这个命令的一个简单应用案例:

`\ExplSyntaxOn`
例 24

```

\keys_define:nn {colorTest}{
  keyA      .tl_set:N      = \l__ztex_keyA_color_tl,
  keyA      .code:n        = { \ztex_color_set:n {#1} },
}
\keys_set:nn {colorTest}{keyA={HTML}{d9d9d9}}
\textcolor{ztex@color@keyA}{This~is~a~test.}
\ExplSyntaxOff

```

This is a test.

6.5 thm 模块

本模块主要用于定理类以及证明类数学环境定制. 本模块提供了丰富的接口以及选项, 与此同时本模块提供了丰富的 Hook, 方便用户直接对环境进行操作.

thm 提供的数学环境主要分为两类:

- 定理类: axiom, definition, theorem, lemma, corollary, proposition, remark;
- 证明类: proof, exercise, example, solution, problem

所以请区分“定理类”和“证明类”两类环境, 以便于正确地使用 thm 提供的各个命令. $\LaTeX$ 的 thm 模块中的部分命令或变量也许没有显式地含有 **theorem** 字样, 但是这些命令或变量仍然是属于“定理类”的; 应用于“证明类”环境的命令或变量均显式地含有 **proof** 字样.

6.5.1 用户接口

 $\backslash$ qedsymbol

Updated: 2024-11-05

 $\backslash$ qedsymbol

此命令用于输出证明环境的结束符号, 默认为 $\square$.

 $\backslash$ zthmlang

Updated: 2025-04-25

 $\backslash$ zthmlang{<lang>}

此命令用于设置定理类环境的语言 (从而会影响到其标题名称), 目前支持 **cn**, **en**, **fr** 三种语言, 仅能在文档的导言区使用.

一个使用样例如下 (此命令仅能在文档的导言区使用, 但为了说明此命令的使用方法, 在本手册中, 此命令的定义被临时改变了):

```
 $\backslash$ begin{theorem}[zthmlang-1]
  This is a chinese zthmlang-1.
 $\backslash$ end{theorem}
 $\backslash$ zthmlang{fr}
 $\backslash$ begin{theorem}[zthmlang-2]
  This is a france zthmlang-2.
 $\backslash$ end{theorem}
 $\backslash$ zthmlang{en}
 $\backslash$ begin{theorem}[zthmlang-3]
  This is a english zthmlang-3.
 $\backslash$ end{theorem}
```

例 25

定理 6.1 (zthmlang-1) This is a chinese zthmlang-1.
Théorème 6.2 (zthmlang-2) This is a france zthmlang-2.
Theorem 6.3 (zthmlang-3) This is a english zthmlang-3.

 $\backslash$ zthmnameset

Updated: 2025-04-25

 $\backslash$ zthmnameset{<lang>}{<key-value>}

此命令用于设置数学环境的名称, 包括“定理类”和“证明类”, 仅能在文档的导言区使用. 预定义的 <lang> 值有: **en**, **cn**, **fr**. 除预定义的这三种语言外, 用户可以使用此命令自行声明 (<lang>), 然后使用命令 $\backslash$ zthmlang{<lang>} 进行切换. **注意:** 此命令需应用于 $\backslash$ zthmlang 命令之前, 否则此命令的相关设置将不会生效; 此命令无法控制由 $\backslash$ zthmenvnew 或 $\backslash$ zthmenvset 创建的环境标题.

下面我们采用键值对的方式对 $\langle \text{key-value} \rangle$ 这一项参数进行描述: `zthmnameset/` 表示它是此 $\langle \text{key-value} \rangle$ 参数的父级命令; 后续为了行文的方便, 我们在描述一个 (父级) 命令之后, 使用 `../` 来表示其缩写形式 (`../` 有时也用于表示任意的键名, 即由用户定义的键名).

注意: 虽然它的设置方法和 `key-value` 这样的数据结构类似, 但是用户不能将 `\keys_define:nn` 这样的命令应用于这类键值对, 而应使用其父级命令 `\zthmnameset` 对其进行设置.

<code>zthmnameset/axiom</code>	<code>axiom</code>	<code>= {<名称>}</code>	初始值: Axiom
<code>zthmnameset/definition</code>	<code>definition</code>	<code>= {<名称>}</code>	初始值: Definition
<code>zthmnameset/theorem</code>	<code>theorem</code>	<code>= {<名称>}</code>	初始值: Theorem
<code>zthmnameset/lemma</code>	<code>lemma</code>	<code>= {<名称>}</code>	初始值: Lemma
<code>zthmnameset/corollary</code>	<code>corollary</code>	<code>= {<名称>}</code>	初始值: Corollary
<code>zthmnameset/proposition</code>	<code>proposition</code>	<code>= {<名称>}</code>	初始值: Proposition
<code>zthmnameset/remark</code>	<code>remark</code>	<code>= {<名称>}</code>	初始值: Remark
<code>zthmnameset/proof</code>	<code>proof</code>	<code>= {<名称>}</code>	初始值: Proof
<code>zthmnameset/exercise</code>	<code>exercise</code>	<code>= {<名称>}</code>	初始值: Exercise
<code>zthmnameset/example</code>	<code>example</code>	<code>= {<名称>}</code>	初始值: Example
<code>zthmnameset/solution</code>	<code>solution</code>	<code>= {<名称>}</code>	初始值: Solution
<code>zthmnameset/problem</code>	<code>problem</code>	<code>= {<名称>}</code>	初始值: Problem

当 $\langle \text{lang} \rangle = \text{en}$ 时, 命令 `\zthmnameset` 中 $\langle \text{key-value} \rangle$ 的设置情况.

<code>../axiom</code>	<code>axiom</code>	<code>= {<名称>}</code>	初始值: Axiome
<code>../definition</code>	<code>definition</code>	<code>= {<名称>}</code>	初始值: Définition
<code>../theorem</code>	<code>theorem</code>	<code>= {<名称>}</code>	初始值: Théorème
<code>../lemma</code>	<code>lemma</code>	<code>= {<名称>}</code>	初始值: Lemme
<code>../corollary</code>	<code>corollary</code>	<code>= {<名称>}</code>	初始值: Corollaire
<code>../proposition</code>	<code>proposition</code>	<code>= {<名称>}</code>	初始值: Proposition
<code>../remark</code>	<code>remark</code>	<code>= {<名称>}</code>	初始值: Remarque
<code>../proof</code>	<code>proof</code>	<code>= {<名称>}</code>	初始值: Preuve
<code>../exercise</code>	<code>exercise</code>	<code>= {<名称>}</code>	初始值: Exercice
<code>../example</code>	<code>example</code>	<code>= {<名称>}</code>	初始值: Exemple
<code>../solution</code>	<code>solution</code>	<code>= {<名称>}</code>	初始值: Solution
<code>../problem</code>	<code>problem</code>	<code>= {<名称>}</code>	初始值: Problème

当 $\langle \text{lang} \rangle = \text{fr}$ 时, 命令 `\zthmnameset` 中 $\langle \text{key-value} \rangle$ 的设置情况.

../axiom	axiom	= {⟨名称⟩}.....	初始值: 公理
../definition	definition	= {⟨名称⟩}.....	初始值: 定义
../theorem	theorem	= {⟨名称⟩}.....	初始值: 定理
../lemma	lemma	= {⟨名称⟩}.....	初始值: 引理
../corollary	corollary	= {⟨名称⟩}.....	初始值: 推论
../proposition	proposition	= {⟨名称⟩}.....	初始值: 命题
../remark	remark	= {⟨名称⟩}.....	初始值: 备注
../proof	proof	= {⟨名称⟩}.....	初始值: 证明
../exercise	exercise	= {⟨名称⟩}.....	初始值: 练习
../example	example	= {⟨名称⟩}.....	初始值: 示例
../solution	solution	= {⟨名称⟩}.....	初始值: 解
../problem	problem	= {⟨名称⟩}.....	初始值: 问题

当 $\langle \text{lang} \rangle = \text{cn}$ 时, 命令 `\zthmnameset` 中 $\langle \text{key-value} \rangle$ 的设置情况.

一个基本的使用案例如下 (此命令仅能在文档的导言区使用, 但为了说明此命令的使用方法, 在本手册中, 此命令的定义被临时改变了):

```
\zthmnameset{cn}{
  theorem=新定理,
  proof=证
}
\zthmlang{cn} % update thm title name
\begin{theorem}[zthmnameset-1]
  This is a theorem zthmnameset-1.
\end{theorem}
\begin{proof}
  This is a proof.
\end{proof}
```

例 26

新定理 6.4 (zthmnameset-1) This is a theorem zthmnameset-1.
证: This is a proof. □

`\zthmenvnew` `\zthmenvnew[⟨type⟩]{⟨key-value⟩}`

New: 2025-09-23

根据第二个参数中的 $\langle \text{key-value} \rangle$ 创建一系列类型为 $\langle \text{type} \rangle$ 的定理环境, 仅可在导言区使用; 如果对应的环境已存在, 则覆盖其原有的定义. $\langle \text{type} \rangle$ 可选 `theorem`, `proof` 两种类型, 默认为 `theorem`. 键值 $\langle \text{key-value} \rangle$ 的语法请参见下面的样例: 例 27.

<code>ztex/thmnew/name</code>	<code>name = <title></code>	初始值: <code><name></code>
<code>ztex/thmnew/color</code>	<code>color = <color></code>	初始值: <code>black</code>
<code>ztex/thmnew/tocsym</code>	<code>tocsym = <code></code>	初始值: <code><name></code>

`<title>` 为新环境对应的名称, 可以省略, 默认以此环境的名称为标题; `<color>` 为合法的 $\LaTeX$ 色彩格式, 可以省略, 默认为 “black”; `<tocsym>` 为该环境在定理目录中的前缀, 若为空或未设置此键, 则默认为此环境的名称. **备注:** 用户也可以使用 `\zthmtocsym` 命令来设置新环境的 `<tocsym>`.

`\zthmenvset` `\zthmenvset{<type>}{<key-value>}`

New: 2025-09-23

此命令和上述的 `\zthmenvnew` 类似, 但此命令会覆盖原有的环境定义, 且可以在导言区或正文中使用.

```

\zthmenvnew{
  Zaxiom,
  Ztheorem={name=Thm, color={HTML}{a0d911}, tocsym=\textbf{ZT}
};},
  Zproposition={name=Prop, color=blue},
}
\zthmenvnew[proof]{
  Zproof={color={cyan}},
  Zexample={name=EXAMPLE, color=red},
  Zsolution={name=Solution},
}
\begin{Zproof}[zthmenvnew-1]
  This is a Zproof zthmenvnew-1.
\end{Zproof}
\begin{Zexample}[zthmenvnew-2]
  This is a Zexample zthmenvnew-2.
\end{Zexample}
\begin{Ztheorem}[zthmenvnew-3]
  This is a Ztheorem zthmenvnew-3
\end{Ztheorem}

```

例 27

Zproof: This is a Zproof zthmenvnew-1.

EXAMPLE: This is a Zexample zthmenvnew-2.

Thm 6.1 (zthmenvnew-3) This is a Ztheorem zthmenvnew-3

 $\backslash$ zthmcnt $\backslash$ zthmcnt{<key-value>}

Updated: 2025-04-25

此命令用于定义数学类环境的计数器, 仅能在导言区使用.

 $\ldots$ /parent parent = <counter> 初始值: section
 $\ldots$ /share share = <true|false> 初始值: false

 <parent> 用于指定定理类环境计数器的父计数器, 默认父计数器为 section; 当父计数器更新时, 此环境的计数器便会重置; <share> 用于控制所有的定理类环境是否共用一个计数器, 默认为 false. **注意:** 若指定所有定理类环境公用计数器, 此时 $\backslash$ cref 对应的共同名称为 “result” 或 “结果”, 具体取决于 $\backslash$ zthmlang 的设置.

 $\backslash$ zthmstyle $\backslash$ zthmstyle{<style>}

Updated: 2025-04-25

 此命令用于设置定理类环境的样式, 仅能在导言区使用. **注意:** 由于技术原因, 当用户需要加载 thm library 时, 必须将命令 $\backslash$ zthmstyle{<style>} 置于 $\backslash$ ztexloadlib{thm} 之前.

 ztex/thm/style/plain plain 不可设置值
 ztex/thm/style/leftbar leftbar 不可设置值
 ztex/thm/style/background background 不可设置值
 ztex/thm/style/fancy fancy 不可设置值

一个基本的使用样例如下 (此命令仅能在文档的导言区使用, 但为了说明此命令的使用方法, 在本手册中, 此命令的定义被临时改变了):

```

\zthmstyle{plain}
\begin{theorem}[zthmstyle-1]
  A `plain' style zthmstyle-1.
\end{theorem}
\zthmstyle{leftbar}
\begin{theorem}[zthmstyle-2]
  A `leftbar' style zthmstyle-2.
\end{theorem}
\zthmstyle{background}
\begin{theorem}[zthmstyle-3]
  A `background' style zthmstyle-3.
\end{theorem}
\zthmstyle{fancy}
\begin{theorem}[zthmstyle-4]

```

例 28

A ‘fancy’ style zthmstyle-4.
`\end{theorem}`

定理 6.5 (zthmstyle-1) A ‘plain’ style zthmstyle-1.

定理 6.6 (zthmstyle-2) A ‘leftbar’ style zthmstyle-2.

定理 6.7 (zthmstyle-3) A ‘background’ style zthmstyle-3.

定理 6.8 (zthmstyle-4) A ‘fancy’ style zthmstyle-4.

<code>\zthmcolorset</code>	<code>\zthmcolorset{⟨key-value⟩}</code>
Updated: 2025-04-25	此命令和 <code>\zcolorset</code> 类似，但其仅用于对数学环境的色彩设置（比如，你不能在此命令中设置 <code>⟨link⟩</code> 对应的色彩），且仅能在导言区使用。此命令仅能用于数学类环境的色彩自定义，如果出现除数学（包括由命令 <code>\zthmnew</code> 所创建的）环境以外色彩设置，那么 ℳT _E X 会抛出错误；
<code>../axiom</code>	<code>axiom</code> = <code>⟨color spec⟩</code> 初始值: <code>ztanblackaxiom</code>
<code>../definition</code>	<code>definition</code> = <code>⟨color spec⟩</code> 初始值: <code>ztanblackdefinition</code>
<code>../theorem</code>	<code>theorem</code> = <code>⟨color spec⟩</code> 初始值: <code>ztanblacktheorem</code>
<code>../lemma</code>	<code>lemma</code> = <code>⟨color spec⟩</code> 初始值: <code>ztanblacklemma</code>
<code>../corollary</code>	<code>corollary</code> = <code>⟨color spec⟩</code> 初始值: <code>ztanblackcorollary</code>
<code>../proposition</code>	<code>proposition</code> = <code>⟨color spec⟩</code> 初始值: <code>ztanblackproposition</code>
<code>../remark</code>	<code>remark</code> = <code>⟨color spec⟩</code> 初始值: <code>ztanblackremark</code>
<code>../proof</code>	...
<code>../exercise</code>	<code>⟨color spec⟩</code> 为一个合法的 ℳT _E X 色彩格式。 “proof, exercise, ..., problem”
<code>../example</code>	的默认值均为 “black”。
<code>../solution</code>	
<code>../problem</code>	

6.5.2 定理目录

本节的命令依赖于 zT_EX 的 zsect 模块, 欲使用本节的命令, 请务必加载 sect 模块. 在加载 sect 模块后, 定理目录样式的定制请参考 sect 模块的文档.

\zthmtoc

Updated: 2026-02-12

\zthmtoc[⟨keyval⟩]

此命令用于打印定理类环境对应的目录, 可选参数 ⟨keyval⟩ 用于自定义定理目录的样式, 配置方法请参见 节 (6.8.4) 的文档说明, 默认为空. **备注:** 定理目录的默认层级为 “subsection”, 证明类环境不会计入定理目录; 此命令依赖于 sect 模块. 该命令的一个简单使用样例如下:

\zthmtoc[		例 29
page.format+ = \texttt,		
hyper.title = true,		
]		
\begin{proposition}[zthmtoc-1]proposition zthmtoc-1		✓
\end{proposition}		
\begin{lemma}[zthmtoc-2]lemma zthmtoc-2		\end{lemma}
\begin{corollary}[zthmtoc-3]corollary zthmtoc-3		\end{corollary}
T	定理 6.1 (zthmlang-1)	42
T	Théorème 6.2 (zthmlang-2)	42
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P	命题 7.1 (thmstyle-obsidian)	209
命题 6.1 (zthmtoc-1) proposition zthmtoc-1		
引理 6.1 (zthmtoc-2) lemma zthmtoc-2		
推论 6.1 (zthmtoc-3) corollary zthmtoc-3		

`\zthmtocadd` `\zthmtocadd[⟨class⟩]{⟨key-value⟩}`

Updated: 2025-04-25

此命令用于向定理类环境目录中添加条目, `⟨class⟩` 表示该条目在目录中的层级, 默认为 “section”. 其它的可选值有: `chapter`, `subsection` 等. **备注:** 此命令依赖于 `sect` 模块.

`../name` `name = {⟨条目名称⟩}.....` 初始值: 无

目前的键仅有 `name`, 后续可能有变动.

一个简单的使用样例如下:

`\zthmtocadd[section]{name=New:Added Thm ITEM}`
例 30

`\zthmtocstop` `\zthmtocstop`

Updated: 2025-09-04

此命令用于停止向定理类环境目录中添加条目, 用户也可以使用 `\ztocstop[lom]` 替代此命令. **备注:** 此命令依赖于 `sect` 模块.

`\zthmtoclevel` `\zthmtoclevel{⟨class⟩}`

Updated: 2025-04-25

此命令用于设置定理目录中各条目的层级, 仅能在导言区使用, `⟨class⟩` 可以为 “chapter, section, subsection, ...” 等. **备注:** 目前暂不支持将不同的定理类环境名设置为不同的层级.

`\zthmtocprefix` `\zthmtocprefix{⟨prefix⟩}`

Updated: 2025-04-25

此命令用于所有定理类环境目录中所有条目的共同前缀, 默认为空.

<code>\zthmtocsym</code>	<code>\zthmtocsym{⟨key-value⟩}</code>
Updated: 2025-04-25	此命令用于分别设置所有定理类环境名在目录中的前缀, 仅能在导言区使用.
<code>../axiom</code>	<code>axiom</code> = ⟨前缀⟩ 初始值: A\
<code>../definition</code>	<code>definition</code> = ⟨前缀⟩ 初始值: D\
<code>../theorem</code>	<code>theorem</code> = ⟨前缀⟩ 初始值: T\
<code>../lemma</code>	<code>lemma</code> = ⟨前缀⟩ 初始值: L\
<code>../corollary</code>	<code>corollary</code> = ⟨前缀⟩ 初始值: C\
<code>../proposition</code>	<code>proposition</code> = ⟨前缀⟩ 初始值: P\
<code>../remark</code>	<code>remark</code> = ⟨前缀⟩ 初始值: R\

其中⟨前缀⟩为任意合法的 $\text{\LaTeX}$ 代码.
此命令的大致使用方法如下:

```
\zthmtocsym{  
  axiom      = AA,  
  definition  = DD,  
  theorem     = TT,  
  lemma      = LL,  
  corollary   = CC,  
  proposition = PP,  
  remark      = RR,  
}
```

例 31

<code>\zthmtocsymrm</code>	此命令用于清除所有由命令 <code>\zthmtocsym</code> 添加在目录中的前缀, 仅能在导言区使用. 注意: 此命令不能清除由 <code>\zthmtocprefix</code> 指定的前缀.
Updated: 2025-04-25	

6.5.3 高级接口

<code>\zthmremoveCJKecglue</code> <code>\zthmrestoreCJKecglue</code> New: 2025-09-03 	命令 <code>\zthmremoveCJKecglue</code> 用于取消原始的 CJKecglue, 命令 <code>\zthmrestoreCJKecglue</code> 用于恢复原始的 CJKecglue 设置.
<code>\zthmnumber</code> $\star$ Updated: 2024-11-05 	此命令表示对应环境的编号, 类似于 amsthm 中的 <code>\thmnumber</code>. 用户不应在除 <code>\zthmttitleformat</code> 外的任何地方使用, 在命令 <code>\zthmttitleformat</code> 之外, 此命令输出的内容无任何实际意义.
<code>\zthmname</code> $\star$ Updated: 2024-11-05 	此命令表示对应环境的名称, 类似于 amsthm 中的 <code>\thmname</code>. 用户不应在除 <code>\zthmttitleformat</code> 外的任何地方使用, 在命令 <code>\zthmttitleformat</code> 之外, 此命令输出的内容无任何实际意义.
<code>\zthmnote</code> $\star$ Updated: 2024-12-05 	<code>\zthmnote{\langle prefix \rangle}{\langle suffix \rangle}</code> 此命令表示对应环境的注释, 类似于 amsthm 中的 <code>\thmnote</code>. 用户不应在除 <code>\zthmttitleformat</code> 外的任何地方使用, 在命令 <code>\zthmttitleformat</code> 之外, 此命令输出的内容无任何实际意义.
<code>\thm@tmp@name</code> $\star$ Updated: 2025-04-25 	此命令用于临时保存定理类环境的名称, 用户可以在自定义定理类环境样式时使用. 注意: 此命令和前述的 <code>\zthmname</code> 不同, 因 <code>\thm@tmp@name</code> 只能取值于合法的定理类环境名称集合, 而 <code>\zthmname</code> 是 <code>\thm@tmp@name</code> 的格式化版本, 可能包含 <code>\bfseries</code>, <code>\sffamily</code> 等格式化命令.
<code>\thm@tmp@color</code> $\star$ <code>\thmproof@tmp@color</code> $\star$ Updated: 2025-04-25 	此二命令用于临时保存定理类环境和证明类环境的色彩, 用于在 <code>\zthmttitleformat</code> 中进行色彩切换. 注意: 普通用户在使用这两个命令时, 请将其置于 <code>\makeatletter</code> 和 <code>\makeatother</code> 之间.
<code>\zthmttitle</code> $\star$ <code>\zthmttitle*</code> $\star$ Updated: 2024-11-05 	<code>\zthmttitle</code> 命令为定理类环境纯文本标题, 包含 <code>\zthmnumber</code>, <code>\zthmname</code>, <code>\zthmnote</code> 三部分以及一些其它文本. <code>\zthmttitle*</code> 为 <code>\zthmttitle</code> 的格式化版本 (取消了 CJKecglue, 且可能包含 <code>\bfseries</code>, <code>\sffamily</code> 等文本格式化命令); 用户在自定义定理类环境样式时应优先使用 <code>\zthmttitle*</code>, 此命令生成的定理类环境标题才能被 <code>\zthmttitleformat</code> 控制. 此二命令中文本的具体格式可以使用后续的 <code>\zthmttitleformat</code> 命令进行修改.

`\zthmtitleswitch`
`\zthmtitleswitch*`

Updated: 2025-04-25

命令 `\zthmtitleswitch` 用于隐藏定理类环境的标题, 命令 `\zthmtitleswitch*` 用于显示标题; 在自定义环境样式时比较有用. 用户不应该在正文中对此命令进行直接的调用.

一个基本的使用案例如下 (此命令仅能在文档的导言区使用, 但为了说明此命令的使用方法, 在本手册中, 此命令的定义被临时改变了):

```
\begin{theorem}[zthmtitleswitch-1]
  A theorem zthmtitleswitch-1.
\end{theorem}
\zthmstylenew{
  ZZZ={begin=, end=, option=\zthmtitleswitch},
}
\zthmstyle{ZZZ}
\begin{theorem}[zthmtitleswitch-2]
  A theorem zthmtitleswitch-2.
\end{theorem}
```

例 32

定理 6.9 (`zthmtitleswitch-1`) A theorem zthmtitleswitch-1.
 A theorem zthmtitleswitch-2.

关于命令 `\zthmstyle` 的使用可以参见下面的说明.

`\zthmtitleformat`
`\zthmtitleformat*`

Updated: 2025-04-25

`\zthmtitleformat[⟨type⟩]{⟨format⟩}`

此命令用于修改类型为 `⟨type⟩` 的数学类环境的标题格式 (即命令 `\zthmtitle*` 中的内容), 仅能在导言区使用. `⟨type⟩` 可选值有 `theorem`, `proof`, 默认值为 `theorem`. 命令 `\zthmtitleformat` 仅应用于之后的第一个 (类型为 `⟨type⟩` 的) 数学类环境标题样式, 而 `\zthmtitleformat*` 则应用于之后的所有 (类型为 `⟨type⟩` 的) 数学类环境. 注意: 如果 `⟨type⟩` 为 `proof`, 那么在 `⟨format⟩` 中仅有 `\zthmname` 和 `\thmproof@tmp@color` 可用.

注意: 此命令默认取消定理类或证明类环境标题中的 `CJKecglue`, 若用户需要保留这些 `glue`, 请在格式化代码最前面加上: “`\zthmrestoreCJKecglue`”.

此命令的一个简单使用案例如下 (此命令仅能在文档的导言区使用, 但为了说明此命令的使用方法, 在本手册中, 此命令的定义被临时改变了):

```
\zthmcolorset{proof=blue!50}
\makeatletter
\zthmtitleformat{\bfseries\color{\thm@tmp@color}\zthmname
\zthmnote{\{}{\}}\zthmnumber\}
```

例 33

```

\zthmttitleformat[proof]{\bfseries\color{\thmproof@tmp@color}}[: ↙
\zthmname :]\_}
\makeatother
\begin{theorem}[zthmttitleformat-1]
  A theorem zthmttitleformat-1.
\end{theorem}
\begin{proof}
  This is a proof.
\end{proof}
-----
定理{zthmttitleformat-1}6.11 A theorem zthmttitleformat-1.
[:证明:] This is a proof. □

```

此外, 还可以参见命令 `\zthmnotemptyTF` 中的使用示例.

`\zthmnotemptyTF` ☆

Updated: 2025-04-29

`\zthmnotemptyTF{⟨true code⟩}{⟨false code⟩}`

此命令用于判断 `\zthmnote` 是否为空, 如果为空则执行 `⟨true code⟩`, 否则执行 `⟨false code⟩`. 这个命令在自定义 `\zthmttitle` 时很有用.

一个使用样例 (zT_EX 内置的 `obsidian` 定理样式对应的大致格式, 具体效果可以参见: 节 (7.5)):

```

\zthmttitleformat*{\bfseries
\zthmname\_ \zthmnumber
\zthmnotemptyTF{}{\}
\zthmnote{}{}}

```

例 34

`\zthmstylenew`

Updated: 2025-04-25

`\zthmstylenew{⟨key-value⟩}`

此命令用于定义新的定理类环境样式, 仅能在导言区使用.

ztex/./begin	begin	= $\langle code \rangle$	初始值: 无
ztex/./end	end	= $\langle code \rangle$	初始值: 无
ztex/./option	option	= $\langle code \rangle$	初始值: 无
ztex/./preamble	preamble	= $\langle code \rangle$	初始值: 无

其中 $\langle code \rangle$ 为任意合法的 L^AT_EX 代码, 这些代码会被置于对应定理类环境的样式代码中. $\langle begin \rangle$ 和 $\langle end \rangle$ 即为这个新样式对应环境的开头和结尾; $\langle option \rangle$ 中的代码在 $\langle begin \rangle$ 之后, 也在环境的开头, 常用于放置一些控制代码; $\langle preamble \rangle$ 中的代码会被 zT_EX 置于文档的导言区, 常用于放置一些用于定理类环境标题格式化的代码.

当用户声明对应的 $\langle style \rangle$ 后, 可以在导言区使用命令: `\zthmstyle{ $\langle style \rangle$ }` 进行加载.

此命令的基本调用格式如下:

`\zthmstylenew`
例 35

```

{
   $\langle style A \rangle$  =
  {
    begin= $\langle begin code 1 \rangle$ ,
    end= $\langle end code 1 \rangle$ ,
    option= $\langle option code 1 \rangle$ ,
    preamble= $\langle preamble code 1 \rangle$ 
  },
   $\langle style B \rangle$  =
  {
    begin= $\langle begin code 2 \rangle$ ,
    end= $\langle end code 2 \rangle$ ,
    option= $\langle option code 2 \rangle$ ,
    preamble= $\langle preamble code 2 \rangle$ 
  },
  ...
}
```

6.5.4 环境钩子

```
ztex/thm-theorem/before
ztex/thm-theorem/begin
ztex/thm-theorem/end
ztex/thm-theorem/after
```

New: 2025-05-05

这四个通用的钩子会将代码添加到所有的 **定理类** 环境中, 更一般的钩子机制请参见后续的 `\zthmhook` 或 `\zthmproofhook` 命令.

```
ztex/thm-proof/before
ztex/thm-proof/begin
ztex/thm-proof/end
ztex/thm-proof/after
```

New: 2025-05-05

这四个通用的钩子会将代码添加到所有的 **证明类** 环境中, 更一般的钩子机制请参见后续的 `\zthmhook` 或 `\zthmproofhook` 命令.

```
ztex/thm-theorem/titleformat
ztex/thm-proof/titleformat
```

New: 2025-05-05

这两个钩子主要用于修改定理 (证明) 类环境的标题样式, 前述的 `\zthmtitleformat` 命令基于这里的 `ztex/thm-theorem/titleformat` 钩子.

```
\zthmhook
```

```
\zthmhook[⟨name⟩]{⟨key-value⟩}
```

```
\zthmhook*
```

```
\zthmhook*[⟨name⟩]{⟨key-value⟩}
```

Updated: 2025-04-25

此命令用于给已有的 (名称为 `⟨name⟩` 的) 定理类环境 Hook 中添加代码, `⟨name⟩` 的默认值为 `theorem`. 已有的 Hook: `⟨ztex/thm/before⟩`, `⟨ztex/thm/begin⟩`, `⟨ztex/thm/end⟩`, `⟨ztex/thm/after⟩`. `\zthmhook` 只应用于下一个定理类环境, `\zthmhook*` 会应用于接下来的所有定理类环境. (`wrapper begin`) 和 (`thm-title`) 相关的钩子请参见 `\zthmbefore` 和 `\zthmtitlebefore` 命令. 各个 Hook 的位置分布如下:

```
(ztex/thm/before) --> (wrapper begin) --> (thm-title)
--> (ztex/thm/begin) -->
      (thm-content)
--> (ztex/thm/end) -->
(wrapper end) --> (ztex/thm/after)
```

这两个命令不支持手动设置 `⟨label⟩`, 针对于 `\zthmhook*`, L^AT_EX 会自动设置 `⟨label⟩`, 其格式为 `thm-hook.⟨Hook Index⟩`.

<code>../before</code>	<code>before = $\langle code \rangle$</code> 初始值: 无
<code>../begin</code>	<code>begin = $\langle code \rangle$</code> 初始值: 无
<code>../end</code>	<code>end = $\langle code \rangle$</code> 初始值: 无
<code>../after</code>	<code>after = $\langle code \rangle$</code> 初始值: 无

其中 $\langle code \rangle$ 为合法的 L^AT_EX 代码片段.

一个简单的使用案例如下:

<pre> \begin{theorem}[zthmhook-1] This is a theorem zthmhook-1. \end{theorem} \zthmhook{before=ZZa_, begin=ZZb_,} \begin{theorem}[zthmhook-2] This is a theorem zthmhook-2. \end{theorem} </pre>	例 36
定理 6.12 (zthmhook-1) This is a theorem zthmhook-1. ZZa 定理 6.13 (zthmhook-2) ZZb This is a theorem zthmhook-2.	

`\zthmproofhook` `\zthmproofhook[ $\langle name \rangle$ ]{ $\langle key-value \rangle$ }`

`\zthmproofhook*` `\zthmproofhook*[ $\langle name \rangle$ ]{ $\langle key-value \rangle$ }`

Updated: 2025-04-25

此命令用于给已有的 (名称为 $\langle name \rangle$ 的) 证明类环境 Hook 中添加代码, $\langle name \rangle$ 的默认值为 `proof`. 已有的 Hook: $\langle ztex/proof/before \rangle$, $\langle ztex/proof/begin \rangle$, $\langle ztex/proof/end \rangle$, $\langle ztex/proof/after \rangle$. `\zthmproofhook` 只应用于下一个证明类环境, `\zthmproofhook*` 会应用于接下来的所有证明类环境. 各个 Hook 的位置分布如下:

<pre> (ztex/proof/before) --> (proof-title) --> (ztex/proof/begin) --> (proof-content) --> (ztex/proof/end) --> (env icon) --> (ztex/proof/after) </pre>
--

和 `\zthmhook`, `\zthmhook*` 类似, 此二命令会自动设置对应的 $\langle label \rangle$, 无需用户手动指定.

<code>../before</code>	<code>before = <code></code>	初始值:	无
<code>../begin</code>	<code>begin = <code></code>	初始值:	无
<code>../end</code>	<code>end = <code></code>	初始值:	无
<code>../after</code>	<code>after = <code></code>	初始值:	无

其中 $\langle code \rangle$ 为合法的 L^AT_EX 代码片段.

一个简单的使用样例如下:

例 37

```

\zthmproofhook*[solution]{
  before=\noindent\textbf{\color{red}BEFORE},
  begin=\textbf{\color{red}BEGIN},
  end=\textbf{\color{red}END},
  after=\textbf{\color{red}AFTER},
}

\begin{proof}
  This is a proof.
\end{proof}

\begin{solution}
  This is solution I.
\end{solution}

\begin{solution}
  This is solution II.
\end{solution}

```

证明: This is a proof. □

BEFORE解: BEGINThis is solution I. **END**

AFTER

BEFORE解: BEGINThis is solution II. **END**

AFTER

`\zthmbefore`

Updated: 2025-04-25

`\zthmbefore[<type>]{<code>}`

此命令用于把 $\langle code \rangle$ 置于每个类别为 $\langle type \rangle$ 的数学环境 (如果 $\langle type \rangle$ 为 theorem, 也就是命令 `\_ztex_thm_warp_start:nnn`; 如果 $\langle type \rangle$ 为 proof, 那么就是 `\_ztex_thm_proof_title:`) 之前. $\langle type \rangle$ 的可选值有: theorem, proof, 默认值为 theorem. $\langle code \rangle$ 默认为 `\par`, 用户可以把 $\langle code \rangle$ 置为空, 或设置为 `\noindent` 以取消段落缩进.

一个简单的使用样例如下:

例 38

```

\zthmbefore{}
Inline item:%
\begin{theorem}[zthmbefore-1]
  This is a theorem.%
\end{theorem}%
\begin{proposition}[zthmbefore-2]
  This is proposition I.
\end{proposition}
\begin{proof}
  This is a proof.
\end{proof}

```

Inline item: **定理 6.14** (zthmbefore-1) This is a theorem. **命题 6.2** (zthmbefore-2) This is proposition I.
证明: This is a proof. □

`\zthmttitlebefore`

Updated: 2025-04-25

`\zthmttitlebefore[⟨type⟩]{⟨code⟩}`

此命令用于把 `⟨code⟩` 置于每个类型为 `⟨type⟩` 的数学环境标题之前. `⟨type⟩` 的可选值有: `theorem`, `proof`, 默认值为 `theorem`. `⟨code⟩` 默认为 `\noindent`, 用户可以把 `⟨code⟩` 置为空以保留段落缩进.

一个简单的使用样例如下:

例 39

```

\zthmttitlebefore[proof]{[PRF-LIKE]}
\begin{solution}
  This is solution zthmttitlebefore.
\end{solution}

```

BEFORE[PRF-LIKE]**解: BEGIN**This is solution zthmttitlebefore. **END AFTER**

6.6 box 模块

本模块封装的命令主要涵盖以下功能：跨页盒子、盒子的线性变换以及内容对齐。其中，盒子的变换与对齐命令依赖于 `ztool` 宏包，跨页盒子的功能则基于 `framed` 与 `framedmulticol` 宏包实现。`box` 模块仅对 `framed` 宏包进行了基础封装，如需更复杂的使用方式，请参考该宏包的官方文档。

NOTE: `framed` 宏包在实际使用中可能会遇到一些问题，比如浮动体、页脚命令、边注命令失效、颜色泄露（参考 `colorframed` 宏包）；而且它无法正确处理分页多栏文本，因此和 `multicol` 等宏包不兼容。这种情况下，可以考虑用本宏集已经加载的 `framedmulticol` 或 `mdframed` 宏包来替代（可参见 Cu_STeX 中的 `Framed` 环境）。本宏集不打算将 `framed` 宏包替换为 `mdframed` 宏包，后续 `framed` 也许会被移除，但 `mdframed` 宏包绝不会进入本宏集。

<code>\getwd</code>	<code>\getwd⟨dim⟩{⟨content⟩}</code>
<code>\getht</code>	<code>\getwd*⟨dim⟩{⟨content⟩}</code>
<code>\getdp</code>	<code>\getht⟨dim⟩{⟨content⟩}</code>
	<code>\getht*⟨dim⟩{⟨content⟩}</code>
New: 2025-07-10	<code>\getdp⟨dim⟩{⟨content⟩}</code>
	<code>\getdp*⟨dim⟩{⟨content⟩}</code>

此系列命令用于获取盒子的尺寸信息，`⟨dim⟩` 为一个 `dim` 寄存器，可以由 `\newdimen` 或 `\newlength` 命令进行声明；带有 “*” 命令的赋值是全局的。

```

\newlength\lenA
\newlength\lenB
\newlength\lenC
\getwd\lenA{XyX} \getht\lenB{XyX} \getdp\lenC{XyX}
\underline{\the\lenA}, \underline{\the\lenB}, \underline{\the\lenC}.
-----
22.20659pt, 7.47885pt, 2.24474pt.

```

例 40

<code>\zraise</code>	<code>\zraise{⟨dim⟩}{⟨content⟩}</code>
<code>\zlower</code>	<code>\zlower{⟨dim⟩}{⟨content⟩}</code>

New: 2025-07-10

这系列命令与原始的 `\raise`, `\lower` 命令类似，但 `\zraise`, `\zlower` 中的 `⟨content⟩` 不必是一个盒子。

```

{\setlength{\fboxsep}{0pt}
raise: \fbox{XXX}\zraise{.5em}{\fbox{XXX}},
lower: \fbox{XXX}\zlower{.5em}{\fbox{XXX}},

```

例 41

```
}
-----
raise: XXXXXXX, lower: XXXXXXX
```

<code>\wscale</code>	<code>\wscale{<dim>}{<content>}</code>
<code>\hscale</code>	<code>\wscale*{<dim>}{<content>}</code>
	<code>\hscale{<dim>}{<content>}</code>
New: 2025-07-10	<code>\hscale*{<dim>}{<content>}</code>

这系列的命令用于盒子的缩放, 当给定的 $\langle dim \rangle$ 大于该 $\langle content \rangle$ 自的 $\langle dim \rangle$ 时, $\langle content \rangle$ 会被原样输出; `\wscale` 调整盒子的宽度, `\hscale` 用于调整盒子的高度; 带有 “*” 的命令仅对盒子的单个维度进行调整, 另一个维度保持不变. 若用户需使用更加复杂的变换, 可以参考后续 `\ztoolboxaffine` 命令. **注意:** 这系列的命令不依赖于 `graphicx` 宏包; 这系列命令不会对盒子的深度进行调整.

```
{\setlength{\fboxsep}{0pt}
  w set:\fbox{XXX}\wscale{1em}{\fbox{XXX}},
  w scale:\fbox{XXX}\wscale*{1em}{\fbox{XXX}}\par
  h scale:\fbox{XXX}\hscale{1em}{\fbox{XXX}},
  h scale:\fbox{XXX}\hscale*{1em}{\fbox{XXX}},
  h scale:\fbox{XXX}\hscale*{.5em}{\fbox{XXX}}\par
}
```

例 42

```
w set:XXXXxxx, w scale:XXXXxxx
h scale:XXXXXXX, h scale:XXXXXXXX, h scale:XXXXxxx
```

<code>\zrotate</code>	<code>\zrotate{<angle>}{<content>}</code>
New: 2025-07-11	

此命令用于旋转盒子, 其并不依赖于 `graphicx` 宏包. 若用户需使用更加复杂的变换, 可以参考后续 `\ztoolboxaffine` 命令.

```
{\setlength{\fboxsep}{0pt}
  \fbox{X}\fbox{\zrotate{90}{X}}\fbox{X}
}
```

例 43

```
XXXX
```

<code>\hidetext</code>	<code>\hidetext[<keyval>]{<content>}</code>
New: 2025-07-10	

此命令用于将 $\langle content \rangle$ 替换为对应的 “方框”, 从而实现文字的隐藏; $\langle keyval \rangle$ 用于设置 “方框” 的样式, 可选值请参见下述说明:

ztex/box/hidetext/map	map	= $\langle \text{tl str} \rangle$	初始值: <code>tl</code>
ztex/box/hidetext/fill	fill	= $\langle \text{color} \rangle$	初始值: <code>black</code>
ztex/box/hidetext/frame	frame	= $\langle \text{color} \rangle$	初始值: <code>black</code>
ztex/box/hidetext/killdp	killdp	= $\langle \text{true false} \rangle$	初始值: <code>false</code>
ztex/box/hidetext/seperator	seperator	= $\langle \text{code} \rangle$	初始值: <code>\-</code>
ztex/box/hidetext/cmd	cmd	= $\langle \text{cmd} \rangle$	初始值: <code>无</code>

$\langle \text{map} \rangle$ 用于指定遍历的方式; $\langle \text{fill} \rangle$ 用于指定填充颜色; $\langle \text{frame} \rangle$ 用于指定边框颜色 (暂时不可用), 用户可以通过指定 `\fboxrule` 来设置 `\fbox` 的边框宽度; $\langle \text{killdp} \rangle$ 用于控制是否忽略盒子的深度 (这样一来, 所有“方框”的底部就对齐了); $\langle \text{seperator} \rangle$ 用于指定“方框”的分割元素, 默认为“`\-`”; $\langle \text{cmd} \rangle$ 用于自定义“方框”格式.

例 44

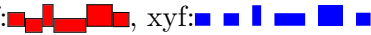
```

{\setlength{\fboxsep}{0pt}
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut
purus elit, vestibulum%
ut, placerat ac, adipiscing vitae, felis. Curabitur dictum
gravida mauris.

xyf:\hidetext[cmd=\fbox{#1}, fill=red]{xyf{xy}{xf}o},
xyf:\hidetext[killdp, fill=blue, separator=\hskip5pt
\relax]{xyf{xy}{xf}o}

\hidetext[map=str]{Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Ut purus elit, vestibulum%
ut, placerat ac, adipiscing vitae, felis. Curabitur dictum
gravida mauris.}
}

```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit,
 vestibulumut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida
 mauris.
 xyf: , xyf: 
 Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum
 ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris.
 ut.

framed

此环境来自 framed 宏包, 用于排版可跨页的带框盒子。

 New: 2025-07-10

`\begin{framed}`

例 45

劳仑衣普桑, 认至将指点效则机, 最你更枝。想极整月正进好志次回总般, 段然取向使张规军证回, 世市总李率英茄持伴。

`\end{framed}`

劳仑衣普桑, 认至将指点效则机, 最你更枝。想极整月正进好志次回总般, 段然取向使张规军证回, 世市总李率英茄持伴。

`\ztextframe`

`\ztextframe[⟨keyval⟩]`

`\ztextframeend`

`\ztextframeend`

 Updated: 2025-04-25

这两个命令基于 framed 宏包, 用于创建可跨页的 (盒子) 环境, 它类似于 Markdown 中的引用环境。⟨keyval⟩ 用于设置该环境的一系列排版参数, 具体方法请参见下述说明:

ztex/box/framed-user/rulewidth	rulewidth = {⟨dim⟩}		初始值: 5pt
ztex/box/framed-user/rulecolor	rulecolor = {⟨color⟩}		初始值: red
ztex/box/framed-user/padding	padding = {⟨dim⟩}		初始值: 5pt
ztex/box/framed-user/bg	bg = {⟨color⟩}		初始值: gray!10
ztex/box/framed-user/adj	adj = {⟨dim⟩}		初始值: 0pt

⟨rulewidth⟩ 用于设置左侧 rule 的宽度; ⟨rulecolor⟩ 左侧 rule 的颜色; ⟨padding⟩ 用于设置左侧的空白间距; ⟨bg⟩ 用于设置右侧文本的背景色; ⟨adj⟩ 用来调整这个盒子的 \hspace, 简单来说: 就是在左右两边各加上 ⟨adj⟩/2, 然后居中排版。

`\NewDocumentEnvironment{envA}{}{}`

例 46
`{\ztextframe[rulewidth=10pt, adj=2cm]}`
`{\ztextframeend}`

`\begin{envA}`

Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Ut purus elit, vestibulum ut,

placerat ac, adipiscing vitae, felis. Curabitur dictum gravida

mauris.

`\end{envA}`

<code>\zmbox</code>	<code>\zmbox{<material>}</code>
New: 2025-09-14	此命令与 L ^A T _E X 2 _ε 中的 <code>\mbox</code> 命令作用相同, 但 <code>\zmbox</code> 的“参数” <code><material></code> 中可以包含 <code>\verb</code> 等命令.
<code>\zfbox</code>	<code>\zfbox[<keyval>]{<material>}</code>
New: 2026-02-05	次命令与 L ^A T _E X 2 _ε 中的 <code>\fbox</code> 命令类似, 但它的“参数” <code><material></code> 中可以包含 <code>\verb</code> 等特殊命令, 且该命令还提供了诸多样式定制接口, 由 <code><keyval></code> 指定. 注意: 与原始的 <code>\fbox</code> 命令类似, <code>\zfbox</code> 命令仅限于单行文本内容, 即不能包含 <code>\par</code> 等命令.

NOTE: 此命令不依赖 PGF/TikZ, 其仅用到了 L^AT_EX 2_ε 原生的 `picture` 环境和 `pict2e` 宏包, 用户只需手动加载 `ztool` 的 `zdraw` 库即可.

该命令的一个简单使用案例:

```
% \documentclass{ztex}
% \ztoolloadlib{zdraw}
AyC-1p3:\fbox{AyC-1p3}:\zfbox{AyC-1p3}:\zfbox[fill=gray!25,
draw=blue, arc=10]{AyC-1p3}\vskip1em\relax

{
  \fboxrule=10pt\relax
  \fboxsep=4pt\relax
  AyC-1p3:%
  \fbox{AyC-1p3}:%
  \zfbox[arc=3, draw=green, fill=pink]{Ayc-1p3}:%
  \zfbox[arc=3, draw=green, fill=pink, text={\sffamily}]{
\verb|% $*&^_#\alpha@=-!`|}
}
```

AyC-1p3:

AyC-1p3:

AyC-1p3:

AyC-1p3

AyC-1p3:

AyC-1p3

Ayc-1p3

% \$*&^_#\alpha@=-!`

例 48

ztex/box/enhance-fbox/draw	draw = $\langle color \rangle$	初始值: black
ztex/box/enhance-fbox/fill	fill = $\langle color \rangle$	初始值: none
ztex/box/enhance-fbox/arc	arc = $\langle \text{浮点数} \rangle$	初始值: 1
ztex/box/enhance-fbox/text	text = $\langle code \rangle$	初始值: 空

上述参数说明: $\langle draw \rangle$ 用于控制边框的颜色, $\langle fill \rangle$ 用于控制填充颜色, $\langle arc \rangle$ 用于指定边框的圆角半径, $\langle text \rangle$ 用于指定 $\langle material \rangle$ 内容的打印样式 (建议将一些字体控制命令置于其中); 除此之外, 边框线的粗细以及边框内容四周的间距由 L^AT_EX 2_ε 中的 `\fboxrule` 和 `\fboxsep` 控制. **注意:** 参数 $\langle arc \rangle$ 不宜过大, 当此值超过矩形边框长宽最小值的一半时, $\langle arc \rangle$ 会自动调整为二者的最小值.

<code>\zboxcollect</code>	<code>\zboxcollect{$\langle material \rangle$}</code>
<code>\zboxcollectcustom</code>	<code>\zboxcollectcustom{$\langle box spec \rangle$}[$\langle pre code \rangle$][$\langle post code \rangle$]{$\langle material \rangle$}</code>
New: 2025-09-13	这两个命令用于将 $\langle material \rangle$ 中的内容搜集到 <code>\ztexcollectbox</code> 这个水平盒子 (<code>\hbox</code>) 中, 然后将其排版出来, 排版方式由 <code>\zboxcollect@typeset</code> 指定. $\langle material \rangle$ 中可以包含 <code>\verb</code> 等命令; AyC-1p3-

命令 `\zboxcollectcustom` 可以通过 $\langle box spec \rangle$ 参数自定义排版样式, $\langle pre code \rangle$ 会被置于 `\ztexcollectbox` 内容的最前面, $\langle post code \rangle$ 会被置于 `\ztexcollectbox` 内容的最后面, $\langle pre code \rangle$ 和 $\langle post code \rangle$ 二者默认为空, 其中一般放置一些与字体控制相关的命令. **备注:** 这两个命令都是健壮的, 这里的 $\langle material \rangle$ 并不是宏参数. 这两个命令的基本使用方法如下:

```
\zboxcollect{\textbf{AAA}-BBB},
\zboxcollectcustom
{\fcolorbox{blue}{gray!30}{\box\ztexcollectbox}}%
{\sffamily CCC-\verb|\beta|}

AAA-BBB, CCC-\beta
```

例 49

<code>\zbox_item_collect:nnnw</code>	<code>\zbox_item_collect:nnnw {$\langle box spec \rangle$}{$\langle pre code \rangle$}{$\langle post code \rangle$}{$\langle material \rangle$}</code>
New: 2025-09-13	此命令和上述的 <code>\zboxcollect</code> 类似, 这里不再重复说明. 备注: 由于 T _E X 的 <code>tokenize</code> 机制, 这里的 $\langle material \rangle$ 并不是宏参数.
Updated: 2026-02-05	

<code>\zboxitemalign</code>	<code>\zboxitemalign[$\langle key-value \rangle$]{$\langle width \rangle$}{$\langle content \rangle$}</code>
Updated: 2025-05-12	此命令用于对盒子内容进行对齐, $\langle width \rangle$ 为排版盒子的宽度, $\langle content \rangle$ 为盒子中的内容. $\langle key-value \rangle$ 用于设置对齐方式与样式. 注意: $\langle content \rangle$ 中的空格会被忽略, 如果需要空格, 请使用 “ <code>_</code> ” 或 “ <code>\~</code> ” 替代.

ztex/box/align/cmd	cmd = $\langle cmd \rangle$ 初始值: 空
ztex/box/align/type	type = $\langle left \textbf{center} right scatter tower \rangle$ 初始值: center
ztex/box/align/custom	custom = $\langle cmd \rangle$ 初始值: 空

$\langle cmd \rangle$ 和 $\langle custom \rangle$ 均为一个命令; 前者可以接受一个参数, 其会应用于 $\langle content \rangle$ 中的每一个 token; 后者须为一个无参数的命令. $\langle type \rangle$ 用于设置对齐方式, 可选值有: left, center, right, scatter. 默认对齐方式为“center(居中对齐)”, scatter 为分散对齐 (此时两端没有空格), tower 对齐方式: content 中每一个 item(token) 对应的对齐参考点为 hc/b, 其横坐标计算方法如下:

$$\langle width \rangle \times \frac{\langle item \ index \rangle}{\langle item \ total \rangle + 1}.$$

在 custom 对应的命令中可以使用 $\backslash total@width$ 来获取 $\langle width \rangle$ 的值, $\backslash align@cmd$ 来获取 $\langle cmd \rangle$ 的内容, $\backslash align@object$ 来获取 $\langle content \rangle$ 的内容, $\backslash align@format$ 来获取 $\langle format \rangle$ 的值. 变量 $\backslash l_ztool_boxitem_seq$ 中保存了 $\langle content \rangle$ 中的所有 token, 其索引从 1 开始.

一个基本的使用案例如下:

例 50

```

\def\blueit#1{\textcolor{blue}{| #1 |}}
\underline{%
  \zboxitemalign[cmd=\blueit,
type=scatter]{15em}{\Tom}{\Amy}{\Jennery}}%
}\par
\underline{%
  \zboxitemalign[cmd=\blueit]{15em}{\Tom}{\Amy}{\Jennery}}%
}

```

```

|Tom|      |Amy|      |Jennery|
|Tom||Amy|| |Jennery|

```

关于 custom 和 tower 的基本使用案例如下:

例 51

```

% 1. 'tower' style
\zboxitemalign[type=tower]{\linewidth}{A}\par
\zboxitemalign[type=tower]{\linewidth}{AA}\par
\zboxitemalign[type=tower]{\linewidth}{AAA}\par

% 2. use 'custom' to achieve 'tower' style
\ExplSyntaxOn\makeatletter

```



```

\def\customType{
  \edef\seqCount{\seq_count:N \l__ztool_boxitem_seq}
  \seq_map_inline:Nn \l__ztool_boxitem_seq
  {
    \edef\item@width{\dim_eval:n {\total@width/(\seqCount+1)}}
    \hskip\item@width\clap{##1}
  }\hskip\item@width\hss
}
\makeatother\ExplSyntaxOff
\def\itemCmd#1{\textcolor{blue}{\sffamily(#1)}}
\dotfill\par
\zboxitemalign[
  type=custom,
  cmd=\itemCmd,
  custom=\customType
]{\linewidth}{AAAAAA}

```

A

A A

A A A

.....

(A) (A) (A) (A) (A) (A)

`\ztoolboxaffine` `\ztoolboxaffine[⟨key-value⟩]{⟨content⟩}{⟨matrix⟩}`

New: 2025-05-12

上述 $\langle content \rangle$ 表示仿射变换作用的对象; $\langle matrix \rangle$ 为一个 2×2 的矩阵, 表示对应的仿射变换矩阵. 若 $\langle matrix \rangle = \{a, b, c, d\}$, 则其对应的仿射变换矩阵 Λ 如下:

$$\Lambda = \begin{bmatrix} a & c \\ b & d \end{bmatrix}.$$

若 $\det \Lambda = 0$, 则此变换无意义, x_YX 会在终端输出一条警告, 最后将 $\langle content \rangle$ 中的内容原样输出到 PDF; $\langle keyval \rangle$ 设置请参见下述 “ztool/affine” 说明.

ztool/affine/debug	debug = $\langle true false \rangle$ 初始值: false
ztool/affine/pole-1	pole-1 = $\langle coffin's\ pole \rangle$ 初始值: 1
ztool/affine/pole-2	pole-2 = $\langle coffin's\ pole \rangle$ 初始值: b
ztool/affine/xoffset	xoffset = $\langle number \rangle$ 初始值: Opt
ztool/affine/yoffset	yoffset = $\langle number \rangle$ 初始值: Opt

$\langle debug \rangle$ 用于调试, 如果设置为 **true**, 则会在 PDF 中输出一些中间变量信息, 用于调试; 其中 $\langle xoffset \rangle$, $\langle yoffset \rangle$ 为水平和垂直方向的偏移量, 默认值均为 **Opt**; $\langle pole-1 \rangle$, $\langle pole-2 \rangle$ 用于设置打印 coffin 时的参考点, 二者必须相交. 关于后四个 $\langle kye \rangle$ 的详细使用方法可以参见 l3coffins 的说明.

命令 `\ztoolboxaffine` 的一些基本使用样例如下:

Original Text: XXX\par 例 52

```

 $\det(A) = 0$ : \ztoolboxaffine{XXX}{0, 0, 0, 2}\par % det(A) = 0
Unit Matrix: \ztoolboxaffine{XXX}{1, 0, 0, 1}\par % unit matrix
Scale Matrix: \ztoolboxaffine[pole-2=vc]{XXX}{2, 0, 0, 2}\par %
scale
 $x$ -scale Matrix: \ztoolboxaffine{XXX}{2, 0, 0, 1}\par % x-scale
 $y$ -scale Matrix: \ztoolboxaffine{XXX}{1, 0, 0, 2}\par % y-scale
 $x$ -shear Matrix: \ztoolboxaffine{XXX}{1, 0, 1, 1}\par % x-shear
 $y$ -shear Matrix: \ztoolboxaffine{XXX}{1, 1, 0, 1}\par % y-shear
Image Test: \rule{2em}{2em}~\ztoolboxaffine{\rule{2em}{2em}}{1,
0, .5, 1}

```

Original Text: XXX

$\det(A) = 0$: XXX

Unit Matrix: XXX

Scale Matrix: XXX

x -scale Matrix: XXXX

y -scale Matrix: XXX

x -shear Matrix: XXX

y -shear Matrix: XXX

Image Test: 

6.7 cmd 模块

该模块提供了一些针对内核命令的 patch 或改进, 涉及到的模块有 l3clist, l3prop, l3token 等. 除此之外, cmd 模块还提供了一些工具宏, 它们会在 ℒ_{TeX} 的各个模块或库中被使用. **注意:** 本模块不是为普通用户准备的, 普通用户建议直接跳过本节内容.

<code>\ztexverb</code>	<code>\ztexverb[⟨format⟩]{⟨item⟩}</code>
Updated: 2025-04-25	此命令和 ℒ _{TeX} 2 _ε 中的 <code>\verb</code> 类似, 用于输出控制序列名称. 和后者类似, 此命令也不能作为任何控制序列的参数. <code>⟨format⟩</code> 用于指定控制序列的打印格式, 默认为 <code>\texttt</code> . 一个基本的使用样例如下:

`\ztexverb{\alpha + \beta}\par`
`\ztexverb[\textsf]{\alpha + \beta}`

`\alpha + \beta`
`\alpha + \beta`

例 53

<code>\zcmd_if_preamble_p: *</code>	<code>\zcmd_if_preamble:TF {⟨true code⟩}{⟨false code⟩}</code>
<code>\zcmd_if_preamble:TF *</code>	此命令用于判断当前是否为导言区, 若为导言区, 则执行 <code>⟨true code⟩</code> , 反之.
New: 2025-09-23	

<code>\zcmd_cs_copy:NN</code>	<code>\zcmd_cs_copy:NN ⟨cmd₁⟩⟨cmd₂⟩</code>
<code>\zcmd_cs_copy:(Nc cN cc)</code>	此命令为 ℒ _{TeX} 中 <code>\let</code> 这一原语的封装, 它使得 <code>⟨cmd₁⟩</code> 指向 ℒ _{TeX} 内部 Hash 表中 <code>⟨cmd₂⟩</code> 所指向的位置; 它的作用是局部的.
New: 2025-06-22	

<code>\zcmd_cs_gcopy:NN</code>	<code>\zcmd_cs_gcopy:NN ⟨cmd₁⟩⟨cmd₂⟩</code>
<code>\zcmd_cs_gcopy:(Nc cN cc)</code>	此命令为 ℒ _{TeX} 中 <code>\let</code> 和 <code>\global</code> 这两个原语的封装, 它使得 <code>⟨cmd₁⟩</code> 指向 ℒ _{TeX} 内部 Hash 表中 <code>⟨cmd₂⟩</code> 所指向的位置; 它的作用是全局的.
New: 2025-06-22	

`\zcmd_clist_to_tl:n` ☆ 此命令会将 `clist` 转为 `token list`.

New: 2025-09-18

TeXhakers note: 返回结果会被包裹在 `\exp_not:n` 中, 这意味着当这些结果出现在 `e-type` 或 `x-type` 参数中时, 返回结果不会进一步展开.

```
\ExplSyntaxOn{\ttfamily
\def\TTTa{TTTa-val}
\detokenize\expandafter
{
  \expanded{\zcmd_clist_to_tl:n {cmd-\TTTa, 101, a0b}}
}
}\ExplSyntaxOff
```

例 54

```
{cmd-\TTTa }{101}{a0b}
```

`\zcmd_ints_to_tl:nn` ☆

`\zcmd_ints_to_tl:(oo|ee)` ☆

New: 2025-09-18

`\zcmd_ints_to_tl:nn` $\{\langle start \rangle\}\{\langle end \rangle\}$

此命令生成从 $\langle start \rangle$ 到 $\langle end \rangle$ 的所有整数, 然后将其转为 `token list`. **备注:** 用户可以将此命令和 `\tl_map_tokens:nn` 结合, 以此模拟 `\int_step_tokens:nnn` 命令的功能. 一个简单的使用案例如下:

```
\ExplSyntaxOn{\ttfamily
\detokenize\expandafter
{
  \expanded{\zcmd_ints_to_tl:nn {9}{21}}
}
}\ExplSyntaxOff
```

例 55

```
{9}{10}{11}{12}{13}{14}{15}{16}{17}{18}{19}{20}{21}
```

`\ztex_int_step_tokens:nn` ☆

`\ztex_int_step_tokens:nnn` ☆

New: 2025-09-21

`\ztex_int_step_tokens:nn` $\{\langle final\ value \rangle\}\{\langle code \rangle\}$

`\ztex_int_step_tokens:nnn` $\{\langle initial\ value \rangle\}\{\langle final\ value \rangle\}\{\langle code \rangle\}$

因为 l3int 中的 `\int_step_tokens:nn` 和 `\int_step_tokens:nnn` 命令在一些比较老的 TeX 发行版中并不存在, 为了兼容性, L^AT_EX 使用 `\tl_map_tokens:nn` 重新实现了它们.

注意: 和 `\tl_map_tokens:nn(\int_step_tokens:nn)` 命令类似, $\langle code \rangle$ 中的脆弱命令需要保护.

```

\zcmd_map_expnot_tl:n ☆ \zcmd_map_expnot_tl:n {\<tl>}
\zcmd_map_expnot_clist:n ☆ \zcmd_map_expnot_clist:n {\<clist>}
\zcmd_map_expnot_sclist:n ☆ \zcmd_map_expnot_sclist:n {\<sclist>}

```

New: 2025-09-03

`\zcmd_map_expnot_tl:n` 会将 $\langle tl \rangle$ 中每个 $\langle token \rangle$ 变为 $\{\backslash exp_not:n \{\langle token \rangle\}\}$ 的形式; 后两者与之类似, 只是它们的作用对象不同.

```

\ExplSyntaxOn{\ttfamily
\edef\TTT{\zcmd_map_expnot_tl:n {A{BC}\fbox}}
\detokenize\expandafter{\TTT}
}\ExplSyntaxOff

```

```

{\exp_not:n {A}}{\exp_not:n {BC}}{\exp_not:n {\fbox }}

```

例 56

```

\zcmd_floats_sort_increase:n ☆ \zcmd_floats_sort_increase:n {\<float list>}
\zcmd_floats_sort_decrease:n ☆ \zcmd_floats_sort_increase:n {\<float tl>}

```

New: 2025-09-18

此而命令用于浮点数排序, 前者升序排列, 后者降序排列. $\langle float list \rangle$ 使用逗号 - “,” 分割, $\langle float tl \rangle$ 使用花括号 - “{ }” 分割. 使用样例如下:

```

\ExplSyntaxOn{\ttfamily
\detokenize\expandafter
{
  \expanded{
    \zcmd_floats_sort_increase:n {3, 01, -2, 5, +1};
    \zcmd_floats_sort_decrease:n {3, 01, -2, 5, +1};
    \zcmd_floats_sort_decrease:n {\{3\}{01}\{-2\}{5\}{+1\}};
  }
}
}\ExplSyntaxOff

```

```

{-2}{+1}{01}{3}{5};{5}{3}{+1}{01}{-2};{5}{3}{+1}{01}{-2};

```

例 57

```

\zcmd_floats_min:n ☆ \zcmd_floats_min:n {\<float list>}
\zcmd_floats_max:n ☆ \zcmd_floats_min:n {\<float tl>}

```

New: 2025-09-18

此二命令基于上述的 `\zcmd_floats_sort_decrease:n` 和 `\zcmd_floats_sort_increase:n` 命令, 返回序列的最小 (大) 值.

```
\prop_item:nn ☆ \prop_item:nn {\prop content}{\key}
\prop_item:(no|ne|Vn|Ve|vn|ve|en|ee) ☆
```

New: 2025-09-12

此命令与 l3prop 中的 `\prop_item:Nn` 类似, 用于取出 `\prop content` 中 `\key` 对应的值.

T_EXhackers note: 返回结果会被包裹在 `\exp_not:n` 中, 这意味着当这些结果出现在 e-type 或 x-type 参数中时, 返回结果不会进一步展开.

```
\ExplSyntaxOn{\ttfamily
\def\TTT{TTT:val}
\detokenize\expandafter{
  \expanded{\prop_item:nn {a=1-\TTT, bb=234}{a}}
}
}\ExplSyntaxOff
```

```
1-\TTT
```

例 58

```
\zcmd_robustify:N
\zcmd_robustify:c
```

New: 2025-09-03

```
\zcmd_if_param_p:N *
\zcmd_if_param_p:c *
\zcmd_if_param:NTF *
\zcmd_if_param:cTF *
```

New: 2025-09-03

```
\zcmd_robustify:N <control sequence>
```

这两个命令封装自 etoolbox 宏包, 可以将脆弱命令变为一个 robust 命令.

```
\zcmd_if_param:NTF <control sequence>{\true code}{\false code}
```

此系列命令封装自 etoolbox 宏包, 如果 <control sequence> 已经定义且需要参数, 则返回 true.

```
\def\TTTa{TTT}
\def\TTTb#1{TTT:#1}
\ExplSyntaxOn
\zcmd_if_param:NTF \TTTa {Have}{NOT~Have};\par
\zcmd_if_param:NTF \TTTb {Have}{NOT~Have};\par
\zcmd_if_param:cTF \TTTa {Have}{NOT~Have};\par
\zcmd_if_param:cTF \TTTb {Have}{NOT~Have}.
\ExplSyntaxOff
```

例 59

```
NOT Have;
Have;
NOT Have;
Have.
```

```
\zcmd_if_protected_p:N *
\zcmd_if_protected_p:c *
\zcmd_if_protected:NTF *
\zcmd_if_protected:cTF *
```

New: 2025-09-03

```
\zcmd_if_protected:NTF <control sequence>{\true code}{\false code}
```

此系列命令封装自 etoolbox 宏包, 如果 <control sequence> 已经定义且在定义时的 prefix 为 \protected, 则返回 true.

```
\zcmd_if_ltxprotect:NTF <control sequence>{\true code}{\false code}
```

此系列命令封装自 etoolbox 宏包, 如果 <control sequence> 已经定义且使用了 L^AT_EX 的 protect 机制 (或使用 \DeclareRobustCommand 声明的命令), 则返回 true. **注意:** 这个命令本身是 robust 的.

```
\ExplSyntaxOn
\def\TTTa{TTTa}
```

例 60

```
\protected\def\TTTb{TTTb}
\DeclareRobustCommand\TTTc{TTTc}
\zcmd_if_protected:NTF \TTTa {Yes}{No};
\zcmd_if_protected:NTF \TTTb {Yes}{No};
\zcmd_if_protected:NTF \TTTc {Yes}{No};
\zcmd_if_ltxprotect:NTF \TTTc {Yes}{No}.
\ExplSyntaxOff
```

No;Yes;No;Yes.

```
\token_if_expandable_p:N ★
\token_if_expandable_p:c ★
\token_if_expandable:NTF ★
\token_if_expandable:cTF ★
```

New: 2025-09-03

此系列命令来自 l3token, 用于测试 $\langle token \rangle$ 是否可展.

```
\token_if_long_macro_p:N ★
\token_if_long_macro_p:c ★
\token_if_long_macro:NTF ★
\token_if_long_macro:cTF ★
```

New: 2025-09-03

此系列命令来自 l3token, 用于测试 $\langle token \rangle$ 是否为一个 long macro.

```
\token_if_primitive_p:N ★
\token_if_primitive_p:c ★
\token_if_primitive:NTF ★
\token_if_primitive:cTF ★
```

New: 2025-09-03

此系列命令来自 l3token, 用于测试 $\langle token \rangle$ 是否为 primitive(原语).

6.7.1 列表补丁

本小节将介绍 cmd 模块提供的一系列 Patch, 它们往往和 clist 中的命令配合使用;

NOTE: 普通用户不应该使用此小节的系列命令, 这系列的命令主要提供给模板的开发者.

<code>\zclist_count:n</code> *	<code>\zclist_count:n {⟨item₁⟩, ..., ⟨item_n⟩}</code>
<code>\zclist_count:(o e f)</code> *	命令 <code>\zclist_count:n</code> 与 <code>\clist_count:n</code> 类似, 但此命令会将空的 <code>⟨item⟩</code> 考虑在内.
New: 2025-06-21	
<code>\zclist_item:nn</code> *	<code>\zclist_item:nn {⟨item₁⟩, ..., ⟨item_n⟩} {⟨index⟩}</code>
<code>\zclist_item:(on en ee)</code> *	命令 <code>\zclist_item:nn</code> 与 <code>\clist_item:nn</code> 类似, 但此命令会将空的 <code>⟨item⟩</code> 考虑在内.
New: 2025-06-21	
<code>\zclist_range:nnn</code> *	<code>\zclist_range:nnn {⟨item₁⟩, ..., ⟨item_n⟩} {⟨start⟩} {⟨end⟩}</code>
<code>\zclist_range:(enn onn)</code> *	命令 <code>\zclist_range:nnn</code> 与 <code>\tl_range:nnn</code> 类似, 但此命令会将空的 <code>⟨item⟩</code> 考虑在内. 注意: 该命令暂时不支持负数索引.
New: 2025-06-21	

下面给出上述 `\zclist_count:n`, `\zclist_item:nn`, `\zclist_range:nnn` 这几个命令的使用案例:

例 61

```

\ExplSyntaxOn
\setlength{\fboxsep}{3pt}
\def\clistA {, 1, 2, }
\zclist_count:o { \clistA };
\fbox{\zclist_item:on { \clistA }{2}}, \fbox{\zclist_item:on {
\clistA }{-1}};
\detokenize\expandafter{\expanded{\zclist_range:onn { \clistA
}{1}{3}}}
\ExplSyntaxOff

```

```

4;[1,□;\scan_stop: ,1,2

```

```
\zcmd_clist_patch:nn      * \zcmd_clist_patch:nn {\replace}{\langle item_1\rangle, \dots, \langle item_n\rangle}
\zcmd_clist_patch:(ne|no) *
```

New: 2025-06-20

Updated: 2025-09-21

该命令会自动将空的 $\langle item \rangle$ 替换为 “ $\langle replace \rangle$ ”。

T_EXhackers note: 返回结果会被包裹在 $\exp_not:n$ 中, 这意味着当这些结果出现在 e-type 或 x-type 参数中时, 返回结果不会进一步展开。

例 62

```
\ExplSyntaxOn
\def\clistA{\zcmd_clist_patch:nn {\scan_stop:}{, a, 2, \TTT, }}
\detokenize\expandafter{\expanded{\clistA}}
\ExplSyntaxOff
```

```
\scan_stop: ,a,2,\TTT ,\scan_stop: ,
```

```
\zcmd_sclist_patch:nn      * \zcmd_sclist_patch:nn {\replace}{\langle item_1\rangle; \dots ; \langle item_n\rangle}
\zcmd_sclist_patch:(ne|no) *
```

New: 2025-06-20

Updated: 2025-09-21

该命令会自动将空的 $\langle item \rangle$ 替换为 “ $\langle replace \rangle$ ”。

T_EXhackers note: 返回结果会被包裹在 $\exp_not:n$ 中, 这意味着当这些结果出现在 e-type 或 x-type 参数中时, 返回结果不会进一步展开。

例 63

```
\ExplSyntaxOn
\def\clistA{\zcmd_sclist_patch:nn {\TTTa}{; a; 2; \TTT; ; }}
\detokenize\expandafter{\expanded{\clistA}}
\ExplSyntaxOff
```

```
\TTTa ;a;2;\TTT ;\TTTa ;\TTTa ;
```

6.7.2 token 命令

本小节主要介绍 L^AT_EX 的 cmd 模块中与 token 判断相关的命令，它们均是完全可展的。

NOTE: 宏包 etl 也提供了本节的大多数命令，具体用法可参见其手册。

<code>\ztexgentoken</code>	★	<code>\ztexgentoken{<charcode>}{<catcode>}</code>
----------------------------	---	---

<code>\ztex_token_gen:nn</code>	★	<code>\ztex_token_gen:nn {<charcode>}{<catcode>}</code>
---------------------------------	---	---

New: 2025-09-19

根据 `<charcode>` 和 `<catcode>` 生成对应的 token。 `\ztexgentoken` 封装自 `\ztex_token_gen:nn`。

<code>\ztex_token_if_eq:NN</code>	★	<code>\ztex_token_if_eq:NN <token₁><token₂></code>
-----------------------------------	---	--

此命令基于原始的 `\ifx` 命令，可以用于一些 implicit token 的判断，如 `\l_peek_token`, `\g_peek_token`。当 `<token1> = <token2>` 时，该命令返回“1”，反之，则返回“0”。

<code>\ztex_tl_if_eq_p:nn</code>	☆	<code>\ztex_tl_if_eq:nnTF {<tl₁>}{<tl₂>}{<true code>}{<false code>}</code>
----------------------------------	---	--

<code>\ztex_tl_if_eq_p:(ne ee)</code>	☆	
---------------------------------------	---	--

<code>\ztex_tl_if_eq:nnTF</code>	☆	
----------------------------------	---	--

<code>\ztex_tl_if_eq:(ne ee)TF</code>	☆	
---------------------------------------	---	--

New: 2025-06-25

此命令与 l3tl 中默认的 `\tl_if_eq:nnTF` 含义相同，但 L^AT_EX 中的 `\ztex_tl_if_eq:nnTF` 是完全可展的。注意：该命令目前还有缺陷（此缺陷也存在于 l3tl 的 `\tl_if_eq:nnTF` 命令中），当 `<tl1>` 与 `<tl2>` 中的 token 数量不一致时，`\ztex_tl_if_eq:nnTF` 会直接返回 `{<false code>}`，比如“`\ztex_tl_if_eq:nnTF {a{aa}}{aaa} {true}{false}`”的返回结果为“false”。

例 64

```

\ExplSyntaxOn
\NewDocumentCommand{\tlifeq}{\mmmm}
{ \ztex_tl_if_eq:nnTF {#1}{#2}{#3}{#4} }
\edef\TTTa{\ztex_tl_if_eq:nnTF {abcdefg}{abcdefgh}{EQ}{NOT~EQ}}
\detokenize\expandafter{\expanded{\TTTa}},~
\edef\TTTb{\ztex_tl_if_eq:nnTF {ab\c_colon_str cd}{ab
cd}{ab:cd}{EQ}{NOT~EQ}}
\detokenize\expandafter{\expanded{\TTTb}},~
\str_set:Nn \l_tmpa_str {:}
\edef\TTTc{\ztex_tl_if_eq:nnTF {ab\c_colon_str cd}{ab
\l_tmpa_str cd}{EQ}{NOT~EQ}}
\detokenize\expandafter{\expanded{\TTTc}}.\par
\ExplSyntaxOff
\tlifeq{a}{a}{EQ}{NOT~EQ},
\tlifeq{a}{b}{EQ}{NOT~EQ},

```

```

\tlifeq{aa}{aa}{EQ}{NOT~EQ},
\tlifeq{aa}{ab}{EQ}{NOT~EQ}.\par

\tlifeq{a{a}}{aa}{EQ}{NOT~EQ},
\tlifeq{aaa}{a{aa}}{EQ}{NOT~EQ},
\tlifeq{aaa}{aaa}{EQ}{NOT~EQ}.\par

```

NOT EQ, NOT EQ, EQ.
EQ, NOT EQ, EQ, NOT EQ.
EQ, NOT EQ, EQ.

```

\ztex_token_if_in_p:nN ☆ \ztex_token_if_in:nNTF {<tl>}{token}{<true code>}{<false code>}
\ztex_token_if_in_p:(oN|eN) ☆
\ztex_token_if_in:nNTF ☆
\ztex_token_if_in:(oN|eN)TF ☆

```

New: 2025-07-13

Updated: 2025-09-21

此命令用于测试 $\langle token \rangle$ 是否存在于 $\langle tl \rangle$ 中, 基于上述的 `\ztex_token_if_eq:nN` 命令. 这里的 $\langle token \rangle$ 可以是 implicit token, 如 `\l_peek_token`, `\g_peek_token`.

```

\ztex_tl_if_in_p:nn ☆ \ztex_tl_if_in:nnTF {<tl1>}{<tl2>}{<true code>}{<false code>}
\ztex_tl_if_in_p:(no|ne|ee) ☆
\ztex_tl_if_in:nnTF ☆
\ztex_tl_if_in:(no|ne|ee)TF ☆

```

New: 2025-06-25

此命令与 `l3tl` 中默认的 `\tl_if_in:nnTF` 含义、用法均相同 (用于测试 $\langle tl_2 \rangle$ 能否在 $\langle tl_1 \rangle$ 中找到), 但 L^AT_EX 中的 `\ztex_tl_if_in:nnTF` 是完全可展的. 注意: 因为此命令基于上述的 `\ztex_tl_if_eq:nn` 命令, 所以该命令目前有缺陷, 该缺陷的详细描述请参见命令 `\ztex_tl_if_eq:nnTF` 的说明.

NOTE: 目前该函数内部采用的字符串匹配算法比较低效, 后续也许会采用 KMP 算法进行重写;

```

\ExplSyntaxOn
\ztex_tl_if_in:nnTF {123456789}{123}{FIND}{NOT~FIND},
\ztex_tl_if_in:nnTF {12x34567x89}{7x89}{FIND}{NOT~FIND},
\edef\TTT{\ztex_tl_if_in:nnTF {1234567x89}{78x9}{FOUND}{NOT~

```

例 65

```
FOUND}}
\detokenize\expandafter{\expanded{\TTT}}
\ExplSyntaxOff
-----
FIND,FIND,NOT FOUND
```

```
\ztex_colon_if_in_p:n    * \ztex_colon_if_in:nTF {\tl}{\true code}{\false code}
\ztex_colon_if_in_p:(e|V) *
\ztex_colon_if_in:nTF    *
\ztex_colon_if_in:(e|V)TF *
```

New: 2025-06-21

此命令用于检测 $\langle tl \rangle$ 中是否含有 “:”。

```
\ztex_head_tail_token_if_eq_p:nnn    * \ztex_head_tail_token_if_eq:nnnTF {\tl}{\head}{\tail}
\ztex_head_tail_token_if_eq_p:(enn|eee) * {\true code}{\false code}
\ztex_head_tail_token_if_eq:nnnTF    *
\ztex_head_tail_token_if_eq:(enn|eee)TF *
```

New: 2025-06-21

该命令用于检测 $\langle tl \rangle$ 的首尾 Token 是否与 $\langle head \rangle$, $\langle tail \rangle$ 相同; 若均相等, 则执行 $\langle true code \rangle$ 对应分支, 反之, 则执行 $\langle false code \rangle$ 对应分支。

```
\ztex_index_token_if_eq_p:nnn    * \ztex_head_tail_token_if_eq:nnnTF {\tl}{\index}{\token}
\ztex_index_token_if_eq_p:(enn|eee) * {\true code}{\false code}
\ztex_index_token_if_eq:nnnTF    *
\ztex_index_token_if_eq:(enn|eee)TF *
```

New: 2025-06-21

该命令用于检测 $\langle tl \rangle$ 内 index 为 $\langle index \rangle$ 的 Token 是否与 $\langle token \rangle$ 相等; 若相等, 则执行 $\langle true code \rangle$ 对应分支, 反之, 则执行 $\langle false code \rangle$ 对应分支。

```
\ztex_tl_pattern_pos:nn ☆ \ztex_tl_pattern_pos:nn {\tl}{\test tl}
```

New: 2025-09-21

此命令会返回 $\langle test tl \rangle$ 在 $\langle tl \rangle$ 中的所有位置, 不同位置使用 “;” 分割。

<code>\ztex_tl_replace_once:nnn</code>	☆	<code>\ztex_tl_replace_once:nnn {<tl>}{<old tokens>}{<new tokens>}</code>
<code>\ztex_tl_replace_once:(onn enn noo nee eee)</code>	☆	

New: 2025-06-25

Updated: 2025-09-21

此命令与 l3tl 中默认的 `\tl_replace_once:nnn` 含义、用法均相同 (用于把 `<tl>` 中第一个匹配到的 `<old tokens>` 替换为 `<new tokens>`), 但 L^AT_EX 中的 `\ztex_tl_replace_once:nnn` 是完全可展的.

T_EXhackers note: 返回结果会被包裹在 `\exp_not:n` 中, 这意味着当这些结果出现在 e-type 或 x-type 参数中时, 返回结果不会进一步展开.

NOTE: 目前该函数内部采用的字符串匹配算法比较低效, 后续也许会采用 KMP 算法进行重写.

<code>\ztex_tl_replace_all:nnn</code>	☆	<code>\ztex_tl_replace_all:nnn {<tl>}{<old tokens>}{<new tokens>}</code>
<code>\ztex_tl_replace_all:(onn enn noo nee eee)</code>	☆	

New: 2025-06-25

Updated: 2025-09-21

此命令与 l3tl 中默认的 `\tl_replace_all:nnn` 含义、用法均相同 (用于把 `<tl>` 中所有的 `<old tokens>` 替换为 `<new tokens>`), 但 L^AT_EX 中的 `\ztex_tl_replace_all:nnn` 是完全可展的.

T_EXhackers note: 返回结果会被包裹在 `\exp_not:n` 中, 这意味着当这些结果出现在 e-type 或 x-type 参数中时, 返回结果不会进一步展开.

NOTE: 目前该函数内部采用的字符串匹配算法比较低效, 后续也许会采用 KMP 算法进行重写.

<code>\ztex_tl_replace_cnt:nnnn</code>	☆	<code>\ztex_tl_replace_cnt:nnnn</code>
<code>\ztex_tl_replace_cnt:(nonn nenn nooo neee eeee)</code>	☆	<code>{<count>}{<tl>}</code>
		<code>{<old tokens>}{<new tokens>}</code>

New: 2025-09-21

此命令与前述的 `\ztex_tl_replace_all(once):nnn` 等命令类似, 不同的是: 此命令可以指定 `<old tokens>` 的替换次数, 次数由 `<count>` 指定.

T_EXhackers note: 返回结果会被包裹在 `\exp_not:n` 中, 这意味着当这些结果出现在 e-type 或 x-type 参数中时, 返回结果不会进一步展开.

NOTE: 目前该函数内部采用的字符串匹配算法比较低效, 后续也许会采用 KMP 算法进行重写.

例 66

```

\ExplSyntaxOn
\edef\TTT{
  \ztex_tl_pattern_pos:nn
  {xxxxabc123def123123fgh123xxx123asdwwzz}
  {123}
}
\edef\TTTa{
  \ztex_tl_replace_once:nnn
  {xxxxabc123def123123fgh123xxx123asdwwzz}
  {123}{|XXX|}
}
\edef\TTTb{
  \ztex_tl_replace_all:nnn
  {xxxxabc123def123123fgh123xxx123asdwwzz}
  {123}{|XXX|}
}
\edef\TTTc{
  \ztex_tl_replace_cnt:nnnn {2}
  {\fbox X\fbox X\fbox x}{\fbox}{\ffbox}
}
\ExplSyntaxOff

Replace index:\detokenize\expandafter{\TTT}\par
Replace once:\detokenize\expandafter{\TTTa}\par
Replace all : \detokenize\expandafter{\TTTb}\par
Replace cnt : \detokenize\expandafter{\TTTc}
-----
Replace index:;8,10;14,16;17,19;23,25;29,31;
Replace once:xxxxabc|XXX|def123123fgh123xxx123asdwwzz
Replace all :xxxxabc|XXX|def|XXX||XXX|fgh|XXX|xxx|XXX|asdwwzz
Replace cnt : \ffbox X \ffbox X \fbox x

```

```
\ztex_token_strip_both:n    *   \ztex_token_strip_both:n {⟨t1⟩}
\ztex_token_strip_both:(e|V) *
```

New: 2025-06-21

此命令会将 ⟨t1⟩ 两侧的 Token 去掉.

```
\ztex_token_strip_left:n    *   \ztex_token_strip_left:n {⟨t1⟩}
\ztex_token_strip_left:(e|V) *
```

New: 2025-06-21

此命令会将 ⟨t1⟩ 左侧的 Token 去掉.

```
\ztex_token_strip_right:n    *   \ztex_token_strip_right:n {⟨t1⟩}
\ztex_token_strip_right:(e|V) *
```

New: 2025-06-21

此命令会将 ⟨t1⟩ 右侧的 Token 去掉.

6.8 sect 模块

6.8.1 序言

XeTeX 的 sect 模块重写了与章节和目录相关的所有命令，其提供了一系列的命令和接口用于章节和目录的自定义；该模块的实现参考了 ctex-headings, titlesec, titletoc, etoc 以及 CuSTeX, CTeX 两个宏集；但 sect 模块并不依赖于以上的任意一个宏包或宏集。在介绍此模块提供命令前，我们做如下的约定：

sect 模块中将章节标题 (title) 分为 “num, name” 两个部分，比如 “1.1 foo” 中 “num = 1.1”, “name = foo”；为后续行文方便，我们在章节标题相关的上下文中：称 “title” 为 “标题”，称 “num” 为 “编号”，称 “name” 为 “名称”。

sect 模块中将章节目录分为 “name, title, leader, page” 四个部分，比如 “1.2 bar ... 1” 中 “name = 1.2”, “title = bar”, “leader=...”, “page = 1”。为后续行文方便，我们在目录相关的上下文中：称 “name” 为 “名称”；称 “title” 为 “标题”，称 “leader” 为 “引导线”，称 “page” 为 “页码”。后文的 “(章节) 目录, 表格目录, 图片目录, ...” 我们统称为 “table”。

sect 模块会阻止 titlesec, titletoc 等宏包的加载；也就是说，当用户加载 sect 模块后，便不能再加载 titlesec, titletoc, etoc 等宏包了，它们与本模块中的部分设置冲突。

sect 模块并不包含类似 titlesec 宏包所提供的那些标题样式，比如 wrap、leftmargin、drop 等。但是它们都可以通过 “explicit” 选项来实现，比如：结合 \hangindent、\hangafter 以及 “explicit” 选项，我们就可以轻松实现 “wrap” 样式。

NOTE:

1. sect 模块还处于早期开发阶段，很多的功能还不够完善：比如 Tagged PDF, 术语 (glossary) 和索引支持等。
2. sect 模块和 ctex 宏包中 heading 相关的设置相冲突。
3. sect 模块和 XeTeX 中的部分模块或库目前处于耦合状态，请慎用 “sect/load=false” 设置。
4. sect 模块中和 glossary 相关的命令和变量暂时不要使用，会有一些潜在的问题：比如其对应的数据文件为 \jobname.log，这会和编译的日志文件冲突 (目前已经将其临时更改为 tog(table of glossary))。

6.8.2 层级/模板

zT_EX 支持动态创建标题, 以及标题层级重定义. zT_EX 会自动处理 Mark, Bookmark 和目录等内容. 除此之外, 本节还收录了部分命令用于修改章节命令或目录模板.

<code>\zseclevelmap</code>	<code>\zseclevelmap{⟨keyval⟩}</code>
New: 2025-08-30	此命令用于调整当前文档的标题层级或新增标题层级, 只能在导言区使用. <code>⟨keyval⟩</code> 中的“键”为标题/目录类型, 比如 “chapter, section, figure, table” 等; <code>⟨keyval⟩</code> 中的“值”为一个整数, 代表该标题/目录类型的层级, 可以为负数. 备注: 在声明一个新的标题层级后, 用户还需要使用 <code>\zsecdefine</code> 命令定义该命令的一些必要参数后才能在后续使用此标题层级. 为了让新的标题层级能够在目录中正常排版, 用户还需要建立其对应的目录 (模板) 实例, 参见 <code>\zsecdefine</code> 命令中的说明.
<code>\zsecTemplateDefaultsEdit</code>	<code>\zsecTemplateDefaultsEdit{⟨keyval⟩}</code>
New: 2025-09-08	此命令用于修改 sect 模块中标题模板的默认值. <code>⟨keyval⟩</code> 的可用键值列表请参见: 节 (6.8.3). 备注: 此命令对已经创建实例的章节命令无效.

6.8.3 章节标题

explicit	explicit = $\langle \text{true} \text{false} \rangle$ 初始值: false
code	code = $\langle \text{code} \rangle$ 初始值: 空

$\langle \text{explicit} \rangle$ 键与 titlesec 宏包的 “explicit” 选项类似, 但在 sect 模块中, 用户可以仅对部分章节命令启用该选项; 当 “explicit = true” 时, 用户需要在 $\langle \text{code} \rangle$ 中指定该章节标题的内容; 此时需借助 \zsecnum, \zsecname, \zsecclass 等宏.

NOTE: 普通用户不建议使用 ‘explicit’ 键, 因为此时需要用户手动处理目录和书签等内容; 为了照顾另一部分用户, 这里给出处理目录和书签需要用到的两个内部函数: _zsect_title_toc_add:nnn, _zsect_bookmark_update:nnn. 这两个命令的参数格式一致, 默认情况下: 第一个参数为 \zsecclass, 第二个参数为 \zsectocname, 第三个参数为 \zsecname.

\zsecclass	★	在 “explicit” 的 $\langle \text{code} \rangle$ 或 \zsecformat 中, 可以使用 “\zsecclass” 表示当前的类型, 使用 “\zsecnum” 表示 $\langle \text{num} \rangle$ 对应的内容, 使用 “\zsecname” 表示 $\langle \text{name} \rangle$ 对应的内容, 使用 \zsectocname 表示当前章节命令写入目录条目对应的 “title(标题)” 值; \zsectocnum 表示当前章节命令写入目录条目的 “name(名称)” 值; \zsectocHnum 表示当前章节命令写入目录条目的超链接目标索引. 一个简单的设置案例如下:
\zsecname	★	
\zsectocname	★	
\zsecnum	★	
\zsectocnum	★	
\zsectocHnum	★	

Updated: 2026-02-15

注意: 用户可以在 “explicit” 的 $\langle \text{code} \rangle$ 或 \zsecformat 中使用这些命令; 但在这些地方之外, 这些命令输出的内容无任何实际意义或不可用; 当用户重定义 \thesection 等宏时, \zsecnum 等宏会同步进行更改 (修改 \theHsection 等宏时也同理).

```

\zsecformat{\section, \subsection,}
{
  explicit=true,
  code={(\zsecclass){\color{blue}\zsecnum}: [\zsecname] \par}
}

```

例 67

type	type = $\langle \text{page} \text{top} \text{normal} \rangle$ 初始值: 空
pagestyle	pagestyle = $\langle \text{style} \rangle$ 初始值: 空

$\langle \text{type} \rangle$ 用于指定该类型章节命令的排版方式: 占据整页 (page), 位于页面顶端 (top), 普通样式 (normal); $\langle \text{pagestyle} \rangle$ 用于指定该类型章节标题所在页面的页面格式, 一般只针对 $\langle \text{type} \rangle$ 为 “page, top” 的章节命令.

hang	hang = <code><false true></code> 初始值: <code>true</code>
break	break = <code><code></code> 初始值: 空
afterindent	afterindent = <code><true false></code> 初始值: <code>false</code>
<code><hang></code> 用于指定该类型章节命令的长标题是否需要悬挂缩进; <code><break></code> 用于控制标题所在位置的分页, 普通用户可以忽略该选项; <code><afterindent></code> 用于指定该类型章节命令后的第一个段落是否首行缩进. “break” 键暂时不可用. NOTE: 若 <code><name></code> 需要换行, 此时 <code><hang></code> 选项应该设置为 <code>false</code>.	
space.before	space.before = <code>{<skip>}</code> 初始值: 空
space.after	space.after = <code>{<skip>}</code> 初始值: 不定
space.left	space.left = <code>{<length>}</code> 初始值: 空
<code><space.before></code> 用于设置标题前的垂直间距; <code><space.after></code> 用于设置标题后的垂直间距, 若 <code>title.inline = true</code>, 则该距离会被转为水平距离; <code><space.left></code> 用于设置标题的左侧距离.	
title.inline	title.inline = <code><true false></code> 初始值: <code>false</code>
title.format	title.format = <code><code></code> 初始值: 空
title.format+	title.format+ = <code><code></code> 初始值: 空
title.before	title.before = <code><code></code> 初始值: 空
title.after	title.after = <code><code></code> 初始值: <code>\par</code>
<code><title.inline></code> 用于控制标题是否换行, 会影响 <code><space.before></code>, <code><space.after></code> 的作用方式 (详情请参见后两者说明); <code><title.format></code> 和 <code><title.format+></code> 会同时作用于 “编号” 和 “名称”, 前者会覆盖原有的样式, 后者会将新的格式代码加入原样式; <code><title.before></code> 置于 “标题” 前, 位于 <code><num.before></code> 之前; <code><title.after></code> 置于 “标题” 后, 位于 <code><name.after></code>(以及 <code><name.sep></code>) 之后.	

num	num	= <code><code></code>	初始值: 空
num.show	num.show	= <code><true false></code>	初始值: true
num.sep	num.sep	= <code><length></code>	初始值: 空
num.width	num.width	= <code><length></code>	初始值: 空
num.format	num.format	= <code><code></code>	初始值: 空
num.format+	num.format+	= <code><code></code>	初始值: 空
num.before	num.before	= <code><code></code>	初始值: 空
num.after	num.after	= <code><code></code>	初始值: 空

`<num>` 用于指定标题编号, 若为空, 则使用 “\the<class>” 的默认值; `<num.show>` 控制这里的所有键, 当 “num.show = true” 时, 这些键才有意义; `<num.sep>` 用于指定标题编号后的额外间距; `<num.width>` 用于指定标题编号宽度, 默认为空, 此时该选项无效 (该项常应用于一些较宽编号); `<num.format>` 用于指定标题编号格式, 会覆盖原有格式; `<num.format+>` 会将新的格式代码加入原代码, 不会覆盖原有格式, 且编号内容可以被格式代码最后的含参命令捕获; `<num.before>` 用于向编号前添加内容; `<num.after>` 会向其后添加内容;

NOTE: x_YTeX 的 sect 模块同时考虑了键 `<num.show>` (, 目录中的 `<name.show>`) 和计数器 “secnumdepth”, “tocdepth”, 三者均会影响 “编号 (名称)” 的显示; 不同的是: `<num.show>` 仅影响标题排版时是否显示 “编号”, 对目录文件没有影响; 与之类似的, 目录设置中的 `<name.show>` 只影响目录在排版时是否显示 “名称”; 但计数器 “secnumdepth” 和 “tocdepth” 会同时影响标题文本和目录文件, 前者会影响 “编号” 的写入, 后者用于决定是否将该层级的章节命令写入目录文件.

name.sep	name.sep	= <code><length></code>	初始值: 空
name.format	name.format	= <code><code></code>	初始值: 空
name.format+	name.format+	= <code><code></code>	初始值: 空
name.before	name.before	= <code><code></code>	初始值: 空
name.after	name.after	= <code><code></code>	初始值: 空

`<name.sep>` 用于指定标题名称后的额外间距; `<name.format>` 用于指定标题名称格式, 会覆盖原有格式; `<name.format+>` 会将新的格式代码加入原代码, 不会覆盖原有格式, 且名称内容可以被格式代码最后的含参命令捕获; `<name.before>` 用于向名称前添加内容; `<name.after>` 会向其后添加内容.

<code>format.num</code>	<code>format</code>	<code>=</code>	<code>\code</code>		初始值:	空
<code>format.num+</code>	<code>format+</code>	<code>=</code>	<code>\code</code>		初始值:	空
<code>format.name</code>	<code>num.format</code>	<code>=</code>	<code>\code</code>		初始值:	空
<code>format.name+</code>	<code>num.format+</code>	<code>=</code>	<code>\code</code>		初始值:	空
<code>format.title</code>	<code>name.format</code>	<code>=</code>	<code>\code</code>		初始值:	空
<code>format.title+</code>	<code>name.format+</code>	<code>=</code>	<code>\code</code>		初始值:	空

`\format.num` 同 `\num.format`; `\format.num+` 同 `\num.format+`; `\format.name` 同 `\name.format`; `\format.name+` 同 `\name.format+`; `\format.title` 同 `\title.format`; `\format.title+` 同 `\title.format+`.

<code>bookmark.num</code>	<code>bookmark.num</code>	<code>=</code>	<code>\true \false</code>		初始值:	<code>false</code>
<code>bookmark.cmd</code>	<code>bookmark.cmd</code>	<code>=</code>	<code>\function</code>		初始值:	<code>#1:-#1</code>
<code>bookmark.before</code>	<code>bookmark.before</code>	<code>=</code>	<code>\code</code>		初始值:	空
<code>bookmark.after</code>	<code>bookmark.after</code>	<code>=</code>	<code>\code</code>		初始值:	空

这两个键用于指定书签中该章节命令对应 `\name` 的前后内容. `\bookmark.num` 为 `true` 时将显示书签前的编号. `\function` 用于自定义书签的样式, 其接受一个参数, 表示当前的书签内容. 关于 `\bookmark.cmd`, 一个简单的使用样例如下 (两种方法效果一样):

<pre>%% wrap bookmark content in '()' for '\section' % and '\subsection' classes. % method I: \def\TTTa#1{(#1)} \zsecformat{\section, \subsection}{bookmark.cmd=\TTTa{##1}} % method II: \zsecformat{\section, \subsection}{bookmark.cmd=(##1)}</pre>	例 68
---	------

<code>\zsectitlestyle</code>	<code>\zsectitlestyle{<style>}</code>
New: 2025-09-10	<code><style></code> 用于指定新建立的章节命令所属的 样式 , 仅能在导言区使用. sect 模块中章节命令的默认样式为 “ltx”, 用户可以使用后续的 <code>\zsecdefine</code> 命令自定义章节命令主题.
<code>\zsectitleOnce</code>	<code>\zsectitleOnce{<class>}{<keyval>}[<toc-content>]{<sec-content>}</code>
New: 2025-09-08	<code>\zsectitleOnce*{<class>}{<keyval>}[<toc-content>]{<sec-content>}</code> 此命令用于排版一次性的标题, 不会影响其它标题格式. <code><class></code> 用于指定此章节命令的层级; <code><keyval></code> 用于指定标题的属性, 可用的键值列表请参见: 节 (6.8.3) ; <code><toc-content></code> 用于指定目录中的内容, 若为空, 此时目录中的该条目使用 <code><sec-content></code> ; <code><sec-content></code> 用于指定标题内容. <code>\zsectitleOnce*</code> 用于排版无编号标题, 不会 写入目录.
<code>\zsecdefine</code>	<code>\zsecdefine<class>{<style>}{<keyval>}</code>
New: 2025-08-29	此命令封装自 <code>\zsect_define_title:Nnn</code> 命令, 用于自定义章节命令格式; <code><style></code> 用于指定新建立的章节命令所属的样式; <code><keyval></code> 中所有的可用键值列表参见节首说明. <code><class></code> 可以是 “ <code>\part</code> , <code>\section</code> , <code>\subsection</code> , ...” 等. 备注: 如果该 <code><class></code> 目录条目的 instance 不存在, 那么用户需手动指定, 参见下面的示例;

```
% note: create a new class 'book' of level 0, but
%       instance 'ztoc/level0' does not exist.
\zseclevelmap{book=0}
\DeclareInstance{ztextoc}{ztoc/level0}{default}
{
  format          = \large\bfseries\color{red},
  name.width      = 1.9em,
  space.before    = 1em\@plus\p@,
  space.hang      = 1.9em,
  space.left      = 1.9em,
  leader.content  = ,
}
```

例 69

`\zsecformat`

Updated: 2025-09-08

`\zsecformat[⟨style⟩]{⟨classes⟩}{⟨keyval⟩}``\zsecformat*[⟨style⟩]{⟨classes⟩}{⟨keyval⟩}`

此命令用于设置类型为 `⟨classes⟩` 的章节命令格式. `⟨style⟩` 用于指定新建立的章节命令所属的样式, 默认值为 “`ltx`”; `⟨classes⟩` 可以是 “`\part`, `\section`, `\subsection`” 等; `⟨keyval⟩` 用于设置其属性; 带有 “*” 的命令用于设置无编号标题的格式 (是指 `\chapter*{}`, `\section*{}` 这样的命令, 而非自身就无编号的 `\paragraph{}` 或 `\subparagraph{}` 命令).

NOTE: 该命令的作用是局部的.

`\zseccaption`

New: 2025-09-07

Updated: 2025-09-10

`\zseccaption{⟨class⟩}[⟨format⟩]{⟨content⟩}`

此命令用于手动添加 caption (即使不在浮动体环境中), 且它会调用 `\zsect_add_⟨class⟩_line:eeee` 进行目录的添加. `⟨class⟩` 可选值有 “`figure`, `table`, `theorem`, `lstlisting`, `algorithm`, `glossary`”; `⟨format⟩` 用于自定义 caption 的格式, 在 `⟨format⟩` 中, 使用 `#1` 表示其前缀 (形如 “`Figure ⟨num⟩`”), 使用 `#2` 表示 caption 的内容, 默认值为 “`#1\IfBlankF{#2}{:}#2`”; `⟨content⟩` 为 caption 的内容, L^AT_EX 会自动为其增加前缀: `\fnum@⟨class⟩`. (当 `⟨content⟩` 为空时, 中间的冒号 – “`:`” 不会显示)

备注: `⟨class⟩` 不能为 “`toc`”; 此命令封装自后续的 `\zsect_caption_use:nnn` 命令; `⟨content⟩` 中的脆弱命令可以不用保护.

6.8.4 章节目录

当目录相关的变量改变或目录数据有更新时, 请编译文档至少两次, 以得到正确的结果.

NOTE: toc 相关设置会覆盖 hyperref 中 linkcolor 的设定, 前者优先级更高.

explicit	explicit = $\langle$ true false $\rangle$ 初始值: false
enhance	enhance = $\langle$ true false $\rangle$ 初始值: false
code	code = $\langle$ code $\rangle$ 初始值: 空
$\langle$ explicit $\rangle$ 键与 titlesec 宏包的 “explicit” 选项类似, 但在 sect 模块中, 用户可以仅对部分章节命令启用该选项; $\langle$ enhance $\rangle$ 设置为 “true” 后, 会启用 \ztocthehilevel, \ztocthecard 等高级宏; 当 “explicit = true” 时, 用户需要在 $\langle$ code $\rangle$ 中指定该章节标题的内容; 此时用户需借助 \ztocdepth, \ztocname, \ztoctitle 和 \ztocpage 命令, 它们的作用描述如下:	

\ztocclass	★ \ztocclass
\ztocstyle	★ ...
\ztocdepth	★ \ztocanchor
\ztocname	★ 这些宏可以在 “explicit” 的 $\langle$ code $\rangle$ 或 \ztocformat 命令中使用: \ztocclass
\ztoctitle	★ 表示当前的目录的类型, “section, chapter” 等; \ztocstyle 表示当前目录的
\ztocpage	★ 样式, 有编号/无编号; 使用 “\ztocdepth” 表示当前的目录条目的深度; 使用
\ztocanchor	★ “\ztocname” 表示当前目录条目 $\langle$ name $\rangle$ 中的内容; 使用 “\ztoctitle” 表示当前
Updated: 2025-09-19	
\ztocanchor 表示目录条目对应其在文档中的 link target.	

备注: 与前述章节命令里 “explicit” 选项相同: 用户可以在 “explicit” 的 $\langle$ code $\rangle$ 或 \ztocformat 命令内部使用这些它们 (比如在 $\langle$ title.after $\rangle$ 键中使用), 但在这些地方之外, 这些命令输出的内容无任何实际意义或不可用.

\ztoclink	\ztoclink{ $\langle$ text $\rangle$ }
Updated: 2025-09-19	
这个宏可以在 “explicit” 的 $\langle$ code $\rangle$ 或 \ztocformat 命令中使用, 用于创建超链接文本. 备注: 在这些地方之外, 此命令的输出内容无实际意义或不可用.	

\ztocifnumbered	★ \ztocifnumbered{ $\langle$ true code $\rangle$ }{ $\langle$ false code $\rangle$ }
Updated: 2025-09-19	
这个宏可以在 “explicit” 的 $\langle$ code $\rangle$ 或 \ztocformat 命令中使用, \ztocifnumbered 用于测试当前条目是否含有编号, 含有编号执行 $\langle$ true code $\rangle$ ($\langle$ name.show $\rangle$ 对此命令没有影响). 备注: 在这些地方之外, 此命令的输出内容无实际意义或不可用.	

<code>\ztoctheindex</code>	☆	当 “ <code>enhance = true</code> ” 时, 这些命令才有效, 且 只能 在 “ <code>explicit</code> ” 的 <code><code></code> 中使用. <code>\ztoctheindex</code> 表示当前目录的 相对索引 ; <code>\ztoctheabsindex</code> 表示当前目录的 绝对索引 ; <code>\ztocthecard</code> 表示当前目录所包含的子目录数量; <code>\ztocthehilevel</code> 和 <code>\ztocthelolevel</code> 分别表示当前所排版目录中 <code><class></code> 的最小值与最大值. 备注: 这些高级宏启用后, 文档编译会花费更多的时间.
<code>\ztoctheabsindex</code>	☆	
<code>\ztocthecard</code>	☆	
<code>\ztocthehilevel</code>	☆	
<code>\ztocthelolevel</code>	☆	

Updated: 2025-09-18

一个简单的目录格式设置案例如下:

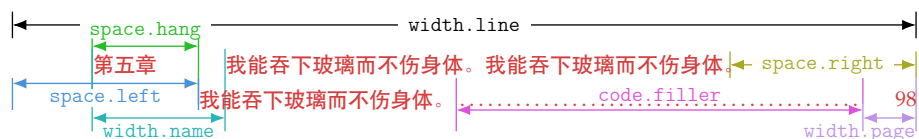
```
\ztocformat{\subsection, \subsubsection}
{ explicit = true,
  code = {
    \hskip2em\rule[1pt]{5pt}{5pt}~
    {\bfseries \ztocname}~\hb@xt@5\ccwd{\ztoctitle}~
    \fbox{\hyper@link{link}{page.\ztocpage}{\ztocpage}}\par
  }
}
```

例 70

`no-parent` `no-parent` = `<true|false>` 初始值: `false`
 若该键设置为 “`true`”, 则当前目录的父级条目会被隐藏; “`no-parent`” 键暂时不可用

`line.end` `line.end` = `<code>` 初始值: `\ztoc@line@end`
`line.width` `line.width` = `<length>` 初始值: 空
`<line.end>` 用于控制每个目录条目结束时的行为, 默认为 `\ztoc@line@end`, 该宏默认定义为 `\par`; `<line.width>` 用于指定当前目录条目的宽度, 该键在处理较长的目录条目时很有用. “`line.width`” 键暂时不可用

`space.before` `space.before` = `<skip>` 初始值: 空
`space.left` `space.left` = `<skip>` 初始值: 空
`space.right` `space.right` = `<skip>` 初始值: `\ztoc@rmargin`
`space.hang` `space.hang` = `<length>` 初始值: 空
`<space.before>` 表示该目录条目前面的垂直间距; `\ztoc@rmargin` 默认为 `\@tocrmarg`; 后面几个长度的含义请参见如下图所示 (此图截取自 CuS_TE_X 宏集手册):



<code>width.name</code>	这几个长度的含义请参见上面的图示 (该图截取自 CusTeX 宏集手册); $\langle width.name \rangle$
<code>width.title</code>	同 $\langle name.width \rangle$; $\langle width.title \rangle$ 同 $\langle title.width \rangle$; $\langle width.page \rangle$ 同 $\langle page.width \rangle$;
<code>width.page</code>	“width.title, width.line” 键暂时不可用
<code>width.line</code>	

<code>name</code>	<code>name</code>	= $\langle code \rangle$	初始值: 空
<code>name.show</code>	<code>name.show</code>	= $\langle true false \rangle$	初始值: true
<code>name.width</code>	<code>name.width</code>	= $\langle length \rangle$	初始值: 空
<code>name.hyper</code>	<code>name.hyper</code>	= $\langle true false \rangle$	初始值: false
<code>name.format</code>	<code>name.format</code>	= $\langle code \rangle$	初始值: 空
<code>name.format+</code>	<code>name.format+</code>	= $\langle code \rangle$	初始值: 空
<code>name.before</code>	<code>name.before</code>	= $\langle code \rangle$	初始值: 空
<code>name.after</code>	<code>name.after</code>	= $\langle code \rangle$	初始值: 空

$\langle name \rangle$ 用于指定目录条目的编号, 若为空, 则使用当前的“名称”; $\langle name.show \rangle$ 用于指定是否显示目录条目“名称”(编号), 若为“true”, 则显示; $\langle name.width \rangle$ 用于指定名称对应的宽度, 当 $\langle name.width \rangle = 0pt$ 时, 将使用“名称”的自然宽度; $\langle name.hyper \rangle$ 用于设置名称是否启用超链接; $\langle name.format \rangle$ 用于指定目录条目名称的格式, 会覆盖原有的格式; $\langle name.format+ \rangle$ 会将新的格式代码加入原代码, 不会覆盖原有的格式, 且名称内容可以被格式代码最后的含参命令捕获; $\langle name.before \rangle$ 用于向名称前添加内容; $\langle name.after \rangle$ 用于向名称后添加内容;

NOTE: x_YTeX 的 sect 模块同时考虑了键 $\langle num.show \rangle$ (, 目录中的 $\langle name.show \rangle$) 和计数器“secnumdepth”, “tocdepth”, 三者均会影响“编号(名称)”的显示; 不同的是: $\langle num.show \rangle$ 仅影响标题排版时是否显示“编号”, 对目录文件没有影响; 与之类似的, 目录设置中的 $\langle name.show \rangle$ 只影响目录在排版时是否显示“名称”; 但计数器“secnumdepth”和“tocdepth”会同时影响标题文本和目录文件, 前者会影响“编号”的写入, 后者用于决定是否将该层级的章节命令写入目录文件。

<code>title.hyper</code>	<code>title.hyper</code>	= $\langle true false \rangle$	初始值: false
<code>title.format</code>	<code>title.format</code>	= $\langle code \rangle$	初始值: 空
<code>title.format+</code>	<code>title.format+</code>	= $\langle code \rangle$	初始值: 空
<code>title.before</code>	<code>title.before</code>	= $\langle code \rangle$	初始值: 空
<code>title.after</code>	<code>title.after</code>	= $\langle code \rangle$	初始值: 空

$\langle title.hyper \rangle$ 用于设置标题是否启用超链接; $\langle title.format \rangle$ 用于指定标题的格式, 会覆盖原有的格式; $\langle title.format+ \rangle$ 会将新的格式代码加入原代码, 不会覆盖原有的格式, 且标题内容可以被格式代码最后的含参命令捕获; $\langle title.before \rangle$ 用于向标题前添加内容; $\langle title.after \rangle$ 用于向标题后添加内容; **“title.width” 键暂时不可用**

<code>hyper.name</code>	<code><hyper.name></code> 同 <code><name.hyper></code> ; <code><hyper.title></code> 同 <code><title.hyper></code> ; <code><hyper.page></code>
<code>hyper.title</code>	同 <code><page.hyper></code> ;
<code>hyper.page</code>	

<code>leader.fill</code>	<code>leader.fill</code>	<code>= <skip></code>		初始值: <code>\hfill</code>
<code>leader.sep</code>	<code>leader.sep</code>	<code>= <length></code>		初始值: <code>\ztoc@leader@sep</code>
<code>leader.raise</code>	<code>leader.raise</code>	<code>= <length></code>		初始值: <code>\ztoc@leader@raise</code>
<code>leader.type</code>	<code>leader.type</code>	<code>= <(<空> x c)></code>		初始值: <code>\ztoc@leader@type</code>
<code>leader.content</code>	<code>leader.content</code>	<code>= <token></code>		初始值: <code>\ztoc@leader@content</code>

这一系列的键用于控制目录中“引导线”的样式；它们可以单独设置，也可以通过设置 `\ztoc@leader@sep`, `\ztoc@leader@raise` 等宏进行全局设置；`<leader.fill>` 用于设置整个引导线的宽度，默认为 `\fill`；`\ztoc@leader@sep` 默认为“4.6pt”，`\ztoc@leader@raise` 默认为“0pt”，`\ztoc@leader@type` 默认为“(<空>”，`\ztoc@leader@content` 默认为“.”。

<code>page.width</code>	<code>page.width</code>	<code>= <length></code>		初始值: <code>\ztoc@page@width</code>
<code>page.hyper</code>	<code>page.hyper</code>	<code>= <true false></code>		初始值: <code>false</code>
<code>page.format</code>	<code>page.format</code>	<code>= <code></code>		初始值: <code>\normalfont\normalcolor</code>
<code>page.format+</code>	<code>page.format+</code>	<code>= <code></code>		初始值: 空
<code>page.before</code>	<code>page.before</code>	<code>= <code></code>		初始值: 空
<code>page.after</code>	<code>page.after</code>	<code>= <code></code>		初始值: 空

`<page.width>` 用于设置页码的宽度。 `<page.hyper>` 用于设置页码是否启用超链接；`<page.format>` 用于指定页码格式，会覆盖原有的格式，默认值为 `\normalfont\normalcolor`；`<page.format+>` 会将新的格式代码加入原代码，不会覆盖原有的格式，且页码内容可以被格式代码最后的含参命令捕获；`<page.before>` 用于向名称前添加内容；`<page.after>` 用于向名称后添加内容；

<code>ignore</code>	<code>ignore</code>	<code>= <true false></code>		初始值: <code>false</code>
<code>ignore.negate</code>	<code>ignore.negate</code>	<code>= <true false></code>		初始值: <code>false</code>
<code>ignore.name</code>	<code>ignore.name</code>	<code>= <clist></code>		初始值: <code>\s_ztoc_ignore_empty_mark</code>
<code>ignore.text</code>	<code>ignore.text</code>	<code>= <t1></code>		初始值: 空
<code>ignore.page</code>	<code>ignore.page</code>	<code>= <clist></code>		初始值: 空

这一系列键用于忽略特定的目录条目, 满足除 `<ignore.negate>` 以外任何一个条件的目录条目将会被忽略; `<ignore>` 为 “true” 时表示忽略该条目, 反之, 则保留; 若当前目录条目的 `<name>` 包含于 `<ignore.name>` 这个逗号分割列表中, 则该目录条目会被忽略; 若当前目录条目的 `<title>` 中包含有 `<ignore.text>` 内的关键词, 则该目录条目会被忽略; 若当前目录条目的 `<page>` 包含于 `<ignore.page>` 中, 则该目录条目会被忽略; `<ignore.negate>` 表示将上述的操作反向, 即, 只保留满足这些 “忽略条件” 的项目.

NOTE:

1. 当 `<ignore.negate>` 为 “true” 时, zT_EX 会依次去判断这些 “忽略条件”, 当找到满足条件的一个目录条目后, 余下的 “忽略条件” 将会被跳过;
2. 在进行比较时, `<ignore.name>` 或 `<ignore.text>` 不需要将匹配内容写完整, 二者均支持部分匹配; 比如目录中的部分 `<title>` 为 “AAA-1, AAA-2”, 那么用户只需要指定 `<ignore.text> = AAA` 即可匹配上述的两个 `<title>`.
3. 这里的比较是基于字符串本身的, 且控制序列后的空格在比较时会被忽略. 比如 “`\ztocformat\subsection{ignore.name={\textbf{T}\;}{XXX}}{YYY}{ZZZ}%`”, 这个设置将会忽略如下的目录条目:

```
\contentsline{subsection}{\textbf{T}\;}{XXX}{YYY}{ZZZ}%
```

我们以后续的定理目录数据作为案例展示 “ignore” 相关键的作用:

```
{
% select subsection with '\textbf {L}\;'
\ztocformat\subsection
{
ignore.negate=true,
ignore.name={\textbf {L}\;}
}
\ztocformat\section{ignore=true} % ignore all section
\zthmtoc
}
```

例 71

L	引理 6.1 (zthmtoc-2)	48
L	引理 7.1 (thmstyle-lapsis)	206

format format+ format.name format.name+ format.title format.title+ format.page format.page+	<code><format></code> 用于控制当前目录条目中 <code><name></code>, <code><title></code> 和 <code><page></code> 的格式; <code><format+></code> 作用和前者作用相同, 但其仅会追加到已有的格式代码中; 备注: <code><format.page></code> 的默认值为 <code>\normalfont\normalcolor</code>. <code><format.name></code> 同 <code><name.format></code>; <code><format.name+></code> 同 <code><name.format+></code>; <code><format.title></code> 同 <code><title.format></code>; <code><format.title+></code> 同 <code><title.format+></code>; <code><format.page></code> 同 <code><page.format></code>; <code><format.page+></code> 同 <code><page.format+></code>;
\ztocenable Updated: 2025-07-06	<code>\ztocenable[<keyval>]</code> 此命令用于启用目录功能, 在导言区添加此命令后 <code>\tableofcontents</code>, <code>\ztocenable</code> 等命令才能正常使用; <code><keyval></code> 用于设置目录类型与来源, 默认值为 <code>toc</code>, 其中可同时填入多个值, 使用逗号分割; 每一项的格式为 “<code><type> = <file></code>”, <code><type></code> 的可选值有 “<code>toc</code>, <code>lof</code>, <code>lot</code>, <code>lom</code>, <code>log</code>, <code>loa</code>”, <code><file></code> 为对应的文件名 (不需要添加后缀), 且 <code><file></code> 可以省略, 默认的文件名为 <code>\jobname</code>, 该文件的后缀为默认的 <code><key></code> 值. 比如 “<code>\ztocenable{lom}</code>”, 它会启用 “定理目录 (<code>lom</code>)”, 其依赖的目录文件为 “<code>\jobname.lom</code>”.
\ztoclineOnce New: 2025-09-20	<code>\ztoclineOnce{<keyval>}{<class>}{<name>}{<title>}{<page>}</code> 此命令用于排版一次性的目录, 不会影响其它目录条目. <code><keyval></code> 的可用键值列表请参见: 节 (6.8.4); <code><class></code> 的可选值有 “<code>section</code>, <code>chapter</code>, <code>figure</code>, <code>theorem</code>, ...” 等; <code><name></code>, <code><title></code> 和 <code><page></code> 的含义在本节已经做过说明, 这里不在复述.
\ztocTemplateDefaultsEdit New: 2025-09-20	<code>\ztocTemplateDefaultsEdit{<keyval>}</code> 此命令用于修改 <code>sect</code> 模块中目录模板的默认值. <code><keyval></code> 的可用键值列表请参见: 节 (6.8.4). 备注: 此命令对已经创建实例的目录命令无效.
\tableofcontents Updated: 2025-07-06	<code>\tableofcontents[<file>]</code> 此命令用于输出文档的全部目录, 当 <code>\ztocenable</code> 启用目录后可用; 和 <code>LaTeX 2_ε</code> 中 <code>\tableofcontents</code> 命令不同的是: 该命令可以在文档中任意位置, 任意次数使用; <code><file></code> 用于指定目录对应的 “<code>*.toc</code>” 文件, 不需要添加文件后缀, 默认为 <code>\jobname</code>.

<code>\multicolcontents</code>	<code>\multicolcontents[⟨file⟩]{⟨column⟩}</code>
<code>Updated: 2025-07-06</code>	此命令将使用多栏布局输出文档的全部目录; <code>⟨file⟩</code> 用于指定目录对应的“*.toc”文件, 不需要 添加文件后缀, 默认为 <code>\jobname</code> . <code>⟨column⟩</code> 表示目录栏数, 接受一个整数.
<code>\ztocstop</code>	<code>\ztocstop[⟨tables⟩]</code>
<code>New: 2025-09-04</code>	此命令会停止目录的收集, 它之后的所有目录数据均不会被记录, 用户应慎用此命令. <code>⟨tables⟩</code> 为一个逗号分割列表, 默认为“toc”.
<code>\ztocset</code>	<code>\zlocaltoc{⟨keyval⟩}</code>
<code>New: 2025-07-10</code>	此命令用于设置目录的格式, 它将作用于所有的目录层级; 可用的键值列表参见下面的说明:

-
- ztex/ztoc/option/rmargin
 - ztex/ztoc/option/ignore.level
 - ztex/ztoc/option/line.end
 - ztex/ztoc/option/page.width
 - ztex/ztoc/option/leader.type
 - ztex/ztoc/option/leader.sep
 - ztex/ztoc/option/leader.raise
 - ztex/ztoc/option/leader.content
-

这些键的具体含义在前文已经做过说明, 这里不再重复.

`\zlocaltoc`

Updated: 2025-09-11

`\zlocaltoc[⟨file⟩][⟨class⟩]{⟨index list⟩}`

此命令用于输出局部目录, 可以在文章中的任意地方, 任意次数使用. `⟨file⟩` 用于指定目录对应的 “*.toc” 文件, **不需要**添加文件后缀, 默认为 `\jobname`; `⟨index list⟩` 为一个逗号分割列表, 其中的每一个元素为一个整数 `⟨index⟩`, 表示 `⟨class⟩` 在目录文中的**相对索引**; 此命令用于输出第 `⟨index⟩` 个 `⟨class⟩` 及其包含的所有子目录. `⟨class⟩` 可以是 “part, section, subsection, figure, table” 等; `⟨index⟩` 从 1 开始计数.

注意: [1.] `⟨index⟩` 并不是 “*.ptoc” 文件中 “name” 后面的值; 举个例子: 比如 *.ptoc 文件中有这么一行内容 “class={subsection}, name={1.3}, ...”, 假如该行的前面还有 4 行含有 subsection(不管它们嵌套在哪个层级中), 此时用户需要将 `⟨index⟩` 置为 “5”. [2.] `\zlocaltoc` 命令能够利用 “class, name, title, page, raw” 所有字段的值, 可以参见命令: `\ztoc_table_filter_key_byclass:nnnNN`; [3.] 如果用户不希望手动计算 `⟨class⟩` 的**相对索引**, 可以使用后续的 `\ztocindex` 命令, 其会根据 `⟨class⟩` 和 `⟨title⟩` 自动返回 **相对索引**.

NOTE: 该命令将得到的结果 (一系列的 `\contentsline`) 保存于 `\g_ztoc_local⟨table⟩_seq` 这个 seq 中, `⟨table⟩` 取决于用户指定的 `⟨class⟩`, 用户可以按照自己的方式操作此 seq.

<pre>{ \ztocformat\subsection{title.after=\P} \zlocaltoc{section}{2} }</pre>		例 72

2 安装使用		2
2.1 在线模板 ¶		2
2.2 本地安装 ¶		2
2.3 快速开始 ¶		3

`\ztocgroupshow`

`\ztocgrouphide`

New: 2025-07-08

`\ztocgroupshow` 命令用于显示局部目录中的插入点 (Hook), 当用户无法确定 `\ztocgroupinsert` 命令中的 `⟨place⟩` 时, 此命令是十分有用的; `\ztocgrouphide` 用于隐藏这些插入点.

NOTE: 这两个命令的作用是局部的.

{

\ztcgroupshow

\zlocaltoc{subsection}{5}

}

例 73

⟨subsection,1,before⟩	
6.2 ref 模块	25
⟨subsection,1,begin⟩	
⟨subsubsection,1,before⟩	
6.2.1 超链接	25
⟨subsubsection,1,begin⟩	
⟨subsubsection,1,end⟩	
⟨subsubsection,1,after⟩	
⟨subsubsection,2,before⟩	
6.2.2 标签与引用	28
⟨subsubsection,2,begin⟩	
⟨subsubsection,2,end⟩	
⟨subsubsection,2,after⟩	
⟨subsubsection,3,before⟩	
6.2.3 杂项	30
⟨subsubsection,3,begin⟩	
⟨subsubsection,3,end⟩	
⟨subsubsection,3,after⟩	
⟨subsection,1,end⟩	
⟨subsection,1,after⟩	

ztoc/tocline/begin

ztoc/tocline/end

在 LaTeX 中, 每条目录的前后放置了这两个钩子; 一般的钩子机制请参见后续的 \ztcgroupinsert 命令.

`\zotocgroupinsert`New: 2025-07-07

`\zotocgroupinsert{⟨place⟩}{⟨code⟩}`

sect 模块对目录进行了分组, 并且在每组目录的前后都放置了一个 Hook(这些 Hook 是根据当前的文档内容动态生成的), 用户可以向这些 Hook 中添加代码(这些添加进去的代码都是一次性的), 从而实现目录的进一步定制; `⟨place⟩` 为 Hook 的名字, 其格式为: “`⟨class⟩,⟨index⟩,⟨before/begin/end/after⟩`”, 它们的位置关系请参见: 段落 (6.8.5.3); 其中 `⟨index⟩` 的计算方法和 `\zlocaltoc` 命令中 `⟨index⟩` 的计算方法不同, 前者 (`\zotocgroupinsert` 命令中的 `⟨index⟩`) 只需考虑当前局部环境内该 `⟨class⟩` 的次序;

NOTE: 如果当前的排版的目录为空, 那么 `⟨code⟩` 会插入到之后目录的 hooks 中.

下面这个示例展示了该命令的基本使用方法, 文件 “./support/data/data.toc” 的内容请参见: 节 (6.8.5).

例 74

```

\zotocgroupinsert{subsection,1,begin}{\fbox{T1-BEGIN}}
\zotocgroupinsert{subsection,1,end}{\fbox{T1-END}\par}
\zotocgroupinsert{subsection,2,begin}{\fbox{T2-BEGIN}}
\zotocgroupinsert{subsection,2,end}{\fbox{T2-END}\par}
\zotocformat\subsection{space.before=.5em}
\zotocformat\subsubsection
{
  explicit = true,
  code = \fcolorbox{red}{gray}{\ztoctitle}\_,
}
\zlocaltoc[./support/data/data]{section}{1}

```

1	AAA-1	1
1.1	BBB-1	1
	T1-BEGIN CCC-1 CCC-2 CCC-3 CCC-4 CCC-5 T1-END	
1.2	BBB-2	1
	T2-BEGIN CCC-6 CCC-7 CCC-8 CCC-9 T2-END	
1.3	BBB-3	1

`\ztocformat`

Updated: 2025-07-06

`\ztocformat{⟨classes⟩}{⟨keyval⟩}`

此命令用于设置类型为 `⟨classes⟩` 的章节命令格式, `⟨classes⟩` 可以是 “`\part`, `\section`, `\subsection`, `figure`, `table`” 等; `⟨keyval⟩` 用于设置其属性.

NOTE: 该命令的作用是局部的.

```

\makeatletter{
  \ztocformat\subsection
  { explicit = true,
    code = {
      \noindent {\bfseries \ztocname~ \ztoctitle}
      \cleaders\hbox{.}\hfill\ztocpage\par
    }
  }
  \ztocformat\subsubsection
  { explicit = true,
    code = {
      \hskip2em\rule[1pt]{5pt}{5pt}~{\bfseries \ztocname}~
      \hb@xt@5\ccwd{\ztoctitle}~
      \fbox{\hyper@link{link}{page.\ztocpage}{\ztocpage}}\par
    }
  }
  \zlocaltoc{subsection}{\ztocindex{subsection}{\pkg{font} 模
块}}
}
\makeatother

```

例 75

6.1 font 模块	14
■ 6.1.1 字体机制	14
■ 6.1.2 默认字体族	17
■ 6.1.3 新建字体族	17
■ 6.1.4 切换字体	19
■ 6.1.5 L ^A T _E X 接口	20
■ 6.1.6 杂项	24

`\ztocotherformat`

New: 2025-09-10

`\ztocotherformat[⟨class⟩]{⟨keyval⟩}`

此命令用于定义除 “toc” 外的其它目录样式. `⟨class⟩` 用于指定设置的目录对象, 可选值有: “figure, table, theorem, lstlisting, algorithm, glossary”, 默认值为 “figure”; `⟨keyval⟩` 格式请参见: 节 (6.8.4). 备注: 该命令也可以使用上述的 `\ztocformat` 命令替代.

`\ztocslot` ★ `\ztocslot`

New: 2025-09-12 此命令返回当前目录条目在当前目录层级中对应的序号, 这个值总是有意义的.

`\ztoccard` ☆ `\ztoccard[⟨table⟩]{⟨class⟩}{⟨title⟩}`

New: 2025-09-18 此命令返回由 `⟨class⟩` 和 `⟨title⟩` 所确定目录的子目录 (card 即 cardinality), 如果由 `⟨class⟩` 和 `⟨title⟩` 所确定目标目录不存在, 其返回 “-1”; `⟨table⟩` 用于指定当前所排版目录对应的 keyval seq 变量, 可选值有: “toc, lot, ...” 等.

`\ztotchilevel` ☆ `\ztotchilevel{⟨table⟩}`

`\ztoclolevel` ☆ `\ztoclolevel{⟨table⟩}`

New: 2025-09-18 这两个命令会返回 `⟨seq⟩` 中的 class level 数据, `⟨seq⟩` 为目录对应的 keyval seq 变量: `\g_ztoc_keyval⟨table⟩_seq`. `⟨seq⟩` 由 `⟨table⟩` 指定, `⟨table⟩` 的可选值有 “toc, lot, lof” 等. `\ztotchilevel` 返回 `⟨seq⟩` 中最小的 level 值; `\ztoclolevel` 返回 `⟨seq⟩` 中最大的 level 值; **备注:** `⟨class⟩` 级别越高, 对应的 level 值越小.

`\ztocindex` ☆ `\ztocindex[⟨table⟩]{⟨class⟩}{⟨title⟩}`

`\ztocindex*[⟨table⟩]{⟨class⟩}{⟨title⟩}`

New: 2025-09-12 命令 `\ztocindex` 用于返回由 `⟨class⟩` 和 `⟨title⟩` 所确定条目在 seq 变量 `\g_ztoc_keyval⟨table⟩_seq` 中的相对索引, `\ztocindex*` 返回绝对索引 (不考虑目录的 `⟨class⟩` 值). `⟨table⟩` 用于指定当前所排版目录对应的 keyval seq 变量, 可选值有: “toc, lot, ...” 等. **备注:** 如果没有满足筛选条件的目录, 此命令总是返回一个小于零的整数.

NOTE: `⟨class⟩` 和 `⟨title⟩` 比较的是字符串本身, 而不是 token 的比较; 所以部分情况下, 用户需要将它们二者提前展开.

`\ztociffirst` ★ `\ztociffirst{⟨true code⟩}{⟨false code⟩}`

New: 2025-09-12 此命令和 `etoc` 宏包提供的 `\tociffirst` 作用相同, 用于判断当前的目录是否为其所在的 level 的第一项. 此命令在 `⟨code⟩` 选项中十分有用, 其用法可以参见: [段落 \(6.8.5.7\)](#) 中的案例. **备注:** 在 e/f-type 中, 此命令可以被完全展开;

`\ztocifmiddle` ☆ `\ztocifmiddle[⟨table⟩]{⟨true code⟩}{⟨false code⟩}`

New: 2025-09-13 此命令用于判断当前目录条目是否位于两条同级目录之间. `⟨table⟩` 默认为 `toc`. **备注:** 实际上是判断当前目录条目之后的条目是否为同级目录.

6.8.5 使用案例

最后附上一些复杂的目录格式定制示例，涵盖目录样式设置以及目录数据提取，可作为用户深度定制的参考。下面这段代码展示了后续样例会用到的一些命令和依赖宏包：

```

1 % \usepackage{tikz} 1
2 % \usepackage{tikz-qtree} 2
3 % \usepackage[log]{tikz-qtree-patch} 3
4 % \usetikzlibrary { mindmap, trees } 4
5 \definecolor{SeaGreen}{rgb}{0.18, 0.55, 0.34} 5
6 \def\customtothemecolor{SeaGreen} 6
7 \ExplSyntaxOn\makeatletter 7
8 \dim_new:N \linewidthfix 8
9 \def\tocupdatelinuxwidthfix{ 9
10   \dim_set:Nn \linewidthfix {\linewidth-2\fbboxsep} 10
11 } 11
12 \def\ornamentcorner#1#2#3{ 12
13   \node[anchor=#1, rotate=#3, outer~sep=0pt, inner~sep=0pt] at (frame.#2) 13
14   {\pgfornament[width=.8cm, color=\customtothemecolor]{10}}; 14
15 \def\ztohyperpage#1{\exp_args:Nne \hyper@link{link}{page.#1}{#1}} 15
16 \makeatother\ExplSyntaxOff 16

```

测试数据 本手册在排版目录时用到的测试数据：“data.toc”，“data_II.toc”，“data.ptoc”。data.ptoc 文件仅用于向读者展示 sect 模块中 keyval seq 的存储结构；当“dump-ptable = true”时，L^AT_EX 会自动生成该文件。

<pre> \contentsline {section}{{1}{AAA-1}}{1}{} \contentsline {subsection}{{1.1}{BBB-1}}{1}{} \contentsline {subsubsection}{{1.1.1}{CCC-1}}{1}{} \contentsline {subsubsection}{{1.1.2}{CCC-2}}{1}{} \contentsline {subsubsection}{{1.1.3}{CCC-3}}{1}{} \contentsline {subsubsection}{{1.1.4}{CCC-4}}{1}{} \contentsline {subsubsection}{{1.1.5}{CCC-5}}{1}{} \contentsline {subsection}{{1.2}{BBB-2}}{1}{} \contentsline {subsubsection}{{1.2.1}{CCC-6}}{1}{} </pre>	data.toc
---	----------

```

\contentsline {subsubsection}{1.2.2}{CCC-7}{1}{}
\contentsline {subsubsection}{1.2.3}{CCC-8}{1}{}
\contentsline {subsubsection}{1.2.4}{CCC-9}{1}{}
\contentsline {subsection}{1.3}{BBB-3}{1}{}

```

```

\contentsline{section}{1}{AAA-1}{1}{}% data_II.toc
\contentsline{subsection}{1}{BBB-1}{1}{}%
\contentsline{subsubsection}{1.1.1}{CCC-1}{1}{}%
\contentsline{subsubsection}{1.1.2}{CCC-2}{2}{}%
\contentsline{subsubsection}{1.1.3}{CCxxxC-3}{3}{}%
\contentsline{subsubsection}{1.1.4}{CCssgda\_shgfeC-4}{100}{}%
\contentsline{subsubsection}{1.1.5}{CsasfCC-5}{123}{}%
\contentsline{subsubsection}{1.1.6}{CasklklCC-x}{999}{}%
\contentsline{subsection}{2}{B BB-2}{1}{}%
\contentsline{subsubsection}{1.2.1}{CCC-6}{1}{}%

```

```

class={section},name={1},title={AAA-1},page={1},anchor={section.1},raw={a,ptoc
\contentsline {section}{1}{AAA-1}{1}{}
class={subsection},name={1.1},title={BBB-1},page={1},anchor={subsection.1.1},
raw={\contentsline {subsection}{1.1}{BBB-1}{1}{}
class={subsubsection},name={1.1.1},title={CCC-1},page={1}, ✓
anchor={subsubsection.1.1.1},raw={\contentsline
{subsubsection}{1.1.1}{CCC-1}{1}{}
class={subsubsection},name={1.1.2},title={CCC-2},page={1}, ✓
anchor={subsubsection.1.1.2},raw={\contentsline
{subsubsection}{1.1.2}{CCC-2}{1}{}
class={subsubsection},name={1.1.3},title={CCC-3},page={1}, ✓
anchor={subsubsection.1.1.3},raw={\contentsline
{subsubsection}{1.1.3}{CCC-3}{1}{}
class={subsubsection},name={1.1.4},title={CCC-4},page={1}, ✓
anchor={subsubsection.1.1.4},raw={\contentsline
{subsubsection}{1.1.4}{CCC-4}{1}{}

```



```

class={subsubsection},name={1.1.5},title={CCC-5},page={1}, ✓
anchor={subsubsection.1.1.5},raw={\contentsline ✓
{subsubsection}{{1.1.5}{CCC-5}}{1}{}}
class={subsection},name={1.2},title={BBB-2},page={1},anchor={subsection.1.2}, ✓
raw={\contentsline {subsection}{{1.2}{BBB-2}}{1}{}}
class={subsubsection},name={1.2.1},title={CCC-6},page={1}, ✓
anchor={subsubsection.1.2.1},raw={\contentsline ✓
{subsubsection}{{1.2.1}{CCC-6}}{1}{}}
class={subsubsection},name={1.2.2},title={CCC-7},page={1}, ✓
anchor={subsubsection.1.2.2},raw={\contentsline ✓
{subsubsection}{{1.2.2}{CCC-7}}{1}{}}
class={subsubsection},name={1.2.3},title={CCC-8},page={1}, ✓
anchor={subsubsection.1.2.3},raw={\contentsline ✓
{subsubsection}{{1.2.3}{CCC-8}}{1}{}}
class={subsubsection},name={1.2.4},title={CCC-9},page={1}, ✓
anchor={subsubsection.1.2.4},raw={\contentsline ✓
{subsubsection}{{1.2.4}{CCC-9}}{1}{}}
class={subsection},name={1.3},title={BBB-3},page={1},anchor={subsection.1.3}, ✓
raw={\contentsline {subsection}{{1.3}{BBB-3}}{1}{}}

```

案例:1 一个很简单的案例:

```

\begin{group
\zotocformat\subsection
{
  format+=\color{teal},
  leader.sep=1pt,
  leader.raise=2.5pt,
  page.width=10pt
}
\zotocgroupinsert{subsection,1,begin}%
{
  \begin{framed}%
  \pgfornament[width = 2cm,color = teal]{67}%
  \quad\rule[-5em]{.5pt}{10em}%
  \begin{minipage}{.75\linewidth}%
  }
\zotocgroupinsert{subsection,1,end}%
{
  \end{minipage}%
  \end{framed}%
}
\zlocaltoc{subsection}{4}
\end{group}

```

例 78

6.1 font 模块.....14



6.1.1	字体机制	14
6.1.2	默认字体族	17
6.1.3	新建字体族	17
6.1.4	切换字体	19
6.1.5	x _Y 接口	20
6.1.6	杂项	24

案例:2 下面这个案例清晰地展示了目录中这些 hook 的位置关系, `BG:<int>` 和 `EG:<int>` 代表目录条目自身局部组的开始和结束位置 (这些局部组目前已被移除 – 2025-09-12).

例 79

```

\ExplSyntaxOn
\int_new:N \g__test_toc_group_int
\AddToHook{ztoc/tocline/begin}
{
  \fbox{BG:\int_use:N \g__test_toc_group_int}
}
\AddToHook{ztoc/tocline/end}
{
  \fbox{EG:\int_use:N \g__test_toc_group_int}
  \int_gincr:N \g__test_toc_group_int
}
\cs_new:Npn \__debug_ztoc_hook:nn #1#2
{ \ztocgroupinsert{#1}{\fbox{\color{#2}\sffamily #1}} }

% begin/end hooks:
\__debug_ztoc_hook:nn {subsection,1,begin}{red}
\__debug_ztoc_hook:nn {subsection,1,end}{red}
\__debug_ztoc_hook:nn {subsubsection,5,begin}{gray}
\__debug_ztoc_hook:nn {subsubsection,5,end}{gray}

% new before/after hooks:
\__debug_ztoc_hook:nn {subsection,1,before}{blue}
\__debug_ztoc_hook:nn {subsection,1,after}{blue}
\__debug_ztoc_hook:nn {subsubsection,1,before}{teal}
\__debug_ztoc_hook:nn {subsubsection,1,after}{teal}
\__debug_ztoc_hook:nn {subsubsection,3,before}{purple}
\__debug_ztoc_hook:nn {subsubsection,3,after}{purple}
\ExplSyntaxOff
\zlocaltoc{subsection}{4}

```

	subsection,1,before	BG:0
6.1	font 模块	14
EG:0	subsection,1,begin	subsubsection,1,before
	subsubsection,1,before	BG:1
6.1.1	字体机制	14
EG:1	subsubsection,1,after	BG:2
6.1.2	默认字体族	17
EG:2	subsubsection,3,before	BG:3
6.1.3	新建字体族	17
EG:3	subsubsection,3,after	BG:4
6.1.4	切换字体	19
EG:4	BG:5	
6.1.5	x _Y 接口	20
EG:5	subsubsection,5,begin	subsubsection,5,end
	subsubsection,5,end	BG:6
6.1.6	杂项	24
EG:6	subsection,1,end	subsection,1,after

案例:3 这个案例相较之上一个案例更加复杂:

例 80

```

\begingroup\makeatletter
\tocupdatelinewidthfix
\def\customtothemecolor{teal}
\ztocformat\subsection
{
  explicit=true,
  code={%
    \noindent\fbcolorbox{white}{\customtothemecolor}{\hb@xt@linewidthfix
      {\bfseries\large\color{white}\ztocname
        \hskip.75em\relax\ztoctitle\hfill\ztocpage}}}%
  }
}
\ztocformat\subsubsection
{
  format+=\color{\customtothemecolor},
  page.format+=\color{\customtothemecolor},
  name.before=\S\;
}
\ztocgroupinsert{subsection,1,begin}%
{%
  \begin{tcolorbox}[
    enhanced, sharp corners,
    boxrule=.5pt, colback=white,
    left=20pt, colframe=\customtothemecolor,
    overlay={
      \ornamentcorner{north west}{north west}{0}
      \ornamentcorner{north west}{south west}{90}
      \ornamentcorner{north west}{south east}{180}
      \ornamentcorner{north west}{north east}{270}
    }
  ]
}

```

```

\pgfornament[width = 2.5cm,color = \customtothemecolor]{8}%
\underline{\kern-6pt\relax\begin{minipage}{.75\linewidth}%
}
\ztcgroupinsert{subsection,1,end}%
{%
\end{minipage}%
\end{tcolorbox}%
}
\zlocaltoc{subsection}{4}
\makeatother\endgroup

```

6.1 font 模块

14



§ 6.1.1 字体机制	14
§ 6.1.2 默认字体族	17
§ 6.1.3 新建字体族	17
§ 6.1.4 切换字体	19
§ 6.1.5 XeTeX 接口	20
§ 6.1.6 杂项	24

案例:4 我们可以使用 `\ztocgroupinsert` 命令复刻 `titletoc` 宏包所提供的样式:

```
\makeatletter
\def\customssstyle#1{%
  \ztocgroupinsert{subsection,#1,begin}
  {\raggedright\hangindent2.3em\relax\noindent\hskip2.3em\relax}{}
  \ztocgroupinsert{subsection,#1,end}{} \par
}
\makeatother
\customssstyle{1} \customssstyle{2}
\customssstyle{3} \customssstyle{4}
{
  \ztocformat\subsubsection{
    explicit=true,
    code={%
      \textit{\ztocname\enskip\ztoctitle\enskip
        (\ztochyperpage{\ztopage})\ztociflast{}{;}\enskip}}
    }
  \ztocformat\subsection{
    explicit=true,
    code = {%
      \S;\ztocname\enskip\ztoctitle
      \hskip1em\ztochyperpage{\ztopage}\par}
    }
  \begin{multicols}{2}
    \zlocaltoc{subsection}{4, 11, 8, 6}
  \end{multicols}
}
```

例 81

§ 6.1 font 模块 14	接口: 目录 (137); 6.8.9 编程接口: 模
【6.1.1 字体机制 (14); 6.1.2 默认字体	板 (150); 6.8.10 编程接口: 杂项 (152)
族 (17); 6.1.3 新建字体族 (17); 6.1.4	】
切换字体 (19); 6.1.5 x _Y TeX 接口 (20);	§ 6.5 thm 模块 41
6.1.6 杂项 (24) 】	【6.5.1 用户接口 (42); 6.5.2 定理目录
§ 6.8 sect 模块 83	(48); 6.5.3 高级接口 (51); 6.5.4 环境
【6.8.1 序言 (83); 6.8.2 层级/模板	钩子 (55) 】
(84); 6.8.3 章节标题 (85); 6.8.4 章节	§ 6.3 page 模块 31
目录 (91); 6.8.5 使用案例 (105);	【6.3.1 页面布局 (31); 6.3.2 页眉页脚
6.8.6 编程接口: 初始化 (131); 6.8.7	(32); 6.3.3 页面水印 (35); 6.3.4 杂项
编程接口: 章节命令 (134); 6.8.8 编程	(37) 】

案例: 5 这个案例相较之上一个案例更加复杂:

例 82

```
\begingroup\makeatletter
% \setcounter{tocdepth}{3}
\def\customtothemecolor{SeaGreen}
\ztocformat\section
{
  explicit=true,
  code={%
    \noindent\hskip1.5em\colorbox{white}{\customtothemecolor}{\hb@xt@
      \dimexpr\linewidth-2\fbboxsep-1.5em\relax
      {\bfseries\large\color{white}第\zhnumber{\ztocname}节
        \hskip.75em\relax\ztoctitle\hfill\ztocpage}}}%
  }
}
\ztocformat\subsection{
  name.before=\S\;, name.width=2.7em,
  format+=\color{\customtothemecolor},
  page.format+=\color{\customtothemecolor},
  leader.sep=2pt, page.width=1em,
  hyper.title=true,
  leader.content=\textcolor{\customtothemecolor}{.},
}
\ztcgroupinsert{section,1,before}%
{%
  \begin{tcolorbox}[
    enhanced, sharp corners,
    boxrule=.5pt, colback=white,
    left=20pt, colframe=\customtothemecolor,
    overlay={
      \ornamentcorner{north west}{north west}{0}
      \ornamentcorner{north west}{south west}{90}
      \ornamentcorner{north west}{south east}{180}
```

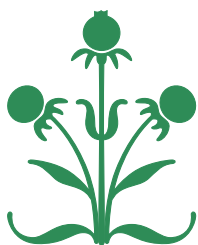
```

\ornamentcorner{north west}{north east}{270}
}
]
\pgfornament[width = 2.5cm,color = \customtothemecolor]{10}%
\begin{minipage}{.77\linewidth}%
}
\ztcgroupinsert{section,1,begin}{%
% \setlength{\columnsep}{0em}
\begin{multicols}{2}}
\ztcgroupinsert{section,1,after}{%
{%
\end{multicols}%
\end{minipage}%
\end{tcolorbox}%
}
\zlocaltoc{section}{6}
\makeatother\endgroup

```

第六节 XeTeX 模块

13



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案例: 6 这个案例复刻了 etoc 中的目录样式 (参见 etoc 宏包第 39 节):

```

1 \zotocgroupinsert{section,1,before} 1
2 {\begin{longtable}{|l|l|r|}} 2
3 \zotocgroupinsert{section,4,after} 3
4 {\ \hline\end{longtable}} 4
5 { 5
6 \zotocformat\section{ 6
7     explicit=true, 7
8     code = {% 8
9         \zotociffirst{\kill\hline\multicolumn{3}{|c|}{\large\bfseries Table of 9
Contents}\ \hline}{\ \hline}%
10         \multicolumn{3}{|c|}{\bfseries 第\; \zotocname\; 节\enskip\ztoctitle}% 10
11     } 11
12 } 12
13 \zotocformat\subsection{ 13
14     explicit=true, 14
15     code = {\ \hline 15
16         \textbf{\S\zotocname} & \zotocname\enskip\ztoctitle & \ztochyperpage{ 16
\zotocpage}
17     } 17
18 } 18
19 \zotocformat\subsubsection{ 19
20     explicit=true, 20
21     code = {\ \ 21
22         & \zotocname\enskip\ztoctitle & \ztochyperpage{\zotocpage} 22
23     } 23
24 } 24
25 \zlocaltoc{section}{1, 2, 7, 9} 25
26 } 26

```

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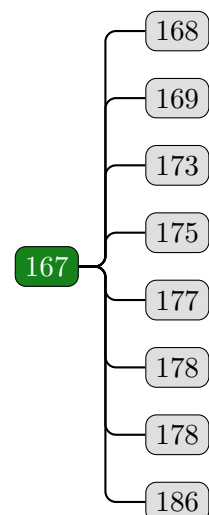
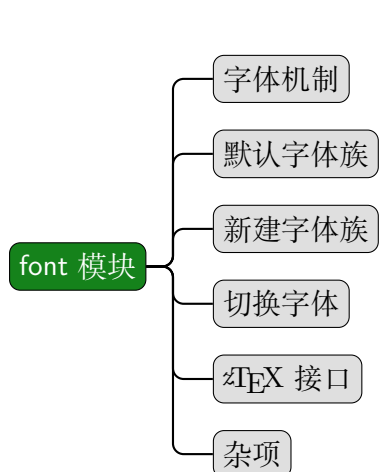
案例: 7 这个案例也是受到 etoc 宏包的启发 – 将目录展示为思维导图的形式:

```

1  \definecolor{Green}{HTML}{15821b}
2  \tikzset{
3    rootKey/.style={draw, rectangle, rounded corners, fill=Green, text=white},
4  }
5  \ExplSyntaxOn
6  \cs_new:Npn \__toc_tree:nn #1#2
7  {
8    \quark_if_no_value:nF {#1}
9    {
10      \Tree[.\node[rootKey]{#1};~#2 ]
11    }
12  }
13  \cs_generate_variant:Nn \__toc_tree:nn {oe}
14  \def\wrapper#1{[.\exp_not:n {#1}}~]}
15  \cs_new:Npn \ztoc_typeset_toc tree:nnn #1#2#3
16  {
17    \ztoc_table_filter_key_byclass:nnnNN {#1}{#2}{#3}
18    \g_ztoc_keyvaltoc_seq \g_tmpb_seq
19    \seq_gpop_left:NN \g_tmpb_seq \secRoot
20    \__toc_tree:oe { \secRoot }
21    { \seq_map_function:NN \g_tmpb_seq \wrapper }
22  }
23  \def\TocTree#1#2#3
24  {
25    \exp_args:Nne \ztoc_typeset_toc tree:nnn
26      {#1}{#2}{#3}
27  }
28  \ExplSyntaxOff
29
30  \noindent\begin{tikzpicture}[
31    grow'=right,
32    level distance=50pt,

```

33	sibling distance=10pt,	33
34	level 1/.style={level distance=25pt},	34
35	every tree node/.style={anchor=west, draw, rectangle, rounded corners, fill=gray!25},	35
36	every level 0 node/.style={anchor=east},	36
37	edge from parent/.style={	37
38	draw, thick, rounded corners,	38
39	edge from parent path={	39
40	(\tikzparentnode.east) -- +(10pt, 0pt)	40
41	- (\tikzchildnode.west)	41
42	}	42
43	},	43
44	]	44
45	\TocTree{subsection}{4}{title}	45
46	\begin{scope}[xshift=12cm]	46
47	\TocTree{subsection}	47
48	{\ztocindex{subsection}{\pkg{alias} 库}}	48
49	{page}	49
50	\end{scope}	50
51	\end{tikzpicture}	51



案例: 8 这个案例相较之上一个案例更加复杂, 它展示了 `\ztoc_group_parser:NNNnnnnn` 命令中 `<step code>` 的使用方法:

```

1 % \usetikzlibrary { mindmap, trees } 1
2 \ExplSyntaxOn 2
3 \int_new:N \g__mindmap_color_int 3
4 \seq_new:N \g__mindmap_data_seq 4
5 \seq_new:N \g__mindmap_keyvaldata_seq 5
6 \tl_new:N \g__mindmap_filter_data_tl 6
7 \def\toclinecmd#1#2#3#4 7
8 { 8
9   % override hyperlink color. 9
10   {\ztexlink{#1.#2}{\textcolor{white}{#3(#4)}}} 10
11 } 11
12 \cs_new:Npn \__ztoc_gen_mindmap_data:nn #1#2 12
13 { 13
14   \tl_gclear:N \g_tmpa_tl 14
15   \ztoc_table_filter_byclass:nnNN 15
16   { #1 }{ #2 } 16
17   \g_ztoc_keyvaltoc_seq 17
18   \g__mindmap_data_seq 18
19   \ztoc_generate_keyvaltable_seq:NN 19
20   \g__mindmap_data_seq 20
21   \g__mindmap_keyvaldata_seq 21
22   \ztoc_group_parser:NNNnnnnn \g__mindmap_keyvaldata_seq 22
23   \toclinecmd \g_tmpa_tl 23
24   { 24
25     child [ concept~color = blue! 25
26       \int_eval:n{\g__mindmap_color_int*6}!green 26
27     ] \bgroup node[concept] 27
28   } 28
29   {}{}{\egroup}{\int_gincr:N \g__mindmap_color_int} 29
30   % replace '\bgroup' and '\egroup' by '{' and '}'. 30
31   \str_clear:N \l_tmpa_str 31

```

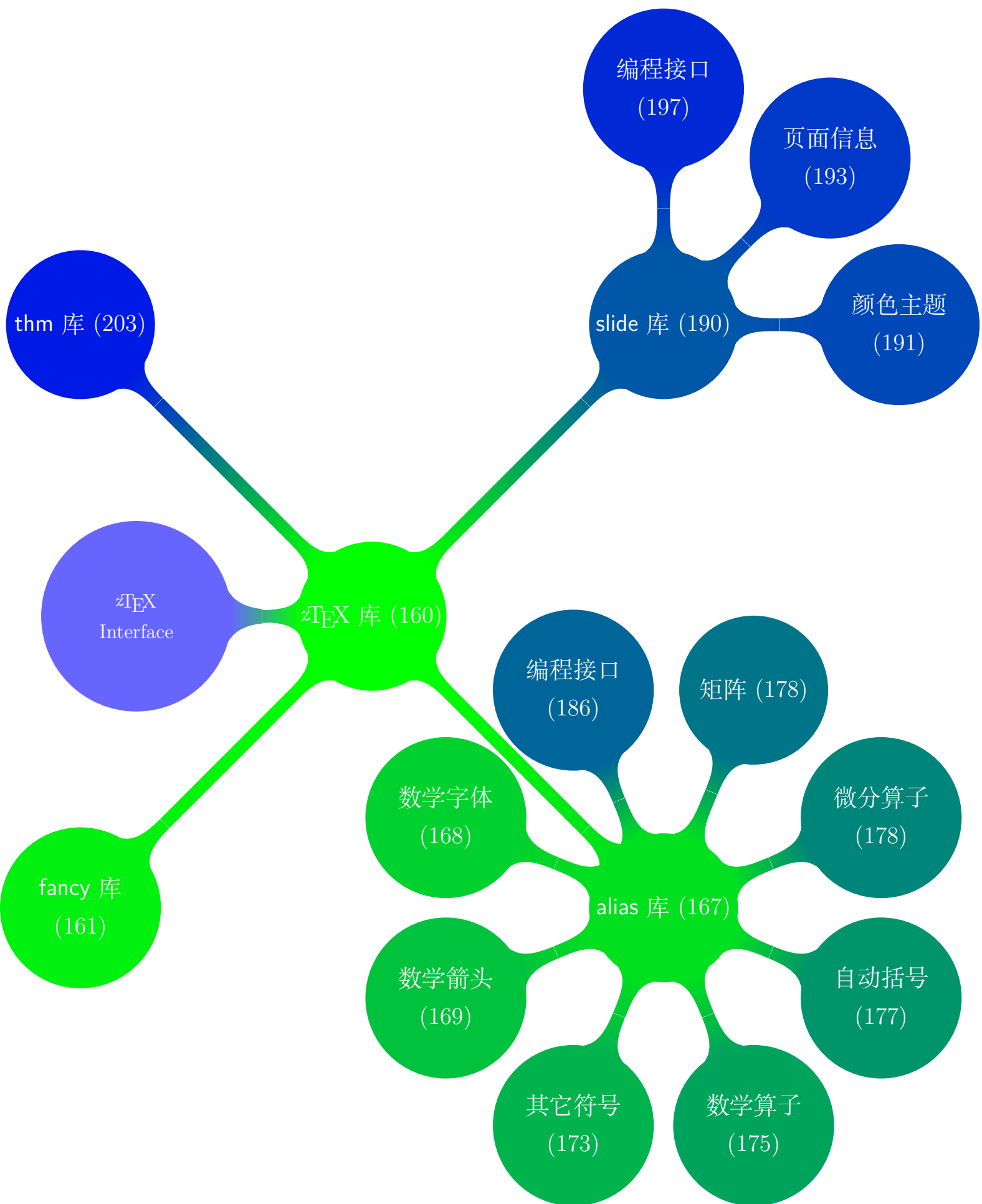
```

32 \exp_args:NNo \str_set:Nn \l_tmpa_str { \g_tmpa_tl } 32
33 \exp_args:NNne \str_replace_all:Nnn \l_tmpa_str 33
34 { \bgroup }{ \char_generate:nn {123}{12} } 34
35 \exp_args:NNne \str_replace_all:Nnn \l_tmpa_str 35
36 { \egroup }{ \char_generate:nn {125}{12} } 36
37 \tl_gset_rescan:Nno \g__mindmap_filter_data_tl 37
38 { \cctab_select:N \c_document_cctab } 38
39 { \l_tmpa_str } 39
40 } 40
41 \cs_new:Npn \__tkz_mind_map:n #1 41
42 { 42
43 \tl_if_empty:nF {#1} 43
44 { 44
45 \node[concept] 45
46 {\color{white}\ztext{}~Interface} 46
47 #1; 47
48 } 48
49 } 49
50 \newcommand{\mindmap}[2] 50
51 { 51
52 \__ztoc_gen_mindmap_data:nn 52
53 { #1 }{ #2 } 53
54 \exp_args:No \__tkz_mind_map:n 54
55 { \g__mindmap_filter_data_tl } 55
56 } 56
57 \ExplSyntaxOff 57
58 \begin{tikzpicture}[ 58
59 mindmap, grow cyclic, 59
60 text width=2cm, align=flush center, 60
61 nodes={concept}, concept color=blue!60, 61
62 root concept/.append style={ 62
63 text width=4cm, font=\Large 63
64 }, 64

```



```
65     level 1 concept/.append style={           65
66         font=\normalsize, color=white         66
67     },                                         67
68     level 1/.append style={                   68
69         level distance=5cm, sibling angle=45,   69
70         text width=3cm, font=\Large           70
71     },                                         71
72     level 2/.append style={                   72
73         level distance=9cm, sibling angle=90,   73
74         text width=3cm, font=\Large           74
75     },                                         75
76     level 3/.append style={                   76
77         level distance=5cm, sibling angle=45,   77
78         text width=3cm, font=\Large           78
79     },                                         79
80 ]                                             80
81 \mindmap{section}{7}                        81
82 \end{tikzpicture}                           82
```



案例: 9 下面这个案例展示了 `\ztociffirst`, `\ztociflast` 以及 `\ztocdepth` 等命令的使用方法:

```

1  \ExplSyntaxOn
2  \newcommand{\nbraket}
3    { \makebox[0pt]{\rule[-7.5pt]{.75pt}{19pt}} }
4  \newcommand{\ubraket}
5    {
6      \makebox[0pt]{
7        \rule[-7pt]{.75pt}{10pt}
8      }
9      \makebox[0pt][l]{
10        \kern-.375pt\raise3pt\hbox{\rule{3pt}{.75pt}}
11      }
12    }
13 \newcommand{\bbraket}
14 {
15   \makebox[0pt]{ \rule[3pt]{.75pt}{10pt} }
16   \makebox[0pt][l]
17     { \kern-.375pt\raise3pt
18       \hbox{\rule{3pt}{.75pt}} }
19 }
20 \newcommand{\showbar}
21 {
22   \prg_replicate:nn {\ztocdepth - 2}
23     { \kern20pt \nbraket }
24   \kern20pt \ztociffirst { \ubraket }
25     { \ztociflast{\bbraket}{\nbraket} }
26 }
27 \ztocformat{\section, \subsection, \subsubsection}
28 {
29   explicit=true,
30   code = {
31     \noindent

```

```
32     \prg_replicate:nn {\ztocdepth-1}          32
33     {                                           33
34         \hskip 20pt\relax                      34
35     }                                           35
36     \llap{\smash{\showbar}}                  36
37     \kern10pt                                  37
38     \ztoctitle\par                             38
39 }                                               39
40 }                                               40
41 \ExplSyntaxOff                                41
42 {                                               42
43     \ztocformat\subsection{ignore.name={10.}} 43
44     \ztocformat\subsubsection{ignore.name={10.}} 44
45     \multicolcontents{2}                      45
46 }
```

基本介绍	sect 模块
安装使用	- 序言 - 层级/模板 - 章节标题 - 章节目录 - 使用案例 - 编程接口: 初始化 - 编程接口: 章节命令 - 编程接口: 目录 - 编程接口: 模板 - 编程接口: 杂项
- 在线模板 - 本地安装 - 快速开始	sclist 模块
基本命令	graphics 模块
文档类选项	counter 模块
状态检测	$\LaTeX$ 库 - fancy 库 - alias 库 - 数学字体 - 数学箭头 - 其它符号 - 数学算子 - 自动括号 - 微分算子 - 矩阵 - 编程接口 - slide 库 - 颜色主题 - 页面信息 - 编程接口 - thm 库
$\LaTeX$ 模块	ztool 宏包
font 模块	TODO
- 字体机制 - 默认字体族 - 新建字体族 - 切换字体 $\LaTeX$ 接口 - 杂项	$\LaTeX$ 源码
ref 模块	索引
- 超链接 - 标签与引用 - 杂项	
page 模块	
- 页面布局 - 页眉页脚 - 页面水印 - 杂项	
color 模块	
thm 模块	
- 用户接口 - 定理目录 - 高级接口 - 环境钩子	
box 模块	
cmd 模块	
- 列表补丁 - token 命令	

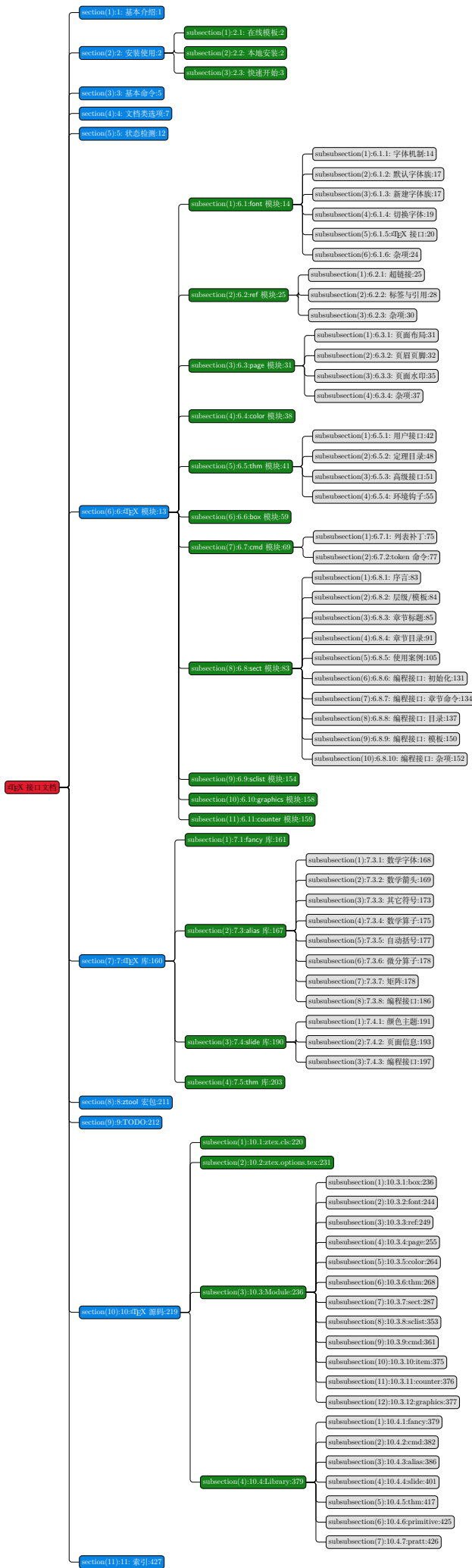
案例: 10 下面这个案例相较之上一个案例更加复杂, 但它展示了 `\ztoc_group_parser:NNNnnnnn` 命令的强大和普适性:

```

1 % \usepackage{tikz}
2 % \usepackage{tikz-qtrees}
3 % \usepackage{tikz-qtrees-patch}
4 \definecolor{Red}{HTML}{e32b2d}
5 \definecolor{Green}{HTML}{15821b}
6 \definecolor{Blue}{HTML}{0984e3}
7 \tikzset{
8   section/.style={draw, rectangle, rounded corners, fill=Red},
9   subsection/.style={draw, rectangle, rounded corners, fill=Blue, text=white},
10  subsubsection/.style={draw, rectangle, rounded corners, fill=Green,
11    text=white},
12 }
13 \ExplSyntaxOn
14 % \def\toclinecmd#1#2#3#4{{#1:#2:#3:#4}~}
15 \def\toclinecmd#1#2#3#4{{\gparserclass(\gparserindex):#2:#3:#4}~}
16 \ztoc_group_parser:NNNnnnnn \g_ztoc_keyvaltoc_seq
17 \toclinecmd \g_tmpa_tl {[.]{~}{~}{~]}~}
18 \def\TocTree
19 {
20   \exp_args:Nno \__toc_tree:nn
21   {\ztex{~接口文档}}
22   {\g_tmpa_tl}
23 }
24 \cs_new:Npn \__toc_tree:nn #1#2
25 {
26   \quark_if_no_value:nF {#1}
27   {
28     \Tree[.\node[section]{#1};~#2 ]
29   }
30 }
31 \ExplSyntaxOff

```

31		31
32	<code>\thispagestyle{empty}</code>	32
33	<code>\begin{tikzpicture}[</code>	33
34	<code>grow'=right, level distance=1cm,</code>	34
35	<code>sibling distance=10pt,</code>	35
36	<code>every tree node/.style = {</code>	36
37	<code>anchor=west, outer sep=0pt, draw, rectangle,</code>	37
38	<code>rounded corners, fill=gray!25</code>	38
39	<code>},</code>	39
40	<code>every level 1 node/.style = {</code>	40
41	<code>subsection</code>	41
42	<code>},</code>	42
43	<code>every level 2 node/.style = {</code>	43
44	<code>subsubsection</code>	44
45	<code>},</code>	45
46	<code>edge from parent/.style={</code>	46
47	<code>draw, thick, rounded corners,</code>	47
48	<code>edge from parent path={</code>	48
49	<code>(\tikzparentnode.east) -- +(.5cm, 0pt)</code>	49
50	<code> - (\tikzchildnode.west)</code>	50
51	<code>},</code>	51
52	<code>},</code>	52
53	<code>]</code>	53
54	<code>\TocTree</code>	54
55	<code>\end{tikzpicture}</code>	55



6.8.6 编程接口: 初始化

本小节描述的命令在章节命令或目录相关的制作过程中都是不可或缺的, 我们在这里对它们进行一个统一规范的描述.

`\c_zsect_level_leagcy_prop`

New: 2025-09-03

该变量记录了原 L^AT_EX 2_ε 中 (常规) 章节的层级信息, 此变量目前未被使用.

`\g_zsect_level_int`
`\c_zsect_level_prop`
`\c_zsect_level_clist`
`\c_zsect_level_tl`

New: 2025-09-03

这些变量记录了文档中的常规章节命令信息 (不包括 “figure, table, ... 等”), 在任何情况下, 用户都不应该修改它们. `\g_zsect_level_int` 表示当前文档中常规章节类型的数量; `\c_zsect_level_prop` 记录了每一个常规章节类型对应的层级.

`\c_ztoc_special_level_prop`
`\c_ztoc_special_level_clist`

New: 2025-09-03

这两个变量记录了文档中的非常规 (章节) 命令层级信息, 比如 figure, table, ... 等.

`\c_ztoc_table_types_clist`
`\c_ztoc_table_types_prop`

New: 2025-09-10

这两个变量记录了非常规 (章节) 目录文件的后缀. `\c_ztoc_table_types_prop` 记录了从 “figure, table, ...” 到目录文件后缀的映射, 其中 “toc” 对应的键 (key) 为 “sect”.

`\zsect_config_sync:nn` `\zsect_config_sync:nn {<class>}{<level>}`

New: 2025-09-03

此命令用于同步章节命令和目录相关的 (初始化) 变量.

`\zsect_class_level_remap:n` `\zsect_class_level_remap:n {<keyval>}`

New: 2025-09-03

此命令封装自上述的 `\zsect_config_sync:nn` 命令, 加入了错误处理, 同步了一些其余的变量. 当用户需要自定义章节或目录类型时, 请使用此命令进行同步 (除非你清楚相关原理, 否则不建议使用 `\zsect_config_sync:nn` 命令进行同步).

NOTE: 用户层面的 `\zsectlevelmap` 基于此命令.

```
\ztoc_get_class_level_expandable:n    * \ztoc_get_class_level_expandable:n {\class}
\ztoc_get_class_level_expandable:(o|e) *
```

New: 2025-09-03

此命令会返回当前文档类中 `\class` 对应的层级, 为一个整数.

```
\ztoc_get_class_level:Nn      \ztoc_get_class_level:Nn <macro> {\class}
\ztoc_get_class_level:(No|Ne)
```

New: 2025-09-03

此命令会返回当前文档类中 `\class` 对应的层级, 为一个整数, 其被置于 `<macro>` 这个宏中.

<code>\g_ztoc_toc_iow</code>	L ^A T _E X 预先创建了这些 stream 接口用于目录文件的写入操作. 它们的含义如下:	
<code>\g_ztoc_lof_iow</code>	• <code>\g_ztoc_toc_iow</code> : (章节) 目录;	• <code>\g_ztoc_lom_iow</code> : 定理目录;
<code>\g_ztoc_lot_iow</code>	• <code>\g_ztoc_lof_iow</code> : 图片目录;	• <code>\g_ztoc_loa_iow</code> : 算法目录;
<code>\g_ztoc_log_iow</code>	• <code>\g_ztoc_lot_iow</code> : 表格目录;	
<code>\g_ztoc_lom_iow</code>	• <code>\g_ztoc_log_iow</code> : 术语目录;	• <code>\g_ztoc_lol_iow</code> : 代码目录.
<code>\g_ztoc_loa_iow</code>		
<code>\g_ztoc_lol_iow</code>		

New: 2025-09-03

NOTE: 除非你清楚相关原理, 否则建议保持这些变量的默认设置.

```
\g_ztoc_toc_iow_bool
\g_ztoc_lof_iow_bool
\g_ztoc_lot_iow_bool
\g_ztoc_log_iow_bool
\g_ztoc_lom_iow_bool
\g_ztoc_loa_iow_bool
\g_ztoc_lol_iow_bool
```

New: 2025-09-03

这些 bool 变量用于检测相应的目录写入接口是否可用, 该类型的目录启用后, 对应的 bool 变量值为 true.

NOTE: 除非你清楚相关原理, 否则建议保持这些变量的默认设置.

```

\g_ztoc_toc_seq
\g_ztoc_lof_seq
\g_ztoc_lot_seq
\g_ztoc_log_seq
\g_ztoc_lom_seq
\g_ztoc_loa_seq
\g_ztoc_lol_seq

```

New: 2025-09-03

这些变量记录了全局目录数据, 用户层面的 `\tableofcontents`, `\multicolcontents` 命令均基于这里的 `\g_ztoc_toc_seq` 变量.

NOTE: 除非你清楚相关原理, 否则建议保持这些变量的默认设置.

```

\g_ztoc_keyvaltoc_seq
\g_ztoc_keyvallof_seq
\g_ztoc_keyvallot_seq
\g_ztoc_keyvallog_seq
\g_ztoc_keyvallom_seq
\g_ztoc_keyvalloa_seq
\g_ztoc_keyvallol_seq

```

New: 2025-09-03

这些变量记录了对应的目录数据, 它们以键值的形式存储, 主要用于目录的筛选. 比如, `\ztoc_table_filter_byclass:nnNN` 命令, 其依赖于这里的 `\g_ztoc_keyvaltoc_seq` 变量.

NOTE: 除非你清楚相关原理, 否则建议保持这些变量的默认设置.

```

\g_ztoc_localtoc_seq
\g_ztoc_locallof_seq
\g_ztoc_locallot_seq
\g_ztoc_locallog_seq
\g_ztoc_locallom_seq
\g_ztoc_localloa_seq
\g_ztoc_locallol_seq

```

New: 2025-09-03

这些变量用于局部目录的排版. 比如, 用户层面的 `\zlocaltoc` 命令, 其筛选得到的目录数据保存于 `\g_ztoc_localtoc_seq` 这个变量中.

NOTE: 除非你清楚相关原理, 否则建议保持这些变量的默认设置.

6.8.7 编程接口: 章节命令

本小节描述的命令均与章节命令相关, 涉及到章节命令的创建, 样式自定义, 层级调整以及章节命令的一些附属命令.

书签管理 考虑到 tagged pdf 支持, 目前和 bookmark 相关的接口命令较少, 后续会有补充.

```
\zsect_bookmark_add:nnn      \zsect_bookmark_add:nnn {\langle level \rangle}{\langle content \rangle}{\langle anchor \rangle}
\zsect_bookmark_add:(ene|eee)
```

New: 2025-09-03

此命令封装自 hyperref 宏包, 用于手动添加书签.

marks 后续会出现 “class” 和 “mark class” 两个词, 这里做一个简短的说明: 前者表示章节命令 class, 可以叫做 “title class”; 后者表示 mark 中的 class 名称.

```
\zsect_mark_new_class_safe:nn \zsect_mark_new_class_safe:nn {\langle bool \rangle}{\langle mark class \rangle}
```

New: 2025-09-03

此命令用于建立新的 mark class, 如果 $\langle \text{mark class} \rangle$ 已存在, 则该命令会抛出错误; $\langle \text{bool} \rangle$ 为 `\c_true_bool` 时会额外创建 “ $\langle \text{mark class} \rangle$ -nonempty” 这个 mark class.

```
\zsect_right_mark_insert:n    \zsect_right_mark_insert:n {\langle content \rangle}
```

New: 2025-09-03

此命令会将 $\langle \text{content} \rangle$ 插入 “ztex-right” 和 “ztex-right-nonempty” (如果 $\langle \text{content} \rangle$ 非空) 这两个 mark class 中.

```
\zsect_mark_insert:nn        \zsect_mark_insert:nn {\langle class \rangle}{\langle content \rangle}
```

New: 2025-09-03

此命令用于更新 mark 列表 (其会自动处理 nonempty mark class 中的内容), 和 L^AT_EX 2_ε 中的 `\markright`, `\markboth` 作用类似. $\langle \text{class} \rangle$ 表示标题的类型, 可选值有 chapter, section, subsection, ... 等; $\langle \text{content} \rangle$ 为自定的 mark 内容.

注意: 在 article 中, `\zsectmarkinsert{section}{content}` 的作用相当于原始的 `\markboth{content}{}` 命令; 但是在 book 文档类中, 其作用相当于 `\markright{content}` 命令.

```
\zsect_mark_user_insert:nn      \zsect_mark_user_insert:nn {\<class>}{\<content>}
```

```
\zsect_mark_user_insert:(Vn|ee)
```

New: 2025-09-03

此命令**只能**用于向**新**建立的 $\langle class \rangle$ 中插入 $\langle content \rangle$, 不会影响原始的几个 mark class(其会自动处理 nonempty mark class 中的内容). 比如, 运行命令 $\backslash zsect_class_level_remap:n \{book=0\}$ 后, L^AT_EX 随即便创建了 “book” 这个 $\langle class \rangle$ 对应的 mark class; 使用 $\backslash zsect_mark_user_insert:nn \{book\}\{XXX\}$ 命令即可向 “book” 对应的这个 mark class 中插入内容 “XXX”.

NOTE: 也可以使用上述的 $\backslash zsect_mark_insert:nn$ 命令向新建的 mark class 中插入内容.

```
\zsect_markclass_lower_empty:n    \zsect_markclass_lower_empty:n {\<class>}
```

```
\zsect_markclass_lower_empty:(V|o|e)
```

New: 2025-09-03

此命令会向其下 (level 值比 $\langle class \rangle$ 大的) 的所有 mark class 中插入空内容. 因为 L^AT_EX 目前维护了 4 个主要的 mark class, 所以它会忽略所有 level 层级大于 4 的 title class.

```
\zsect_robust_left_mark:    这两个命令相较之原始的 \leftmark 和 \rightmark 命令更加的实用和健壮.
```

```
\zsect_robust_right_mark:
```

New: 2025-09-03

章节命令 考虑到 tagged pdf 支持, 目前和章节命令相关的接口命令较少, 后续会有补充.

```
\zsect_once_title:nnnnn      \zsect_once_title:nnnnn {\<keyval>}
```

```
\zsect_once_title:(nonnn|nennn|ooooo|eeeeee)    {\<class>}{\<bool>}{\<toc-content>}{\<sec-content>}
```

New: 2025-09-08

此命令用于排版一次性的标题, 不会影响其它标题格式. $\langle keyval \rangle$ 用于指定标题的属性, 可用的键值列表请参见: 节 (6.8.3); $\langle class \rangle$ 用于指定标题的类别, 可选值有 “section, chapter, ...” 等 (所有位于 $\backslash c_zsect_level_clist$ 中的 $\langle class \rangle$ 均可); $\langle bool \rangle$ 用于指定该标题是否需要编号, 当其为 $\backslash c_true_bool$ 时, 不显示编号, 为 $\backslash c_false_bool$ 时显示编号; $\langle toc-content \rangle$ 用于指定目录中的内容, 可以置为空, 此时目录中的该条目使用 $\langle sec-content \rangle$; $\langle sec-content \rangle$ 用于指定标题内容.

```
\zsect_define_title:Nnn          \zsect_define_title:Nnn <class> {\<style>}{\<keyval>}
\zsect_define_title:(Non|Nen|cnn|con|cen)
```

New: 2025-07-06

此命令用于定义标题, 所有可用的键值列表参见节首说明: 节 (6.8.1). `<class>` 可以是 “`\part`, `\section`, `\subsection`” 等. `<style>` 用于指定新建立的章节命令所属的样式. **备注:** 如果该 `<class>` 目录条目的 instance 不存在, 此时用户需要手动指定, 请参见 `\zsecdefine` 命令中的设置样例.

```
\zsect_caption_use:nnn          \zsect_caption_use:nnn {\<class>}{\<format>}{\<content>}
\zsect_caption_use:(ooo|eee)
```

New: 2025-09-07

Updated: 2025-09-10

此命令用于手动添加 caption, 它会主动调用 `\zsect_add_<class>_line:eeee` 命令添加目录. `<class>` 可选值有 “figure, table, theorem, lstlisting, algorithm, glossary”; `<format>` 用于指定 caption 的格式, 使用 #1 表示其前缀, 使用 #2 表示 caption 的内容; `<content>` 为 caption 的内容, L^AT_EX 会自动为其增加前缀: `\fnum@<class>`. (当 `<content>` 为空时, 中间的冒号 – “:” 不会显示). **备注:** `<content>` 中的脆弱命令可以不用保护.

6.8.8 编程接口: 目录

本小节描述目录相关的编程接口, 它们与目录的: 启用, 更新以及格式化相关. 基于这些命令, 用户可以更加方便地去控制目录的生成与排版.

NOTE: sect 模块重定义了原始的 `\numberline` 和 `\contentsline` 命令.

<code>\ztoc_enable_table:nn</code>	<code>\ztoc_enable_table:nn <{file}>{<table>}</code>
<code>\ztoc_enable_table_swap:nn</code>	<code>\ztoc_enable_table_swap:nn {<table>}{<file>}</code>

New: 2025-09-03

此二命令用于控制启用的目录类型, 只有当 `<table>` 类型的目录启用后, 用户才能在后文中使用 `<table>` 类型的目录; `<file>` (不能添加文件后缀) 和 `<table>` 的格式和含义与 `\ztoc_generate_table_seq:nn` 一致.

NOTE: 此命令基于前述的 `\ztoc_generate_table_seq:nn` 命令.

<code>\ztoc_stop_table:n</code>	<code>\ztoc_stop_table:n {<table>}</code>
---------------------------------	---

New: 2025-09-03

此命令会停止目录的收集, 它之后的所有目录数据均不会被记录, 用户应慎用此命令. `<table>` 可选值有 “toc, lof, lot” 等.

<code>\@dottedtocline</code>	<code>\@dottedtocline</code> <code>{<level>}{<indent>}{<name width>}{<title>}{<page>}</code>
------------------------------	---

New: 2025-09-06

此命令和 L^AT_EX 2_ε 中的 `\@dottedtocline` 定义和作用完全一致. **备注:** 更推荐用户使用后续的 `\dottedtocline:nnnnn` 命令.

<code>\dottedtocline:nnnnn</code>	<code>\dottedtocline:nnnnn</code> <code>{<level>}{<indent>}{<name width>}{<title>}{<page>}</code>
-----------------------------------	--

New: 2025-09-03

此命令基于后续的 `\zddottedtocline:nnnnnnnnnn` 命令, 用于输出目录条目. 如果 `<title>` 中含有 `\numberline` 命令, 此时 `<name width>` 用于调整 `<name>` 的宽度; 如果没有, 那么 `<name width>` 会变为当前条目的第 2 行到最后一行的 `\leftskip` 值.

注意: 在 L^AT_EX 2_ε 的原始 `\dottedtocline` 命令中, `<name width>` 被称为 “num width”.

NOTE: 该命令之后可能会使用 `l3galley` 进行重写.

`\zdottoctocline:nnnnnnnnnn`

New: 2025-09-03

`\zdottoctocline:nnnnnnnnnn``{\langle level \rangle}{\langle indent \rangle}{\langle tocrmarg \rangle}``{\langle name width \rangle}{\langle title \rangle}{\langle leader \rangle}``{\langle page spec \rangle}{\langle endl ine \rangle}{\langle before skip \rangle}`

相较于上述的 `\dottedtoctocline:nnnnn` 命令, 此命令更加灵活. `\langle page spec \rangle` 用于指定 leader 后的内容; `\langle endl ine \rangle` 会插入到行尾, 可以用于控制目录条目是否换行; `\langle before skip \rangle` 用于设置目录前的垂直间距.

注意: 在 L^AT_EX 2_ε 的原始 `\dottedtoctocline` 命令中, `\langle name width \rangle` 被称为 “num width”.

NOTE: 该命令之后可能会使用 `l3galley` 进行重写.

下面这个案例展示了 `\dottedtocline:nnnnn` 和 `\zdottedtocline:nnnnnnnnnn` 两个命令的使用方法:

例 83

```
% \usepackage{lipsum}
\ExplSyntaxOn\makeatletter
\let\TTTa\dottedtocline:nnnnn
\let\TTTb\zdottedtocline:nnnnnnnnnn
\def\ltxtocleader{
  \leaders\hbox
    { $ \m@th
      \mkern \@dotsep mu
      \hbox{.}
      \mkern \@dotsep mu
    } \hfill
}
\def\tocsepline{\par\noindent\dotfill\par}
\makeatother\ExplSyntaxOff

\tocsepline
\TTTa{2}{2em}{8em}{Test Title \fbox{A}}{99}
\TTTa{2}{2em}{6em}{\lipsum[1][1-3]}{100}
\TTTa{2}{2em}{4em}{\numberline{1.1.3}\lipsum[1][1-3]}{111}

\tocsepline
\TTTb{2}{2em}{3em}{6em}{\lipsum[1][1-3]}\ltxtocleader
{100}{\par}{2em plus 10pt}
\TTTb{2}{2em}{6em}{6em}{\numberline{1.1.3}\lipsum[1][1-3]}{
\ltxtocleader}
{\fcolorbox{blue}{gray}{111}}{\par}{2em plus 10pt}
\tocsepline
```

Test Title A		99
--	-----------	----

	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris.	100
1.1.3	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris.	111
.....		
	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris.	100
1.1.3	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris.	111
.....		

<code>\ztoc_once_tocline:nnnn</code>	<code>\ztoc_once_tocline:nnnn {<keyval>}</code>
<code>\ztoc_once_tocline:(nooo neee oooo eeee)</code>	<code>{<toc depth>}{<name>}{<title>}{<page>}</code>

New: 2025-09-08

此命令用于排版一次性的目录, 不会影响其它目录条目. `<keyval>` 的可用键值列表请参见: 节 (6.8.4); `<toc depth>` 用于指定该条目所处的目录层级; `<name>`, `<title>` 和 `<page>` 的含义在之前已经做过说明, 这里不在复述.

NOTE: 此命令不能直接使用, 其依赖于 `\ztoc@current@class` 这个变量; 用户需首先将 `\ztoc@current@class` 定义为 “section, figure, ...” 等 `<class>` 后, 才能正常使用此命令.

<code>\zsect_add_toc_line:nnnn</code>	<code>\zsect_add_toc_line:nnnn</code>
<code>\zsect_add_toc_line:(oooo neee nooe eeoe eeee)</code>	<code>{<class>}{<name>}{<title>}</code> <code>{<page>}{<anchor>}</code>

New: 2025-09-03

该命令用于向文件 `\jobname.toc` 中添加一条目录条目, 如果 toc 没有启用, 则该命令不会进行任何操作; 上述各个参数的含义请参见 节 (6.8.1), 这里不再复述.

<code>\zsect_add_table_line:nnnn</code>	<code>\zsect_add_table_line:nnnn</code>
<code>\zsect_add_table_line:(oooo eeee)</code>	<code>{<class>}{<name>}{<title>}</code> <code>{<page>}{<anchor>}</code>

New: 2025-09-03

此命令与 `\zsect_add_toc_line:nnnn` 命令类似, 但写入的目录文件为: `\jobname.lot`.

<code>\zsect_add_figure_line:nnnn</code>	<code>\zsect_add_figure_line:nnnn</code>
<code>\zsect_add_figure_line:(oooo eeee)</code>	<code>{<class>}{<name>}{<title>}</code> <code>{<page>}{<anchor>}</code>

New: 2025-09-03

此命令与 `\zsect_add_toc_line:nnnn` 命令类似, 但写入的目录文件为: `\jobname.lof`.

<code>\zsect_add_theorem_line:nnnn</code>	<code>\zsect_add_theorem_line:nnnn</code>
<code>\zsect_add_theorem_line:(oooo eeee)</code>	<code>{<class>}{<name>}{<title>}</code> <code>{<page>}{<anchor>}</code>

New: 2025-09-06

此命令与 `\zsect_add_toc_line:nnnn` 命令类似, 但写入的目录文件为: `\jobname.lot`.

<code>\zsect_add_lstlisting_line:nnnn</code>	<code>\zsect_add_lstlisting_line:nnnn</code>
<code>\zsect_add_lstlisting_line:(oooo eeee)</code>	<code>{<class>}{<name>}{<title>}</code> <code>{<page>}{<anchor>}</code>

New: 2025-09-06

此命令与 `\zsect_add_toc_line:nnnn` 命令类似, 但写入的目录文件为: `\jobname.lol`.

<code>\zsect_add_glossary_line:nnnn</code>	<code>\zsect_add_glossary_line:nnnn</code>
<code>\zsect_add_glossary_line:(oooo eeee)</code>	<code>{\langle class \rangle}{\langle name \rangle}{\langle title \rangle}</code>
	<code>{\langle page \rangle}{\langle anchor \rangle}</code>

New: 2025-09-06

此命令与 `\zsect_add_toc_line:nnnn` 命令类似, 但写入的目录文件为: `\jobname.log`.

NOTE: 此命令现在还未完全适配, 用户暂时不要使用.

<code>\zsect_add_algorithm_line:nnnn</code>	<code>\zsect_add_algorithm_line:nnnn</code>
<code>\zsect_add_algorithm_line:(oooo eeee)</code>	<code>{\langle class \rangle}{\langle name \rangle}{\langle title \rangle}</code>
	<code>{\langle page \rangle}{\langle anchor \rangle}</code>

New: 2025-09-06

此命令与 `\zsect_add_toc_line:nnnn` 命令类似, 但写入的目录文件为: `\jobname.loa`.

<code>\zsect_add_to_table:nn</code>	<code>\zsect_add_to_table:nn {\langle table \rangle}{\langle content \rangle}</code>
<code>\zsect_add_to_table:(no oo ne ee)</code>	

New: 2025-09-03

此命令和上述的 `\zsect_add_toc_line:nnnn` 命令类似, 如果类型为 `\langle table \rangle` 的目录没有启用, 则该命令不会进行任何操作; `\langle content \rangle` 为写入的内容, 形式通常为:

$$\backslash\mathrm{contentsline}\langle\mathrm{class}\rangle\{\langle\mathrm{name}\rangle\}\{\langle\mathrm{title}\rangle\}\{\langle\mathrm{page}\rangle\}\{\langle\mathrm{anchor}\rangle\}$$

`\langle table \rangle` 为 `\langle table \rangle` 类型, 可选值有 “toc, lot, lof, ...” 等.

<code>\ztoc_lcmd_setup:nn</code>	<code>\ztoc_lcmd_setup:nn {\langle class \rangle}{\langle level \rangle}</code>
<code>\ztoc_lcmd_setup:(oo ee)</code>	

New: 2025-09-03

此命令用于创建目录中相应的 `\l@{\langle class \rangle}` 命令, 应用于 chapter, section 这样的常规 class; `\langle class \rangle` 中不包含 “\”, `\langle level \rangle` 为一个整数.

<code>\ztoc_special_lcmd_setup:nn</code>	<code>\ztoc_special_lcmd_setup:nn {\langle class \rangle}{\langle level \rangle}</code>
<code>\ztoc_special_lcmd_setup:(oo ee)</code>	

New: 2025-09-03

此命令用于创建目录中相应的 `\l@{\langle class \rangle}` 命令, 应用于 figure, table 这样的非常规 class; `\langle class \rangle` 中不包含 “\”, `\langle level \rangle` 为一个整数.

<code>\ztoc_generate_table_seq:nn</code>	<code>\ztoc_generate_table_seq:nn {\langle file \rangle}{\langle table \rangle}</code>
--	--

New: 2025-09-03

此命令用于生成目录对应的 seq 数据 (变量 `\g_ztoc_{\langle table \rangle}_seq` 和变量 `\g_ztoc_keyval_{\langle table \rangle}_seq`). `\langle file \rangle` 为文件名, 不能包含文件的扩展名; `\langle table \rangle` 为目录类型, 所有可选值为: “toc, lot, lof, lom, loa, log, lol”, 分别代表 “(章节) 目录, 表格目录, 图片目录, 定理目录, 算法目录, 术语目录, 代码目录”.

```
\ztoc_generate_keyvaltable_seq:NN \ztoc_generate_keyvaltable_seq:NN <ori seq> <seq>
\ztoc_generate_keyvaltable_seq:cc
```

New: 2025-09-19

此命令用于从 $\langle ori\ seq \rangle$ 生成 keyval seq 变量 $\langle seq \rangle$ (即, 变量 $\backslash g_ztoc_keyval\langle table \rangle_seq$), 且其赋值是全局的. $\langle ori\ seq \rangle$ 可以使用后续的目录筛选命令生成, 如: $\backslash ztoc_table_filter_byclass:nnNN$; 也可以使用 ztool 提供的 $\backslash ztool_read_file_as_seq:nnN$ 命令生成.

```
\ztoc_generate_keyvaltable_seq:nN \ztoc_generate_keyvaltable_seq:nN {\file} <seq>
\ztoc_generate_keyvaltable_seq:(nc|oN|eN)
```

New: 2025-09-03

此命令用于从 $\langle file \rangle$ 生成 keyval seq 数据 (变量 $\backslash g_ztoc_keyval\langle table \rangle_seq$), 且其赋值是全局的; $\langle file \rangle$ 中 **必须** 包含文件拓展名, 该文件中保存的目录类型可以是: “(章节) 目录, 表格目录, 图片目录, 定理目录, 算法目录, 术语目录”; 生成 keyval(seq) 数据保存于 $\langle seq \rangle$ 变量中.

NOTE: 前述的 $\backslash ztoc_generate_table_seq:nn$ 命令便是基于此命令和 ztool 宏包中的 $\backslash ztool_gread_file_as_seq:nnN$ 命令

```
\ztoc_table_filter_key_byclass:nnnNN \ztoc_table_filter_key_byclass:nnnNN
\ztoc_table_filter_key_byclass:(oooNN|eeeNN|nnncc|eeec) {\class}{\index}{\key}
\ztoc_table_filter_key_byclass:(oooNN|eeeNN|nnncc|eeec) <seq> <filter seq>
```

New: 2025-09-12

此命令会从 $\langle seq \rangle$ 变量中筛选目录条目, 并将筛选结果保存在 $\langle filter\ seq \rangle$ 中 (该赋值是全局的). 筛选规则由 $\langle class \rangle$ 和 $\langle index \rangle$ 指定: 筛选目录条目 $\langle key \rangle$ 字段的值 (筛选该条目及其包含的所有子目录); $\langle class \rangle$ 的可选值有 “chapter, section, ...” 等; $\langle index \rangle$ 为一个整数, 表示所有 $\langle class \rangle$ 条目的中该条目的绝对次序 (更加详细的介绍可以参见 $\backslash zlocaltoc$ 命令); $\langle key \rangle$ 的可选值有: “class, name, title, page, anchor, raw”.

<code>\ztoc_table_filter_byclass:nnNN</code>	<code>\ztoc_table_filter_byclass:nnNN</code>
<code>\ztoc_table_filter_byclass:(nncc neNN enNN eeNN eccc)</code>	<code>{\langle class \rangle}{\langle index \rangle}</code>
New: 2025-09-03	<code>\langle seq \rangle \langle filter seq \rangle</code>
Updated: 2025-09-12	

此命令基于上述的 `\ztoc_table_filter_key_byclass:nnnNN` 命令, 用于从 `\langle seq \rangle` 变量中筛选目录条目, 并将筛选结果保存在 `\langle filter seq \rangle` 中 (该赋值是全局的). 筛选规则由 `\langle class \rangle` 和 `\langle index \rangle` 指定: 筛选的是 “raw” 字段对应的值; `\langle class \rangle` 的可选值有 “chapter, section, ...” 等; `\langle index \rangle` 为一个整数, 表示所有 `\langle class \rangle` 条目的中该条目的绝对次序 (更加详细的介绍可以参见 `\zlocaltoc` 命令).

<code>\ztoc_table_filter_byname:nnNN</code>	<code>\ztoc_table_filter_byname:nnNN</code>
<code>\ztoc_table_filter_byname:(eeNN nncc eccc)</code>	<code>{\langle name \rangle}{\langle title \rangle}</code>
New: 2025-09-12	<code>\langle seq \rangle \langle filter prop \rangle</code>

此命令会从 `\langle seq \rangle` 变量中筛选目录条目, 并将筛选结果保存在 `\langle filter prop \rangle` 中 (该赋值是全局的). `\langle name \rangle` 和 `\langle title \rangle` 同时匹配的条目才会被筛选; `\langle filter prop \rangle` 表示筛选结果对应的键值 (prop) 变量, 如果不存在则返回空的键值变量; `\langle filter prop \rangle` 提供的键值可以参见: 段落 (6.8.5.1) 中 `data.ptoc` 文件.

NOTE: `\langle name \rangle` 和 `\langle title \rangle` 比较的是字符串本身, 而不是 token 的比较; 所以部分情况下, 用户需要将它们二者提前展开.

<code>\ztoc_tocline_index_byclasstitle:nnnN</code>	☆ <code>\ztoc_tocline_index_byclasstitle:nnnN</code>
<code>\ztoc_tocline_index_byclasstitle:(nnnc eeen eeec)</code>	☆ <code>{\langle class \rangle}{\langle title \rangle}</code>
New: 2025-09-15	<code>{\langle bool \rangle}{\langle seq \rangle}</code>

此命令用于确定目录条目在 `\langle seq \rangle` 变量中索引, 并且在 e-type 中可以被完全展开. 目录条目由 `\langle class \rangle` 和 `\langle title \rangle` 确定, 二者需要同时匹配; 如果 `\langle seq \rangle` 中没有匹配的条目, 此命令总是返回一个小于零的整数值; 当 `\langle bool \rangle` 为 `\c_true_bool` 时, 此命令返回**相对**索引, 即该条目在所有相同 `\langle class \rangle` 目录条目中的索引; 当 `\langle bool \rangle` 为 `\c_false_bool` 时, 此命令返回**绝对**索引, 即该条目在整个目录中的所用, 不考虑 `\langle class \rangle`.

<code>\ztoc_tocline_index_byname</code> <code>title:nnN</code>	☆	<code>\ztoc_tocline_index_byname</code> <code>title:nnN</code>
<code>\ztoc_tocline_index_byname</code> <code>title:(nnc eeN eec)</code>	☆	<code>{\langle name \rangle}{\langle title \rangle} \langle seq \rangle</code>

New: 2025-09-12

此命令用于确定目录条目在 `\langle seq \rangle` 变量中**绝对索引**, 并且在 `e-type` 中可以被完全展开. 目录条目由 `\langle name \rangle` 和 `\langle title \rangle` 确定, 二者需要同时匹配; 如果 `\langle seq \rangle` 中没有匹配的条目, 此命令总是返回一个小于零的整数. **备注:** 此命令无法返回**相对索引**.

NOTE: `\langle name \rangle` 和 `\langle title \rangle` 比较的是字符串本身, 而不是 token 的比较; 所以部分情况下, 用户需要将它们二者提前展开.

<code>\ztoc_tocline_level_byname</code> <code>title:Nnnn</code>	☆	<code>\ztoc_tocline_level_byname</code> <code>title:Nnnn</code>
		<code>\langle seq \rangle{\langle name \rangle}{\langle title \rangle}{\langle offset \rangle}</code>

New: 2025-09-15

此命令首先通过 `\langle name \rangle` 和 `\langle title \rangle` 确定初始目录条目的索引; 当确定此条目后, 可以通过偏移值 (此为一个整数) `\langle offset \rangle` 指定一条 (新) 目录条目, 返回 (新) 指定条目的 class level. **备注:** 此命令基于上述的 `\ztoc_tocline_index_byname``title:nnN` 命令.

<code>\ztoc_tocline_level_compare_p:Nnnnnn</code>	☆	<code>\ztoc_tocline_level_compare:NnnnnnTF</code>
<code>\ztoc_tocline_level_compare_p:(cnnnnn Neennn ceennn)</code>	☆	<code>\langle seq \rangle{\langle name \rangle}{\langle title \rangle}</code>
<code>\ztoc_tocline_level_compare:NnnnnnTF</code>	☆	<code>{\langle offset_1 \rangle}{\langle offset_2 \rangle}{\langle relation \rangle}</code>
<code>\ztoc_tocline_level_compare:(cnnnnn Neennn ceennn)TF</code>	☆	<code>{\langle true code \rangle}{\langle false code \rangle}</code>

New: 2025-09-15

此命令首先通过 `\langle name \rangle` 和 `\langle title \rangle` 确定目标目录在 `\langle seq \rangle` 中的索引, 在此基础上, `\langle offset_1 \rangle` 和 `\langle offset_2 \rangle` 指定了其余的两条目录条目; 随后比较这两条目录条目的 class level, 比较方法由 `\langle relation \rangle` 指定, 可以选择: “<, =, <, <=, >=” 等. **备注:** 此命令基于上述的 `\ztoc_tocline_index_byname``title:nnN` 命令; 此命令在 `f`-型参数中**无法**被完全展开.

<code>\ztoc_tocline_card:nnN</code>	☆	<code>\ztoc_tocline_card:nnN</code>
<code>\ztoc_tocline_card:(ooN eeN nnc eec)</code>	☆	<code>{\langle class \rangle}{\langle title \rangle} \langle seq \rangle</code>

New: 2025-09-18

此命令返回由 `\langle class \rangle` 和 `\langle title \rangle` 所确定目录的**子目录** (card 即 cardinality), 如果由 `\langle class \rangle` 和 `\langle title \rangle` 所确定目标目录不存在, 其返回 “-1”; `\langle seq \rangle` 为目录对应的 keyval `seq` 变量, 即 `\g_ztoc_keyval\langle table \rangle_seq`.

```

\ztoc_tocline_level_data:N ☆ \ztoc_tocline_level_data:N <seq>
\ztoc_tocline_level_data:c ☆ \ztoc_tocline_hilevel:N <seq>
\ztoc_tocline_hilevel:N ☆ \ztoc_tocline_lolevel:N <seq>
\ztoc_tocline_hilevel:c ☆
\ztoc_tocline_lolevel:N ☆
\ztoc_tocline_lolevel:c ☆

```

New: 2025-09-18

这系列的命令返回 $\langle seq \rangle$ 中的 class level 数据, $\langle seq \rangle$ 为目录对应的 keyval seq 变量. $\backslash ztoc_tocline_level_data:N$ 会返回所有目录条目的 level 值; 而 $\backslash ztoc_tocline_hilevel:N$ 或 $\backslash ztoc_tocline_lolevel:N$ 只返回 $\langle seq \rangle$ 中最小的 level 值或最大的 level 值 (因为 $\langle class \rangle$ 级别越高, 对应的 level 值越小).

```

\ztoc_if_first_tocline_p: ☆ \ztoc_if_first_tocline:TF {\true code}{\false code}
\ztoc_if_first_tocline:TF ☆

```

New: 2025-09-12

此命令和 etoc 宏包提供的 $\backslash tociffirst$ 作用相同, 用于判断当前的目录是否为其所在的 level 的第一项.

```

\ztoc_if_middle_tocline_p:Nnn ☆ \ztoc_if_middle_tocline:NnnTF
\ztoc_if_middle_tocline_p:(Nee|cee) ☆ \langle seq \rangle \langle name \rangle \langle title \rangle
\ztoc_if_middle_tocline:NnnTF ☆ {\true code}{\false code}
\ztoc_if_middle_tocline:(Nee|cee)TF ☆

```

New: 2025-09-12

此命令用于判断当前目录条目是否位于两条同级目录之间. $\langle seq \rangle$ 用于指定当前所排版目录对应的 keyval seq 变量. **备注:** 实际上是判断当前目录条目之后的条目是否为同级目录.

```

\ztoc_if_last_tocline_p:Nnn ☆ \ztoc_if_last_tocline:NnnTF
\ztoc_if_last_tocline_p:(Nee|cee) ☆ \langle seq \rangle \langle name \rangle \langle title \rangle
\ztoc_if_last_tocline:NnnTF ☆ {\true code}{\false code}
\ztoc_if_last_tocline:(Nee|cee)TF ☆

```

New: 2025-09-12

此命令用于判断当前目录是否为其所在 level 的最后一项, $\langle seq \rangle$ 用于指定当前所排版目录对应的 keyval seq 变量. **备注:** 在 f-type 中, 此命令不能被完全展开.

NOTE: 当最后的目录条目为 “ $\backslash contentsline\{subsection\}\{\}\{\}$ ”,
 $\backslash contentsline\{subsection\}\{\}\{\}$ ” 时, 此命令对于上述 “section” 条目的判断会出错, 此时用户可以手动添加对应的代码.

```
\ztoc_maintable_seq_save:n      \ztoc_maintable_seq_save:n {\table}
\ztoc_maintable_seq_restore:n   \ztoc_maintable_seq_restore:n {\table}
```

New: 2025-09-03

Updated: 2025-09-15

命令 `\ztoc_maintable_seq_save:n` 会将当前的目录数据保存于两个临时 seq 中; 命令 `\ztoc_maintable_seq_restore:n` 用于将原始的目录数据还原, 并清空上述的两个临时 seq; `\table` 的描述请参见命令 `\ztoc_generate_table_seq:nn` 的说明.

```
\ztoc_table_typeset_before:
\ztoc_table_typeset_after:
```

New: 2025-09-05

这两个命令常常在排版目录时使用, 用于初始化目录解析 (hook) 相关的变量.

```
\ztoc_save_main_table:n      \ztoc_save_main_table:n {\file}
\ztoc_restore_main_table:n   \ztoc_restore_main_table:n {\file}
```

New: 2025-09-07

这两个命令用于保存和恢复当前文档的主目录数据. 当用户需要自定义目录文件时, 可以使用这两个命令. `\file` 为文件名, 不包含文件后缀.

```
\ztoc_table_typeset:Nn      \ztoc_table_typeset:Nn <seq>{\separator}
\ztoc_table_typeset:(Ne|cn|ce)
```

New: 2025-09-05

此命令用于排版目录, `<seq>` 表示目录对应的数据文件, `<separator>` 表示相邻目录条目之间的分隔符 (常常置为空). **注意:** 用户如果想直接使用 `\g_ztoc_toc_seq` 等变量输出目录, 请在其前后分别加上 `\ztoc_table_typeset_before:` 和 `\ztoc_table_typeset_after:` 命令.

<code>\ztoc_normal_format_set:Nn</code>	<code>\ztoc_normal_format_set:Nn <class> {<format>}</code>
<code>\ztoc_special_format_set:nn</code>	<code>\ztoc_special_format_set:nn {<class>} {<format>}</code>
<code>\ztoc_all_format_set:nn</code>	<code>\ztoc_all_format_set:nn {<classes>} {<format>}</code>

New: 2025-09-03

这三个命令用于设置目录的格式。 `\ztoc_normal_format_set:Nn` 用于设置“part, chapter, section, ...”等常规目录条目的样式, `<class>` 的可选值有 `\part`, `\chapter`, ... 等; `\ztoc_special_format_set:nn` 用于设置“figure, table, algorithm, ...”等特殊目录条目的样式, `<class>` 的可选值有 `figure`, `table`, `algorithm`, ... 等; 命令 `\ztoc_all_format_set:nn` 的设置对象涵盖了以上两者, `<classes>` 为一个逗号分割列表, 其可选值有 `\part`, `\chapter`, `figure`, `table`, ... 等。

<code>\ztoc_group_parser:NNNnnnnn</code>	<code>\ztoc_group_parser:NNNnnnnn</code>
<code>\ztoc_group_parser:(cccnnnnn NNNneeee ccceeeee)</code>	<code><seq><cmd><t1></code>
	<code>{<before>}{<begin>}{<end>}{<after>}</code>
	<code>{<step code>}</code>

New: 2025-09-15

Updated: 2025-09-19

此命令按照 `<cmd>` 对数据 `<seq>` 进行格式化, 格式化后的结果保存在 `<t1>` 中。 `<seq>` 为目录对应的 keyval `seq` 变量; `<cmd>` 接受 4 个参数, 用于指定目录条目的格式化方式, `<cmd>` 中各个参数的含义如下:

- #1: 目录条目中的 `<class>` 值;
- #2: 目录条目中的 `<name>` 值;
- #3: 目录条目中的 `<title>` 值;
- #4: 目录条目中的 `<page>` 值;

`<before>`, `<begin>`, `<end>` 以及 `<after>` 在写入 `<t1>` 时会被执行 e-型展开, 所以请将脆弱命令使用 `\noexpand` 或 `\unexpanded` 保护起来; `<step code>` 在每一条目处理完成后执行, 可以置为空。 **备注:** 变量 `<t1>` 的赋值是全局的; 此命令比后续的 `\ztoc_group_hook_create:Nnnn` 更加灵活, 用户完全可以使用此命令实现后续的 `\ztoc_group_hook_create:Nnnn` 命令。

<code>\ztoc_group_hook_create:Nnnn</code>	<code>\ztoc_group_hook_create:Nnnn <cmd></code>
<code>\ztoc_group_hook_create:cnnn</code>	<code>{<toc dpeth>}{<name>}{<title>}{<page>}</code>

New: 2025-09-08

此命令用于解析目录数据 (一次处理一条数据), 然后增加对应的 hooks. 这些 hooks 的名称可参见 `\ztoc_group_hook_add:nn` 中的说明. `<cmd>` 为生成目录条目的命令, 命令必须为 `\<cmd>:nnn` 的形式, 它们用于接受后续的 3 个参数; 后续 3 个参数的意义请参见命令 `\zsect_add_toc_line:nnnn` 的说明.

<code>\ztoc_group_hook_add:nn</code>	<code>\ztoc_group_hook_add:nn {<hook>}{<code>}</code>
--------------------------------------	---

New: 2025-09-03

此命令用于将 `<code>` 加入目录的 `<hook>` 中 (这些 hook 可以由 `\ztoc_group_hook_create:Nnnn` 命令创建). `<hook>` 的格式如下:

`<class>, <int>, before|begin|end|after`

用户可以在 `<code>` 中添加对应的格式化代码.

NOTE: 上述命令 `<hook>` 参数中的内容会执行 'e'-型展开.

6.8.9 编程接口: 模板

章节命令模板和目录模板均由下面的命令建立, 且下述命令建立模板的 $\langle template \rangle$ $\langle type \rangle$ 中均含有 “ztex” 前缀, 其它参数目前没有变动. 普通用户建议直接跳过本节之后的内容, 后续命令的使用难度较大.

<code>\ztex_new_templatetype:nn</code>	<code>\ztex_new_templatetype:nn {$\langle template \rangle$} {$\langle no. of args \rangle$}</code>
New: 2025-09-17	此命令用于创建模板, 接受的参数个数由 $\langle no. of args \rangle$ 指定.

<code>\ztex_declare_template_interface:nnnn</code>	<code>\ztex_declare_template_interface:nnnn</code>
New: 2025-09-17	<code>{$\langle type \rangle$} {$\langle template \rangle$} {$\langle no. of args \rangle$}</code> <code>{$\langle key list \rangle$}</code>

此命令用于声明模板的接口 (声明一系列键以及键的属性), 接口参数个数必须和 $\langle type \rangle$ 一致.

<code>\ztex_declare_template_code:nnnnn</code>	<code>\ztex_declare_template_code:nnnnn</code>
New: 2025-09-17	<code>{$\langle type \rangle$} {$\langle template \rangle$} {$\langle no. of args \rangle$}</code> <code>{$\langle key binding \rangle$} {$\langle template code \rangle$}</code>

此命令用于声明模板的具体内容 (将键绑定到特定变量, 在 $\langle code \rangle$ 中使用该变量), 由 $\langle template code \rangle$ 指定, 接口参数个数必须和 $\langle type \rangle$ 一致.

<code>\ztex_declare_template_copy:nnn</code>	<code>\ztex_declare_template_copy:nnn</code>
New: 2025-09-17	<code>{$\langle type \rangle$} {$\langle template_2 \rangle$} {$\langle template_1 \rangle$}</code>

将 $\langle type \rangle$ 的模板 $\langle template_1 \rangle$ 复制到 $\langle template_2 \rangle$. 在之后, $\langle template_2 \rangle$ 将独立于 $\langle template_1 \rangle$.

<code>\ztex_declare_instance:nnnn</code>	<code>\ztex_declare_instance:nnnn</code>
New: 2025-09-17	<code>{$\langle type \rangle$} {$\langle instance \rangle$} {$\langle template \rangle$}</code> <code>{$\langle parameters \rangle$}</code>

此命令用于建立模板实例, 在 $\langle parameters \rangle$ 中可以对 $\langle template \rangle$ 提供的键进行设置.

<code>\ztex_declare_instance_copy:nnn</code>	<code>\ztex_declare_instance_copy:nnn</code>
New: 2025-09-17	<code>{$\langle type \rangle$} {$\langle instance_2 \rangle$} {$\langle instance_1 \rangle$}</code>

将 $\langle type \rangle$ 中 $\langle instance_1 \rangle$ 的值复制到 $\langle instance_2 \rangle$. 在之后, $\langle instance_2 \rangle$ 将独立于 $\langle instance_1 \rangle$.

<code>\ztex_use_instance:nn</code>	<code>\ztex_use_instance:nn {$\langle type \rangle$} {$\langle instance \rangle$} $\langle arguments \rangle$</code>
New: 2025-09-17	使用 $\langle type \rangle$ 中的 $\langle instance \rangle$ 实例, $\langle arguments \rangle$ 应该根据 $\langle type \rangle$ 的定义提供.

`\ztex_use_template:nnn`

New: 2025-09-17

`\ztex_use_template:nnn`
`{\langle type \rangle}{\langle template \rangle}{\langle settings \rangle}`
`\langle arguments \rangle`

使用 $\langle type \rangle$ 中的 $\langle template \rangle$ 模板. 在 $\langle settings \rangle$ 中可以对 $\langle template \rangle$ 提供的键进行设置; $\langle arguments \rangle$ 应该根据 $\langle type \rangle$ 的定义提供. **备注:** 如果某个键可接受一个具体的 instance, 那么此命令也可以作为这个键的值.

`\ztex_edit_instance:nnn`

New: 2025-09-17

`\ztex_edit_instance:nnn` $\{\langle type \rangle\}\{\langle instance \rangle\}\{\langle new values \rangle\}$

此命令用于对已存在的实例进行修改, $\langle new values \rangle$ 是一个键值列表.

`\ztex_edit_template_defaults:nnn`

New: 2025-09-17

`\ztex_edit_template_defaults:nnn`
`{\langle type \rangle}{\langle template \rangle}{\langle new defaults \rangle}`

此命令用于对已声明的模板进行修改, $\langle new defaults \rangle$ 是一个键值列表.

模板/实例信息 下面的命令用于显示模板或实例的内部信息, 常常在调试模板时使用:

`\ztex_show_instance_values:nn`

New: 2025-09-20

`\ztex_show_instance_values:nn` $\{\langle type \rangle\}\{\langle instance \rangle\}$

显示 $\langle type \rangle$ 中 $\langle instance \rangle$ 实例的值.

`\ztex_show_template_code:nn`

New: 2025-09-20

`\ztex_show_template_code:nn` $\{\langle type \rangle\}\{\langle template \rangle\}$

显示 $\langle type \rangle$ 中 $\langle template \rangle$ 模板的代码实现.

`\ztex_show_template_defaults:nn`

New: 2025-09-20

`\ztex_show_template_defaults:nn` $\{\langle type \rangle\}\{\langle template \rangle\}$

显示 $\langle type \rangle$ 中 $\langle template \rangle$ 模板的默认值.

`\ztex_show_template_interface:nn`

New: 2025-09-20

`\ztex_show_template_interface:nn` $\{\langle type \rangle\}\{\langle template \rangle\}$

显示 $\langle type \rangle$ 中 $\langle template \rangle$ 模板的接口, 包括其提供的键和键属性.

`\ztex_show_template_variables:nn`

New: 2025-09-20

`\ztex_show_template_variables:nn` $\{\langle type \rangle\}\{\langle template \rangle\}$

显示 $\langle type \rangle$ 中 $\langle template \rangle$ 模板所提供键对应的绑定变量信息.

6.8.10 编程接口: 杂项

```
\zsect_instance_set_fallback:nn \zsect_instance_set_fallback:nn {\original}{\fallback}
```

New: 2025-09-10

此命令用于给 Instance 设置 fallback. **NOTE: 此命令目前不可用.**

```
\zsect_restore_protect: \zsect_unexpand_protect:
```

New: 2025-08-21

这两个命令用于重定义 L^AT_EX 中的 `\protect` 宏, 前者将其重定义为 `\relax`, 后者将其重定义为 `\noexpand`.

注意: 当 `section` 等命令中含有其它命令时, 请使用 `\protect` 将其保护起来, 如 `\section{xxx-\protect\cmd-yyy}`; 但是“`\caption` 命令”或“数学环境名称 (`\zthmname`)”中, 其内的命令可以不添加 `\protect`, 因其原本就被 `\exp_not:n` 或 `\exp_not:N` 保护.

```
\zsect_leaders:nnnnn \zsect_leaders:nnnnn {\type}{\repeat}{\width}{\raise}{\skip}
```

New: 2025-09-03

此命令用于引导线等内容的排版, 其本身为一个 glue. `<type>` 表示 leader 的类型, 可选值有: “`<空>`, `c`, `x`”; `<repeat>` 表示基本的重复元素; `<width>` 表示每一个 `<repeat>` 所占的宽度; `<raise>` 表示 `<repeat>` 在垂直方向上的位移, 向上为正; `<skip>` 表示该 leader 所占的总宽度, 接受一个 `skip` 值.

```
\zsect_ltx_float_setup:nnnnnn \zsect_ltx_float_setup:nnnnnn
{\counter}{\counter form}{\position}
{\level}{\ext}{\num}
```

New: 2025-09-07

Updated: 2025-09-20

此命令用于创建与 L^AT_EX 2_ε 中 `figure`, `table` 等环境类似的浮动体环境. `<counter>` 为计数器名称; `<counter form>` 为计数器的格式; `<position>` 为浮动体的放置位置; `<level>` 为目录层级; `<ext>` 为待写入的目录文件后缀名; `<num>` 为 `\caption` 命令中的编号.

```
\zsect_ltx_float_begin:n \zsect_ltx_float_begin:n {\class}
\zsect_ltx_float_end:
```

New: 2025-09-07

这两个命令用于开启和关闭 `<class>` 对应的浮动环境.

下面这个使用样例展示了如何创建一个具有浮动机制的 `algorithmX` 环境:

```
\ExplSyntaxOn\makeatletter
% create float env: algorithmX
\zsect_ltx_float_setup:nnnnnn
{algorithmX}{\@arabic}{tbp}{3}{loa}
{AlgorithmX\nobreakspace\thealgorithmX}
```

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```
\NewDocumentEnvironment{algorithmX}{{}  
  { \zsect_ltx_float_begin:n {algorithmX}  
  { \zsect_ltx_float_end: }  
\makeatother\ExplSyntaxOff  
  
% use this env(support optional argument):  
\begin{algorithmX}[!htb]  
\caption{algorithmX Example}  
\end{algorithmX}
```

6.9 sclist 模块

Semicolon (comma) list(简称为 `sclist`) 与 `expl3` 中的 “`clist`” 类似, 第一层分隔符为 “;”, 第二层分隔符可以为 “,”; x_YTeX 创建此模块是为了在处理以 “;” 划分的数据时维持相关操作的 “可展性”; x_YTeX 的 `sclist` 库提供了以下的一些命令:

<code>\sclist_new:N</code>	<code>\sclist_new:N <sclist var></code>
----------------------------	---

<code>\sclist_new:c</code>	该命令与原始的 <code>\clist_new:N</code> 命令类似.
----------------------------	---

New: 2025-06-20

<code>\sclist_const:Nn</code>	<code>\sclist_const:Nn <sclist var> {<semicolon list>}</code>
-------------------------------	---

<code>\sclist_const:(Ne cn ce)</code>	该命令与原始的 <code>\clist_cont:Nn</code> 命令类似.
---------------------------------------	---

New: 2025-06-20

<code>\sclist_clear:N</code>	<code>\sclist_clear:N <sclist var></code>
------------------------------	---

<code>\sclist_clear:c</code>	该命令与原始的 <code>\clist_clear:N</code> 命令类似.
------------------------------	---

<code>\sclist_gclear:N</code>	
-------------------------------	--

<code>\sclist_gclear:c</code>	
-------------------------------	--

New: 2025-06-20

<code>\sclist_clear_new:N</code>	<code>\sclist_clear_new:N <sclist var></code>
----------------------------------	---

<code>\sclist_clear_new:c</code>	该命令与原始的 <code>\clist_clear_new:N</code> 命令类似.
----------------------------------	---

<code>\sclist_gclear_new:N</code>	
-----------------------------------	--

<code>\sclist_gclear_new:c</code>	
-----------------------------------	--

New: 2025-06-20

<code>\sclist_set_eq:NN</code>	<code>\sclist_set_eq:NN <sclist var₁> <sclist var₂></code>
--------------------------------	--

<code>\sclist_set_eq:(cN Nc cc)</code>	该命令与原始的 <code>\clist_set_eq:NN</code> 命令类似.
--	---

<code>\sclist_gset_eq:NN</code>	
---------------------------------	--

<code>\sclist_gset_eq:(cN Nc cc)</code>	
---	--

New: 2025-06-20

<code>\sclist_set:Nn</code>	<code>\sclist_set:Nn <sclist var> {<item₁>; ...; <item_n>}</code>
-----------------------------	--

<code>\sclist_set:(NV Ne No cn cV ce co)</code>	
---	--

<code>\sclist_gset:Nn</code>	
------------------------------	--

<code>\sclist_gset:(NV Ne No cn cV ce co)</code>	
--	--

New: 2025-06-20

该命令与原始的 `\clist_set:Nn` 命令类似.

<code>\sclist_if_empty_p:N</code> *	<code>\sclist_if_empty_p:N</code> $\langle sclist\ var \rangle$
<code>\sclist_if_empty_p:c</code> *	<code>\sclist_if_empty:N</code> $\langle sclist\ var \rangle$ $\{\langle true\ code \rangle\}$ $\{\langle false\ code \rangle\}$
<code>\sclist_if_empty:N</code> <u>TF</u> *	该命令与原始的 <code>\clist_if_empty:N</code> 命令类似.
<code>\sclist_if_empty:c</code> <u>TF</u> *	

New: 2025-06-20

<code>\sclist_if_empty_p:N</code> *	<code>\sclist_if_empty_p:n</code> $\langle sclist\ var \rangle$
<code>\sclist_if_empty_p:c</code> *	<code>\sclist_if_empty:n</code> $\langle semicolon\ list \rangle$ $\{\langle true\ code \rangle\}$ $\{\langle false\ code \rangle\}$
<code>\sclist_if_empty:N</code> <u>TF</u> *	该命令与原始的 <code>\clist_if_empty:n</code> 命令类似.
<code>\sclist_if_empty:c</code> <u>TF</u> *	

New: 2025-06-20

<code>\sclist_map_function:NN</code> ☆	<code>\sclist_map_function:NN</code> $\langle sclist\ var \rangle$ $\langle function \rangle$
<code>\sclist_map_function:cN</code> ☆	此系列命令与原始的 <code>\clist_map_function:NN</code> 命令类似.
<code>\sclist_map_function:nN</code> ☆	
<code>\sclist_map_function:eN</code> ☆	

New: 2025-06-20

<code>\sclist_map_tokens:Nn</code> ☆	<code>\sclist_map_tokens:Nn</code> $\langle sclist\ var \rangle$ $\{\langle code \rangle\}$
<code>\sclist_map_tokens:cn</code> ☆	此系列命令与原始的 <code>\clist_map_tokens:Nn</code> 命令类似.
<code>\sclist_map_tokens:nn</code> ☆	

New: 2025-06-20

<code>\sclist_count:N</code> *	<code>\sclist_count:N</code> $\langle sclist\ var \rangle$
<code>\sclist_count:c</code> *	该命令与原始的 <code>\clist_count:N</code> 命令类似.
<code>\sclist_count:n</code> *	
<code>\sclist_count:e</code> *	

New: 2025-06-20

<code>\sclist_item:Nn</code> *	<code>\sclist_item:Nn</code> $\langle sclist\ var \rangle$ $\{\langle int\ expr \rangle\}$
<code>\sclist_item:cn</code> *	该命令与原始的 <code>\clist_item:Nn</code> 命令类似.
<code>\sclist_item:nn</code> *	
<code>\sclist_item:en</code> *	

New: 2025-06-20

<code>\sclist_show:N</code>	<code>\sclist_show:N</code> $\langle sclist\ var \rangle$
<code>\sclist_show:c</code>	该命令与原始的 <code>\clist_show:N</code> 命令类似.

New: 2025-06-20

<code>\sclist_show:n</code>	<code>\sclist_show:n {<tokens>}</code>
<code>New: 2025-06-20</code>	该命令与原始的 <code>\clist_show:n</code> 命令类似.
<code>\sclist_log:N</code>	<code>\sclist_log:N <sclist var></code>
<code>\sclist_log:c</code>	该命令与原始的 <code>\clist_log:N</code> 命令类似.
<code>New: 2025-06-20</code>	
<code>\sclist_log:n</code>	<code>\sclist_log:n {<tokens>}</code>
<code>New: 2025-06-20</code>	该命令与原始的 <code>\clist_log:n</code> 命令类似.

下面这个案例展示了 sclist 中的 `\sclist_map_tokens:nn` 和 `\sclist_map_tokens:Nn` 两个命令的基本使用方法:

```
\ExplSyntaxOn
\sclist_new:N \l_tmpc_sclist
\sclist_set:Nn \l_tmpc_sclist {1;23;456;}
\cs_set:Npn \__test_sclist_map:nn #1#2 {[#1](#2)|}
\def\TTTa{
  \sclist_map_tokens:nn {a;bc;def}
  { \__test_sclist_map:nn {XX} }
}
\def\TTTb{
  \sclist_map_tokens:Nn \l_tmpc_sclist
  { \__test_sclist_map:nn {YY} }
}
\detokenize\expandafter{\expanded{\TTTa}}\par
\detokenize\expandafter{\expanded{\TTTb}}
\ExplSyntaxOff
```

例 85

```
[XX](a)|[XX](bc)|[XX](def)|
[YY](1)|[YY](23)|[YY](456)|
```

6.10 graphics 模块

本模块主要设置图片相关的选项，一部分的命令来自 graphicx，一部分命令来自 l3graphics。

<code>\graphicspath</code>	<code>\graphicspath{<path>}</code>
New: 2024-11-05	此命令用于指定图片的搜索路径，此命令来自 graphicx 宏包，默认搜索的路径包括：./figure/，./figures/，./image/，./images/，./pictures/，./picture/，./pics/，./pics/，./graphics/，./graphic/。若用户需要增加额外的路径，请按照如下方式添加：

```
\graphicspath{
  {./Fig/}{./Img/}
}
```

例 86

<code>\zgraphicsinclude</code>	<code>\zgraphicsinclude[<options>]{<file>}</code>
New: 2025-09-20	此命令封装自 l3graphics 宏包，用于插入图片，图片由 <file> 指定。<file> 的搜索路径请参见 \graphicspath 的说明；<options> 用于设置图片属性，包括：“type, page, draft, pagebox”等。

<code>\includegraphicsifexist</code>	<code>\includegraphicsifexist[<img-spec>][<sub-img>]{}</code>
New: 2026-02-16	此命令用于插入图片，其会检测 是否存在；若 不存在，则插入 <sub-img>，<sub-img> 默认为 “example-image-a”；<img-spec> 用于指定图片的排版格式，默认为 “width=.75\linewidth”。

<code>\zgraphicspagecnt</code>	<code>\zgraphicspagecnt{<file>}</code>
New: 2025-09-20	此命令用于计算 PDF 图片的页数，计算结果保存于 \zgraphicspageint 这个宏中。

<code>\zgraphicspageint</code> *	命令 \zgraphicspagecnt 的结果保存于 \zgraphicspageint 这个宏中，这个宏是完全可展的。
New: 2025-09-20	

6.11 counter 模块

本模块主要设置计数器相关的选项, 目前提供的命令较少.

 $\backslash$ zcntnew

 New: 2025-09-23

 $\backslash$ zcntnew{<counter>}[<parent>]

此命令和 L^AT_EX 2_ε 中的 $\backslash$ newcounter 命令语法一致, 用于声明一个新的计数器. 不同的是: 当计数器 counter 已存在时, 该命令不会执行任何操作. **备注:** 此命令封装自 $\backslash$ zcnt_safe_new:nn 命令.

 $\backslash$ zcnt_safe_new:nn

 New: 2025-09-23

 $\backslash$ zcnt_safe_new:nn {<counter>}{<parent>}

此命令用于声明一个新的计数器, 当计数器 counter 已存在时, 该命令不会执行任何操作. <parent> 可以置为空, 当 <parent> 非空时, 其会变为 <counter> 的父计数器.

 $\backslash$ ztextcntwith

 $\backslash$ counterwithin

 Updated: 2025-04-25

 $\backslash$ ztextcntwith{<child>}{<parent>}

 $\backslash$ counterwithin{<child>}{<parent>}

这两个命令作用相同, 均用于给指定的 <child> 计数器添加一个父计数器 <parent>. 当 <parent> 计数器增加时, <child> 计数器会自动重置, 二者均为原始命令 $\backslash$ @addtoreset 的封装.

7 zT_EX 库

本节主要介绍 zT_EX 中的各类库 (library): 这些库旨在优化 L^AT_EX 文档的撰写与呈现效果, 其中有些还对 zT_EX 的原始功能进行了进一步扩展. 但在加载这些库后, 文档的编译速度势必会减慢, 请酌情加载 zT_EX 提供的库.

zT_EX 中任何一个库均不会自动加载, 用户需要使用 `\ztexloadlib{<library name>}` 手动加载, 所有可用的 `<library name>` 列表如下:

- `ztex.library.fancy.tex`
- `ztex.library.alias.tex`
- `ztex.library.slide.tex`
- `ztex.library.thm.tex`
- `ztex.library.cmd.tex`
- `ztex.library.pratt.tex`
- `ztex.library.primitive.tex`

zT_EX 所提供库的加载方式如下:

```
% \documentclass{ztex}  
\ztexloadlib{fancy}  
\ztexloadlib{alias}  
\ztexloadlib{slide}  
\ztexloadlib{thm}  
\ztexloadlib{cmd}  
\ztexloadlib{pratt}  
\ztexloadlib{primitive}
```

例 87

7.1 fancy 库

此 library 用于章节的格式化以及部分的宏包加载，目前仅对 \chapter 进行了重定义. 加载此库后, ℒ_{TEX} 会自动载入 anyfontsize 宏包.

ztex/fancy	fancy = $\langle true false \rangle$ 初始值: false 此选项可以用于加载 fancy library, 默认为 false. 注意: 在加载 fancy 库的同时, ℒ _{TEX} 会同时加载 tcolorbox, tikz 以及 tikz 的 calc 库.
\thmark	\thmark{ $\langle number \rangle$ }
Updated: 2025-04-25	此命令用于数字序号格式化, 其中 $\langle number \rangle$ 为任意整数. 一个简单的使用样例如下:

$\backslash\text{thmark}\{1\}, \backslash\text{thmark}\{2\}, \backslash\text{thmark}\{25\}$

例 88

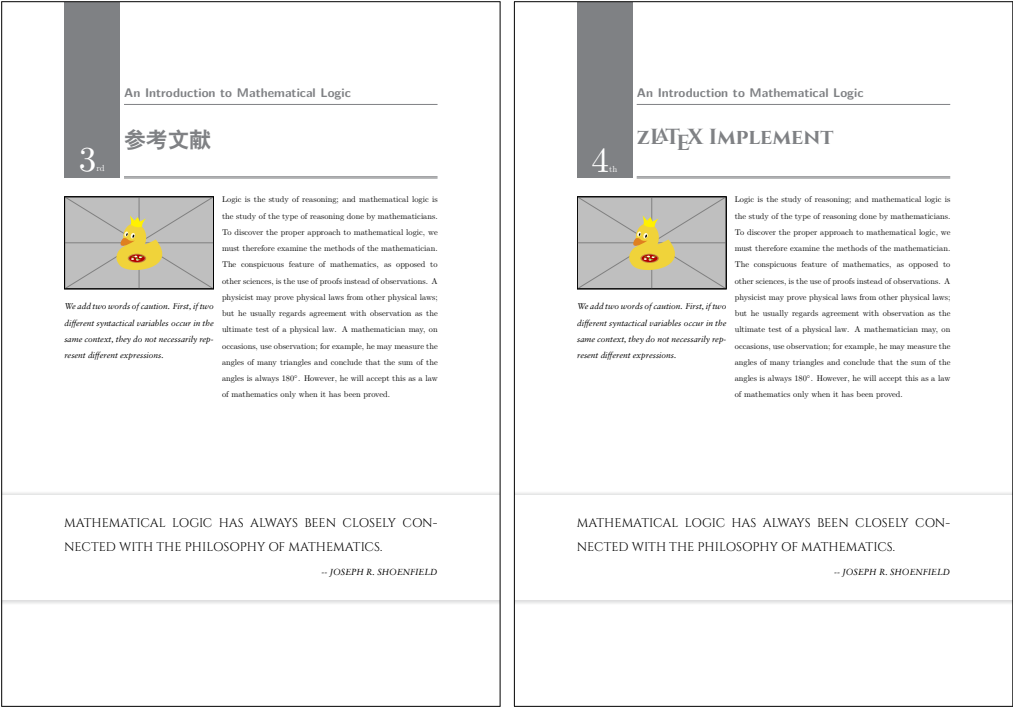
st, nd, th

\zfancychapset	\zfancychapset{ $\langle keyval \rangle$ }
New: 2025-07-10	此命令用于设置 \chapter 也的一些基本信息, $\langle keyval \rangle$ 列表请参见下述说明:

ztex/fancy/chap/text/subtitle	subtitle = { $\langle content \rangle$ }		初始值: SUBTITLE
ztex/fancy/chap/text/saying	saying = { $\langle content \rangle$ }		初始值: SAYING
ztex/fancy/chap/text/sayauthor	sayauthor = { $\langle content \rangle$ }		初始值: SAY-AUTHOR
ztex/fancy/chap/text/rcontent	lcontent = { $\langle content \rangle$ }		初始值: L-CONTENT
ztex/fancy/chap/text/lcontent	rcontent = { $\langle content \rangle$ }		初始值: R-CONTENT

$\langle subtitle \rangle$ 用于设置章节的副标题; $\langle saying \rangle$ 用于设置单元引言; $\langle sayauthor \rangle$ 用于指定引言作者; $\langle lcontent \rangle$, $\langle rcontent \rangle$ 用于指定页面左右两个的内容.

加载 fancy 库后, \chapter 页的样式大致如下:



cmd 库

L^AT_EX 的 cmd 库提供了一种新的命令定义方式：类似 Python 中的 `def fun(<arg-spec>) {<code>}`；该库目前很不成熟，请谨慎使用。

`\znewcmd`
`\zsetcmd`
`\zgsetcmd`

New: 2025-06-19

Updated: 2025-09-30

`\znewcmd<cmd>{<arg-spec>}{<code>}`

用户可以使用这三个命令创建控制序列，新创建的控制序列名称为 *<cmd>*。*<arg-spec>* 的格式为：*<var>*:*<type>*=*<default>*；其中 *<var>* 为局部变量的名称，可以使用数字，下划线（但此时需使用 `\zcmdvar` 命令进行引用）；*<type>* 用于指定变量 *<var>* 的类型，可以省略；目前 *<type>* 的可选值有 “`tl`”, “`str`”, “`int`”, “`fp`”, “`clist`”, “`dim`”, “[*<type>*]”，其中 “`tl`” 为默认类型， “[*<type>*]” 用于表示数组，数组中元素的类型均为（元素类型必须相同）*<type>*；*<default>* 用于指定变量 *<var>* 的默认值，可以省略；*<code>* 即为函数体。

备注：在函数体中，所有的局部变量均为完全可展的。

`\ztex_cmd_create:nnnnn`
New: 2025-09-30

`\ztex_cmd_create:nnnnn`
`{<cmd>}{<arg-parser>}{<code>}`
`{<setup>}{<boolean>}`

此命令用于创建类似 `\znewcmd`, `\zsetcmd` 的命令。*<cmd>* 为函数名，不包含 “\”；*<arg-spec>* 用于解析命令参数；*<code>* 为函数的具体执行内容，可以在其中使用已声明的命令参数；*<setup>* 表示 *<cmd>* 的创建方式，可以使用 “`cs_new:Npn`”, “`cs_set:Npn`” 等；*<boolean>* 为一个布尔值，用于控制命令参数是否需要计算。

备注：此命令的使用方法可以参考下述 `\znewcmd` 的定义：

```

\cs_set_protected:Npn \znewcmd #1#2#3
{
  \cs_if_exist:NT {#1}
  {
    \ztex_msg_set:nn {znewcmd@exist}
    {
      command~\string#1~already~exists!
    }
    \ztex_msg_error:n {znewcmd@exist}
  }
  \exp_args:Ne \ztex_cmd_create:nnnnn {\cs_to_str:N #1}{#2}
  {
    #3
    {\cs_new:Npn}{\c_false_bool}
  }
}

```

例 89

<code>\fpuse</code>	★	<code>\fpuse{⟨var⟩}</code>
<code>\intuse</code>	★	<code>\intuse{⟨var⟩}</code>
<code>\dimuse</code>	★	<code>\dimuse{⟨var⟩}</code>
<code>\clistuse</code>	★	<code>\clistuse⟨var⟩{⟨index⟩}</code>

New: 2025-06-19 在 `\znewcmd`, `\zsetcmd`, `\zgsetcmd` 所定义控制序列对应的 `⟨code⟩` 中, 部分的变量并不能直接使用, 需要使用 `\fpuse`, `\dimuse` 等命令进行引用.

<code>\cmdvar</code>	★	<code>\cmdvar{⟨var⟩}</code>
----------------------	---	-----------------------------

New: 2025-12-02 此命令用于引用已经声明的变量, 如果被引用的变量含有数字, 横线, 下划线等特殊字符, 建议使用该命令.

最后，我们在这里给出命令 `\znewcmd` 的一些使用案例，以供用户参考：

例 90

```

\ExplSyntaxOn
\cs_set_eq:NN \tlEQNnTF \tl_if_eq:NnTF
\ExplSyntaxOff
% new command
\znewcmd\CMDA{argA=argA-val, argB:str=argB-val, argC}
{
  \tlEQNnTF \argA {argA-val}{argA~EQUALS}{argA~not~EQUALS}\par
  \tlEQNnTF \argB {argB-val}{argB~EQUALS}{argB~not~EQUALS}\par
  \string\argC=\argC\par
}
\CMDA{argB=argB-val-new}
% set command
\dotfill\par
\zsetcmd\CMDB{
  argA = {\`Group variable range Test'},
  argF:fp = 3.1415926,
  argG:int = 100,
  argH:dim = 12pt+1em,
  argI:clist = {AA, BB, CC},
}{
  \string\argF=\fpuse{\argF};
  \string\argG=\intuse\argG;
  \string\argH=\dimuse\argH;
  \string\argI=\clistuse\argI{2}.
  \par\dotfill\par
  Argument of \string\CMDA(local variable test):
  \string\argA=\argA\par
}
\CMDB{argF=6.2830178, argG=200}
% group test
\dotfill\par

```

```

\begingroup
\zsetcmd\CMDA{arg-1=aaa}{CODE=\cmdvar{arg-1}}
INNER: \CMDA{};
\endgroup
OUTER: \CMDA{}
% vector type
\dotfill\par
\znewcmd\CMDD{argA:[int]={1, 2, 3, 4}, argB:[str], argC:[t1]}
{
  CODE 1=(\argA{1}), (\argA{4})\par
  CODE 2=(\argB{1}), (\argB{-1})\par
  CODE 3=(\argC{1})
}
\CMDD{argA={5.55, 6, 7, 8}, argB={AAA, BBB, CCC}}

```

argA EQUALS

argB not EQUALS

\argC=zCMD@EMPTY

\argF=6.2830178; \argG=200; \argH=22.95pt; \argI=BB.

Argument of \CMDA(local variable test): \argA="Group variable range Test"

INNER: CODE=aaa; OUTER: argA EQUALS

argB not EQUALS

\argC=zCMD@EMPTY

CODE 1=(5.55), (8)

CODE 2=(AAA), (CCC)

CODE 3=(zCMD@EMPTY)

7.3 alias 库

alias 库为一系列命令定义了别名, 用于简化用户在数学环境中的命令输入, 后文称此为 alias. 此 library 默认加载 amssymb, mathrsfs, mathtools 三个宏包; alias 库建立了以下几个方面的 alias:

- 数学字体命令
- 各类箭头
- 各类数学算符
- 其余常见符号
- 自动括号命令 (试验阶段)
- (偏) 微分算子
- 矩阵

对于自动括号命令, 目前还很不成熟, 如果不清楚该命令的原理, 还请不要使用. 针对此特性, 推荐用户使用 physics2 宏包. 除此之外, alias 库并没有对 mathtools 中的 `\mathclap`, `\mathllap` 等命令进行封装.

WARNING: 尽管 XeTeX 已经可以把所有的 alias 限制于一个局部组内, 但由于 alias 库自定义的命令数量实在庞大, 所以仍然可能会与部分已有命令冲突.

`\zaliasOn`
`\zaliasOff`

Updated: 2026-02-12

`\zaliasOn[⟨prefix⟩]` 初始值: **OLD**
 此二命令用于临时启用或关闭 XeTeX 的 alias 库中的命令别名; `⟨prefix⟩` 用于设置当前文档中已存在的 (外部) 命令前缀, 默认为 “OLD”; 如果在此二命令之外使用 alias 库中的别名命令, 那么 XeTeX 会抛出错误; **注意:** 此二命令并不会产生编组.

```
% \usepackage{ascii} % for \FF{}
\FF{} from 'ascii' package, \S{} from \LaTeX{};
\zaliasOn[XXX]
Inline math $\B{Q} \cong \B{Z}$;
\begin{align*}
  \int \FF{o(x)} \cdot a^{h(x)} \dd x \cdot \XXXhom(\S{F}(x)) \checkmark
\XXXdiv g(x) \dd x \\
  \dd y \dd x = \text{\XXXFF} = \text{\XXXS}
\end{align*}
```

例 91

`\zaliasOff`

¶ from ‘ascii’ package, § from L^AT_EX; Inline math $\mathbb{Q} \cong \mathbb{Z}$;

$$\int \mathbf{o}(\mathbf{x}) \cdot a^{h(x)\mathrm{d}x} \cdot \mathrm{hom}(\mathcal{F}(x)) \div g(x) \mathrm{d}x$$

$$\mathrm{d}y/\mathrm{d}x = \text{¶} = \text{§}$$

zalias

Updated: 2025-04-25

`\begin{zalias}[\langle prefix \rangle] ... \end{zalias}`

此环境与上述的 `\zaliasOn` 和 `\zaliasOff` 命令作用类似, 其会形成一个局部组. 在这个组中, 所有的 alias 均有效; `\langle prefix \rangle` 用于设置当前文档中已存在的 (外部) 命令前缀, 默认为 “OLD”; **注意:** 在正文中可以多次或嵌套使用此环境, 但必须成对出现, 否则将会导致编组不匹配, 编译出错.

`\begin{zalias}`

例 92

`\$\B{Q} \cong \B{Z} \OLDdiv 1 = 0$`

`\end{zalias}`

$$\mathbb{Q} \cong \mathbb{Z} \div 1 = 0$$

NOTE: 为了本节后续行文的简洁性, 我们默认所有示例代码中的别名命令均位于上述的 `\zaliasOn` 和 `\zaliasOff` 命令之间亦或者是 `zalias` 环境中.

7.3.1 数学字体

`\F`

`\R`

`\K`

`\C`

`\B`

`\S`

`\FF`

Updated: 2024-12-05

`\F{\langle tokens \rangle}`

`\R{\langle tokens \rangle}`

`\K{\langle tokens \rangle}`

`\C{\langle tokens \rangle}`

`\B{\langle tokens \rangle}`

`\S{\langle tokens \rangle}`

`\FF{\langle tokens \rangle}`

以上各命令的原始定义: `\F` 为 `\boldsymbol`, `\R` 为 `\mathrm`, `\K` 为 `\mathfrak`, `\C` 为 `\mathcal`, `\B` 为 `\mathbb`, `\S` 为 `\mathscr`, `\FF` 为 `\mathbf`.

Normal Version: `$\mathbf{A} + \mathrm{A} + \mathfrak{a} + \mathcal{A} + \mathbb{A} + \mathscr{A} + \mathbf{A}$ \`

Alias Version: `$\F{A} + \R{A} + \K{a} + \C{A} + \B{A} + \S{A} + \FF{A}$`

例 93

Normal Version:	$\mathbf{A} + \mathbf{A} + \mathfrak{a} + \mathcal{A} + \mathbb{A} + \mathscr{A} + \mathbf{A}$
Alias Version:	$\mathbf{A} + \mathbf{A} + \mathfrak{a} + \mathcal{A} + \mathbb{A} + \mathscr{A} + \mathbf{A}$

7.3.2 数学箭头

此 library 定义的一系列箭头命令遵循如下的规则:

- 首字母重复表示对应箭头的加长,
- 首字母大写表示对应箭头的双线版本,
- 前置 n 或 N 表示对应箭头的否定.

<code>\ma</code>
<code>\mma</code>
Updated: 2024-12-05

以上各命令的原始定义: `\ma` 为 `\mapsto`, `\mma` 为 `\longmapsto`. 注意: 此命令及其后续类似命令均表示该命令在未来可能会有改动, 比如未来其可能会接受参数.

Normal Version:	<code>\$\mapsto b, a\longmapsto b\$ \</code>	例 94
Alias Version:	<code>\$\ma b, a\mma b\$</code>	
Normal Version:	$a \mapsto b, a \longmapsto b$	
Alias Version:	$a \mapsto b, a \longmapsto b$	

<code>\la</code>
<code>\La</code>
<code>\nla</code>
<code>\Nla</code>
<code>\lla</code>
<code>\Lla</code>
Updated: 2024-12-05

以上各命令的原始定义: `\la` 为 `\leftarrow`, `\La` 为 `\Leftarrow`, `\nla` 为 `\nleftarrow`, `\Nla` 为 `\nLeftarrow`, `\lla` 为 `\longleftarrow`, `\Lla` 为 `\Longleftarrow`.

Normal Version:	<code>\$\leftarrow b, a\Leftarrow b, a\nleftarrow b, a\nLeftarrow b, a\longleftarrow b, a\Longleftarrow b\$ \</code>	例 95
Alias Version:	<code>\$\la b, a\La b, a\nla b, a\Nla b, a\lla b, a\Lla b\$.</code>	
Normal Version:	$a \leftarrow b, a \Leftarrow b, a \nleftarrow b, a \nLeftarrow b, a \longleftarrow b, a \Longleftarrow b$	
Alias Version:	$a \leftarrow b, a \Leftarrow b, a \nleftarrow b, a \nLeftarrow b, a \longleftarrow b, a \Longleftarrow b$	

Updated: 2024-12-05

Normal Version: $a \rightarrow b, a \Rightarrow b, a \nrightarrow b, a \not\Rightarrow b, a \longrightarrow b, a \Longrightarrow b$
Alias Version: $a \rightarrow b, a \Rightarrow b, a \nrightarrow b, a \not\Rightarrow b, a \longrightarrow b, a \Longrightarrow b$.

Updated: 2024-12-05

Normal Version: $a \leftrightarrow b, a \Leftrightarrow b, a \nleftrightarrow b, a \nLeftrightarrow b, a \longleftrightarrow b, a \Longleftrightarrow b$
Alias Version: $a \leftrightarrow b, a \Leftrightarrow b, a \nleftrightarrow b, a \nLeftrightarrow b, a \longleftrightarrow b, a \Longleftrightarrow b$.

<code>\xla</code>	<code>\xla[⟨above⟩](⟨below⟩)</code>
<code>\xla*</code>	<code>\xla*[⟨above⟩](⟨below⟩)</code>
<code>\Xla</code>	<code>\Xla[⟨above⟩](⟨below⟩)</code>
<code>\Xla*</code>	<code>\Xla*[⟨above⟩](⟨below⟩)</code>
<code>\xxla</code>	<code>\xxla[⟨above⟩](⟨below⟩)</code>
<code>\xxla*</code>	<code>\xxla*[⟨above⟩](⟨below⟩)</code>
<code>\xra</code>	<code>\xra[⟨above⟩](⟨below⟩)</code>
<code>\xra*</code>	<code>\xra*[⟨above⟩](⟨below⟩)</code>
<code>\Xra</code>	<code>\Xra[⟨above⟩](⟨below⟩)</code>
<code>\Xra*</code>	<code>\Xra*[⟨above⟩](⟨below⟩)</code>
<code>\xxra</code>	<code>\xxra[⟨above⟩](⟨below⟩)</code>
<code>\xxra*</code>	<code>\xxra*[⟨above⟩](⟨below⟩)</code>

Updated: 2024-12-05

以上所有带有 * 命令中的 ⟨above⟩ 和 ⟨below⟩ 参数均会被放入 \text 命令中, 以上命令的原始定义: \xla 为 \xleftarrow, \Xla 为 \Xleftarrow, \xxla 为 \xLongleftarrow, \xra 为 \xrightarrow, \Xra 为 \Xrightarrow, \xxra 为 \xLongrightarrow. 使用示例如下:

Normal Version:	<code>\$\xleftarrow[b]{a} + \Xleftarrow[b]{a} + \xLongleftarrow[b]{a} + \xrightarrow[b]{a} + \Xrightarrow[b]{a} + \xLongrightarrow[b]{a}\$ \\</code>	例 98
Alias Version:	<code>\$\xla[a](b) + \Xla[a](b) + \xxla[a](b) + \xra[a](b) + \Xra[a](b) + \xxra[a](b)\$ \\</code>	
Alias Text Version:	<code>\$\xla*[a](b) + \Xla*[a](b) + \xxla*[a](b) + \xra*[a](b) + \Xra*[a](b) + \xxra*[a](b)\$ \\</code>	
Normal Version:	$\frac{a}{b} + \frac{a}{b} + \frac{a}{b} + \frac{a}{b} + \frac{a}{b} + \frac{a}{b}$	
Alias Version:	$\frac{a}{b} + \frac{a}{b} + \frac{a}{b} + \frac{a}{b} + \frac{a}{b} + \frac{a}{b}$	
Alias Text Version:	$\frac{a}{b} + \frac{a}{b} + \frac{a}{b} + \frac{a}{b} + \frac{a}{b} + \frac{a}{b}$	

<code>\hla</code>	<code>\hla[⟨above⟩](⟨below⟩)</code>
<code>\hla*</code>	<code>\hla*[⟨above⟩](⟨below⟩)</code>
<code>\hra</code>	<code>\hra[⟨above⟩](⟨below⟩)</code>
<code>\hra*</code>	<code>\hra*[⟨above⟩](⟨below⟩)</code>

Updated: 2024-12-05

以上所有带有 * 命令中的 ⟨above⟩ 和 ⟨below⟩ 参数均会被放入 \text 命令中, 以上命令的原始定义: \hla 为 \xhookleftarrow, \hra 为 \xhookrightarrow.

Normal Version:	<code>\$\xhookleftarrow[b]{a} + \xhookrightarrow[b]{a}\$ \\</code>	例 99
-----------------	--	------

Alias Version: $\hla[a](b) + \hra[a](b)$ \\	
Alias Text Version: $\hla*[a](b) + \hra*[a](b)$	

Normal Version: $\overset{a}{\underset{b}{\leftarrow}} + \overset{a}{\underset{b}{\rightarrow}}$	
Alias Version: $\overset{a}{\underset{b}{\leftarrow}} + \overset{a}{\underset{b}{\rightarrow}}$	
Alias Text Version: $\overset{a}{\underset{b}{\leftarrow}} + \overset{a}{\underset{b}{\rightarrow}}$	

7.3.3 其它符号

\A

\E

Updated: 2024-12-05

以上两个命令分别表示“任意 (∀)”和“存在 (∃)”符号.

Normal Version:	<code>\$\forall \varepsilon > 0, \exists \delta\$</code>	<code>\</code>	例 100
Alias Version:	<code>\A \varepsilon > 0, \E \delta\$</code>		

Normal Version:	$\forall \varepsilon > 0, \exists \delta$		
Alias Version:	$\forall \varepsilon > 0, \exists \delta$		

\ns

\se

\sse

Updated: 2024-12-05

以上三个命令的原始定义: `\ns` 为 `\varnothing`, `\se` 为 `\backsim`, `\sse` 为 `\cong`.

Normal Version:	<code>\$\varnothing, \backsim, \cong\$</code>	<code>\</code>	例 101
Alias Version:	<code>\ns, \se, \sse\$</code>		

Normal Version:	$\varnothing, \backsim, \cong$		
Alias Version:	$\varnothing, \backsim, \cong$		

\dd

Updated: 2024-12-05

此命令主要用于替代默认的 `\mathrm{d}`, 与此同时, 其会自动处理左右间隔, 更加规范的处理可以参见 `fixdiff`.

Normal Version:	<code><u>\$\displaystyle\int x\mathrm{d}x = x^{\int}</u></code>	例 102
	<code>\mathrm{d} x } = \frac{1}{2} x^2 + \mathrm{C}\$</code>	
Alias Version:	<code><u>\$\displaystyle\int x\dd x = x^{\int}</u></code>	✓
	<code>\frac{1}{2} x^2 + \mathrm{C}\$.</code>	

<code>\begin{align*}</code>		
	<code>\int \FF{o(x)}\cdot a^{h(x)}\dd x\cdot\OLDhom(\S{F}(x))</code>	✓
	<code>\OLDdiv g(x)\dd x\</code>	
	<code>\dd y/\dd x</code>	

<code>\end{align*}</code>

Normal Version: $\int x \, \mathrm{d}x = x^{\int x \mathrm{d}x} = \frac{1}{2}x^2 + \mathrm{C}$
Alias Version: $\int x \, \mathrm{d}x = x^{\int x \mathrm{d}x} = \frac{1}{2}x^2 + \mathrm{C}.$
$\int \mathbf{o}(\mathbf{x}) \cdot a^{h(x)\mathrm{d}x} \cdot \mathrm{hom}(\mathcal{F}(x)) \div g(x) \, \mathrm{d}x$
$\mathrm{d}y/\mathrm{d}x$

<code>\CC</code>	<code>\CC</code>
<code>\RR</code>	<code>\RR</code>
<code>\NN</code>	<code>\NN</code>
<code>\ZZ</code>	<code>\ZZ</code>
Updated: 2024-12-05	以上四个命令分别表示复数域，实数域，自然数集以及整数集.

Normal Version: <code>$\mathbb{C}$</code> , <code>$\mathbb{R}$</code> , <code>$\mathbb{N}$</code> , <code>$\mathbb{Z}$</code>	↙ 例 103
Alias Version: <code>$\mathbb{C}$</code> , <code>$\mathbb{R}$</code> , <code>$\mathbb{N}$</code> , <code>$\mathbb{Z}$</code>	

Normal Version: $\mathbb{C}, \mathbb{R}, \mathbb{N}, \mathbb{Z}$	
Alias Version: $\mathbb{C}, \mathbb{R}, \mathbb{N}, \mathbb{Z}$	

<code>../alt</code>	<code>alt</code>	<code>=</code>	<code>\langle name \rangle</code>		初始值: <code>alt</code>
<code>../rot</code>	<code>rot</code>	<code>=</code>	<code>\langle name \rangle</code>		初始值: <code>rot</code>
<code>../div</code>	<code>div</code>	<code>=</code>	<code>\langle name \rangle</code>		初始值: <code>div</code>
<code>../curl</code>	<code>curl</code>	<code>=</code>	<code>\langle name \rangle</code>		初始值: <code>curl</code>
<code>../grad</code>	<code>grad</code>	<code>=</code>	<code>\langle name \rangle</code>		初始值: <code>grad</code>
<code>../id</code>	<code>id</code>	<code>=</code>	<code>\langle name \rangle</code>		初始值: <code>Id</code>
<code>../im</code>	<code>im</code>	<code>=</code>	<code>\langle name \rangle</code>		初始值: <code>Im</code>
<code>../ker</code>	<code>ker</code>	<code>=</code>	<code>\langle name \rangle</code>		初始值: <code>Ker</code>
<code>../cok</code>	<code>cok</code>	<code>=</code>	<code>\langle name \rangle</code>		初始值: <code>Cok</code>
<code>../hom</code>	<code>hom</code>	<code>=</code>	<code>\langle name \rangle</code>		初始值: <code>Hom</code>
<code>../supp</code>	<code>supp</code>	<code>=</code>	<code>\langle name \rangle</code>		初始值: <code>supp</code>
<code>../sign</code>	<code>sign</code>	<code>=</code>	<code>\langle name \rangle</code>		初始值: <code>sign</code>
<code>../trace</code>	<code>trace</code>	<code>=</code>	<code>\langle name \rangle</code>		初始值: <code>trace</code>

上述为 $\LaTeX$ 默认定义的数学算子, 用户可以修改 `\langle name \rangle` 的值来修改其形式.

一个简单的使用样例如下 (此命令仅能在文档的导言区使用, 但为了说明此命令的使用方法, 在本手册中, 此命令的定义被临时改变了):

```
\[ \alt, \im \]
\zaliasopset{alt=ALT, im=IM}
\[ \alt, \im \]
```

例 105

`alt, Im`

`ALT, IM`

7.3.5 自动括号

`\zab`

Updated: 2025-07-13

`\zab[⟨size⟩]⟨type⟩⟨content⟩⟨type⟩`

此命令用于处理括号的自动缩放, `⟨size⟩` 用于控制括号的大小, 可选值有 “`\big`, `\Big`, `\bigg`, `\Bigg`, `*`”, “`*`” 表示不对括号进行缩放; `⟨type⟩` 用于表示括号的类型, 可选值有: “`()`, `[]`, `{}`, `||`, `<>`, `\|\|`”. **注意:** 该命令目前处于实验阶段, 可能存在一些潜在问题, 请谨慎使用. 一个简单的使用样例如下:

例 106

```

\begin{align*}
\zab(\frac{1}{2}) = 0, & \quad \& \zab*(\frac{1}{2}) = 0, \& \\
\zab\big(\frac{1}{2}) = 0, & \quad \& \zab\Bigg(\frac{1}{2}) = 0. \& \\
\zab[\frac{1}{2}] = 0, & \quad \& \zab*(\frac{1}{2}) = 0, \& \\
\zab\Big<\frac{1}{2}> = 0, & \quad \& \zab<\frac{1}{2}> = 0. \& \\
\zab\{\frac{1}{2}\} = 0, & \quad \& \zab|\frac{1}{2}| = 0, \& \\
\zab\|\frac{1}{2}\| = 0, & \quad \& \zab\Bigg\|\frac{1}{2}\| = 0. \\
\end{align*}
\begin{align*}
\zab{1+\int_0^1 (1+x)^2dx} \& \\
\zab\bigg\|1+\int_0^1 \|1+x\|^2dx\| \\
\end{align*}

```

$$\begin{array}{cccc}
 \left(\frac{1}{2}\right) = 0, & \left(\frac{1}{2}\right) = 0, & \left(\frac{1}{2}\right) = 0, & \left(\frac{1}{2}\right) = 0. \\
 \left[\frac{1}{2}\right] = 0, & \left(\frac{1}{2}\right) = 0, & \left\langle\frac{1}{2}\right\rangle = 0, & \left\langle\frac{1}{2}\right\rangle = 0. \\
 \left\{\frac{1}{2}\right\} = 0, & \left|\frac{1}{2}\right| = 0, & \left\|\frac{1}{2}\right\| = 0, & \left\|\frac{1}{2}\right\| = 0.
 \end{array}$$

$$\left\{1 + \int_0^1 (1+x)^2 dx\right\} \qquad \left\|1 + \int_0^1 \|1+x\|^2 dx\right\|$$

NOTE: 该命令无法处理 “`|a|b+c|d|`” 这种情况, 其只能解析到 “`a+`”, 后续的 `tokens` 将会被忽略; 可以将此命令写为 `\zab|{a|b+c|d}|`, 这样便能保证参数被正确解析.

7.3.6 微分算子

<code>\dv</code>	<code>\dv{⟨fun⟩, ⟨var-1⟩, ⟨var-2⟩, ...}</code>
<code>\pdv</code>	<code>[⟨ord-1⟩, ⟨ord-2⟩, ...]</code>
<code>\dv*</code>	<code>\pdv</code> 命令的用法与 <code>\dv</code> 命令相同, 含有 “*” 的命令将采用 “a/b” 的格式排版.
<code>\pdv*</code>	

New: 2025-06-19

% \dv exampels:

例 107

`\begin{align*}`
`\dv{, xx, y, \textsf{ww}}[zz, \mathbf{g}, \B{X}]`
`& = \dv{, x, y, z}[, +++\alpha+1, +\xi+3+, \eta+2] \\\`
`\dv{, x} + \dv{, t}[2] = \dv*{f, \xi}`
`& = \dv{\varphi, x, y, z, \tau}[2, 2, 2, 1] \\\`
`\dv{, x, y, z}[1, \xi, \eta+2]`
`& = \dv{, (x^1), (x^2), (x^3)}[1, 3, 1]`
`\end{align*}`

% \pdv exampels:

`\begin{align*}`
`\pdv{, x} + \pdv{, t}[2] = \pdv*{f, \xi}`
`& = \pdv{\varphi, x, y, z, \tau}[2, 2, 2, 1] \\\`
`\pdv{, x, y, z}[1, \xi, \eta+2]`
`& = \pdv{, (x^1), (x^2), (x^3)}[1, 3, 1]`
`\end{align*}`

$$\frac{\mathrm{d}^{zz+\mathbf{g}+\mathbb{X}}}{\mathrm{d}x x^{zz}\mathrm{d}y^{\mathbf{g}}\mathrm{d}w w^{\mathbb{X}}} = \frac{\mathrm{d}^{\alpha+\xi+\eta+6}}{\mathrm{d}x\mathrm{d}y^{+++ \alpha+1}\mathrm{d}z^{+\xi+3+}}$$

$$\frac{\mathrm{d}}{\mathrm{d}x} + \frac{\mathrm{d}^2}{\mathrm{d}t^2} = \mathrm{d}f/\mathrm{d}\xi = \frac{\mathrm{d}^7\varphi}{\mathrm{d}x^2\mathrm{d}y^2\mathrm{d}z^2\mathrm{d}\tau}$$

$$\frac{\mathrm{d}^{\xi+\eta+3}}{\mathrm{d}x\mathrm{d}y^{\xi}\mathrm{d}z^{\eta+2}} = \frac{\mathrm{d}^5}{\mathrm{d}(x^1)\mathrm{d}(x^2)^3\mathrm{d}(x^3)}$$

$$\frac{\partial}{\partial x} + \frac{\partial^2}{\partial t^2} = \partial f/\partial \xi = \frac{\partial^7\varphi}{\partial x^2\partial y^2\partial z^2\partial \tau}$$

$$\frac{\partial^{\xi+\eta+3}}{\partial x\partial y^{\xi}\partial z^{\eta+2}} = \frac{\partial^5}{\partial (x^1)\partial (x^2)^3\partial (x^3)}$$

7.3.7 矩阵

alias 中的这一部分矩阵命令依赖于 `\int_step_tokens:nn` 和 `\int_step_tokens:nnn` 命令, 但它们在 2025-01-15 之后才正式被添加到 l3int 中. 如果这些命令不可用, 则它们会被替换为 x_{TeX} 实现的 `\ztex_int_step_tokens:nn` 或 `\ztex_int_step_tokens:nnn` 命令.

`\mat`

New: 2025-06-20

Updated: 2026-02-14

```
\mat{
  \langle item_{11} \rangle, ..., \langle item_{1n} \rangle;
  ...
  \langle item_{m1} \rangle, ..., \langle item_{mn} \rangle;
}
```

此命令用于生成矩阵或表格数据 (包含 “&, \” 等 token), 其维度为 $m \times n$; 用户可以将此命令嵌套在其它的矩阵或表格环境中, **不可单独使用**; 一些简单的使用案例如下 (`\zmat` 命令的使用方法请参见后续说明):

```
\[ \begin{matrix}
  \mat{1, 2; 3, 4}
\end{matrix} = \begin{vmatrix}
  \mat{1, , 3, 4; 4, 5, 6}
\end{vmatrix} \]
```

```
\begin{center}
  \begin{tabular}{|c|c|c|}
    \mat{1, 2, 3; 4, 5, 6; 7, 8, 9}
  \end{tabular} VS
  \begin{tabular}{|c|c|c|}
    \zmat[i]{3}
  \end{tabular}
\end{center}
```

例 108

$$\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} = \begin{vmatrix} 1 & 3 & 4 \\ 4 & 5 & 6 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} \text{ VS } \begin{vmatrix} 0 & & \\ & 0 & \\ & & 0 \end{vmatrix}$$

<code>\omat</code>	<code>\omat{</code>
<code>\pmat</code>	<code>\langle item_{11} \rangle & \dots & \langle item_{1n} \rangle \\</code>
<code>\bmat</code>	<code>\dots</code>
<code>\Bmat</code>	<code>\langle item_{m1} \rangle & \dots & \langle item_{mn} \rangle \\</code>
<code>\vmat</code>	<code>}</code>
<code>\Vmat</code>	<code>\omat*{</code>
	<code>\langle item_{11} \rangle, \dots, \langle item_{1n} \rangle;</code>
	<code>\dots</code>
	<code>\langle item_{m1} \rangle, \dots, \langle item_{mn} \rangle;</code>
	<code>}</code>

New: 2025-06-20

Updated: 2026-02-14

这系列命令用于输出排版矩阵, 其维度为 $m \times n$; alias 库提供的这系列矩阵排版命令有两种语法: (1) 传统的表格/矩阵语法 - 不添加 “*” 的命令, 使用 `&`, `\\` 进行行列分割; (2) 简写语法 - 添加 “*” 的命令, 使用 “,” 和 “;” 进行行列分割. 此系列命令可以与后续的 `\imat`, `\zmat`, `\hmat` 等命令配合使用, alias 库也鼓励用户这样去使用.

`\omat`(original matrix) 对应 `matrix` 环境; “p” 的含义与 `amsmath` 宏包中 `\pmatrix` 命令内的 “p” 含义相同, “b, v” 等参数的含义同理.

NOTE: 尽管含有 “*” 的命令有时可以兼容两种语法 (当参数内部的普通 ‘,’ 和 ‘;’ 已被 “{}” 包裹时), 但仍然不建议用户混用!

下面这个示例展示了两种新/旧语法的使用区别:

<pre>% correct syntax: $\pmatrix{1, 2, 3; 4, 5, 6}$ VS $\pmatrix{1 & 2 & 3 \\ 4 & 5 & 6}$ \vskip.75em\relax % wrong syntax(do NOT use): $\pmatrix{1 & 2 & 3 \\ 4 & 5{,;,} & 6}$ VS $\pmatrix{1, 2, 3; 4, 5, 6}$</pre>	例 109
<pre> $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \text{ VS } \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5,;, & 6 \end{pmatrix} \text{ VS } (1, 2, 3; 4, 5, 6)$</pre>	

此系列命令对应的各种矩阵示意:

```

\begin{align*}
\text{\text{omat}} = \text{\omat*}{ 1, , 3; 4, 5, ; , 8, 9 } \text{\quad}
& \text{\text{mat}} = \text{\begin{Vmatrix}\mat{1, , 3; 4,5,; ,8,9}} \quad \swarrow
\end{Vmatrix}}
\text{\text{pmat}} = \text{\pmat*}{ 1, , 3; 4, 5, ; , 8, 9 } \text{\quad}
& \text{\text{bmat}} = \text{\bmat*}{ 1, , 3; 4, 5, ; , 8, 9 } \text{\quad}
\text{\text{Bmat}} = \text{\Bmat*}{ 1, , 3; 4, 5, ; , 8, 9 } \text{\quad}
& \text{\text{vmat}} = \text{\vmat*}{ 1\& \& 3 \text{\quad} 4\& 5\& \text{\quad} \& 7\& 8 } \text{\quad}
\text{\text{Vmat-1}} = \text{\Vmat*}{ 1, , 3; 40.102, 55, ; , 8, 9 } \text{\quad}
& \text{\text{Vmat-2}} = \text{\Vmat*}{ 1, , 3; \textsf{xxx}, \mathbb{XX},; , 8,9}
\end{align*}

```

$$\begin{array}{lcl}
\text{omat} = \begin{array}{cc} 1 & 3 \\ 4 & 5 \\ & 8 \ 9 \end{array} & \text{mat} = & \left\| \begin{array}{cc} 1 & 3 \\ 4 & 5 \\ & 8 \ 9 \end{array} \right\| \\
\text{pmat} = \begin{pmatrix} 1 & 3 \\ 4 & 5 \\ & 8 \ 9 \end{pmatrix} & \text{bmat} = & \left[\begin{array}{cc} 1 & 3 \\ 4 & 5 \\ & 8 \ 9 \end{array} \right] \\
\text{Bmat} = \left\{ \begin{array}{cc} 1 & 3 \\ 4 & 5 \\ & 8 \ 9 \end{array} \right\} & \text{vmat} = & \left| \begin{array}{cc} 1 & 3 \\ 4 & 5 \\ & 7 \ 8 \end{array} \right| \\
\text{Vmat-1} = \left\| \begin{array}{cc} 1 & 3 \\ 40.102 & 55 \\ & 8 \ 9 \end{array} \right\| & \text{Vmat-2} = & \left\| \begin{array}{cc} 1 & 3 \\ \textsf{xxx} & \mathbb{XX} \\ & 8 \ 9 \end{array} \right\|
\end{array}$$

`\imat` `\imat {<filler>}{<item1>, ..., <itemn>}`

`\admat` `\admat {<filler>}{<item1>, ..., <itemn>}`

New: 2025-06-20

此二命令用于生成对角矩阵或反对角矩阵, 其维度为 $n \times n$; `<filler>` 用于指定非对角线元素, `<item>` 中空值默认为“1”. **注意:** 此命令不能单独使用, 用户需将此命令置于一个矩阵或表格环境中, 或将其置于前述 `\mat`, `\pmat` 等命令中.

```

\begin{align*}
\text{\omat{\imat{0}{1, ,3}}} =
\text{\pmat{\admat{}{1, 2, , 4, 5}}} =
\end{align*}

```

```
\vmat{\imat{\cdot}{1,,2}}
\end{align*}
```

$$\begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{matrix} = \begin{pmatrix} & & & 1 \\ & & 2 & \\ & 1 & & \\ 4 & & & \end{pmatrix} = \begin{vmatrix} 1 & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & 1 & \cdot \\ \cdot & \cdot & \cdot & 2 \end{vmatrix}$$

<code>\zmat</code>	<code>\zmat[⟨type⟩]{⟨n⟩}</code>
New: 2025-06-20	此命令用于输入零矩阵, 其维度为 $n \times n$; <code>⟨type⟩</code> 用于设置该矩阵的样式, 默认为“i”, 可选值有“i, a, z”. 注意: 此命令不能单独使用, 用户需将此命令置于一个矩阵或表格环境中, 或将其置于前述 <code>\mat</code> , <code>\pmat</code> 等命令中.

```
\begin{align*}
\omat{\zmat{4}} =
\vmat{\zmat[z]{5}} =
\pmat{\zmat[a]{4}}
\end{align*}
```

$$\begin{matrix} 0 & & & & \\ & 0 & & & \\ & & 0 & & \\ & & & 0 & \\ & & & & 0 \end{matrix} = \begin{vmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{vmatrix} = \begin{pmatrix} & & & & 0 \\ & & & 0 & \\ & & 0 & & \\ & 0 & & & \\ 0 & & & & \end{pmatrix}$$

例 112

\jmat

New: 2025-06-20

Updated: 2026-02-14

```

\jmat[⟨style⟩]{
  ⟨dep1⟩, ..., ⟨depm⟩;
  ⟨indep2⟩, ..., ⟨indepn⟩
}

\jmat*[⟨keyval⟩]{
  ⟨dep1⟩, ..., ⟨depm⟩;
  ⟨indep2⟩, ..., ⟨indepn⟩
}

```

此命令用于输入 Jacobian 矩阵, 其维度为 $m \times n$; $\langle dep_i \rangle$ 表示第 i 个因变量, $\langle indep_i \rangle$ 表示第 i 个自变量; (1) 针对 `\jmat` 命令, $\langle style \rangle$ 用于指定每个元素的排版样式, 默认为 “textstyle”, **此时不可单独使用**, 需嵌套于表格或矩阵环境中. (2) 针对 `\jmat*` 命令, $\langle keyval \rangle$ 用于指定 (矩阵的) 排版样式, 用法请参见后续的键值说明, **此时可以单独使用**, 且通常情况下不应将其嵌套于表格或矩阵环境中;

NOTE: 部分情况下, 用户需手动调整 `\arraystretch` 的值.

\hmat

New: 2025-06-20

Updated: 2026-02-14

```

\hmat[⟨style⟩]{
  ⟨dep⟩;
  ⟨indep1⟩, ..., ⟨indepn⟩
}

\hmat*[⟨keyval⟩]{
  ⟨dep⟩;
  ⟨indep1⟩, ..., ⟨indepn⟩
}

```

此命令用于输入 Hessian 矩阵, 其维度为 $1 \times n$; $\langle dep \rangle$ 表示因变量, $\langle indep_i \rangle$ 表示第 i 个自变量; (1) 针对 `\hmat` 命令, $\langle style \rangle$ 用于指定每个元素的排版样式, 默认为 “textstyle”, **此时不可单独使用**, 需嵌套于表格或矩阵环境中. (2) 针对 `\hmat*` 命令, $\langle keyval \rangle$ 用于指定 (矩阵的) 排版样式, 用法请参见后续的键值说明, **此时可以单独使用**, 且通常情况下不应将其嵌套于表格或矩阵环境中;

NOTE: 部分情况下, 用户需手动调整 `\arraystretch` 的值.

ztex/zalias/jhmat/b

ztex/zalias/jhmat/c

ztex/zalias/jhmat/s

```

b = {\border}..... 初始值: 空
c = {\command}..... 初始值: textstyle
s = {\float}..... 初始值: 1.25

```

$\langle b \rangle$ 用于指定矩阵的 delimiter 样式, 可选值有: “b, p, B, v, V”; $\langle c \rangle$ 用于设置矩阵中每个公式的显示方式, 默认为 “textstyle”; $\langle s \rangle$ 用于设置 `\arraystretch` 这个值, 默认为 “1.25”.

```
% \jmat examples:
```

例 113

```
\begin{align*}
\jmat*{f_1, f_2; x, y} =
\jmat*[c=displaystyle, b=V, s=2]{f, g, h; \textsf{x},
\mathbb{Y}, \F{z}} =
\bm{at}{\jmat{f, g; x, y, z}}
\end{align*}
```

```
% \hmat examples:
```

```
\begin{align*}
\hmat*[c=displaystyle, s=2.5]{;x,y,z, {w\textbf{w}}} =
{ \renewcommand{\arraystretch}{1.5}
\vmat{\hmat{g;\textsf{x},\mathbb{K},z}} }
\end{align*}
```

$$\begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix} = \begin{vmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial Y} & \frac{\partial f}{\partial z} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial Y} & \frac{\partial g}{\partial z} \\ \frac{\partial h}{\partial x} & \frac{\partial h}{\partial Y} & \frac{\partial h}{\partial z} \end{vmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} & \frac{\partial g}{\partial z} \end{bmatrix}$$

$$\begin{pmatrix} \frac{\partial^2}{\partial x^2} & \frac{\partial^2}{\partial x \partial y} & \frac{\partial^2}{\partial x \partial z} & \frac{\partial^2}{\partial x \partial w \mathbf{w}} \\ \frac{\partial^2}{\partial y \partial x} & \frac{\partial^2}{\partial y^2} & \frac{\partial^2}{\partial y \partial z} & \frac{\partial^2}{\partial y \partial w \mathbf{w}} \\ \frac{\partial^2}{\partial z \partial x} & \frac{\partial^2}{\partial z \partial y} & \frac{\partial^2}{\partial z^2} & \frac{\partial^2}{\partial z \partial w \mathbf{w}} \\ \frac{\partial^2}{\partial w \mathbf{w} \partial x} & \frac{\partial^2}{\partial w \mathbf{w} \partial y} & \frac{\partial^2}{\partial w \mathbf{w} \partial z} & \frac{\partial^2}{\partial w \mathbf{w}^2} \end{pmatrix} = \begin{vmatrix} \frac{\partial^2 g}{\partial x^2} & \frac{\partial^2 g}{\partial x \partial \mathbb{K}} & \frac{\partial^2 g}{\partial x \partial z} \\ \frac{\partial^2 g}{\partial \mathbb{K} \partial x} & \frac{\partial^2 g}{\partial \mathbb{K}^2} & \frac{\partial^2 g}{\partial \mathbb{K} \partial z} \\ \frac{\partial^2 g}{\partial z \partial x} & \frac{\partial^2 g}{\partial z \partial \mathbb{K}} & \frac{\partial^2 g}{\partial z^2} \end{vmatrix}$$

<code>\gmat</code>	<code>\gmat {⟨v₁⟩, ..., ⟨v_n⟩}</code>
New: 2025-06-20	此命令用于生成 Gram 矩阵, 其维度为 $n \times n$; 此命令仅为后续 <code>\xmat</code> 命令的一个特例. 注意: 此命令不能单独使用, 用户需将此命令置于一个矩阵或表格环境中, 或将其置于前述 <code>\mat</code> , <code>\pmat</code> 等命令中.
<code>\xmat</code>	<code>\xmat {m, n, \⟨matcmd⟩}</code>
New: 2025-06-20	此命令可以自定义矩阵的生成方式, 其维度为 $m \times n$; 矩阵元素由 <code>\⟨matcmd⟩</code> 指定, <code>\⟨matcmd⟩</code> 接受两个参数, 分别表示该元素的横坐标与纵坐标. 注意: 此命令不能单独使用, 用户需将此命令置于一个矩阵或表格环境中, 或将其置于前述 <code>\mat</code> , <code>\pmat</code> 等命令中; 命令 <code>\⟨matcmd⟩</code> 内部的脆弱命令应被保护.

NOTE: 1. 此处的 `\xmat` 命令与 `physics2` 宏包中的 `\xmat` 命令不同;
 2. 此命令的实现在之后可能会被重写.

```

\protected\def\cmdA#1#2{g^{#1#2}}
\begin{align*}
\begin{bmatrix}
\xmat{3, 4, \cmdA}
\end{bmatrix} =
\begin{bmatrix}
\gmat{v_1, v_2, v_3, v_4}
\end{bmatrix}
\end{align*}

```

例 114

$$\begin{bmatrix} g^{11} & g^{12} & g^{13} & g^{14} \\ g^{21} & g^{22} & g^{23} & g^{24} \\ g^{31} & g^{32} & g^{33} & g^{34} \end{bmatrix} = \begin{bmatrix} \langle v_1, v_1 \rangle & \langle v_1, v_2 \rangle & \langle v_1, v_3 \rangle & \langle v_1, v_4 \rangle \\ \langle v_2, v_1 \rangle & \langle v_2, v_2 \rangle & \langle v_2, v_3 \rangle & \langle v_2, v_4 \rangle \\ \langle v_3, v_1 \rangle & \langle v_3, v_2 \rangle & \langle v_3, v_3 \rangle & \langle v_3, v_4 \rangle \\ \langle v_4, v_1 \rangle & \langle v_4, v_2 \rangle & \langle v_4, v_3 \rangle & \langle v_4, v_4 \rangle \end{bmatrix}$$

7.3.8 编程接口

L^AT_EX 的 alias 库除了给普通用户提供一系列的命令（接口）外，还为熟悉 L^AT_EX 编程的用户提供了编程接口。

```
\zalias_make_cmd_robust:n      \zalias_make_cmd_robust:n <command>
\zalias_make_cmd_robust:(e|o|f)
```

New: 2025-06-22

此命令用于将命令 `\<command>` 变为一个 Robust 命令，`<command>` 为该命令的名称，不包含“\”。**注意：**原始的 `\<command>` 仅在 `zalias` 环境或 `\zaliasOn` 与 `\zaliasOff` 内被重定义为 Robust，在此范围之外，该命令将恢复为其原始定义。此命令与 `etoolbox` 提供的 `\robustify` 命令不同，请勿混用。

```
\ztex_mathalias_set:nn      \ztex_mathalias_set <inner>{<outer>}
\ztex_mathalias_set:(ee|oo)
```

New: 2025-06-22

此命令用于设置 `zalias` 环境，或 `\zaliasOn` 与 `\zaliasOff` 内命令的别名；`<outer>` 是用户在外部的声明的命令，`<inner>` 为用户在内部使用的命令，二者均不包含“\”；在此范围之外，`<outer>` 将恢复为其原始定义。

```
\zalias_matrix_from_list:n    *   \zalias_matrix_from_list:n <list>
\zalias_matrix_from_list:(e|o|f) *
```

New: 2025-06-22

此命令会根据 `<list>` 生成对应的矩阵数据，是上述 `\mat`，`\pamt` 等命令的基础；且此命令完全可展，所以该命令可以与 `tabularray` 之类的宏包结合使用。

```
\z@mat@plain *   \z@mat@plain <list>
```

New: 2025-06-22

此命令即为上述的 `\zalias_matrix_from_list:n` 命令。

`\ExplSyntaxOn`
例 115

```

\edef\MatDataA{\zalias_matrix_from_list:n {1, 2.00, , 4, ; , 6,
7.00, 9, 10 ; , 12, 13.00, , }}
\ExplSyntaxOff
\SetTblrOuter{expand=\MatDataA}
\begin{tblr}
{
  rowspec = {
    |[2pt,green7]Q|[teal7]Q|[green7]Q|[2pt, green6]

```



```

Q|[green5]Q|[green4]Q|[green3]Q|[3pt,teal7]
}
}
\MatDataA
\end{tblr}

```

1	2.00	4
6	7.00	9 10
12	13.00	

```

\zalias_diag_mat_data:nnnn * \zalias_diag_mat_data:nnnn {<bool>}{<other default>}
\zalias_diag_mat_data:nnne * {<diag default>}{<list>}

```

New: 2025-06-22

此命令会根据 $\langle list \rangle$ 生成对应的矩阵数据, 是上述 `\imat`, `\adamt`, `\zmat` 三个命令的基础; $\langle bool \rangle$ 用于指定对角矩阵的类型, $\langle bool \rangle$ 为 `\c_false_bool` 时, 为反对角矩阵; $\langle other\ default \rangle$ 用于指定非对角元素的默认值, $\langle diag\ default \rangle$ 用于指定对角线上元素的默认值; 且此命令完全可展, 所以该命令可以与 `tabularray` 之类的宏包结合使用.

```

\ExplSyntaxOn
\edef\MatDataB{\zalias_diag_mat_data:nnnn {
\c_true_bool}{?}{*}{1.00, , 2, 3, , 5}}
\edef\MatDataC{\zalias_diag_mat_data:nnnn {
\c_false_bool}{@}{*}{1.00, , 2, 3, , 5}}
\ExplSyntaxOff
\SetTblrOuter{expand={\MatDataB, \MatDataC}}
\begin{tblr}{ hlines, vlines }
\MatDataB
\end{tblr}
\quad = \quad
\begin{tblr}{ hlines, vlines }
\MatDataC
\end{tblr}

```

例 116

1.00	?	?	?	?	?		@	@	@	@	@	1.00
?	*	?	?	?	?		@	@	@	@	*	@
?	?	2	?	?	?		@	@	@	2	@	@
?	?	?	3	?	?	=	@	@	3	@	@	@
?	?	?	?	*	?		@	*	@	@	@	@
?	?	?	?	?	5		5	@	@	@	@	@

```
\zalias_jmat_data:nn      * \zalias_jmat_data:nn {<style>}{<list>}
```

```
\zalias_jmat_data:(ne|no) * \zalias_hmat_data:nn {<style>}{<list>}
```

```
\zalias_hmat_data:nn      *
```

```
\zalias_hmat_data:(ne|no) *
```

New: 2025-06-22

此二命令会根据 $\langle list \rangle$ 生成对应的 Jacobian 或 Hessian 矩阵数据, 是上述 $\backslash jmat$, $\backslash hmat$ 两个命令的基础; $\langle style \rangle$ 用于指定 Hessian 矩阵中每一项的排版样式, $\langle style \rangle$ 中不包含 “\”; 且此命令完全可展, 所以该命令可以与 `tabularray` 之类的宏包结合使用.

```
\ExplSyntaxOn
\edef\MatDataD{\zalias_jmat_data:nn {displaystyle}{f, g; x, y,
z}}
\edef\MatDataE{\zalias_hmat_data:nn {textstyle}{g; \textsf{x},
\mathbb{K}, z}}
\ExplSyntaxOff
\SetTblrOuter{expand={\MatDataD, \MatDataE}}

jmat =
\begin{tblr}{ hlines, vlines, cells={mode=math} }
\MatDataD
\end{tblr}, \qqquad
hmat =
\begin{tblr}{ hlines, vlines, cells={mode=math} }
\MatDataE
\end{tblr}
```

例 117

$$\text{jmat} = \begin{array}{|c|c|c|} \hline \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \\ \hline \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} & \frac{\partial g}{\partial z} \\ \hline \end{array}, \quad \text{hmat} = \begin{array}{|c|c|c|} \hline \frac{\partial^2 g}{\partial x^2} & \frac{\partial^2 g}{\partial x \partial \mathbb{K}} & \frac{\partial^2 g}{\partial x \partial z} \\ \hline \frac{\partial^2 g}{\partial \mathbb{K} \partial x} & \frac{\partial^2 g}{\partial \mathbb{K}^2} & \frac{\partial^2 g}{\partial \mathbb{K} \partial z} \\ \hline \frac{\partial^2 g}{\partial z \partial x} & \frac{\partial^2 g}{\partial z \partial \mathbb{K}} & \frac{\partial^2 g}{\partial z^2} \\ \hline \end{array}$$

```
\zalias_xmat_data:nn      * \zalias_xmat_data:nn {\cmd}{m, n}
```

```
\zalias_xmat_data:(ne|no) *
```

New: 2025-06-22

此命令会根据 `\langle cmd \rangle` 自动生成对应的矩阵数据, 其维度为 $m \times n$; 该命令是上述 `\gmat`, `\xmat` 两个命令的基础; `\langle cmd \rangle` 接受两个参数, 分别代表矩阵中该元素的横坐标与纵坐标; `m` 为矩阵的行数, `n` 为矩阵的列数; 且此命令完全可展, 所以该命令可以与 `tabularray` 之类的宏包结合使用.

\ExplSyntaxOn

例 118

$$\backslash\mathrm{protected}\backslash\mathrm{def}\backslash\mathrm{cmdA}\#1\#2\{\mathrm{g}^{\#1\#2}\}$$

```
\edef\MatDataF{\zalias_xmat_data:nn {\cmdA}{3, 4}}
```

```
\ExplSyntaxOff
```

```
\SetTblrOuter{expand=\MatDataF}
```

$$\mathbf{xmat} =$$

```
\begin{tblr}{ hlines, vlines, cells={mode=math} }
```

\MatDataF

`\end{tblr}`

$$\text{xmat} = \begin{array}{|c|c|c|c|} \hline g^{11} & g^{12} & g^{13} & g^{14} \\ \hline g^{21} & g^{22} & g^{23} & g^{24} \\ \hline g^{31} & g^{32} & g^{33} & g^{34} \\ \hline \end{array}$$

7.4 slide 库

此 library 用于将文档切换到 slide 模式, 无需用户对文档源码进行大的改动, 仅需在导言区加载此 library 即可, L^AT_EX 会自动处理文档的分页, 浮动体等细节.

由于此 library 内部 patch 了很多的 L^AT_EX 内部命令, 所以请谨慎加载. 另外, 加载此 library 并不会牺牲太多的编译速度.

zslide 中的坐标系统: 在不另加说明的情况下, zslide 中的坐标系统均以当前页面的左上角为原点, 取向上向右为正方向. 这就意味着你的纵坐标往往为负值, 横坐标往往为正值.

NOTE:slide 库 Patch 了大量的原始命令, 可能与部分宏包中的设置相冲突; slide 库中没有使用 fancyhdr 定义任何的页面样式.

slide 库的使用方法是简单的, 一个基本的使用样例如下:

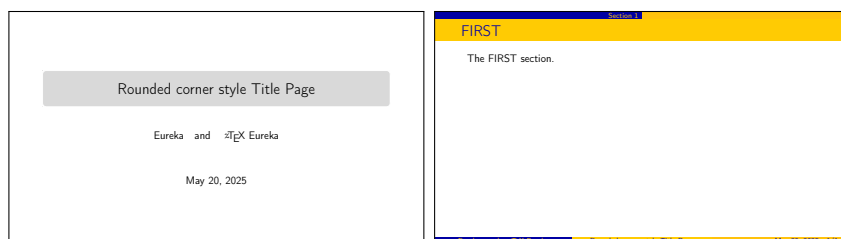
例 119

```

\documentclass[
  layout={slide, aspect=16|9},
]{ztex}
\title{Rounded corner style Title Page}
\author{Eureka\quad and \quad \ztex{} Eureka}
\date{\today}
\begin{document}
\maketitle
\section{FIRST}
The FIRST section.
\end{document}

```

上述代码的编译产生的 slide 结果如下:



7.4.1 颜色主题

<code>\zslidethemeuse</code>	<code>\zslidethemeuse[⟨key-value⟩]{⟨name⟩}</code>
Updated: 2025-04-25	此命令仅能在导言区使用, 其会根据 <code>⟨spec⟩</code> 对颜色主题 <code>⟨name⟩</code> 中的部分配置进行重写, 然后再应用 <code>⟨name⟩</code> 这一 slide 主题. <code>⟨key-value⟩</code> 列表请参见后续 <code>\zslideset</code> 命令.
注意: 为了编译速度考虑, L^AT_EX 仅加载一个主题; 所以用户应在加载 <code>ztex</code> 时便通过键 <code>⟨theme⟩</code> 指定 slide 的主题. 且命令 <code>\zslidethemeuse</code> 更大程度上是出于方便用户修改预定义主题中的某一特定项目这一目的而提供的.	
<code>\zslidethemenew</code>	<code>\zslidethemenew{⟨name⟩}{⟨key-value⟩}</code>
Updated: 2025-04-25	此命令会按照 <code>⟨key-value⟩</code> 创建名为 <code>⟨name⟩</code> 的 slide 颜色主题, 仅可在导言区使用. 具体的可调整选项请参见命令 <code>\zslideset</code> 中的 <code>⟨key-value⟩</code> 参数说明.
<code>AnnArborDefault</code>	<code>\documentclass[layout={slide, theme=AnnArborDefault}]{ztex}</code> <code>\zslidethemeuse[⟨spec⟩]{AnnArborDefault}</code>
Updated: 2024-11-05	可以在加载文档类时选择此主题, 还可以使用命令 <code>\zslidethemenew</code> 根据 <code>⟨spec⟩</code> 对此主题进行部分配置进行重定义. 本主题具体效果请前往 Beamer Theme Matrix 查看.
<code>AnnArborBeaver</code>	<code>\documentclass[layout={slide, theme=AnnArborBeaver}]{ztex}</code> <code>\zslidethemeuse[⟨key-value⟩]{AnnArborBeaver}</code>
Updated: 2024-11-05	可以在加载文档类时选择此主题, 还可以使用上述命令根据 <code>⟨key-value⟩</code> 对此主题进行部分配置进行重定义. 本主题具体效果请前往 Beamer Theme Matrix 查看.
<code>AnnArborAlbatross</code>	<code>\documentclass[layout={slide, theme=AnnArborAlbatross}]{ztex}</code> <code>\zslidethemeuse[⟨key-value⟩]{AnnArborAlbatross}</code>
Updated: 2024-11-05	可以在加载文档类时选择此主题, 还可以使用上述命令根据 <code>⟨key-value⟩</code> 对此主题进行部分配置进行重定义. 本主题具体效果请前往 Beamer Theme Matrix 查看.
<code>AnnArborSeahorse</code>	<code>\documentclass[layout={slide, theme=AnnArborSeahorse}]{ztex}</code> <code>\zslidethemeuse[⟨key-value⟩]{AnnArborSeahorse}</code>
Updated: 2024-11-05	可以在加载文档类时选择此主题, 还可以使用上述命令根据 <code>⟨key-value⟩</code> 对此主题进行部分配置进行重定义. 本主题具体效果请前往 Beamer Theme Matrix 查看.

AnnArborSpruce

Updated: 2024-12-05

```
\documentclass[layout={slide, theme=AnnArborSpruce}]{ztex}
```

```
\zslidethemeuse[⟨key-value⟩]{AnnArborSpruce}
```

可以在加载文档类时选择此主题, 还可以使用上述命令根据 *⟨key-value⟩* 对此主题进行部分配置进行重定义. 本主题具体效果请前往 Beamer Theme Matrix 查看.

7.4.2 页面信息

<code>\zslideset</code>	<code>\zslideset[⟨key⟩]{⟨spec⟩}</code>
Updated: 2025-04-25	在加载 slide 库后, 此命令用于调整 X _Y 关于 slide 的默认配置. ⟨key⟩ 表示 X _Y 中属于 zslide 库的键名, 默认为空, 此时即为根目录.
<code>ztex/./zslide/doc</code>	<code>doc = {⟨key-value⟩}</code>
<code>ztex/./zslide/sec</code>	<code>sec = {⟨key-value⟩}</code>
<code>ztex/./zslide/UL</code>	<code>UL = {⟨key-value⟩}</code>
<code>ztex/./zslide/UR</code>	<code>...</code>
<code>ztex/./zslide/BL</code>	<code>BR = {⟨key-value⟩}</code>
<code>ztex/./zslide/BC</code>	<code>toc = {⟨key-value⟩}</code>
<code>ztex/./zslide/BR</code>	上述的每一个键均为元键 (Meta Key), 接受的值需为键值对.
<code>ztex/./zslide/toc</code>	
<code>ztex/./doc/bg-color</code>	<code>bg-color = ⟨颜色⟩</code>初始值: white
<code>ztex/./doc/text-color</code>	<code>text-color = ⟨颜色⟩</code>初始值: black
<code>ztex/./doc/text-style</code>	<code>text-style = ⟨rmdefault sfdefault ttdefault⟩</code>初始值: sfdefault
	⟨bg-color⟩ 和 ⟨text-color⟩ 分别表示背景色和文本颜色, 默认情况下分别为 white, black; ⟨text-style⟩ 表示 slide 里文本的样式, 其可选值为: rmdefault, sfdefault, ttdefault.
<code>ztex/./sec/bg</code>	<code>fg = ⟨颜色⟩</code>初始值: Ann-default-I
<code>ztex/./sec/fg</code>	<code>bg = ⟨颜色⟩</code>初始值: Ann-default-II
<code>ztex/./sec/prefix</code>	<code>prefix = ⟨文本⟩</code>初始值: 空
<code>ztex/./sec/suffix</code>	<code>suffix = ⟨文本⟩</code>初始值: 空
	⟨fg⟩ 和 ⟨bg⟩ 分别表示 section 栏的文本颜色和背景色, 默认情况下分别为 Ann-default-I, Ann-default-II; ⟨文本⟩ 用于设置 slide 页面中 section 标题的前后缀.
<code>ztex/./UL/bg</code>	<code>fg = ⟨颜色⟩</code>初始值: Ann-default-II
<code>ztex/./UL/fg</code>	<code>bg = ⟨颜色⟩</code>初始值: Ann-default-I
<code>ztex/./UL/text</code>	<code>text = ⟨文本⟩</code>初始值: \zslideUL
	⟨fg⟩ 和 ⟨bg⟩ 分别表示 slide 页面中 UL 的文本颜色和背景色, 默认情况下分别为 Ann-default-II, Ann-default-I; ⟨text⟩ 用于设置 slide 左上角 (Upper Left) 导航栏对应的文本, 默认为 \zslideUL. UR, BL, BC, BR 这几个元键的属性完全一致, 这里不再一一说明.

ztex/./toc/label	label	=	{⟨key-value⟩}
ztex/./toc/suffix	suffix	=	{⟨key-value⟩}
ztex/./toc/leftmargin	leftmargin	=	{⟨key-value⟩}

上述的每一个键均为元键, 需要用接受的值也为键值对; ⟨label⟩ 表示目录页各层级的 label 格式设置; ⟨suffix⟩ 中的内容将追加到表示目录条目尾部; ⟨leftmargin⟩ 表示不同层级距离页边距的距离. 因为三者的属性完全类似, 所以我们这里只对 ⟨leftmargin⟩ 这个元键加以说明.

ztex/./leftmargin/chapter	chapter	=	{⟨长度⟩}..... 初始值: 1.9em
ztex/./leftmargin/section	section	=	{⟨长度⟩}..... 初始值: 1.5em
ztex/./leftmargin/subsection	subsection	=	{⟨长度⟩}..... 初始值: 3.8em

这三个距离中的 ⟨长度⟩ 接受一个长度参数, 其默认值分别为 1.9em, 1.5em, 3.8em.

注意: 此系列键值在处理不同文档类时兼容性不太好, 而且该设置是全局的; 因它们由 \ztocformat 命令提供, 所以建议用户直接使用 \ztocformat 命令进行目录格式定制;

在特定的子目录, 如 ⟨key⟩=doc 或 ⟨key⟩=toc/leftmargin 时, 一个设置样例如下:

```
\zslideset[doc]{
  bg-color=yellow!20,
  text-color=red
}
\zslideset[toc/leftmargin]{
  chapter=1em,
  section=4em,
}
```

例 120

\zslidelogo	\zslidelogo[⟨key-value⟩]{⟨picture⟩}
-------------	-------------------------------------

Updated: 2025-04-25

此命令用于设置 slide 的 logo 图标, 仅可在导言区使用.

ztex/slide/logo/position	width	=	⟨长度⟩..... 初始值: 2.5em
ztex/slide/logo/width	exclude	=	⟨逗号分割列表⟩..... 初始值: 1
ztex/slide/logo/exclude	position	=	(⟨长度 1, 长度 2⟩). 初始值: (\paperwidth-\ztex_quad_dim, 1.5em)

⟨position⟩ 表示 logo 图标在页面中的位置, 默认为右上角; ⟨width⟩ 表示 logo 图标的宽度, 默认为 2.5em; ⟨exclude⟩ 表示 logo 图标在 slide 页面中排除的页码范围, 默认为 1.

<code>\zslideframetitle</code>	<code>\zslideframetitle{<title>}</code>
New: 2025-05-09	此命令用于在没有 <code>\section</code> 命令出现时手动创建 slide 页面对应的标题, 和 beamer 中的 <code>\frametitle</code> 命令类似.
	注意: 此命令会自动换页, 即自动插入 <code>\newpage</code> 命令.
<code>\zslidetitle</code>	此三个命令用于分别保存导言区 <code>\@title</code> , <code>\@author</code> , <code>\@date</code> 三个变量的值, 用户可以在正文部分使用此三个变量.
<code>\zslideauthor</code>	
<code>\zslidedate</code>	注意: 如果在 slide 模式下未定义这三个变量, 那么 L ^A T _E X 会抛出错误.
Updated: 2025-04-25	
<code>\zslidedocolor</code>	<code>\zslidedocolor[<layer>]{<color>}</code>
Updated: 2025-04-25	此命令用于覆盖原本的 slide 文本或背景色, <code><layer></code> 可选值有: <code>fg</code> , <code>bg</code> ; <code><bg></code> 默认的 <code><color></code> 为 <code>white</code> , <code><fg></code> 默认的 <code><color></code> 为 <code>black</code> .
	注意: 一次只能设置一个 <code><layer></code> , 且用户不应该滥用此命令.
<code>\zslideUL</code>	这三个命令分别表示 slide 模式下, UL, UR, BR 位置处默认的文本信息.
<code>\zslideUR</code>	
<code>\zslideBR</code>	
Updated: 2025-04-25	
<code>zslide:titlepage</code>	<code>\pageref{zslide:titlepage}</code>
<code>zslide:lastpage</code>	<code>\pageref{zslide:lastpage}</code>
Updated: 2025-04-25	引用当前文档的最后一页, 用于 slide 制作时的页码引用. 使用样例如下:
<code>zslide@titlepage</code>	<code>\hyper@link{<context>}{zslide@titlepage}{<link text>}</code>
<code>zslide@lastpage</code>	<code>\hyper@link{<context>}{zslide@lastpage}{<link text>}</code>
Updated: 2024-11-05	上述两 Targets 由命令 <code>\hyper@anchor</code> 设置, 分别应用于引用当前文档的第一页和最后一页, 在 zslide 中, 标题页的页码为 0.
	注意: 普通用户不应该直接使用这两个 Targets, 此二 Targets 主要提供给模板的开发者, 用户应使用位于首页和尾页的 <code>zslide:titlepage</code> 和 <code>zslide:lastpage</code> 两 label.
<code>zslide@title@color</code>	<code>\color{zslide@title@color}<item></code>
Updated: 2025-04-25	<code>\textcolor{zslide@title@color}{<item>}</code>
	此颜色用于设置 slide 模式下 title 的背景色, 默认为: HTML:d9d9d9(即 ).

<code>\zslideframeind</code>	<code>\zslideframeind</code>
Updated: 2025-04-25	用户可以在自定义导航栏时使用此命令, 此命令在每一页 Frame 中会返回其在这个 section 中对应的 Frame Index. 比如在某个 section 中第 1 页, 其返回的 Frame Index 为 1.
<code>\zslideframeall</code>	<code>\zslideframeall{<name>}</code>
Updated: 2025-04-25	用户可以在自定义导航栏时使用此命令, 此命令可以根据 <code><name></code> 来获取 <code>\jobname.aux</code> 中变量 <code>\zsec@<name>@cnt</code> 的值. <code><name></code> 一般为大写罗马数字: I, II, III, ... 等, 其默认返回当前 section 下的 Frame 总数; 第一次编译亦或者是变量 <code>\zsec@<name>@cnt</code> 不存在时, 命令 <code>\zslideframeall</code> 将会返回 ??.
<code>\zslidenavsym</code>	<code>\zslidenavsym[<target symbol>][<other symbol>]</code>
Updated: 2025-04-25	此命令为内部命令 <code>\zslide_nav_sym:nnnn</code> 的一个具体实现. <code><target symbol></code> 默认为 ●, <code><other symbol></code> 默认为 ○. 这两个 symbol 的详细说明请参见后续的 <code>\zslide_nav_sym:nnnn</code> 命令.
<code>\zslidepageTF</code>	<code>\zslidepageTF{<formula>}{<true code>}{<false code>}</code>
Updated: 2025-04-25	此命令此命令在自定义 slide 的元信息时很有用, 其会自动比较当前页码与 <code><formula></code> 的关系, 然后执行对应的分支. 一个使用样例如下:

```

\zslidethemeuse[
  UR={text=\zslidepageTF{=1}{\zslideUR:\_ \zslidenavsym}},
]{AnnArborSpruce}

```

例 121

7.4.3 编程接口

`\zslide_nav_sym:nnnn`

Updated: 2025-04-25

`\zslide_nav_sym:nnnn {<range>}{<target>}{<target symbol>}{<other symbol>}`

此命令用于创建 slide 中的导航栏, `<range>` 接受一个正整数, 表示 frame 的总数; `<target>` 为接受一个在 $0 \sim \langle range \rangle$ 内的正整数, 表示选定的编号. `<target symbol>` 为选定的编号的符号, `<other symbol>` 为其它编号的符号.

注意: 此命令需放入 `shipout/background` 或 `shipout/foreground` 这两个 Hook 中; 普通用户不应该直接调用此命令, 此命令主要提供给模板的开发者.

`\zslide_framecnt_aux:nn`

Updated: 2025-04-25

`\zslide_framecnt_aux:nn {<name>}{<number>}`

此命令会向文件 `\jobname.aux` 中写入一个变量, 其名称为: `\zsec@<name>@cnt`, 其值为: `<number>`; `<name>` 一般为一大写罗马数字, 如 I, II, III, IV 等. 此命令在制作进度条或向后搜集文档内容时是十分有用的.

`\zslide_status_bar:nnnn`

Updated: 2025-04-25

`\zslide_status_bar:nnnn {<type>}{<coordinate>}{<width>}{<height>}`

此命令用于创建 slide 的页面背景色块, 为方便叙述, 我们称其为 `<BOX>`. 其中 `<coordinate>` 表示 `<BOX>` 左下角坐标, 形如 `(10pt, -.1\paperwidth)`, 以当前页面的左上角为原点, 取向上向右为正方向; `<type>` 为状态栏类型, 目前所有可选值有: UR, UL, BL, BC, BR, sec; `<width>` 为宽度, 接受一个浮点数, 默认以 `\paperwidth` 为单位. `<height>` 为状态栏的高度, 接受一个合法的 dim 类型值, 如 10pt, 2em 等.

注意: 此命令需放入 `shipout/background` 或 `shipout/foreground` 这两个 Hook 中; 普通用户不应该直接调用此命令, 此命令主要提供给模板的开发者.

`\zslide_status_info:nnnn`

Updated: 2025-04-25

`\zslide_status_info:nnnn {<type>}{<coordinate>}{<width>}{<content>}`

此命令用于创建 slide 的页面元信息, 其被置于一个 box 中, 为方便叙述, 我们称其为 `<BOX>`. 其中 `<type>` 表示 `<BOX>` 在页面上的位置, 可选值有: foot, head; `\g_zslide_status_info_head_B_dim` 和 `\g_zslide_status_info_foot_B_dim` 两个寄存器存放了 head 和 foot 中文字基线的纵坐标. `<coordinate>` 表示 `<BOX>` 的左下角坐标, 接受一个浮点数, 以 `\paperwidth` 为单位. 此参数以当前页面的左上角为原点, 取向上向右为正方向; `<width>` 为当前 `<BOX>` 的 (弹性) 宽度, 接受一个浮点数, 以 `\paperwidth` 为单位. `<content>` 表示 `<BOX>` 中存放的文本或图片内容.

注意: 此命令需放入 `shipout/background` 或 `shipout/foreground` 这两个 Hook 中; 普通用户不应该直接调用此命令, 此命令主要提供给模板的开发者.

<code>\zslide_meta:n</code>	<code>\zslide_meta:n {<key>}</code>
-----------------------------	---

Updated: 2025-04-25

此命令可以根据 `<key>` 获取 slide 的 status info 中对应的元信息.**注意:** 普通用户不应该直接调用此命令, 此命令主要提供给模板的开发者.

<code>\g_zslide_status_info_sec_L_dim</code>	<code>\g_zslide_status_info_sec_L_dim</code> 初始值: <code>1cm</code>
<code>\g_zslide_status_info_sec_C_dim</code>	<code>\g_zslide_status_info_sec_C_dim</code> 初始值: <code>-1.7em</code>
<code>\g_zslide_status_info_head_C_dim</code>	<code>\g_zslide_status_info_head_C_dim</code> 初始值: <code>-0.35em</code>
<code>\g_zslide_status_info_foot_C_dim</code>	<code>\g_zslide_status_info_foot_C_dim</code> 初始值: <code>-\zph+0.35em</code>

New: 2025-01-14

`\g_zslide_status_info_sec_L_dim` 中存放了 section 文本距离页面左边界的距离, 默认值为 `1cm`; `\g_zslide_status_info_sec_C_dim` 中存放了 section 文本竖直方向对称轴的纵坐标, 默认值为 `-1.7em`. 最后两个寄存器存放了 head 和 foot 中文本竖直方向对称轴的纵坐标, 前者的默认值为 `-0.35em`, 后者的默认值为 `-\paperheight+0.35em`.

注意: 普通用户不应该直接修改此系列寄存器, 此命令主要提供给模板的开发者.

<code>\g_zslide_status_bar_head_H_dim</code>	<code>\g_zslide_status_bar_head_H_dim</code> 初始值: <code>.7em</code>
<code>\g_zslide_status_bar_foot_H_dim</code>	<code>\g_zslide_status_bar_foot_H_dim</code> 初始值: <code>.7em</code>
<code>\g_zslide_status_bar_sec_H_dim</code>	<code>\g_zslide_status_bar_sec_H_dim</code> 初始值: <code>2em</code>
<code>\g_zslide_status_bar_sec_B_dim</code>	<code>\g_zslide_status_bar_sec_B_dim</code> 初始值: <code>-2.7em</code>

New: 2025-01-14

前两个寄存器存放了 slide 中 head 和 foot 对应背景色块的高度, 默认值均为 `.7em`, 其对应的背景矩形色块底边的纵坐标均为 `.7em`; `\g_zslide_status_bar_sec_H_dim` 中存放了 section 的背景色块的高度, 默认值为 `2em`; `\g_zslide_status_bar_sec_B_dim` 中存放了 section 的背景矩形色块底边对应的纵坐标, 默认值为 `-2.7em`; 当改变此三个寄存器的值时, 对应色块的基线保持不变, 其高度会做出相应的改变.

注意: 普通用户不应该直接修改此系列寄存器, 此命令主要提供给模板的开发者.

最后, 我们展示如何利用上述的编程接口复刻 beamer 主题: XiaoShan. 复刻效果如下:

xiaoshan beamer theme Eureka September 10, 2025	Fisrt Section Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. 1/4
Second Section Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. 2/4	Third Section 3.1 First subsection Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. 3/4

```

1 % 'xiaoshan' beamer theme 1
2 \documentclass[layout={slide=true, theme=AnnArborSpruce}]{ztex} 2
3 \usepackage{pgfornament-han} 3
4 \usepackage{lipsum} 4
5 \zslidethemeuse[ 5
6   UL={bg=white, text=}, 6
7   UR={bg=white, text=}, 7
8   BL={bg=white, text=}, 8
9   BC={bg=white, text=}, 9
10  BR={bg=white, text=\thepage/\pageref{zslide:lastpage}}, 10
11 ]{AnnArborSpruce} 11
12 \ExplSyntaxOn\makeatletter 12
13 \zsecformat\subsection 13
14 { 14
15   explicit = true, 15
16   code = { 16
17     \vskip.5em\relax 17
18     \par\noindent 18
19     {\bfseries \zsecnum\kern2pt\relax \zsecname} 19
20     \par\vskip.5em\relax 20
21   }, 21
22 } 22
23 % dim config 23
24 \dim_gset:Nn \g_zslide_status_info_sec_C_dim {-1em} 24
25 \dim_gset:Nn \g_zslide_status_bar_sec_B_dim {-2em} 25
26 \dim_gset:Nn \g_zslide_status_info_head_C_dim {-2.25em} 26
27 \dim_gset:Nn \g_zslide_status_info_foot_C_dim { -\zph+0.75em } 27
28 % navigator 28
29 \AddToHook{shipout/foreground} 29
30 { 30
31   \zslide_status_info:nnnn {head}{0}{1} 31
32   { 32
33     { \color{green!25!gray}\rule[-3pt]{\paperwidth}{.51em} } 33

```

66	<code>\section{Fourth Section}</code>	66
67	<code>\lipsum[3][1-4]</code>	67
68	<code>\subsection{Second subsection}</code>	68
69	<code>\lipsum[3][1-4]</code>	69
70	<code>\end{document}</code>	70

7.5 thm 库

本 library 中定义了一系列的定理类主题以及环境图标 (icon), 在加载 theme library 的同时, 会自动导入 tcolorbox, tikz 和 pifont 三个宏包. 同时也会加载 tikz 的 fadings, calc 两个库. 如此数量的宏包导入必然会拖慢整个文档的编译, 请酌情考虑加载此 library.

NOTE:

1. 由于技术原因, 当用户需要加载 thm 库时, 必须将命令 `\zthmstyle{<style>}` 置于 `\ztexloadlib{thm}` 之前;
2. 若用户在自定义定理类环境样式时需要更改 zT_EX 的默认配色, 请将 `\ztex_keys_set:nn` 或其它基于 `\keys_set:nn` 的命令放置于命令 `\zthmstylenew` 对应样式的 `<preamble>` 中而非 `<option>` 中, 否则 zT_EX 中的一系列与 `\zcolorset` 相关的函数将失去对新定义数学类环境样式的色彩控制能力.

 $\backslash\text{zthmiconset}$ $\backslash\text{zthmiconset}\{\langle\text{key-value}\rangle\}$

Updated: 2025-04-25

此命令用于设置定理类环境的图标, 仅能在导言区使用.

$\backslash\text{axiom}$	axiom	$= \langle\text{icon}\rangle$		初始值:	❖
$\backslash\text{definition}$	definition	$= \langle\text{icon}\rangle$		初始值:	♣
$\backslash\text{theorem}$	theorem	$= \langle\text{icon}\rangle$		初始值:	♥
$\backslash\text{lemma}$	lemma	$= \langle\text{icon}\rangle$		初始值:	♣
$\backslash\text{corollary}$	corollary	$= \langle\text{icon}\rangle$		初始值:	♣
$\backslash\text{proposition}$	proposition	$= \langle\text{icon}\rangle$		初始值:	♠
$\backslash\text{remark}$	remark	$= \langle\text{icon}\rangle$		初始值:	✱
	proof	$= \langle\text{icon}\rangle$		初始值:	无
	exercise	$= \langle\text{icon}\rangle$		初始值:	无
	example	$= \langle\text{icon}\rangle$		初始值:	无
	solution	$= \langle\text{icon}\rangle$		初始值:	无
	problem	$= \langle\text{icon}\rangle$		初始值:	无

上述键值配置为 $\langle\text{style}\rangle=\text{paris}$ 时的样式, 其中 $\langle\text{icon}\rangle$ 为一个合法的图标 (文字).

一个基本的使用样例如下 (此命令仅能在文档的导言区使用, 但为了说明此命令的使用方法, 在本手册中, 此命令的定义被临时改变了):

 $\backslash\text{zthmiconset}$

例 122

```

{
  axiom      = \ding{118},
  definition = \ding{168},
  theorem    = \(\heartsuit\),
  lemma      = \ding{68},
  corollary  = \ding{168},
  proposition = \(\spadesuit\),
  remark     = \ding{102},
}

```

 $\backslash\text{zthmiconuse}$ $\backslash\text{zthmiconuse}\{\langle\text{thm env name}\rangle\}$

Updated: 2025-04-25

此命令用于使用定理类环境的图标, $\langle\text{thm env name}\rangle$ 即为所有预定义的定理类环境名. 此命令在自定义定理环境样式时比较有用, 不推荐用户于正文中使用.

一个基本的使用样例如下 (此命令仅能在文档的导言区使用, 但为了说明此命令的使用方法, 在本手册中, 此命令的定义被临时改变了):

`\zthmiconuse{theorem}`

例 123

`\zthmiconuse{lemma}`

♡ ♣

<code>\zthmiconrm</code>	<code>\zthmiconrm</code>
Updated: 2025-04-25	此命令会清除所有定理类环境的图标, 不推荐用户在正文中使用.
<code>shadow</code>	<code>\zthmstyle{shadow}</code>
Updated: 2024-12-05	加载此 library 后即可应用上述样式, 样式预览如下:

`% \ztexloadlib{alias}`

例 124

`\begin{remark}[thmstyle-shadow]`

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves;

`\begin{align}`

`\underset{}{\mathbf{v}}` `\bigotimes` `\mathbf{w}`

`&= \sum_{i=1}^3 \underline{\text{left}}(a_{i1}u^iv^1+a_{i2}u^iv^2+a_{i3}u^iv^3`

`\right)` `\text{right}`

`&= \int x \, dx = \frac{1}{2}x^2 + \mathbf{C}`

`\end{align}`

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves;%

`\end{remark}`

注记 7.1 (thmstyle-shadow)

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves;

$$\mathbf{v} \otimes \mathbf{w} = \sum_{i=1}^3 (a_{i1}u^i v^1 + a_{i2}u^i v^2 + a_{i3}u^i v^3) \tag{7.1}$$
$$= \int x \, dx = \frac{1}{2}x^2 + C \tag{7.2}$$

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves;

paris

Updated: 2024-12-05

`\zthmstyle{paris}`
加载此 library 后即可应用上述样式, 样式预览如下:

`% \ztexloadlib{alias}`

例 125

`\begin{axiom}[thmstyle-paris]`

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves;

`\begin{align}`

`\underset{}{\mathbf{v}}` `\bigotimes` `\mathbf{w}`

`& = \sum_{i=1}^3 \underline{\text{left}}``(a_{i1}u^{i1}+a_{i2}u^{i2}+a_{i3}u^{i3}`
`\text{right}) \\\`

`& = \int x \, dx = \frac{1}{2}x^2 + R{C}`

`\end{align}`

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves;%

`\end{axiom}`

公理 7.1 (thmstyle-paris)

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves;

$$\mathbf{v} \bigotimes \mathbf{w} = \sum_{i=1}^3 (a_{i1}u^i v^1 + a_{i2}u^i v^2 + a_{i3}u^i v^3) \tag{7.3}$$
$$= \int x \, dx = \frac{1}{2}x^2 + C \tag{7.4}$$

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; ❖

lapsis

Updated: 2024-12-05

`\zthmstyle{lapsis}`
加载此 library 后即可应用上述样式, 样式预览如下:

`% \ztexloadlib{alias}`

例 126

`\begin{lemma}[thmstyle-lapsis]`

As any dedicated reader can clearly see, the Ideal of practical

```
reason is a representation of, as far as I know, the things in
themselves;
\begin{align}
\underset{}{\mathbf{v}} \mathrel{\mathop{\times}} \mathbf{w}
&= \sum_{i=1}^3 \underline{\text{left}}(a_{i1}u^iv^1+a_{i2}u^iv^2+a_{i3}u^iv^3
\underline{\text{right}}) \\\
&= \int x \, \mathrm{d} x = \frac{1}{2} x^2 + \mathsf{R}\{C\}
\end{align}
As any dedicated reader can clearly see, the Ideal of practical
\tcblower
\begin{align}
\int x \, \mathrm{d} x = \frac{1}{2} x^2 + \mathsf{R}\{C\}
\end{align}
reason is a representation of, as far as I know, the things in
themselves;%
\end{lemma}
```

引理 7.1

thmstyle-

lapsis

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves;

$$\mathbf{v} \otimes \mathbf{w} = \sum_{i=1}^3 (a_{i1}u^i v^1 + a_{i2}u^i v^2 + a_{i3}u^i v^3) \tag{7.5}$$
$$= \int x \, \mathrm{d} x = \frac{1}{2} x^2 + C \tag{7.6}$$

As any dedicated reader can clearly see, the Ideal of practical

$$\int x \, \mathrm{d} x = \frac{1}{2} x^2 + C \tag{7.7}$$

reason is a representation of, as far as I know, the things in themselves; ♥

elegant

Updated: 2024-12-05

\zthmstyle{elegant}

加载此 library 后即可应用上述样式, 样式预览如下:

```
% \ztexloadlib{alias}
\begin{definition}[thmstyle-elegant]
As any dedicated reader can clearly see, the Ideal of practical
```

例 127

reason is a representation of, as far as I know, the things in themselves;

```
\begin{align}
\underset{}{\mathbf{v}} \mathbin{\bigotimes} \mathbf{w}
&= \sum_{i=1}^3 \left( a_{i1} u^i v^1 + a_{i2} u^i v^2 + a_{i3} u^i v^3 \right) \\
&= \int x \, dx = \frac{1}{2} x^2 + R[C]
\end{align}
```

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves;%

```
\end{definition}
```

定义 7.1 (thmstyle-elegant)

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves;

$$\mathbf{v} \mathbin{\bigotimes} \mathbf{w} = \sum_{i=1}^3 (a_{i1} u^i v^1 + a_{i2} u^i v^2 + a_{i3} u^i v^3) \quad (7.8)$$

$$= \int x \, dx = \frac{1}{2} x^2 + C \quad (7.9)$$

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; 

tcb

\zthmstyle{tcb}

New: 2025-06-29

加载此 library 后即可应用上述样式, 样式预览如下:

```
% \ztextloadlib{alias}
\begin{theorem}[thmstyle-tcb]
As any dedicated reader can clearly see, the Ideal of practical
reason is a representation of, as far as I know, the things in
themselves;
\begin{align}
\underset{}{\mathbf{v}} \mathbin{\bigotimes} \mathbf{w}
&= \sum_{i=1}^3 \left( a_{i1} u^i v^1 + a_{i2} u^i v^2 + a_{i3} u^i v^3 \right) \\
&= \int x \, dx = \frac{1}{2} x^2 + R[C]
\end{align}
```

例 128

“命题:7.1

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves;

$$\mathbf{v} \otimes \mathbf{w} = \sum_{i=1}^3 (a_{i1}u^i v^1 + a_{i2}u^i v^2 + a_{i3}u^i v^3) \tag{7.12}$$

$$= \int x \, \mathrm{d}x = \frac{1}{2}x^2 + \mathrm{C} \tag{7.13}$$

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves;

8 ztool 宏包

本宏集已独立实现了一个 ztool 宏包, 此模块中包含原来已被废弃的 l3sys-shell 中的所有命令. 除此之外, ztool 提供了 box 操作, 文件 IO 以及基本图形绘制相关的函数. 在 ztool 的协助下, $\LaTeX$ 能够避免或减少命令行 `-shell-escape` 参数或其它相关宏包的调用 (如 robust-externalize 宏包).

ztool 宏包的详细使用方法请参见其[用户手册](#).

9 TODO

$\mathcal{X}\text{TeX}$ 的开发还远远没有结束，还有很多功能需要完善，这里列出部分将来可能会完善的功能 (☐ – 未完成; ☒ – 已完成; ☒ – 不考虑该功能):

- ☐ 封装 `geometry` 宏包的相关接口，使得用户可以通过 $\mathcal{X}\text{TeX}$ 的接口来设置页面布局和纸张大小等参数.
- ☒ 2025-07-06-已完成:在独立实现 `titlesec` 和 `titletoc` 之前，先暂时把这两个宏包的接口封装一下，放入 $\mathcal{X}\text{TeX}$ 中.
- ☒ 2025-08-20-已完成:使用 `new marker mechanism` 重写 $\text{L}\text{A}\text{T}\text{E}\text{X} 2_{\epsilon}$ 的 `mark` 机制，方便和 `fancyhdr` 宏包配合使用.
- ☒ (使用 `\IfMarksEqualTF[page]{ztex-right}{top}{first}` 进行比较即可)有时当前页面的 `ztex-right` 本应是空，但 `\FirstMark{ztex-right}` 却返回了上一节 `ztex-right` 中最后一个 `mark`. 能否提供一个一般性的命令或测试?
- ☐ 重写 `\zpagestyle` 和 `\zpage_set_style:nnn` 的接口.
- ☒ 2025-04-27-已完成:自定义 `syntax` 环境，用于排版代码. (比如给出相关命令的 `<key>` 或 `<key>` 的默认值).
- ☒ 2025-05-12-已完成:把自己修改的那个 Euler Math 变体配置进 $\mathcal{X}\text{TeX}$ ，命名为 `var-euler`，然后把相关配置写入 `fontcfg` module.
- ☐ 给 `\zpagemask` 命令增加一个 `<transparent>` `key` 以适配不同的对象 (文本，图片) 以及引擎.
- ☒ 2025-02-04-已完成:添加一个证明类环境的 `\zthmProofTitleFormat` 接口，用于设置证明类环境的标题格式.
- ☐ 完善 `Metropolis zslide` 主题，实现 `zslide` 中的 `\zslidethemeuse` 和 `\zslideColorUse` 接口，包括二者的自由组合.
- ☒ (使用 `\thepage` 命令足矣)添加一个真正的 `\zslideframeall` 命令，并把现在的 `\zslideframeall` 命令重命名为 `\zslideFrameSecTotal`.
- ☒ 2025-04-22-已完成:完善 `thm` module 的 `icon` 接口 (类似 `ElegantL\text{A}\text{T}\text{E}\text{X}` 系列)，但此接口仅在用户加载 `theme library` 时才可用.
- ☒ 2025-04-22-已完成:完善 `thm` module 中 `paris` 主题的分页样式.

- ☒ 2025-05-12-已完成:使用 `ztool` 缩放 `thm` module 中 `obsidian` 样式标题中的 `icon`.
- ☐ 重新实现部分 `xcoffins` 宏包中的命令, 目标为: 实现 `\parbox` 的功能, 并且比之更加的易用.
- ☐ 封装 `PlainTeX` 中的 `\parshape` 及其相关命令, 使之更加的易用.
- ☐ 封装 `\lastbox` 相关命令, 实现段落的分割和盒子的跨页需求.
- ☒ (使用 `CuSTeX` 中的 `framedmulticol` 宏包)在实现跨页盒子的基础上, 手动实现 `framed` 宏包的功能, 在替代该宏包原有功能的基础上, 提供更加易用的接口.
- ☐ 将 `framed` 宏包替换为 `mdframed` 宏包: 1.`framed` 宏包在 `thm` 模块内部也有使用 2. `mdframed` 宏包修复了 `framed` 宏包的一系列问题, 并且速度也(可以)很快.
- ☒ 2025-05-12-已完成:增加一个基于任意变换矩阵的盒子(内容)操作命令, 也许是依赖 `l3draw` ?? 或许增加一个 `\ztool_set_to_wd_ht:nnn` 或 `\ztool_set_wd_ht_plus_dp:nnnn` 命令???
- ☐ 提供列表设置的相关命令, 目标是成为宏包 `enumerate` 的一个可选替代. (直接从原始的 `list` 环境出发?? 未来会把这部分命令抽离到一个新的单独模块)
- ☐ 实现 `\hyper@icon` 接口, 用于设置文档中的超链接图标. (没有 `icon` 的超链接未免过于单调)
- ☒ 2025-02-05-已完成:优化 `module` 和 `library` 的加载检测机制, 完善相关变量的检测设置, 如在 `alias` 这一 `library` 中将变量 `\g__ztex_math_alias_bool` 显式的设置为 `true`.
- ☒ 2025-04-20-已完成:创建 `\zaliasOn`, `\zaliasOff` 两命令用于限制 `alias` `library` 中命令的使用范围.
- ☒ 2025-06-15-已完成:修复 `alias` 库中别名与已知命令冲突的问题.
- ☒ 2025-06-15-已完成:参考 `fixdif` 宏包, 修复了 `alias` 库中 `\dd` 命令的一系列间距问题.
- ☒ 2025-05-12-已完成:在部分 `ℒTeX` 内置命令的实现中增加 `\__ztex_plus_-key_aux:nnn` 命令, 用于在保留原内容的基础上增加内容.

- ☒ 2025-05-12-已完成:修复 `\zthmtocadd` 增加的定理条目超链接跳转异常这一问题.
- ☒ 2025-04-28-已完成:增加分散对齐命令 `\zboxitemalign`.
- ☒ 2025-04-28-已完成:重新制作 $\text{\LaTeX}$ 的 logo.
- ☒ 2025-05-12-已完成:增加 `\appmatter` 和 `\backmatter` 的定义.
- ☒ 2025-09-03-已完成:增加键值: 默认的 CMR 和 CMM 字体定义, 用于切换回默认字体.
- ☒ 2025-08-13-已完成:考虑西文字体的所有 Font Feature, 然后将其加入到 font 模块.
- ☒ 2025-09-24-已完成:修复 font/doc 这个键内的配置在 $\text{\XeTeX}$ 下的适配问题.
- ☐ 在 slide 库中增加类似 `\step`, `\pause` 这样的 beamer 命令;
- ☒ (此需求不适合 $\text{\LaTeX}$)更进一步, 在 slide 库中实现动画接口.
- ☐ 在 font 模块中配置 unicode-math 宏包的相关命令.
- ☒ 2025-05-09-已完成:修复 slide 下 section 标题文本基线在 $\langle lang \rangle = \text{en/cn}$ 下无法同时垂直对齐的问题.
- ☒ (此为中英文字体本身的问题)修复 slide 模式下当 section 标题为中英混排时基线不一致的问题.
- ☐ (难) 增加浮动体控制相关的接口.
- ☐ (难) 增加 output routine 相关的操作接口.
- ☐ 部分 `\ztex_label_hook_preamble_last` 或 `\ztex_hook_preamble_last` 存在滥用的情况, 需要清理.
- ☐ 实现部分直接操作 PDF 的接口, 比如 OCG, 图层/蒙版, 亦或者是透明度之类的, 可以参考 PDF Reference Manual.
- ☒ 2025-05-12-已完成:针对同一个仿射变换矩阵, 比如 $\Lambda = \begin{pmatrix} 1 & 0 & .5 & 1 \end{pmatrix}$ 时, `\ztoolboxaffine` 和 `\pdfsetmatrix` 的输出不一致; 但是当 $\Lambda = \begin{pmatrix} 1 & 0 & 1 & 1 \end{pmatrix}$ 时, 二者的结果是一致的; 什么原因呢? 似乎是基本单位不一致?
- ☒ 2025-05-15-已完成:`.initial:n` 在 `.inherit:n` 后会报错, 需要修复.

- ☒ 2025-09-03(已经增加了 `primitive` 库)-已完成:部分引擎对应的 `primitive` 的封装, 比如 `pdfTeX` 中的 `\pdfsetmatrix`, `XYTeX` 中的 `\ifprimitive` 等.
- ☐ 阅读引擎/驱动手册, 对 `\special` 命令进行介绍 (或是封装), 比如一些直接操作 PDF 的命令?
- ☒ 2025-06-25-已完成:能否定义一个完全可展的 `token replace` 命令, 在文件读写过程中可能会有用.
- ☒ 2025-06-25-已完成:实现类似 Python 中那样的自定义命令接口 – 关键点为参数类型标注以及默认值标注, 似乎用 `xtemplate` 也能做?
- ☐ 实现类似 `luacode` 或 `pythontex` 宏包所提供命令类似的接口, 统一管理一系列的 `shell escape`.
- ☒ 2026-02-14-已完成:`alias` 库中与矩阵相关的 “`\mat`, `\pmat`, ...” 命令并没有很好的实现内容 (数据) 和 (排版) 格式的分离, 它们这几个命令应该仅用于矩阵的排版, 而非数据的生成.
- ☐ `alias` 库中矩阵相关的命令, 能否实现自动设置 `\arraystretch` 的值??
- ☐ 修复 `\qedsymbol` 位置不正确的问题, 或者参考 `amsthm` 宏包直接写一个新的 `\zqedhere` 命令.
- ☐ 把原始的 `LATEX 2ε` 中的 `\label`, `\ref` 和 `\pageref` 命令使用 `ltproperty` 进行重写;(这样或许还能解决页面元素绝对定位的问题?)
- ☐ 修复 `LuaTeX` 和 `XYTeX` 下中文字体高度不一致的问题.
- ☐ 使用 KMP 算法重写 `\ztex_tl_if_in:nnTF` 函数, 同时需保证其是完全可展的.
- ☒ 2025-08-20-已完成:完善 `\listoffigures`, `\listoftables`, `\listofalgorithms` 等命令, 它们暂时无法使用.
- ☐ 补充 Tagged PDF 相关的代码, 且目前 `\DocumentMetadata` 和 `bookmark` 相关的 `link targets` 冲突.
- ☐ 在 `page` 模块中实现一个增强的 `\marginpar` 命令, 目的是成为 `sidenotes` 宏包的一个可选替代. (目前 `marginnote` 宏包存在一系列的问题, 且维护情况也不乐观), 可以参考 `luatodonotes`, `chuushaku`, `marginnote` 等宏包.
- ☒ 2025-08-31-已完成:`\ztocgroupinsert` 与 `\zlocaltoc` 中的 `<index>` 不一致?

- ☒ 2025-07-06-已完成:处理两个相邻 `\section` 和 `\subsection` 之间多余的垂直间距.
- ☒ 2025-07-06-已完成:添加 `sect` 模块后, `thm` 模块中的 `\zthmtoc` 命令失效.
- ☒ (真正的原因是: `\paragraph` 后的垂直间距被忽略, 因为它被转为了水平间距)`\subparagraph` 前的垂直间距丢失了?
- ☒ 2025-08-20-已完成:现在的 `sect` 模块无法处理 `\texorpdfstring` 宏, 因其与 `\ignore` 相关的键冲突.
- ☒ 2025-09-01-已完成:由 “*.toc” 文件自动生成 “*.ptoc” 文件.(这需要对目录数据进行解析, 涉及到的命令比较多, 暂时不考虑)
- ☒ (使用 `\UseTemplate` 命令即可实现一次性的标题样式)添加 `\EditNextInstance` 命令, 作用: 仅修改下一个章节命令的格式.
- ☒ 2025-08-30-已完成:命令 `\zsect_define_title:Nn` 中的 `\class` 参数只能是当前文档类中已有的标题级别 (如 `part`, `section`, `subsection` 等), 不能为新增的自定义级别.
- ☒ 2025-09-02(并没有添加 `\zlocaltocenable` 命令)-已完成:`\ztocenabletable` 命令会改变之后所有与目录相关的变量, 从而所有目录相关命令的输出均不符合预期, 可以考虑增加一个 `\zlocaltocenable` 命令.
- ☐ 让 `\zlocaltoc` 以及其背后的命令支持解析 $\text{\LaTeX 2}_{\epsilon}$ 中原始目录语法 (相关的 `\__ztoc_extract_name:w` 和 `\__ztoc_extract_title:w` 命令都需要重新设计; 目前它们和 `\use_i:nn`, `\use_ii:nn` 混用).
- ☐ 在目录组划分时所加入的 `hooks`, 也许应该换为 `\UseHookWithArguments`.
- ☒ 2025-09-21-已完成:命令 `\ztex_tl_replace_all:nnn {[ab[c]d]{[]}{(}` 运行结果为 “c]d”, 需要修复.
- ☐ 完成术语 (`glossary`) 和索引 (`index`) 相关的接口 (`listing` 和 `algorithm` 的接口目前已经可用).
- ☐ 实现 `\zsect_instance_set_fallback:nn` 命令 (或许还应该增加一系列的模板调试命令?).
- ☒ 2025-09-21-已完成:为了兼容性, 使用 `\tl_map_tokens:nn` 命令替代 `cmd` 模块, `alias` 库, ... 中的 `\int_step_tokens:nnn` 命令.

- ☒ 2025-09-23(重新实现为 `\zthmenvnew` 和 `\zthmenvset`, 但它不需要实现额外的功能; 数学环境样式理应由 `\zthmstylenew` 命令管理)-已完成:重写 `\zthmnew` 命令
- ☒ 2025-08-30-已完成:部分情况下页码不正确: `\part` 命令在目录中的页码与其真正所在的页面相差 1.
- ☐ 如果当前排版的目录数据为空, 那么添加的 `hooks` 会应用到下一个非空目录, 如何避免这个问题?(Hack `\hook_use:n ?`)
- ☐ 使用 `l3galley` 提供的命令重写目录格式相关的接口 (这部分代码目前采用 Plain $\text{\TeX}$ 实现).
- ☐ 也许我应该去除 `article`, `book` 等基础文档类的依赖, 重新实现一个基础的 `ztex-base` 文档类?
- ☐ 在 `font` 模块中增加: 对 NFSS 字体机制的详细介绍; 在 $\text{\pdfTeX}$ 下调用 (Open)TrueType 字体的方法; 介绍 `*.fd`, `*.map` 等文件的编写规则.
- ☐ 增加“虚拟字体”相关的接口.
- ☒ 2025-08-31(更新后, 所有的 `template code` 均被置于一个局部组中)-已完成: `sect` 模块的“章节命令”接口中, 部分变量需要手动清空 (需要进行充分详细的测试).
- ☐ 在 $\text{\zTeX}$ 宏集中再开发一个 `mindmap` 宏包, 提供以下的功能: 将对应的 `tikz` 对应的 `tl` 变量 `dump` 到一个外部文件中; 使用 `\levelA`, `\levelB`, ... 等命令替代原始的 `\section`, `\subsecction`, ... 等命令; 在 `mindmap` 模式下, 仅保留 `\section`, `\subsection`, ... 等命令向目录文件输出条目的功能 (似乎可以直接在 `\zsecformat` 命令中设置 `explicit=true`, `code={}`).
- ☐ 加载 `mindmap` 宏包后, 需要给每一个脑图节点增加一个或多个用于定位和连接使用的 TikZ `anchor`.
- ☐ 提供一个 `\zsecpdfstringcmds` 命令, 用于将部分命令在 PDF String 环境中替换为用户的自定义命令 (似乎 `hyperref` 宏包已经提供了对应的接口?).
- ☐ 在 `page` 模块中增加一些类似 Koma-Scripts 宏集中 `typearea` 宏包所提供的接口: 例如临时改变 PDF 纸张大小 (`\pdfpagewidth`, `\pdfpageheight`) 的接口, 临时改变页面布局的接口 (能否实现不换页更改页面样式?), ...

- ☐ 增加一个 `\ztool_box_clip:Nnn` 函数, `adjust` 宏包已经做了相关工作, 可以参见: <https://tex.stackexchange.com/questions/22731/how-to-clip-a-tex-box-using-low-level-ps-commands>, 并且 `interface3` 中已经有了 `\box_set_trim:Nnnnn` 等函数了, 不用自己从驱动层面封装.

- ☒ 2026-02-14(需要完全可展的实现时, 用户需自行参见后续的编程接口)-已完成:重构 `alias` 库中的所有矩阵命令 - 加 “*” 的 `\mat` 等命令表示新的语法. 也许得考虑展开问题, 毕竟有时需令 “&”, “\” 等符号提前被识别到.

- ☒ 2026-09-23-已完成:`\zfbox` 命令还是有问题, 当作用于包含几个大段落的 `\parbox` 时, 基线不对.

- ☒ (目前缺少 `lap` 相关命令的实现, 原始的 “`set to`” 系命令的实现并无问题)`ztool` 中的 “`set to ht`” 和 “`scale to ht`” 其实作用相同, 与 `lap` 对比 `scale` 不同! 也许可以使用 `\vbox to` 和 `\hbox to` 两种方法来重新实现前者.

- ☒ 2026-02-12-已完成:把 `alias` 中 `\zaliasOn/\zaliasOff` 内部的 `group` 去掉, 增加一个 `zaliascope` 环境.

- ☒ 2026-02-12-已完成:在 `sect` 模块中, 部分低级目录在多行时, 左侧缩进有问题. (这是正常的, 因为目录条目的 “`name`” 部分没有显示, 用户自行手动调整即可)

- ☐ `sect` 模块已经是许多模块的依赖了, 可以考虑去掉 “`sect/load`” 这个文档类选项了?

- ☐ 添加 `\ztexifmodload` 和 `\ztexiflibload`, 以及 `\ztexiffileload` 命令?

- ☐ 添加 `\zcmdmaplist` 命令, 参数形式为 `[times]{cmd}[list separator]{list}`, 此时 `cmd` 模块中的 `\zcmd_map_exptot_tl:n` 等命令也需要重构.

- ☐ 考虑增加一个 `\zcmdfilterlist` 命令, 或许可以再做一个 `functional` 库, 其中包含函数式编程的常见操作, 如 `map`, `reduce`, `fold`, `unfold`, `take`, `drop`, `filter`, `zip`, `compose`, `pipe` 等.

10 $\text{\LaTeX}$ 源码

本节列出 $\text{\LaTeX}$ 宏集的源码, 供有需要的用户参考, 点击页码即可跳转至相应位置. 另, 本节所有打印源码的对应行号已经过特殊处理, 用户可直接从 PDF 中复制源码样例, 加以修改使用.

10.1	ztex.cls	220	10.3.10	item	375
10.2	ztex.options.tex	231	10.3.11	counter	376
10.3	Module	236	10.3.12	graphics	377
10.3.1	box	236	10.4	Library	379
10.3.2	font	244	10.4.1	fancy	379
10.3.3	ref	249	10.4.2	cmd	382
10.3.4	page	255	10.4.3	alias	386
10.3.5	color	264	10.4.4	slide	401
10.3.6	thm	268	10.4.5	thm	417
10.3.7	sect	287	10.4.6	primitive	425
10.3.8	sclist	353	10.4.7	pratt	426
10.3.9	cmd	361			

1	%%%	1
2	%% ztex.cls	2
3	%% Copyright 2024, 2025, 2026 Zongping Ding.	3
4	%	4
5	% This work may be distributed and/or modified under the conditions of the	5
6	% LaTeX Project Public License, either version 1.3 of this license or any	6
7	% later version.	7
8	% The latest version of this license is in	8
9	% http://www.latex-project.org/lppl.txt	9
10	% and version 1.3 or later is part of all distributions of LaTeX	10
11	% version 2005/12/01 or later.	11
12	%	12
13	% This work has the LPPL maintenance status `maintained'.	13
14	%	14
15	% The Current Maintainer of this work is Zongping Ding.	15
16	%	16
17	% This work consists of the files ztex.cls,	17
18	% the modules: ztex.module.box.tex,	18
19	% ztex.module.cmd.tex,	19
20	% ztex.module.color.tex,	20
21	% ztex.module.counter.tex,	21
22	% ztex.module.font.tex,	22
23	% ztex.module.graphics.tex,	23
24	% ztex.module.item.tex,	24
25	% ztex.module.page.tex,	25
26	% ztex.module.ref.tex,	26
27	% ztex.module.sclist.tex,	27
28	% ztex.module.sect.tex,	28
29	% ztex.module.thm.tex,	29
30	% and the libraries: ztex.library.alias.tex,	30
31	% ztex.library.cmd.tex,	31
32	% ztex.library.slide.tex,	32
33	% ztex.library.thm.tex,	33
34	% ztex.library.primitive.tex,	34
35	% ztex.library.fancy.tex.	35
36	% ztex.library.pratt.tex.	36
37	%%%	37
38	\ExplSyntaxOn	38
39	\NeedsTeXFormat{LaTeX2e}	39
40	\gdef\ztex@d@te{2026/02/16}	40
41	\gdef\ztex@versi@n{1.3}	41
42	\tl_const:Nn \c__ztex_class_name_tl {ztex}	42
43	\tl_const:Nn \c__ztex_class_version_tl {\ztex@versi@n}	43
44	\tl_const:Nn \c__ztex_class_date_tl {\ztex@d@te}	44
45	\clist_const:Nn \c__ztex_lang_support_clist {en, cn}	45
46	\tl_const:Nn \c__ztex_class_description_tl	46

```

47 {
48     A~pre-release~universal~document~class~for~article,~book,~and~slides;~
49     Support~languages:\clist_use:Nn \c__ztex_lang_support_clist{,~}
50 }
51 \ProvidesExplClass{\c__ztex_class_name_tl}          % Class name
52               {\c__ztex_class_date_tl}              % Class Date updated
53               {\c__ztex_class_version_tl}           % Class Version latest
54               {\c__ztex_class_description_tl}       % Class Description
55
56
57
58 % -----
59 %                               class module and library
60 % -----
61 \clist_new:N \g__ztex_module_library_loaded_clist
62 \clist_gclear:N \g__ztex_module_library_loaded_clist
63 \cs_new_nopar:Npn \__ztex_load_module_library:nn #1#2
64 {
65     \clist_map_inline:nn {#2}
66     {
67         \clist_if_in:NnTF \g__ztex_module_library_loaded_clist
68         { #1:##1 }
69         {
70             \msg_set:nnn {ztex} {#1-loaded}
71             {
72                 ztex~#1~"##1"~already~loaded,give~
73                 up~loading~again.\msg_line_context:
74             }
75             \msg_warning:nnn {ztex} {#1-loaded} {##1}
76         }{
77             \file_if_exist:nTF {#1/ztex.#1.##1.tex}
78             {
79                 \clist_gput_right:Nn
80                 \g__ztex_module_library_loaded_clist
81                 { #1:##1 }
82                 \makeatletter\file_input:n {#1/ztex.#1.##1.tex}
83             }{
84                 \msg_set:nnn {ztex} {#1-not-found}
85                 { ztex~#1~`##1'~not~found. }
86                 \msg_error:nnn {ztex} {#1-not-found}
87                 { ##1 }
88             }
89         }
90     }
91 }
92 \NewDocumentCommand\ztexloadmod{m}
93 {
94     \__ztex_load_module_library:nn {module}{#1}

```

95	\ExplSyntaxOff	95
96	}	96
97	\NewDocumentCommand\ztexloadlib{m}	97
98	{	98
99	__ztex_load_module_library:nn {library}{#1}	99
100	\ExplSyntaxOff	100
101	}	101
102		102
103		103
104		104
105	% -----	105
106	% class tools	106
107	% -----	107
108	% ztex hook interface	108
109	\RequirePackage[box]{ztool}	109
110	\cs_new_protected:Npn \ztex_hook_preamble_last:n #1	110
111	{	111
112	\hook_gput_code:nnn {env/document/before}	112
113	{ztex/preamble/last}{#1}	113
114	}	114
115	\cs_new_protected:Npn \ztex_label_hook_preamble_last:nn #1#2	115
116	{	116
117	\hook_gput_code:nnn {env/document/before}	117
118	{#1}{#2}	118
119	}	119
120	\cs_new_protected:Npn \ztex_hook_doc_begin:n #1	120
121	{	121
122	\hook_gput_code:nnn {begindocument}	122
123	{ztex/doc/begin}{#1}	123
124	}	124
125	\cs_new_protected:Npn \ztex_hook_doc_end:n #1	125
126	{	126
127	\hook_gput_code:nnn {enddocument}	127
128	{ztex/doc/end}{#1}	128
129	}	129
130		130
131	% ztex key-value setup interface	131
132	\cs_new_protected:Npn \ztex_option_keys_define:n	132
133	{ \keys_define:nn { ztex / option } }	133
134	\cs_new_protected:Npn \ztex_keys_define:nn #1	134
135	{ \keys_define:nn { ztex / #1 } }	135
136	\cs_new_protected:Npn \ztex_keys_set:nn #1	136
137	{ \keys_set:nn { ztex / #1 } }	137
138	\cs_new:Npn __ztex_plus_key_aux:nnn #1#2#3	138
139	{% #1:var; #2:p-key; #3:s-key	139
140	#2 / #3 .tl_set:N = \exp_not:c { #1 } ,	140
141	#2 / #3 + .code:n = { \tl_put_right:Nn \exp_not:c { #1 } { ##1 } } ,	141
142	#2 / #3 ~ + .code:n = { \tl_put_right:Nn \exp_not:c { #1 } { ##1 } }	142

143	}	143
144		144
145		145
146		146
147	% -----	147
148	% ztex Message system	148
149	% -----	149
150	\prop_gput:Nnn \g_msg_module_type_prop	150
151	{ ztex } { Class }	151
152	\cs_new_protected:Npn \ztex_msg_set:nn #1#2	152
153	{	153
154	\msg_if_exist:nnTF { ztex }{ #1 }	154
155	{ \msg_set:nnn { ztex }{ #1 }{ #2 } }	155
156	{ \msg_new:nnn { ztex }{ #1 }{ #2 } }	156
157	}	157
158	\cs_new_protected:Npn \ztex_metakey_msg_warning:nn #1#2	158
159	{	159
160	\ztex_msg_set:nn { #1 }	160
161	{	161
162	You~use~an~invalid~key~"\l_keys_path_str"~or~key~	162
163	assign~for~it~in~the~meta~key~"#1",~Valid~options~	163
164	are:#2;~Assignment~Ignored~and~zLaTeX~default~"#1"~	164
165	settings~of~this~key~substitute.	165
166	}	166
167	\ztex_msg_warn:n { #1 }	167
168	}	168
169	\prop_const_from_keyval:Nn \c__ztex_msg_type_prop	169
170	{	170
171	info = info,	171
172	warn = warning,	172
173	error = error,	173
174	fatal = fatal,	174
175	}	175
176	\prop_map_inline:Nn \c__ztex_msg_type_prop	176
177	{	177
178	\cs_new_protected:cpn { ztex_msg_#1:n } ##1	178
179	{	179
180	\use:c { msg_#2:nn }	180
181	{ ztex }{ ##1 }	181
182	}	182
183	\cs_new_protected:cpn { ztex_msg_#1:nn } ##1##2	183
184	{	184
185	\use:c { msg_#2:nnn }	185
186	{ ztex }{ ##1 }{ ##2 }	186
187	}	187
188	\cs_new_protected:cpn { ztex_msg_#1:nnn } ##1##2##3	188
189	{	189
190	\use:c { msg_#2:nnnn }	190

```

191         { ztex }{ ##1 }{ ##2 }{ ##3 }
192     }
193 }
194
195 % ztex class options message
196 \ztex_msg_set:nn {option-unknown}
197 {
198     You~use~an~unknown~class~option~key:~'\l_keys_path_str'.~
199     Valid~options~are:~lang,~hyper,~fancy,~class,~classOption(<clist>),~
200     toc(<key-value>),~font(<key-value>),~layout(<key-value>),~
201     section(<key-value>),~mathSpec(<key-value>),~bib_index(<key-value>).~
202     Assignment~Ignored~and~LaTeX~default~settings~substitute.
203 }
204 \ztex_msg_set:nn {option-language}
205 {
206     Current~invalid~language~option~is:~'\g__ztex_lang_str',~
207     ztex~only~support~'en(english)',~and~'cn(chinese)'.~till~now.
208 }
209
210
211
212 % -----
213 %                               class option
214 % -----
215 % package options passing
216 \cs_new:Npn \ztex_package_options_pass:nn #1#2
217 {
218     \PassOptionsToPackage{#2}{#1}
219 }
220 \cs_new:Npn \ztex_package_options_pass_deprecate:n #1
221 {
222     \ztex_msg_set:nn {package-option}
223     {
224         No~options~were~passed~to~package:~#1,~
225         Deprecated~this~option(s)~for~package~#1.
226     }
227     \ztex_msg_warn:n {package-option}
228 }
229 \ztex_msg_set:nn { metakey@file@missing }
230 {
231     file~'ztex.options.tex'~is~missing~
232     from~the~ztex~bundle...
233 }
234 % setup class options
235 \keys_define:nn { ztex }{
236     % basic options
237     lang .str_gset:N = \g__ztex_lang_str,
238     lang .initial:n = { en },

```

239	lang	.usage:n	= load,	239
240	hyper	.bool_gset:N	= \g__ztex_hyperref_bool,	240
241	hyper	.initial:n	= { false },	241
242	hyper	.default:n	= { true },	242
243	hyper	.usage:n	= load,	243
244	hyper-suppress	.clist_gset:N	= \g__ztex_hyper_suppress_clist,	244
245	hyper-suppress	.initial:n	= { toc },	245
246	hyper-suppress	.usage:n	= load,	246
247	fancy	.bool_gset:N	= \g__ztex_fancy_bool,	247
248	fancy	.initial:n	= { false },	248
249	fancy	.default:n	= { true },	249
250	fancy	.usage:n	= load,	250
251	cref-backend	.str_gset:N	= \g__ztex_cref_backend_str,	251
252	cref-backend	.initial:n	= { zref-clever },	252
253	% sub class and meta key			253
254	class	.str_gset:N	= \g__ztex_subclass_type_str,	254
255	class	.initial:n	= { article },	255
256	class	.usage:n	= load,	256
257	classOption	.clist_gset:N	= \g__ztex_subclass_option_clist,	257
258	classOption	.initial:n	= { oneside, 12pt },	258
259	classOption	.usage:n	= load,	259
260	packageOption	.code:n	= {	260
261	\keyval_parse:NNn			261
262	\ztex_package_options_pass_deprecate:n			262
263	\ztex_package_options_pass:nn {#1}			263
264	},			264
265	packageOption	.usage:n	= load,	265
266	% ztex options meta key			266
267	sect	.meta:nn	= { ztex / sect }{#1},	267
268	sect	.usage:n	= load,	268
269	font	.meta:nn	= { ztex / font }{#1},	269
270	layout	.meta:nn	= { ztex / layout }{#1},	270
271	layout	.usage:n	= load,	271
272	mathSpec	.meta:nn	= { ztex / mathSpec }{#1},	272
273	bib_index	.meta:nn	= { ztex / bib_index }{#1},	273
274	unknown	.code:n	= {	274
275	\ztex_msg_warn:n {option-unknown}			275
276	}			276
277	}			277
278	% sub(meta) key implementation			278
279	\file_if_exist_input:nF { ztex.options.tex }			279
280	{ \ztex_msg_fatal:n { metakey@file@missing } }			280
281				281
282	% ztex options setup			282
283	\ProcessKeyOptions [ztex]			283
284	\NewDocumentCommand{\ztexset}{m}			284
285	{			285
286	\keys_set:nn {ztex}{#1}			286

```
287 } 287
288 \newcommand{\ztexoption} 288
289 { 289
290     \str_use:N \g__ztex_lang_str {~,~} 290
291     \clist_use:Nn \g__ztex_subclass_option_clist 291
292     { ~,~ } 292
293 } 293
294 294
295 295
296 296
297 % ----- 297
298 %             subClass and package Option 298
299 % ----- 299
300 % pass clist options main subclass: 'article', 'book', 'ctexbook' 300
301 \ztex_msg_set:nn {option-subclass} 301
302 { 302
303     subclass~option:"\g__ztex_subclass_type_str"~is~ 303
304     not~accessible,~Valid~options~are:article,~book,~ 304
305     ctexbook,~l3doc~and~l3dox. 305
306 } 306
307 \str_case:VnF \g__ztex_subclass_type_str 307
308 { 308
309     {article}{ 309
310         \PassOptionsToClass{\g__ztex_subclass_option_clist} 310
311         { article } 311
312         \LoadClass{article} 312
313     } 313
314     {book}{ 314
315         \PassOptionsToClass{\g__ztex_subclass_option_clist} 315
316         { book } 316
317         \LoadClass{book} 317
318     } 318
319     {ctexbook}{ 319
320         \str_set:Nn \g__ztex_lang_str {cn} 320
321         \PassOptionsToClass{\g__ztex_subclass_option_clist} 321
322         { ctexbook } 322
323         \PassOptionsToPackage{quiet}{fontspec} 323
324         \LoadClass{ctexbook} 324
325     } 325
326     {l3doc}{ 326
327         \PassOptionsToClass{\g__ztex_subclass_option_clist} 327
328         { l3doc } 328
329         \LoadClass{l3doc} 329
330     } 330
331 }{ 331
332     \ztex_msg_error:n {option-subclass} 332
333 } 333
334 334
```


335	% baisc document class and packages option	335
336	\tl_set_rescan:NnV \l_tmpa_tl	336
337	{ \cctab_select:N \c_code_cctab }	337
338	\g__ztex_lang_str	338
339	\clist_if_in:NVF \c__ztex_lang_support_clist \l_tmpa_tl	339
340	{ \ztex_msg_error:n {option-language} }	340
341	\str_case:VnF \g__ztex_lang_str	341
342	{	342
343	{en} {	343
344	\sys_if_engine_xetex:T	344
345	{	345
346	\ztex_hook_preamble_last:n	346
347	{	347
348	\bool_if:NF \g__ztex_sysfont_cfg_bool	348
349	{	349
350	\ztex_msg_set:nn {compile-engine-pdftex}	350
351	{	351
352	Current~compile~engine~is~XETEX,~For~better~	352
353	output,~use~PDFTEX~instead.	353
354	}	354
355	\ztex_msg_warn:n {compile-engine-pdftex}	355
356	}	356
357	}	357
358	}	358
359	\RequirePackage[T1]{fontenc}	359
360	}	360
361	{cn} {	361
362	\sys_if_engine_pdfTeX:T	362
363	{	363
364	\ztex_msg_set:nn {compile-engine-xetex}	364
365	{	365
366	Current~compile~engine~is~PDFTEX,~For~	366
367	chinese~material,~use~XETEX~instead.	367
368	}	368
369	\ztex_msg_error:n {compile-engine-xetex}	369
370	}	370
371	\PassOptionsToPackage{quiet}{fontspec}	371
372	\PassOptionsToPackage{no-math}{fontspec}	372
373	\str_if_eq:VnF \g__ztex_subclass_type_str	373
374	{ ctexbook }	374
375	{	375
376	\RequirePackage[UTF8, scheme=plain]{ctex}	376
377	\linespread{1.3}	377
378	}	378
379	}	379
380	}{	380
381	\ztex_msg_error:n {option-language}	381
382	}	382

383		383
384		384
385		385
386	% -----	386
387	% ztex module	387
388	% -----	388
389	_ztex_load_module_library:nn {module}{sclist}	389
390	_ztex_load_module_library:nn {module}{cmd}	390
391	_ztex_load_module_library:nn {module}{box}	391
392	_ztex_load_module_library:nn {module}{counter}	392
393	_ztex_load_module_library:nn {module}{sect}	393
394	_ztex_load_module_library:nn {module}{thm}	394
395	_ztex_load_module_library:nn {module}{ref}	395
396	_ztex_load_module_library:nn {module}{color}	396
397	_ztex_load_module_library:nn {module}{font}	397
398	_ztex_load_module_library:nn {module}{graphics}	398
399	_ztex_load_module_library:nn {module}{item}	399
400	_ztex_load_module_library:nn {module}{page}	400
401		401
402		402
403		403
404	% -----	404
405	% ztex library	405
406	% -----	406
407	\bool_if:NT \g_ztex_math_alias_bool	407
408	{	408
409	_ztex_load_module_library:nn {library}{alias}	409
410	}	410
411	\bool_if:NTF \g_ztex_slide_bool	411
412	{	412
413	_ztex_load_module_library:nn {library}{slide}	413
414	}{ \newcommand\zslideset[1]{ } }	414
415	\bool_if:NT \g_ztex_fancy_bool	415
416	{ _ztex_load_module_library:nn {library}{fancy} }	416
417		417
418		418
419		419
420	% -----	420
421	% module/library checker	421
422	% -----	422
423	\bool_new:N \g_ztex_thm_lib_load_bool	423
424	\bool_gset_false:N \g_ztex_thm_lib_load_bool	424
425	\newcommand\ztexhyperTF[2]	425
426	{	426
427	\bool_if:NTF \g_ztex_hyperref_bool	427
428	{ #1 }{ #2 }	428
429	}	429
430	\newcommand\ztexfancyTF[2]	430

431	{	431
432	\bool_if:NTF \g__ztex_fancy_bool	432
433	{ #1 }{ #2 }	433
434	}	434
435	\newcommand\ztexmarginTF[2]	435
436	{	436
437	\bool_if:NTF \g__ztex_margin_bool	437
438	{ #1 }{ #2 }	438
439	}	439
440	\newcommand\ztexslideTF[2]	440
441	{	441
442	\bool_if:NTF \g__ztex_slide_bool	442
443	{ #1 }{ #2 }	443
444	}	444
445	\newcommand\ztexsysfontTF[2]	445
446	{	446
447	\bool_if:NTF \g__ztex_sysfont_cfg_bool	447
448	{ #1 }{ #2 }	448
449	}	449
450	\newcommand\ztexaliasTF[2]	450
451	{	451
452	\bool_if:NTF \g__ztex_math_alias_bool	452
453	{ #1 }{ #2 }	453
454	}	454
455	\newcommand\ztexbibindTF[2]	455
456	{	456
457	\bool_if:NTF \g__ztex_bib_index_load_bool	457
458	{ #1 }{ #2 }	458
459	}	459
460	\newcommand\ztethmlibTF[2]	460
461	{	461
462	\bool_if:NTF \g__ztex_thm_lib_load_bool	462
463	{ #1 }{ #2 }	463
464	}	464
465		465
466		466
467		467
468	% -----	468
469	% ztex logo	469
470	% -----	470
471	\NewDocumentCommand\zTeX{s}	471
472	{	472
473	\IfBooleanTF{#1}	473
474	{	474
475	__ztool_leave_vmode:	475
476	\underline{\raise0.0894ex\hbox{z}}	476
477	\underline{\kern-0.4645ex\hbox{\TeX}}	477
478	}{	478

479	\ztool_scale_to_wd_and_ht:nnn	479
480	{.9ex}{1.3ex}	480
481	{	481
482	\ztool_rotate:nn {89}{\(\aleph\)}	482
483	} \kern-0.3423ex\hbox{\TeX}	483
484	}	484
485	}	485
486	\NewDocumentCommand\zLaTeX{s}	486
487	{	487
488	\IfBooleanTF{#1}	488
489	{	489
490	__ztool_leave_vmode:	490
491	\raise0.1ex\hbox{z}	491
492	\kern-0.3ex\hbox{\LaTeX}	492
493	}{	493
494	L\kern -.375em	494
495	{	495
496	\lower-.4ex \hbox {	496
497	\ztool_scale_to_wd_and_ht:nnn	497
498	{.9ex}{1.3ex}	498
499	{	499
500	\ztool_rotate:nn {88}{\(\aleph\)}	500
501	}	501
502	}	502
503	} \kern -.05em\TeX	503
504	}	504
505	}	505
506	\let\ztex\zTeX	506
507	\let\zlatex\zLaTeX	507
508	\protected\def\HoLogo@zTeX#1{\zTeX}	508
509	\protected\def\HoLogo@ztex#1{\zTeX}	509
510	\protected\def\HoLogo@zLaTeX#1{\zLaTeX}	510
511	\protected\def\HoLogo@zlatex#1{\zLaTeX}	511

```

1 % ==> sect options
2 \ztex_keys_define:nn { sect }{
3   load      .bool_gset:N = \g__ztex_sect_load_bool,
4   load      .initial:n   = { true },
5   load      .default:n   = { true },
6   load      .usage:n     = load,
7   style     .tl_gset:N   = \g__zsect_title_style_tl,
8   style     .initial:n   = { ltx },
9   dump-ptable .bool_gset:N = \g__zsect_dump_ptable_bool,
10  dump-ptable .initial:n   = { false },
11  dump-ptable .default:n   = { true },
12  dump-ptable .usage:n     = load,
13 }
14
15
16 % ==> font options
17 \ztex_keys_define:nn { font }{
18   sysfont    .bool_gset:N = \g__ztex_sysfont_cfg_bool,
19   sysfont    .initial:n   = { false },
20   sysfont    .default:n   = { true },
21   doc        .choice:,
22   doc / ptmx .code:n      = {
23     \ztex_label_hook_preamble_last:nn
24     { ztex-docfont-ptmx }
25     {
26       \RequirePackage{mathptmx}
27       \RequirePackage{newtxtext}
28     }
29     \DeclareSymbolFont{letters}{OML}{ntxmi}{m}{it}
30     \DeclareMathAlphabet{\mathbf}{OT1}{ntxtlf}{b}{it}
31     \DeclareSymbolFont{CMMletters}{OML}{cmm}{m}{it}
32     \DeclareSymbolFont{CMMsymbols}{OMS}{cmsy}{m}{n}
33     \DeclareSymbolFont{CMMlargesymbols}{OMX}{cmex}{m}{n}
34     \DeclareMathSymbol{\new@pi}{0}{CMMletters}{"19}
35     \DeclareMathSymbol{\new@jmath}{0}{CMMletters}{"7C}
36     \DeclareMathSymbol{\new@amalg}{0}{CMMsymbols}{"71}
37     \DeclareMathSymbol{\new@coprod}{1}{CMMlargesymbols}{"61}
38   \ztex_hook_doc_begin:n
39   {
40     \let\pi\new@pi
41     \let\jmath\new@jmath
42     \let\amalg\new@amalg
43     \let\coprod\new@coprod
44   }
45 },
46 doc / newtx .code:n      = {

```

```

47 \ztex_label_hook_preamble_last:nn
48 { ztex-docfont-newtex }
49 {
50 \RequirePackage{newtxtext}
51 \RequirePackage{newtxmath}
52 }
53 },
54 doc / lmm .code:n = {
55 \sys_if_engine_pdftex:TF
56 {
57 \RequirePackage{lmodern}
58 \RequirePackage{fixcmex}
59 }{
60 \ztex_msg_set:nn {lmm-font-pdftex}
61 {
62 The~default~font~for~XeTeX/LuaTeX~is~latin~
63 modern,~there~is~no~need~to~load~lmodern~again.
64 }
65 \ztex_msg_warn:n {lmm-font-pdftex}
66 }
67 },
68 text .choice:,
69 text / times .code:n = {
70 \ztex_label_hook_preamble_last:nn
71 { ztex-textfont-newtex }
72 {
73 \RequirePackage{newtxtext}
74 }
75 },
76 doc / texgyre .code:n = { }, % TODO: implement it !!
77 math .choice:,
78 math / newtx .code:n = {
79 \ztex_hook_preamble_last:n
80 {
81 \RequirePackage{newtxmath}
82 }
83 },
84 math / mtpro2 .code:n = {
85 \ztex_hook_preamble_last:n
86 {
87 \RequirePackage[
88 lite, subscriptcorrection,
89 slantedGreek, nofontinfo
90 ]{mtpro2}
91 }
92 },
93 math / euler .code:n = {
94 \ztex_hook_preamble_last:n

```

```

95     {
96         \RequirePackage[OT1, euler-digits]{eulervm}
97     }
98 },
99 math / var-euler .code:n      = {
100     \usepackage[OT1]{eulervm}
101     \DeclareSymbolFont{cmmlargesymbols}{OMX}{cmex}{m}{n}
102     \DeclareSymbolFont{greekletters}{OML}{cmm}{m}{it}
103     \DeclareMathDelimiter{\new@int}{\mathop}{cmmlargesymbols}{"52}{cmmlargesymbols}{"5A}
104     \DeclareMathDelimiter{\new@sum}{\mathop}{cmmlargesymbols}{"50}{cmmlargesymbols}{"58}
105     \ztex_hook_doc_begin:n
106     {
107         \DeclareRobustCommand\int {\new@int}
108         \DeclareMathSymbol{\kappa}{\mathord}{greekletters}{"14}
109         \DeclareMathSymbol{ \tau }{\mathord}{greekletters}{"1C}
110         \DeclareMathSymbol{\omega}{\mathord}{greekletters}{"21}
111     }
112 },
113 math / ptmx .code:n          = {
114     \ztex_msg_set:nn {option-font-math}
115     {To~use~ptmx~math~font,use~the~'doc=ptmx'~setting~instead.}
116     \ztex_msg_warn:n {option-font-math}
117 },
118 math / mathpazo .code:n      = {
119     \ztex_hook_preamble_last:n
120     {
121         \let\cmrrmdefault\rmdefault
122         \let\cmrsfdefault\sfdefault
123         \let\cmrttdefault\ttdefault
124         \RequirePackage{mathpazo}
125         \let\rmdefault\cmrrmdefault
126         \let\sfdefault\cmrsfdefault
127         \let\ttdefault\cmrttdefault
128     }
129 },
130 math / unknown .code:n       = {
131     \ztex_metakey_msg_warning:nn
132     {option-mathSpec-font}
133     {newtx, mtp2, euler, mathpazo}
134 },
135 unknown .code:n              = {
136     \ztex_metakey_msg_warning:nn
137     { option-font }
138     {
139         sysfont(<bool>:false),
140         doc(<choice>:newtx,ptmx),
141         text(<choice>:times),
142         math(<choice>:newtx,mtp2,euler,mathpazo)

```

```

143 }
144 }
145 }
146
147
148 % ==> page(layout) options
149 \ztex_keys_define:nn { layout }{
150     margin          .bool_gset:N = \g__ztex_margin_bool,
151     margin          .initial:n   = { false },
152     margin          .default:n   = { true },
153     slide           .bool_gset:N = \g__ztex_slide_bool,
154     slide           .initial:n   = { false },
155     slide           .default:n   = { true },
156     aspect          .tl_gset:N   = \g__ztex_aspectratio_tl,
157     aspect          .initial:n   = { 12|9 },
158     theme           .str_gset:N   = \g__ztex_slide_theme_str,
159     theme           .initial:n   = { AnnArborDefault },
160     unknown         .code:n      = {
161         \ztex_metakey_msg_warning:nn {option-layout}
162         {margin(<bool>:false), slide, aspect, theme}
163     }
164 }
165
166
167 % ==> thm(mathspec) options
168 \ztex_keys_define:nn { mathSpec }{
169     alias           .bool_gset:N = \g__ztex_math_alias_bool,
170     alias           .initial:n   = { false },
171     alias           .default:n   = { true },
172     envStyle        .tl_gset:N   = \g__ztex_thm_style_tl,
173     envStyle        .initial:n   = { plain },
174     font            .choice:,
175     font / newtx    .meta:nn     = { ztex / font / math }{#1},
176     font / mtpro2   .meta:nn     = { ztex / font / math }{#1},
177     font / euler    .meta:nn     = { ztex / font / math }{#1},
178     font / var-euler .meta:nn     = { ztex / font / math }{#1},
179     font / mathpazo .meta:nn     = { ztex / font / math }{#1},
180     unknown         .code:n      = {
181         \ztex_metakey_msg_warning:nn {option-mathSpec}
182         {alias(<bool>:false), envStyle, font(<choice>:newtx,mtpro2,euler,mathpazo)}
183     }
184 }
185
186
187 % ==> bib/index options
188 \ztex_keys_define:nn { bib_index }{
189     load            .bool_gset:N = \g__ztex_bib_index_load_bool,
190     source          .str_gset:N  = \g__ztex_bib_source_str,

```


191	source	.initial:n	= { ref.bib },	191
192	backend	.str_gset:N	= \g__ztex_bib_backend_str,	192
193	backend	.initial:n	= { biber },	193
194	unknown	.code:n	= {	194
195		\ztex_metakey_msg_warning:nn	{option-bib_index}	195
196		{load(<bool>:false), source, backend}		196
197		}		197
198		}		198

10.3 Module

10.3.1 box

```
1 \ProvidesExplFile{ztex.module.box.tex} 1
2 {2026/09/23}{\ztex@version} 2
3 {box~module~for~ztex} 3
4 4
5 5
6 %%%% box module for ztex %%%% 6
7 \RequirePackage{framed} 7
8 \RequirePackage{framedmulticol} 8
9 9
10 10
11 % ==> dimension / tmp variables 11
12 \dim_new:N \c_ztex_quad_dim 12
13 \ztool_gget_wd:Nn \c_ztex_quad_dim {\quad} 13
14 \dim_new:N \l__zbox_tmpa_dim 14
15 \dim_new:N \l__zbox_tmpb_dim 15
16 \dim_new:N \l__zbox_tmpc_dim 16
17 \box_new:N \l__zbox_tmpa_box 17
18 \box_new:N \l__zbox_tmpb_box 18
19 \box_new:N \l__zbox_tmpc_box 19
20 \cs_new:Npn \zbox_ltx_rule:nnn #1#2#3 20
21 { \rule[#1]{#2}{#3} } 21
22 22
23 23
24 % ==> better '\mbox' and '\fbox'(\verb can be used inside) 24
25 % REF: https://texhacks.blogspot.com/2011/01/better-mbox-and-fbox.html?m=1 25
26 % enhance \mbox 26
27 \def\zmbbox{\leavevmode\hbox} 27
28 28
29 % universe box-content collect interface 29
30 \newbox\ztexcollectbox 30
31 \def\zboxcollectpre{} 31
32 \def\zboxcollect@pre 32
33 { 33
34 \bgroup 34
35 \zboxcollectpre 35
36 \aftergroup\zboxcollect@post 36
37 } 37
38 \def\zboxcollectpost{} 38
39 \def\zboxcollect@post 39
40 { 40
41 \zboxcollectpost 41
42 \egroup 42
43 \zboxcollect@typeset 43
44 } 44
45 \def\zboxcollect@typeset 45
```

```

46 {
47   \box\ztexcollectbox\relax
48 }
49 \cs_set_protected:Npn \zbox_item_collect:nnnw #1#2#3
50 {% #1:box typeset spec; #2/#3:box material pre/post
51 % NOTE: box typeset is after 'post'.
52 \cs_set:Npn \zboxcollect@typeset
53 { #1 }
54 \def\zboxcollectpre{#2}
55 \def\zboxcollectpost{#3}
56 \zboxcollect
57 }
58
59 % users' interface
60 \protected\long\def\zboxcollect
61 {
62   \leavevmode
63   \afterassignment\zboxcollect@pre
64   \setbox\ztexcollectbox\hbox
65 }
66 \NewDocumentCommand{\zboxcollectcustom}{mO{}O{}}
67 { \zbox_item_collect:nnnw { #1 } { #2 } { #3 } }
68
69 % enhance \fbox
70 \ztex_keys_define:nn { box/enhance-fbox }
71 {
72   draw .tl_set:N = \l__zbox_fbox_draw_tl,
73   draw .initial:n = { black },
74   fill .tl_set:N = \l__zbox_fbox_fill_tl,
75   fill .initial:n = { none },
76   arc .fp_set:N = \l__zbox_fbox_arc_fp,
77   arc .initial:n = { 1 },
78   text .tl_set:N = \l__zbox_fbox_text_sty_tl,
79   text .initial:n = {},
80 }
81 \NewDocumentCommand\zfbox{o}
82 {
83   \IfValueT{#1}{\ztex_keys_set:nn { box/enhance-fbox } {#1}}
84   \zbox_item_collect:nnnw
85   {
86     \begin{zpic} [
87       unit=1pt, % \wd, \ht and \dp returns dim in 'pt', thus we must set 'unit=1pt'.
88       xoffset=\dim_to_decimal:n { \fboxrule/2 },
89       yoffset=-\dim_to_decimal:n
90       { \dp\ztexcollectbox + \fboxsep + \fboxrule/2 },
91       width=\dim_to_decimal:n
92       { \wd\ztexcollectbox + 2\fboxsep + 2\fboxrule },
93       height=\dim_to_decimal:n

```

```

94 { \ht\ztexcollectbox + \fboxsep + \fboxrule },
95 depth=\dim_to_decimal:n
96 { \dp\ztexcollectbox + \fboxsep + \fboxrule },
97 ]
98 \zrectangle[
99 arc=\l__zbox_fbox_arc_fp, width=\fboxrule,
100 fill=\l__zbox_fbox_fill_tl, draw=\l__zbox_fbox_draw_tl
101 ]
102 ( 0, 0 )
103 (
104 \dim_to_decimal:n { \wd\ztexcollectbox + 2\fboxsep + \fboxrule },
105 \dim_to_decimal:n
106 {
107 \ht\ztexcollectbox + \dp\ztexcollectbox + 2\fboxsep +
108 \fboxrule
109 }
110 )
111 \put
112 (
113 \dim_to_decimal:n
114 { \wd\ztexcollectbox / 2 + \fboxsep + \fboxrule / 2 },
115 \dim_to_decimal:n
116 { \dp\ztexcollectbox + \fboxsep + \fboxrule / 2 }
117 ) { \clap {\box_use_drop:N \ztexcollectbox} }
118 \end{zpic}
119 % reset fbox style options
120 \ztex_keys_set:nn { box/enhance-fbox }
121 {
122 draw = black, fill = none,
123 arc = 1, text = ,
124 }
125 \relax
126 }
127 { \l__zbox_fbox_text_sty_tl }
128 {}
129 }
130
131
132 % ==> 'framed' env for user
133 \ztex_keys_define:nn { box/framed-user }
134 {
135 rulewidth .dim_set:N = \l__zbox_frameduser_rulewd_dim,
136 rulewidth .initial:n = { 5pt },
137 rulecolor .tl_set:N = \l__zbox_frameduser_rulecolor_tl,
138 rulecolor .initial:n = { red },
139 padding .dim_set:N = \l__zbox_frameduser_padding_dim,
140 padding .initial:n = { 5pt },
141 bg .tl_set:N = \l__zbox_frameduser_bgcolor_tl,

```

142	bg	.initial:n = { gray!10 },	142
143	adj	.dim_set:N = \l__zbox_frameduser_boxadj_dim, % width adjust	143
144	adj	.initial:n = { 0pt },	144
145	}		145
146	\cs_new:Nn __ztex_make_frame_options_restore:		146
147	{		147
148	\dim_set:Nn \l__zbox_frameduser_rulewd_dim	{ 5pt }	148
149	\tl_set:Nn \l__zbox_frameduser_rulecolor_tl	{ red }	149
150	\dim_set:Nn \l__zbox_frameduser_padding_dim	{ 5pt }	150
151	\tl_set:Nn \l__zbox_frameduser_bgcolor_tl	{ gray!10 }	151
152	\dim_set:Nn \l__zbox_frameduser_boxadj_dim	{ 0pt }	152
153	}		153
154	\cs_new_protected:Npn \ztex_make_frame_begin:nnnnn	#1#2#3#4#5	154
155	{%	#1:rule color; #2:rule width; #3:padding; #4:bg color; #5:box width adjust	155
156	\def\FrameCommand		156
157	{		157
158	{ \color{#1} \vrule width #2 }	% leftbar	158
159	{ \color{#4} \vrule width #3 }	% padding	159
160	\colorbox{#4}		160
161	}		161
162	\MakeFramed		162
163	{		163
164	% NOTE: \width = padding + rulewidth		164
165	\dim_set:Nn \l__zbox_tmpa_dim	{ -\width + #5 }	165
166	\advance\hsize \l__zbox_tmpa_dim	\relax	166
167	\FrameRestore		167
168	}		168
169	}		169
170	\cs_generate_variant:Nn \ztex_make_frame_begin:nnnnn	{ eeeee, ooooo }	170
171	\cs_new_protected:Npn \ztex_make_frame_end:		171
172	{ \endMakeFramed }		172
173	\NewDocumentCommand\ztexframe{o}		173
174	{		174
175	\IfValueT { #1 }		175
176	{		176
177	\ztex_keys_set:nn { box/framed-user }	{#1}	177
178	}		178
179	\ztex_make_frame_begin:eeeee		179
180	{ \l__zbox_frameduser_rulecolor_tl }		180
181	{ \l__zbox_frameduser_rulewd_dim }		181
182	{ \l__zbox_frameduser_padding_dim }		182
183	{ \l__zbox_frameduser_bgcolor_tl }		183
184	{ \l__zbox_frameduser_boxadj_dim }		184
185	}		185
186	\NewDocumentCommand\ztexframeend{}		186
187	{		187
188	\ztex_make_frame_end:		188
189	__ztex_make_frame_options_restore:		189

190	}	190
191		191
192		192
193	% ==> box info, scale, raise/lower	193
194	% get dim info	194
195	\NewDocumentCommand{\getwd}{smm}	195
196	{	196
197	\IfBooleanTF{#1}	197
198	{	198
199	\ztool_gget_wd:Nn #2{#3}	199
200	}{	200
201	\ztool_get_wd:Nn #2{#3}	201
202	}	202
203	}	203
204	\NewDocumentCommand{\getht}{smm}	204
205	{	205
206	\IfBooleanTF{#1}	206
207	{	207
208	\ztool_gget_ht:Nn #2{#3}	208
209	}{	209
210	\ztool_get_ht:Nn #2{#3}	210
211	}	211
212	}	212
213	\NewDocumentCommand{\getdp}{smm}	213
214	{	214
215	\IfBooleanTF{#1}	215
216	{	216
217	\ztool_gget_dp:Nn #2{#3}	217
218	}{	218
219	\ztool_get_dp:Nn #2{#3}	219
220	}	220
221	}	221
222	% scale box	222
223	\NewDocumentCommand{\wscale}{smm}	223
224	{	224
225	\IfBooleanTF{#1}	225
226	{	226
227	\ztool_scale_to_wd:nn {#2}{#3}	227
228	}{	228
229	\ztool_set_to_wd:nn {#2}{#3}	229
230	}	230
231	}	231
232	\NewDocumentCommand{\hscale}{smm}	232
233	{	233
234	\IfBooleanTF{#1}	234
235	{	235
236	\ztool_scale_to_ht:nn {#2}{#3}	236
237	}{	237

238	\ztool_set_to_ht:nn {#2}{#3}	238
239	}	239
240	}	240
241	\NewDocumentCommand{\zrotate}{mm}	241
242	{	242
243	\ztool_rotate:nn {#1}{#2}	243
244	}	244
245	% raise box	245
246	\NewDocumentCommand{\zraise}{mm}	246
247	{	247
248	\box_move_up:nn {#1}{\hbox:n {#2}}	248
249	}	249
250	\NewDocumentCommand{\zlower}{mm}	250
251	{	251
252	\box_move_down:nn {#1}{\hbox:n {#2}}	252
253	}	253
254		254
255		255
256	% ==> hide text	256
257	\tl_new:N \l__zbox_hidetext_map_tl	257
258	\ztex_keys_define:nn { box/hidetext }	258
259	{	259
260	map .choice:,	260
261	map / tl .code:n = { \tl_set:Nn \l__zbox_hidetext_map_tl { tl } },	261
262	map / str .code:n = { \tl_set:Nn \l__zbox_hidetext_map_tl { str } },	262
263	map / unknown .code:n = {	263
264	\ztex_msg_set:nn { zbox@hidetext@maptype }	264
265	{ map~type~must~be~'tl'~or~'str',~but~you~entered~'#1' }	265
266	\ztex_msg_error:n { zbox@hidetext@maptype } },	266
267	fill .tl_set:N = \l__zbox_hidetext_fill_tl,	267
268	fill .initial:n = { black },	268
269	frame .tl_set:N = \l__zbox_hidetext_frame_tl,	269
270	frame .initial:n = { black },	270
271	killdp .bool_set:N = \l__zbox_hidetext_killdp_bool,	271
272	killdp .initial:n = { false },	272
273	killdp .default:n = { true },	273
274	separator .tl_set:N = \l__zbox_hidetext_separator_tl,	274
275	separator .initial:n = { _ }, % to allow line break	275
276	% separator .initial:n = { \discretionary{}{}{} }, % to allow line break	276
277	cmd .cs_set:Np = __zbox_hidetext_cmd:n #1,	277
278	cmd .initial:n = { #1 },	278
279	}	279
280	\NewDocumentCommand{\hidetext}{om}	280
281	{	281
282	\group_begin:	282
283	\tl_set:Nn \l__zbox_hidetext_map_tl { tl }	283
284	\IfValueT { #1 }	284
285	{	285

```

286 \ztex_keys_set:nn { box/hidetext }{ #1 }
287 }
288 \use:c { \l__zbox_hidetext_map_tl _map_inline:nn }{ #2 }
289 {
290   \hbox_set:Nn \l__zbox_tmpa_box { ##1 }
291   \bool_if:NTF \l__zbox_hidetext_killdp_bool
292     { \dim_set:Nn \l__zbox_tmpa_dim { Opt } }
293     { \dim_set:Nn \l__zbox_tmpa_dim { - \box_dp:N \l__zbox_tmpa_box } }
294   \__zbox_hidetext_cmd:n
295     { \textcolor { \l__zbox_hidetext_fill_tl }
296       {
297         \zbox_ltx_rule:nnn
298           { \dim_use:N \l__zbox_tmpa_dim }
299           { \box_wd:N \l__zbox_tmpa_box }
300           { \box_ht:N \l__zbox_tmpa_box }
301       }
302     }
303   \l__zbox_hidetext_separator_tl
304 }
305 \group_end:
306 }
307
308
309 % ==> box item align
310 \ztex_msg_set:nn {boxitem-align}
311 {
312   Valid~align~options~for~\string\zboxitemalign~are:
313   'left',~'center',~'right',~'scatter',~'tower'~and~'custom'.
314 }
315 \ztex_keys_define:nn { box / align }
316 {
317   cmd .tl_set:N = \l__ztex_boxitem_align_cmd_tl,
318   cmd .initial:n = { \use:n },
319   type .tl_set:N = \l__ztex_boxitem_align_type_tl,
320   type .initial:n = { center },
321   custom .tl_set:N = \l__ztex_boxitem_align_custom_tl,
322   custom .initial:n = { \align@object },
323 }
324 % NOTE: any explicit blank space in 'object' will be absorded.
325 \NewDocumentCommand{\zboxitemalign}{omm}
326 {% #1:cmd, #2:width; #3:object
327   \group_begin:
328   \ztex_keys_set:nn { box / align }{#1}
329   \tl_if_in:nVF {left, center, right, scatter, tower, custom}
330     \l__ztex_boxitem_align_type_tl
331     { \ztex_msg_error:n {boxitem-align} }
332   \ztool_box_item_align:Nnno
333     \l__ztex_boxitem_align_cmd_tl

```


334	{ #2 }{ #3 }	334
335	{ \l__ztex_boxitem_align_type_tl }	335
336	\group_end:	336
337	}	337

10.3.2 font

```
1 \ProvidesExplFile{ztex.module.font.tex}
2 {2025/09/30}{\ztex@versi@n}
3 {font~module~for~ztex}
4
5
6 %%%% font module for ztex %%%%
7 \bool_if:NT \g__ztex_sysfont_cfg_bool
8 {
9   \RequirePackage{fontspec}
10 }
11 \cs_set_protected:Npn \ztex_font_set:n #1
12 {
13   \ztex_keys_set:nn { font }{#1}
14 }
15 \NewDocumentCommand{\zfontset}{m}
16 {
17   \ztex_font_set:n {#1}
18 }
19 % reset text font to the default computer modern
20 \NewDocumentCommand{\resetfont}{0{cm}}
21 {
22   \str_case:nnT { #1 }
23   {
24     {cm}{
25       \renewcommand\rmdefault{cmr}
26       \renewcommand\sfddefault{cmss}
27       \renewcommand\ttdefault{cmtt}
28     }
29     {lm}{
30       \renewcommand\rmdefault{lmr}
31       \renewcommand\sfddefault{lmss}
32       \renewcommand\ttdefault{lmtt}
33     }
34   }{ \normalfont }
35 }
36
37
38 % ==> patches:
39 % symbols:
40 \DeclareMathSymbol{\blacktriangleright}{\mathrel}{AMSa}{"49}
41 \cs_new:Nn \__ztex_text_symbol_patch:
42 {
43   \let\oldtextbullet\textbullet
44   \DeclareTextFontCommand{\zslideCmsyOms}
45     {\fontfamily{cmsy}\fontencoding{OMS}\selectfont}
46   \DeclareRobustCommand{\textbullet}
```

```

47      {\zslideCmsy0ms\oldtextbullet}
48  }
49  % cancel CJKecglue(use it inside a group)
50  % NOTE: default 'CJKecglue' is one space.
51  \cs_new_protected:Nn \zfont_cancel_CJKecglue:
52  {
53      \str_if_eq:VnT \g__ztex_lang_str { cn }
54      {
55          \xeCJKsetup
56          {
57              CJKecglue = {\hskip Opt plus 0.08\baselineskip}
58          }
59      }
60  }
61  \cs_new_protected:Nn \zfont_restore_CJKecglue:
62  {
63      \str_if_eq:VnT \g__ztex_lang_str { cn }
64      {
65          \xeCJKsetup
66          {
67              CJKecglue = { \tex_space:D }
68          }
69      }
70  }
71  \cs_new_protected:Npn \removeCJKecglue
72  { \zfont_cancel_CJKecglue: }
73  \cs_new_protected:Npn \restoreCJKecglue
74  { \zfont_restore_CJKecglue: }
75
76
77  % ==> using system fonts
78  %%% NOTE %%%
79  % 1. MOST FONTS only have a limited set of FEATURES
80  % 2. MOST CJK fonts' features are not equal to western fonts.
81  \ztex_keys_define:nn { fontcfg / new }
82  {
83      name      .tl_set:N = \l__ztex_fontcfg_new_name_tl, % font name / file name
84      path      .tl_set:N = \l__ztex_fontcfg_new_path_tl,
85      path      .initial:n = { },
86      feat      .meta:nn = { ztex / fontcfg / new / feat }{#1},
87      feat / ext      .tl_set:N = \l__ztex_fontcfg_new_ext_tl,
88      feat / Extension .meta:n = { feat / ext = #1 },
89      feat / ext      .initial:n = { }, % extension
90      feat / up        .tl_set:N = \l__ztex_fontcfg_new_up_tl,
91      feat / UprightFont .meta:n = { feat / up = #1 },
92      feat / up        .initial:n = { * }, % *-regular
93      feat / sl        .tl_set:N = \l__ztex_fontcfg_new_sl_tl,
94      feat / SlantedFont .meta:n = { feat / sl = #1 },

```

```

95     feat / sl          .initial:n    = { * }, % *-slant          95
96     feat / sc          .tl_set:N     = \l__ztex_fontcfg_new_sc_tl,  96
97     feat / SmallCapsFont .meta:n    = { feat / sc = #1 },      97
98     feat / sc          .initial:n    = { * }, % *-smallcaps    98
99     feat / bd          .tl_set:N     = \l__ztex_fontcfg_new_bd_tl,  99
100    feat / BoldFont     .meta:n      = { feat / bd = #1 },        100
101    feat / bd          .initial:n    = { * }, % *-bold          101
102    feat / it          .tl_set:N     = \l__ztex_fontcfg_new_it_tl,  102
103    feat / ItalicFont   .meta:n      = { feat / it = #1 },        103
104    feat / it          .initial:n    = { * }, % *-italic        104
105    feat / bdit         .tl_set:N     = \l__ztex_fontcfg_new_bdit_tl, 105
106    feat / BoldItalicFont .meta:n    = { feat / bdit = #1 },      106
107    feat / bdit         .initial:n    = { * }, % *-bolditalic    107
108    feat / bds1         .tl_set:N     = \l__ztex_fontcfg_new_bds1_tl, 108
109    feat / BoldSlantedFont .meta:n   = { feat / bds1 = #1 },      109
110    feat / bds1         .initial:n    = { * }, % *-boldslant     110
111    feat / other        .tl_set:N     = \l__ztex_fontcfg_new_other_tl, 111
112    feat / other        .initial:n    = { }, % other fontspec config 112
113 } 113
114 114
115 \ztex_msg_set:nn { fontcfg / lang } 115
116 { Current~font~type~supported~are:'en',~'CJK'. } 116
117 \cs_set:Npn \__ztex_fontcfg_newfamily_copy:nnnnn #1#2#3#4#5 117
118 {% #1:font cmd; #2:font file path(format 'Path=xxx,'); 118
119 % #3:font file name; #4:font feat; #5:en/CJK 119
120 \str_case:nnF {#5} 120
121 { 121
122     {en}{ 122
123         \exp_args:Nc \setfontfamily{zfont@#1}{#3}[#2 #4] 123
124         \exp_args:Nc \NewDocumentCommand { #1 }{} 124
125         { 125
126             \use:c {zfont@#1} 126
127         } 127
128     } 128
129     {CJK}{ 129
130         \setCJKfamilyfont{zfont@#1}{#3}[#2 #4] 130
131         \exp_args:Nc \NewDocumentCommand { #1 }{} 131
132         { 132
133             \CJKfamily{zfont@#1} 133
134         } 134
135     } 135
136     {en/CJK}{ 136
137         \setCJKfamilyfont{zfont@#1}{#3}[#2 #4] 137
138         \exp_args:Nc \setfontfamily{zfont@#1}{#3}[#2 #4] 138
139         \exp_args:Nc \NewDocumentCommand { #1 }{} 139
140         { 140
141             \use:c {zfont@#1} 141
142             \CJKfamily{zfont@#1} 142

```

```
143 }
144 }
145 }{
146   \ztex_msg_error:n { fontcfg / new }
147 }
148 }
149 \cs_generate_variant:Nn \__ztex_fontcfg_newfamily_copy:nnnnn
150 { ooooo, eeeee }
151 \cs_new_protected:Npn \__ztex_sysfont_new:nnn #1#2#3
152 {% #1:en/cn; #2:cmd; #3:key-value(font cfg args)
153   \ztex_keys_set:nn { fontcfg / new } {#3}
154   \__ztex_fontcfg_newfamily_copy:eeeeee
155   { #2 }
156   {
157     \tl_if_empty:NF \l__ztex_fontcfg_new_path_tl
158     { Path=\l__ztex_fontcfg_new_path_tl, }
159   }
160   { \l__ztex_fontcfg_new_name_tl }
161   {
162     \tl_if_empty:NF \l__ztex_fontcfg_new_ext_tl
163     { Extension = \l__ztex_fontcfg_new_ext_tl, }
164     UprightFont = \l__ztex_fontcfg_new_up_tl,
165     BoldFont = \l__ztex_fontcfg_new_bd_tl,
166     ItalicFont = \l__ztex_fontcfg_new_it_tl,
167     SlantedFont = \l__ztex_fontcfg_new_sl_tl,
168     SmallCapsFont = \l__ztex_fontcfg_new_sc_tl,
169     BoldItalicFont = \l__ztex_fontcfg_new_bdit_tl,
170     BoldSlantedFont = \l__ztex_fontcfg_new_bdsl_tl,
171     \l__ztex_fontcfg_new_other_tl % other fontspec config
172   }{ #1 }
173   % Reset key value, '\cs{group_end:}' conflict with '\cs{newfontfamily}',
174   % See also: https://tex.stackexchange.com/q/729765/294585.
175   \ztex_keys_set:nn { fontcfg / new }
176   {
177     path = ,
178     feat / ext = ,
179     feat / up = *,
180     feat / bd = *,
181     feat / it = *,
182     feat / sl = *,
183     feat / sc = *,
184     feat / bdsm = *,
185     feat / bdit = *,
186     feat / other = ,
187   }
188 }
189 \bool_if:NTF \g__ztex_sysfont_cfg_bool
190 {
```

191	__ztex_sysfont_new:nnn {en}{cinzel}	191
192	{	192
193	name = Cinzel-Regular.ttf,	193
194	feat / bd = Cinzel-Bold,	194
195	feat / it = ParsiMatn-Italic,	195
196	}	196
197	}{ \def\cinzel{\relax} }	197
198	\NewDocumentCommand{\zfontfamilynew}{0{en/CJK}mm}	198
199	{	199
200	__ztex_sysfont_new:nnn	200
201	{ #1 }	201
202	{ \cs_to_str:N #2 }	202
203	{ #3 }	203
204	}	204
205		205
206		206
207	% set document main fontfamily	207
208	\NewDocumentCommand{\zfontfamilyset}{0{en}m}	208
209	{	209
210	\clist_map_inline:nn { #1 }	210
211	{	211
212	\keyval_parse:nnn	212
213	{ \use_none:n }	213
214	{ __ztex_fontcfg_setfamily_aux:nnn {##1} }	214
215	{ #2 }	215
216	}	216
217	}	217
218	\ztex_msg_set:nn { fontcfg / family }	218
219	{ Valid~family~options~are:'main',~'sans'~and~'mono'. }	219
220	\cs_set:Npn __ztex_fontcfg_setfamily_aux:nnn #1#2#3	220
221	{% #1:lang, #2:family, #3:{font}{font features}	221
222	\tl_if_in:nnF {en, CJK}{#1}	222
223	{ \ztex_msg_error:n { fontcfg / lang } }	223
224	\tl_if_in:nnF {main, sans, mono}{#2}	224
225	{ \ztex_msg_error:n { fontcfg / family } }	225
226	\ltx_main_fontfamily_set:eee	226
227	{ \str_case:nn {#1}{\en}{\CJK}{CJK} #2 }	227
228	{ \use_i:nn #3 }	228
229	{ \use_ii:nn #3 }	229
230	}	230
231	\cs_new_protected:Npn \ltx_main_fontfamily_set:nnn #1#2#3	231
232	{	232
233	\cs:w set #1 font\cs_end:	233
234	{ #2 }[#3]	234
235	}	235
236	\cs_generate_variant:Nn \ltx_main_fontfamily_set:nnn	236
237	{ eee }	237

```

1 \ProvidesExplFile{ztex.module.ref.tex}
2 {2025/09/20}{\ztex@versi@n}
3 {ref~module~for~ztex}
4
5
6 %%%% ref module for ztex %%%%
7 % ==> package loading
8 \bool_if:NT \g__ztex_bib_index_load_bool {
9   \RequirePackage{indextools}
10  \str_case:Vn \g__ztex_bib_backend_str {
11    {bibtex}{\RequirePackage[backend=bibtex]{biblatex}}
12    {biber}{\RequirePackage[backend=biber]{biblatex}}
13  }
14  \exp_args:Nx \addbibresource{\str_use:N \g__ztex_bib_source_str}
15 }
16 \bool_if:NT \g__ztex_hyperref_bool
17 {
18   \clist_map_inline:Nn \g__ztex_hyper_suppress_clist
19   {
20     \exp_after:wN \def
21     \cs:w hyper@nopatch@#1 \cs_end: { }
22   }
23   \RequirePackage{hyperref}
24   \SetLinkTargetFilter{ztex@\jobname @#1}
25 }
26
27
28 % ==> provide hyper command
29 \ProvideDocumentCommand\hypersetup{m}{}
30 \ProvideDocumentCommand\hyper@anchor{m}{}
31 \ProvideDocumentCommand\hyper@link{mmm}{#3}
32 \ProvideDocumentCommand\hyper@linkstart{mm}{}
33 \ProvideDocumentCommand\hyper@linkend{}{}
34 \ProvideDocumentCommand\hyper@linkfile{mmm}{}
35 \ProvideDocumentCommand\MakeLinkTarget{sO{m}}{}
36 \ProvideDocumentCommand\LinkTargetOn{}{}
37 \ProvideDocumentCommand\LinkTargetOff{}{}
38 \ProvideDocumentCommand\NextLinkTarget{m}{}
39 \ProvideDocumentCommand\SetLinkTargetFilter{m}{}
40 \ProvideDocumentCommand\pdfbookmark{omm}{}
41 \ProvideDocumentCommand\texorpdfstring{mm}{#1}
42 \cs_new:Npn \ztex_make_link_target:n #1
43 {
44   \MakeLinkTarget*{#1}
45 }
46 \cs_generate_variant:Nn \ztex_make_link_target:n { e }

```

```
47 \NewDocumentCommand{\zsetHcnt}{mm}
48 {
49   \exp_after:wN \def\cs:w theH #1\cs_end:
50   { #2 }
51 }
52 \NewDocumentCommand{\ztexlink}{sm}
53 {
54   \IfBooleanTF{#1}{
55     \zboxcollectcustom
56     {
57       \hyper@link{link}{#2}{\box\ztexcollectbox}
58     }
59   }{
60     \hyper@link{link}{#2}
61   }
62 }
63
64 % link to title class
65 \ztex_msg_set:nn { linkclass@nonexist }
66 {
67   link~title:'#2'~of~class~'#1'~does~NOT~
68   exsit,~check~it~again.
69 }
70 \NewDocumentCommand{\ztexlinkclass}{smm}
71 {
72   \edef\ztexlinkclass@curtoclineindex
73   {
74     \ztoc_tocline_index_byclasstitle:nnnN
75     { #2 }{ #3 }{ \c_false_bool }
76     \g_ztoc_keyvaltoc_seq
77   }
78   \exp_args:No \int_compare:nNnTF
79   {
80     \ztexlinkclass@curtoclineindex
81     } < { 0 }
82   {
83     \ztex_msg_warn:nnn
84     { linkclass@nonexist }
85     { #2 }{ #3 }
86   }{
87     \edef\ztexlinkclass@curtocline
88     {
89       \seq_item:Ne \g_ztoc_keyvaltoc_seq
90       {
91         \ztexlinkclass@curtoclineindex
92       }
93     }
94     \IfBooleanTF{#1}
```



```
95     { \exp_args:NNe \ztexlink* }
96     { \exp_args:Ne \ztexlink }
97     {
98         \exp_args:No \prop_item:nn
99         { \ztexlinkclass@curtocline }{ anchor }
100     }
101 }
102 }
103
104 % display style of the link page:
105 % \pdf_destination:nn
106
107
108 % ==> clever reference for sections, figure and table
109 \cs_set:Npn \cref@pl@suffix {\str_if_eq:VnF \g__ztex_lang_str {cn}{s}}
110 \str_case:VnF \g__ztex_cref_backend_str
111 {
112     {cleveref}{
113         \RequirePackage[nameinlink]{cleveref}
114         \str_case:VnF \g__ztex_lang_str {
115             {en}{
116                 \IfClassLoadedTF{book}{
117                     \crefname{part}{part}{parts}
118                     \crefname{chapter}{chapter}{chapters}
119                 }{\relax}
120                 \crefname{section}{section}{sections}
121                 \crefname{subsection}{subsection}{subsections}
122                 \crefname{figure}{figure}{figures}
123                 \crefname{table}{table}{tables}
124                 \crefname{equation}{equation}{equations}
125                 \crefname{ztex@thm@sharecnt}{Result}{Results}
126             }
127             {cn}{
128                 \IfClassLoadedTF{book}{
129                     \crefname{part}{部分}{部分}
130                     \crefname{chapter}{章}{章}
131                 }{\relax}
132                 \crefname{section}{节}{节}
133                 \crefname{subsection}{小节}{小节}
134                 \crefname{figure}{图}{图}
135                 \crefname{table}{表}{表}
136                 \crefname{equation}{方程}{方程}
137                 \crefname{ztex@thm@sharecnt}{结果}{结果}
138             }
139         }{\ztex_msg_error:n {option-language}}
140         \creflabelformat{ztex@thm@sharecnt}{#2(#1)#3}
141         \cs_new:Npn \__ztex_cref_math_env:n #1 {
142             \exp_args:Nnff \crefname{#1}
```

143	<code>{\prop_item:cn {g__ztex_thm_name_prop}{#1}}</code>	143
144	<code>{\prop_item:cn {g__ztex_thm_name_prop}{#1}\cref@pl@suffix}</code>	144
145	<code>\creflabelformat{#1}{##2(##1)##3}</code>	145
146	<code>% Arg-spec of command \cs{creflabelformat}:</code>	146
147	<code>% ##1: the counter, like '2.1';</code>	147
148	<code>% ##2/##3: hyperlink scope identifier</code>	148
149	<code>}</code>	149
150	<code>}</code>	150
151	<code>{zref-clever}{</code>	151
152	<code>\RequirePackage{zref-clever}</code>	152
153	<code>\exp_args:Ne \zcsetup{</code>	153
154	<code> nameinlink,</code>	154
155	<code> lang = \g__ztex_lang_str,</code>	155
156	<code> typeset = both,</code>	156
157	<code> refbounds = { ,(), },</code>	157
158	<code>}</code>	158
159	<code>% Pre-defined Language files:</code>	159
160	<code>% English, German, French,</code>	160
161	<code>% Portuguese, and Spanish.</code>	161
162	<code>\zcDeclareLanguageAlias{en}{english}</code>	162
163	<code>\zcLanguageSetup{english}{</code>	163
164	<code> type = ztex@thm@sharecnt,</code>	164
165	<code> name-sg = Result,</code>	165
166	<code> Name-sg = Result,</code>	166
167	<code> name-pl = Results,</code>	167
168	<code> Name-pl = Results,</code>	168
169	<code>}</code>	169
170	<code>\zcDeclareLanguage{chinese}</code>	170
171	<code>\zcDeclareLanguageAlias{cn}{chinese}</code>	171
172	<code>\zcLanguageSetup{chinese}{</code>	172
173	<code> type = part,</code>	173
174	<code> name-sg = 部分,</code>	174
175	<code> Name-sg = 部分,</code>	175
176	<code> name-pl = 部分,</code>	176
177	<code> Name-pl = 部分,</code>	177
178	<code> type = chapter,</code>	178
179	<code> name-sg = 章,</code>	179
180	<code> Name-sg = 章,</code>	180
181	<code> name-pl = 章,</code>	181
182	<code> Name-pl = 章,</code>	182
183	<code> type = section,</code>	183
184	<code> name-sg = 节,</code>	184
185	<code> Name-sg = 节,</code>	185
186	<code> name-pl = 节,</code>	186
187	<code> Name-pl = 节,</code>	187
188	<code> type = subsection,</code>	188
189	<code> name-sg = 小节,</code>	189
190	<code> Name-sg = 小节,</code>	190

```
191         name-pl = 小节,
192         Name-pl = 小节,
193     type = paragraph,
194         name-sg = 段落,
195         Name-sg = 段落,
196         name-pl = 段落,
197         Name-pl = 段落,
198     type = figure,
199         name-sg = 图,
200         Name-sg = 图,
201         name-pl = 图,
202         Name-pl = 图,
203     type = table,
204         name-sg = 表,
205         Name-sg = 表,
206         name-pl = 表,
207         Name-pl = 表,
208     type = equation,
209         name-sg = 方程,
210         Name-sg = 方程,
211         name-pl = 方程,
212         Name-pl = 方程,
213     type = ztex@thm@sharecnt,
214         name-sg = 结果,
215         Name-sg = 结果,
216         name-pl = 结果,
217         Name-pl = 结果,
218 }
219 \cs_new:Npn \__ztex_cref_math_env:n #1
220 {
221     \zcRefTypeSetup {#1}
222     {
223         name-sg = \prop_item:cn {g__ztex_thm_name_prop}{#1},
224         Name-sg = \prop_item:cn {g__ztex_thm_name_prop}{#1},
225         name-pl = \prop_item:cn {g__ztex_thm_name_prop}{#1}\cref@pl@suffix,
226         Name-pl = \prop_item:cn {g__ztex_thm_name_prop}{#1}\cref@pl@suffix,
227     }
228 }
229 \let\cref\zceref
230 }
231 }{
232 \ztex_msg_set:nn {option-backend}{
233     option-backend~invalid,~ztex~currently~only~support~'cleveref'
234     ~or~'zref-clever'~for~option-backend.
235 }
236 \ztex_msg_error:n {option-backend}
237 }
238 }
```

239		239
240	% ==> save position	240
241	<code>\NewDocumentCommand\zrefsavepos{m}</code>	241
242	{	242
243	<code>\tex_savepos:D</code>	243
244	<code>\property_record:nn {ztexref@#1@pos}{xpos, ypos}</code>	244
245	<code>\tex_savepos:D</code>	245
246	}	246
247	<code>\NewExpandableDocumentCommand\zrefgetpos{mm}</code> % return in 'pt'	247
248	{	248
249	<code>\fp_eval:n</code>	249
250	<code>{ \property_ref:nn {ztexref@#1@pos}{#2}/65536 }</code>	250
251	}	251

```

1 \ProvidesExplFile{ztex.module.page.tex}
2 {2026/02/01}{\ztex@versi@n}
3 {page~module~for~ztex}
4
5
6 %%%% page module for ztex %%%%
7 \RequirePackage{geometry}
8 % TODO: replace 'sidenotes' by 'minipage'
9 \cs_set:Npn \__ztex_layout_setup:n #1
10 { \geometry{#1} }
11
12
13 % ==> document mode
14 \if@twoside
15 \bool_if:NTF \g__ztex_margin_bool {
16 \__ztex_layout_setup:n {
17 a4paper,
18 left=2.5cm, right=7.5cm,
19 bottom=3.5cm, top=3.2cm,
20 headsep=.3cm, footskip=1.5cm,
21 marginparsep=2em
22 }
23 \dim_gset:Nn \marginparwidth{14em}
24 }{
25 \__ztex_layout_setup:n {
26 a4paper,
27 left=3cm, right=5.5cm,
28 bottom=3.5cm, top=3.2cm,
29 headsep=.3cm, footskip=1.5cm,
30 marginparsep=1em
31 }
32 \ztex_msg_set:nn {option-page-margin}
33 {No~margin~option~is~only~accessible~in~oneside~layout,
34 ~margin~option~is~now~enabled~by~default.}
35 \ztex_msg_warn:n {option-page-margin}
36 }
37 \else
38 \bool_if:NTF \g__ztex_margin_bool {
39 \__ztex_layout_setup:n {
40 a4paper,
41 left=2.5cm, right=7.5cm,
42 bottom=3.5cm, top=3.2cm,
43 headsep=.3cm, footskip=1.5cm,
44 marginparsep=2em
45 }
46 \dim_gset:Nn \marginparwidth{14em}

```

47	}{	47
48	__ztex_layout_setup:n {	48
49	a4paper,	49
50	left=3cm, right=3cm,	50
51	bottom=3.5cm, top=3.2cm,	51
52	headsep=.3cm, footskip=1.5cm,	52
53	marginparsep=1em	53
54	}	54
55	\renewcommand{\marginpar}[1]{\leftbar\underline{noindent}#1\endleftbar}	55
56	}	56
57	\fi	57
58		58
59		59
60	% ==> backmatter and appmatter	60
61	\IfClassLoadedTF{book}	61
62	{	62
63	\renewcommand{\backmatter}	63
64	{	64
65	\cleardoublepage	65
66	\@mainmattertrue	66
67	\pagestyle{plain}	67
68	}	68
69	\newcommand{\appmatter}	69
70	{	70
71	\cleardoublepage	71
72	\@mainmattertrue	72
73	\pagestyle{plain}	73
74	\setcounter{chapter}{0}	74
75	\def\thechapter{\Alph{chapter}}	75
76	\renewcommand\theHchapter{Appendix-\thechapter}	76
77	}	77
78	}{}	78
79		79
80		80
81	% ==> title page	81
82	\let\ori@maketitle\maketitle	82
83	\bool_if:NTF \g__ztex_slide_bool	83
84	{	84
85	\newcommand\ztex@maketitle	85
86	{	86
87	\bool_if:NT \g__ztex_hyperref_bool	87
88	{	88
89	\phantomsection	89
90	\hypertarget{zslide:titlepage}{}	90
91	}	91
92	\newgeometry{margin=1cm}	92
93	\null\vfill\begin{center}	93
94	\begin{tabular}{c}	94

```

95         \begin{zpic}[unit=\textwidth]
96         \zrectangle[arc=.01, draw=white, fill=zslide@title@color](-0.48, -.05)(.48,
.05)
97         \put(-.425, -.018){\hb@xt@.85\textwidth{\hss\Large\zslidetitle\hss}}
98         \end{zpic}\\[3.5em]
99         \zslideauthor\\[3em]
100        \zslidedate
101        \end{tabular}
102        \end{center}\vfill\null
103        \thispagestyle{empty}\setcounter{page}{0}
104        \restoregeometry
105    }
106    }{
107        \cs_generate_variant:Nn \ztool_get_ht:Nn {No}
108        \long\def\format@title{{\huge\bfseries\@title}}
109        \long\def\format@author{{\Large\bfseries\@author}}
110        \long\def\format@date{{\Large\textcolor{gray}\@date}}
111        \newcommand\title@upper@box[2][0pt]
112        {
113            \parbox[b][#2][r]{\l_tmpa_dim}{
114                {\format@title}\\[#1]
115                {\format@author}
116            }
117        }
118        \newcommand\ztex@maketitle
119        {
120            \thispagestyle{empty}
121            % calc max width/height, add '1pt' for right padding in case of wrong line break
122            \ztool_get_wd:Nn \l_tmpa_dim {\hbox:n {\format@title}}
123            \ztool_get_wd:Nn \l_tmpb_dim {\hbox:n {\format@author}}
124            \dim_set:Nn \l_tmpa_dim {
125                \dim_min:nn {
126                    \dim_max:nn {\l_tmpa_dim}{\l_tmpb_dim}
127                }{.8\textwidth} + 1pt} % the max title width
128            \ztool_get_ht_plus_dp:Nn \l_tmpb_dim {\title@upper@box{}}
129            \dim_set:Nn \l_tmpb_dim {\dim_max:nn {80pt}{\l_tmpb_dim}} % the total title height
130            % typeset info
131            \vfill\vspace*{20pt}\begin{center}
132                \rule{6pt}{\l_tmpb_dim}\enskip
133                \title@upper@box[\fill]{\l_tmpb_dim}
134                \par\vfill\format@date
135            \end{center}\newpage
136        }
137    }
138    \RenewDocumentCommand{\maketitle}{so}
139    {
140        \IfBooleanTF{#1}{\ori@maketitle}
141        {

```

142	\IfNoValueTF{#2}	142
143	{ \ztex@maketitle }	143
144	{	144
145	\newgeometry{margin=#2}	145
146	\ori@maketitle	146
147	\restoregeometry	147
148	}	148
149	}	149
150	}	150
151		151
152		152
153	% ==> fancyhdr setup	153
154	\cs_new_protected:Npn \zpage_set_style:nnn #1#2#3	154
155	{	155
156	\tl_if_empty:nTF { #2 }	156
157	{ \fancypagestyle{#1}{#3} }	157
158	{ \fancypagestyle{#1}[#2]{#3} }	158
159	}	159
160	\NewDocumentCommand{\zpagestyleset}{0{ }mm}	160
161	{	161
162	\zpage_set_style:nnn {#2}{#1}{#3}	162
163	% \fancypagestyleassign {#1}{#1@new}	163
164	% ERROR: '! TeX capacity exceeded,'	164
165	}	165
166	\bool_if:NF \g__ztex_slide_bool	166
167	{	167
168	\RequirePackage{fancyhdr}	168
169	\zpage_set_style:nnn { fancy }{ }	169
170	{	170
171	\fancyhf{ }	171
172	\dim_gset:Nn \headheight{15pt}	172
173	\renewcommand{\headrule}{\hrule width\textwidth}	173
174	\if@twoside	174
175	\fancyhead[EL]{\ztexleftmark}	175
176	\fancyhead[ER]{\thepage}	176
177	\fancyhead[OL]{\thepage}	177
178	\fancyhead[OR]{\ztexrightmark}	178
179	<u>\else</u>	179
180	\IfClassLoadedTF{book}{	180
181	\fancyhead[L]{\thepage}	181
182	\fancyhead[R]{\ztexrightmark}	182
183	}{	183
184	\fancyhead[L]{\thepage}	184
185	\bool_if:NTF \g__ztex_sect_load_bool	185
186	{	186
187	\fancyhead[R]	187
188	{	188
189	\IfMarksEqualTF{ztex-left}{first}{last}	189

190		190
191	{ \FirstMark{ztex-left} }	191
192	}	192
193	}{	193
194	\fancyhead[R]	194
195	{	195
196	\IfMarksEqualTF{2e-left}{first}{last}	196
197	{ \FirstMark{2e-left} }	197
198	{ \FirstMark{2e-left} \;\(\rightarrow\)\; \LastMark{2e-left} }	198
199	}	199
200	}	200
201	}	201
202	<u>\fi</u>	202
203	}	203
204	\zpage_set_style:nnn { plain }{}	204
205	{	205
206	\fancyhf{}	206
207	\renewcommand{\headrulewidth}{0pt}	207
208	\renewcommand{\headrule}{}{}	208
209	\fancyfoot[C]{\thepage}	209
210	}	210
211	}	211
212		212
213		213
214	% ==> front/main matter	214
215	\IfClassLoadedTF{book}{	215
216	\renewcommand\frontmatter{	216
217	\cleardoublepage	217
218	\pagestyle{plain}	218
219	\@mainmatterfalse	219
220	\pagenumbering{Roman}	220
221	}	221
222	\renewcommand\mainmatter{	222
223	\cleardoublepage	223
224	\pagestyle{fancy}	224
225	\@mainmattertrue	225
226	\pagenumbering{arabic}	226
227	}	227
228	}{	228
229	\bool_if:NF \g__ztex_slide_bool	229
230	{ \pagestyle{fancy} }	230
231	}	231
232		232
233		233
234	% ==> page annotation	234
235	% #1: fore/background; #2: position;	235
236	% #3: anchor; #4: object;	236
237	% #5: hook range.	237

```

238 \dim_const:Nn \zph {\paperheight}
239 \dim_const:Nn \zpw {\paperwidth}
240 \cs_generate_variant:Nn \hook_gput_code:nnn {nne}
241 \cs_new_protected:Npn \ztex_page_annotate:nnnnn #1#2#3#4#5
242 {
243   \tl_if_empty:eTF {#5}
244   {
245     \hook_gput_code:nnn {shipout/#1}
246       { ztex@page@mask-\l__ztex_page_mask_label_tl }
247       { \put#2{\makebox(0, 0)[#3]{#4}} }
248     % tikz/pgf layer and pagemask rule:
249     \exp_args:Nno \hook_gset_rule:nnnn {shipout/#1}
250       { ztex@page@mask-\l__ztex_page_mask_label_tl }
251       { < }{ pgfrcs }
252   }{
253     \hook_gput_next_code:nn {shipout/#1}
254       {\put#2{\makebox(0, 0)[#3]{#4}}}
255   }
256 }
257 % general pagemask rule:
258 \hook_gset_rule:nnnn {shipout/foreground}
259 { . }{ < }{ pgfrcs }
260 \hook_gset_rule:nnnn {shipout/background}
261 { . }{ < }{ pgfrcs }
262 \ztex_keys_define:nn { page/mask }
263 {
264   layer .choice:,
265   layer / foreground .code:n =
266     { \tl_set:Nn \l__ztex_page_mask_layer_tl {#1} },
267   layer / background .code:n =
268     { \tl_set:Nn \l__ztex_page_mask_layer_tl {#1} },
269   layer / unknown .code:n =
270     {
271       \ztex_msg_set:nn {pagemask@layer@wrong}
272       {layer~only~accepts~values:'foreground'~or~'background'}.}
273       \ztex_msg_error:n {pagemask@layer@wrong}
274     },
275   layer .initial:n = background,
276   position .tl_set:N = \l__ztex_page_mask_position_tl,
277   position .initial:n = {( .5\zpw, .5\zph )},
278   anchor .tl_set:N = \l__ztex_page_mask_anchor_tl,
279   anchor .initial:n = c,
280   label .tl_set:N = \l__ztex_page_mask_label_tl,
281   label .initial:n = { DEFAULT },
282 }
283 \cs_generate_variant:Nn \ztex_page_annotate:nnnnn {eee}
284 \cs_new:Npn \__page_mask_pos_parse:w (#1, #2)
285 {(

```

```
286 \dim_to_decimal:n {#1} pt,
287 \dim_to_decimal:n {#2-\paperheight} pt
288 )}
289 \ztex_msg_set:nn {pageinfo}
290 {
291   Only~star~version~of~\string\zpagemask\
292   is~label~allowed.
293 }
294 \prop_gclear_new:N \g__zpage_mask_label_prop
295 \NewDocumentCommand{\zpagemask}{so+m}
296 {
297   \group_begin:
298   \IfValueT{ #2 }
299   {
300     \ztex_keys_set:nn { page/mask }{#2}
301   }
302   \IfBooleanTF{#1}{\gdef\@once@hook@sign{}}
303   {
304     \gdef\@once@hook@sign{*}
305     \tl_if_eq:enF { \l__ztex_page_mask_label_tl }
306     { DEFAULT }
307     { \ztex_msg_warn:n {pageinfo} }
308   }
309   \exp_args:Neee \hook_gset_rule:nnnn
310   { shipout/\l__ztex_page_mask_layer_tl }
311   { ztex@page@mask-\l__ztex_page_mask_label_tl }
312   { < }{ pgfrcs }
313   \exp_args:NNe \prop_gput_from_keyval:Nn \g__zpage_mask_label_prop
314   { \l__ztex_page_mask_label_tl = \l__ztex_page_mask_layer_tl }
315   \ztex_page_annotate:eeenn
316   { \l__ztex_page_mask_layer_tl }
317   { \exp_after:wN \__page_mask_pos_parse:w \l__ztex_page_mask_position_tl }
318   { \l__ztex_page_mask_anchor_tl }
319   { #3 }{ \@once@hook@sign }
320   \group_end:
321 }
322 \ztex_msg_set:nn {zpage@masklabel@nonexist}
323 {
324   Page~mask~of~label:~'#1'~does~NOT~exist,~
325   i'll~give~up~this~operation.~Current~page~
326   mask~label~listed~as~follows(without~outer~braces):
327   \prop_map_function:NN \g__zpage_mask_label_prop
328   \__prop_typeout:nn
329 }
330 \NewDocumentCommand{\zpagemaskrm}{ m }
331 {
332   \prop_if_in:NnTF \g__zpage_mask_label_prop
333   { #1 }
```

334	{	334
335	\exp_args:Ne \hook_gremove_code:nn	335
336	{ shipout/\prop_item:Nn \g__zpage_mask_label_prop {#1} }	336
337	{ ztex@page@mask-#1 }	337
338	\prop_gpop:NnN \g__zpage_mask_label_prop	338
339	{ #1 } \l_tmpa_tl	339
340	}{	340
341	\ztex_msg_warn:nn	341
342	{ zpage@masklabel@nonexist }	342
343	{ #1 }	343
344	}	344
345	}	345
346	\cs_new:Npn __prop_typeout:nn #1#2	346
347	{ \{\{#1\}~~~=>~~~{#2} }	347
348	\NewDocumentCommand{\zpagemaskrule}{mmm}	348
349	{	349
350	\prop_if_in:NnTF \g__zpage_mask_label_prop {#1}	350
351	{	351
352	\prop_if_in:NnTF \g__zpage_mask_label_prop {#3}	352
353	{	353
354	\exp_args:Neee \hook_gset_rule:nnnn	354
355	{ shipout/\prop_item:Nn \g__zpage_mask_label_prop {#1} }	355
356	{ ztex@page@mask-#1 }	356
357	{ #2 }	357
358	{ ztex@page@mask-#3 }	358
359	}{	359
360	\ztex_msg_warn:nn	360
361	{ zpage@masklabel@nonexist }	361
362	{ #2 }	362
363	}	363
364	}{	364
365	\ztex_msg_warn:nn	365
366	{ zpage@masklabel@nonexist }	366
367	{ #1 }	367
368	}	368
369	}	369
370		370
371		371
372	% ==> page target	372
373	\AddToHook{shipout/firstpage}{	373
374	\label{ztex:titlepage}	374
375	\hyper@anchor{ztex@titlepage}	375
376	}	376
377	\AddToHook{shipout/lastpage}{	377
378	\label{ztex:lastpage}	378
379	\hyper@anchor{ztex@lastpage}	379
380	}	380
381		381

382		382
383		383
384	% ==> doc info	384
385	\gdef \@abspage@last{-1}	385
386	% TODO: consider this new implement ?	386
387	\ztex_msg_set:nn { pageall@wrong }	387
388	{	388
389	Current~total~page~number~is~wrong,~	389
390	Rerun~to~correct~this.	390
391	}	391
392	% \ztex_hook_doc_end:n	392
393	% {	393
394	% \int_compare:nNnF	394
395	% { \@abspage@last } = { \g_shipout_readonly_int }	395
396	% {	396
397	% \ztex_msg_warn:n { pageall@wrong }	397
398	% }	398
399	% }	399
400	\newcommand {\ztexpageall}	400
401	{	401
402	\bool_lazy_any:nTF	402
403	{	403
404	{ \int_compare_p:n {\@abspage@last > 100000} }	404
405	{ \int_compare_p:n {\@abspage@last < 0} }	405
406	}{ \c_empty_tl }{\@abspage@last }	406
407	}	407
408	\ztex_hook_preamble_last:n	408
409	{	409
410	\let \ztextitle\@title	410
411	\let \ztexauthor\@author	411
412	\let \ztexdate\@date	412
413	}	413

10.3.5 color

```
1 \ProvidesExplFile{ztex.module.color.tex} 1
2 {2025/09/24}{\ztex@versi@n} 2
3 {color~module~for~ztex} 3
4 4
5 5
6 %%%% color module for ztex %%%% 6
7 \RequirePackage{xcolor} 7
8 8
9 9
10 % ==> color setup 10
11 % dynamic color setup 11
12 \regex_new:N \l__ztex_color_mode_regex 12
13 \regex_set:Nn \l__ztex_color_mode_regex 13
14 { (\cB..\cE.)\cE. } 14
15 \cs_new:Npn \ztex_color_set:n #1 15
16 { 16
17 \regex_match:NnTF \l__ztex_color_mode_regex {#1} 17
18 { 18
19 \definecolor{ztex@color@\l_keys_key_str}#1 19
20 }{ 20
21 \colorlet{ztex@color@\l_keys_key_str}{#1} 21
22 } 22
23 \tl_set:ce 23
24 { \l__ztex_\l_keys_key_str _color_tl } 24
25 { ztex@color@\l_keys_key_str } 25
26 } 26
27 27
28 % all colors 28
29 % NOTE: 29
30 % 1. to make use of '\ztex_color_set:n', color 30
31 % name must be 'ztex@color@<key>'. 31
32 % 2. this module is highly experimental. 32
33 \definecolor{ztex@color@royalred}{RGB}{157, 16, 45} 33
34 \definecolor{ztex@color@axiom}{HTML}{000000} 34
35 \definecolor{ztex@color@definition}{HTML}{bdc3c7} 35
36 \definecolor{ztex@color@theorem}{HTML}{27ae60} 36
37 \definecolor{ztex@color@lemma}{HTML}{2980b9} 37
38 \definecolor{ztex@color@corollary}{HTML}{8e44ad} 38
39 \definecolor{ztex@color@proposition}{HTML}{f39c12} 39
40 \definecolor{ztex@color@remark}{HTML}{c92a2a} 40
41 41
42 % slide and structure color 42
43 \definecolor{zslide@title@color}{HTML}{d9d9d9} 43
44 \definecolor{ztex@color@fancychap}{HTML}{7f8184} 44
45 45
46 46
```

```

47 % ==> structure theme
48 \ztex_keys_define:nn {color}{
49   fancychap      .tl_set:N    = \l__ztex_fancy_chap_color_tl,
50   fancychap      .initial:n    = { ztex@color@fancychap },
51   fancychap      .code:n       = { \ztex_color_set:n {#1} },
52 }
53
54
55 % ==> index and ref theme
56 \ztex_keys_define:nn {color}{
57   link           .tl_set:N    = \l__ztex_link_color_tl,
58   link           .initial:n    = { purple },
59   link           .code:n       = { \ztex_color_set:n {#1} },
60   cite           .tl_set:N    = \l__ztex_cite_color_tl,
61   cite           .initial:n    = { blue },
62   cite           .code:n       = { \ztex_color_set:n {#1} },
63   url            .tl_set:N    = \l__ztex_url_color_tl,
64   url            .initial:n    = { ztex@color@royalred },
65   url            .code:n       = { \ztex_color_set:n {#1} },
66 }
67
68
69 % ==> thm env themecolor
70 \ztex_keys_define:nn {color}{
71   % theorem-like envs (numbered)
72   axiom          .tl_set:N    = \l__ztex_axiom_color_tl,
73   axiom          .initial:n    = { ztex@color@axiom },
74   axiom          .code:n       = { \ztex_color_set:n {#1} },
75   definition     .tl_set:N    = \l__ztex_definition_color_tl,
76   definition     .initial:n    = { ztex@color@definition },
77   definition     .code:n       = { \ztex_color_set:n {#1} },
78   theorem        .tl_set:N    = \l__ztex_theorem_color_tl,
79   theorem        .initial:n    = { ztex@color@theorem },
80   theorem        .code:n       = { \ztex_color_set:n {#1} },
81   lemma          .tl_set:N    = \l__ztex_lemma_color_tl,
82   lemma          .initial:n    = { ztex@color@lemma },
83   lemma          .code:n       = { \ztex_color_set:n {#1} },
84   corollary      .tl_set:N    = \l__ztex_corollary_color_tl,
85   corollary      .initial:n    = { ztex@color@corollary },
86   corollary      .code:n       = { \ztex_color_set:n {#1} },
87   proposition    .tl_set:N    = \l__ztex_proposition_color_tl,
88   proposition    .initial:n    = { ztex@color@proposition },
89   proposition    .code:n       = { \ztex_color_set:n {#1} },
90   remark         .tl_set:N    = \l__ztex_remark_color_tl,
91   remark         .initial:n    = { ztex@color@remark },
92   remark         .code:n       = { \ztex_color_set:n {#1} },
93   % proof-like envs (unnumbered)
94   proof          .tl_set:N    = \l__ztex_proof_color_tl,

```

95

proof

.initial:n

=

{ black },

95

96

proof

.code:n

=

{ \ztex_color_set:n {#1} },

96

97

exercise

.tl_set:N

=

\l__ztex_exercise_color_tl,

97

98

exercise

.initial:n

=

{ black },

98

99

exercise

.code:n

=

{ \ztex_color_set:n {#1} },

99

100

example

.tl_set:N

=

\l__ztex_example_color_tl,

100

101

example

.initial:n

=

{ black },

101

102

example

.code:n

=

{ \ztex_color_set:n {#1} },

102

103

solution

.tl_set:N

=

\l__ztex_solution_color_tl,

103

104

solution

.initial:n

=

{ black },

104

105

solution

.code:n

=

{ \ztex_color_set:n {#1} },

105

106

problem

.tl_set:N

=

\l__ztex_problem_color_tl,

106

107

problem

.initial:n

=

{ black },

107

108

problem

.code:n

=

{ \ztex_color_set:n {#1} },

108

109

}

109

110

110

111

111

112

% ==> unknown color key

112

113

\ztex_keys_define:nn {color}

113

114

{

114

115

unknown

.code:n

=

115

116

{

116

117

\ztex_metakey_msg_warning:nn {color}

117

118

{

118

119

link, cite, url, chapter, chapter-rule,

119

120

axiom, definition, theorem, lemma, corollary,

120

121

proposition, remark

121

122

}

122

123

}

123

124

}

124

125

125

126

126

127

% ==> init color theme

127

128

\DeclareHookRule{env/document/before}

128

129

{ztex-themecolor-setup-user}

129

130

{>}

130

131

{ztex-thmptheorem-setup-inner}

131

132

\DeclareHookRule{env/document/before}

132

133

{ztex-themecolor-setup-user}

133

134

{>}

134

135

{ztex-thmpproof-setup-inner}

135

136

\NewDocumentCommand{\zcolorset}{m}

136

137

{

137

138

\ztex_label_hook_preamble_last:nn {ztex-themecolor-setup-user}

138

139

{

139

140

\ztex_keys_set:nn {color}{#1}

140

141

\bool_if:NT \g__ztex_hyperref_bool

141

142

{

142

266

143	<code>\hypersetup</code>	143
144	<code>{</code>	144
145	<code>colorlinks = true,</code>	145
146	<code>urlcolor = \tl_use:N \l__ztex_url_color_tl,</code>	146
147	<code>linkcolor = \tl_use:N \l__ztex_link_color_tl,</code>	147
148	<code>citecolor = \tl_use:N \l__ztex_cite_color_tl,</code>	148
149	<code>}</code>	149
150	<code>}</code>	150
151	<code>}</code>	151
152	<code>}</code>	152
153	<code>\@onlypreamble\zcolorset</code>	153
154		154
155	<code>% init color setup:</code>	155
156	<code>\zcolorset</code>	156
157	<code>{</code>	157
158	<code>link=purple,</code>	158
159	<code>cite=blue,</code>	159
160	<code>url=ztex@color@royalred,</code>	160
161	<code>}</code>	161

10.3.6 thm

```
1 \ProvidesExplFile{ztex.module.thm.tex}
2 {2026/02/12}{\ztex@versi@n}
3 {thm~module~for~ztex}
4
5
6 %%%%      thm module for ztex      %%%%
7 % basic packages
8 \RequirePackage{amsfonts, amsmath}
9 \RequirePackage{esint}
10
11
12 % ==> module init
13 \cs_new_protected:Npn \zthmremoveCJKecglue
14 { \zfont_cancel_CJKecglue: }
15 \cs_new_protected:Npn \zthmrestoreCJKecglue
16 { \zfont_restore_CJKecglue: }
17
18 \clist_gclear:N \g__ztex_thm_theorem_clist
19 \clist_gclear:N \g__ztex_thm_proof_clist
20 \cs_new_protected:Npn \ztex_thm_create:nn #1#2
21 {
22   \clist_gput_right:cn {g__ztex_thm_#1_clist}{#2}
23 }
24 \cs_generate_variant:Nn \ztex_thm_create:nn { ne }
25 \ztex_thm_create:nn {theorem}
26 {
27   axiom, definition, theorem, lemma,
28   corollary, proposition, remark,
29 }
30 \ztex_thm_create:nn {proof}
31 {
32   proof, exercise, example, solution, problem,
33 }
34 \ztex_msg_set:nn {thm-name}
35 {
36   An~unexpected~math~env~name~in~multichoice~key:
37   '\l_keys_key_str',~there~is~no~internal~config~for~it.
38 }
39
40 % thm title definition
41 \cs_new_protected:Npn \ztex_thm_name_set:nn #1#2
42 {
43   \prop_gset_from_keyval:cn
44     {g__ztex_thm_name_#1_prop}{#2}
45 }
46 \ztex_thm_name_set:nn {en}{
```

```
47 axiom = Axiom, 47
48 definition = Definition, 48
49 theorem = Theorem, 49
50 lemma = Lemma, 50
51 corollary = Corollary, 51
52 proposition = Proposition, 52
53 remark = Remark, 53
54 proof = Proof, 54
55 exercise = Exercise, 55
56 example = Example, 56
57 solution = Solution, 57
58 problem = Problem, 58
59 } 59
60 \ztex_thm_name_set:nn {cn}{ 60
61 axiom = 公理, 61
62 definition = 定义, 62
63 theorem = 定理, 63
64 lemma = 引理, 64
65 corollary = 推论, 65
66 proposition = 命题, 66
67 remark = 注记, 67
68 proof = 证明, 68
69 exercise = 练习, 69
70 example = 示例, 70
71 solution = 解, 71
72 problem = 问题, 72
73 } 73
74 \ztex_thm_name_set:nn {fr}{ 74
75 axiom = Axiome, 75
76 definition = Définition, 76
77 theorem = Théorème, 77
78 lemma = Lemme, 78
79 corollary = Corollaire, 79
80 proposition = Proposition, 80
81 remark = Remarque, 81
82 proof = Preuve, 82
83 exercise = Exercice, 83
84 example = Exemple, 84
85 solution = Solution, 85
86 problem = Problème, 86
87 } 87
88 \tl_if_exist:NF \g__ztex_lang_math_tl 88
89 { 89
90 \tl_set_eq:cc 90
91 { g__ztex_lang_math_tl } 91
92 { g__ztex_lang_str } 92
93 } 93
94 \NewDocumentCommand{\zthmnameset}{mm} 94
```

```
95 {
96     \prop_gput_from_keyval:cn
97     { g__ztex_thm_name_#1_prop }
98     { #2 }
99 }
100
101
102 % ==> thm module tools
103 \NewDocumentCommand{\zthmlang}{m}
104 {
105     \tl_gset:Nn \g__ztex_lang_math_tl {#1}
106     \prop_set_eq:cc
107     { g__ztex_thm_name_prop }
108     { g__ztex_thm_name_\g__ztex_lang_math_tl _prop }
109 }
110 % \@onlypreamble\zthmlang
111 \prop_new:c {g__ztex_thm_name_prop}
112 \prop_gclear:c {g__ztex_thm_name_prop}
113 \ztex_hook_preamble_last:n
114 {
115     \prop_set_eq:cc
116     { g__ztex_thm_name_prop }
117     { g__ztex_thm_name_\g__ztex_lang_math_tl _prop }
118 }
119 \tl_new:N \g__ztex_thm_theorem_title_tl
120 \protected\def\zthmtitle{\@ifstar\@zthmtitle\@@zthmtitle}
121 \def\@zthmtitle{\__ztex_thm_theorem_title:}
122 \def\@@zthmtitle{\tl_use:N \g__ztex_thm_theorem_title_tl}
123 \bool_new:N \g__ztex_thm_title_inline_bool
124 \NewDocumentCommand{\zthmtitleswitch}{s}
125 {
126     \IfBooleanTF{#1}
127     { \bool_gset_true:N \g__ztex_thm_title_inline_bool }
128     { \bool_gset_false:N \g__ztex_thm_title_inline_bool }
129 }
130 \cs_new:Npn \__ztex_thm_color_set_check:nn #1#2
131 {
132     \clist_clear:N \l_tmpa_clist
133     \clist_put_right:NV \l_tmpa_clist \g__ztex_thm_theorem_clist
134     \clist_put_right:NV \l_tmpa_clist \g__ztex_thm_proof_clist
135     \clist_if_in:NnF \l_tmpa_clist {#1}
136     { \ztex_msg_error:n {thm-color-set} }
137 }
138 \ztex_msg_set:nn {thm-color-set}
139 {
140     Your~color~spec~key~'#1'~is~not~in~the~
141     thm~env~list,~please~check~it~again.
142 }
```

```
143 \NewDocumentCommand{\zthmcolorset}{m} 143
144 { 144
145     \ztex_label_hook_preamble_last:nn 145
146     { ztex-thmcolor-setup-user } 146
147     { 147
148         \keyval_parse:nnn 148
149         { \use_none:n } 149
150         { \__ztex_thm_color_set_check:nn } 150
151         { #1 } 151
152         \ztex_keys_set:nn {color}{#1} 152
153     } 153
154 } 154
155 \DeclareHookRule{env/document/before} 155
156 {ztex-thmcolor-setup-user} 156
157 {>} 157
158 {ztex-thmenv-setup-user} 158
159 \@onlypreamble\zthmcolorset 159
160 160
161 161
162 % create new thm env(init related variables) 162
163 \ztex_keys_define:nn { thmnew } 163
164 { 164
165     name .tl_set:N = \l__zthm_new_name_tl, 165
166     color .tl_set:N = \l__zthm_new_color_tl, 166
167     tocsym .tl_set:N = \l__zthm_new_tocsym_tl, 167
168 } 168
169 \cs_new:Nn \__zthm_new_keyval_clear: 169
170 { 170
171     \tl_clear:N \l__zthm_new_name_tl 171
172     \tl_clear:N \l__zthm_new_color_tl 172
173     \tl_clear:N \l__zthm_new_tocsym_tl 173
174 } 174
175 \cs_new:Npn \__ztex_thm_color_set:nn #1#2 175
176 { 176
177     \tl_if_empty:eTF { #2 } 177
178     { \ztex_keys_set:nn {color}{#1=black} } 178
179     { \ztex_keys_set:nn {color}{#1=#2} } 179
180 } 180
181 \cs_new:Npn \__ztex_color_keyval_add:n #1 181
182 { 182
183     \ztex_keys_define:nn { color } 183
184     { 184
185         #1 .tl_set:c = { l__ztex_#1_color_tl }, 185
186         #1 .initial:n = { black }, 186
187         #1 .code:n = { \ztex_color_set:n {##1} }, 187
188     } 188
189 } 189
190 \cs_new:Npn \__zthm_envnew_var_init:Nnn #1#2#3 190
```

```

191 {
192     \ztex_thm_create:nn {#2}{#3}
193     \__ztex_color_keyval_add:n {#3}
194     \prop_gput:Nnn #1 { #3 }{ #3 }
195     \prop_gput:Nnn \g_ztex_thm_toc_symbols_prop
196         { #3 }{ #3 }
197 }
198 \cs_new:Npn \__zthm_envnew_var_init:Nnnn #1#2#3#4
199 {
200     \ztex_thm_create:nn {#2}{#3}
201     \__ztex_color_keyval_add:n {#3}
202     \ztex_keys_set:nn { thmnew }{ #4 }
203     \exp_args:Nno \__ztex_thm_color_set:nn
204         { #3 }{ \l_zthm_new_color_tl }
205     \tl_if_empty:NTF \l_zthm_new_name_tl
206     {
207         \prop_gput:Nnn #1
208             { #3 }{ #3 }
209     }{
210         \prop_gput:Nno #1
211             { #3 }{ \l_zthm_new_name_tl }
212     }
213     \tl_if_empty:NTF \l_zthm_new_tocsym_tl
214     {
215         \prop_gput:Nnn \g_ztex_thm_toc_symbols_prop
216             { #3 }{ #3 }
217     }{
218         \prop_gput:Nno \g_ztex_thm_toc_symbols_prop
219             { #3 }{ \l_zthm_new_tocsym_tl }
220     }
221     \__zthm_new_keyval_clear:
222 }
223 \cs_new_protected:Npn \zthm_envnew_var_init:Nnn #1#2#3
224 {
225     \keyval_parse:nnn
226         { \__zthm_envnew_var_init:Nnn #1{#2} }
227         { \__zthm_envnew_var_init:Nnnn #1{#2} }
228         { #3 }
229 }
230 \NewDocumentCommand{\zthmenvnew}{0{theorem}m}
231 {
232     \ztex_label_hook_preamble_last:nn
233         { ztex-thmenv-setup-user }
234     {
235         \zthm_envnew_var_init:Nnn
236             \g__ztex_thm_name_prop
237             { #1 }{ #2 }
238     }

```

239	}	239
240	\@onlypreamble\zthmenvnew	240
241		241
242		242
243	% ==> new thm style interface	243
244	\NewDocumentCommand{\zthmstylenew}{+m}	244
245	{	245
246	\keyval_parse:nnn	246
247	{ \use_none:n }	247
248	{ __ztex_thm_new_style:nn }	248
249	{ #1 }	249
250	}	250
251	\cs_new_protected:Npn __ztex_thm_new_style:nn #1#2	251
252	{	252
253	\ztex_keys_define:nn { thm/style }	253
254	{	254
255	#1 .meta:nn = { ztex/thm/style/#1 }{##1},	255
256	#1 / begin .tl_gset:c = { g__ztex_thm_style_#1_begin_tl },	256
257	#1 / end .tl_gset:c = { g__ztex_thm_style_#1_end_tl },	257
258	#1 / option .tl_gset:c = { g__ztex_thm_style_#1_option_tl },	258
259	#1 / preamble .code:n = {	259
260	% NOTE:	260
261	% 1. thm preamble can be only set by one style	261
262	% 2. '\g__ztex_thm_style_tl' need to be set	262
263	% before '\ztexloadlib{theme}'	263
264	\tl_if_eq:cnT {g__ztex_thm_style_tl}	264
265	{ #1 }{ ##1 }	265
266	},	266
267	}	267
268	\ztex_keys_set:nn { thm/style }{ #1={#2} }	268
269	}	269
270	\NewDocumentCommand{\zthmstyle}{m}	270
271	{	271
272	\tl_gset:Nn \g__ztex_thm_style_tl {#1}	272
273	}	273
274	% title switch and tcb warning, create thm styles	274
275	\cs_new:Npn __ztex_thm_title_inline:n #1	275
276	{	276
277	\tl_if_eq:nnTF {#1}{T}	277
278	{\bool_gset_true:N \g__ztex_thm_title_inline_bool}	278
279	{\bool_gset_false:N \g__ztex_thm_title_inline_bool}	279
280	}	280
281	% tcolorbox and tikz warning if missing	281
282	% when create new thm style	282
283	\ztex_msg_set:nn {mathEnv-dependency}	283
284	{	284
285	MathEnv~style:'\g__ztex_thm_style_tl'~requires~	285
286	package~'tcolorbox'~and~'tikz',~and~either~of~	286

```
287 which~hasn't~been~loaded~in~your~preamble.~ 287
288 Reset~to~default~'plain'~style~now. 288
289 } 289
290 \cs_new:Nn \__ztex_thm_tcolorbox_warning: 290
291 { 291
292   \@ifpackageloaded{tcolorbox}{\relax} 292
293   { 293
294     \ztex_msg_warn:n {mathEnv-dependency} 294
295     \tl_gset:Nn \g__ztex_thm_style_tl {plain} 295
296   } 296
297 } 297
298 \cs_set:Npn \__ztex_thm_frame_make:n #1 298
299 { 299
300   \vspace{-.75em}\def\FrameCommand{#1} 300
301   \MakeFramed{\advance\hsize-\width \FrameRestore} 301
302 } 302
303 \zthmstylenew 303
304 { 304
305   plain = { 305
306     begin =, 306
307     end =, 307
308     option = \__ztex_thm_title_inline:n {T} 308
309   }, 309
310   leftbar = { 310
311     begin = { 311
312       \__ztex_thm_frame_make:n 312
313       { 313
314         {\color{\thm@tmp@color}\vrule~ width~ 3pt} 314
315         \hspace{5pt} 315
316       } 316
317     }, 317
318     end = {\endMakeFramed\vspace{-.75em}}, 318
319     option = { \__ztex_thm_title_inline:n {T} } 319
320   }, 320
321   background = { 321
322     begin = { 322
323       \__ztex_thm_frame_make:n {\colorbox{\thm@tmp@color}} 323
324     }, 324
325     end = {\endMakeFramed\vspace{-.75em}}, 325
326     option = { \__ztex_thm_title_inline:n {T} } 326
327   }, 327
328   fancy = { 328
329     begin = { 329
330       \__ztex_thm_frame_make:n 330
331       { 331
332         {\color{\thm@tmp@color}\vrule~ width~ 3pt} 332
333         \colorbox{\thm@tmp@color!10} 333
334       } 334
```



```

335 },
336 end = {\endMakeFramed\vspace{-.75em}},
337 option = { \_ztex_thm_title_inline:n {T} }
338 },
339 }
340
341
342 % ==> thm format and style setup
343 \ztex_msg_set:nn {mathEnv-style}
344 {
345   You~use~an~incorrect~MathEnv~style:~'\g__ztex_thm_style_tl',~
346   All~of~the~pre-defined~styles~are:'plain',~'leftbar',~
347   'background',~'fancy';And~styles~defined~inside~'thm'~
348   library(load~it~by~command~'\token_to_str:N\ztexloadlib{thm}')~are:~
349   'shadow',~'paris',~'lapsis',~'tcb',~'obsidian',~and~'elegant'.
350 }
351 % thm counter
352 \bool_new:N \g__ztex_thm_cntshare_bool
353 \ztex_keys_define:nn {thm/cnt}
354 {
355   share      .bool_gset:N = \g__ztex_thm_cntshare_bool,
356   share      .initial:n   = { false },
357   share      .default:n   = { true  },
358   parent     .tl_gset:N   = \g__ztex_thm_cntparent_tl,
359   parent     .initial:n   = { section },
360 }
361 \NewDocumentCommand{\zthmcnt}{m}
362 {
363   \group_begin:
364     \ztex_keys_set:nn {thm/cnt}{#1}
365   \group_end:
366 }
367 \@onlypreamble\zthmcnt
368 % thm env warper
369 \cs_new:Npn \_ztex_thm_wrap_start:nnn #1#2#3
370 {
371   \def\thm@tmp@name{#1}
372   \def\thm@tmp@color{\tl_use:c {l__ztex_#1_color_tl}}
373   \_ztex_thm_theorem_title_gen:nnn {#1}{#2}{#3}
374   \tl_if_exist:cTF
375     { g__ztex_thm_style_\g__ztex_thm_style_tl _option_tl }
376     { \tl_use:c {g__ztex_thm_style_\g__ztex_thm_style_tl _option_tl} }
377     { \ztex_msg_error:n {mathEnv-style} }
378   \tl_if_exist:cTF
379     { g__ztex_thm_style_\g__ztex_thm_style_tl _begin_tl }
380     { \tl_use:c {g__ztex_thm_style_\g__ztex_thm_style_tl _begin_tl} }
381     { \ztex_msg_error:n {mathEnv-style} }
382 }

```

383	<code>\tl_new:N \l__ztex_thm_toc_prefix_tl</code>	383
384	<code>\newcommand\zthmtocprefix[1]</code>	384
385	<code>{</code>	385
386	<code>\tl_set:Nn \l__ztex_thm_toc_prefix_tl</code>	386
387	<code>{ \exp_not:n {#1} }</code>	387
388	<code>}</code>	388
389	<code>\@onlypreamble\zthmtocprefix</code>	389
390	<code>\cs_new:Npn __ztex_thm_wrap_end:n #1</code>	390
391	<code>{</code>	391
392	<code>\tl_if_exist:cTF</code>	392
393	<code>{ g__ztex_thm_style_\g__ztex_thm_style_tl _end_tl }</code>	393
394	<code>{ \tl_use:c {g__ztex_thm_style_\g__ztex_thm_style_tl _end_tl} }</code>	394
395	<code>{ \ztex_msg_error:n {mathEnv-style} }</code>	395
396	<code>\zsect_add_theorem_line:eeoe</code>	396
397	<code>{ \g__ztex_thm_toc_level_tl }</code>	397
398	<code>{</code>	398
399	<code>{</code>	399
400	<code>\exp_not:N \l__ztex_thm_toc_prefix_tl</code>	400
401	<code>\exp_not:n</code>	401
402	<code>{</code>	402
403	<code>\prop_item:Nn \g_ztex_thm_toc_symbols_prop</code>	403
404	<code>{ #1 }</code>	404
405	<code>}</code>	405
406	<code>}</code>	406
407	<code>{ \g__ztex_thm_theorem_title_tl }</code>	407
408	<code>}</code>	408
409	<code>{ \thepage }</code>	409
410	<code>{ zthm@#1.\zthmnumber }</code>	410
411	<code>}</code>	411
412		412
413	<code>% thm theorem title interface</code>	413
414	<code>\NewHook{ztex/thm-theorem/titleformat}</code>	414
415	<code>\cs_new:Npn __ztex_thm_theorem_title_gen:nnn #1#2#3</code>	415
416	<code>{% #1:env-name; #2:note; #3:separator</code>	416
417	<code>\cs_set:Npn \zthmname</code>	417
418	<code>{</code>	418
419	<code>{\prop_item:cn {g__ztex_thm_name_prop}{#1}}</code>	419
420	<code>}</code>	420
421	<code>\cs_set:Npn \zthmnote ##1##2</code>	421
422	<code>{</code>	422
423	<code>\tl_if_empty:nF {#2}</code>	423
424	<code>{ ##1</code>	424
425	<code>\exp_not:n {\exp_not:n {#2}}</code>	425
426	<code>##2 }</code>	426
427	<code>}</code>	427
428	<code>\bool_if:NTF \g__ztex_thm_cntshare_bool</code>	428
429	<code>{</code>	429
430	<code>\cs_set:Npn \zthmnumber</code>	430

431	{	431
432	\cs:w the\g__ztex_thm_cntparent_tl	432
433	\cs_end:.\arabic{ztex@thm@sharecnt}	433
434	}	434
435	\refstepcounter{ztex@thm@sharecnt}	435
436	}{	436
437	\cs_set:Npn \zthmnumber	437
438	{	438
439	\cs:w the\g__ztex_thm_cntparent_tl	439
440	\cs_end:.\arabic{#1}	440
441	}	441
442	\refstepcounter{#1}	442
443	}	443
444	\tl_gset:Nn \g__ztex_thm_theorem_title_tl	444
445	{	445
446	\zthmname #3 \zthmnumber	446
447	\tl_if_empty:eF {\zthmnote{}}{}}{#3}	447
448	\zthmnote{({})} #3	448
449	}	449
450	\UseHook{ztex/thm-theorem/titleformat}	450
451	}	451
452	\cs_new:Npn __ztex_thm_theorem_title:	452
453	{	453
454	\group_begin:	454
455	\zfont_cancel_CJKecglue:	455
456	\noindent\bfseries	456
457	\tl_use:N \g__ztex_thm_theorem_title_tl	457
458	\group_end:	458
459	}	459
460	% thm proof title interface	460
461	\tl_new:N \g__ztex_thm_proof_title_tl	461
462	\NewHook{ztex/thm-proof/titleformat}	462
463	\cs_new:Npn __ztex_thm_proof_title_gen:nn #1#2	463
464	{% #1:env-name; #2:separator	464
465	\cs_set:Npn \zthmname	465
466	{	466
467	{\prop_item:cn {g__ztex_thm_name_prop}}{#1}}	467
468	}	468
469	\def\thmproof@tmp@color	469
470	{ \tl_use:c {l__ztex_#1_color_tl} }	470
471	\tl_gset:Nn \g__ztex_thm_proof_title_tl	471
472	{	472
473	\zthmname #2	473
474	}	474
475	\UseHook{ztex/thm-proof/titleformat}	475
476	}	476
477	\cs_new:Npn __ztex_thm_proof_title:	477
478	{	478

```

479     \group_begin:
480         \noindent\bfseries\color{\thmproof@tmp@color}
481         \tl_use:N \g__ztex_thm_proof_title_tl : \,
482     \group_end:
483 }
484 % users' interface of thm title format
485 \tl_new:N \g__ztex_thm_proof_title_before_tl
486 \tl_new:N \g__ztex_thm_theorem_title_before_tl
487 \tl_gset:Nn \g__ztex_thm_proof_title_before_tl
488 { \noindent }
489 \tl_gset:Nn \g__ztex_thm_theorem_title_before_tl
490 { \noindent }
491 \NewDocumentCommand{\zthmtitlebefore}{0{theorem}m}
492 {
493     \tl_gset:cn {g__ztex_thm_#1_title_before_tl}
494     { #2 }
495 }
496 \newcommand{\zthm@title@before}[1]
497 {
498     \tl_use:c {g__ztex_thm_#1_title_before_tl}
499 }
500 \NewDocumentCommand{\zthmtitleformat}{s0{theorem}m}
501 {
502     \IfBooleanTF{#1}
503     {
504         \AddToHook{ztex/thm-#2/titleformat}
505         {
506             \cs_set:cpn {__ztex_thm_#2_title:}
507             {
508                 \group_begin:
509                 \zfont_cancel_CJKecglue:
510                 #3
511                 \group_end:
512             }
513         }
514     }{
515         \AddToHookNext{ztex/thm-#2/titleformat}
516         {
517             \cs_set:cpn {__ztex_thm_#2_title:}
518             {
519                 \group_begin:
520                 \zfont_cancel_CJKecglue:
521                 #3
522                 \group_end:
523             }
524         }
525     }
526 }

```

527	\@onlypreamble\zthmtitleformat	527
528	\newcommand\zthmnotemptyTF[2]	528
529	{	529
530	\tl_if_empty:eTF {\zthmnote{}}{}}	530
531	{ #1 }	531
532	{ #2 }	532
533	}	533
534		534
535		535
536	% ==> Thm tocline interface	536
537	% NOTE: thm tocline relies on 'sect' module	537
538	\NewDocumentCommand\zthmtocstop{}	538
539	{	539
540	\ztoc_stop_table:n { lom }	540
541	}	541
542	\cs_if_exist:NF \zsect_add_theorem_line:nnnn	542
543	{	543
544	\cs_set_protected:Nn \zsect_add_theorem_line:nnnn	544
545	{ }	545
546	}	546
547	\cs_generate_variant:Nn \zsect_add_theorem_line:nnnn	547
548	{ eeoe, nnee, nnoe }	548
549	\ztex_keys_define:nn { thm/add }	549
550	{	550
551	name .tl_set:N = \l__ztex_add_thm_toc_name_tl,	551
552	name .initial:n = { },	552
553	title .tl_set:N = \l__ztex_add_thm_toc_title_tl,	553
554	title .initial:n = { },	554
555	}	555
556	\int_new:N \g_zthm_added_toc_target_int	556
557	\NewDocumentCommand{\zthmtocadd}{0{section}m}	557
558	{	558
559	\int_incr:N \g_zthm_added_toc_target_int	559
560	\edef\zthmtoc@tmp@target	560
561	{	561
562	zthm@toc-add.	562
563	\int_use:N \g_zthm_added_toc_target_int	563
564	}	564
565	\MakeLinkTarget*{\zthmtoc@tmp@target}	565
566	\group_begin:	566
567	\ztex_keys_set:nn {thm/add}{#2}	567
568	\zsect_add_theorem_line:nnoe {#1}	568
569	{	569
570	{ \l__ztex_add_thm_toc_name_tl }	570
571	{ \l__ztex_add_thm_toc_title_tl }	571
572	}	572
573	{ \thepage }	573
574	{ \zthmtoc@tmp@target }	574

```
575 \group_end: 575
576 } 576
577 577
578 \tl_new:N \g__ztex_thm_toc_level_tl 578
579 \tl_set:Nn \g__ztex_thm_toc_level_tl {subsection} 579
580 \NewDocumentCommand{\zthmtoclevel}{m} 580
581 { 581
582 \tl_gset:Nn \g__ztex_thm_toc_level_tl {#1} 582
583 } 583
584 \@onlypreamble\zthmtoclevel 584
585 \NewDocumentCommand{\zthmtoc}{0{}} 585
586 { 586
587 \group_begin: 587
588 \exp_after:wN \exp_args:Nc \exp_after:wN \ztocformat 588
589 \exp_after:wN { \g__ztex_thm_toc_level_tl }{ #1 } 589
590 \ztoc_table_typeset:Nn \g_ztoc_lom_seq { } 590
591 \group_end: 591
592 } 592
593 % thm toc symbols 593
594 \prop_new:N \g_ztex_thm_toc_symbols_prop 594
595 \prop_set_from_keyval:Nn \g_ztex_thm_toc_symbols_prop 595
596 { 596
597 axiom = { \textbf{A}\; }, 597
598 definition = { \textbf{D}\; }, 598
599 theorem = { \textbf{T}\; }, 599
600 lemma = { \textbf{L}\; }, 600
601 corollary = { \textbf{C}\; }, 601
602 proposition = { \textbf{P}\; }, 602
603 remark = { \textbf{R}\; }, 603
604 } 604
605 \NewDocumentCommand{\zthmtocsym}{m} 605
606 { 606
607 \ztex_label_hook_preamble_last:nn 607
608 { ztex-thmtocsym-setup-user } 608
609 { 609
610 \prop_gput_from_keyval:Nn 610
611 \g_ztex_thm_toc_symbols_prop 611
612 { #1 } 612
613 } 613
614 } 614
615 \NewDocumentCommand{\zthmtocsymrm}{} 615
616 { 616
617 \prop_gclear:N \g_ztex_thm_toc_symbols_prop 617
618 } 618
619 \@onlypreamble\zthmtocsym 619
620 \@onlypreamble\zthmtocsymrm 620
621 621
622 622
```

```

623 % ==> thm env definition
624 % theorem-like env
625 \cs_new_protected:Npn \zthm_theorem_env_setup:n #1
626 { % #1:env name; #2:env title separator
627   \zcnt_safe_new:oo { #1 }
628   { \g__ztex_thm_cntparent_tl }
629   \exp_after:wN \def\cs:w the#1\cs_end:
630   {
631     \cs:w the\g__ztex_thm_cntparent_tl\cs_end:
632     .\arabic{#1}
633   }
634   \__ztex_cref_math_env:n { #1 }
635   \DeclareDocumentEnvironment{#1}{0{}}
636   {
637     \UseHook{ztex/thm-theorem/before}
638     \UseHook{ztex/thm-theorem-#1/before}
639     \__ztex_thm_wrap_start:nnn {#1}{##1}{\ }
640     \MakeLinkTarget*{zthm@#1.\zthmnumber}
641     \bool_if:NT \g__ztex_thm_title_inline_bool
642     {
643       \group_begin:
644         \zthm@title@before{theorem}
645         \__ztex_thm_theorem_title:
646       \group_end:
647     }
648     \UseHook{ztex/thm-theorem/begin}
649     \UseHook{ztex/thm-theorem-#1/begin}
650     \tl_trim_spaces:n
651     }{
652       \UseHook{ztex/thm-theorem/end}
653       \UseHook{ztex/thm-theorem-#1/end}
654       \__ztex_thm_wrap_end:n {#1}
655       \UseHook{ztex/thm-theorem/after}
656       \UseHook{ztex/thm-theorem-#1/after}
657     }
658   }
659   \ztex_label_hook_preamble_last:nn
660   { ztex-thmptheorem-setup-inner }
661   {
662     \zcnt_safe_new:oo
663     { ztex@thm@sharecnt }
664     { \g__ztex_thm_cntparent_tl }
665     \def\theztex@thm@sharecnt
666     {
667       \cs:w the\g__ztex_thm_cntparent_tl\cs_end:
668       .\arabic{ztex@thm@sharecnt}
669     }
670     \clist_map_inline:Nn \g__ztex_thm_theorem_clist

```

```

671 {
672     \zthm_theorem_env_setup:n {#1}
673 }
674 }
675
676 % proof-like env
677 \newcommand{\qedsymbol}{\ensuremath{\square}}
678 \cs_new_protected:Npn \zthm_proof_env_setup:n #1
679 {% #1:env name; #2:env title separator
680     \DeclareDocumentEnvironment{#1}{O{}}
681     {
682         \UseHook{ztex/thm-proof/before}
683         \UseHook{ztex/thm-proof-#1/before}
684         \__ztex_thm_proof_title_gen:nn {#1}{\,,}
685         \group_begin:
686             \zthm@title@before{proof}
687             \__ztex_thm_proof_title:
688         \group_end:
689         \UseHook{ztex/thm-proof/begin}
690         \UseHook{ztex/thm-proof-#1/begin}
691         \tl_set:Nn \l__thm_proof_name_tl {#1}
692         \tl_trim_spaces:n
693     }{
694         \UseHook{ztex/thm-proof/end}
695         \UseHook{ztex/thm-proof-#1/end}
696         \str_if_eq:NnTF \l__thm_proof_name_tl
697             { proof }
698             { \hfill\qedsymbol\par }
699             { \par }
700         \UseHook{ztex/thm-proof/after}
701         \UseHook{ztex/thm-proof-#1/after}
702     }
703 }
704 \ztex_label_hook_preamble_last:nn
705 { ztex-thmproof-setup-inner }
706 {
707     \clist_map_inline:Nn \g__ztex_thm_proof_clist
708     {
709         \zthm_proof_env_setup:n { #1 }
710     }
711 }
712
713 % create thm env manually
714 \cs_new:Nn \__zthm_env_setup:nn
715 {
716     \exp_args:Nno \use:c
717         { zthm_#1_env_setup:n }
718         { #2 }

```


719	}	719
720	\cs_new:Nn __zthm_env_setup:nnn	720
721	{	721
722	% NOTE: '#3' has been dropped.	722
723	\exp_args:Nno \use:c	723
724	{ zthm_#1_env_setup:n }	724
725	{ #2 }	725
726	}	726
727	\NewDocumentCommand{\zthmenvset}{mm}	727
728	{	728
729	\zcmd_if_preamble:TF	729
730	{	730
731	\exp_args:Nc \zthm_envnew_var_init:Nnn	731
732	{ g__ztex_thm_name_\g__ztex_lang_math_tl _prop }	732
733	{ #1 }{ #2 }	733
734	}{	734
735	\zthm_envnew_var_init:Nnn	735
736	\g__ztex_thm_name_prop	736
737	{ #1 }{ #2 }	737
738	}	738
739	\keyval_parse:nnn	739
740	{ __zthm_env_setup:nn {#1} }	740
741	{ __zthm_env_setup:nnn {#1} }	741
742	{ #2 }	742
743	}	743
744		744
745		745
746	% ==> thm theorem-like env hook interface	746
747	% general thm hook	747
748	\NewHook{ztex/thm-theorem/before}	748
749	\NewHook{ztex/thm-theorem/begin}	749
750	\NewReversedHook{ztex/thm-theorem/end}	750
751	\NewReversedHook{ztex/thm-theorem/after}	751
752	\NewHook{ztex/thm-proof/before}	752
753	\NewHook{ztex/thm-proof/begin}	753
754	\NewReversedHook{ztex/thm-proof/end}	754
755	\NewReversedHook{ztex/thm-proof/after}	755
756	\int_new:N \g__ztex_thm_proof_hook_index_int	756
757	\int_new:N \g__ztex_thm_theorem_hook_index_int	757
758	\int_gzero:N \g__ztex_thm_proof_hook_index_int	758
759	\int_gzero:N \g__ztex_thm_theorem_hook_index_int	759
760		760
761	% specific thm hook	761
762	\clist_map_inline:nn {theorem, proof}	762
763	{	763
764	\clist_map_inline:cn { g__ztex_thm_#1_clist }	764
765	{	765
766	\NewHook{ztex/thm-#1-##1/before}	766

```

767 \NewHook{ztex/thm-#1-##1/begin}
768 \NewReversedHook{ztex/thm-#1-##1/end}
769 \NewReversedHook{ztex/thm-#1-##1/after}
770 \int_new:c {g__ztex_thm_#1_##1_hook_index_int}
771 \int_gzero:c {g__ztex_thm_#1_##1_hook_index_int}
772 }
773 }
774 \cs_generate_variant:Nn \hook_gput_code:nnn {ne}
775 \cs_new_protected:Npn \__ztex_thm_hook_parser_setup:nnn #1#2#3
776 {% #1:boolean; #2:hook-type(theorem/proof); #3:env name
777 \bool_if:nTF { #1 }
778 {
779 \cs_set:Npn \__ztex_thm_hook_parser:nn ##1##2
780 {
781 \tl_if_empty:nTF { #3 }
782 {
783 \int_gincr:c {g__ztex_thm_#2_hook_index_int}
784 \hook_gput_code:nen
785 { ztex/thm-#2/##1 }
786 {
787 thm-#2-hook.
788 \int_use:c {g__ztex_thm_#2_hook_index_int}
789 }{ ##2 }
790 }{
791 \int_gincr:c {g__ztex_thm_#2_#3_hook_index_int}
792 \hook_gput_code:nen
793 { ztex/thm-#2-#3/##1 }
794 {
795 thm-#2-#3-hook.
796 \int_use:c {g__ztex_thm_#2_#3_hook_index_int}
797 }{ ##2 }
798 }
799 }
800 }{
801 \cs_set:Npn \__ztex_thm_hook_parser:nn ##1##2
802 {
803 \tl_if_empty:nTF { #3 }
804 {
805 \int_gincr:c {g__ztex_thm_#2_hook_index_int}
806 \hook_gput_next_code:nn {ztex/thm-#2/##1}{##2}
807 }{
808 \int_gincr:c {g__ztex_thm_#2_#3_hook_index_int}
809 \hook_gput_next_code:nn {ztex/thm-#2-#3/##1}{##2}
810 }
811 }
812 }
813 }

```

```

815 % users' interface of thm hook
816 \NewDocumentCommand{\zthmhook}{s0{theorem}m}
817 {
818   \exp_args:Neee \__ztex_thm_hook_parser_setup:nnn
819   { \IfBooleanTF{#1}{\c_true_bool}{\c_false_bool} }
820   { theorem }
821   { \IfValueT{#2}{#2} }
822   \keyval_parse:NNn
823   \use_none:n
824   \__ztex_thm_hook_parser:nn
825   { #3 }
826 }
827 \NewDocumentCommand{\zthmproofhook}{s0{proof}m}
828 {
829   \exp_args:Neee \__ztex_thm_hook_parser_setup:nnn
830   { \IfBooleanTF{#1}{\c_true_bool}{\c_false_bool} }
831   { proof }
832   { \IfValueT{#2}{#2} }
833   \keyval_parse:NNn
834   \use_none:n
835   \__ztex_thm_hook_parser:nn
836   { #3 }
837 }
838 \hook_gput_code:nnn {ztex/thm-theorem/before}
839 {thm-theorem-before-par}{\par}
840 \hook_gput_code:nnn {ztex/thm-proof/before}
841 {thm-proof-before-par}{\par}
842 \NewDocumentCommand{\zthmbefore}{0{theorem}+m}
843 {
844   \hook_gremove_code:nn {ztex/thm-#1/before}
845   {thm-#1-before-par}
846   \hook_gput_code:nnn {ztex/thm-#1/before}
847   {thm-#1-before}{#2}
848 }
849 \@onlypreamble\zthmbefore
850
851
852 % ==> ztex thm hooks seq order
853 \DeclareHookRule{env/document/before}
854 {ztex-thmenv-setup-user}
855 {<}
856 {ztex-thmptheorem-setup-inner}
857 \DeclareHookRule{env/document/before}
858 {ztex-thmenv-setup-user}
859 {<}
860 {ztex-thmproof-setup-inner}
861 \DeclareHookRule{env/document/before}
862 {ztex-thmenv-setup-user}

```


1	%%%	1
2	%% This is file 'ztex.module.sect.tex' of ztex bundle.	2
3	%% The following code is experimental and will likely change.	3
4	%% Copyright 2024, 2025, 2026 Zongping Ding.	4
5	%%	5
6	%% This work may be distributed and/or modified under the	6
7	%% conditions of the LaTeX Project Public License, either	7
8	%% version 1.3 of this license or any later version.	8
9	%% The latest version of this license is in	9
10	%% http://www.latex-project.org/lppl.txt	10
11	%% and version 1.3 or later is part of all distributions of LaTeX	11
12	%% version 2005/12/01 or later.	12
13	%%	13
14	%% This work has the LPPL maintenance status `maintained'.	14
15	%%	15
16	%% The Current Maintainer of this work is Zongping Ding.	16
17	%%%	17
18	\ProvidesExplFile{ztex.module.sect.tex}	18
19	{2026/03/03}{\ztex@versi@n}	19
20	{sect~module~for~ztex}	20
21		21
22		22
23	%%% sect module for ztex %%%%	23
24	%% REFERENCE:	24
25	% 1. https://github.com/Sophanatprime/cus/blob/main/module/cus.module.struct.tex	25
26	% 2. https://github.com/CTeX-org/ctex-kit/blob/master/ctex/ctex.dtx	26
27	% 3. https://github.com/jbezos/titlesec	27
28	% 4. https://github.com/jfbu/etoc	28
29		29
30		30
31	%%% disable 'sect' module scope begin %%%%	31
32	% ==> disable 'section' module	32
33	\bool_if:NF \g__ztex_sect_load_bool	33
34	{ \file_input_stop: }	34
35		35
36		36
37	% ==> disable 'titlesec', 'titletoc', 'etoc' etc ...	37
38	\ztex_msg_set:nn { zsect@disable }	38
39	{	39
40	You~can~NOT~use~'sect'~module~together~with~	40
41	'titlesec',~'titletoc',~'titleps',~'sectsty',~	41
42	'tocloft',~'etoc',~'multitoc'~'minitoc'~'cus',~	42
43	'koma-script~sub-packages'~etc~...	43
44	}	44
45	\cs_new:Npn __zsect_package_disable_error:	45
46	{	46

47	\msg_fatal:nn { ztex } { zsect@disable }	47
48	\file_input_stop:	48
49	}	49
50	\cs_new:Npn \zsect_package_disable_error:	50
51	{	51
52	\@ifpackageloaded{ scrextend}{ __zsect_package_disable_error: }{}	52
53	\@ifpackageloaded{ titlesec }{ __zsect_package_disable_error: }{}	53
54	\@ifpackageloaded{ titletoc }{ __zsect_package_disable_error: }{}	54
55	\@ifpackageloaded{ titleps }{ __zsect_package_disable_error: }{}	55
56	\@ifpackageloaded{ sectsty }{ __zsect_package_disable_error: }{}	56
57	\@ifpackageloaded{ tocloft }{ __zsect_package_disable_error: }{}	57
58	\@ifpackageloaded{ multitoc }{ __zsect_package_disable_error: }{}	58
59	\@ifpackageloaded{ minitoc }{ __zsect_package_disable_error: }{}	59
60	\@ifpackageloaded{ etoc }{ __zsect_package_disable_error: }{}	60
61	\@ifpackageloaded{ cus }{ __zsect_package_disable_error: }{}	61
62	}	62
63	\ztex_hook_preamble_last:n	63
64	{	64
65	\zsect_package_disable_error:	65
66	}	66
67		67
68		68
69		69
70	% -----	70
71	% sect module init	71
72	% -----	72
73	__ztool_load_library:n { file-io }	73
74		74
75	\newdimen\zsect@dim@a	75
76	\newdimen\zsect@dim@b	76
77	\newdimen\zsect@dim@c	77
78	\box_new:N \l__ztoc_name_box	78
79	\box_new:N \l__ztoc_title_box	79
80	\box_new:N \@zsectmpboxa	80
81	\scan_new:N \s__ztoc_ignore_empty_mark	81
82		82
83	\tl_new:N \l__zsect_level_keyval_tl	83
84	\tl_new:N \l__zsect_level_clist_tl	84
85	\tl_new:N \l__zsect_level_tl	85
86	\int_new:N \g_zsect_level_int	86
87	\int_new:N \l__zsect_class_type_int	87
88	\int_set:Nn \l__zsect_class_type_int { 0 }	88
89	\prop_new:N \c_ztoc_special_level_prop	89
90	\clist_new:N \l__zsect_potential_class_clist	90
91	\clist_new:N \g__zsect_mark_user_clist	91
92	\cs_generate_variant:Nn \cs_set:Npn { Npo }	92
93		93
94		94

```

95 % ==> generate sect title/toc class data
96 % * \g_zsect_level_int (total level amount, for future use)
97 % * \c_zsect_level_prop
98 % * \c_ztoc_special_level_prop / \c_ztoc_special_level_clist
99 % * \c_zsect_level_clist
100 % * \c_zsect_level_tl
101 \clist_set:Nn \l__zsect_potential_class_clist
102 {
103     volume, book, part, chapter,
104     section, subsection, subsubsection,
105     paragraph, subparagraph,
106 }
107 \cs_new:Npn \__zsect_getcur_title_class:n #1
108 {
109     \cs_if_exist:cT { #1 }
110     {
111         \int_incr:N \l__zsect_class_type_int
112         \tl_put_right:Ne \l__zsect_level_keyval_tl
113             { #1 = \int_use:N \l__zsect_class_type_int, }
114         \tl_put_right:Ne \l__zsect_level_clist_tl { #1, }
115         \tl_put_right:Ne \l__zsect_level_tl { {#1} }
116     }
117 }
118 \clist_map_function:NN \l__zsect_potential_class_clist
119     \__zsect_getcur_title_class:n
120 % sec class prop
121 \exp_args:NNo \prop_const_from_keyval:Nn \c_zsect_level_prop
122 {
123     \l__zsect_level_keyval_tl
124 }
125 % sec class clist
126 \clist_const:Ne \c_zsect_level_clist { \l__zsect_level_clist_tl }
127 % sec class tl
128 \tl_const:Ne \c_zsect_level_tl { \l__zsect_level_tl }
129 % sec class int
130 \int_gset:Nn \g_zsect_level_int { \l__zsect_class_type_int }
131 % special sec class prop
132 \clist_const:Nn \c_ztoc_special_level_clist
133 {
134     figure, table, theorem,
135     algorithm, glossary, lstlisting,
136 }
137 % NOTE: these special class name is depending on
138 %       the '@capttype' defined by '\caption'.
139 \prop_gset_from_keyval:Nn \c_ztoc_special_level_prop
140 {
141     figure      = 3,
142     table       = 3,

```

```

143     theorem      = 3,
144     algorithm    = 3,
145     glossary     = 3,
146     lstlisting   = 3,
147 }
148 \clist_const:Nn \c_ztoc_table_types_clist
149 {
150     toc, % table of contents
151     lot, % list of tables
152     lof, % list of figures
153     lom, % list of theorems
154     tog, % table of glossary('.log' is already used.)
155     loa, % list of algorithm
156     lol, % list of listings
157 }
158 \prop_set_from_keyval:Nn \c_ztoc_table_types_prop
159 {
160     sect      = toc,
161     figure    = lof,
162     table     = lot,
163     theorem   = lom,
164     algorithm = loa,
165     glossary  = tog,
166     lstlisting = lol,
167 }
168 % leagcy sec class prop
169 \prop_const_from_keyval:Nn \c_zsect_level_leagcy_prop
170 {
171     volume      = -3,
172     book        = -2,
173     part        = -1,
174     chapter     = 0,
175     section     = 1,
176     subsection  = 2,
177     subsubsection = 3,
178     paragraph   = 4,
179     subparagraph = 5,
180 }
181 % --> section class path map (for future use)
182 % TODO: generate these path Dynamically
183 \prop_const_from_keyval:Nn \g__ztoc_class_pathmap_prop
184 {
185     subparagraph = part/chapter/section/subsection/subsubsection/paragraph/,
186     paragraph    = part/chapter/section/subsection/subsubsection/,
187     subsubsection = part/chapter/section/subsection/,
188     subsection   = part/chapter/section/,
189     section      = part/chapter/,
190     chapter      = part/,

```



```

191 }
192
193
194 % ==> users' custom class level interface
195 % TODO: user may modify this constant variable,
196 %      is there a better way ?
197 \cs_new_protected:Npn \zsect_class_level_remap:n #1
198 {
199     % title class data: clist/tl/int
200     % NOTE: this catch of 'empty prop' exception should come
201     %      before prop assignment.
202     \keyval_parse:NNn
203         \__zsect_level_prop_empty_handle:n
204         \zsect_config_sync:nn
205         { #1 }
206     % title class level prop
207     \prop_gput_from_keyval:Nn \c_zsect_level_prop
208         { #1 }
209 }
210 \cs_new_protected:Npn \zsect_config_sync:nn #1#2
211 {
212     % sync const
213     \clist_if_in:NnF \c_zsect_level_clist
214         { #1 }
215         { \int_gincr:N \g_zsect_level_int }
216     \clist_gput_right:Nn \c_zsect_level_clist
217         { {#1} }
218     \tl_gput_right:Nn \c_zsect_level_tl
219         { {#1} }
220     % sync mark class
221     \__zsect_mark_user_sync:n { #1 }
222     % sync cmds like '\l@section'
223     \ztoc_lcmd_setup:nn { #1 } { #2 }
224     \clist_if_in:NnT \c_ztoc_special_level_clist { #1 }
225     {
226         \ztoc_special_lcmd_setup:nn { #1 } { #2 }
227     }
228     % sync toc group parser variables
229     \__ztoc_gparser_var_setup:n { #1 }
230     % sync toc filter int
231     \int_if_exist:cF { g__toc_filter_#1_int }
232     {
233         \int_new:c { g__toc_filter_#1_int }
234     }
235     % sync special toc class prop
236     \clist_if_in:NnT \c_ztoc_special_level_clist { #1 }
237     {
238         \prop_gput_from_keyval:Nn \c_ztoc_special_level_prop

```

239	{ #1=#2 }	239
240	}	240
241	}	241
242	\cs_new:Npn __zsect_level_prop_empty_handle:n #1	242
243	{	243
244	\ztex_msg_set:nn {zsect@levelprop@empty}	244
245	{ Level~re-mapping~invalid:~there~is~no~value~for~'#1'. }	245
246	\ztex_msg_error:n {zsect@levelprop@empty}	246
247	}	247
248	\cs_new:Npn __zsect_mark_user_sync:n #1	248
249	{	249
250	\clist_gput_right:Nn \g__zsect_mark_user_clist	250
251	{ #1 }	251
252	\zsect_mark_new_class_safe:nn	252
253	{ \c_true_bool }	253
254	{ ztex-user-#1 }	254
255	}	255
256	\NewDocumentCommand{\zseclevelmap}{m}	256
257	{	257
258	\zsect_class_level_remap:n {#1}	258
259	}	259
260	\@onlypreamble\zseclevelmap	260
261		261
262		262
263		263
264	% -----	264
265	% new NewTemplateType from scratch	265
266	% -----	266
267	% NOTE:	267
268	% 1. only 'type' has prefix 'ztex'.	268
269	% 2. private cmds like '__template_use_instance:nn' may be used in the future.	269
270	\cs_new_protected:Npn \ztex_new_templatetype:nn #1#2	270
271	{	271
272	\NewTemplateType{ztex#1}{#2}	272
273	}	273
274	\cs_new_protected:Npn \ztex_declare_template_interface:nnnn #1#2#3#4	274
275	{	275
276	\DeclareTemplateInterface{ztex#1}{#2}	276
277	{#3}{#4}	277
278	}	278
279	\cs_new_protected:Npn \ztex_declare_template_code:nnnnn #1#2#3#4#5	279
280	{	280
281	\DeclareTemplateCode{ztex#1}{#2}	281
282	{#3}{#4}{#5}	282
283	}	283
284	\cs_new_protected:Npn \ztex_declare_template_copy:nnn #1#2#3	284
285	{	285
286	\DeclareTemplateCopy{ztex#1}	286

```

287     {#2}{#3}
288 }
289 \cs_new_protected:Npn \ztex_declare_instance:nnnn #1#2#3#4
290 {
291     \DeclareInstance{ztex#1}{#2}
292     {#3}{#4}
293 }
294 \cs_new_protected:Npn \ztex_declare_instance_copy:nnn #1#2#3
295 {
296     \DeclareInstanceCopy{ztex#1}{#2}{#3}
297 }
298 \cs_new_protected:Npn \ztex_use_instance:nn #1#2
299 {
300     \UseInstance{ztex#1}{#2}
301     % <argument>
302 }
303 \cs_new_protected:Npn \ztex_use_template:nnn #1#2#3
304 {
305     \UseTemplate{ztex#1}{#2}
306     { #3 }
307     % <argument>
308 }
309 \cs_new_protected:Npn \ztex_edit_instance:nnn #1#2#3
310 {
311     \EditInstance{ztex#1}{#2}{#3}
312 }
313 \cs_new_protected:Npn \ztex_edit_template_defaults:nnn #1#2#3
314 {
315     \EditTemplateDefaults{ztex#1}
316     {#1}{#3}
317 }
318
319 % get templates & instances info(debug)
320 \cs_new_protected:Npn \ztex_show_instance_values:nn #1#2
321 {
322     \ShowInstanceValues{ztex#1}{#2}
323 }
324 \cs_new_protected:Npn \ztex_show_template_code:nn #1#2
325 {
326     \ShowTemplateCode{ztex#1}{#2}
327 }
328 \cs_new_protected:Npn \ztex_show_template_defaults:nn #1#2
329 {
330     \ShowTemplateDefaults{ztex#1}{#2}
331 }
332 \cs_new_protected:Npn \ztex_show_template_interface:nn #1#2
333 {
334     \ShowTemplateInterface{ztex#1}{#2}

```

```
335 } 335
336 \cs_new_protected:Npn \ztex_show_template_variables:nn #1#2 336
337 { 337
338     \ShowTemplateVariables{ztex#1}{#2} 338
339 } 339
340 340
341 341
342 342
343 % ----- 343
344 % bookmark interface 344
345 % ----- 345
346 % TODO: '\DocumentMetadata' will make all bookmark targets wrong. 346
347 \cs_new_protected:Npn \zsect_bookmark_add:nnn #1#2#3 347
348 { 348
349     \pdfbookmark[#1]{#2}{#3} 349
350 } 350
351 \cs_generate_variant:Nn \zsect_bookmark_add:nnn 351
352 { ene, eee } 352
353 353
354 354
355 355
356 % ----- 356
357 % toc interface variables init 357
358 % c / t / f / m / g / a / l 358
359 % ----- 359
360 % ==> table related variables setup 360
361 \seq_gclear_new:N \g__ztoc_table_enabled_seq 361
362 \clist_map_inline:Nn \c_ztoc_table_types_clist 362
363 { 363
364     %%% public: 364
365     % iow and iow checker: 365
366     \iow_new:c { g_ztoc_#1_iow } 366
367     \bool_new:c { g_ztoc_#1_iow_bool } 367
368     % global table seq: 368
369     \seq_gclear_new:c { g_ztoc_#1_seq } 369
370     % local table seq(relies on keyval seq): 370
371     \seq_gclear_new:c { g_ztoc_local#1_seq } 371
372     % keyval table seq: 372
373     \seq_gclear_new:c { g_ztoc_keyval#1_seq } 373
374 374
375     %%% private: 375
376     % stopped keyval table seq: 376
377     \seq_gclear_new:c { g__ztoc_stopped_keyval#1_seq } 377
378 } 378
379 379
380 380
381 % ==> leagcy toc interface 381
382 % NOTE: 382
```

```
383 % 1. re-def these commands at last to prevent them from being modified; 383
384 % 2. '\numberline' has been deprecated in 'zsect'. 384
385 \ztex_hook_preamble_last:n 385
386 { 386
387     \cs_set_protected:Npn \numberline #1 387
388     { 388
389         \hb@xt@{zsect@dim@a{#1}\hfil} 389
390     } 390
391 % latex2e '\contentsline' syntax as follows: 391
392 %   \contentsline{<type>}{<entry>}{<page>}{<anchor>} 392
393 % 'entry' looks like: 393
394 %   \numberline {<toc name>}<toc title> 394
395 \protected\def\contentsline #1#2#3#4 395
396 { 396
397     \gdef\@contentsline@destination 397
398     { #4 } 398
399     \gdef\ztoc@current@class 399
400     { #1 } 400
401     \csname l@#1\endcsname {#2}{#3} 401
402 } 402
403 } 403
404 \cs_new_protected:Npn \zsect_hang_from:n #1 404
405 { 405
406     \setbox\@zsectmpboxa\hbox{#{#1}} 406
407     \hangindent\wd\@zsectmpboxa 407
408     \noindent\box\@zsectmpboxa 408
409 } 409
410 \cs_new_protected:Npn \zsec_normal_text:n #1 410
411 { 411
412     {\normalsize\normalfont\normalcolor #1} 412
413 } 413
414 \cs_new_protected:Npn \zsect_leaders:nnnnn #1#2#3#4#5 414
415 {% #1:type, #2:repeat, #3:width, #4:raise, #5:skip 415
416     \cs:w #1leaders\cs_end: \hbox:n { 416
417         \box_move_up:nn { #4 } 417
418         { 418
419             \hbox_to_wd:nn {#3}{\hss #2 \hss} 419
420         } 420
421     } \hskip #5\relax 421
422 } 422
423 \protected\def\@dottedtocline #1#2#3#4#5 423
424 {% 424
425     \ifnum #1>\c@tocdepth \else 425
426         \vskip \z@ \@plus.2\p@ 426
427         {\leftskip #2\relax \rightskip \@tocrmarg \parfillskip -\rightskip 427
428         \parindent #2\relax\@afterindenttrue 428
429         \interlinepenalty\@M 429
430         \leavevmode 430
```

431	<code>\@tempdima #3\relax</code>	431
432	<code>\advance\leftskip \@tempdima \null\nobreak\hskip -\leftskip</code>	432
433	<code>{#4}\nobreak</code>	433
434	<code>\leaders\hbox{\${\m@th</code>	434
435	<code>\mkern \@dotsep mu\hbox{.}\mkern \@dotsep</code>	435
436	<code>mu\$}\hfill</code>	436
437	<code>\nobreak\hb@xt@\@pnumwidth{\hfil #5%</code>	437
438	<code>\kern-\p@\kern\p@}%</code>	438
439	<code>\par}%</code>	439
440	<code>\fi</code>	440
441	<code>}</code>	441
442	<code>% TODO: refactor '\zdottoctocline:nnnnnnnnn' to recieve more than</code>	442
443	<code>% 9 args using '\def\cmda#1{#1\cldb} \def\cldb#1{#1}'.</code>	443
444	<code>\cs_new_protected:Npn \zdottoctocline:nnnnnnnnn #1#2#3#4#5#6#7#8#9</code>	444
445	<code>{% #3:('num' width) or (the extra rightskip for lines [2:-1]).</code>	445
446	<code>\ifnum #1 > \c@tocdepth \else</code>	446
447	<code>\vskip #9 \relax</code>	447
448	<code>{</code>	448
449	<code>\leftskip #2 \relax</code>	449
450	<code>\rightskip #3 \parfillskip -\rightskip</code>	450
451	<code>\parindent #2 \relax\@afterindenttrue</code>	451
452	<code>\interlinepenalty\@M</code>	452
453	<code>\leavevmode</code>	453
454	<code>\zsect@dim@a #4 \relax</code>	454
455	<code>\advance\leftskip \zsect@dim@a</code>	455
456	<code>\null\nobreak \hskip -\leftskip</code>	456
457	<code>{ #5 } \nobreak</code>	457
458	<code>#6 % leaders</code>	458
459	<code>\nobreak #7 #8</code>	459
460	<code>}</code>	460
461	<code>\fi</code>	461
462	<code>}</code>	462
463	<code>\cs_new_protected:Npn \dottedtocline:nnnnn #1#2#3#4#5</code>	463
464	<code>{</code>	464
465	<code>\zdottoctocline:nnnnnnnnn</code>	465
466	<code>{#1}{#2}{\@tocrmarg}</code>	466
467	<code>{#3}{#4}</code>	467
468	<code>{</code>	468
469	<code>\leaders\hbox</code>	469
470	<code>{\${\m@th</code>	470
471	<code>\mkern \@dotsep mu</code>	471
472	<code>\hbox{.}</code>	472
473	<code>\mkern \@dotsep mu</code>	473
474	<code>\$}\hfill</code>	474
475	<code>}</code>	475
476	<code>{ \hb@xt@\@pnumwidth{\hfil\normalfont \normalcolor #5} }</code>	476
477	<code>{ \par }{ \z@ \@plus.2\p@ }</code>	477
478	<code>}</code>	478

```
479
480
481 % ==> ztoc tocline adding interface
482 \cs_new_protected:Npn \zsect_restore_protect:
483   { \let\protect\relax }
484 \cs_new_protected:Npn \zsect_unexpand_protect:
485   { \let\protect\noexpand }
486 \cs_new_protected:Npn \zsect_add_to_table:nn #1#2
487   {% #1:table type; #2:content
488     \bool_if:cT { g_ztoc_#1_iow_bool }
489     {
490       \zsect_unexpand_protect:
491       \iow_now:ce { g_ztoc_#1_iow }
492         { #2 }
493       \zsect_restore_protect:
494     }
495   }
496 \cs_generate_variant:Nn \zsect_add_to_table:nn
497   { no, oo, ne, ee }
498
499 % >> normal toc
500 \cs_new_protected:Npn \zsect_add_toc_line:nnnn #1#2#3#4
501   {
502     \bool_if:NT \g_ztoc_toc_iow_bool
503     {
504       \zsect_unexpand_protect:
505       \iow_now:Ne \g_ztoc_toc_iow
506         {
507           \token_to_str:N \contentsline{#1}{#2}{#3}{#4}
508           \c_percent_str
509         }
510       \zsect_restore_protect:
511     }
512   }
513 \cs_generate_variant:Nn \zsect_add_toc_line:nnnn
514   { oooo, eeee, eeoe, nnee, nooe }
515
516 % >> other contents(reley on '\caption' commands)
517 % NOTE:
518 % 1. theorem toc is handled in 'thm' module
519 % 2. if '\@captive' undefined, an ERROR will occur, 'figure'
520 %    and 'table' env define '\@captive' to 'figure' or 'table'.
521 % 3. package 'algorithm2e' redefine '\@caption' command
522 %    inside env and restore it outside.
523 % TODO: replace hook with sockets for contents
524 %       patch hook -- {zsect/caption/contents},
525 %       for that we only need one plug to recieve
526 %       toc args spec to generate table.
```

```

527 \hook_new_with_args:nn {zsect/caption/contents}{3}
528 % add contents hook to normal '\@caption':
529 \hook_gput_code_with_args:nnn
530 { cmd/@caption/before }
531 { zsect@caption@patches }
532 {
533   \hook_use:nnw {zsect/caption/contents}
534   {3}{#1}{#2}{#1}
535 }
536 % add contents hook to '\@caption' in 'algorithm2e':
537 \ztex_hook_preamble_last:n
538 {
539   \@ifpackageloaded{ algorithm2e }
540   {
541     \hook_gput_code_with_args:nnn
542     { cmd/algocf@latexcaption/before }
543     { zsect@algocaption@patches }
544     {
545       \hook_use:nnw {zsect/caption/contents}
546       {3}{algorithm}{#2}{algocf}
547     }
548     }{ \relax }
549   }
550 % add code to the above contents hooks:
551 \hook_gput_code_with_args:nnn
552 { zsect/caption/contents }
553 { zsect@caption@contents }
554 {
555   \use:c { zsect_add_#1_line:eeee }
556   { #1 }
557   { {\use:c {the#3}}{\ignorespaces \exp_not:n {#2}} }
558   { \thepage }
559   { \ztexhyperTF {#1}.\use:c {the#3}}{} }
560 }
561 \cs_new_protected:Npn \zsect_add_table_line:nnnn #1#2#3#4
562 {
563   \zsect_add_to_table:nn { lot }
564   {
565     \token_to_str:N \contentsline{#1}{#2}{#3}{#4}
566     \c_percent_str
567   }
568 }
569 \cs_new_protected:Npn \zsect_add_figure_line:nnnn #1#2#3#4
570 {
571   \zsect_add_to_table:nn { lof }
572   {
573     \token_to_str:N \contentsline{#1}{#2}{#3}{#4}
574     \c_percent_str

```


575	}	575
576	}	576
577	\cs_new_protected:Npn \zsect_add_theorem_line:nnnn #1#2#3#4	577
578	{	578
579	\zsect_add_to_table:nn { lom }	579
580	{	580
581	\token_to_str:N \contentsline{#1}{#2}{#3}{#4}	581
582	\c_percent_str	582
583	}	583
584	}	584
585	\cs_new_protected:Npn \zsect_add_lstlisting_line:nnnn #1#2#3#4	585
586	{	586
587	\zsect_add_to_table:nn { lol }	587
588	{	588
589	\token_to_str:N \contentsline{#1}{#2}{#3}{#4}	589
590	\c_percent_str	590
591	}	591
592	}	592
593	\cs_new_protected:Npn \zsect_add_glossary_line:nnnn #1#2#3#4	593
594	{	594
595	\zsect_add_to_table:nn { tog }	595
596	{	596
597	\token_to_str:N \contentsline{#1}{#2}{#3}{#4}	597
598	\c_percent_str	598
599	}	599
600	}	600
601	\cs_new_protected:Npn \zsect_add_algorithm_line:nnnn #1#2#3#4	601
602	{	602
603	\zsect_add_to_table:nn { loa }	603
604	{	604
605	\token_to_str:N \contentsline{#1}{#2}{#3}{#4}	605
606	\c_percent_str	606
607	}	607
608	}	608
609	\cs_generate_variant:Nn \zsect_add_table_line:nnnn	609
610	{ oooo, eeee }	610
611	\cs_generate_variant:Nn \zsect_add_figure_line:nnnn	611
612	{ oooo, eeee }	612
613	\cs_generate_variant:Nn \zsect_add_theorem_line:nnnn	613
614	{ oooo, eeee }	614
615	\cs_generate_variant:Nn \zsect_add_lstlisting_line:nnnn	615
616	{ oooo, eeee }	616
617	\cs_generate_variant:Nn \zsect_add_glossary_line:nnnn	617
618	{ oooo, eeee }	618
619	\cs_generate_variant:Nn \zsect_add_algorithm_line:nnnn	619
620	{ oooo, eeee }	620
621		621
622		622

```

623 % ==> float env for special class(use ltx interface)
624 % special class caption name
625 \str_case:VnF \g__ztex_lang_str
626 {
627   { en }
628   {
629     \ztex_hook_preamble_last:n
630     {
631       \cs_set_nopar:Npn \figurename{Figure}
632       \cs_set_nopar:Npn \tablename{Table}
633       \cs_set_nopar:Npn \lstlistingname{Listing}
634     }
635   }
636   { cn }
637   {
638     \ztex_hook_preamble_last:n
639     {
640       \cs_set_nopar:Npn \figurename{图}
641       \cs_set_nopar:Npn \tablename{表}
642       \cs_set_nopar:Npn \lstlistingname{代码}
643     }
644   }
645 }{ \relax }
646
647 % special '\caption' to record contents
648 \cs_new_protected:Npn \zsect_caption_use:nnn #1#2#3
649 {% #1:special class; #2:format; #3:caption
650   \clist_if_in:NnTF \c_ztoc_special_level_clist
651     { #1 }{ \def\@capttype{#1} }
652     {
653       \ztex_msg_set:nn {wrong@caption@type}
654       {
655         caption~type~'#1'~is~not~permitted,~
656         use~'figure',~'table',~etc~instead.
657       }
658       \ztex_msg_error:n {wrong@caption@type}
659     }
660   \__zsect_title_cnt_refstep:nn
661     { \g__ztex_hyperref_bool }
662     { #1 }
663   % caption format
664   \cs_set:Npn \__zsect_caption_format:nn ##1##2
665     { #2 }
666   % sync contents
667   \use:c { zsect_add_#1_line:eeee }
668     { #1 }
669     {
670       \use:c {the#1} }

```

671	{	671
672	\tl_if_empty:nTF {#3}	672
673	{ \ignorespaces \exp_not:c {fnum@#1} }	673
674	{	674
675	\ignorespaces	675
676	\exp_not:n {	676
677	__zsect_caption_format:nn	677
678	{ \use:c {fnum@#1} }	678
679	{ #3 }	679
680	}	680
681	}	681
682	}	682
683	}	683
684	{ \thepage }	684
685	{ \ztexhyperTF {#1.\use:c {the#1}}{} }	685
686	% typeset caption	686
687	__zsect_caption_format:nn	687
688	{ \use:c {fnum@#1} }	688
689	{ #3 }	689
690	}	690
691	\cs_generate_variant:Nn \zsect_caption_use:nnn	691
692	{ ooo, eee }	692
693	\NewDocumentCommand{\zseccaption}	693
694	{m O{##1}\IfBlankF{##2}{:}##2} +m}	694
695	{	695
696	\zsect_caption_use:nnn	696
697	{ #1 }{ #2 }{ #3 }	697
698	}	698
699		699
700	% ltx float env setup interface	700
701	\cs_new_protected:Npn \zsect_ltx_float_setup:nnnnnn #1#2#3#4#5#6	701
702	{% #1:counter; #2:cnt form; #3:position;	702
703	% #4:level; #5:ext; #6:num.	703
704	\zcmt_safe_new:nn {#1}{}	704
705	\cs_set:cpn {the#1}{#2\use:c {c@#1}}	705
706	\cs_set:cpn {fps@#1}{#3}	706
707	\cs_set:cpn {ftype@#1}{#4}	707
708	\cs_set:cpn {ext@#1}{#5}	708
709	\cs_set:cpn {fnum@#1}{#6}	709
710	}	710
711	\cs_new_protected:Npn \zsect_ltx_float_begin:n #1	711
712	{ \@float{#1} }	712
713	\cs_new_protected:Npn \zsect_ltx_float_end:	713
714	{ \end@float }	714
715		715
716		716
717	% ==> toc template declare	717
718	% NOTE: toc = name + title + leaders + page	718

719	<code>\gdef\ztoc@leader@type{}</code>	719
720	<code>\gdef\ztoc@leader@content{.}</code>	720
721	<code>\long\gdef\ztoc@line@end{\par}</code>	721
722	<code>\def\ztoc@ignore@level{}</code>	722
723		723
724	<code>\newlength{\ztoc@rmargin}</code>	724
725	<code>\newlength{\ztoc@page@width}</code>	725
726	<code>\newlength{\ztoc@leader@sep}</code>	726
727	<code>\newlength{\ztoc@leader@raise}</code>	727
728	<code>\setlength{\ztoc@rmargin}{\@tocrmarg}</code>	728
729	<code>\setlength{\ztoc@leader@sep}{4.5pt}</code>	729
730	<code>\setlength{\ztoc@leader@raise}{0pt}</code>	730
731	<code>\setlength{\ztoc@page@width}{\@pnumwidth}</code>	731
732		732
733	<code>\ztex_new_templatetype:nn {toc}{3}</code>	733
734	<code>\ztex_declare_template_interface:nnnn {toc}{default}{3}</code>	734
735	<code>{</code>	735
736	<code>no-parent : boolean,</code>	736
737		737
738	<code>ignore : boolean = { false },</code>	738
739	<code>ignore.negate : boolean = { false },</code>	739
740	<code>ignore.text : tokenlist = \s__ztoc_ignore_empty_mark,</code>	740
741	<code>ignore.name : commalist = { },</code>	741
742	<code>ignore.page : commalist = { },</code>	742
743		743
744	<code>hyper.name : boolean = { false },</code>	744
745	<code>hyper.title : boolean = { false },</code>	745
746	<code>hyper.page : boolean = { true },</code>	746
747		747
748	<code>line.end : tokenlist = \ztoc@line@end,</code>	748
749	<code>line.width : length,</code>	749
750		750
751	<code>name : tokenlist = { },</code>	751
752	<code>name.show : boolean = { true },</code>	752
753	<code>name.width : length,</code>	753
754	<code>name.format : tokenlist,</code>	754
755	<code>name.format+ : tokenlist = { },</code>	755
756	<code>name.before : tokenlist = { },</code>	756
757	<code>name.after : tokenlist = { },</code>	757
758	<code>name.hyper : boolean = \KeyValue { hyper.name },</code>	758
759		759
760	<code>title.width : length,</code>	760
761	<code>title.format : tokenlist = { },</code>	761
762	<code>title.format+ : tokenlist = { },</code>	762
763	<code>title.before : tokenlist = { },</code>	763
764	<code>title.after : tokenlist = { },</code>	764
765	<code>title.hyper : boolean = \KeyValue { hyper.title },</code>	765
766		766

```

767 page.format      : tokenlist = \normalfont\normalcolor,
768 page.format+    : tokenlist = { },
769 page.before     : tokenlist = { },
770 page.after      : tokenlist = { },
771 page.width      : length     = \ztoc@page@width,
772 page.hyper      : boolean    = \KeyValue { hyper.page },
773
774 format          : tokenlist = { },
775 format+         : tokenlist = { },
776 format.name     : tokenlist = \KeyValue { name.format },
777 format.name+    : tokenlist = \KeyValue { name.format+ },
778 format.title    : tokenlist = \KeyValue { title.format },
779 format.title+   : tokenlist = \KeyValue { title.format+ },
780 format.page     : tokenlist = \KeyValue { page.format },
781 format.page+    : tokenlist = \KeyValue { page.format+ },
782
783 width.name      : length     = \KeyValue { name.width },
784 width.title     : length,
785 width.page      : length     = \KeyValue { page.width },
786 width.line      : length     = \KeyValue { line.width },
787
788 space.before    : skip,
789 space.left      : skip,
790 space.right     : skip      = \ztoc@rmargin,
791 space.hang      : length     = \KeyValue { name.width },
792
793 leader.fill     : skip      = { \fill },
794 leader.sep      : length     = \ztoc@leader@sep,
795 leader.raise    : length     = \ztoc@leader@raise,
796 leader.type     : tokenlist = \ztoc@leader@type,
797 leader.content  : tokenlist = \ztoc@leader@content,
798
799 explicit       : boolean    = { false },
800 enhance        : boolean    = { false },
801 code           : tokenlist = { },
802 }
803 \ztex_declare_template_code:nnnnn {toc}{default}{3}
804 {
805   no-parent     = \l__ztoc_no_parent_bool, % TODO: handle it in local toc
806
807   ignore        = \l__ztoc_ignore_bool,
808   ignore.text    = \l__ztoc_ignore_text_tl,
809   ignore.name    = \l__ztoc_ignore_name_clist,
810   ignore.page    = \l__ztoc_ignore_page_clist,
811   ignore.negate  = \l__ztoc_ignore_negate_bool,
812
813   line.end      = \l__ztoc_line_end_tl,
814   line.width     = \l__ztoc_width_line_dim, % TODO: handle this key in the future

```

815			815
816	hyper.name	= \l__ztoc_hyper_name_bool,	816
817	hyper.title	= \l__ztoc_hyper_title_bool,	817
818	hyper.page	= \l__ztoc_hyper_page_bool,	818
819			819
820	format	= \l__ztoc_format_tl,	820
821	format+	= \l__ztoc_format_p_tl,	821
822	format.name	= \l__ztoc_name_format_tl,	822
823	format.name+	= \l__ztoc_name_format_p_tl,	823
824	format.title	= \l__ztoc_title_format_tl,	824
825	format.title+	= \l__ztoc_title_format_p_tl,	825
826	format.page	= \l__ztoc_page_format_tl,	826
827	format.page+	= \l__ztoc_page_format_p_tl,	827
828			828
829	name	= \l__ztoc_name_tl,	829
830	name.show	= \l__ztoc_name_show_bool,	830
831	name.width	= \l__ztoc_width_name_dim,	831
832	name.format	= \l__ztoc_name_format_tl,	832
833	name.format+	= \l__ztoc_name_format_p_tl,	833
834	name.before	= \l__ztoc_name_before_tl,	834
835	name.after	= \l__ztoc_name_after_tl,	835
836	name.hyper	= \l__ztoc_hyper_name_bool,	836
837			837
838	title.width	= \l__ztoc_width_title_dim,	838
839	title.format	= \l__ztoc_title_format_tl,	839
840	title.format+	= \l__ztoc_title_format_p_tl,	840
841	title.before	= \l__ztoc_title_before_tl,	841
842	title.after	= \l__ztoc_title_after_tl,	842
843	title.hyper	= \l__ztoc_hyper_title_bool,	843
844			844
845	page.format	= \l__ztoc_page_format_tl,	845
846	page.format+	= \l__ztoc_page_format_p_tl,	846
847	page.before	= \l__ztoc_page_before_tl,	847
848	page.after	= \l__ztoc_page_after_tl,	848
849	page.width	= \l__ztoc_width_page_dim,	849
850	page.hyper	= \l__ztoc_hyper_page_bool,	850
851			851
852	width.name	= \l__ztoc_width_name_dim,	852
853	width.title	= \l__ztoc_width_title_dim, % TODO: handle this key in the future	853
854	width.page	= \l__ztoc_width_page_dim,	854
855	width.line	= \l__ztoc_width_line_dim, % TODO: handle this key in the future	855
856			856
857	space.before	= \l__ztoc_space_before_skip,	857
858	space.left	= \l__ztoc_space_left_skip,	858
859	space.right	= \l__ztoc_space_right_skip,	859
860	space.hang	= \l__ztoc_space_hang_dim,	860
861			861
862	leader.fill	= \l__ztoc_leader_fill_skip,	862

```

863 leader.sep = \l__ztoc_leader_sep_dim,
864 leader.raise = \l__ztoc_leader_raise_dim,
865 leader.type = \l__ztoc_leader_sep_tl,
866 leader.content = \l__ztoc_leader_content_tl,
867
868 explicit = \l__ztoc_explicit_bool,
869 enhance = \l__ztoc_public_macro_enhance_bool,
870 code = \l__ztoc_code_tl,
871 }{
872 % Instance args spec:
873 % #1: toc depth(int)
874 % #2: {name}{title}
875 % #3: page
876 %%% NOTE:
877 % 1. use '\ztoc@current@class' to get current class.
878 % 2. group wrapper has been removed.
879 \AssignTemplateKeys
880 \__ztoc_public_macro_setup:nnn
881 { #1 }{ #2 }{ #3 }
882 \ztoc_group_hook_create:Nnnn
883 \__ztoc_dotted_tocline:nnn
884 { #1 }{ #2 }{ #3 }
885 }
886
887 % ==> toc public macros setup
888 % NOTE: these basic macros do NOT cost too much time
889 \cs_new:Nn \__ztoc_public_macro_setup:nnn
890 {
891 \cs_gset_nopar:Npn \ztocdepth { #1 }
892 \cs_gset_nopar:Npe \ztocanchor { \@contentsline@destination }
893 \cs_gset_nopar:Npe \ztoctype { numbered } % TODO: handle this macro.
894 \cs_gset_nopar:Npe \ztocclass { \ztoc@current@class }
895 \cs_gset_nopar:Npn \ztocpage { #3 }
896 \cs_gset_protected:Npn \ztoclink ##1
897 {
898 \exp_args:No \ztexlink
899 { \@contentsline@destination }
900 { ##1 }
901 }
902 \cs_gset:Npn \ztocifnumbered ##1##2
903 {
904 \tl_if_empty:eTF
905 { \__ztoc_extract_name:w #2\scan_stop: }
906 { ##2 }{ ##1 }
907 }
908 \edef\ztoc@expnot@nametitle
909 { \zcmd_map_expnot_tl:n {#2} }
910 \exp_args:NNe \cs_gset:Npn \ztocname

```

911	{	911
912	\exp_after:wN __ztoc_extract_name:w	912
913	\ztoc@expnot@nametitle \scan_stop:	913
914	}	914
915	\exp_args:NNe \cs_gset:Npn \ztoctitle	915
916	{	916
917	\exp_after:wN __ztoc_extract_title:w	917
918	\ztoc@expnot@nametitle \scan_stop:	918
919	}	919
920	}	920
921	% NOTE: these advance macro only available in 'explicit' code,	921
922	% for that they will cost too much time if call them	922
923	% for every tocline. We use an bool to control it.	923
924	\cs_new:Nn __ztoc_public_macro_advance_setup:nnn	924
925	{	925
926	\cs_gset_nopar:Npe \ztoctheindex	926
927	{	927
928	\exp_args:Nee \ztocindex{\ztoc@current@class}	928
929	{	929
930	__ztoc_extract_title:w	930
931	#2 \scan_stop:	931
932	}	932
933	}	933
934	\cs_gset_nopar:Npe \ztoctheabsindex	934
935	{	935
936	\exp_args:Nnee \ztocindex*{\ztoc@current@class}	936
937	{	937
938	__ztoc_extract_title:w	938
939	#2 \scan_stop:	939
940	}	940
941	}	941
942	\cs_gset_nopar:Npe \ztocthecard	942
943	{	943
944	\exp_args:Nee \ztoccard	944
945	{ \ztoc@current@class }	945
946	{	946
947	__ztoc_extract_title:w	947
948	#2 \scan_stop:	948
949	}	949
950	}	950
951	\cs_gset_nopar:Npe \ztocthehilevel	951
952	{ \ztochilevel{toc} }	952
953	\cs_gset_nopar:Npe \ztocthelolevel	953
954	{ \ztoclolevel{toc} }	954
955	}	955
956		956
957	% ==> add hooks to tocline	957
958	% toc group hook parser implement	958


```
1007 % group hook aux functions:
1008 \cs_new:Npn \__ztoc_gparser_var_setup:n #1
1009 {
1010     \bool_if_exist:cF { g__toc_#1_in_bool }
1011     { \bool_new:c { g__toc_#1_in_bool } }
1012     \bool_gset_false:c { g__toc_#1_in_bool }
1013     \int_if_exist:cF { g__toc_group_#1_int }
1014     { \int_new:c { g__toc_group_#1_int } }
1015     \int_set:cn { g__toc_group_#1_int }{ 0 }
1016 }
1017 \exp_args:Ne \clist_map_inline:nn
1018 { \c_zsect_level_clist, \c_ztoc_special_level_clist }
1019 {
1020     \__ztoc_gparser_var_setup:n { #1 }
1021 }
1022 \cs_new:Npn \__ztoc_reset_toc_group_int:
1023 {
1024     \clist_map_inline:Nn \c_zsect_level_clist
1025     {
1026         \int_gset:cn { g__toc_group_##1_int }
1027         { 0 }
1028     }
1029 }
1030 \cs_new:Npn \__ztoc_step_toc_group_int:n #1
1031 {
1032     \int_gincr:c { g__toc_group_#1_int }
1033     \__ztoc_reset_class_below_int:nn { #1 }{0}
1034 }
1035 \cs_new:Npn \__ztoc_use_toc_group_int:n #1
1036 {
1037     \int_use:c { g__toc_group_#1_int }
1038 }
1039 \cs_generate_variant:Nn \__ztoc_use_toc_group_int:n { e }
1040 \cs_new:Npn \__ztoc_reset_class_below_int:nn #1#2
1041 {
1042     \ztoc_get_class_level:Nn \ztoc@tmplevel@int
1043     { #1 }
1044     \prop_map_inline:Nn \c_zsect_level_prop
1045     {
1046         \int_compare:nNnT { ##2 } > { \ztoc@tmplevel@int }
1047         {
1048             \int_gset:cn { g__toc_group_##1_int }{ #2 }
1049         }
1050     }
1051 }
1052
1053
1054 % ==> toc ignore interface setup
```

```
1055 \cs_set:Npn \__ztoc_dotted_tocline:nnn #1#2#3
1056 {
1057     \bool_if:NTF \l__ztoc_ignore_negate_bool
1058     {
1059         \__ztoc_ignore_negate_parser:nnn #{1}{2}{3}
1060     }{
1061         \__ztoc_ignore_parser:nnn #{1}{2}{3}
1062     }
1063 }
1064 \cs_new:Npn \__ztoc_ignore_parser:nnn #1#2#3
1065 {
1066     \clist_if_in:NnF \ztoc@ignore@level { #1 }
1067     {
1068         \bool_if:NF \l__ztoc_ignore_bool
1069         {
1070             % NOTE: '#3' can NOT be warpped in any command, for
1071             %         example, '#3' can not be '\hyperlink{page.3}{3}'.
1072             \clist_if_in:NnF \l__ztoc_ignore_page_clist { #3 }
1073             {
1074                 % NOTE: compare string instead of tokenlist, for that
1075                 %         'title/name' may be formatted as '\textbf{xxx}'.
1076                 \clist_if_empty:NTF \l__ztoc_ignore_name_clist
1077                 {
1078                     \exp_args:NNo \str_set:Nn \l_tmpb_str {\use_ii:nn #2}
1079                     \exp_args:NNo \str_if_in:NnF \l_tmpb_str
1080                     { \l__ztoc_ignore_text_tl }
1081                     {
1082                         \__ztoc_dotted_tocline_raw:nnn #{1}{2}{3}
1083                     }
1084                 }{
1085                     \clist_map_inline:Nn \l__ztoc_ignore_name_clist
1086                     {
1087                         \exp_args:NNo \str_set:Nn \l_tmpa_str {\use_i:nn #2}
1088                         \exp_args:NNo \str_set:Nn \l_tmpb_str {\use_ii:nn #2}
1089                         \str_if_in:NnF \l_tmpa_str { ##1 }           % check 'name'
1090                         {
1091                             \exp_args:NNo \str_if_in:NnF \l_tmpb_str % check 'title'('text')
1092                             { \l__ztoc_ignore_text_tl }
1093                             {
1094                                 \__ztoc_dotted_tocline_raw:nnn #{1}{2}{3}
1095                             }
1096                         }
1097                     }
1098                 }
1099             }
1100         }
1101     }
1102 }
```

```
1103 \cs_new:Npn \__ztoc_ignore_negate_parser:nnn #1#2#3 1103
1104 { 1104
1105     \clist_if_in:NnT \ztoc@ignore@level { #1 } 1105
1106     { 1106
1107         \__ztoc_dotted_tocline_raw:nnn {#1}{#2}{#3} 1107
1108         \prg_map_break:Nn \__ztoc_ignore_negate_break: {} 1108
1109     } 1109
1110     \clist_if_in:NnT \l__ztoc_ignore_page_clist { #3 } 1110
1111     { 1111
1112         \__ztoc_dotted_tocline_raw:nnn {#1}{#2}{#3} 1112
1113         \prg_map_break:Nn \__ztoc_ignore_negate_break: {} 1113
1114     } 1114
1115     \exp_args:NNo \str_set:Nn \l_tmpa_str {\use_i:nn #2} 1115
1116     \clist_map_inline:Nn \l__ztoc_ignore_name_clist 1116
1117     { 1117
1118         \str_if_in:NnT \l_tmpa_str { ##1 } 1118
1119         { 1119
1120             \__ztoc_dotted_tocline_raw:nnn {#1}{#2}{#3} 1120
1121             \prg_map_break:Nn \__ztoc_ignore_negate_break: {} 1121
1122         } 1122
1123     } 1123
1124     %% name ignore parser begin -> 1124
1125     \exp_args:NNo \str_set:Nn \l_tmpb_str {\use_ii:nn #2} 1125
1126     \exp_args:NNo \str_if_in:NnT \l_tmpb_str 1126
1127     { \l__ztoc_ignore_text_tl } 1127
1128     { 1128
1129         \__ztoc_dotted_tocline_raw:nnn {#1}{#2}{#3} 1129
1130     } 1130
1131     %% <- name ignore parser end 1131
1132     \prg_break_point:Nn \__ztoc_ignore_negate_break: {} 1132
1133 } 1133
1134 1134
1135 % NOTE: The implementation of '\__ztoc_dotted_tocline_raw:nnn' 1135
1136 % comes afterward. 1136
1137 \cs_new:Npn \__ztoc_ignore_negate_break: 1137
1138 { \prg_map_break:Nn \__ztoc_ignore_negate_break: { } } 1138
1139 1139
1140 1140
1141 % ==> commands to get class level, including types: 1141
1142 % Type 1: 'chapter', 'section' etc 1142
1143 % Type 2: 'figure', 'table' etc (current private) 1143
1144 % NOTE: anytime when 'special toc class' maybe included, 1144
1145 % replace '\prop_item:Nn \c_zsect_level_prop {#1}' 1145
1146 % by '\ztoc_get_class_level_expandable:n {#1}' 1146
1147 \cs_new:Npn \ztoc_get_class_level_expandable:n #1 1147
1148 { 1148
1149     \prop_if_in:NnT \c_zsect_level_prop { #1 } 1149
1150     { 1150
```

```
1151 \prop_item:Nn \c_zsect_level_prop {#1} 1151
1152 } 1152
1153 \prop_if_in:NnT \c_ztoc_special_level_prop { #1 } 1153
1154 { 1154
1155 \prop_item:Nn \c_ztoc_special_level_prop {#1} 1155
1156 } 1156
1157 } 1157
1158 \cs_new_protected:Npn \ztoc_get_class_level:Nn #1#2 1158
1159 { 1159
1160 \prop_if_in:NnT \c_zsect_level_prop { #2 } 1160
1161 { 1161
1162 \edef #1 1162
1163 { \prop_item:Nn \c_zsect_level_prop {#2} } 1163
1164 } 1164
1165 \prop_if_in:NnT \c_ztoc_special_level_prop { #2 } 1165
1166 { 1166
1167 \edef #1 1167
1168 { \prop_item:Nn \c_ztoc_special_level_prop {#2} } 1168
1169 } 1169
1170 \bool_lazy_any:nT 1170
1171 { 1171
1172 { !(\cs_if_exist_p:N #1) } 1172
1173 { \tl_if_empty_p:N #1 } 1173
1174 }{ 1174
1175 \ztex_msg_set:nn {ztoc@invalid@class} 1175
1176 { 1176
1177 \tl_if_empty:nTF {#2} 1177
1178 { current~toc~class~is~empty. } 1178
1179 { '#2'~is~an~invalid~toc~class~for~ztex. } 1179
1180 } 1180
1181 \ztex_msg_error:n {ztoc@invalid@class} 1181
1182 } 1182
1183 } 1183
1184 \cs_generate_variant:Nn \ztoc_get_class_level:Nn 1184
1185 { No, Ne } 1185
1186 \cs_generate_variant:Nn \ztoc_get_class_level_expandable:n 1186
1187 { o, e } 1187
1188 1188
1189 1189
1190 % ---> inner components of toc below <---- 1190
1191 \hook_new_pair:nn 1191
1192 { ztoc/tocline/begin } 1192
1193 { ztoc/tocline/end } 1193
1194 \cs_new:Npn \__ztoc_dotted_tocline_raw:nnn #1#2#3 1194
1195 { 1195
1196 \ifnum #1 > \c@tocdepth \else 1196
1197 \edef\ztoc@tmpa@skip 1197
1198 { 1198
```

1199	\skip_eval:n {	1199
1200	\l__ztoc_space_left_skip -	1200
1201	\l__ztoc_space_hang_dim	1201
1202	}	1202
1203	}	1203
1204	\hook_use:n { ztoc/tocline/begin }	1204
1205	\bool_if:NTF \l__ztoc_explicit_bool	1205
1206	{	1206
1207	\bool_if:NT \l__ztoc_public_macro_enhance_bool	1207
1208	{	1208
1209	__ztoc_public_macro_advance_setup:nnn	1209
1210	{ #1 }{ #2 }{ #3 }	1210
1211	}	1211
1212	\l__ztoc_code_tl	1212
1213	\bool_set_false:N \l__ztoc_public_macro_enhance_bool	1213
1214	}{	1214
1215	{	1215
1216	\vskip \l__ztoc_space_before_skip \relax	1216
1217	\leftskip \ztoc@tmpa@skip \relax	1217
1218	\skip_if_finite:nF { \l__ztoc_leader_fill_skip }	1218
1219	{	1219
1220	\rightskip \l__ztoc_space_right_skip \parfillskip -\rightskip	1220
1221	}	1221
1222	\parindent \ztoc@tmpa@skip \relax\@afterindenttrue	1222
1223	\interlinepenalty\@M	1223
1224	\leavevmode	1224
1225	\zsect@dim@a \l__ztoc_space_hang_dim \relax	1225
1226	\advance\leftskip \zsect@dim@a	1226
1227	\null\nobreak \hskip -\leftskip	1227
1228	{	1228
1229	__ztoc_name_title_typeset:nn { #2 }	1229
1230	{ \@contentsline@destination }	1230
1231	} \nobreak	1231
1232	__ztoc_dottedline_leader_set: \nobreak % leaders	1232
1233	__ztoc_dottedline_page_set:nn { #3 }{ page.#3 }	1233
1234	\l__ztoc_line_end_tl	1234
1235	}	1235
1236	}	1236
1237	\hook_use:n { ztoc/tocline/end }	1237
1238	\skip_set:Nn \l__ztoc_space_before_skip {\z@ \@plus.2\p@}	1238
1239	\fi	1239
1240	}	1240
1241	% toc line internal items	1241
1242	\cs_new:Npn __ztoc_name_title_typeset:nn #1#2	1242
1243	{	1243
1244	\bool_lazy_all:nT	1244
1245	{	1245
1246	{ \l__ztoc_name_show_bool }	1246

1247	{	1247
1248	! \tl_if_empty_p:e	1248
1249	{ __ztoc_extract_name:w #1\scan_stop: }	1249
1250	}	1250
1251	}{	1251
1252	__ztoc_item_hyper_begin_aux:nn {name}{ #2 }	1252
1253	\exp_args:Nf __ztoc_dottedline_name_set:n	1253
1254	{ __ztoc_extract_name:w #1\scan_stop: }	1254
1255	__ztoc_item_hyper_end_aux:n {name}	1255
1256	}	1256
1257	__ztoc_item_hyper_begin_aux:nn {title}{ #2 }	1257
1258	\exp_args:Nf __ztoc_dottedline_title_set:n	1258
1259	{ __ztoc_extract_title:w #1\scan_stop: }	1259
1260	__ztoc_item_hyper_end_aux:n {title}	1260
1261	}	1261
1262	\cs_new:Npn __ztoc_item_hyper_begin_aux:nn #1#2	1262
1263	{	1263
1264	\bool_if:cT { l__ztoc_hyper_#1_bool }	1264
1265	{	1265
1266	\hyper@linkstart{link}{#2}	1266
1267	}	1267
1268	}	1268
1269	\cs_new:Npn __ztoc_item_hyper_end_aux:n #1	1269
1270	{	1270
1271	\bool_if:cT { l__ztoc_hyper_#1_bool }	1271
1272	{ \hyper@linkend }	1272
1273	}	1273
1274	\cs_new:Npn __ztoc_dottedline_name_set:n #1	1274
1275	{	1275
1276	\hbox_set:Nn \l__ztoc_name_box	1276
1277	{	1277
1278	\l__ztoc_format_tl	1278
1279	\l__ztoc_format_p_tl	1279
1280	\l__ztoc_name_format_tl	1280
1281	\l__ztoc_name_format_p_tl	1281
1282	{	1282
1283	\l__ztoc_name_before_tl	1283
1284	\tl_if_empty:NTF \l__ztoc_name_tl	1284
1285	{ #1 }{ \l__ztoc_name_tl }	1285
1286	\l__ztoc_name_after_tl	1286
1287	}	1287
1288	}	1288
1289	\dim_compare:nTF { \l__ztoc_width_name_dim = 0pt }	1289
1290	{	1290
1291	\box_use_drop:N \l__ztoc_name_box	1291
1292	}{	1292
1293	\hb@xt@ \l__ztoc_width_name_dim	1293
1294	{ \box_use_drop:N \l__ztoc_name_box \hss }	1294

1295 } 1295

1296 } 1296

1297 \cs_new:Npn __ztoc_dottedline_title_set:n #1 1297

1298 { 1298

1299 % \hb@xt@ \l__ztoc_width_title_dim 1299

1300 { 1300

1301 \l__ztoc_format_tl 1301

1302 \l__ztoc_format_p_tl 1302

1303 \l__ztoc_title_format_tl 1303

1304 \l__ztoc_title_format_p_tl 1304

1305 { 1305

1306 \l__ztoc_title_before_tl 1306

1307 #1 1307

1308 \l__ztoc_title_after_tl 1308

1309 } 1309

1310 } 1310

1311 } 1311

1312 \cs_new:Npn __ztoc_dottedline_leader_set: 1312

1313 { 1313

1314 \zsect_leaders:nnnnn { \l__ztoc_leader_sep_tl } 1314

1315 { \l__ztoc_leader_content_tl } 1315

1316 { \dim_eval:n {\l__ztoc_leader_sep_dim*2} } 1316

1317 { \l__ztoc_leader_raise_dim } 1317

1318 { \l__ztoc_leader_fill_skip } 1318

1319 } 1319

1320 \cs_new:Npn __ztoc_dottedline_page_set:nn #1#2 1320

1321 { 1321

1322 __ztoc_item_hyper_begin_aux:nn {page}{ #2 } 1322

1323 \hb@xt@\l__ztoc_width_page_dim 1323

1324 { 1324

1325 \hss 1325

1326 \l__ztoc_format_tl 1326

1327 \l__ztoc_format_p_tl 1327

1328 \l__ztoc_page_format_tl 1328

1329 \l__ztoc_page_format_p_tl 1329

1330 { 1330

1331 \l__ztoc_page_before_tl 1331

1332 #1 1332

1333 \l__ztoc_page_after_tl 1333

1334 } 1334

1335 } 1335

1336 __ztoc_item_hyper_end_aux:n {page} 1336

1337 } 1337

1338 1338

1339 % \cs_new:Npn \ltx_ztoc_extract_name:w \numberline #1\scan_stop: 1339

1340 % { \tl_item:nn {#1}{1} } 1340

1341 % \cs_new:Npn \ltx_ztoc_extract_title:w \numberline #1\scan_stop: 1341

1342 % { \tl_range:nnn {#1}{2}{-1} } 1342


```
1343 \cs_new:Npn \__ztoc_extract_name:w #1\scan_stop:
1344 { \tl_item:nn {#1}{1} }
1345 \cs_new:Npn \__ztoc_extract_title:w #1\scan_stop:
1346 { \tl_item:nn {#1}{-1} }
1347 % --->      inner components of toc above      <----
1348
1349
1350 % ==> declare '\l@<class>' in an abstract level
1351 \ztex_declare_instance:nnnn {toc}{ztoc/level 1}{default}
1352 {
1353     format          = \large\bfseries,
1354     name.width      = 1.9em,
1355     space.before    = 1em\@plus\p@,
1356     space.hang      = 1.9em,
1357     space.left      = 1.9em,
1358     leader.content  = ,
1359 }
1360 \ztex_declare_instance:nnnn {toc}{ztoc/level 2}{default}
1361 {
1362     format          = \bfseries,
1363     name.width      = 1.5em,
1364     space.before    = 1em\@plus\p@,
1365     space.hang      = 1.5em,
1366     space.left      = 1.5em,
1367     leader.content  = ,
1368 }
1369 \ztex_declare_instance:nnnn {toc}{ztoc/level 3}{default}
1370 {
1371     name.width      = 2.3em,
1372     space.hang      = 2.3em,
1373     space.left      = 3.8em,
1374 }
1375 \ztex_declare_instance:nnnn {toc}{ztoc/level 4}{default}
1376 {
1377     name.width      = 3.2em,
1378     space.hang      = 3.2em,
1379     space.left      = 7em,
1380 }
1381 \ztex_declare_instance:nnnn {toc}{ztoc/level 5}{default}
1382 {
1383     name.width      = 4.1em,
1384     space.hang      = 4.1em,
1385     space.left      = 11.1em,
1386 }
1387 \ztex_declare_instance:nnnn {toc}{ztoc/level 6}{default}
1388 {
1389     name.width      = 5em,
1390     space.hang      = 5em,
```

```
1391     space.left      = 16.2em,
1392   }
1393   \ztex_declare_instance:nnnn {toc}{ztoc/level 7}{default}
1394   {
1395     name.width      = 6em,
1396     space.hang      = 6em,
1397     space.left      = 22.25em,
1398   }
1399
1400
1401   % ==> setup '\l@<class>' command in toc.
1402   % NOTE: for that packages like 'listings' maybe redefine
1403   %       '\l@<class>', we define them at preamble last.
1404   \cs_new_protected:Npn \ztoc_lcmd_setup:nn #1#2
1405   {% #1:class; #2:level(int)
1406     \ztex_label_hook_preamble_last:nn
1407     { ztex/sect/lcmd }
1408     {
1409       \cs_set:cpn {l@#1} ##1##2
1410       {
1411         \ztex_use_instance:nn
1412         { toc } { ztoc/level #2 }
1413         { #2 }{ ##1 }{ ##2 }
1414       }
1415     }
1416   }
1417   \cs_new_protected:Npn \ztoc_special_lcmd_setup:nn #1#2
1418   {% #1:class; #2:level(int)
1419     \ztex_declare_instance_copy:nnn { toc }
1420     { ztoc/#1 }{ ztoc/level #2 }
1421     \ztex_label_hook_preamble_last:nn
1422     { ztex/sect/lcmd@special }
1423     {
1424       \cs_set:cpn {l@#1} ##1##2
1425       {
1426         \ztex_use_instance:nn
1427         { toc }{ ztoc/#1 }
1428         { #2 }{ ##1 }{ ##2 }
1429       }
1430     }
1431   }
1432   \cs_generate_variant:Nn \ztoc_lcmd_setup:nn
1433   { oo, ee }
1434   \cs_generate_variant:Nn \ztoc_special_lcmd_setup:nn
1435   { oo, ee }
1436   \prop_map_inline:Nn \c_zsect_level_prop
1437   {
1438     \ztoc_lcmd_setup:nn
```

1439	{ #1 }{ #2 }	1439
1440	}	1440
1441	\clist_map_inline:Nn \c_ztoc_special_level_clist	1441
1442	{	1442
1443	\exp_args:Nne \ztoc_special_lcmd_setup:nn { #1 }	1443
1444	{	1444
1445	\prop_item:Nn \c_ztoc_special_level_prop	1445
1446	{ #1 }	1446
1447	}	1447
1448	}	1448
1449		1449
1450		1450
1451	% ==> 'toc line add' for 'sec' part	1451
1452	\cs_new_protected:Npn __zsect_title_toc_add:nnn #1#2#3	1452
1453	{	1453
1454	\zsect_unexpand_protect:	1454
1455	\exp_args:Ne \int_compare:nT	1455
1456	{ \c@tocdepth >= \prop_item:Nn \c_zsect_level_prop {#1} }	1456
1457	{	1457
1458	\zsect_add_toc_line:nnnn	1458
1459	{ #1 }	1459
1460	{	1460
1461	{	1461
1462	\int_compare:nNnT	1462
1463	{ \number\c@secnumdepth + 1 }	1463
1464	>	1464
1465	{ \prop_item:Nn \c_zsect_level_prop {#1} }	1465
1466	{ \zsect@tocnum }	1466
1467	}	1467
1468	{	1468
1469	% NOTE: to support command '\texorpdfstring' in '#2', the	1469
1470	% original '\exp_not:n {#2}' has been removed, you	1470
1471	% need to add macro '\protect' manually !	1471
1472	\exp_args:Nne \tl_if_empty:oTF { #2 } % add 'o'-expand to test if ' ✓	1472
	\zsectocname' etc are empty	
1473	{ #3 }{ #2 }	1473
1474	}	1474
1475	}	1475
1476	{ \thepage }	1476
1477	{ \ztexhyperTF {#1.\zsect@tocHnum}{ } }	1477
1478	}	1478
1479	\zsect_restore_protect:	1479
1480	}	1480
1481		1481
1482		1482
1483	% ==> user interface for 'toc' part:	1483
1484	%% programming layer	1484
1485	% >> toc group parser	1485

```
1486 \cs_new_protected:Nn \ztoc_group_parser:NNNnnnnnn 1486
1487 {% #1:keyval seq; #2:cmd; #3:tl(saved in); 1487
1488 % #4:before; #5:begin: #6:end; #7:after; 1488
1489 % #8:step code. 1489
1490 \seq_map_inline:Nn #1 1490
1491 { 1491
1492 \edef\ztoc@gparser@curclass 1492
1493 { 1493
1494 \prop_item:nn {##1}{ class } 1494
1495 } 1495
1496 \exp_args:No \__ztoc_step_toc_group_int:n 1496
1497 { \ztoc@gparser@curclass } 1497
1498 \ztoc_get_class_level:No \ztoc@gparser@newclass@level 1498
1499 { \ztoc@gparser@curclass } 1499
1500 \cs_set_eq:NN \gparserclass\ztoc@gparser@curclass 1500
1501 \cs_set:Npe \gparseranchor 1501
1502 { 1502
1503 \prop_item:nn {##1}{ class } 1503
1504 . 1504
1505 \prop_item:nn {##1}{ name } 1505
1506 } 1506
1507 \cs_set:Npn \gparserindex 1507
1508 { 1508
1509 \__ztoc_use_toc_group_int:e 1509
1510 { \ztoc@gparser@curclass } 1510
1511 } 1511
1512 \bool_while_do:nn 1512
1513 { 1513
1514 ( ! \seq_if_empty_p:N \g__ztoc_gparser_curstack_seq ) && 1514
1515 ( 1515
1516 \int_compare_p:n 1516
1517 { 1517
1518 ( 1518
1519 \ztoc_get_class_level_expandable:e 1519
1520 { 1520
1521 \seq_item:Nn \g__ztoc_gparser_curstack_seq {1} 1521
1522 } + 0 1522
1523 ) 1523
1524 >= \ztoc@gparser@newclass@level 1524
1525 } 1525
1526 ) 1526
1527 }{ 1527
1528 \seq_gpop:NN \g__ztoc_gparser_curstack_seq \l__ztoc_gparser_prev_tl 1528
1529 \tl_gput_right:Ne #3 {#6} % end 1529
1530 \tl_gput_right:Ne #3 {#7} % after 1530
1531 } 1531
1532 \tl_gput_right:Ne #3 {#4} % before 1532
1533 \edef\ztoc@gpaser@cmdargs 1533
```

1534	{	1534
1535	{	1535
1536	\ztoc@gparser@curclass	1536
1537	}	1537
1538	{	1538
1539	\prop_item:nn { ##1 }{ name }	1539
1540	}	1540
1541	{	1541
1542	\prop_item:nn { ##1 }{ title }	1542
1543	}	1543
1544	{	1544
1545	\prop_item:nn { ##1 }{ page }	1545
1546	}	1546
1547	}	1547
1548	\edef\ztoc@gpaser@cmdargs	1548
1549	{	1549
1550	\exp_args:No \zcmd_map_expnot_tl:n	1550
1551	{ \ztoc@gpaser@cmdargs }	1551
1552	}	1552
1553	\exp_after:wN \def \exp_after:wN	1553
1554	\ztoc@gparser@line \exp_after:wN	1554
1555	{	1555
1556	\exp_after:wN #2	1556
1557	\ztoc@gpaser@cmdargs	1557
1558	}	1558
1559	\exp_args:NNe \tl_gput_right:Nn #3	1559
1560	{ \ztoc@gparser@line }	1560
1561	% insert step code	1561
1562	#8	1562
1563	\tl_gput_right:Ne #3 {#5} % begin	1563
1564	\seq_gpush:Ne \g__ztoc_gparser_curstack_seq	1564
1565	{ \ztoc@gparser@curclass }	1565
1566	}	1566
1567	% clear stack data:	1567
1568	\seq_map_inline:Nn \g__ztoc_gparser_curstack_seq	1568
1569	{	1569
1570	\tl_gput_right:Ne #3 {#6} % end	1570
1571	\tl_gput_right:Ne #3 {#7} % after	1571
1572	}	1572
1573	}	1573
1574	\cs_generate_variant:Nn \ztoc_group_parser:NNNnnnnn	1574
1575	{ ccc, NNNeeeeee, ccceeeee }	1575
1576		1576
1577	% >> group hooks add	1577
1578	\cs_new_protected:Npn \ztoc_group_hook_create:Nnnn #1#2#3#4	1578
1579	{	1579
1580	\exp_args:No __ztoc_step_toc_group_int:n	1580
1581	{ \ztoc@current@class }	1581

1582	\ztoc_get_class_level:No \ztoc@newclass@level	1582
1583	{ \ztoc@current@class }	1583
1584	\bool_while_do:nn	1584
1585	{	1585
1586	(! \seq_if_empty_p:N \g__ztoc_gparser_curstack_seq) &&	1586
1587	(	1587
1588	\int_compare_p:n	1588
1589	{	1589
1590	(	1590
1591	\ztoc_get_class_level_expandable:e	1591
1592	{	1592
1593	\clist_item:en	1593
1594	{ \seq_item:Nn \g__ztoc_gparser_curstack_seq {1} }	1594
1595	{ 1 }	1595
1596	} + 0	1596
1597	)	1597
1598	>= \ztoc@newclass@level	1598
1599	}	1599
1600	)	1600
1601	}{	1601
1602	\seq_gpop:NN \g__ztoc_gparser_curstack_seq \l__ztoc_gparser_prev_tl	1602
1603	\ztoc_group_hook_add:n {\l__ztoc_gparser_prev_tl,end}	1603
1604	\ztoc_group_hook_add:n {\l__ztoc_gparser_prev_tl,after}	1604
1605	}	1605
1606	\ztoc_group_hook_add:n	1606
1607	{	1607
1608	\ztoc@current@class,	1608
1609	__ztoc_use_toc_group_int:e {\ztoc@current@class},	1609
1610	before	1610
1611	}	1611
1612	#1 {#2}{#3}{#4}	1612
1613	\ztoc_group_hook_add:n	1613
1614	{	1614
1615	\ztoc@current@class,	1615
1616	__ztoc_use_toc_group_int:e {\ztoc@current@class},	1616
1617	begin	1617
1618	}	1618
1619	\seq_gpush:Ne \g__ztoc_gparser_curstack_seq	1619
1620	{	1620
1621	\ztoc@current@class	1621
1622	,\int_eval:n { __ztoc_use_toc_group_int:e {\ztoc@current@class} }	1622
1623	}	1623
1624	}	1624
1625	\cs_generate_variant:Nn \ztoc_group_hook_create:Nnnn { c }	1625
1626	% >> control table iow	1626
1627	\cs_new_protected:Npn \ztoc_stop_table:n #1	1627
1628	{	1628
1629	% stop iow of table	1629

```
1630 \bool_gset_false:c { g_ztoc_#1_iow_bool } 1630
1631 % dump keyval seq to file (after the seq2file in '\tocenable') 1631
1632 \seq_gset_eq:cc { g_ztoc_stopped_keyval#1_seq } 1632
1633 { g_ztoc_keyval#1_seq } 1633
1634 \bool_if:NT \g_zsect_dump_ptable_bool 1634
1635 { 1635
1636 \ztex_hook_doc_end:n 1636
1637 { 1637
1638 \ztool_write_seq_to_file:nce { \c_true_bool } 1638
1639 { g_ztoc_stopped_keyval#1_seq } 1639
1640 { \c_sys_jobname_str.p#1 } 1640
1641 } 1641
1642 } 1642
1643 } 1643
1644 % >> table typeset (include toc group hooks) 1644
1645 \cs_new_protected:Npn \ztoc_save_main_table:n #1 1645
1646 { 1646
1647 % NOTE: read '\jobname.toc' in the second time, you 1647
1648 % will get nothing, i.e. variable '\g_ztoc_toc_seq' 1648
1649 % will be set to empty. 1649
1650 \str_if_eq:eeF { #1 }{ \c_sys_jobname_str } 1650
1651 { 1651
1652 \ztoc_maintable_seq_save:n { toc } 1652
1653 \ztoc_generate_table_seq:nm { #1 }{ toc } 1653
1654 } 1654
1655 } 1655
1656 \cs_new_protected:Npn \ztoc_restore_main_table:n #1 1656
1657 { 1657
1658 \str_if_eq:eeF { #1 }{ \c_sys_jobname_str } 1658
1659 { 1659
1660 \ztoc_maintable_seq_restore:n { toc } 1660
1661 } 1661
1662 } 1662
1663 % table typeset before/after 1663
1664 \cs_new_protected:Npn \ztoc_table_typeset_before: 1664
1665 { 1665
1666 \__ztoc_reset_toc_group_int: 1666
1667 } 1667
1668 \cs_new_protected:Npn \ztoc_table_typeset_after: 1668
1669 { 1669
1670 \seq_map_inline:Nn \g_ztoc_gparser_curstack_seq 1670
1671 { 1671
1672 \seq_gpop:NN 1672
1673 \g_ztoc_gparser_curstack_seq 1673
1674 \l_ztoc_gparser_prev_tl 1674
1675 \ztoc_group_hook_add:n 1675
1676 { \l_ztoc_gparser_prev_tl,end } 1676
1677 \ztoc_group_hook_add:n 1677
```

```
1678      {\l__ztoc_gparser_prev_tl,after }
1679    }
1680  }
1681 % NOTE: anytime you need to typeset a table, use the
1682 %       following command, or the hooks may be wrong.
1683 \cs_new_protected:Npn \ztoc_table_typeset:Nn #1#2
1684   {% #1:table seq; #2:separator.
1685     \ztoc_table_typeset_before:
1686       \seq_use:Nn #1 { #2 }
1687     \ztoc_table_typeset_after:
1688   }
1689 \cs_generate_variant:Nn \ztoc_table_typeset:Nn
1690   { Ne, cn, ce }
1691
1692 % >> toc format setup interface
1693 \cs_new_protected:Npn \ztoc_normal_format_set:Nn #1#2
1694   {
1695     \token_if_cs:NF #1
1696     {
1697       \ztex_msg_set:nn {ztoc@format@notcs}
1698       { current~toc~class~argument::~'#1'~is~not~a~control~sequence. }
1699       \ztex_msg_error:n {ztoc@format@notcs}
1700     }
1701     \prop_if_in:NnF \c_zsect_level_prop { \cs_to_str:N #1 }
1702     {
1703       \ztex_msg_set:nn {ztoc@instancenormal@nonexist}
1704       {
1705         Instance~'ztoc/level\prop_item:Nn \c_zsect_level_prop {\cs_to_str:N #1}'~
1706         for~\tl_to_str:n {#1} of~type~'ztextoc'~do~NOT~exist.
1707       }
1708       \ztex_msg_error:n {ztoc@instancenormal@nonexist}
1709     }
1710     \exp_args:Nne \ztex_edit_instance:nnn
1711     { toc }
1712     { ztoc/level
1713       \prop_item:Nn \c_zsect_level_prop
1714       { \cs_to_str:N #1 }
1715     }{ #2 }
1716   }
1717 \cs_new_protected:Npn \ztoc_special_format_set:nn #1#2
1718   {
1719     \prop_if_in:NnF \c_ztoc_special_level_prop { #1 }
1720     {
1721       \ztex_msg_set:nn {ztoc@instancespecial@nonexist}
1722       {
1723         Instance~'ztoc/#1'~for~'#1'~of
1724         ~type~'ztextoc'~do~NOT~exist.
1725       }
1726     }
```



```
1726 \ztex_msg_error:n {ztoc@instancespecial@nonexist} 1726
1727 } 1727
1728 \exp_args:Nne \ztex_edit_instance:nnn 1728
1729 { toc }{ ztoc/#1 }{ #2 } 1729
1730 } 1730
1731 \cs_new_protected:Npn \ztoc_all_format_set:nn #1#2 1731
1732 { 1732
1733 \clist_map_inline:nn { #1 } 1733
1734 { 1734
1735 \prop_if_in:NnTF \c_ztoc_special_level_prop { ##1 } 1735
1736 { 1736
1737 \ztoc_special_format_set:nn { ##1 }{ #2 } 1737
1738 \prg_map_break:Nn \__ztoc_format_user_break: {} 1738
1739 }{ 1739
1740 \ztoc_normal_format_set:Nn { ##1 }{ #2 } 1740
1741 \prg_map_break:Nn \__ztoc_format_user_break: {} 1741
1742 } 1742
1743 \prg_break_point:Nn \__ztoc_format_user_break: {} 1743
1744 } 1744
1745 } 1745
1746 \cs_new:Npn \__ztoc_format_user_break: 1746
1747 { \prg_map_break:Nn \__ztoc_format_user_break: {} } 1747
1748 1748
1749 % >> table data setup interface 1749
1750 \cs_new_protected:Npn \ztoc_generate_table_seq:nn #1#2 1750
1751 {% #1:file(no ext); #2:type(toc, lof, etc) 1751
1752 \bool_lazy_and:nnT 1752
1753 { ! \str_if_eq_p:ee { #1 }{ \c_sys_jobname_str } } 1753
1754 { ! \file_if_exist_p:n { #1.#2 } } 1754
1755 { 1755
1756 \ztex_msg_set:nn { tablefile@nonexist } 1756
1757 { 1757
1758 custom~table~file~'#1.#2'~does~NOT~exist,~ 1758
1759 check~the~file~name~again. 1759
1760 } 1760
1761 \ztex_msg_error:n { tablefile@nonexist } 1761
1762 } 1762
1763 % generate global toc seq 1763
1764 \ztoc_gread_file_as_seq:nnc { \c_false_bool } 1764
1765 { #1.#2 } 1765
1766 { g_ztoc_#2_seq } 1766
1767 % generate keyval toc seq 1767
1768 \ztoc_generate_keyvaltable_seq:nc { #1.#2 } 1768
1769 { g_ztoc_keyval#2_seq } 1769
1770 } 1770
1771 \cs_new_protected:Npn \ztoc_enable_table:nn #1#2 1771
1772 {% #1:file; #2:toc, lom, etc. 1772
1773 \clist_map_inline:nn { #2 } 1773
```

```
1774 {
1775     % generate (keyval) toc seq
1776     \ztoc_generate_table_seq:nn { #1 }{ ##1 }
1777     % dump '*.ptoc, *.plot' for debugging
1778     \bool_if:NT \g__zsect_dump_ptable_bool
1779     {
1780         \ztex_hook_doc_end:n
1781         {
1782             \ztool_write_seq_to_file:nce { \c_true_bool }
1783             { g_ztoc_keyval##1_seq }
1784             { \c_sys_jobname_str.p##1 }
1785         }
1786     }
1787     % stream for writing table file
1788     \str_if_eq:eeT { #1 }{ \c_sys_jobname_str }
1789     {
1790         \bool_gset_true:c { g_ztoc_##1_iow_bool }
1791         \exp_args:Nne \iow_open:cn { g_ztoc_##1_iow }
1792         { \c_sys_jobname_str.##1 }
1793     }
1794 }
1795 }
1796 \cs_new:Npn \ztoc_enable_table_swap:nn #1#2
1797 { \ztoc_enable_table:nn { #2 }{ #1 } }
1798
1799 % >> save and restore table data seq
1800 \seq_new:N \g__ztoc_maintable_tmp_seq
1801 \seq_gclear:N \g__ztoc_maintable_tmp_seq
1802 \seq_new:N \g__ztoc_maintable_keyvaltmp_seq
1803 \seq_gclear:N \g__ztoc_maintable_keyvaltmp_seq
1804 % NOTE: the 'save' function should before '\ztoc_generate_table_seq'
1805 \cs_new_protected:Npn \ztoc_maintable_seq_save:n #1
1806 {
1807     \seq_gset_eq:Nc \g__ztoc_maintable_tmp_seq
1808     { g_ztoc_#1_seq }
1809     \seq_gset_eq:Nc \g__ztoc_maintable_keyvaltmp_seq
1810     { g_ztoc_keyval#1_seq }
1811 }
1812 \cs_new_protected:Npn \ztoc_maintable_seq_restore:n #1
1813 {
1814     \seq_gset_eq:cN { g_ztoc_#1_seq }
1815     \g__ztoc_maintable_tmp_seq
1816     \seq_gset_eq:cN { g_ztoc_keyval#1_seq }
1817     \g__ztoc_maintable_keyvaltmp_seq
1818     \seq_gclear:N \g__ztoc_maintable_tmp_seq
1819     \seq_gclear:N \g__ztoc_maintable_keyvaltmp_seq
1820 }
1821
```

```
1822 % >> generate keyval toc seq variables, e.g.,
1823 %     '\g_ztoc_keyval<toc|lof|...>_seq' from a
1824 %     user-specified toc, lof, ... file.
1825 \seq_new:N \l__ztoc_keyvaltable_tmp_seq
1826 % keyval seq from raw seq
1827 \cs_new_protected:Npn \ztoc_generate_keyvaltable_seq:NN #1#2
1828 {
1829     \seq_map_inline:Nn #1
1830     {
1831         \__ztoc_keyval_table_parser:Nw
1832         #2 ##1 \scan_stop:
1833     }
1834 }
1835 \cs_generate_variant:Nn \ztoc_generate_keyvaltable_seq:NN
1836 { cc }
1837 % keyval seq from table file
1838 \cs_new_protected:Npn \ztoc_generate_keyvaltable_seq:nN #1#2
1839 {% #1:file with ext; #2:seq
1840     \seq_gclear:N #2
1841     \ztool_gread_file_as_seq_keep_spaces:nnN
1842     { \c_false_bool }{ #1 }
1843     \l__ztoc_keyvaltable_tmp_seq
1844     \seq_map_inline:Nn \l__ztoc_keyvaltable_tmp_seq
1845     {
1846         \__ztoc_keyval_table_parser:Nw
1847         #2 ##1 \scan_stop:
1848     }
1849 }
1850 \cs_generate_variant:Nn \ztoc_generate_keyvaltable_seq:nN
1851 { nc, oN, eN }
1852 \cs_new:Npn \__ztoc_keyval_table_parser:Nw #1\contentsline #2#3#4#5\scan_stop:
1853 {
1854     \zsect_unexpand_protect:
1855     \edef\ztoc@keyvalparser@nametitle
1856     {
1857         \zcmd_map_expnot_tl:n
1858         { #3 }
1859     }
1860     \seq_gput_right:Ne #1
1861     {
1862         class = { \exp_not:n {#2} },
1863         name  = { \exp_after:wN \use_i:nn \ztoc@keyvalparser@nametitle },
1864         title = { \exp_after:wN \use_ii:nn \ztoc@keyvalparser@nametitle },
1865         page  = { \exp_not:n {#4} },
1866         anchor = { #5 },
1867         raw   = { \exp_not:N \contentsline
1868                 { \exp_not:n {#2} }
1869                 { \exp_not:n {#3} }
```

```
1870         {\exp_not:n {#4} }
1871         { \exp_not:n {#5} }
1872     }
1873 }
1874 \cs_undefine:N \ztoc@keyvalparser@nametitle
1875 \zsect_restore_protect:
1876 }
1877
1878 % >> extract toc in local region (relies on keyval toc seq)
1879 % 'byclass' filter variables setup:
1880 \int_new:N \l__ztoc_tmpa_int
1881 \prop_new:N \l__ztoc_tmpa_prop
1882 \prop_new:N \l__ztoc_tmpb_prop
1883 \bool_new:N \l__ztoc_filter_item_found_bool
1884 \exp_args:Ne \clist_map_inline:nn
1885 { \c_zsect_level_clist, \c_ztoc_special_level_clist }
1886 {
1887     \int_new:c { g__toc_filter_#1_int }
1888 }
1889 \cs_new:Npn \__ztoc_reset_filter_int:
1890 {
1891     \exp_args:Ne \clist_map_inline:nn
1892     { \c_zsect_level_clist, \c_ztoc_special_level_clist }
1893     {
1894         \int_gset:cn { g__toc_filter_##1_int }
1895         { 0 }
1896     }
1897 }
1898 \cs_new:Npn \__ztoc_step_filter_int:n #1
1899 {
1900     \int_gincr:c { g__toc_filter_#1_int }
1901 }
1902 \cs_new:Npn \__ztoc_use_filter_int:n #1
1903 {
1904     \int_use:c { g__toc_filter_#1_int }
1905 }
1906 \cs_generate_variant:Nn \ztoc_table_filter_byclass:nnnN
1907 { ne, en, ee, nccc, eecc }
1908
1909 % only count sub-tocline by key(similar to filter but expandable)
1910 \cs_new:Npn \ztoc_tocline_card:nnN #1#2#3
1911 {
1912     \int_compare:nNnTF
1913     {
1914         \ztoc_tocline_index_byclasstitle:nnnN
1915         { #1 } { #2 } { \c_false_bool } #3
1916     } < { 0 } { -1 }
1917     { \int_eval:n {
```

1918	\exp_args:Ne \tl_map_tokens:nn	1918
1919	{	1919
1920	\zcmd_ints_to_tl:ee	1920
1921	{	1921
1922	\ztoc_tocline_index_byclasstitle:nnnN	1922
1923	{ #1 }{ #2 }{ \c_false_bool }	1923
1924	#3	1924
1925	}{ \seq_count:N #3 }	1925
1926	}{	1926
1927	__ztoc_tocline_card_byclasstitle:nnnn	1927
1928	{ #1 }{ #2 }{ #3 }	1928
1929	} + 0	1929
1930	}	1930
1931	}	1931
1932	}	1932
1933	\cs_new:Nn __ztoc_tocline_card_byclasstitle:nnnn	1933
1934	{	1934
1935	\int_compare:nNnTF	1935
1936	{	1936
1937	\ztoc_get_class_level_expandable:e	1937
1938	{	1938
1939	\prop_item:en	1939
1940	{	1940
1941	\seq_item:Ne #3	1941
1942	{	1942
1943	\ztoc_tocline_index_byclasstitle:nnnN	1943
1944	{ #1 }{ #2 }{ \c_false_bool } #3	1944
1945	}	1945
1946	}{ class }	1946
1947	} + 0	1947
1948	}	1948
1949	<	1949
1950	{	1950
1951	\ztoc_get_class_level_expandable:e	1951
1952	{	1952
1953	\prop_item:en	1953
1954	{ \seq_item:Nn #3 {#4+1} }	1954
1955	{ class }	1955
1956	} + 0	1956
1957	}{ +1 }{ \tl_map_break: }	1957
1958	}	1958
1959	\cs_generate_variant:Nn \ztoc_tocline_card:nnN	1959
1960	{ oo, ee, nnc, eec }	1960
1961		1961
1962	% high/low level int of current table	1962
1963	\cs_new:Npn \ztoc_tocline_hilevel:N #1	1963
1964	{	1964
1965	\exp_args:Ne \zcmd_floats_min:n	1965

1966	{	1966
1967	\ztoc_tocline_level_data:N #1	1967
1968	}	1968
1969	}	1969
1970	\cs_new:Npn \ztoc_tocline_lolevel:N #1	1970
1971	{	1971
1972	\exp_args:Ne \zcmd_floats_max:n	1972
1973	{	1973
1974	\ztoc_tocline_level_data:N #1	1974
1975	}	1975
1976	}	1976
1977	\cs_new:Npn \ztoc_tocline_level_data:N #1	1977
1978	{	1978
1979	\seq_map_function:NN #1	1979
1980	__ztoc_tocline_hilo_level_aux:n	1980
1981	}	1981
1982	\cs_generate_variant:Nn \ztoc_tocline_level_data:N	1982
1983	{ c }	1983
1984	\cs_generate_variant:Nn \ztoc_tocline_hilevel:N	1984
1985	{ c }	1985
1986	\cs_generate_variant:Nn \ztoc_tocline_lolevel:N	1986
1987	{ c }	1987
1988	\cs_new:Nn __ztoc_tocline_hilo_level_aux:n	1988
1989	{	1989
1990	{\ztoc_get_class_level_expandable:e	1990
1991	{	1991
1992	\prop_item:nn { #1 }{ class }	1992
1993	}}	1993
1994	}	1994
1995		1995
1996	% extract custom key by class	1996
1997	\cs_new_protected:Npn \ztoc_table_filter_key_byclass:nnnNN #1#2#3#4#5	1997
1998	{% #1:class; #2:index; #3:type; #4:raw seq; #5:res seq	1998
1999	\seq_gclear:N #5	1999
2000	\bool_set_false:N \l__ztoc_filter_item_found_bool	2000
2001	\seq_map_inline:Nn #4	2001
2002	{	2002
2003	\prop_set_from_keyval:Nn \l__ztoc_tmpa_prop { ##1 }	2003
2004	\exp_args:Ne __ztoc_step_filter_int:n	2004
2005	{ \prop_item:Nn \l__ztoc_tmpa_prop {class} }	2005
2006	\exp_args:Ne \int_compare:nNnT	2006
2007	{ __ztoc_use_filter_int:n {#1} } = {#2+1}	2007
2008	{ \seq_map_break: }	2008
2009	\bool_if:NT \l__ztoc_filter_item_found_bool	2009
2010	{	2010
2011	\exp_args:Ne \int_compare:nNnT	2011
2012	{ \prop_item:Nn \c_zsect_level_prop {#1} }	2012
2013	>	2013

```
2014         { \exp_args:NNe \prop_item:Nn \c_zsect_level_prop
2015             { \prop_item:Nn \l__ztoc_tmpa_prop {class} }
2016         }{ \seq_map_break: }
2017     }
2018     \exp_args:Ne \int_compare:nNnT
2019     { \__ztoc_use_filter_int:n {#1} } = {#2}
2020     {
2021         \bool_set_true:N \l__ztoc_filter_item_found_bool
2022         \seq_gput_right:Ne #5
2023         { \prop_item:Nn \l__ztoc_tmpa_prop {#3} }
2024     }
2025 }
2026 \__ztoc_reset_filter_int:
2027 }
2028 \cs_generate_variant:Nn \ztoc_table_filter_key_byclass:nnnNN
2029 { ooo, eee, nnncc, eeec }
2030
2031 % extract 'raw' by class
2032 \cs_new_protected:Npn \ztoc_table_filter_byclass:nnNN #1#2#3#4
2033 { % #1:class; #2:index; #3:raw seq; #4:res seq
2034     \ztoc_table_filter_key_byclass:nnnNN
2035     { #1 }{ #2 }{ raw }
2036     #3 #4
2037 }
2038
2039 % extract by 'name' & 'title'(i.e., {1.2.4}{AAA})
2040 % NOTE: name and title need to be expanded before grabbed.
2041 \cs_new_protected:Npn \ztoc_table_filter_byname_title:nnNN #1#2#3#4
2042 { % #1:name; #2:title; #3:ori seq; #4:prop
2043     \prop_gclear:N #4
2044     \seq_map_inline:Nn #3
2045     {
2046         \prop_set_from_keyval:Nn \l__ztoc_tmpb_prop { ##1 }
2047         \int_incr:N \l__ztoc_tmpa_int
2048         \bool_lazy_all:nT
2049         {
2050             { \exp_args:Ne \str_if_eq_p:nn
2051                 { \prop_item:Nn \l__ztoc_tmpb_prop {name} }
2052                 { #1 }
2053             }
2054             { \exp_args:Ne \str_if_eq_p:nn
2055                 { \prop_item:Nn \l__ztoc_tmpb_prop {title} }
2056                 { #2 }
2057             }
2058         }{
2059             \prop_gset_eq:NN #4 \l__ztoc_tmpb_prop
2060         }
2061     }
```

```
2062 }
2063 \cs_generate_variant:Nn \ztoc_table_filter_byname title:nnNN
2064 { ee, mncc, eecc }
2065
2066 % get index of current contents line (expandable)
2067 % NOTE: if not found, an negative value will be returned
2068 \cs_new:Npn \__special_seq_map_break:
2069 { \prg_map_break:Nn \__special_seq_map_break: { } }
2070
2071 % get index by class & name
2072 \cs_new:Npn \ztoc_tocline_index_byclasstitle:nnnN #1#2#3#4
2073 {% #1:class; #2:title; #3:bool; #4:ori seq
2074   \int_eval:n {
2075     (
2076       \seq_map_tokens:Nn #4
2077       {
2078         \__ztoc_tocline_index_byclasstitle_aux:nnnn
2079         { #1 } { #2 } { #3 }
2080       } + 0
2081     )
2082     \prg_break_point:Nn \__special_seq_map_break: {}
2083   }
2084 }
2085 \cs_new:Nn \__ztoc_tocline_index_byclasstitle_aux:nnnn
2086 {
2087   \bool_lazy_all:nTF
2088   {
2089     { \exp_args:Ne \str_if_eq_p:nn
2090       { \prop_item:nn { #4 } {class} }
2091       { #1 }
2092     }
2093     { \exp_args:Ne \str_if_eq_p:nn
2094       { \prop_item:nn { #4 } {title} }
2095       { #2 }
2096     }
2097   } {
2098     -1 \prg_map_break:Nn \__special_seq_map_break:
2099     { }*(-1) }
2100   } {
2101     \bool_if:nTF { #3 }
2102     {
2103       \exp_args:Ne \str_if_eq:nnTF
2104       { \prop_item:nn { #4 } {class} }
2105       { #1 } { -1 } { +0 }
2106     } { -1 }
2107   }
2108 }
2109 \cs_generate_variant:Nn \ztoc_tocline_index_byclasstitle:nnnN
```


2110	{ nnc, eec, eeN }	2110
2111		2111
2112	% get index by name & title	2112
2113	\cs_new:Npn \ztoc_tocline_index_byname:nnN #1#2#3	2113
2114	{% #1:name; #2:title; #3:ori seq	2114
2115	\int_eval:n {	2115
2116	(	2116
2117	\seq_map_tokens:Nn #3	2117
2118	{	2118
2119	__ztoc_tocline_index_byname:nnn	2119
2120	{ #1 }{ #2 }	2120
2121	} + 0	2121
2122	)	2122
2123	\prg_break_point:Nn __special_seq_map_break: {}	2123
2124	}	2124
2125	}	2125
2126	\cs_new:Npn __ztoc_tocline_index_byname:nnn #1#2#3	2126
2127	{	2127
2128	\bool_lazy_all:nTF	2128
2129	{	2129
2130	{ \exp_args:Ne \str_if_eq_p:nn	2130
2131	{ \prop_item:nn { #3 }{name} }	2131
2132	{ #1 }	2132
2133	}	2133
2134	{ \exp_args:Ne \str_if_eq_p:nn	2134
2135	{ \prop_item:nn { #3 }{title} }	2135
2136	{ #2 }	2136
2137	}	2137
2138	}{	2138
2139	-1 \prg_map_break:Nn __special_seq_map_break:	2139
2140	{ }*(-1) }	2140
2141	}{ -1 }	2141
2142	}	2142
2143	\cs_generate_variant:Nn \ztoc_tocline_index_byname:nnN	2143
2144	{ nnc, eec, eeN }	2144
2145		2145
2146	% get class level by name & title(expandable)	2146
2147	\cs_new:Npn \ztoc_tocline_level_byname:Nnnn #1#2#3#4	2147
2148	{% #1:ori seq; #2:name; #3:title; #4:offset(int)	2148
2149	\int_compare:nNnT	2149
2150	{ \ztoc_tocline_index_byname:nnN {#2}{#3}#1 }	2150
2151	<	2151
2152	{ 0 }{ \relax } % TODO: handle non-exist case	2152
2153	\ztoc_get_class_level_expandable:e	2153
2154	{	2154
2155	\prop_item:en	2155
2156	{	2156
2157	\seq_item:Ne #1	2157

```

2158         { \ztoc_tocline_index_bynametitle:nnN {#2}{#3}#1 + (#4) }
2159     }
2160     { class }
2161 } + 0
2162 }
2163
2164 % >> tocline index & position
2165 % get current toc index(expandable)
2166 \newcommand{\ztocslot}
2167 {
2168     \__ztoc_use_toc_group_int:e
2169     { \ztoc@current@class }
2170 }
2171 \NewExpandableDocumentCommand{\ztocindex}{s0{toc}mm}
2172 {
2173     \IfBooleanTF{#1}
2174     {
2175         \ztoc_tocline_index_byclasstitle:nnnc
2176         { #3 }{ #4 }{ \c_false_bool }
2177         { g_ztoc_keyval#2_seq }
2178     }
2179     {
2180         \ztoc_tocline_index_byclasstitle:nnnc
2181         { #3 }{ #4 }{ \c_true_bool }
2182         { g_ztoc_keyval#2_seq }
2183     }
2184 }
2185 \NewExpandableDocumentCommand{\ztoccard}{0{toc}mm}
2186 {
2187     \ztoc_tocline_card:nnc
2188     { #2 }{ #3 }
2189     { g_ztoc_keyval#1_seq }
2190 }
2191 \NewExpandableDocumentCommand{\ztotchilevel}{m}
2192 {
2193     \ztoc_tocline_hilevel:c
2194     { g_ztoc_keyval#1_seq }
2195 }
2196 \NewExpandableDocumentCommand{\ztoclolevel}{m}
2197 {
2198     \ztoc_tocline_lolevel:c
2199     { g_ztoc_keyval#1_seq }
2200 }
2201
2202 % tocline level compare (expandable)
2203 \prg_new_conditional:Npnn \ztoc_tocline_level_compare:Nnnnnn #1#2#3#4#5#6
2204 { p, T, F, TF }
2205 {% #1:ori seq; #2:name; #3:title;

```

2206	% #4:index_1 offset; #5:index_2 offset;	2206
2207	% #6:operator	2207
2208	\int_compare:nNnT	2208
2209	{ \ztoc_tocline_index_byname:nnN {#2}{#3}#1 }	2209
2210	<	2210
2211	{ 0 }	2211
2212	{ \prg_return_false: }	2212
2213	\int_compare:nTF	2213
2214	{	2214
2215	(	2215
2216	\ztoc_tocline_level_byname:Nnnn	2216
2217	#1 {#2}{#3}{#4}	2217
2218	)	2218
2219	#6	2219
2220	(	2220
2221	\ztoc_tocline_level_byname:Nnnn	2221
2222	#1 {#2}{#3}{#5}	2222
2223	)	2223
2224	}	2224
2225	{ \prg_return_true: }	2225
2226	{ \prg_return_false: }	2226
2227	}	2227
2228	\prg_generate_conditional_variant:Nnn \ztoc_tocline_level_compare:Nnnnnn	2228
2229	{ c, Nee, cee }{ p, T, F, TF }	2229
2230	\prg_new_conditional:Npnn \ztoc_tocline_level_curnext_compare:Nnnn #1#2#3#4	2230
2231	{ p, T, F, TF }	2231
2232	{% #1:ori seq; #2:name; #3:title; #4:operator	2232
2233	\ztoc_tocline_level_compare:NnnnnnTF #1 {#2}{#3}{0}{1}{#4}	2233
2234	{ \prg_return_true: }	2234
2235	{ \prg_return_false: }	2235
2236	}	2236
2237	% check if first/middle/last tocline(expandable)	2237
2238	\prg_new_conditional:Npnn \ztoc_if_first_tocline: {p, T, F, TF}	2238
2239	{	2239
2240	\int_compare:nNnTF	2240
2241	{ __ztoc_use_toc_group_int:e {\ztoc@current@class} }	2241
2242	= { 1 }	2242
2243	{ \prg_return_true: }	2243
2244	{ \prg_return_false: }	2244
2245	}	2245
2246	\prg_new_conditional:Npnn \ztoc_if_middle_tocline:Nnn #1#2#3	2246
2247	{ p, T, F, TF }	2247
2248	{% #1:ori seq; #2:name; #3:title	2248
2249	\ztoc_tocline_level_curnext_compare:NnnnTF #1	2249
2250	{ #2 }{ #3 }{ = }	2250
2251	{ \prg_return_true: }	2251
2252	{ \prg_return_false: }	2252
2253	}	2253

```

2254 \prg_generate_conditional_variant:Nnn \ztoc_if_middle_tocline:Nnn
2255 { c, Nee, cee }{ p, T, F, TF }
2256 \prg_new_conditional:Npnn \ztoc_if_last_tocline:Nnn #1#2#3
2257 { p, T, F, TF }
2258 {% #1:ori seq; #2:name; #3:title
2259 \ztoc_tocline_level_curnext_compare:NnnnTF #1
2260 { #2 }{ #3 }{ > }
2261 { \prg_return_true: }
2262 { \prg_return_false: }
2263 }
2264 \prg_generate_conditional_variant:Nnn \ztoc_if_last_tocline:Nnn
2265 { c, Nee, cee }{ p, T, F, TF }
2266
2267 % user interface of tocline position check:
2268 \newcommand{\ztociffirst}[2]
2269 {
2270 \ztoc_if_first_tocline:TF
2271 { #1 }{ #2 }
2272 }
2273 \NewExpandableDocumentCommand{\ztocifmiddle}{0{toc}mm}
2274 {
2275 \ztoc_if_middle_tocline:ceeTF { g_ztoc_keyval#1_seq }
2276 { \ztocname }{ \ztoctitle }
2277 { #2 }{ #3 }
2278 }
2279 \NewExpandableDocumentCommand{\ztociflast}{0{toc}mm}
2280 {
2281 \ztoc_if_last_tocline:ceeTF { g_ztoc_keyval#1_seq }
2282 { \ztocname }{ \ztoctitle }
2283 { #2 }{ #3 }
2284 }
2285
2286
2287 %% document layer
2288 % >> format toc data
2289 \tl_gclear:N \ztocgroupparserOutput
2290 \NewDocumentCommand{\ztocgroupparser}{0{toc}0{ }mmmmm}
2291 {
2292 \exp_args:Nc \ztoc_group_parser:NNNnnnnn
2293 { g_ztoc_keyval#1_seq }
2294 #3 \ztocgroupparserOutput
2295 {#4}{#5}{#6}{#7}{#2}
2296 }
2297
2298 % >> template defaults & one-time toc
2299 \NewDocumentCommand{\ztocTemplateDefaultsEdit}{m}
2300 {
2301 \ztex_edit_template_defaults:nnn

```

230223032304230523062307230823092310231123122313231423152316231723182319232023212322232323242325232623272328232923302331233223332334233523362337233823392340234123422343234423452346234723482349

{ toc }{ default }{ #1 }

}

\cs_new_protected:Npn \ztoc_once_tocline:nnnn #1#2#3#4

{

\ztex_use_template:nnn {toc}

{ default }{ #1 }

{ #2 }{ #3 }{ #4 }

}

\cs_generate_variant:Nn \ztoc_once_tocline:nnnn

{ nooo, neee, oooo, eeee }

\NewDocumentCommand{\ztoclineOnce}{\mmmm}

{

\gdef\ztoc@current@class { #2 }

\exp_args:Nne \ztoc_once_tocline:nnnn

{ #1 } % keyval settings

{ \ztoc_get_class_level_expandable:n {#2} } % toc depth

{ #3 } % {name}{title}

{ #4 } % page

\cs_undefine:N \ztoc@current@class

}

% >> enable/disable table interface

\NewDocumentCommand{\ztocenable}{0{toc} }

{

\seq_gset_from_clist:Nn \g__ztoc_table_enabled_seq

{ #1 }

\keyval_parse:nnn

{ \ztoc_enable_table:nn {\c_sys_jobname_str} }

{ \ztoc_enable_table_swap:nn }

{ #1 }

}

\NewDocumentCommand{\ztocstop}{0{toc}}

{

\clist_map_inline:nn { #1 }

{

\ztoc_stop_table:n { ##1 }

}

}

% >> toc format setup

\ztex_keys_define:nn { ztoc/option }

{

rmargin .code:n = { \setlength\ztoc@rmargin {#1} },

ignore.level .code:n = { \gdef\ztoc@ignore@level {#1} },

line.end .code:n = { \long\gdef\ztoc@line@end {#1} },

page.width .code:n = { \setlength\ztoc@page@width{#1} },

}

335

230223032304230523062307230823092310231123122313231423152316231723182319232023212322232323242325232623272328232923302331233223332334233523362337233823392340234123422343234423452346234723482349

2350	leader.type	.code:n = { \gdef\ztoc@leader@type{#1} },	2350
2351	leader.sep	.code:n = { \setlength\ztoc@leader@sep {#1} },	2351
2352	leader.raise	.code:n = { \setlength\ztoc@leader@raise{#1} },	2352
2353	leader.content	.code:n = { \def\ztoc@leader@content{#1} },	2353
2354	}		2354
2355	\NewDocumentCommand{\ztocset}{ m }		2355
2356	{		2356
2357	\ztex_keys_set:nn { ztoc/option }		2357
2358	{ #1 }		2358
2359	}		2359
2360	\NewDocumentCommand{\ztocformat}{m+m}		2360
2361	{		2361
2362	\ztoc_all_format_set:nn {#1}{#2}		2362
2363	}		2363
2364	\NewDocumentCommand{\ztocgroupinsert}{m+m}		2364
2365	{		2365
2366	\hook_gput_next_code:nn {#1}{#2}		2366
2367	}		2367
2368	\NewDocumentCommand{\ztocgroupshow}{}{		2368
2369	{ \bool_set_true:N \l_ztoc_show_hooks_bool }		2369
2370	\NewDocumentCommand{\ztocgrouphide}{}{		2370
2371	{ \bool_set_false:N \l_ztoc_show_hooks_bool }		2371
2372			2372
2373	% >> commands to print toc		2373
2374	% global toc interface		2374
2375	\DeclareDocumentCommand{\tableofcontents}{ 0{\jobname} }		2375
2376	{		2376
2377	\ztoc_save_main_table:n { #1 }		2377
2378	\ztoc_table_typeset:Nn \g_ztoc_toc_seq {}		2378
2379	\ztoc_restore_main_table:n { #1 }		2379
2380	}		2380
2381	\DeclareDocumentCommand{\multicolcontents}{ 0{\jobname}m }		2381
2382	{		2382
2383	\ztoc_save_main_table:n { #1 }		2383
2384	\begin{multicols}{#2}		2384
2385	\ztoc_table_typeset:Nn \g_ztoc_toc_seq {}		2385
2386	\end{multicols}		2386
2387	\ztoc_restore_main_table:n { #1 }		2387
2388	}		2388
2389	% local toc interface		2389
2390	\NewDocumentCommand{\zlocaltoc}{0{\jobname}mm}		2390
2391	{		2391
2392	\ztoc_save_main_table:n { #1 }		2392
2393	\ztoc_table_typeset_before:		2393
2394	\edef\zlocaltoc@curext		2394
2395	{		2395
2396	\exp_args:NNe \prop_item:Nn \c_ztoc_table_types_prop		2396
2397	{		2397

2398	\prop_if_in:NnTF \c_zsect_level_prop { #2 }	2398
2399	{ sect }{ #2 }	2399
2400	}	2400
2401	}	2401
2402	\exp_args:Ne \clist_map_inline:nn { #3 }	2402
2403	{	2403
2404	\ztoc_table_filter_byclass:nccc	2404
2405	{ #2 }{ ##1 }	2405
2406	{ g_ztoc_keyval\zlocaltoc@curext _seq }	2406
2407	{ g_ztoc_local \zlocaltoc@curext _seq }	2407
2408	\seq_use:cn	2408
2409	{ g_ztoc_local\zlocaltoc@curext _seq }	2409
2410	{ }	2410
2411	\ztoc_table_typeset_after:	2411
2412	}	2412
2413	\cs_undefine:N \zlocaltoc@curext	2413
2414	\ztoc_restore_main_table:n { #1 }	2414
2415	}	2415
2416		2416
2417		2417
2418		2418
2419	% -----	2419
2420	% list of tables, figures, etc	2420
2421	% -----	2421
2422	% >> special class contents format	2422
2423	\NewDocumentCommand{\ztocotherformat}{0{figure}m}	2423
2424	{	2424
2425	\prop_if_in:NnTF \c_ztoc_special_level_prop { #1 }	2425
2426	{	2426
2427	\exp_args:Nne \ztex_edit_instance:nnn	2427
2428	{ toc }{ ztoc/#1 }{#2}	2428
2429	}{	2429
2430	\ztex_msg_set:nn {ztoc@instancespecial@nonexist}	2430
2431	{ Instance~'ztoc/#1'~of~type~'ztextoc'~do~NOT~exist. }	2431
2432	\ztex_msg_error:n {ztoc@instancespecial@nonexist}	2432
2433	}	2433
2434	}	2434
2435		2435
2436	% >> print contents	2436
2437	\DeclareDocumentCommand{\listoffigures}{}{	2437
2438	{	2438
2439	\ztoc_table_typeset:Nn \g_ztoc_lof_seq	2439
2440	{}	2440
2441	}	2441
2442	\DeclareDocumentCommand{\listoftables}{}{	2442
2443	{	2443
2444	\ztoc_table_typeset:Nn \g_ztoc_lot_seq	2444
2445	{}	2445

2446	}	2446
2447	\DeclareDocumentCommand{\listoftheorems}{}{	2447
2448	{	2448
2449	\ztoc_table_typeset:Nn \g_ztoc_lom_seq	2449
2450	{}	2450
2451	}	2451
2452	\DeclareDocumentCommand{\listoflistings}{}{	2452
2453	{	2453
2454	\ztoc_table_typeset:Nn \g_ztoc_lol_seq	2454
2455	{}	2455
2456	}	2456
2457	% NOTE: package 'algorithm2e' define '\listofalgorithms'.	2457
2458	\ztex_hook_preamble_last:n	2458
2459	{	2459
2460	\DeclareDocumentCommand{\listofalgorithms}{}{	2460
2461	{	2461
2462	\ztoc_table_typeset:Nn \g_ztoc_loa_seq	2462
2463	{}	2463
2464	}	2464
2465	}	2465
2466	% NOTE: make this change to command name	2466
2467	% according to 'glossaries' package.	2467
2468	\DeclareDocumentCommand{\printglossaries}{}{	2468
2469	{	2469
2470	\ztoc_table_typeset:Nn \g_ztoc_tog_seq	2470
2471	{}	2471
2472	}	2472
2473	\DeclareCommandCopy{\listofglossaries}	2473
2474	{\printglossaries}	2474
2475		2475
2476		2476
2477		2477
2478	% -----	2478
2479	% section title interface	2479
2480	% -----	2480
2481	% ==> mark interface	2481
2482	% maker class setup	2482
2483	\cs_new_protected:Npn \zsect_mark_new_class_safe:nn #1#2	2483
2484	{% #1:bool; #2:class	2484
2485	\seq_if_in:NnTF \g__mark_classes_seq { #2 }	2485
2486	{	2486
2487	\ztex_msg_set:nn { mark-override }	2487
2488	{ mark~class:'#2'~already~exists,~you~can~NOT~override~it }	2488
2489	\ztex_msg_error:n { mark-override }	2489
2490	}{	2490
2491	\mark_new_class:n { #2 }	2491
2492	\bool_if:nT { #1 }	2492
2493	{	2493


```
2494 \mark_new_class:n { #2-nonempty }
2495 }
2496 }
2497 }
2498 \zsect_mark_new_class_safe:nn { \c_false_bool }{ ztex-1st }
2499 \zsect_mark_new_class_safe:nn { \c_false_bool }{ ztex-left }
2500 \zsect_mark_new_class_safe:nn { \c_true_bool }{ ztex-right }
2501 \zsect_mark_new_class_safe:nn { \c_false_bool }{ ztex-4th }
2502
2503 % marker form
2504 \tl_new:N \g__zsect_marker_form_tl
2505 \tl_gset:Nn \g__zsect_marker_form_tl
2506 { \zsecname }
2507 \cs_new:Npn \zsect_marker_form:n #1
2508 {
2509 \tl_gset:Nn \g__zsect_marker_form_tl
2510 { #1 }
2511 }
2512 \cs_generate_variant:Nn \zsect_marker_form:n
2513 { o, e }
2514
2515 % insert mark
2516 \cs_new_protected:Npn \zsect_mark_insert:nn #1#2
2517 {
2518 \exp_args:Ne \int_case:nn { \prop_item:Nn \c_zsect_level_prop {#1} } {
2519 {
2520 {1}{ \mark_insert:nn { ztex-1st } {#2} }
2521 {2}{ \mark_insert:nn { ztex-left} {#2} }
2522 {3}{ \zsect_right_mark_insert:n {#2} }
2523 {4}{ \mark_insert:nn { ztex-4th } {#2} }
2524 }
2525 \clist_if_in:NnT \g__zsect_mark_user_clist {#1}
2526 {
2527 \zsect_mark_user_insert:nn { #1 }{ #2 }
2528 }
2529 }
2530 \cs_new_protected:Npn \zsect_mark_user_insert:nn #1#2
2531 {
2532 \mark_insert:nn {ztex-user-#1}{#2}
2533 \tl_if_empty:oF {#2}
2534 {
2535 \mark_insert:nn {ztex-user-#1-nonempty}{#2}
2536 }
2537 }
2538 \cs_new_protected:Npn \zsect_right_mark_insert:n #1
2539 {
2540 \mark_insert:nn {ztex-right}{#1}
2541 \tl_if_empty:oF {#1}
```

254225432544254525462547254825492550255125522553255425552556255725582559256025612562256325642565256625672568256925702571257225732574257525762577257825792580258125822583258425852586258725882589

```

{
  \mark_insert:nn {ztex-right-nonempty}{#1}
}
}

\cs_new_protected:Npn \zsect_markclass_lower_empty:n #1
{
  \prop_map_inline:Nn \c_zsect_level_prop
  {
    % NOTE:only have 4 mark class at now.
    \int_compare:nT { ##2 > 4 }
      { \prop_map_break: }
    \int_compare:nNnT { ##2 } >
      { \prop_item:Nn \c_zsect_level_prop {#1} }
      { \zsect_mark_insert:nn {##1}{#1} }
  }
}

\cs_generate_variant:Nn \zsect_mark_insert:nn
{ Vn, ee }

\cs_generate_variant:Nn \zsect_markclass_lower_empty:n
{ V, o, e }

\cs_new:Npn \__zsect_ltx_mark_end:
{ \if@nobreak\ifvmode\nobreak\fi\fi }

% user mark interface
\NewDocumentCommand{\zsectmarkform}{m}
{
  \zsect_marker_form:n { #1 }
}

\@onlypreamble\zsectmarkform

\NewDocumentCommand{\zsectmarkinsert}{mm}
{
  \zsect_mark_insert:nn {#1}{#2}
}

\DeclareDocumentCommand{\markright}{m}
{
  \zsect_right_mark_insert:n {#1}
  \__zsect_ltx_mark_end:
}

\DeclareDocumentCommand{\markboth}{mm}
{
  \mark_insert:nn { ztex-left } {#1}
  \zsect_right_mark_insert:n {#2}
  \__zsect_ltx_mark_end:
}

\IfClassLoadedTF{book}
{
  \DeclareDocumentCommand{\chaptermark}{m}
  { \zsect_mark_insert:nn {chapter}{#1} }

```

254225432544254525462547254825492550255125522553255425552556255725582559256025612562256325642565256625672568256925702571257225732574257525762577257825792580258125822583258425852586258725882589

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```

2590 \DeclareDocumentCommand{\sectionmark}{m}
2591 { \zsect_mark_insert:nn {section}{#1} }
2592 }{
2593 \DeclareDocumentCommand{\sectionmark}{m}
2594 { \zsect_mark_insert:nn {section}{#1} }
2595 \DeclareDocumentCommand{\subsectionmark}{m}
2596 { \zsect_mark_insert:nn {subsection}{#1} }
2597 }
2598 \cs_new_protected:Npn \zsect_robust_left_mark:
2599 { \mark_use_first:nn {page}{ztex-left} }
2600 \cs_new_protected:Npn \zsect_robust_right_mark:
2601 {
2602 \mark_if_eq:nnnnTF {page}{ztex-right-nonempty}{top}{first}
2603 {
2604 \(\ \langle\mbox{empty}\rangle \)
2605 \ztex_msg_warn:n { right-mark-empty }
2606 }{ \mark_use_first:nn {page}{ztex-right-nonempty} }
2607 }
2608 \cs_set:Npn \leftmark { \mark_use_last:nn {page}{ztex-left} }
2609 \cs_set:Npn \rightmark { \mark_use_first:nn {page}{ztex-right} }
2610 \cs_set_eq:NN \ztexleftmark \zsect_robust_left_mark:
2611 \cs_set_eq:NN \ztexrightmark \zsect_robust_right_mark:
2612 \cs_set_eq:NN \robustleftmark \zsect_robust_left_mark:
2613 \cs_set_eq:NN \robustrightmark \zsect_robust_right_mark:
2614 \ztex_msg_set:nn { right-mark-empty }
2615 { right-mark-is-empty-at~page:\thepage. }
2616
2617
2618 % ==> title interface (title = num + name)
2619 \ztex_new_templatetype:nn {sect}{4}
2620 \ztex_declare_template_interface:nnnn {sect}{default}{4}
2621 {
2622 type : tokenlist,
2623 hang : boolean = { true },
2624 break : tokenlist,
2625 pagestyle : tokenlist,
2626 afterindent : boolean = { false },
2627
2628 space.before : skip,
2629 space.after : skip,
2630 space.left : length,
2631
2632 title.inline : boolean = { false },
2633 title.format : tokenlist,
2634 title.format+ : tokenlist = { },
2635 title.before : tokenlist = { },
2636 title.after : tokenlist = { \par },
2637

```

2638	name.sep	: length	= { Opt },	2638
2639	name.before	: tokenlist	= { },	2639
2640	name.after	: tokenlist	= { },	2640
2641	name.format	: tokenlist	= { },	2641
2642	name.format+	: tokenlist	= { },	2642
2643				2643
2644	num	: tokenlist	= { },	2644
2645	num.show	: boolean	= { true },	2645
2646	num.sep	: length,		2646
2647	num.with	: tokenlist	= { },	2647
2648	num.format	: tokenlist	= { },	2648
2649	num.format+	: tokenlist	= { },	2649
2650	num.before	: tokenlist	= { },	2650
2651	num.after	: tokenlist	= { },	2651
2652				2652
2653	format.num	: tokenlist	= \KeyValue { num.format },	2653
2654	format.num+	: tokenlist	= \KeyValue { num.format+ },	2654
2655	format.name	: tokenlist	= \KeyValue { name.format },	2655
2656	format.name+	: tokenlist	= \KeyValue { name.format+ },	2656
2657	format.title	: tokenlist	= \KeyValue { title.format },	2657
2658	format.title+	: tokenlist	= \KeyValue { title.format+ },	2658
2659				2659
2660	explicit	: boolean	= { false },	2660
2661	code	: tokenlist	= { },	2661
2662				2662
2663	bookmark.num	: boolean	= false,	2663
2664	bookmark.before	: tokenlist,		2664
2665	bookmark.after	: tokenlist,		2665
2666	bookmark.cmd	: function{1}	= #1,	2666
2667	}			2667
2668	\ztex_declare_template_code:nnnnn	{sect}{default}{4}		2668
2669	{			2669
2670	type	= \l__zsect_title_type_tl,		2670
2671	hang	= \l__zsect_title_hang_bool,		2671
2672	break	= \l__zsect_title_break_tl,	% TODO: implement it !	2672
2673	pagestyle	= \l__zsect_title_pagestyle_tl,		2673
2674	afterindent	= \l__zsect_title_afterindent_bool,		2674
2675				2675
2676	space.before	= \l__zsect_title_spbef_skip,		2676
2677	space.after	= \l__zsect_title_spaf_skip,		2677
2678	space.left	= \l__zsect_title_left_dim,		2678
2679				2679
2680	format.num	= \l__zsect_title_num_format_tl,		2680
2681	format.num+	= \l__zsect_title_num_format_p_tl,		2681
2682	format.name	= \l__zsect_title_name_format_tl,		2682
2683	format.name+	= \l__zsect_title_name_format_p_tl,		2683
2684	format.title	= \l__zsect_title_format_tl,		2684
2685	format.title+	= \l__zsect_title_format_p_tl,		2685

2686		2686
2687	title.inline = \l__zsect_title_inline_bool,	2687
2688	title.format = \l__zsect_title_format_tl,	2688
2689	title.format+ = \l__zsect_title_format_p_tl,	2689
2690	title.before = \l__zsect_title_before_tl,	2690
2691	title.after = \l__zsect_title_after_tl,	2691
2692		2692
2693	name.sep = \l__zsect_title_name_sep_dim,	2693
2694	name.format = \l__zsect_title_name_format_tl,	2694
2695	name.format+ = \l__zsect_title_name_format_p_tl,	2695
2696	name.before = \l__zsect_title_name_before_tl,	2696
2697	name.after = \l__zsect_title_name_after_tl,	2697
2698		2698
2699	num = \l__zsect_title_num_tl,	2699
2700	num.show = \l__zsect_title_num_show_bool,	2700
2701	num.sep = \l__zsect_title_num_sep_dim,	2701
2702	num.with = \l__zsect_title_num_width_tl, % TODO: implement it !	2702
2703	num.format = \l__zsect_title_num_format_tl,	2703
2704	num.format+ = \l__zsect_title_num_format_p_tl,	2704
2705	num.before = \l__zsect_title_num_before_tl,	2705
2706	num.after = \l__zsect_title_num_after_tl,	2706
2707		2707
2708	explicit = \l__zsect_title_explicit_bool,	2708
2709	code = \l__zsect_title_code_tl,	2709
2710		2710
2711	bookmark.num = \l__zsect_title_bookmark_num_bool,	2711
2712	bookmark.before = \l__zsect_title_bookmark_before_tl,	2712
2713	bookmark.after = \l__zsect_title_bookmark_after_tl,	2713
2714	bookmark.cmd = __zsect_title_bookmark_cmd:n,	2714
2715	}{	2715
2716	% Instance args spec:	2716
2717	% #1: class name	2717
2718	% #2: bool:\c_true_bool(numberless) \c_false_bool;	2718
2719	% #3: toc-content	2719
2720	% #4: sec-content	2720
2721	% NOTE:	2721
2722	% 1. sectioning hooks will be added by 'lthooks.dtx'.	2722
2723	% 2. these groups aim to set TemplateKeys-related variables locally.	2723
2724	% 3. '\refstepcounter' can NOT be inside a group, and	2724
2725	% should be after '__zsect_title_pagespec_before:'.	2725
2726	\group_begin:	2726
2727	\AssignTemplateKeys	2727
2728	__zsect_title_pagespec_before:	2728
2729	\group_end:	2729
2730	__zsect_title_cnt_refstep:nn { #2 }{ #1 }	2730
2731	\group_begin:	2731
2732	\AssignTemplateKeys	2732
2733	__zsec_public_macro_setup:nnnn	2733

2734 { #1 }{ #2 }{ #3 }{ #4 } 2734

2735 % start typesetting sec title: 2735

2736 \exp_args:Nno __zsect_mark_update:nn 2736

2737 { #1 }{ \g__zsect_marker_form_tl } 2737

2738 __zsect_title_typeset:nnnn { #1 }{ #2 }{ #3 }{ #4 } 2738

2739 __zsect_title_pagespec_after: 2739

2740 \group_insert_after:N \g__zsect_after_group_tl 2740

2741 \group_end: 2741

2742 \tl_gclear:N \g__zsect_after_group_tl 2742

2743 } 2743

2744 2744

2745 2745

2746 % ---> inner components of title below <---- 2746

2747 % setup sec title alias 2747

2748 \cs_new:Nn __zsec_public_macro_setup:nnnn 2748

2749 { 2749

2750 % alias for internal usage: 2750

2751 \edef\zsect@num 2751

2752 { 2752

2753 \tl_if_empty:NTF \l__zsect_title_num_tl 2753

2754 { \cs:w the#1 \cs_end: } 2754

2755 { \l__zsect_title_num_tl } 2755

2756 } 2756

2757 \edef\zsect@tocnum {\cs:w the #1 \cs_end:} 2757

2758 \edef\zsect@tocHnum 2758

2759 { 2759

2760 \ztexhyperTF 2760

2761 { \cs:w theH#1 \cs_end: } 2761

2762 { \cs:w the #1 \cs_end: } 2762

2763 } 2763

2764 % alias for users' interface: 2764

2765 \cs_set_nopar:Npn \zsecclass { #1 } 2765

2766 \cs_set_nopar:Npn \zsectocname { #3 } 2766

2767 \cs_set_nopar:Npn \zsecname { #4 } 2767

2768 \cs_set_nopar:Npn \zsecnum { \zsect@num } 2768

2769 \cs_set_nopar:Npn \zsectocnum { \zsect@tocnum } 2769

2770 \cs_set_nopar:Npn \zsectocHnum { \zsect@tocHnum } 2770

2771 } 2771

2772 \cs_new:Npn __zsect_title_typeset:nnnn #1#2#3#4 2772

2773 { 2773

2774 \bool_if:NTF \l__zsect_title_explicit_bool 2774

2775 { 2775

2776 \l__zsect_title_code_tl 2776

2777 }{ 2777

2778 \bool_if:nF { #2 } 2778

2779 { 2779

2780 % 1. set '\Hy@pdfstring' true to support command '\texorpdfstring'; 2780

2781 % 2. non-empty toc name(#1) comes before the sec name(#2); 2781

2782 % 3. command or math shift may occur in '#1'(i.e. toc name), thus 2782

2783 % '\texorpdfstring' need to work well in toc name. 2783

2784 __zsect_bookmark_update:nnn { #1 }{ #3 }{ #4 } 2784

2785 __zsect_title_toc_add:nnn { #1 }{ #3 }{ #4 } 2785

2786 } 2786

2787 __zsect_title_space_before: 2787

2788 __zsect_title_space_left: 2788

2789 __zsect_title_body:nnn { #1 }{ #2 }{ #4 } 2789

2790 __zsect_title_space_after: 2790

2791 } 2791

2792 } 2792

2793 \cs_new:Npn __zsect_mark_update:nn #1#2 2793

2794 { 2794

2795 \zsect_mark_insert:nn { #1 }{ #2 } 2795

2796 % \zsect_markclass_lower_empty:n { #1 } 2796

2797 } 2797

2798 \cs_new:Npn __zsect_bookmark_update:nnn #1#2#3 2798

2799 { 2799

2800 \bool_if:NT \g__ztex_hyperref_bool 2800

2801 { 2801

2802 \group_begin: 2802

2803 \Hy@pdfstringtrue 2803

2804 % NOTE: do NOT apply '\zsect_unexpand_protect:' 2804

2805 % to command '\pdfbookmark'. 2805

2806 \zsect_unexpand_protect: 2806

2807 \xdef\zsect@temp@a 2807

2808 { 2808

2809 __zsect_title_bookmark_cmd:n 2809

2810 { 2810

2811 \tl_if_empty:oTF {#2}{#3}{#2} 2811

2812 } 2812

2813 } 2813

2814 \zsect_restore_protect: 2814

2815 \exp_after:wN __zsect_title_bookmark_add:nn 2815

2816 \exp_after:wN { \zsect@temp@a }{ #1 } 2816

2817 \group_end: 2817

2818 \cs_undefine:N \zsect@temp@a 2818

2819 } 2819

2820 } 2820

2821 % ----- 2821

2822 \cs_new:Npn __zsect_title_cnt_refstep:nn #1#2 2822

2823 { 2823

2824 \bool_if:nTF { #1 } 2824

2825 { \stepcounter{#2} } 2825

2826 { \refstepcounter{#2} } 2826

2827 } 2827

2828 \cs_new:Npn __zsect_title_bookmark_add:nn #1#2 2828

2829 {% #2:class; #1:can only have valid pdfstring !!! 2829

2830 \zsect_bookmark_add:eee 2830

2831 { 2831

2832 \prop_item:Nn \c_zsect_level_prop 2832

2833 { #2 } 2833

2834 }{ 2834

2835 \l__zsect_title_bookmark_before_tl 2835

2836 \bool_if:NT \l__zsect_title_bookmark_num_bool 2836

2837 { \zsect@tocnum\u } 2837

2838 \exp_not:n { #1 } 2838

2839 \l__zsect_title_bookmark_after_tl 2839

2840 } 2840

2841 { #2.\zsect@tocHnum } 2841

2842 } 2842

2843 % page spec 2843

2844 \cs_new:Npn __zsect_title_pagespec_before: 2844

2845 { 2845

2846 \bool_if:NF \l__zsect_title_explicit_bool 2846

2847 { 2847

2848 __zsect_title_type_spec:nn { page, top } 2848

2849 { \newpage\hspace{0pt} } 2849

2850 \tl_if_empty:NF \l__zsect_title_pagestyle_tl 2850

2851 { 2851

2852 \exp_args:No \thispagestyle 2852

2853 { \l__zsect_title_pagestyle_tl } 2853

2854 } 2854

2855 } 2855

2856 } 2856

2857 \cs_new:Npn __zsect_title_pagespec_after: 2857

2858 { 2858

2859 \bool_if:NF \l__zsect_title_explicit_bool 2859

2860 { 2860

2861 __zsect_title_type_spec:nn { page } 2861

2862 { \hspace{0pt}\newpage } 2862

2863 } 2863

2864 } 2864

2865 \cs_new:Npn __zsect_title_type_spec:nn #1#2 2865

2866 { 2866

2867 \exp_args:Nne \clist_if_in:nnT { #1 } 2867

2868 { \l__zsect_title_type_tl } 2868

2869 { #2 } 2869

2870 } 2870

2871 % space spec 2871

2872 \cs_new:Nn __zsect_title_space_before: 2872

2873 { 2873

2874 \exp_args:Nne \clist_if_in:nnTF {page, top}{\l__zsect_title_type_tl} 2874

2875 { \vskip\l__zsect_title_spbf_skip\relax } 2875

2876 { 2876

2877 \if@noskipsec \leavevmode \fi \par 2877

2878	\zsect@dim@b \l__zsect_title_spb_f_skip\relax	2878
2879	\ifdim \zsect@dim@b < \z@	2879
2880	\zsect@dim@b -\zsect@dim@b\relax	2880
2881	\fi	2881
2882	\if@nobreak	2882
2883	\everypar{}	2883
2884	\else	2884
2885	\addpenalty \@secpenalty	2885
2886	\addvspace \zsect@dim@b	2886
2887	\fi	2887
2888	}	2888
2889	}	2889
2890	\tl_gclear_new:N \g__zsect_after_group_tl	2890
2891	\cs_new:Nn __zsect_title_space_after:	2891
2892	{	2892
2893	\bool_if:NTF \l__zsect_title_inline_bool	2893
2894	{ \hskip \l__zsect_title_spaf_skip\relax }	2894
2895	{	2895
2896	\vskip \l__zsect_title_spaf_skip\relax	2896
2897	\bool_if:NTF \l__zsect_title_afterindent_bool	2897
2898	{	2898
2899	\tl_gset:Nn \g__zsect_after_group_tl	2899
2900	{\@afterindenttrue }	2900
2901	}{	2901
2902	\tl_gset:Nn \g__zsect_after_group_tl	2902
2903	{\@afterindentfalse}	2903
2904	}	2904
2905	\tl_gput_right:Nn \g__zsect_after_group_tl	2905
2906	{ \@afterheading }	2906
2907	}	2907
2908	}	2908
2909	\cs_new:Nn __zsect_title_space_left:	2909
2910	{	2910
2911	\noindent\hspace*{\l__zsect_title_left_dim}	2911
2912	}	2912
2913	% title body	2913
2914	\cs_new:Npn __zsect_title_body:nnn #1#2#3	2914
2915	{% #1:class; #2:bool; #3:name	2915
2916	\group_begin: % this group make these formats act locally.	2916
2917	\l__zsect_title_format_tl	2917
2918	\l__zsect_title_format_p_tl	2918
2919	\l__zsect_title_before_tl	2919
2920	\bool_if:nT { #2 }	2920
2921	{ \bool_set_false:N \l__zsect_title_num_show_bool }	2921
2922	\bool_lazy_all:nT	2922
2923	{	2923
2924	{ \l__zsect_title_num_show_bool }	2924
2925	{	2925

2926 \int_compare_p:nNn 2926

2927 { \number\c@secnumdepth + 1 } 2927

2928 > 2928

2929 { \prop_item:Nn \c_zsect_level_prop {#1} } 2929

2930 } 2930

2931 }{ 2931

2932 \bool_if:NT \l__zsect_title_hang_bool { \zsect_hang_from:n } 2932

2933 { 2933

2934 \l__zsect_title_num_format_tl 2934

2935 \l__zsect_title_num_format_p_tl 2935

2936 { 2936

2937 \l__zsect_title_num_before_tl 2937

2938 { \zsect@num } 2938

2939 \l__zsect_title_num_after_tl 2939

2940 } 2940

2941 \hskip \l__zsect_title_num_sep_dim\relax 2941

2942 } 2942

2943 } 2943

2944 { 2944

2945 \l__zsect_title_name_format_tl 2945

2946 \l__zsect_title_name_format_p_tl 2946

2947 { 2947

2948 \l__zsect_title_name_before_tl 2948

2949 { #3 } 2949

2950 \l__zsect_title_name_after_tl 2950

2951 } 2951

2952 \hskip \l__zsect_title_name_sep_dim\relax 2952

2953 \zsec_normal_text:n { \l__zsect_title_after_tl } 2953

2954 % NOTE: add this '\par' to make current paragraph be processed, 2954

2955 % thus, '\centering'-like commands will make sense. 2955

2956 \bool_if:NF \l__zsect_title_inline_bool {\par} 2956

2957 } 2957

2958 \group_end: 2958

2959 } 2959

2960 % ---> inner components of title above <---- 2960

2961 2961

2962 2962

2963 % ==> define title 2963

2964 \cs_new:Npn \zsect_define_title:Nnn #1#2#3 2964

2965 {% #1:cmd; #2:style; #3:keyval 2965

2966 % setup instance 2966

2967 \zcnt_safe_new:ee { \cs_to_str:N #1 }{} 2967

2968 \exp_args:Ne __zsect_setup_instance:nn 2968

2969 { \cs_to_str:N #1 @#2 }{ #3 } 2969

2970 % TODO: check if instance valid: 2970

2971 % \edef\valid@sect@instance 2971

2972 % { 2972

2973 % \zsect_instance_set_fallback:nn 2973

2974 % {#1} 2974

2975 % {#2} 2975

2976 % } 2976

2977 % use instance: 2977

2978 \DeclareDocumentCommand{ #1 }{sO{ }m} 2978

2979 { 2979

2980 \IfBooleanTF{##1} 2980

2981 { 2981

2982 \exp_args:Neee \ztex_use_instance:nn {sect} 2982

2983 { \cs_to_str:N #1 @\g__zsect_title_style_tl-numberless } 2983

2984 { \cs_to_str:N #1 } 2984

2985 { \c_true_bool } 2985

2986 { ##2 }{ ##3 } 2986

2987 }{ 2987

2988 \exp_args:Neee \ztex_use_instance:nn {sect} 2988

2989 { \cs_to_str:N #1 @\g__zsect_title_style_tl } 2989

2990 { \cs_to_str:N #1 } 2990

2991 { \c_false_bool } 2991

2992 { ##2 }{ ##3 } 2992

2993 } 2993

2994 } 2994

2995 } 2995

2996 \cs_new:Npn \zsect_instance_set_fallback:nn #1#2 2996

2997 { 2997

2998 \IfInstanceExistsTF{#1} 2998

2999 { #1 } 2999

3000 { #2 } 3000

3001 } 3001

3002 \cs_generate_variant:Nn \zsect_define_title:Nnn 3002

3003 { No, Ne, c, co, ce } 3003

3004 \cs_new_protected:Npn __zsect_setup_instance:nn #1#2 3004

3005 { 3005

3006 \ztex_declare_instance:nnnn 3006

3007 { sect }{ #1 } 3007

3008 { default }{ #2 } 3008

3009 \ztex_declare_instance_copy:nnn 3009

3010 { sect } 3010

3011 { #1-numberless } 3011

3012 { #1 } 3012

3013 } 3013

3014 \zsect_define_title:Nnn \part { ltx } 3014

3015 { 3015

3016 type = page, 3016

3017 pagestyle = empty, 3017

3018 hang = false, 3018

3019 space.before = 0pt \@plus .7fill, 3019

3020 space.after = 0pt \@plus 1fill, 3020

3021 title.format = \huge\bfseries\centering, 3021

3022 num = \Roman{part}\par, 3022

3023 num.before = {PART~}, 3023

3024 num.sep = 0pt, 3024

3025 } 3025

3026 \zssect_define_title:Nnn \chapter { ltx } 3026

3027 { 3027

3028 type = top, 3028

3029 pagestyle = plain, 3029

3030 space.before = 50pt, 3030

3031 space.after = 40pt, 3031

3032 title.format = \normalfont\huge\bfseries\centering, 3032

3033 num = \Roman{chapter}, 3033

3034 num.before = {CHAP~}, 3034

3035 num.sep = 15pt, 3035

3036 } 3036

3037 \zssect_define_title:Nnn \section { ltx } 3037

3038 { 3038

3039 type = normal, 3039

3040 space.left = 0pt, 3040

3041 space.before = -3.5ex \@plus -1ex \@minus -.2ex, 3041

3042 space.after = 2.3ex \@plus .2ex, 3042

3043 title.format = \normalfont\Large\bfseries, 3043

3044 num.sep = 18pt, 3044

3045 } 3045

3046 \zssect_define_title:Nnn \subsection { ltx } 3046

3047 { 3047

3048 type = normal, 3048

3049 space.left = 0pt, 3049

3050 space.before = -3.25ex\@plus -1ex \@minus -.2ex, 3050

3051 space.after = 1.5ex \@plus .2ex, 3051

3052 title.format = \normalfont\large\bfseries, 3052

3053 num.sep = 15pt, 3053

3054 } 3054

3055 \zssect_define_title:Nnn \subsubsection { ltx } 3055

3056 { 3056

3057 type = normal, 3057

3058 space.left = 0pt, 3058

3059 space.before = -3.25ex\@plus -1ex \@minus -.2ex, 3059

3060 space.after = 1.5ex \@plus .2ex, 3060

3061 title.format = \normalfont\normalsize\bfseries, 3061

3062 num.sep = 13pt, 3062

3063 } 3063

3064 \zssect_define_title:Nnn \paragraph { ltx } 3064

3065 { 3065

3066 type = normal, 3066

3067 title.inline = true, 3067

3068 title.after = , 3068

3069 space.left = 0pt, 3069

```
3070     space.before = 3.25ex \@plus 1ex \@minus .2ex,
3071     space.after  = -1em,
3072     title.format = \normalfont\normalsize\bfseries,
3073     num.show     = false,
3074     name.sep     = 18pt,
3075 }
3076 \zsect_define_title:Nnn \subparagraph { ltx }
3077 {
3078     type          = normal,
3079     title.inline  = true,
3080     title.after   = ,
3081     space.left    = 18pt,
3082     space.before  = 3.25ex \@plus 1ex \@minus .2ex,
3083     space.after   = -1em,
3084     title.format  = \normalfont\normalsize\bfseries,
3085     num.show      = false,
3086     name.sep      = 19pt,
3087 }
3088 \NewDocumentCommand{\zsecdefine}{mmm}
3089 {
3090     \zsect_define_title:Nnn #1
3091     { #2 }{ #3 }
3092 }
3093
3094
3095 % ==> custom interface for user
3096 % sec title style
3097 \NewDocumentCommand{\zsecttitlestyle}{m}
3098 {
3099     \tl_gset:Nn \g__zsect_title_style_tl
3100     { #1 }
3101 }
3102 \@onlypreamble\zsecttitlestyle
3103
3104 % sec title format
3105 \NewDocumentCommand{\zsecformat}{s O{ ltx } m +m}
3106 {
3107     \clist_map_inline:nn { #3 }
3108     {
3109         \edef\zsect@tmp@instance@name
3110         {
3111             \cs_to_str:N ##1 @#2
3112             \IfBooleanT{#1}{-numberless}
3113         }
3114         \IfInstanceExistsTF {ztexsect}
3115         { \zsect@tmp@instance@name }
3116         {
3117             \exp_args:Nne \ztex_edit_instance:nnn
```

3118	{ sect }	3118
3119	{ \zsect@tmp@instance@name }	3119
3120	{ #4 }	3120
3121	}{	3121
3122	\ztex_msg_set:nn {zsect@instance@nonexist}	3122
3123	{	3123
3124	Instance~'\zsect@tmp@instance@name'~of~	3124
3125	TemplateType~'ztexsect'~does~NOT~exist.	3125
3126	}	3126
3127	\ztex_msg_error:n {zsect@instance@nonexist}	3127
3128	}	3128
3129	}	3129
3130	\cs_undefine:N \zsect@tmp@instance@name	3130
3131	}	3131
3132		3132
3133	% template defaults & one-time title	3133
3134	\NewDocumentCommand{\zsectTemplateDefaultsEdit}{m}	3134
3135	{	3135
3136	\ztex_edit_template_defaults:nnn	3136
3137	{ sect }{ default }{ #1 }	3137
3138	}	3138
3139	\cs_new_protected:Npn \zsect_once_title:nnnnn #1#2#3#4#5	3139
3140	{	3140
3141	\ztex_use_template:nnn { sect }	3141
3142	{ default }{ #1 }	3142
3143	{ #2 }{ #3 }{ #4 }{ #5 }	3143
3144	}	3144
3145	\cs_generate_variant:Nn \zsect_once_title:nnnnn	3145
3146	{ no, ne, ooooo, eeeee }	3146
3147	\NewDocumentCommand{\zsectitleOnce}{smmO{}m}	3147
3148	{	3148
3149	\zsect_once_title:nnnnn	3149
3150	{ #3 } % keyval settings	3150
3151	{ #2 } % class	3151
3152	{ \IfBooleanTF{#1}{\c_true_bool}{\c_false_bool} }	3152
3153	{ #4 } % toc content	3153
3154	{ #5 } % sec content	3154
3155	}	3155

```

1 \ProvidesExplFile{ztex.module.sclist.tex}
2 {2026/02/01}{\ztex@versi@n}
3 {sclist~module~for~ztex}
4
5
6 %%%% semicolon list interface for ztex %%%%
7 % NOTE: the purpose of the 'sclist' module is to
8 % support extensibility for semicolon list.
9 % REF: https://github.com/latex3/latex3/blob/develop/l3kernel/l3clist.dtx
10 % ==> scan marks, sclist map break
11 \scan_new:N \s__sclist_mark
12 \scan_new:N \s__sclist_stop
13 \cs_new:Npn \__sclist_use_none_delimit_by_s_mark:w #1 \s__sclist_mark { }
14 \cs_new:Npn \__sclist_use_none_delimit_by_s_stop:w #1 \s__sclist_stop { }
15 \cs_new:Npn \__sclist_use_i_delimit_by_s_stop:nw #1 #2 \s__sclist_stop {#1}
16 \cs_new_protected:Npn \__sclist_tmp:w { }
17
18
19 % ==> '\__sclist_sanitiz:n' and '\sclist_if_empty:n(N)(pTF)'
20 \prg_new_eq_conditional:NNn \sclist_if_empty:N \tl_if_empty:N
21 { p , T , F , TF }
22 \prg_new_eq_conditional:NNn \sclist_if_empty:c \tl_if_empty:c
23 { p , T , F , TF }
24 \prg_new_conditional:Npnn \sclist_if_empty:n #1 { p , T , F , TF }
25 {
26 \__sclist_if_empty_n:w ? #1
27 ; \s__sclist_mark \prg_return_false:
28 ; \s__sclist_mark \prg_return_true:
29 \s__sclist_stop
30 }
31 \cs_new:Npn \__sclist_if_empty_n:w #1 ,
32 {
33 \tl_if_empty:oTF { \use_none:nn #1 ? }
34 { \__sclist_if_empty_n:w ? }
35 { \__sclist_if_empty_n:wNw }
36 }
37 \cs_new:Npn \__sclist_if_empty_n:wNw #1 \s__sclist_mark #2#3 \s__sclist_stop {#2}
38 \cs_new:Npn \__sclist_trim_next:w #1 ;
39 {
40 \tl_if_empty:oTF { \use_none:nn #1 ? }
41 { \__sclist_trim_next:w \prg_do_nothing: }
42 { \tl_trim_spaces_apply:oN {#1} \exp_end: }
43 }
44 \cs_new:Npn \__sclist_sanitiz:n #1
45 {
46 \exp_after:wN \__sclist_sanitiz:Nn \exp_after:wN \c_empty_tl

```

47	\exp:w _\sclist_trim_next:w \prg_do_nothing:	47
48	#1 ; \s__sclist_stop \prg_break: ; \prg_break_point:	48
49	}	49
50	\cs_new:Npn _\sclist_sanitizе:Nn #1#2	50
51	{	51
52	_\sclist_use_none_delimit_by_s_stop:w #2 \s__sclist_stop	52
53	#1 _\sclist_wrap_item:w #2 ;	53
54	\exp_after:wN _\sclist_sanitizе:Nn \exp_after:wN ;	54
55	\exp:w _\sclist_trim_next:w \prg_do_nothing:	55
56	}	56
57	\prg_new_conditional:Npnn _\sclist_if_wrap:n #1 { TF }	57
58	{	58
59	\tl_if_empty:oTF	59
60	{	60
61	_\sclist_if_wrap:w	61
62	\s__sclist_mark ? #1 ~ \s__sclist_mark ? ~ #1	62
63	\s__sclist_mark ; ~ \s__sclist_mark #1 ;	63
64	}	64
65	{	65
66	\tl_if_head_is_group:nTF { #1 { } }	66
67	{	67
68	\tl_if_empty:nTF {#1}	68
69	{ \prg_return_true: }	69
70	{	70
71	\tl_if_empty:oTF { \use_none:n #1}	71
72	{ \prg_return_true: }	72
73	{ \prg_return_false: }	73
74	}	74
75	}	75
76	{ \prg_return_false: }	76
77	}	77
78	{ \prg_return_true: }	78
79	}	79
80	\cs_new:Npn _\sclist_if_wrap:w #1 \s__sclist_mark ? ~ #2 ~ \s__sclist_mark #3 ; { }	80
81	\cs_new:Npn _\sclist_wrap_item:w #1 ;	81
82	{ _\sclist_if_wrap:nTF {#1} { \exp_not:n { {#1} } } { \exp_not:n {#1} } }	82
83		83
84		84
85	% ==> '\sclist_new:N' and '\sclist_(g)set:Nn'	85
86	\cs_new_eq:NN \sclist_new:N \tl_new:N	86
87	\cs_new_eq:NN \sclist_new:c \tl_new:c	87
88	\cs_new_eq:NN \sclist_set_eq:NN \tl_set_eq:NN	88
89	\cs_new_eq:NN \sclist_set_eq:Nc \tl_set_eq:Nc	89
90	\cs_new_eq:NN \sclist_set_eq:cN \tl_set_eq:cN	90
91	\cs_new_eq:NN \sclist_set_eq:cc \tl_set_eq:cc	91
92	\cs_new_eq:NN \sclist_gset_eq:NN \tl_gset_eq:NN	92
93	\cs_new_eq:NN \sclist_gset_eq:Nc \tl_gset_eq:Nc	93
94	\cs_new_eq:NN \sclist_gset_eq:cN \tl_gset_eq:cN	94


```
95 \cs_new_eq:NN \sclist_gset_eq:cc \tl_gset_eq:cc 95
96 \cs_new_protected:Npn \sclist_const:Nn #1#2 96
97 { \tl_const:Ne #1 { \__sclist_sanitiz:n {#2} } } 97
98 \cs_new_protected:Npn \sclist_set:Nn #1#2 98
99 { \__kernel_tl_set:Nx #1 { \__sclist_sanitiz:n {#2} } } 99
100 \cs_new_protected:Npn \sclist_gset:Nn #1#2 100
101 { \__kernel_tl_gset:Nx #1 { \__sclist_sanitiz:n {#2} } } 101
102 \cs_generate_variant:Nn \sclist_const:Nn { Ne , c , ce } 102
103 \cs_generate_variant:Nn \sclist_const:Nn { Nx , cx } 103
104 \cs_generate_variant:Nn \sclist_set:Nn { NV , Ne , c , cV , ce } 104
105 \cs_generate_variant:Nn \sclist_set:Nn { No , Nx , co , cx } 105
106 \cs_generate_variant:Nn \sclist_gset:Nn { NV , Ne , c , cV , ce } 106
107 \cs_generate_variant:Nn \sclist_gset:Nn { No , Nx , co , cx } 107
108 \cs_new_eq:NN \sclist_clear:N \tl_clear:N 108
109 \cs_new_eq:NN \sclist_clear:c \tl_clear:c 109
110 \cs_new_eq:NN \sclist_gclear:N \tl_gclear:N 110
111 \cs_new_eq:NN \sclist_gclear:c \tl_gclear:c 111
112 \cs_new_eq:NN \sclist_clear_new:N \tl_clear_new:N 112
113 \cs_new_eq:NN \sclist_clear_new:c \tl_clear_new:c 113
114 \cs_new_eq:NN \sclist_gclear_new:N \tl_gclear_new:N 114
115 \cs_new_eq:NN \sclist_gclear_new:c \tl_gclear_new:c 115
116 116
117 117
118 118
119 % ==> '\sclist_map_function:NN' and '\sclist_map_function:nN' 119
120 \cs_new:Npn \sclist_map_function:NN #1#2 120
121 { 121
122     \sclist_if_empty:NF #1 122
123     { 123
124         \exp_after:wN \__sclist_map_function:Nw \exp_after:wN #2 #1 ; 124
125         \s__sclist_stop ; \s__sclist_stop ; \s__sclist_stop ; \s__sclist_stop ; 125
126         \s__sclist_stop ; \s__sclist_stop ; \s__sclist_stop ; \s__sclist_stop ; 126
127         \prg_break_point:Nn \sclist_map_break: { } 127
128     } 128
129 } 129
130 \cs_new:Npn \__sclist_map_function:Nw #1 #2; #3; #4; #5; #6; #7; #8; #9; 130
131 { 131
132     \__sclist_use_none_delimit_by_s_stop:w 132
133     #9 \__sclist_map_function_end:w \s__sclist_stop 133
134     #1 {#2} #1 {#3} #1 {#4} #1 {#5} #1 {#6} #1 {#7} #1 {#8} #1 {#9} 134
135     \__sclist_map_function:Nw #1 135
136 } 136
137 \cs_new:Npn \__sclist_map_function_end:w \s__sclist_stop #1#2 137
138 { 138
139     \__sclist_use_none_delimit_by_s_stop:w #2 \sclist_map_break: \s__sclist_stop 139
140     #1 {#2} 140
141     \__sclist_map_function_end:w \s__sclist_stop 141
142 } 142
```

```

143 \cs_generate_variant:Nn \sclist_map_function:NN { c }
144 \cs_new:Npn \sclist_map_function:nN #1#2
145 {
146     \exp_after:wN \__sclist_map_function_n:Nn \exp_after:wN #2
147     \exp:w \__sclist_trim_next:w \prg_do_nothing: #1 ;
148     \s__sclist_stop \sclist_map_break: ;
149     \prg_break_point:Nn \sclist_map_break: { }
150 }
151 \cs_new:Npn \__sclist_map_function_n:Nn #1 #2
152 {
153     \__sclist_use_none_delimit_by_s_stop:w #2 \s__sclist_stop
154     \__sclist_map_unbrace:wn #2 ; #1
155     \exp_after:wN \__sclist_map_function_n:Nn \exp_after:wN #1
156     \exp:w \__sclist_trim_next:w \prg_do_nothing:
157 }
158 \cs_new:Npn \__sclist_map_unbrace:wn #1; #2 { #2 {#1} }
159 \cs_generate_variant:Nn \sclist_map_function:nN { e }
160
161 % '\sclist_map_tokens:Nn' and '\sclist_map_tokens:nn'
162 \cs_new:Npn \sclist_map_tokens:Nn #1#2
163 {
164     \sclist_if_empty:NF #1
165     {
166         \exp_last_unbraced:Nno \__sclist_map_tokens:nw {#2} #1 ;
167         \s__sclist_stop ; \s__sclist_stop ; \s__sclist_stop ; \s__sclist_stop ;
168         \s__sclist_stop ; \s__sclist_stop ; \s__sclist_stop ; \s__sclist_stop ;
169         \prg_break_point:Nn \sclist_map_break: { }
170     }
171 }
172 \cs_new:Npn \__sclist_map_tokens:nw #1 #2; #3; #4; #5; #6; #7; #8; #9;
173 {
174     \__sclist_use_none_delimit_by_s_stop:w
175     #9 \__sclist_map_tokens_end:w \s__sclist_stop
176     \use:n {#1} {#2} \use:n {#1} {#3} \use:n {#1} {#4} \use:n {#1} {#5}
177     \use:n {#1} {#6} \use:n {#1} {#7} \use:n {#1} {#8} \use:n {#1} {#9}
178     \__sclist_map_tokens:nw {#1}
179 }
180 \cs_new:Npn \__sclist_map_tokens_end:w \s__sclist_stop \use:n #1#2
181 {
182     \__sclist_use_none_delimit_by_s_stop:w #2 \sclist_map_break: \s__sclist_stop
183     #1 {#2}
184     \__sclist_map_tokens_end:w \s__sclist_stop
185 }
186 \cs_generate_variant:Nn \sclist_map_tokens:Nn { c }
187 \cs_new:Npn \sclist_map_tokens:nn #1#2
188 {
189     \__sclist_map_tokens_n:nw {#2}
190     \prg_do_nothing: #1 ; \s__sclist_stop \sclist_map_break: ;

```

```

191 \prg_break_point:Nn \sclist_map_break: { }
192 }
193 \cs_new:Npn \__sclist_map_tokens_n:nw #1#2 ;
194 {
195     \tl_if_empty:oF { \use_none:nn #2 ? }
196     {
197         \__sclist_use_none_delimit_by_s_stop:w #2 \s__sclist_stop
198         \tl_trim_spaces_apply:oN {#2} \use_ii_i:nn
199         \__sclist_map_unbrace:wn ; {#1}
200     }
201     \__sclist_map_tokens_n:nw {#1} \prg_do_nothing:
202 }
203 \cs_new:Npn \sclist_map_break:
204 { \prg_map_break:Nn \sclist_map_break: { } }
205 \cs_new:Npn \sclist_map_break:n
206 { \prg_map_break:Nn \sclist_map_break: }
207
208
209 % ==> '\sclist_count:n' and '\sclist_count:N'
210 \cs_new:Npn \sclist_count:N #1
211 {
212     \int_eval:n
213     {
214         0
215         \sclist_map_function:NN #1 \__sclist_count:n
216     }
217 }
218 \cs_generate_variant:Nn \sclist_count:N { c }
219 \cs_new:Npn \__sclist_count:n #1 { + 1 }
220 \cs_set_protected:Npn \__sclist_tmp:w #1
221 {
222     \cs_new:Npn \sclist_count:n ##1
223     {
224         \int_eval:n
225         {
226             0
227             \__sclist_count:w #1
228             ##1 ; \s__sclist_stop \prg_break: ; \prg_break_point:
229         }
230     }
231     \cs_new:Npn \__sclist_count:w ##1 ;
232     {
233         \__sclist_use_none_delimit_by_s_stop:w ##1 \s__sclist_stop
234         \tl_if_blank:nF {##1} { + 1 }
235         \__sclist_count:w #1
236     }
237 }
238 \exp_args:No \__sclist_tmp:w \c_space_tl

```

239	<code>\cs_generate_variant:Nn \sclist_count:n { e }</code>	239
240		240
241		241
242	<code>% ==> '\sclist_item:nn' and '\sclist_item:Nn'</code>	242
243	<code>\cs_new:Npn \sclist_item:Nn #1#2</code>	243
244	<code>{</code>	244
245	<code> __sclist_item:ffoN</code>	245
246	<code> { \sclist_count:N #1 }</code>	246
247	<code> { \int_eval:n {#2} }</code>	247
248	<code> #1</code>	248
249	<code> __sclist_item_N_loop:nw</code>	249
250	<code>}</code>	250
251	<code>\cs_new:Npn __sclist_item:nnnN #1#2#3#4</code>	251
252	<code>{</code>	252
253	<code> \int_compare:nNnTF {#2} < 0</code>	253
254	<code> {</code>	254
255	<code> \int_compare:nNnTF {#2} < { - #1 }</code>	255
256	<code> { __sclist_use_none_delimit_by_s_stop:w }</code>	256
257	<code> { \exp_args:Nf #4 { \int_eval:n { #2 + 1 + #1 } } }</code>	257
258	<code> }</code>	258
259	<code> {</code>	259
260	<code> \int_compare:nNnTF {#2} > {#1}</code>	260
261	<code> { __sclist_use_none_delimit_by_s_stop:w }</code>	261
262	<code> { #4 {#2} }</code>	262
263	<code> }</code>	263
264	<code> { } ; #3 ; \s__sclist_stop</code>	264
265	<code>}</code>	265
266	<code>\cs_generate_variant:Nn __sclist_item:nnnN { ffo, ff }</code>	266
267	<code>\cs_new:Npn __sclist_item_N_loop:nw #1 #2;</code>	267
268	<code>{</code>	268
269	<code> \int_compare:nNnTF {#1} = 0</code>	269
270	<code> { __sclist_use_i_delimit_by_s_stop:nw { \exp_not:n {#2} } }</code>	270
271	<code> { \exp_args:Nf __sclist_item_N_loop:nw { \int_eval:n { #1 - 1 } } }</code>	271
272	<code>}</code>	272
273	<code>\cs_generate_variant:Nn \sclist_item:Nn { c }</code>	273
274	<code>\cs_new:Npn \sclist_item:nn #1#2</code>	274
275	<code>{</code>	275
276	<code> __sclist_item:ffnN</code>	276
277	<code> { \sclist_count:n {#1} }</code>	277
278	<code> { \int_eval:n {#2} }</code>	278
279	<code> { #1 }</code>	279
280	<code> __sclist_item_n:nw</code>	280
281	<code>}</code>	281
282	<code>\cs_generate_variant:Nn \sclist_item:nn { e }</code>	282
283	<code>\cs_new:Npn __sclist_item_n:nw #1</code>	283
284	<code> { __sclist_item_n_loop:nw {#1} \prg_do_nothing: }</code>	284
285	<code>\cs_new:Npn __sclist_item_n_loop:nw #1 #2;</code>	285
286	<code>{</code>	286

```

287 \exp_args:No \tl_if_blank:nTF {#2}
288 { \__sclist_item_n_loop:nw {#1} \prg_do_nothing: }
289 {
290 \int_compare:nNnTF {#1} = 0
291 { \exp_args:No \__sclist_item_n_end:n {#2} }
292 {
293 \exp_args:Nf \__sclist_item_n_loop:nw
294 { \int_eval:n { #1 - 1 } }
295 \prg_do_nothing:
296 }
297 }
298 }
299 \cs_new:Npn \__sclist_item_n_end:n #1 #2 \s__sclist_stop
300 { \tl_trim_spaces_apply:nN {#1} \__sclist_item_n_strip:n }
301 \cs_new:Npn \__sclist_item_n_strip:n #1 { \__sclist_item_n_strip:w #1 ; }
302 \cs_new:Npn \__sclist_item_n_strip:w #1 ; { \exp_not:n {#1} }
303
304
305 % ==> debug sclist
306 \msg_new:nnn { sclist } { show }
307 {
308 The~semicolon~list~ \tl_if_empty:nF {#1} { #1 ~ }
309 \tl_if_empty:nTF {#2}
310 { is~empty \\>~ . }
311 { contains~the~items~(without~outer~braces): #2 . }
312 }
313 \cs_new_protected:Npn \sclist_show:N { \__sclist_show:NN \msg_show:nneeee }
314 \cs_generate_variant:Nn \sclist_show:N { c }
315 \cs_new_protected:Npn \sclist_log:N { \__sclist_show:NN \msg_log:nneeee }
316 \cs_generate_variant:Nn \sclist_log:N { c }
317 \cs_new_protected:Npn \__sclist_show:NN #1#2
318 {
319 \__kernel_chk_tl_type:NnnT #2 { sclist } { \exp_not:o #2 }
320 {
321 \int_compare:nNnTF { \sclist_count:N #2 }
322 = { \exp_args:No \sclist_count:n #2 }
323 {
324 #1 { sclist } { show }
325 { \token_to_str:N #2 }
326 { \sclist_map_function:NN #2 \msg_show_item:n }
327 { } { }
328 }
329 {
330 \msg_error:nnee { sclist } { non-sclist }
331 { \token_to_str:N #2 } { \tl_to_str:N #2 }
332 }
333 }
334 }

```

335	\cs_new_protected:Npn \sclist_show:n { __sclist_show:Nn \msg_show:nneeee }	335
336	\cs_new_protected:Npn \sclist_log:n { __sclist_show:Nn \msg_log:nneeee }	336
337	\cs_new_protected:Npn __sclist_show:Nn #1#2	337
338	{	338
339	#1 { sclist } { show }	339
340	{ } { \sclist_map_function:nN {#2} \msg_show_item:n } { } { }	340
341	}	341
342		342
343		343
344	% ==> scratch variables	344
345	\sclist_new:N \l_tmpa_sclist	345
346	\sclist_new:N \l_tmpb_sclist	346
347	\sclist_new:N \g_tmpa_sclist	347
348	\sclist_new:N \g_tmpb_sclist	348

10.3.9 cmd

```

1 \ProvidesExplFile{ztex.module.cmd.tex}
2 {2025/09/23}{\ztex@versi@n}
3 {cmd~module~for~ztex}
4
5
6 %%%% cmd module for ztex %%%%
7 \RequirePackage{etoolbox}
8 \NewDocumentCommand\ztexverb{0{\texttt}v}
9 { #1{#2} }
10
11
12 % ==> preamble check
13 \prg_new_conditional:Npnn \zcmd_if_preamble:
14 { p, T, F, TF }
15 {
16 \ifx\@onlypreamble\@notprerr
17 \prg_return_false:
18 \else
19 \prg_return_true:
20 \fi
21 }
22
23
24 % ==> tex command copy
25 \cs_new:Npn \__zcmd_cs_copy:NN #1#2
26 {
27 \tex_let:D #1#2
28 }
29 \cs_new:Npn \__zcmd_cs_gcopy:NN #1#2
30 {
31 \tex_global:D \tex_let:D #1#2
32 }
33 \cs_set_eq:NN \zcmd_cs_copy:NN \__zcmd_cs_copy:NN
34 \cs_set_eq:NN \zcmd_cs_gcopy:NN \__zcmd_cs_gcopy:NN
35 \cs_generate_variant:Nn \zcmd_cs_copy:NN { cc, cN, Nc }
36 \cs_generate_variant:Nn \zcmd_cs_gcopy:NN { cc, cN, Nc }
37 \cs_generate_variant:Nn \__zcmd_cs_copy:NN { cc, cN, Nc }
38 \cs_generate_variant:Nn \__zcmd_cs_gcopy:NN { cc, cN, Nc }
39
40
41 % ==> '\exp_not:N(n)' wrapper for tl/clist/sclist
42 % NOTE: these macros mainly for '\.._map_tokens..'
43 \cs_new:Npn \__zcmd_map_expnot_tl_wrap:n #1
44 {
45 { \exp_not:N \exp_not:n
46 { \exp_not:n {#1} } } }

```

47	}	47
48	\cs_new:Npn __zcmd_map_expnot_clist_wrap:n #1	48
49	{	49
50	\exp_not:N \exp_not:n	50
51	{ \exp_not:n {#1} } ,	51
52	}	52
53	\cs_new:Npn __zcmd_map_expnot_sclist_wrap:n #1	53
54	{	54
55	\exp_not:N \exp_not:n	55
56	{ \exp_not:n {#1} } ;	56
57	}	57
58	\cs_new:Npn \zcmd_map_expnot_tl:n #1	58
59	{	59
60	\tl_map_function:nN {#1}	60
61	__zcmd_map_expnot_tl_wrap:n	61
62	}	62
63	\cs_new:Npn \zcmd_map_expnot_clist:n #1	63
64	{	64
65	\clist_map_function:nN {#1}	65
66	__zcmd_map_expnot_clist_wrap:n	66
67	}	67
68	\cs_new:Npn \zcmd_map_expnot_sclist:n #1	68
69	{	69
70	\sclist_map_function:nN {#1}	70
71	__zcmd_map_expnot_sclist_wrap:n	71
72	}	72
73		73
74		74
75	% ==> command protect and other utils, most	75
76	% of them are from 'etoolbox' package.	76
77	% ltx commands utils:	77
78	\cs_new_protected:Npn \zcmd_robustify:N #1	78
79	{ \robustify #1 }	79
80	\cs_generate_variant:Nn \zcmd_robustify:N { c }	80
81	\prg_new_conditional:Npnn \zcmd_if_param:N #1	81
82	{ p, T, F, TF }	82
83	{	83
84	\ifdefparam #1	84
85	{ \prg_return_true: }	85
86	{ \prg_return_false: }	86
87	}	87
88	\prg_new_conditional:Npnn \zcmd_if_protected:N #1	88
89	{ p, T, F, TF }	89
90	{	90
91	\ifdefprotected #1	91
92	{ \prg_return_true: }	92
93	{ \prg_return_false: }	93
94	}	94


```

95 % NOTE: '\ifdefltxprotect' is not expandable.
96 \prg_new_protected_conditional:Npnn \zcmd_if_ltxprotect:N #1
97 { T, F, TF }
98 {
99     \ifdefltxprotect #1
100     { \prg_return_true: }
101     { \prg_return_false: }
102 }
103 \prg_generate_conditional_variant:Nnn \zcmd_if_param:N
104 { c }{ p, T, F, TF }
105 \prg_generate_conditional_variant:Nnn \zcmd_if_protected:N
106 { c }{ p, T, F, TF }
107 \prg_generate_conditional_variant:Nnn \zcmd_if_ltxprotect:N
108 { c }{ T, F, TF }
109
110 % other utils based on expl3:
111 \prg_generate_conditional_variant:Nnn \token_if_expandable:N
112 { c }{ p, T, F, TF }
113 \prg_generate_conditional_variant:Nnn \token_if_long_macro:N
114 { c }{ p, T, F, TF }
115 \prg_generate_conditional_variant:Nnn \token_if_primitive:N
116 { c }{ p, T, F, TF }
117
118
119 % ==> integer/floats tools
120 \cs_new:Npn \int_step_break:
121 { \prg_map_break:Nn \int_step_break: { } }
122
123 % NOTE: for compatibility, we do NOT use '\int_step_tokens:nnn'
124 % we conver 'int_map' to 'tl_map' instead.
125 \cs_new:Npn \zcmd_ints_to_tl:nn #1#2
126 {
127     \int_step_function:nnN
128     { #1 }{ #2 }
129     \__zcmd_range_int_wrapper:n
130 }
131 \cs_new:Npn \__zcmd_range_int_wrapper:n #1
132 { {#1} }
133 \cs_generate_variant:Nn \zcmd_ints_to_tl:nn
134 { oo, ee }
135
136 % sort a given clist floats
137 \cs_new:Npn \zcmd_floats_sort_decrease:n #1
138 {
139     \ztex_token_if_in:nNTF {#1},
140     {
141         \exp_args:Ne \tl_sort:nN { \zcmd_clist_to_tl:n {#1} }
142         \__zcmd_fp_sort_decrease_compare:nnTF

```

143	}{	143
144	\tl_sort:nN { #1 }	144
145	__zcmd_fp_sort_decrease_compare:nnTF	145
146	}	146
147	}	147
148	\cs_new:Npn \zcmd_floats_sort_increase:n #1	148
149	{	149
150	\ztex_token_if_in:nNTF {#1},	150
151	{	151
152	\exp_args:Ne \tl_sort:nN { \zcmd_clist_to_tl:n {#1} }	152
153	__zcmd_fp_sort_increase_compare:nnTF	153
154	}{	154
155	\tl_sort:nN { #1 }	155
156	__zcmd_fp_sort_increase_compare:nnTF	156
157	}	157
158	}	158
159	\prg_new_conditional:Npnn __zcmd_fp_sort_decrease_compare:nn #1#2	159
160	{ TF }	160
161	{	161
162	\fp_compare:nNnTF {#1} > {#2}	162
163	{ \prg_return_true: }	163
164	{ \prg_return_false: }	164
165	}	165
166	\prg_new_conditional:Npnn __zcmd_fp_sort_increase_compare:nn #1#2	166
167	{ TF }	167
168	{	168
169	\fp_compare:nNnTF {#1} < {#2}	169
170	{ \prg_return_true: }	170
171	{ \prg_return_false: }	171
172	}	172
173		173
174	% find max or min floats	174
175	\cs_new:Npn \zcmd_floats_max:n #1	175
176	{	176
177	\exp_args:Ne \tl_head:n	177
178	{ \zcmd_floats_sort_decrease:n {#1} }	178
179	}	179
180	\cs_new:Npn \zcmd_floats_min:n #1	180
181	{	181
182	\exp_args:Ne \tl_head:n	182
183	{ \zcmd_floats_sort_increase:n {#1} }	183
184	}	184
185		185
186	% reimplement '\int_step_tokens:nn(nnn)' if they are unavailable	186
187	\cs_new:Npn \ztex_int_step_tokens:nnn #1#2	187
188	{	188
189	\exp_args:Ne \tl_map_tokens:nn	189
190	{ \zcmd_ints_to_tl:nn {#1}{#2} }	190

```

191 }
192 \cs_new:Npn \ztex_int_step_tokens:nn #1
193 { \ztex_int_step_tokens:nnn {1}{#1} }
194 \cs_if_exist:NF \int_step_tokens:nn
195 {
196     \cs_gset_eq:NN \int_step_tokens:nn \ztex_int_step_tokens:nn
197     \cs_gset_eq:NN \int_step_tokens:nnn \ztex_int_step_tokens:nnn
198 }
199
200
201 % ==> clist and sclist tool
202 \cs_generate_variant:Nn \clist_use:nn { en }
203 % clist to tl
204 \cs_new:Npn \zcmd_clist_to_tl:n #1
205 {
206     \clist_map_function:nN {#1}
207     \__zcmd_clist_tl_wrapper:n
208 }
209 \cs_new:Npn \__zcmd_clist_tl_wrapper:n #1
210 { \exp_not:n {#{#1}} }
211
212 % clist and sclist patch
213 \scan_new:N \s__clist_patch_stop
214 \scan_new:N \s__sclist_patch_stop
215 \cs_new:Npn \__zcmd_clist_patch:nw #1 #2,
216 {%#1:replace; #2:current
217     \tl_if_blank:nTF { #2 }
218     {
219         \exp_not:n { #1 },
220         \__zcmd_clist_patch:nw {#1}
221     }{
222         \bool_lazy_or:nnT
223         { \int_compare_p:nNn {\tl_count:n {#2}} > {1} }
224         { ! \tl_if_eq_p:NN #2\s__sclist_patch_stop }
225         {
226             \exp_not:n { #2 },
227             \__zcmd_clist_patch:nw {#1}
228         }
229     }
230 }
231 \cs_new:Npn \zcmd_clist_patch:nn #1#2
232 {
233     \__zcmd_clist_patch:nw {#1} #2
234     , \s__clist_patch_stop ,
235 }
236 \cs_new:Npn \__zcmd_sclist_patch:nw #1 #2;
237 {%#1:replace; #2:current
238     \tl_if_blank:nTF { #2 }

```

239	{	239
240	\exp_not:n { #1 };	240
241	__zcmd_sclist_patch:nw {#1}	241
242	}{	242
243	\bool_lazy_or:nnT	243
244	{ \int_compare_p:nNn {\tl_count:n {#2}} > {1} }	244
245	{ ! \tl_if_eq_p:NN #2\s__sclist_patch_stop }	245
246	{	246
247	\exp_not:n { #2 };	247
248	__zcmd_sclist_patch:nw {#1}	248
249	}	249
250	}	250
251	}	251
252	\cs_new:Npn \zcmd_sclist_patch:nn #1#2	252
253	{	253
254	__zcmd_sclist_patch:nw {#1} #2	254
255	; \s__sclist_patch_stop ;	255
256	}	256
257	\cs_new:Npn \zclist_item:nn #1#2	257
258	{	258
259	\int_compare:nNnTF {#2} < 0	259
260	{	260
261	\int_compare:nNnTF {#2} < { - \zclist_count:n {#1} }	261
262	{ }	262
263	{	263
264	\clist_item:en {\zcmd_clist_patch:nn {\scan_stop:}{#1}}	264
265	{ \int_eval:n { #2 + 1 + \zclist_count:n {#1} } }	265
266	}	266
267	}{	267
268	\int_compare:nNnTF {#2} > {\zclist_count:n {#1}}	268
269	{ }	269
270	{	270
271	\clist_item:en {\zcmd_clist_patch:nn {\scan_stop:}{#1}}	271
272	{ #2 }	272
273	}	273
274	}	274
275	}	275
276	\cs_new:Npn \zclist_count:n #1	276
277	{	277
278	\clist_count:e	278
279	{	279
280	\zcmd_clist_patch:nn {\scan_stop:}{#1}	280
281	}	281
282	}	282
283	\cs_generate_variant:Nn \zcmd_clist_patch:nn	283
284	{ ne, no }	284
285	\cs_generate_variant:Nn \zcmd_sclist_patch:nn	285
286	{ ne, no }	286

287		287
288	% clist item/count/range	288
289	% TODO: support negative index	289
290	\cs_new:Npn \zclist_range:nnn #1#2#3	290
291	{	291
292	\exp_args:Ne \clist_use:nn	292
293	{	293
294	\int_step_tokens:nnn {#2}{#3}	294
295	{ __zclist_range_item_aux:nnn {#1}{,} }	295
296	}{,}	296
297	}	297
298	\cs_new:Nn __zclist_range_item_aux:nnn	298
299	{	299
300	\zclist_item:nn {#1}{#3}#2	300
301	}	301
302	\cs_generate_variant:Nn \zclist_item:nn	302
303	{ on, en, ee }	303
304	\cs_generate_variant:Nn \zclist_count:n	304
305	{ e, o, f }	305
306	\cs_generate_variant:Nn \zclist_range:nnn	306
307	{ e, o }	307
308		308
309		309
310	% ==> token check and manipulations tools	310
311	% NOTE: all of these macros are expandable	311
312	% token generate	312
313	\cs_new:Npn \ztex_token_gen:nn #1#2	313
314	{	314
315	\char_generate:nn {#1}{#2}	315
316	}	316
317	\newcommand{\ztexgentoken}[2]	317
318	{	318
319	\ztex_token_gen:nn {#1}{#2}	319
320	}	320
321		321
322	% tl strip left / right / both:	322
323	\cs_generate_variant:Nn \tl_tail:n {e}	323
324	\cs_new:Npn \ztex_token_strip_both:n #1	324
325	{	325
326	\tl_range:nnn {#1}{2}{-2}	326
327	}	327
328	\cs_new_eq:NN \ztex_token_strip_left:n \tl_tail:n	328
329	\cs_new:Npn \ztex_token_strip_right:n #1	329
330	{	330
331	\tl_range:nnn {#1}{1}{-2}	331
332	}	332
333	\cs_generate_variant:Nn \ztex_token_strip_both:n	333
334	{ e, V }	334

335	\cs_generate_variant:Nn \ztex_token_strip_left:n	335
336	{ e, V }	336
337	\cs_generate_variant:Nn \ztex_token_strip_right:n	337
338	{ e, V }	338
339		339
340	% token if chinese check	340
341	% REF: https://tex.stackexchange.com/q/156792/294585	341
342	\prg_new_conditional:Npnn \zcmd_token_if_chinese:n #1	342
343	{ p, T, F, TF }	343
344	{	344
345	\int_compare:nNnTF { `#1 } > { 19968 }	345
346	{ \prg_return_true: }	346
347	{ \prg_return_false: }	347
348	}	348
349		349
350		350
351	% >> token(s) equal check:	351
352	% single token if eq check -- 'nn'-signature:	352
353	\prg_new_conditional:Npnn __ztex_token_if_eq:nn #1#2	353
354	{ T, F, TF }	354
355	{	355
356	\bool_xor:nnT	356
357	{ \tl_if_empty_p:n {#1} }	357
358	{ \tl_if_empty_p:n {#2} }	358
359	{ \prg_return_false: }	359
360	\exp_args:Ne \bool_lazy_any:nT	360
361	{	361
362	{ \int_compare_p:n {\tl_count:n {#1}>1} }	362
363	{ \int_compare_p:n {\tl_count:n {#2}>1} }	363
364	}{	364
365	\ztex_msg_set:nn {zcmd@token@check}	365
366	{	366
367	Either~of~the~tokens~is~not~single,~	367
368	input~tokens~are(without~outer~brace):	368
369	\iow_newline:\#1(target)={\exp_not:n {#1}},	369
370	\iow_newline:\#2(test)={\exp_not:n {#2}}.	370
371	}	371
372	\ztex_msg_error:n {zcmd@token@check}	372
373	}	373
374	\tl_if_eq:NNTF #1#2	374
375	{ \prg_return_true: }	375
376	{ \prg_return_false: }	376
377	}	377
378	\prg_generate_conditional_variant:Nnn __ztex_token_if_eq:nn	378
379	{ e, ne, ee }{ T, F, TF }	379
380		380
381	% token of index if eq	381
382	\prg_new_conditional:Npnn \ztex_index_token_if_eq:nnn #1#2#3	382

```

383 { p, T, F, TF }
384 {% #1:tl; #2:index; #3:token
385   \__ztex_token_if_eq:neTF {#3}{\tl_item:nn {#1}{#2}}
386   {
387     \prg_return_true:
388   }{
389     \prg_return_false:
390   }
391 }
392
393 % head/tail tokens if eq
394 \prg_new_conditional:Npnn \ztex_head_tail_token_if_eq:nnn #1#2#3
395 { p, T, F, TF }
396 {% #1:tl; #2:head; #3:tail
397   \__ztex_token_if_eq:neTF {#2}{\tl_item:nn {#1}{1}}
398   {
399     \__ztex_token_if_eq:neTF {#3}{\tl_item:nn {#1}{-1}}
400     { \prg_return_true: }
401     { \prg_return_false: }
402   }
403   { \prg_return_false: }
404 }
405 \prg_generate_conditional_variant:Nnn \ztex_head_tail_token_if_eq:nnn
406 { e, nee, eee }{ p, T, F, TF }
407 \prg_generate_conditional_variant:Nnn \ztex_index_token_if_eq:nnn
408 { e, nee, eee }{ p, T, F, TF }
409
410 % multi-tokens if eq check:
411 \prg_new_conditional:Npnn \ztex_tl_if_eq:nn #1#2
412 { p, T, F, TF }
413 {
414   \exp_args:Ne \int_compare:nTF {\tl_count:n {#1} = \tl_count:n {#2}}
415   {
416     \exp_args:Ne \int_compare:nTF {
417       \exp_not:N \int_from_bin:n { \__ztex_tl_if_eq_aux:nn {#1}{#2} }
418       =
419       \exp_not:N \int_from_bin:n { \prg_replicate:nn {\tl_count:n {#1}}{1} }
420     }{ \prg_return_true: }{ \prg_return_false: }
421   }{ \prg_return_false: }
422 }
423 \cs_new:Npn \__ztex_tl_if_eq_aux:nn #1#2
424 {
425   \exp_args:Ne \int_compare:nTF {\tl_count:n {#1} = \tl_count:n {#2}}
426   {
427     \int_step_tokens:nn {\tl_count:n {#1}}
428     {
429       \__ztex_tl_if_eq_aux_iii:nnnnn {#1}{#2}
430       { 1 } { 0 }

```

```

431         }
432     }{ 0 }
433 }
434 \prg_new_conditional:Npnn \__ztex_tl_if_eq_aux_ii:nnn #1#2#3
435 { T, F, TF }
436 {
437     \exp_args:Nee \__ztex_token_if_eq:nnTF
438     {\tl_item:nn {#1}{#3}}{\tl_item:nn {#2}{#3}}
439     { \prg_return_true: }
440     { \prg_return_false: }
441 }
442 \cs_new:Npn \__ztex_tl_if_eq_aux_iii:nnnnn #1#2#3#4#5
443 {
444     \__ztex_tl_if_eq_aux_ii:nnnTF {#1}{#2}{#5}{#3}{#4}
445 }
446 \prg_generate_conditional_variant:Nnn \ztex_tl_if_eq:nn
447 { e, ne, ee }{ p, T, F, TF }
448
449
450 % >> token if in check
451 % single token if in:
452 % NOTE: including implicit token '\l_peek_token'
453 \prg_new_conditional:Npnn \ztex_token_if_in:nN #1#2 { p, T, F, TF }
454 {
455     \exp_args:Nee \int_compare:nNnTF
456     {
457         \exp_args:Ne \int_from_bin:n
458         {
459             \tl_map_tokens:nn { #1 }
460             { \ztex_token_if_eq:NN #2 }
461         }
462     } = { 0 }
463     { \prg_return_false: }
464     { \prg_return_true: }
465 }
466 \cs_new:Npn \ztex_token_if_eq:NN #1#2
467 { \tex_ifx:D #1 #2 1 \else: 0 \fi: }
468 \prg_generate_conditional_variant:Nnn \ztex_token_if_in:nN
469 { o, e }{ p, T, F, TF }
470
471 % multi-tokens if in:
472 \prg_new_conditional:Npnn \ztex_tl_if_in:nn #1#2
473 { p, T, F, TF }
474 {
475     \exp_args:Ne \int_step_tokens:nn { \tl_count:n {#1}}-\tl_count:n {#2}+1 }
476     {
477         \__ztex_tl_if_in_aux:nnnn { #1 }{ #2 }
478         {

```



```

479         \prg_map_break:Nn \int_step_break:
480         { \prg_return_true: }
481     }
482 }
483 \prg_return_false:
484 \prg_break_point:Nn \int_step_break: { }
485 }
486 \cs_new:Npn \__ztex_tl_if_in_aux:nnnn #1#2#3#4
487 {
488     \exp_args:Ne \ztex_tl_if_eq:nnTF
489     { \tl_range:nnn {#1}{#4}{#4+\tl_count:n {#2}-1} }{ #2 }
490     { #3 }{ }
491 }
492 \prg_generate_conditional_variant:Nnn \ztex_tl_if_in:nn
493 { no, ne, ee }{ p, T, F, TF }
494
495 % special ':' token check:
496 \prg_new_conditional:Npnn \ztex_colon_if_in:n #1 {p, T, F, TF}
497 {
498     \ztex_tl_if_in:nnTF {#1}{:}
499     { \prg_return_true: }
500     { \prg_return_false: }
501 }
502 \prg_generate_conditional_variant:Nnn \ztex_colon_if_in:n
503 {e, V}{T, F, TF}
504
505
506 % >> tl replace (which is expandable)
507 \cs_generate_variant:Nn \tl_range:nnn
508 { nne, nen, nee }
509 \cs_new:Npn \ztex_tl_pattern_pos:nn #1#2
510 {
511     \exp_args:Ne \int_step_tokens:nn
512     { \tl_count:n {#1}-\tl_count:n {#2}+1 }
513     {
514         \__ztex_tl_pattern_pos:nnn { #1 }{ #2 }
515     };
516 }
517 \cs_new:Npn \__ztex_tl_pattern_pos:nnn #1#2#3
518 {
519     \exp_args:Ne \ztex_tl_if_eq:nnTF
520     { \tl_range:nnn {#1}{#3}{#3+\tl_count:n {#2}-1} }
521     { #2 }
522     { ;#3, \int_eval:n {#3+\tl_count:n {#2}-1} }
523     { }
524 }
525
526 % extract tokens by pattern positions sclist, e.g.,

```

```
527 % positions = ';1,2;5,6;' 527
528 \cs_new:Npn \ztex_tl_replace:nnnn #1#2#3#4 528
529 { % #1:ori tl; #2:pattern pos sclist; 529
530 % #3:replacement; #4:index in pattern pos 530
531 % very proceeding tokens: 531
532 \int_compare:nNnT { #4 } = { 1 } 532
533 { 533
534 \tl_range:nnn { #1 } { 1 } 534
535 { \clist_item:en {\sclist_item:nn {#2}{#4}}{1}-1 } 535
536 } 536
537 % replacement: 537
538 \exp_not:n { #3 } 538
539 % trailing tokens: 539
540 \tl_if_empty:eTF { \sclist_item:nn {#2}{#4+1} } 540
541 { 541
542 \int_compare:nNnF 542
543 { 543
544 \clist_item:en {\sclist_item:nn {#2}{#4}}{2}+1 544
545 } > { \tl_count:n {#1} } 545
546 { 546
547 \tl_range:nee { #1 } 547
548 { 548
549 \clist_item:en {\sclist_item:nn {#2}{#4}}{2}+1 549
550 } { -1 } 550
551 } 551
552 } { 552
553 \tl_range:nee { #1 } 553
554 { 554
555 \clist_item:en {\sclist_item:nn {#2}{#4}}{2}+1 555
556 } { 556
557 \clist_item:en {\sclist_item:nn {#2}{#4+1}}{1}-1 557
558 } 558
559 } 559
560 } 560
561 \cs_new:Npn \ztex_tl_replace_once:nnn #1#2#3 561
562 { 562
563 \exp_args:Nne \ztex_tl_replace:nnnn { #1 } 563
564 { 564
565 \sclist_item:en { \ztex_tl_pattern_pos:nn {#1}{#2} } 565
566 { 1 } 566
567 } { #3 } { 1 } 567
568 } 568
569 \cs_new:Npn \ztex_tl_replace_all:nnn #1#2#3 569
570 { 570
571 \int_step_tokens:nn 571
572 { 572
573 \sclist_count:e {\ztex_tl_pattern_pos:nn {#1}{#2}} 573
574 } { 574
```

```
575 \exp_args:Nne \ztex_tl_replace:nnnn {#1} 575
576 { 576
577 \ztex_tl_pattern_pos:nn {#1}-{#2} 577
578 }{ #3 } 578
579 } 579
580 } 580
581 \cs_new:Npn \ztex_tl_replace_cnt:nnnn #1#2#3#4 581
582 { 582
583 \int_step_tokens:nn { #1 } 583
584 { 584
585 \exp_args:Nne \ztex_tl_replace:nnnn {#2} 585
586 { 586
587 \int_step_tokens:nn { #1 } 587
588 { 588
589 ; \sclist_item:en 589
590 { \ztex_tl_pattern_pos:nn {#2}-{#3} } 590
591 } ; 591
592 }{ #4 } 592
593 } 593
594 } 594
595 \cs_generate_variant:Nn \ztex_tl_replace_once:nnn 595
596 { o, e, noo, nee, eee } 596
597 \cs_generate_variant:Nn \ztex_tl_replace_all:nnn 597
598 { o, e, noo, nee, eee } 598
599 \cs_generate_variant:Nn \ztex_tl_replace_cnt:nnnn 599
600 { no, ne, nooo, neee, eeee } 600
601 601
602 602
603 % ==> implement an expandable '\prop_item:nn' 603
604 \cs_new:Npn \prop_item:nn #1#2 604
605 { 605
606 \use:e { 606
607 \_zcmd_prop_item_aux_i:nn {#1}-{#2} 607
608 } 608
609 } 609
610 \cs_new:Npn \_zcmd_prop_item_aux_i:nn #1#2 610
611 { 611
612 \keyval_parse:nnn 612
613 { \use_none:n } 613
614 { \_zcmd_prop_item_aux_ii:nnn {#2} } 614
615 { #1 } 615
616 } 616
617 \cs_new:Npn \_zcmd_prop_item_aux_ii:nnn #1#2#3 617
618 { 618
619 \str_if_eq:nnT { #1 }{ #2 } 619
620 { \exp_not:n {#3} } 620
621 } 621
622 \cs_generate_variant:Nn \prop_item:nn 622
```

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10.3.10 item

1	<code>\ProvidesExplFile{ztex.module.item.tex}</code>	1
2	<code>{2025/07/05}{\ztex@versi@n}</code>	2
3	<code>{item~module~for~ztex}</code>	3
4		4
5		5
6	<code>%%%% item module for ztex %%%%</code>	6
7	<code>\renewcommand{\labelitemii}{\(\circ\)}</code>	7
8	<code>\renewcommand{\labelitemiii}{\(\diamond\)}</code>	8

10.3.11 counter

```
1 \ProvidesExplFile{ztex.module.counter.tex} 1
2 {2025/09/23}{\ztex@version} 2
3 {counter~module~for~ztex} 3
4 4
5 5
6 %%%% counter module for ztex %%%% 6
7 \NewDocumentCommand{\ztxcntwith}{mm}{\@addtoreset{#1}{#2}} 7
8 8
9 9
10 % ==> counter spec 10
11 % \setcounter{secnumdepth}{3} 11
12 \setcounter{tocdepth}{3} 12
13 \setcounter{secnumdepth}{3} 13
14 \counterwithin{equation}{section} 14
15 15
16 16
17 % ==> counter form 17
18 \cs_new:Npn \zcnt_to_arabic:N #1 18
19 { 19
20 \exp_after:wN \def \cs:w the#1 \cs_end: 20
21 { \exp_args:Ne \arabic{#1} } 21
22 } 22
23 \cs_generate_variant:Nn \zcnt_to_arabic:N { c } 23
24 24
25 % create counter safely 25
26 \cs_new_protected:Npn \zcnt_safe_new:nn #1#2 26
27 { 27
28 \cs_if_exist:cF { c@#1 } 28
29 { 29
30 \tl_if_empty:nTF { #2 } 30
31 { 31
32 \newcounter{ #1 } 32
33 }{ 33
34 \newcounter{ #1 }[ #2 ] 34
35 } 35
36 } 36
37 } 37
38 \cs_generate_variant:Nn \zcnt_safe_new:nn 38
39 { oo, ee } 39
40 \NewDocumentCommand{\zcntnew}{mO{}} 40
41 { 41
42 \zcnt_safe_new:nn {#1}{#2} 42
43 } 43
```

10.3.12 graphics

```
1 \ProvidesExplFile{ztex.module.graphics.tex} 1
2 {2026/02/16}{\ztex@versi@n} 2
3 {graphics~module~for~ztex} 3
4 4
5 5
6 %%%% graphics module for ztex %%%% 6
7 \RequirePackage{graphicx} 7
8 \cs_if_exist:NF \graphics_include:nn 8
9 { 9
10 \RequirePackage{l3graphics} 10
11 } 11
12 12
13 13
14 % graphics path 14
15 \clist_const:Nn \c__ztex_graphics_path_clist 15
16 { 16
17 ./picture/, ./pictures/, 17
18 ./graphic/, ./graphics/, 18
19 ./figure/, ./figures/, 19
20 ./image/, ./images/, 20
21 ./pic/, ./pics/, 21
22 } 22
23 \clist_map_inline:Nn \c__ztex_graphics_path_clist 23
24 { 24
25 \graphicspath{{#1}} 25
26 \seq_put_right:Nn 26
27 \l_graphics_search_path_seq 27
28 { #1 } 28
29 } 29
30 30
31 % include graphics(expl3) 31
32 \NewDocumentCommand{\zgraphicsinclude}{0+{m}} 32
33 { 33
34 \graphics_include:nn 34
35 { #1 } 35
36 { #2 } 36
37 } 37
38 38
39 % count pdf graphics pages 39
40 \tl_new:N \zgraphicspageint 40
41 \NewDocumentCommand{\zgraphicspagecnt}{m} 41
42 { 42
43 \graphics_get_pagecount:nN {#1} 43
44 \zgraphicspageint 44
45 } 45
46 46
```

47	% if exsit graphics include	47
48	\NewDocumentCommand\includegraphicsifexsit{0{width=.75\linewidth}0{example-image-a}m}	48
49	{	49
50	\file_if_exist:nTF {#3}	50
51	{ \includegraphics[#1]{#3} }	51
52	{ \includegraphics[#1]{#2} }	52
53	}	53

10.4 Library

10.4.1 fancy

```
1 \ProvidesExplFile{ztex.library.fancy.tex} 1
2 {2025/09/24}{\ztex@version} 2
3 {fancy~library~for~ztex} 3
4 4
5 5
6 %%%% fancy library for ztex %%%% 6
7 \RequirePackage{anyfontsize} 7
8 \bool_gset_true:N \g__ztex_fancy_bool 8
9 \newif\ifloadtikz 9
10 \bool_if:NTF \g__ztex_fancy_bool 10
11 { 11
12 \RequirePackage[many]{tcolorbox} 12
13 \loadtikztrue 13
14 }{ \loadtikzfalse } 14
15 \ExplSyntaxOff\ifloadtikz 15
16 \RequirePackage{tikz} 16
17 \usetikzlibrary{calc} 17
18 \fi\ExplSyntaxOn 18
19 19
20 20
21 % ==> fancy chapter 21
22 \zsecformat\chapter 22
23 { 23
24 explicit = true, 24
25 code = { 25
26 \newpage \thispagestyle{empty} 26
27 \begin{tikzpicture}[overlay, remember~ picture] 27
28 % mark nodes (need 'calc' library) 28
29 \coordinate (A) at ($(current~ page.north~ west)+(.125\paperwidth, 0pt)$); 29
30 \coordinate (stripES) at ($(A)+(5em, -.25\paperheight)$); 30
31 % chapter head 31
32 \fill[\l__ztex_fancy_chap_color_tl] (A) rectangle (stripES); 32
33 \draw[draw=\l__ztex_fancy_chap_color_tl] (stripES)++(.25em, 4em) -- ++(.75 33
\paperwidth-3.25em, 0pt);
34 \draw[draw=\l__ztex_fancy_chap_color_tl] (stripES)++(.25em, 1.5pt) -- ++(.75 34
\paperwidth-3.25em, 0pt);
35 \draw[draw=\l__ztex_fancy_chap_color_tl] (stripES)++(.25em, 0em) -- ++(.75 35
\paperwidth-3.25em, 0pt);
36 % chapter title and index 36
37 \node[anchor=south, color=white] at ($(stripES)+(-2.5em, 0em)$) 37
38 { 38
39 \normalsize\scalebox{4}{\arabic{chapter}} 39
40 \exp_args:Ne \thmark{\thechapter} 40
41 }; 41
42 \node[anchor=south~ west, inner~ sep=0pt, 42
```

```

43         yshift=4.25em, xshift=.25em,
44         color=\l__ztex_fancy_chap_color_tl,
45         font=\Large\bfseries,
46     ] at (stripES) {\l__zfancy_chap_subtitle_tl};
47 \node[anchor=south~ west, inner~ sep=0pt,
48       yshift=1.25em, xshift=.25em,
49       color=\l__ztex_fancy_chap_color_tl,
50       font=\cinzel\Huge\bfseries,
51     ] at (stripES) {\zsecname};
52 % parbox insert
53 \node[anchor=north~ west, inner~ sep=0pt] at ($(stripES)+(-5em, -1em)$)
54 {
55     \parbox[t]{.3\paperwidth}{\fontsize{10pt}{15pt}
56       \selectfont\cinzel\itshape\l__zfancy_chap_lcontent_tl}
57 };
58 \node[anchor=north~ west, inner~ sep=0pt] at ($(stripES)+(-5em+.45em+.3\paperwidth,
-1em)$)
59 {
60     \parbox[t]{\dimeval{.45\paperwidth-.45em}}{
61       \fontsize{10pt}{15pt}\selectfont\l__zfancy_chap_rcontent_tl}
62 };
63 % saying block
64 \coordinate (sayingWN) at ($(current~ page.south~ west)+(0, .3\paperheight)$);
65 \shade[top~ color=white, bottom~ color=\l__ztex_fancy_chap_color_tl!25] (sayingWN)
66   rectangle ++(1\paperwidth, 5pt);
67 \shade[top~ color=\l__ztex_fancy_chap_color_tl!25, bottom~ color=white]
68   ($(sayingWN)+(0em, -.15\paperheight)$)
69   rectangle ++(1\paperwidth, -5pt);
70 \node at ($(sayingWN)+(.5\paperwidth, -0.075\paperheight)$)
71 {
72     \parbox[t]{}[r]{.75\paperwidth}
73     {
74         \fontsize{15pt}{22.5pt}\selectfont
75         \MakeUppercase{\cinzel\l__zfancy_chap_saying_tl\
76         \hspace*{\fill}{\itshape\normalsize\l__zfancy_chap_sayauthor_tl}}
77     }
78 };
79 \end{tikzpicture}
80 \newpage
81 }
82 }
83 \prop_new:N \g_arabic_suffix_prop
84 \prop_set_from_keyval:Nn \g_arabic_suffix_prop
85 {
86     0=th, 1=st, 2=nd, 3=rd,
87     11=th, 12=th, 13=th, _=th,
88 }
89 \NewDocumentCommand\thmark{m}

```

90	{	90
91	\int_compare:nTF { 11 <= #1 <= 13 }	91
92	{ \prop_item:Ne \g_arabic_suffix_prop {#1} }	92
93	{	93
94	\int_compare:nTF {\int_mod:nn {#1}{10} > 3}	94
95	{\prop_item:Ne \g_arabic_suffix_prop {_}}	95
96	{\prop_item:Ne \g_arabic_suffix_prop {\int_mod:nn {#1}{10}}}	96
97	}	97
98	}	98
99		99
100	% fancy chapter page text	100
101	\ztex_keys_define:nn { fancy/chap/text }	101
102	{	102
103	subtitle .tl_set:N = \l_zfancy_chap_subtitle_tl,	103
104	subtitle .initial:n = { SUBTITLE },	104
105	saying .tl_set:N = \l_zfancy_chap_saying_tl,	105
106	saying .initial:n = { SAYING },	106
107	sayauthor .tl_set:N = \l_zfancy_chap_sayauthor_tl,	107
108	sayauthor .initial:n = { SAY-AUTHOR },	108
109	rcontent .tl_set:N = \l_zfancy_chap_rcontent_tl,	109
110	rcontent .initial:n = { R-CONTENT },	110
111	lcontent .tl_set:N = \l_zfancy_chap_lcontent_tl,	111
112	lcontent .initial:n = { L-CONTENT },	112
113	}	113
114	\NewDocumentCommand{\zfancychapset}{m}	114
115	{	115
116	\ztex_keys_set:nn { fancy/chap/text }{ #1 }	116
117	}	117

10.4.2 cmd

```

1 \ProvidesExplFile{ztex.library.cmd.tex}
2 {2025/09/30}{\ztex@versi@n}
3 {cmd~library~for~ztex}
4
5
6 % scratch variables setup:
7 \tl_new:N \l__ztex_cmd_args_tl
8 \int_new:N \l__ztex_cmd_argcnt_int
9 \str_new:N \l__ztex_cmd_name_str
10
11 % ztex cmd kernel
12 \cs_set_eq:NN \fpuse \fp_to_tl:n
13 \cs_set_eq:NN \intuse \int_eval:n
14 \cs_set_eq:NN \dimuse \dim_eval:n
15 \cs_set_eq:NN \cmdvar \use:c
16 % TODO: implement negative index for list argument.
17 \cs_set:Npn \clistuse #1#2
18 { \clist_item:Nn #1{#2} }
19 \cs_new_protected:Npn \ztex_cmd_create:nnnnn #1#2#3#4#5
20 {% #1:cmd name; #2:arg-spec(default as 'tl'); #3:code;
21 % #4:cmd-type; #5:evaluate bool.
22 % parse arg-spec
23 \int_set:Nn \l__ztex_cmd_argcnt_int {\clist_count:n {#2}}
24 \str_set:Nn \l__ztex_cmd_name_str {#1}
25 % create cmd
26 \cs_generate_from_arg_count:ccnn {#1}{#4}{1}
27 {
28 \group_begin:
29 \keyval_parse:nnn
30 { \__ztex_cmd_extract_var:nn {#5} }
31 { \__ztex_cmd_extract_var_default:nnn {#5} }
32 { #2 }
33 \keys_set:nn { ztex/cmd/#1 }{ ##1 }
34 #3
35 \group_end:
36 }
37 }
38 \cs_generate_variant:Nn \cs_generate_from_arg_count:NNnn {ccnn}
39 \cs_set:Npn \__ztex_cmd_extract_var:nn #1#2
40 {
41 % \exp_after:wN \def\cs:w#1\cs_end:{
42 \__ztex_cmd_arg_type_check:n { #2 }
43 \__ztex_cmd_keys_parser:eenn
44 { \exp_after:wN \__ztex_cmd_arg_name:w \l__ztex_cmd_args_tl \scan_stop: }
45 { \exp_after:wN \__ztex_cmd_arg_type:w \l__ztex_cmd_args_tl \scan_stop: }
46 { zCMD@EMPTY }{ #1 }

```

```

47 }
48 \cs_set:Npn \__ztex_cmd_extract_var_default:nnn #1#2#3
49 {% #1=<name>:<type>
50 \__ztex_cmd_arg_type_check:n { #2 }
51 \__ztex_cmd_keys_parser:eenn
52 { \exp_after:wN \__ztex_cmd_arg_name:w \l__ztex_cmd_args_tl \scan_stop: }
53 { \exp_after:wN \__ztex_cmd_arg_type:w \l__ztex_cmd_args_tl \scan_stop: }
54 { #3 }{ #1 }
55 }
56 \cs_new:Npn \__ztex_cmd_arg_type_check:n #1
57 {
58 \tl_set_rescan:Nne \l__ztex_cmd_args_tl
59 {
60 \cctab_select:N \c_document_cctab
61 \char_set_catcode_letter:n { 58 }
62 }{ #1 }
63 \tl_set:No \l__ztex_cmd_args_tl
64 {
65 \l__ztex_cmd_args_tl
66 }
67 \tl_put_right:Ne \l__ztex_cmd_args_tl
68 { \ztex_colon_if_in:eF {\l__ztex_cmd_args_tl}{:tl} }
69 }
70 \cs_new:Npn \__ztex_cmd_arg_name:w #1:#2\scan_stop:
71 { #1 }
72 \cs_new:Npn \__ztex_cmd_arg_type:w #1:#2\scan_stop:
73 { #2 }
74 \cs_generate_variant:Nn \clist_map_function:nN { nc, vc }
75 \cs_new:Npn \__ztex_cmd_keys_parser:nnnn #1#2#3#4
76 {% #1:key-name; #2:type; #3:default; #4:evaluate bool.
77 \exp_args:Nee \keys_define:nn { ztex/cmd/\l__ztex_cmd_name_str }
78 {
79 \ztex_head_tail_token_if_eq:ennTF {#2}{[]{} }
80 {
81 #1 .code:n =
82 {
83 \cs_set:Npe \exp_not:c {#1} ####1
84 {
85 \bool_if:nTF { #4 }
86 {
87 \exp_not:N \exp_not:N \exp_not:N \clist_item:nn
88 {
89 \exp_not:N \__zcmd_list_arg_handle:nn
90 { \zcmd_map_expnot_clist:n {##1} }
91 { \ztex_token_strip_both:n {#2} }
92 }{ ####1 }
93 }
94 {

```

```

95         \exp_not:N \exp_not:N \exp_not:N
96         \clist_item:nn
97         { \zcmd_map_expnot_clist:n {##1} }
98         { ####1 }
99     }
100 }
101 },
102 }
103 {
104     #1 .#2_set:c = { #1 },
105 }
106 % NOTE: we do NOT use '\zcmd_map_expnot_clist:n' here.
107 #1 .initial:n = { \exp_not:n {#3} },
108 }
109 }
110 \cs_generate_variant:Nn \__ztex_cmd_keys_parser:nnnn
111 { ee }
112 % vector(list) syntax for ztexpd arg-spec
113 \cs_set:Npn \__zcmd_list_arg_handle:nn #1#2
114 { % #1:list; #2:type
115     \exp_args:No \clist_map_function:nc { #1 }
116     {
117         __zcmd_list_arg_#2 :n
118     }
119 }
120 \cs_set:Npn \__zcmd_list_arg_int:n #1
121 { \int_eval:n {#1}, }
122 \cs_set:Npn \__zcmd_list_arg_fp:n #1
123 { \fp_eval:n {#1}, }
124 \cs_set:Npn \__zcmd_list_arg_str:n #1
125 { \tl_to_str:n {#1}, }
126 \cs_set:Npn \__zcmd_list_arg_dim:n #1
127 { \dim_eval:n {#1}, }
128 \cs_set:Npn \__zcmd_list_arg_tl:n #1
129 { \exp_not:n {#1}, }
130
131
132 % ==> users' interface
133 % TOTAL 8 types in theory -->
134 % (set, new) x (fragile, robust)
135 % x (long, short) x (local, global);
136 % NOTE: all of the commands defined by '\ztexdef' is
137 % 1. robust,
138 % 2. long,
139 \cs_set_protected:Npn \znewcmd #1#2#3
140 {
141     \cs_if_exist:NT {#1}
142     {

```

143	\ztex_msg_set:nn {znewcmd@exist}	143
144	{	144
145	command~\string#1~already~exsits!	145
146	}	146
147	\ztex_msg_error:n {znewcmd@exist}	147
148	}	148
149	\exp_args:Ne \ztex_cmd_create:nnnnn {\cs_to_str:N #1}{#2}	149
150	{	150
151	#3	151
152	}{cs_new:Npn}{\c_false_bool}	152
153	}	153
154	\cs_set_protected:Npn \zsetcmd #1#2#3	154
155	{	155
156	\exp_args:Ne \ztex_cmd_create:nnnnn {\cs_to_str:N #1}{#2}	156
157	{	157
158	#3	158
159	}{cs_set:Npn}{\c_false_bool}	159
160	}	160
161	\cs_set_protected:Npn \zgsetcmd #1#2#3	161
162	{	162
163	\exp_args:Ne \ztex_cmd_create:nnnnn {\cs_to_str:N #1}{#2}	163
164	{	164
165	#3	165
166	}{cs_gset:Npn}{\c_false_bool}	166
167	}	167

10.4.3 alias

```
1 \ProvidesExplFile{ztex.library.alias.tex} 1
2 {2026/02/14}{\ztex@versi@n} 2
3 {alias~library~for~ztex} 3
4 4
5 5
6 %%%% alias library for ztex %%%% 6
7 \bool_gset_true:N \g__ztex_math_alias_bool 7
8 \RequirePackage{amssymb, mathrsfs} 8
9 \RequirePackage{mathtools} 9
10 10
11 11
12 \ztex_msg_set:nn { expl-too-old@alias } 12
13 { 13
14 *~Matrix-related~aliases~from~the~'alias'~library~are~not~ 14
15 available~in~your~TeX~distribution.  \ 15
16 *~Please~install~an~TeX~distribution~up~to~'January~15,~2025'~or~ 16
17 update~using~your~TeX~package~manager~or~from~CTAN~to~use~it.  \ 17
18 *~See~zTeX~documentation.~Loading~matrix-related~aliases~from~ 18
19 'alias'~library~will~abort! 19
20 } 20
21 21
22 22
23 % ==> copy the original cs from hash table 23
24 \__zcmd_cs_copy:cc {z@ltx@s}{S} 24
25 \__zcmd_cs_copy:cc {z@ltx@div}{div} 25
26 \__zcmd_cs_copy:cc {z@ltx@hom}{hom} 26
27 27
28 28
29 % ==> Alias switch on/off 29
30 \bool_new:N \g__ztex_math_alias_switch_bool % for future use 30
31 \bool_gset_false:N \g__ztex_math_alias_switch_bool 31
32 \seq_new:N \g__ztex_mathalias_user_seq 32
33 \seq_new:N \g__ztex_mathalias_internal_seq 33
34 \seq_new:N \g__ztex_mathalias_protected_seq 34
35 \seq_gclear:N \g__ztex_mathalias_user_seq 35
36 \seq_gclear:N \g__ztex_mathalias_internal_seq 36
37 \seq_gclear:N \g__ztex_mathalias_protected_seq 37
38 38
39 \cs_set_protected:Npn \__zalias_init: 39
40 { 40
41 \char_set_mathcode:nn {"2F}{ "413D} % for '/' in 'fixdif' 41
42 \bool_gset_true:N \g__ztex_math_alias_switch_bool 42
43 \seq_gset_from_clist:NN \g__ztex_mathalias_user_seq 43
44 \g__ztex_mathalias_user_clist 44
45 \seq_gset_from_clist:NN \g__ztex_mathalias_internal_seq 45
46 \g__ztex_mathalias_internal_clist 46
```



```
47 }
48 \cs_set_protected:Npn \__zalias_cmd_create:n #1
49 {
50   \seq_map_indexed_inline:Nn \g__ztex_mathalias_user_seq
51   {
52     \cs_if_exist:cT {##2}
53     {
54       \seq_gput_right:Nn \g__ztex_mathalias_protected_seq {##2}
55       \__zcmd_cs_copy:cc {z@ltx@##2}{##2} % store the original
56       \__zcmd_cs_copy:cc {#1##2}{##2} % for tmp usage
57     }
58     \cs_set_protected:cpe {##2}
59     {
60       \exp_not:N \cs:w
61       \seq_item:Nn \g__ztex_mathalias_internal_seq
62       { ##1 }
63       \exp_not:N \cs_end:
64     }
65   }
66 }
67 \cs_set_protected:Nn \__zalias_delete:
68 {
69   \seq_map_inline:Nn \g__ztex_mathalias_user_seq
70   {
71     \seq_if_in:NnF \g__ztex_mathalias_protected_seq {##1}
72     {
73       \cs_undefine:c {##1}
74     }
75   }
76 }
77
78 \NewDocumentCommand{\zaliasError}{}
79 {
80   \ztex_msg_set:nn {math-alias-cmd}{
81     Math~alias~related~commands~only~available~
82     between~'\zaliasOn'~and~'\zaliasOff'~
83     or~in~the~environment~'zalias'
84   }
85   \ztex_msg_error:n {math-alias-cmd}
86 }
87 \NewDocumentCommand{\zaliasOn}{0{OLD}}
88 {
89   \__zalias_init:
90   \__zalias_cmd_create:n {#1}
91 }
92 \NewDocumentCommand{\zaliasOff}{}
93 {
94   \__zalias_delete:
```

95	\bool_gset_false:N \g__ztex_math_alias_switch_bool	95
96	}	96
97	\NewDocumentEnvironment{zalias}{0{OLD}}	97
98	{	98
99	\group_begin:	99
100	\bool_gset_true:N \g__ztex_math_alias_switch_bool	100
101	__zalias_cmd_create:n { #1 }	101
102	}{	102
103	\bool_gset_false:N \g__ztex_math_alias_switch_bool	103
104	\group_end:	104
105	}	105
106		106
107		107
108	% ==> mathalias commands setup interface	108
109	\clist_new:N \g__ztex_mathalias_user_clist	109
110	\clist_new:N \g__ztex_mathalias_internal_clist	110
111	\clist_gclear:N \g__ztex_mathalias_user_clist	111
112	\clist_gclear:N \g__ztex_mathalias_internal_clist	112
113	\cs_new:Npn \ztex_mathalias_set:nn #1#2	113
114	{% #1:the users' interface; #2: the internal interface	114
115	\clist_put_right:Nn \g__ztex_mathalias_user_clist {#1}	115
116	\clist_put_right:Nn \g__ztex_mathalias_internal_clist {#2}	116
117	}	117
118	\cs_generate_variant:Nn \ztex_mathalias_set:nn { ee, oo }	118
119		119
120		120
121	% ==> make text and math commands robust	121
122	\cs_new:Npn \zalias_make_cmd_robust:n #1	122
123	{	123
124	__zcmd_cs_copy:cc {z@ltx@#1}{#1}	124
125	\ztex_mathalias_set:nn {#1}{z@ltx@#1}	125
126	% \exp_after:wN \tex_protected:D \exp_after:wN	126
127	% \def\cs:w #1\cs_end:{\cs:w z@ltx@#1\cs_end:} % --> works	127
128	}	128
129	\cs_generate_variant:Nn \zalias_make_cmd_robust:n { e, o, f }	129
130	\zalias_make_cmd_robust:n {mathrm}	130
131	\zalias_make_cmd_robust:n {mathbf}	131
132	\zalias_make_cmd_robust:n {mathfrak}	132
133	\zalias_make_cmd_robust:n {mathcal}	133
134	\zalias_make_cmd_robust:n {mathscr}	134
135	\zalias_make_cmd_robust:n {mathbb}	135
136	\zalias_make_cmd_robust:n {textrm}	136
137	\zalias_make_cmd_robust:n {textbf}	137
138	\zalias_make_cmd_robust:n {textsf}	138
139	\zalias_make_cmd_robust:n {textsc}	139
140	\zalias_make_cmd_robust:n {textsl}	140
141	\zalias_make_cmd_robust:n {textit}	141
142		142

```

143 % ==> Math Font
144 % ==> Math Font
145 \DeclareRobustCommand{\z@R}[1]{\ensuremath{\mathrm{#1}}}
146 \DeclareRobustCommand{\z@K}[1]{\ensuremath{\mathfrak{#1}}}
147 \DeclareRobustCommand{\z@C}[1]{\ensuremath{\mathcal{#1}}}
148 \DeclareRobustCommand{\z@B}[1]{\ensuremath{\mathbb{#1}}}
149 \DeclareRobustCommand{\z@S}[1]{\ensuremath{\mathscr{#1}}}
150 \DeclareRobustCommand{\z@F}[1]{\ensuremath{\boldsymbol{#1}}}
151 \DeclareRobustCommand{\z@FF}[1]{\ensuremath{\mathbf{#1}}}
152 \ztex_mathalias_set:nn
153 { R, K, C, B, S, F, FF }
154 { z@R, z@K, z@C, z@B, z@S, z@F, z@FF }
155
156
157 % ==> Math Arrow
158 % simple arrow
159 \prop_new:N \g_ztex_math_simple_arrow_prop
160 \prop_set_from_keyval:Nn \g_ztex_math_simple_arrow_prop
161 { % 1.double:long; 2.capital:double line;
162   % 3.neg:negation; 4.No '\cs{nlongleftarrow}', '\cs{nLongleftarrow}' etc.
163   ma = \mapsto,
164   mma = \longmapsto,
165   % left arrow
166   la = \leftarrow,
167   La = \Leftarrow,
168   nla = \nleftarrow,
169   Nla = \nLeftarrow,
170   lla = \longleftarrow,
171   Lla = \Longleftarrow,
172   % right arrow
173   ra = \rightarrow,
174   Ra = \Rightarrow,
175   nra = \nrightarrow,
176   Nra = \nRightarrow,
177   rra = \longrightarrow,
178   Rra = \Longrightarrow,
179   % bidirectional arrow
180   da = \leftrightharpoonup,
181   Da = \Leftrightarrow,
182   nda = \nleftrightharpoonup,
183   Nda = \nLeftrightarrow,
184   dda = \longleftrightarrow,
185   Dda = \Longleftrightarrow,
186 }
187 \prop_map_inline:Nn \g_ztex_math_simple_arrow_prop
188 {
189   \cs_new_protected:cpn {z@#1}{#2}
190 }

```

```

191 \ztex_mathalias_set:nn
192 { ma, mma, la, La, nla, Nla,
193   lla, Lla, ra, Ra, nra, Nra,
194   rra, Rra, da, Da, nda, Nda,
195   dda, Dda }
196 { z@ma, z@mma, z@la, z@La, z@nla, z@Nla,
197   z@lla, z@Lla, z@ra, z@Ra, z@nra, z@Nra,
198   z@rra, z@Rra, z@da, z@Da, z@nda, z@Nda,
199   z@dda, z@Dda }
200 % extend text arrow
201 \cs_new:Npn \ext_arrow_set:nn #1#2
202 { \exp_args:Nee \NewDocumentCommand{\use:c {z@#1}}{s0{}D(){} }
203   {
204     \IfBooleanTF{##1}
205       {#2[\text{##3}]{\text{##2}}}
206       {#2[##3]{##2}}
207   }
208 }
209 \keyval_parse:NNn \use_none:n \ext_arrow_set:nn
210 {
211   xla = \xleftarrow,
212   Xla = \xLeftarrow,
213   xxla = \xLongleftarrow,
214   xra = \xrightarrow,
215   Xra = \xRightarrow,
216   xxra = \xLongrightarrow,
217   hla = \xhookleftarrow,
218   hra = \xhookrightarrow,
219 }
220 \ztex_mathalias_set:nn
221 { xla, Xla, xxla, xra, Xra, xxra, hla, hra }
222 { z@xla, z@Xla, z@xxla, z@xra, z@Xra, z@xxra, z@hla, z@hra }
223
224
225 % ==> Math Operator and symbols
226 % REF: 1. https://en.wikipedia.org/wiki/List\_of\_mathematical\_abbreviations
227 %      2. https://tex.stackexchange.com/a/289946/294585
228 \DeclareRobustCommand{\z@A}{\ensuremath{\forall}}
229 \DeclareRobustCommand{\z@E}{\ensuremath{\exists}}
230 \DeclareRobustCommand{\z@ns}{\ensuremath{\varnothing}}
231 \DeclareRobustCommand{\z@se}{\ensuremath{\backsimeq}}
232 \DeclareRobustCommand{\z@sse}{\ensuremath{\cong}}
233 \DeclareRobustCommand{\z@CC}{\ensuremath{\mathbb{C}}}
234 \DeclareRobustCommand{\z@RR}{\ensuremath{\mathbb{R}}}
235 \DeclareRobustCommand{\z@ZZ}{\ensuremath{\mathbb{Z}}}
236 \DeclareRobustCommand{\z@NN}{\ensuremath{\mathbb{N}}}
237 \DeclareRobustCommand{\z@dd}{\mathinner{\mathrm{d}}\zalias@mu@p}
238 \def\zalias@mu@p{\mathchoice{\mskip-\thinmuskip}{\mskip-\thinmuskip}{-}{-}}

```

```

239 \ztex_mathalias_set:nn
240 { A, E, ns, se, sse, CC, RR, ZZ, NN, dd }
241 { z@A, z@E, z@ns, z@se, z@sse, z@CC, z@RR, z@ZZ, z@NN, z@dd }
242 % math operator alias setup
243 \prop_set_from_keyval:Nn \g_ztex_math_op_prop
244 {
245     alt = alt,
246     rot = rot,
247     div = div,
248     curl = curl,
249     grad = grad,
250     id = Id,
251     im = Im,
252     ker = Ker,
253     cok = Cok,
254     hom = Hom,
255     supp = supp,
256     sign = sign,
257     trace = trace,
258 }
259 \prop_map_inline:Nn \g_ztex_math_op_prop
260 {
261     \exp_args:Ne \DeclareRobustCommand{\use:c {z@#1}}
262     {
263         \operatorname{\prop_item:Nn \g_ztex_math_op_prop {#1}}
264         \peek_after:Nw \ztex_op_check:
265     }
266 }
267 \tl_const:Nn \c_ztex_math_ops_tl { \cdot \wedge \times \oplus \otimes }
268 \cs_new_protected:Nn \ztex_op_check: {
269     \tl_map_inline:Nn \c_ztex_math_ops_tl {
270         \token_if_eq_meaning:NNT \l_peek_token ##1 { \tl_map_break:n {{\!}} }
271     }
272 }
273 \ztex_mathalias_set:nn
274 { alt, rot, div, curl, grad, id,
275     im, ker, cok, hom, supp, sign, trace }
276 { z@alt, z@rot, z@div, z@curl, z@grad, z@id,
277     z@im, z@ker, z@cok, z@hom, z@supp, z@sign, z@trace }
278 \NewDocumentCommand\zaliasopset{m}
279 {
280     \prop_put_from_keyval:Nn \g_ztex_math_op_prop {#1}
281 }
282 \@onlypreamble\zaliasopset
283
284
285 % ==> pyhsics package commands
286 % '\ab'(auto brace) command

```

```
287 \cs_new_protected:Npn \z@ab 287
288 { 288
289 \peek_after:Nw \__zab_case_match: 289
290 } 290
291 \cs_new_protected:Npn \__zab_case_match: 291
292 { 292
293 \token_case_meaning:NnF \l_peek_token 293
294 { 294
295 {\big}{\__zab_case_match_remove_next:nn {\big} {\big} } 295
296 {\Big}{\__zab_case_match_remove_next:nn {\Big} {\Big} } 296
297 {\bigg}{\__zab_case_match_remove_next:nn {\bigg} {\bigg} } 297
298 {\Bigg}{\__zab_case_match_remove_next:nn {\Bigg} {\Bigg} } 298
299 { * }{\__zab_case_match_remove_next:nn {}{}} 299
300 { ( }{\__zab_case_match_auto: } 300
301 { [ }{\__zab_case_match_auto: } 301
302 { < }{\__zab_case_match_auto: } 302
303 { | }{\__zab_case_match_auto: } 303
304 { \l }{\__zab_case_match_auto: } 304
305 {\c_group_begin_token}{\__zab_case_match_auto: } 305
306 }{ 306
307 \ztex_msg_warn:n { zab@invalid@deli@size } 307
308 \ztex_msg_warn:n { zab@invalid@deli@type } 308
309 \msg_fatal:nn { ztex } { zab@syntax@invalid } 309
310 % TODO: improve the error message !!! 310
311 % \ztex_token_if_in:nNTF {\c_group_begin_token(<|\|} \l_peek_token 311
312 % { 312
313 % \__zab_case_match_auto: 313
314 % }{ 314
315 % \ztex_msg_error:n { zab@invalid@deli@type } 315
316 % } 316
317 } 317
318 } 318
319 \cs_new:Npn \__zab_case_match_remove_next:nn #1#2 319
320 { 320
321 \tl_set:Nn \l__zalias_deli_modifer_left_tl { #1 } 321
322 \tl_set:Nn \l__zalias_deli_modifer_right_tl { #2 } 322
323 \exp_after:wN \peek_after:Nw \exp_after:wN 323
324 \__zab_case_match_kernel: \use_none:n 324
325 } 325
326 \cs_new:Npn \__zab_case_match_auto: 326
327 { 327
328 \tl_set:Nn \l__zalias_deli_modifer_left_tl { \left } 328
329 \tl_set:Nn \l__zalias_deli_modifer_right_tl { \right } 329
330 \peek_after:Nw \__zab_case_match_kernel: 330
331 } 331
332 % TODO: This piece of code might be redundant 332
333 \cs_new:Npn \__zab_case_match_kernel: 333
334 { 334
```

335	\token_case_meaning:NnF \l_peek_token	335
336	{	336
337	{\c_group_begin_token}{ \z@ab@curly }	337
338	{()}{\z@ab@round }	338
339	{[]}{\z@ab@square}	339
340	{< >}{\z@ab@angle }	340
341	{ }{\z@ab@vert }	341
342	{\ }{\z@ab@Vert }	342
343	{\ztex_msg_error:n { zab@invalid@deli@type } }	343
344	}	344
345	\ztex_msg_set:nn { zab@syntax@invalid }{ \string\zab\space syntax~error }	345
346	\ztex_msg_set:nn { zab@invalid@deli@type }	346
347	{	347
348	The~delimiter~types~\string\zab\space support~	348
349	are:'{}',~'()' ,~'[]' ,~'<>' ,~' ' ,~'\ \ ' ;~	349
350	but~you~enter~'\token_to_meaning:N \l_peek_token'.	350
351	}	351
352	\ztex_msg_set:nn { zab@invalid@deli@size }	352
353	{	353
354	The~delimiter~sizes~\string\zab\space support~	354
355	are:'\string\big' ,~'\string\Big' ,~'\string\bigg' ,~	355
356	'\string\Bigg' ;~	356
357	but~you~enter~'\token_to_meaning:N \l_peek_token'.	357
358	}	358
359	\tl_new:N \l__zalias_deli_modifer_left_tl	359
360	\tl_new:N \l__zalias_deli_modifer_right_tl	360
361	\protected\def\zab@left	361
362	{	362
363	\l__zalias_deli_modifer_left_tl	363
364	}	364
365	\protected\def\zab@right	365
366	{	366
367	\l__zalias_deli_modifer_right_tl	367
368	}	368
369	\NewDocumentCommand\z@ab@curly {m} {\zab@left \{#1 \zab@right\}}	369
370	\NewDocumentCommand\z@ab@round {r()}{\zab@left (#1 \zab@right)}	370
371	\NewDocumentCommand\z@ab@square{r[]}{\zab@left [#1 \zab@right]}	371
372	\NewDocumentCommand\z@ab@vert {r }{\zab@left #1 \zab@right }	372
373	\NewDocumentCommand\z@ab@Vert{r\ \ }{\zab@left \ #1 \zab@right\ } % double line	373
374	\NewDocumentCommand\z@ab@angle {r<>}{\zab@left \lang{#1 \zab@right}\rangle}	374
375	\ztex_mathalias_set:nn { zab }{ z@ab }	375
376		376
377		377
378	% '\dv' and '\p dv' command	378
379	\seq_new:N \l__zalias_num_rest_seq	379
380	\tl_new:N \l__zalias_num_extract_tl	380
381	\seq_new:N \l__zalias_num_extract_seq	381
382	\regex_set:Nn \l__zalias_num_extract_tl { -(?:\d+\.\d* \.\d+ \d+) }	382

```
383 \cs_new:Npn \__zalias_extract_num:nNN #1#2#3
384 {
385     \regex_extract_all:NnN \l__zalias_num_extract_tl
386     { #1 } #2
387     \exp_args:NNe \regex_split:NnN \l__zalias_num_extract_tl
388     { \clist_use:nn {#1}{+} } #3
389 }
390 \cs_new:Npn \__zalias_expr_format:N #1
391 { }
392
393 \tl_new:N \l__zalias_dv_order_tl
394 \tl_new:N \l__zalias_dv_frac_over_tl
395 \tl_new:N \l__zalias_dv_frac_lower_tl
396 \cs_set:Npn \__zalias_derivative:nnnn #1#2#3#4
397 {% #1:start check; #2:over; #3:below; #4:'\dd'/'\partial'
398     \__zalias_extract_num:nNN {#3}
399     \l__zalias_num_extract_seq
400     \l__zalias_num_rest_seq
401     \tl_set:Ne \l__zalias_dv_order_tl
402     {
403         \seq_use:Nn \l__zalias_num_rest_seq {}
404     }
405     \tl_regex_replace_all:Nnn \l__zalias_dv_order_tl {\+[2,]}{+}
406     \tl_set:Ne \l__zalias_dv_order_tl
407     {
408         \ztex_index_token_if_eq:ennTF {\l__zalias_dv_order_tl}{1}{+}
409         { \tl_tail:N \l__zalias_dv_order_tl }
410         { \l__zalias_dv_order_tl }
411         \tl_if_empty:VF \l__zalias_dv_order_tl
412         {
413             \seq_if_empty:NF \l__zalias_num_extract_seq
414             {
415                 \ztex_index_token_if_eq:ennF {\l__zalias_dv_order_tl}{-1}{+}
416                 { + }
417             }
418         }
419     }
420     \tl_set:Ne \l__zalias_dv_frac_over_tl
421     {
422         #4~{
423             \l__zalias_dv_order_tl
424             \seq_if_empty:NF \l__zalias_num_extract_seq
425             {
426                 \fp_eval:n
427                 {
428                     \seq_use:Nn \l__zalias_num_extract_seq {+}
429                 }
430             }
```



```
431 } 431
432 \zclist_item:nn {#2}{1} 432
433 } 433
434 \tl_set:Nn \l__zalias_dv_frac_lower_tl 434
435 { 435
436 \int_step_inline:nnn {2} 436
437 { \zclist_count:e {#2} } 437
438 { 438
439 #4\zclist_item:nn {#2}{##1} 439
440 ^{ 440
441 \tl_if_eq:neF {1} 441
442 { \zclist_item:nn {#3}{##1-1} } 442
443 { \zclist_item:nn {#3}{##1-1} } 443
444 } 444
445 } 445
446 } 446
447 \IfBooleanTF{#1} 447
448 { 448
449 \l__zalias_dv_frac_over_tl/ 449
450 \l__zalias_dv_frac_lower_tl 450
451 }{ 451
452 \frac{\l__zalias_dv_frac_over_tl} 452
453 {\l__zalias_dv_frac_lower_tl} 453
454 } 454
455 } 455
456 \NewDocumentCommand{\z@dv}{sm0{}} 456
457 { 457
458 \__zalias_derivative:nnnn {#1}{#2}{#3}{\mathrm{d}} 458
459 } 459
460 \NewDocumentCommand{\z@pdv}{sm0{}} 460
461 { 461
462 \__zalias_derivative:nnnn {#1}{#2}{#3}{\partial} 462
463 } 463
464 \ztex_mathalias_set:nn { dv, pdv }{ z@dv, z@pdv } 464
465 465
466 466
467 % matrix commands 467
468 \seq_new:N \l__zalias_matrix_a_seq 468
469 \seq_new:N \l__zalias_matrix_b_seq 469
470 \cs_new:Npn \zalias_matrix_from_list:n #1 470
471 { 471
472 \sclist_map_tokens:nn {#1} 472
473 { 473
474 \__zalias_mat_generate_row:n 474
475 } 475
476 } 476
477 \cs_new:Npn \__zalias_mat_generate_row:n #1 477
478 { 478
```

```
479 \clist_use:en
480 {
481   \exp_args:Ne \clist_map_tokens:nn
482     { \zcmd_clist_patch:nn {\scan_stop:}{#1} }
483   {
484     \__zalias_mat_item_cmd:n
485   }
486   }{ & } \\
487 }
488 \cs_new:Npn \__zalias_mat_item_cmd:n #1
489 { #1, }
490 \cs_generate_variant:Nn \zalias_matrix_from_list:n {e, o, f}
491 % NOTE: do NOT nest other mat cmd in '\mat' or '\pmat' ...
492 \cs_set_eq:NN \z@mat \zalias_matrix_from_list:n
493 \cs_set_eq:NN \z@mat@plain \zalias_matrix_from_list:n
494 \NewEnvironmentCopy{omatrix}{matrix}
495 % NOTE:
496 % 1. star      -> new ztex syntax
497 % 2. no star   -> traditional syntax
498 % 3. sometimes -> both syntaxes are compatible in star version
499 %               if there is no open ',' or ';' inside
500 \cs_new_protected:Npn \__zalias_mat_typeset:n #1
501 {
502   \exp_args:Nc \NewDocumentCommand{z@#1mat}{sm}
503   {
504     \IfBooleanTF{##1}
505       { \begin{#1matrix} \z@mat@plain{##2} \end{#1matrix} }
506       { \begin{#1matrix} ##2 \end{#1matrix} }
507   }
508 }
509 \clist_map_inline:nn { o, p, b, B, v, V }
510 { \__zalias_mat_typeset:n {#1} }
511 \ztex_mathalias_set:nn
512 { mat,   omat,   pmat,   bmat,   Bmat,   vmat,   Vmat }
513 { z@mat, z@omat, z@pmat, z@bmat, z@Bmat, z@vmat, z@Vmat }
514
515 % '\imat', '\admat' and '\zmat'
516 \cs_new:Npn \zalias_diag_mat_data:nnnn #1#2#3#4
517 {
518   \exp_args:Ne \int_step_tokens:nn {\zclist_count:n {#4}}
519   {
520     \__zalias_diag_mat_aux:nnen
521     { #1 }{ #2 }
522     { \zcmd_clist_patch:nn {#3}{#4} }
523   }
524 }
525 \cs_new:Npn \__zalias_diag_mat_aux:nnnn #1#2#3#4
526 {
```

```

527 \bool_if:nTF {#1}
528 {
529     \prg_replicate:nn { #4-1 }{ #2 & }
530 }{
531     \prg_replicate:nn { \clist_count:n {#3} - #4 }
532         { #2 & }
533 }
534 \clist_item:nn { #3 }{#4}
535 \bool_if:nTF {!#1}
536 {
537     \prg_replicate:nn { #4-1 }{ & #2 }
538 }{
539     \prg_replicate:nn { \clist_count:n {#3} - #4 }
540         { & #2 }
541 }
542 \int_compare:nNnF {#4}={\clist_count:n {#3}}{\}
543 }
544 \cs_generate_variant:Nn \_zalias_diag_mat_aux:nnnn { nne }
545 \cs_generate_variant:Nn \zalias_diag_mat_data:nnnn { nnne }
546 \cs_set:Npn \z@imat #1#2 { \zalias_diag_mat_data:nnnn {\c_true_bool}{#1}{1}{#2} }
547 \cs_set:Npn \z@admat #1#2 { \zalias_diag_mat_data:nnnn {\c_false_bool}{#1}{1}{#2} }
548 \NewDocumentCommand{\z@zmat}{ O{i} m }
549 {
550     \str_case:nnF {#1}
551     {
552         {i}{
553             \zalias_diag_mat_data:nnne
554             { \c_true_bool }{ }{ 0 }
555             { \prg_replicate:nn {#2-1}{0,} }
556         }
557         {a}{
558             \zalias_diag_mat_data:nnne
559             { \c_false_bool }{ }{ 0 }
560             { \prg_replicate:nn {#2-1}{,} }
561         }
562         {z}{
563             \zalias_diag_mat_data:nnne
564             { \c_true_bool }{ 0 }{ 0 }
565             { \prg_replicate:nn {#2-1}{,} }
566         }
567     }{
568         \ztex_msg_set:nn {zalias@zmat}
569         { '\string\zmat'~only~support~'i',~'a'~and~'z'~type,~but~you~enter~'#1'.}
570         \ztex_msg_error:n {zalias@zmat}
571     }
572 }
573 \ztex_mathalias_set:nn { imat, admat, zmat }{ z@imat, z@admat, z@zmat }
574

```

```
575 % '\jmat' and '\hmat' 575
576 \cs_new:Npn \zalias_jmat_data:nn #1#2 576
577 { 577
578     \exp_args:Ne \clist_map_tokens:nn { \sclist_item:nn {#2}{1} } 578
579     { 579
580         \exp_args:Ne \__zalias_jmat_row:nnn 580
581         { #1 } 581
582         { \sclist_item:nn {#2}{2} } 582
583     } 583
584 } 584
585 \cs_new:Npn \__zalias_jmat_row:nnn #1#2#3 585
586 { 586
587     \clist_use:en 587
588     { 588
589         \exp_args:Ne \clist_map_tokens:nn { #2 } 589
590         { \__zalias_frac_partial:nnn {#1}{#3} }, 590
591     }{ & } \\\ 591
592 } 592
593 \cs_new:Npn \__zalias_frac_partial:nnn #1#2#3 593
594 { 594
595     \exp_not:c {#1} \exp_not:N \frac 595
596     { \exp_not:N \mathstrut \exp_not:N \partial #2 } 596
597     { \exp_not:N \mathstrut \exp_not:N \partial #3 } , 597
598 } 598
599 \cs_generate_variant:Nn \zalias_jmat_data:nn { ne, no } 599
600 \ztex_keys_define:nn { zalias/jhmat } 600
601 { 601
602     b .tl_set:N = \l__zalias_jhmat_border_tl, 602
603     b .initial:n = { p }, 603
604     c .tl_set:N = \l__zalias_jhmat_cmd_tl, 604
605     c .initial:n = { textstyle }, 605
606     s .fp_set:N = \l__zalias_jhmat_stretch_fp, 606
607     s .initial:n = { 1.25 }, 607
608 } 608
609 \NewDocumentCommand{\z@jmat}{s0{}m} 609
610 { 610
611     \IfBooleanTF{#1} 611
612     { 612
613         \group_begin: 613
614         \ztex_keys_set:nn { zalias/jhmat }{ #2 } 614
615         \renewcommand{\arraystretch}{\fp_use:N \l__zalias_jhmat_stretch_fp} 615
616         \exp_args:No \begin{\l__zalias_jhmat_border_tl matrix} 616
617         \exp_args:No \zalias_jmat_data:nn {\l__zalias_jhmat_cmd_tl}{#3} 617
618         \exp_args:No \end{\l__zalias_jhmat_border_tl matrix} 618
619         \group_end: 619
620     }{ 620
621         \exp_args:Ne \zalias_jmat_data:nn 621
622         {\tl_if_empty:nTF {#2}{textstyle}{#2}}{#3} 622
```

```

623     }
624 }
625 \cs_new:Npn \zalias_hmat_data:nn #1#2
626 {
627     \exp_args:Ne \clist_map_tokens:nn { \sclist_item:en {\zcmd_sclist_patch:nn {
628         \scan_stop:}{#2}}{2} }
629     {
630         \exp_args:Neee \__zalias_hmat_row:nnnn { #1 }
631         { \sclist_item:en {\zcmd_sclist_patch:nn {\hbox{}}{#2}}{1} }
632         { \sclist_item:en {\zcmd_sclist_patch:nn {\scan_stop:}{#2}}{2} }
633     }
634 \cs_new:Npn \__zalias_hmat_row:nnnn #1#2#3#4
635 {
636     \clist_use:en
637     {
638         \clist_map_tokens:nn {#3}
639         {
640             \__zalias_hmat_item:nnnn {#1}{#2}{#4}
641         }
642     }{&} \
643 }
644 \cs_new:Npn \__zalias_hmat_item:nnnn #1#2#3#4
645 {
646     \tl_if_eq:nnTF {#3}{#4}
647     {
648         {\exp_not:c {#1} \z@pdv{#2,#4}[2]}
649     }{
650         {\exp_not:c {#1} \z@pdv{#2,#3,#4}[1, 1]}
651     } ,
652 }
653 \cs_generate_variant:Nn \zalias_hmat_data:nn { ne, no }
654 \NewDocumentCommand{\z@hmat}{sO{}m}
655 {
656     \IfBooleanTF{#1}
657     {
658         \group_begin:
659         \ztex_keys_set:nn { zalias/jhmat }{ #2 }
660         \renewcommand{\arraystretch}{\fp_use:N \l__zalias_jhmat_stretch_fp}
661         \exp_args:No \begin{\l__zalias_jhmat_border_tl matrix}
662         \exp_args:No \zalias_hmat_data:nn {\l__zalias_jhmat_cmd_tl}{#3}
663         \exp_args:No \end{\l__zalias_jhmat_border_tl matrix}
664         \group_end:
665     }{
666         \exp_args:Ne \zalias_hmat_data:nn
667         {\tl_if_empty:nTF {#2}{textstyle}{#2}}{#3}
668     }
669 }

```

```

670 \ztex_mathalias_set:nn { jmat, hmat }{ z@jmat, z@hmat }
671
672 % '\xmat'
673 \cs_new:Npn \zalias_xmat_data:nn #1#2
674 {
675     \exp_args:Ne \int_step_tokens:nn { \clist_item:nn {#2}{1} }
676     {
677         \exp_args:Nne \__zalias_xmat_row:nnn { #1 }
678         { \clist_item:nn {#2}{2} }
679     }
680 }
681 \cs_new:Npn \__zalias_xmat_row:nnn #1#2#3
682 {% #1:cmd; #2:x-range; #3:y-coor
683     \clist_use:en
684     {
685         \exp_args:Ne \int_step_tokens:nn { #2 }
686         { ,#1 {#3} }
687     }{ & } \\
688 }
689 \cs_new:Npn \z@xmat #1
690 {
691     \zalias_xmat_data:nn {\clist_item:nn {#1}{-1}}
692     {
693         \clist_item:nn {#1}{1},
694         \clist_item:nn {#1}{2}
695     }
696 }
697 \cs_generate_variant:Nn \zalias_xmat_data:nn { ne, no }
698 \ztex_mathalias_set:nn { xmat }{ z@xmat }
699
700 % \gmat
701 \cs_new:Npn \z@gmat #1
702 {
703     \z@xmat
704     {
705         \zclist_count:n {#1},
706         \zclist_count:n {#1},
707         \__zalias_gmat_item:nnn {#1}
708     }
709 }
710 \cs_new:Npn \__zalias_gmat_item:nnn #1#2#3
711 {
712     \langle
713     \zclist_item:nn {#1}{#2} ,
714     \zclist_item:nn {#1}{#3}
715     \rangle
716 }
717 \ztex_mathalias_set:nn { gmat }{ z@gmat }

```

```

1 \ProvidesExplFile{ztex.library.slide.tex}
2 {2025/09/10}{\ztex@versi@n}
3 {slide~library~for~ztex}
4
5
6 %%%%      slide library for ztex      %%%%
7 \__ztool_load_library:n { zdraw }
8 \bool_gset_true:N \g__ztex_slide_bool
9 \exp_args:NNnx \seq_set_split:Nnn \l_tmpa_seq
10 { | }{ \g__ztex_aspectratio_tl }
11 \geometry
12 {
13   papersize={\seq_item:Nn \l_tmpa_seq {1}cm, \seq_item:Nn \l_tmpa_seq {2}cm},
14   hmargin=1.25cm, top=.8cm, includefoot, bottom=5.5pt,
15   footskip=\dim_eval:n {1.25em + 5pt}
16 }
17 \cs_generate_variant:Nn \dim_set:Nn { Ne }
18 \dim_set:Ne \zpw {\seq_item:Nn \l_tmpa_seq {1}cm}
19 \dim_set:Ne \zph {\seq_item:Nn \l_tmpa_seq {2}cm}
20
21
22 % ==> marker and commands patches
23 \mark_new_class:n {zslide-left}
24 \mark_new_class:n {zslide-right}
25 \IfClassLoadedTF{book}{
26   \let\cleardoublepage\clearpage
27   \renewcommand\chaptermark[1]{ \mark_insert:nn {zslide-left}{#1} }
28   \renewcommand\thesection{\arabic{section}}
29   \ztex_hook_preamble_last:n
30   {
31     \renewcommand\mainmatter{}
32     \renewcommand\frontmatter{}
33   }
34   \zsecformat\part
35   {
36     type          = page,
37     space.before  = 0pt plus .8fill,
38     space.after   = 0pt plus 1fill,
39     pagestyle     = empty,
40     title.format+ = \centering,
41   }
42   \zsecformat\chapter
43   {
44     type          = page,
45     space.before  = 0pt plus .8fill,
46     space.after   = 0pt plus 1fill,

```

```

47     pagestyle      = empty,
48     title.format+  = \centering,
49 }
50 }{ \relax }
51 \dim_new:N \g_zslide_status_info_sec_C_dim % vertical axis of symmetry
52 \dim_new:N \g_zslide_status_info_sec_L_dim
53 \dim_gset:Nn \g_zslide_status_info_sec_C_dim {-1.7em}
54 \dim_gset:Nn \g_zslide_status_info_sec_L_dim {1cm}
55 \renewcommand\sectionmark[1]{\mark_insert:nn {zslide-left}{#1}}
56 \renewcommand\subsectionmark[1]{\mark_insert:nn {zslide-right}{\thesubsection\_#1}}
57 \coffin_new:N \g_zslide_status_info_sec_text_coffin
58 \cs_new:Npn \__zslide_status_info_sec_coffin_typeset:n #1
59 {
60     \hcoffin_gset:Nn \g_zslide_status_info_sec_text_coffin
61     { \textcolor{\tl_use:N \l_ztex_slide_sec_fg_tl}{#1} }
62     \__zslide_frame_title_info:n
63     {
64         \tl_use:N \l__ztex_slide_sec_prefix_tl
65         \coffin_typeset:Nnnnn \g_zslide_status_info_sec_text_coffin
66         { l }{ vc }
67         { Opt }{ Opt }
68         \tl_use:N \l_ztex_slide_sec_suffix_tl
69     }
70 }
71 \cs_new:Npn \__zslide_frame_title_info:n #1
72 {
73     \AddToHookNext{ shipout / foreground }
74     {
75         \put(
76             \dim_use:c {g_zslide_status_info_sec_L_dim},
77             \dim_use:c {g_zslide_status_info_sec_C_dim}
78         ){ #1 }
79     }
80 }
81 \cs_generate_variant:Nn \__zslide_status_info_sec_coffin_typeset:n {o}
82 \bool_new:N \g_new_manual_sec_bool
83 \bool_gset_false:N \g_new_manual_sec_bool
84 \NewDocumentCommand{\zslideframetitle}{m}
85 {
86     \newpage
87     % backgroud status bar
88     \bool_gset_true:N \g_new_manual_sec_bool
89     \AddToHook{shipout/background}
90     {
91         \bool_if:NT \g_new_manual_sec_bool
92         {
93             \zslide_status_bar:nnnn {sec}
94             {(0, \dim_use:c {g_zslide_status_bar_sec_B_dim})}

```



```
95         {1}
96         {\dim_use:c {g_zslide_status_bar_sec_H_dim}}
97     }
98 }
99 % foreground status info
100 \hcoffin_gset:Nn \g__zslide_status_info_sec_text_coffin
101 { \Large\textcolor{\tl_use:N \l__ztex_slide_sec_fg_tl}{#1} }
102 \__zslide_frame_title_info:n
103 {
104     \tl_use:N \l__ztex_slide_sec_prefix_tl
105     \coffin_typeset:Nnnnn \g__zslide_status_info_sec_text_coffin
106     { l }{ vc }
107     { Opt }{ Opt }
108     \tl_use:N \l__ztex_slide_sec_suffix_tl
109 }
110 % after vspace
111 \vspace*{.5em}
112 }
113 \zsecformat\section
114 {
115     explicit = true,
116     code      = {
117         \__zslide_status_info_sec_coffin_typeset:o { \Large\zsecname }
118         \bool_gset_true:N \g_new_sec_bool
119         \int_gset:Nn \g__ztex_slide_framecnt_int {1}
120         \vspace*{.7em}
121     },
122 }
123 \hook_gput_code:nnn {cmd/tableofcontents/before}
124 {zslide-toc-leftmark}
125 {
126     \mark_insert:nn {zslide-left}{contents}
127 }
128
129
130 % ==> status rule bar and metadata-item
131 \bool_new:N \g_new_sec_bool
132 \int_new:N \g__ztex_slide_framecnt_int
133 \int_gset:Nn \g__ztex_slide_framecnt_int {1}
134 \cs_new:Npn \zslide_framecnt_aux:nn #1#2 {
135     \iow_now:Nn \@auxout {
136         \unexpanded{\global\@namedef{zsec@#1@cnt}{#2}}
137     }
138 }
139 \cs_generate_variant:Nn \zslide_framecnt_aux:nn {ee}
140 \AddToHook{cmd/chapter/before}{\newpage}
141 \AddToHook{cmd/tableofcontents/before}
142 {\renewcommand{\contentsname}{Outline}}
```

```

143 \AddToHook{cmd/section/before}{
144     \newpage\int_gdecr:N \g__ztex_slide_framecnt_int
145     \ifnum\arabic{section}=0\else
146         \zslide_framecnt_aux:ee
147         {\Roman{section}}
148         {\int_use:N \g__ztex_slide_framecnt_int}
149     \fi
150 }
151 \AddToHook{shipout/firstpage}{
152     \setcounter{page}{0}
153     \label{zslide:titlepage}
154     \hyper@anchor{zslide@titlepage}
155 }
156 \AddToHook{shipout/lastpage}{
157     \label{zslide:lastpage}
158     \hyper@anchor{zslide@lastpage}
159     \zslide_framecnt_aux:ee
160     {\Roman{section}}
161     {\int_use:N \g__ztex_slide_framecnt_int}
162 }
163 \AddToHook{shipout/after}{
164     \bool_gset_false:N \g_new_sec_bool
165     \bool_gset_false:N \g_new_manual_sec_bool
166     \int_gincr:N \g__ztex_slide_framecnt_int
167 }
168 \hook_gput_code:nnn {shipout/background}{zslide-background}
169 {
170     \put(0, -\paperheight){\textcolor
171         {\tl_use:N \l__ztex_slide_doc_bgcolor_tl}
172         {\rule{1\paperwidth}{1\paperheight}}}}
173 }
174
175 % interface for status bar and metadata
176 \dim_new:N \g_zslide_status_bar_head_H_dim
177 \dim_new:N \g_zslide_status_bar_foot_H_dim
178 \dim_new:N \g_zslide_status_bar_sec_H_dim
179 \dim_new:N \g_zslide_status_bar_sec_B_dim
180 \dim_gset:Nn \g_zslide_status_bar_head_H_dim {.7em}
181 \dim_gset:Nn \g_zslide_status_bar_foot_H_dim {.7em}
182 \dim_gset:Nn \g_zslide_status_bar_sec_H_dim {2em}
183 \dim_gset:Nn \g_zslide_status_bar_sec_B_dim {-2.7em}
184 \AddToHook{shipout/background}{
185     \zslide_status_bar:nnnn {UL}{(0, -\dim_use:c {g_zslide_status_bar_head_H_dim})}
186     {.5}{\dim_use:c {g_zslide_status_bar_head_H_dim}}
187     \zslide_status_bar:nnnn {UR}{(.5\paperwidth, -\dim_use:c {g_zslide_status_bar_head_H_dim})}
188     {.5}{\dim_use:c {g_zslide_status_bar_head_H_dim}}
189     \zslide_status_bar:nnnn {BL}{(0,
190         -.5\paperheight)}
191     {.33}{\dim_use:c {g_zslide_status_bar_foot_H_dim}}

```

```

191 \zslide_status_bar:nnnn {BC}{(.33\paperwidth,-\paperheight)} 191
192 { .34}{\dim_use:c {g_zslide_status_bar_foot_H_dim}} 192
193 \zslide_status_bar:nnnn {BR}{(.67\paperwidth,-\paperheight)} 193
194 { .33}{\dim_use:c {g_zslide_status_bar_foot_H_dim}} 194
195 \bool_if:NT \g_new_sec_bool { 195
196 \zslide_status_bar:nnnn {sec} 196
197 {(0, \dim_use:c {g_zslide_status_bar_sec_B_dim})} 197
198 {1} 198
199 {\dim_use:c {g_zslide_status_bar_sec_H_dim}} 199
200 } 200
201 } 201
202 \AddToHook{shipout/foreground}{ 202
203 \zslide_status_info:nnnn {head}{ 0 }{.5 }{ \hfill\zslide_meta:n {UL}\_ } 203
204 \zslide_status_info:nnnn {head}{.5 }{.5 }{ \_ \zslide_meta:n {UR}\hfill } 204
205 \zslide_status_info:nnnn {foot}{ 0 }{.33}{ \hfill\zslide_meta:n {BL}\hfill } 205
206 \zslide_status_info:nnnn {foot}{.33}{.34}{ \hfill\zslide_meta:n {BC}\hfill } 206
207 \zslide_status_info:nnnn {foot}{.67}{.33}{ \hfill\zslide_meta:n {BR}\quad } 207
208 \exp_args:Ne \hyper@anchor{zslide@\FirstMark{zslide-left}.\int_use:N ✓ 208
\g__ztex_slide_framecnt_int}
209 } 209
210 \cs_new_protected:Npn \zslide_status_bar:nnnn #1#2#3#4 { 210
211 \ifnum\thepage=0\else 211
212 \put#2 {\textcolor{\tl_use:c {l__ztex_slide_#1_bg_tl}}{\rule{#3\paperwidth}{#4}}} 212
213 \fi 213
214 } 214
215 \dim_new:N \g_zslide_status_info_head_C_dim % vertical axis of symmetry 215
216 \dim_new:N \g_zslide_status_info_foot_C_dim 216
217 \dim_gset:Nn \g_zslide_status_info_head_C_dim {-0.35em} % 0.3483ex=1.5pt 217
218 \dim_gset:Nn \g_zslide_status_info_foot_C_dim {-\paperheight+0.35em} % 1.5pt 218
219 \coffin_new:N \g__zslide_status_info_coffin 219
220 \cs_new_protected:Npn \zslide_status_info:nnnn #1#2#3#4 220
221 {% #1:head/foot; #2:start-$x$; #3:width; #4:content; 221
222 \hcoffin_gset:Nn \g__zslide_status_info_coffin 222
223 { \hbox~ to~ #3\paperwidth {#4} } 223
224 \ifnum\thepage=0\else 224
225 \put(#2\paperwidth, \dim_use:c {g_zslide_status_info_#1_C_dim}) 225
226 { 226
227 \coffin_typeset:Nnnnn \g__zslide_status_info_coffin 227
228 { l }{ vc } 228
229 { Opt }{ Opt } 229
230 } 230
231 \fi 231
232 } 232
233 \cs_set:Npn \zslide_nav_sym:nnnn #1#2#3#4 { 233
234 \int_step_inline:nnn {1}{#1}{ 234
235 \int_compare:nNnTF {#2} = {##1} 235
236 {\bool_if:NTF \g__ztex_hyperref_bool 236
237 {\hyper@link{link}{zslide@\FirstMark{zslide-left}.\##1}{#3}} 237

```

```

238     {#3}
239 }
240 {\bool_if:NTF \g__ztex_hyperref_bool
241   {\hyper@link{link}{zslide@\FirstMark{zslide-left}}.##1}{#4}}
242   {#4}
243 }
244 }
245 }
246 % zslide metadata key-value
247 \ztex_hook_preamble_last:n {
248   \let\zslidetitle\@title
249   \let\zslideauthor\@author
250   \let\zslidedate\@date
251 }
252 \ztex_keys_define:nn { slide }{
253   % theme related keys
254   doc .meta:nn = { ztex / slide / doc }{#1},
255   doc / bg-color .tl_set:N = \l__ztex_slide_doc_bgcolor_tl,
256   doc / text-color .tl_set:N = \l__ztex_slide_doc_textcolor_tl,
257   doc / text-style .tl_set:N = \l__ztex_slide_doc_textstyle_tl,
258   sec .meta:nn = { ztex / slide / sec }{#1},
259   sec / prefix .tl_set:N = \l__ztex_slide_sec_prefix_tl,
260   sec / suffix .tl_set:N = \l__ztex_slide_sec_suffix_tl,
261   sec / bg .tl_set:N = \l__ztex_slide_sec_bg_tl,
262   sec / fg .tl_set:N = \l__ztex_slide_sec_fg_tl,
263   UL .meta:nn = { ztex / slide / UL }{#1},
264   UL / text .tl_set:N = \l__ztex_slide_UL_text_tl,
265   UL / bg .tl_set:N = \l__ztex_slide_UL_bg_tl,
266   UL / fg .tl_set:N = \l__ztex_slide_UL_fg_tl,
267   UR .meta:nn = { ztex / slide / UR }{#1},
268   UR / text .tl_set:N = \l__ztex_slide_UR_text_tl,
269   UR / bg .tl_set:N = \l__ztex_slide_UR_bg_tl,
270   UR / fg .tl_set:N = \l__ztex_slide_UR_fg_tl,
271   BL .meta:nn = { ztex / slide / BL }{#1},
272   BL / text .tl_set:N = \l__ztex_slide_BL_text_tl,
273   BL / bg .tl_set:N = \l__ztex_slide_BL_bg_tl,
274   BL / fg .tl_set:N = \l__ztex_slide_BL_fg_tl,
275   BC .meta:nn = { ztex / slide / BC }{#1},
276   BC / text .tl_set:N = \l__ztex_slide_BC_text_tl,
277   BC / bg .tl_set:N = \l__ztex_slide_BC_bg_tl,
278   BC / fg .tl_set:N = \l__ztex_slide_BC_fg_tl,
279   BR .meta:nn = { ztex / slide / BR }{#1},
280   BR / text .tl_set:N = \l__ztex_slide_BR_text_tl,
281   BR / bg .tl_set:N = \l__ztex_slide_BR_bg_tl,
282   BR / fg .tl_set:N = \l__ztex_slide_BR_fg_tl,
283   % toc related keys
284   toc .meta:nn = { ztex / slide / toc }{#1},
285   toc / leftmargin .meta:nn = { ztex / slide / toc / leftmargin }{#1},

```

```
286 toc / leftmargin / chapter .dim_set:N = \l__ztex_slide_toc_leftmargin_chapter_dim, 286
287 toc / leftmargin / chapter .initial:n = { 1.9em }, 287
288 toc / leftmargin / section .dim_set:N = \l__ztex_slide_toc_leftmargin_section_dim, 288
289 toc / leftmargin / section .initial:n = { 1.5em }, 289
290 toc / leftmargin / subsection .dim_set:N = \l__ztex_slide_toc_leftmargin_subsection_dim, 290
291 toc / leftmargin / subsection .initial:n = { 3.8em }, 291
292 toc / label .meta:nn = { ztex / slide / toc / label }{#1}, 292
293 toc / label / chapter .tl_set:N = \l__ztex_slide_toc_label_chapter_tl, 293
294 toc / label / chapter .initial:n = { }, 294
295 toc / label / section .tl_set:N = \l__ztex_slide_toc_label_section_tl, 295
296 toc / label / section .initial:n = { }, 296
297 toc / label / subsection .tl_set:N = \l__ztex_slide_toc_label_subsection_tl, 297
298 toc / label / subsection .initial:n = { }, 298
299 toc / suffix .meta:nn = { ztex / slide / toc / suffix }{#1}, 299
300 toc / suffix / chapter .tl_set:N = \l__ztex_slide_toc_suffix_chapter_tl, 300
301 toc / suffix / chapter .initial:n = { }, 301
302 toc / suffix / section .tl_set:N = \l__ztex_slide_toc_suffix_section_tl, 302
303 toc / suffix / section .initial:n = { }, 303
304 toc / suffix / subsection .tl_set:N = \l__ztex_slide_toc_suffix_subsection_tl, 304
305 toc / suffix / subsection .initial:n = { }, 305
306 toc / unknown .code:n = { 306
307 \ztex_metakey_msg_warning:nn {slide-toc}{ 307
308 leftmargin(<key-value>:chapter[<dim>:2em], section[<dim>:4em], subsection[<dim>:6em]), 308
~
309 label(<key-value>:chapter[<tl>:thechapter;hbox:1em], section[<tl>:thesection;hbox:1em], 309
310 subsection[<tl>:thesubsection;hbox:2em]),~ 310
311 after(<key-value>:chapter[tl:<empty>], section[tl:<empty>], subsection[tl:<empty>]) 311
312 } 312
313 }, 313
314 unknown .code:n = { 314
315 \ztex_metakey_msg_warning:nn {slide}{ 315
316 sec(<key-value>:prefix, suffix, bg, fg),~ 316
317 UL(<key-value>:text, bg, fg), UR(<key-value>:text, bg, fg),~ 317
318 BL(<key-value>:text, bg, fg), BC(<key-value>:text, bg, fg),~ 318
319 BR(<key-value>:text, bg, fg) 319
320 } 320
321 } 321
322 } 322
323 \cs_new_protected:Npn \zslide_meta:n #1 { 323
324 \tl_if_eq:nnT {#1}{BC}{ \bool_if:NT \g__ztex_hyperref_bool 324
325 { \hyper@link{link}{zslide@titlepage} }} 325
326 { \scriptsize\textcolor{\tl_use:c {l__ztex_slide_#1_fg_tl} } 326
327 { \tl_use:c {l__ztex_slide_#1_text_tl} } } 327
328 } 328
329 329
330 330
331 % ==> zslide custom interface 331
332 % zslide users' tools 332
```

```

333 \NewDocumentCommand{\zslideframeall}{m}
334 {
335     \cs_if_exist:cTF {zsec@#1@cnt}
336     {\cs:w zsec@#1@cnt\cs_end:}
337     {??}
338 }
339 \NewDocumentCommand{\zslideframeind}{}{
340     \int_use:N \g__ztex_slide_framecnt_int
341 }
342 \NewDocumentCommand{\zslidenavsym}{0{\(\bullet\)}0{\(\circ\)}}
343 {
344     \cs_if_exist:cTF {zsec@\Roman{section}@cnt}
345     {\zslide_nav_sym:nnnn
346      {\zslideframeall{\Roman{section}}}
347      {\zslideframeind}
348      {\textcolor{\l__ztex_slide_UR_fg_tl}{#1}}
349      {\textcolor{\l__ztex_slide_UR_fg_tl}{#2}}
350      }{??}
351 }
352 \ztex_keys_define:nn { slide / logo }
353 {
354     position .tl_gset:N = \g__ztex_slide_logo_position_tl,
355     position .initial:n = { (\paperwidth-\c_ztex_quad_dim, 1.5em) },
356     width .dim_gset:N = \g__ztex_slide_logo_width_dim,
357     width .initial:n = { 2.5em },
358     exclude .clist_gset:N = \g__ztex_slide_logo_exclude_clist,
359     exclude .initial:n = { 0 },
360 }
361 \NewDocumentCommand{\zslidelogo}{om}
362 {
363     \IfValueT{#1}{\ztex_keys_set:nn { slide / logo }{#1}}
364     \ztex_page_annotate:eeenn
365     {background}
366     {\exp_after:wN \__page_mask_pos_parse:w \g__ztex_slide_logo_position_tl}
367     {rb}{
368         \edef\current@page{\thepage}
369         \clist_if_in:NVF \g__ztex_slide_logo_exclude_clist\current@page
370         {\includegraphics[width=\g__ztex_slide_logo_width_dim]{#2}}
371     }{}
372 }
373 \@onlypreamble\zslidelogo
374
375 \clist_map_inline:nn { chapter, section, subsection }{
376     \clist_if_in:NnT \c_zsect_level_clist { #1 }
377     {
378         \exp_args:Nc \ztocformat { #1 }
379         {
380             name.before = \tl_use:c { l__ztex_slide_toc_label_#1_tl },

```

```

381         title.after = \tl_use:c { l__ztex_slide_toc_suffix_#1_tl },
382         space.left = \dim_use:c { l__ztex_slide_toc_leftmargin_#1_dim },
383     }
384 }
385 }
386 \gdef\zslidetoc@sicon
387 {
388     \box_move_up:nn {2pt}
389     {
390         \hbox:n {\ztool_set_to_wd:nn
391             {6pt}{\(\blacktriangleright\)}}
392     }
393 }
394 \gdef\zslidetoc@ssicon{\rule[2pt]{3pt}{3pt}}
395 % slide mode setup interface
396 \NewDocumentCommand{\zslideset}{om}{
397     \IfNoValueTF {#1}{
398         \ztex_keys_set:nn { slide }{#2}
399     }{
400         \ztex_keys_set:nn { slide / #1 }{#2}
401     }
402 }
403
404
405 % ==> slide theme create interface
406 \clist_new:N \g__zslide_all_themes_clist
407 \clist_gclear:N \g__zslide_all_themes_clist
408 \cs_new_protected:Npn \__zslide_theme_create:nn #1#2 {
409     \tl_new:c {g__zslide_theme_#1_tl}
410     \clist_gput_right:Nn \g__zslide_all_themes_clist {g__zslide_theme_#1_tl}
411     \keys_precompile:nnN { ztex/slide }{#2}\l_tmpa_tl
412     \tl_set_eq:cc {g__zslide_theme_#1_tl} {l_tmpa_tl}
413 }
414 \str_new:N \g__zslide_theme_current_str
415 \cs_new_protected:Npn \__zslide_theme_use:nn #1#2 {
416     \tl_use:c {g__zslide_theme_#1_tl}
417     \IfNoValueF{#2}{
418         \ztex_keys_set:nn { slide }{#2}
419     }
420 }
421 \cs_generate_variant:Nn \color_select:n {e}
422 \cs_new_protected:Npn \zslide_set_doc_text_color:n #1
423 {
424     \color{#1}\global\let\default@color\current@color % xcolor
425     \color_select:e {#1} % l3color
426 }
427 \NewDocumentCommand{\zslidethemenew}{mm}{
428     \__zslide_theme_create:nn {#1}{#2}

```



```

429 }
430 \NewDocumentCommand{\zslidethemeuse}{om}{
431   \_zslide_theme_use:nn {#2}{#1}
432 }
433 \NewDocumentCommand\zslidedocolor{0{fg}m}{
434   \str_case:nnF {#1}{
435     { fg }{ \zslide_set_doc_text_color:n {#2} }
436     { bg }{ \tl_set:Nn \l__ztex_slide_doc_bgcolor_tl {#2} }
437   }{
438     \ztex_metakey_msg_warning:nn {slide-theme-doc}
439     { bg(<color>:white), fg(<color>:black) }
440   }
441 }
442 % page check interface
443 \prg_new_conditional:Npnn \zslide_if_page:n #1 {p, T, F, TF}
444 {
445   \int_compare:nTF {\thepage#1}
446     { \prg_return_true: }
447     { \prg_return_false: }
448 }
449 \prg_generate_conditional_variant:Nnn \zslide_if_page:n {e} { T, F, TF }
450 \NewDocumentCommand{\zslidepageTF}{mmm}
451 {
452   \zslide_if_page:nTF {#1}
453     {#2}{#3}
454 }
455 % BUG: if no subsection, mark-'zslide-right' added manually will be lost
456 \NewDocumentCommand{\zslideUL}{}
457 {
458   \ifnum\arabic{section}=0\else Section\ \thesection\fi
459 }
460 \NewDocumentCommand{\zslideUR}{}
461 {
462   \mark_if_eq:nnnnTF {page}{zslide-right}{first}{last}
463     {\ifnum\arabic{subsection}=0\else\FirstMark{zslide-right}\fi}
464     {\ifnum\arabic{subsection}=0\else\FirstMark{zslide-right}\,--\,\LastMark{zslide-right}
465 \fi}
466 }
467 \NewDocumentCommand{\zslideBR}{}
468 {
469   \zslidedate\quad
470   \thepage/\bool_if:NT \g__ztex_hyperref_bool
471     {\hyper@link{link}{zslide@lastpage}}{
472       \textcolor{\l__ztex_slide_BR_fg_tl}
473       {\pageref*{zslide:lastpage}}
474     }
475 }

```



```

476 % ==> pre-defined slide theme: 'theme'-'color'
477 % slide theme: AnnArbor-default
478 \str_case:NnF \g__ztex_slide_theme_str {
479     {AnnArborDefault}{
480         \definecolor{Ann-default-I}{HTML}{0000a3} % blue
481         \definecolor{Ann-default-II}{HTML}{ffc20c} % light yellow
482         \definecolor{Ann-default-III}{HTML}{ffcb03}
483         \zslide_theme_create:nn {AnnArborDefault}{
484             doc = {
485                 bg-color = white,
486                 text-color = black,
487                 text-style = sfdefault
488             },
489             UL = {
490                 bg    = Ann-default-I,
491                 fg    = Ann-default-II,
492                 text  = {\zslideUL}
493             },
494             UR = {
495                 bg    = Ann-default-II,
496                 fg    = Ann-default-I,
497                 text  = {\zslideUR}
498             },
499             BL = {
500                 bg    = Ann-default-I,
501                 fg    = Ann-default-III,
502                 text  = \zslideauthor
503             },
504             BC = {
505                 bg    = Ann-default-III,
506                 fg    = Ann-default-I,
507                 text  = \zslidetitle
508             },
509             BR = {
510                 bg    = Ann-default-II,
511                 fg    = Ann-default-I,
512                 text  = \zslideBR
513             },
514             sec = {
515                 fg    = Ann-default-I,
516                 bg    = Ann-default-III,
517                 prefix = {},
518                 suffix = {}
519             }
520         }
521     }
522 }
523

```

524	% slide theme: AnnArbor-beaver	524
525	{AnnArborBeaver}{	525
526	\definecolor{Ann-bea-I}{HTML}{a30000}	526
527	\definecolor{Ann-bea-II}{HTML}{e0e0e0}	527
528	\definecolor{Ann-bea-III}{HTML}{f0f0f0}	528
529	__zslide_theme_create:nn {AnnArborBeaver}{	529
530	doc = {	530
531	bg-color = white,	531
532	text-color = black,	532
533	text-style = sfdefault	533
534	},	534
535	UL = {	535
536	bg = Ann-bea-I,	536
537	fg = Ann-bea-II,	537
538	text = {\zslideUL}	538
539	},	539
540	UR = {	540
541	bg = Ann-bea-II,	541
542	fg = Ann-bea-I,	542
543	text = {\zslideUR}	543
544	},	544
545	BL = {	545
546	bg = Ann-bea-I,	546
547	fg = Ann-bea-II,	547
548	text = \zslideauthor	548
549	},	549
550	BC = {	550
551	bg = Ann-bea-III,	551
552	fg = Ann-bea-I,	552
553	text = \zslidetitle	553
554	},	554
555	BR = {	555
556	bg = Ann-bea-II,	556
557	fg = Ann-bea-I,	557
558	text = \zslideBR	558
559	},	559
560	sec = {	560
561	fg = Ann-bea-I,	561
562	bg = Ann-bea-III,	562
563	prefix = {},	563
564	suffix = {}	564
565	}	565
566	}	566
567	}	567
568		568
569	% slide theme: AnnArbor-Albatross	569
570	{AnnArborAlbatross}{	570
571	\definecolor{Ann-alb-I}{HTML}{000039} % UL bg	571

572	<code>\definecolor{Ann-alb-II}{HTML}{bfbfff}</code> % UL fg	572
573	<code>\definecolor{Ann-alb-III}{HTML}{00005f}</code> % UR bg	573
574	<code>\definecolor{Ann-alb-IV}{HTML}{00004c}</code> % BC bg	574
575	<code>\definecolor{Ann-alb-V}{HTML}{00007f}</code> % doc bg	575
576	<code>\definecolor{Ann-alb-VI}{HTML}{ffe700}</code> % doc text color	576
577	<code>_zslide_theme_create:nn {AnnArborAlbatross}{</code>	577
578	<code>doc = {</code>	578
579	<code>bg-color = Ann-alb-V,</code>	579
580	<code>text-color = Ann-alb-VI,</code>	580
581	<code>text-style = sfdefault</code>	581
582	<code>},</code>	582
583	<code>UL = {</code>	583
584	<code>bg = Ann-alb-I,</code>	584
585	<code>fg = Ann-alb-II,</code>	585
586	<code>text = {\zslideUL}</code>	586
587	<code>},</code>	587
588	<code>UR = {</code>	588
589	<code>bg = Ann-alb-III,</code>	589
590	<code>fg = Ann-alb-II,</code>	590
591	<code>text = {\zslideUR}</code>	591
592	<code>},</code>	592
593	<code>BL = {</code>	593
594	<code>bg = Ann-alb-I,</code>	594
595	<code>fg = Ann-alb-II,</code>	595
596	<code>text = \zslideauthor</code>	596
597	<code>},</code>	597
598	<code>BC = {</code>	598
599	<code>bg = Ann-alb-IV,</code>	599
600	<code>fg = Ann-alb-II,</code>	600
601	<code>text = \zslidetitle</code>	601
602	<code>},</code>	602
603	<code>BR = {</code>	603
604	<code>bg = Ann-alb-III,</code>	604
605	<code>fg = Ann-alb-II,</code>	605
606	<code>text = \zslideBR</code>	606
607	<code>},</code>	607
608	<code>sec = {</code>	608
609	<code>bg = Ann-alb-IV,</code>	609
610	<code>fg = Ann-alb-II,</code>	610
611	<code>prefix = {},</code>	611
612	<code>suffix = {}</code>	612
613	<code>}</code>	613
614	<code>}</code>	614
615	<code>}</code>	615
616		616
617	<code>% slide theme: AnnArbor-seahorse</code>	617
618	<code>{AnnArborSeahorse}{</code>	618
619	<code>\definecolor{Ann-sea-I}{HTML}{c2c2e8}</code> % UL bg	619

```
\definecolor{Ann-sea-II}{HTML}{d7d7f0} % UR bg
\definecolor{Ann-sea-III}{HTML}{cccccc} % BC bg
\__zslide_theme_create:nn {AnnArborSeahorse}{
  doc = {
    bg-color = white,
    text-color = black,
    text-style = sfdefault
  },
  UL = {
    bg    = Ann-sea-I,
    fg    = black,
    text  = {\zslideUL}
  },
  UR = {
    bg    = Ann-sea-II,
    fg    = black,
    text  = {\zslideUR}
  },
  BL = {
    bg    = Ann-sea-I,
    fg    = black,
    text  = \zslideauthor
  },
  BC = {
    bg    = Ann-sea-III,
    fg    = black,
    text  = \zslidetitle
  },
  BR = {
    bg    = Ann-sea-II,
    fg    = black,
    text  = \zslideBR
  },
  sec = {
    fg    = black,
    bg    = Ann-sea-III,
    prefix = {},
    suffix = {}
  }
}

% slide theme: AnnArbor-Spruce
{AnnArborSpruce}{
  \definecolor{Ann-spr-I}{HTML}{005128} % UL bg
  \definecolor{Ann-spr-II}{HTML}{d8e8e0} % UR bg
  \definecolor{Ann-spr-III}{HTML}{99c1ad} % BC bg
  \definecolor{Ann-spr-IV}{HTML}{7fb298} % UL/BL fg
```

```

668 \definecolor{Ann-spr-V}{HTML}{e5efea} % sec bg
669 \_zslide_theme_create:nn {AnnArborSpruce}{
670     doc = {
671         bg-color = white,
672         text-color = black,
673         text-style = sfdefault
674     },
675     UL = {
676         bg    = Ann-spr-I,
677         fg    = Ann-spr-IV,
678         text = {\zslideUL}
679     },
680     UR = {
681         bg    = Ann-spr-II,
682         fg    = Ann-spr-I,
683         text = {\zslideUR}
684     },
685     BL = {
686         bg    = Ann-spr-I,
687         fg    = Ann-spr-IV,
688         text = \zslideauthor
689     },
690     BC = {
691         bg    = Ann-spr-III,
692         fg    = Ann-spr-I,
693         text = \zslidetitle
694     },
695     BR = {
696         bg    = Ann-spr-II,
697         fg    = Ann-spr-I,
698         text = \zslideBR
699     },
700     sec = {
701         fg    = Ann-spr-I,
702         bg    = Ann-spr-V,
703         prefix = {},
704         suffix = {}
705     }
706 }
707 }
708 }{
709 \ztex_metakey_msg_warning:nn {slide-theme}{
710     AnnArborDefault(default), AnnArborBeaver,
711     AnnArborAlbatross, AnnArborSeahorse
712 }
713 \str_set:Nn \g__ztex_slide_theme_str {AnnArborDefault}
714 }
715

```

716		716
717	% ==> slide mode init options	717
718	_zslide_theme_use:nn { \str_use:N \g_ztex_slide_theme_str }{}	718
719	\ztex_hook_preamble_last:n	719
720	{	720
721	\pagestyle{empty}	721
722	_ztex_text_symbol_patch:	722
723	\zslide_set_doc_text_color:n { \tl_use:N \l_ztex_slide_doc_textcolor_tl }	723
724	\renewcommand{\familydefault}{ \tl_use:c {\l_ztex_slide_doc_textstyle_tl} }	724
725	\str_case:VnF \g_ztex_lang_str {	725
726	{cn} {\renewcommand{\CJKfamilydefault}{ \tl_use:c {CJK\l_ztex_slide_doc_textstyle_tl}	726
	}}	
727	{fr} {}	727
728	{\relax}	728
729	}	729

10.4.5 thm

```

1 \ProvidesExplFile{ztex.library.thm.tex}
2 {2026/09/23}{\ztex@versi@n}
3 {thm~library~for~ztex}
4
5
6 %%%% thm library for ztex %%%%
7 \bool_gset_true:N \g__ztex_thm_lib_load_bool
8 %% ==> preamble
9 \RequirePackage[many]{tcolorbox}
10 \RequirePackage{adjustbox}
11 \RequirePackage{tikz}
12 \RequirePackage{etoolbox}
13 \ExplSyntaxOff
14 \usetikzlibrary{fadings, calc}
15 \ExplSyntaxOn
16 \RequirePackage{pifont}
17
18
19
20 %% ==> thm icon interface
21 \prop_new:N \g__ztex_thm_icon_prop
22 \prop_gclear:N \g__ztex_thm_icon_prop
23 \cs_new_protected:Npn \__ztex_thm_icon_set:n #1
24 {
25     \prop_gput_from_keyval:cn {g__ztex_thm_icon_prop}{#1}
26 }
27 \cs_new_protected:Npn \__ztex_thm_icon_use:n #1
28 {% #1: thm env type name
29     \prop_item:cn {g__ztex_thm_icon_prop}{#1}
30 }
31 \cs_generate_variant:Nn \__ztex_thm_icon_use:n {o, e}
32 \NewDocumentCommand{\zthmiconset}{m}
33 {
34     \__ztex_thm_icon_set:n {#1}
35 }
36 \NewDocumentCommand{\zthmiconuse}{m}
37 {
38     \__ztex_thm_icon_use:n {#1}
39 }
40 \NewDocumentCommand{\zthmiconrm}{}
41 {
42     \prop_gclear:N \g__ztex_thm_icon_prop
43 }
44 \@onlypreamble\zthmiconset
45
46

```

47		47
48	%% ==> thm additional theme	48
49	\zthmstylenew {	49
50	% theme shadow: copy from an old book	50
51	shadow = {	51
52	begin =	52
53	{	53
54	\begin{tcolorbox}	54
55	[	55
56	enhanced~ jigsaw, breakable,	56
57	top=1.5pt, bottom=1.5pt,	57
58	left=3pt, right=3pt,	58
59	boxrule=0pt, sharp~corners,	59
60	drop~fuzzy~shadow,	60
61	colback={\thm@tmp@color!10},	61
62	borderline~west={3pt}{0pt}{\thm@tmp@color}	62
63	]	63
64	},	64
65	end = { \end{tcolorbox} },	65
66	option =	66
67	{	67
68	__ztex_thm_title_inline:n { T }	68
69	__ztex_thm_tcolorbox_warning:	69
70	}	70
71	},	71
72	% tcolorbox default	72
73	tcb = {	73
74	begin =	74
75	{	75
76	\begin{tcolorbox}	76
77	[	77
78	enhanced, breakable,	78
79	top=1.5pt, bottom=1.5pt,	79
80	left=3pt, right=3pt,	80
81	sharp~corners, boxrule=0.8pt,	81
82	colback=\thm@tmp@color!10,	82
83	colframe=\thm@tmp@color,	83
84	title=\zthmtitle*,	84
85	]	85
86	},	86
87	end = { \end{tcolorbox} },	87
88	option =	88
89	{	89
90	__ztex_thm_title_inline:n { F }	90
91	__ztex_thm_tcolorbox_warning:	91
92	},	92
93	preamble =	93
94	{	94


```

95     \ztex_keys_set:nn {color}
96     {
97         axiom      = {HTML}{2c3e50},
98         remark     = purple!55!black,
99         definition  = orange!55!black,
100        theorem    = blue!55!black,
101        lemma      = green!55!black,
102        corollary   = green!55!black,
103        proposition = {RGB}{0, 173, 247},
104    },
105 },
106 },
107 % theme paris from: An internet sketch book
108 paris = {
109     begin =
110     {
111         \begin{tcolorbox}
112         [
113             enhanced,    breakable,
114             top=1.5pt,   bottom=1.5pt,
115             left=3pt,    right=3pt,
116             boxrule=0pt, sharp~corners,
117             colback=gray!5, drop~fuzzy~shadow,
118             overlay~unbroken =
119             {
120                 \draw[\thm@tmp@color, line-width=0.2pt] (frame.north~west)--(frame.north~
121                 \draw[\thm@tmp@color, line-width=3pt] ([yshift=1.5pt]frame.north~
122                 \node[anchor=south~east, outer~sep=0pt, text=\thm@tmp@color]
123                 at (\linewidth-width, 1.5pt) { \_ztex_thm_icon_use:o {\thm@tmp@name} };
124             },
125             overlay~first =
126             {
127                 \draw[\thm@tmp@color, line-width=0.2pt] (frame.north~west)--(frame.north~
128                 \draw[\thm@tmp@color, line-width=3pt] ([yshift=1.5pt]frame.north~
129             },
130             overlay~last =
131             {
132                 \node[anchor=south~east, outer~sep=0pt, text=\thm@tmp@color]
133                 at (\linewidth-width, 1.5pt) { \_ztex_thm_icon_use:o {\thm@tmp@name} };
134             },
135         ]
136     },
137     end = { \end{tcolorbox} },
138     option =

```

```

139 {
140     \_ztex_thm_title_inline:n {T}
141     \_ztex_thm_tcolorbox_warning:
142 },
143 preamble =
144 {
145     \_ztex_thm_icon_set:n
146     {
147         axiom      = \ding{118},
148         definition = \ding{168},
149         theorem    = \(\heartsuit\),
150         lemma      = \ding{68},
151         corollary  = \ding{168},
152         proposition = \(\spadesuit\),
153         remark     = \ding{102} ,
154         proof      = ,
155         exercise   = ,
156         example    = ,
157         solution   = ,
158         problem    = ,
159     }
160 },
161 },
162 % elegant theme from: ElegantLaTeX Project
163 elegant = {
164     begin =
165     {
166         \begin{tcolorbox}
167         [
168             enhanced,    breakable,
169             top=8pt,      bottom=1.5pt,
170             left=3pt,     right=3pt,
171             arc=3pt,      boxrule=0.5pt,
172             before~upper*={\setlength{\parindent}{1em}},
173             fontupper=\rmfamily,    fonttitle=\bfseries,
174             lower~separated=false, separator~sign={.},
175             attach~boxed~title~to~top~left={yshift=-0.11in, xshift=0.15in},
176             boxed~title~style={boxrule=0pt, colframe=white, arc=0pt, outer~arc=0pt},
177             title = \zthmtitle*,
178             coltitle = white,          colbacktitle = \thm@tmp@color,
179             colframe = \thm@tmp@color, colback  = \thm@tmp@color!5,
180             overlay~unbroken~and~last = {
181                 \node[anchor=south~east, outer~sep=0pt, text=\thm@tmp@color]
182                     at (\linewidth-width, 1.5pt) { \_ztex_thm_icon_use:o {\thm@tmp@name} };
183             },
184         ]
185     },
186     end = { \end{tcolorbox} },

```

```

187 option =
188 {
189     \_ztex_thm_title_inline:n {F}
190     \_ztex_thm_tcolorbox_warning:
191 },
192 preamble =
193 {
194     % color
195     \ztex_keys_set:nn {color}{
196         axiom      = {HTML}{2c3e50},
197         definition  = {RGB}{0, 166, 82},
198         theorem     = {RGB}{255, 134, 23},
199         lemma       = {RGB}{255, 134, 23},
200         corollary   = {RGB}{255, 134, 23},
201         proposition = {RGB}{0, 173, 247},
202     }
203     % icon
204     \_ztex_thm_icon_set:n
205     {
206         axiom      = \ding{118},
207         definition  = \ding{168},
208         theorem     = \(\heartsuit\),
209         lemma       = \ding{68},
210         corollary   = \ding{168},
211         proposition = \(\spadesuit\),
212         remark      = \ding{102} ,
213         proof       = ,
214         exercise    = ,
215         example     = ,
216         solution    = ,
217         problem     = ,
218     }
219 },
220 % obsidian theme from: obsidian plug 'Callouts'
221 obsidian = {
222     begin =
223     {
224         \begin{tcolorbox}
225         [
226             enhanced,    breakable,
227             top=5pt,      bottom=8pt,
228             left=10pt,    right=10pt,
229             arc=3pt,      frame~hidden,
230             colback = \thm@tmp@color!20,
231             ] { \zthmtitle* }\par
232     },
233     end = { \end{tcolorbox} },
234

```

```

235 preamble =
236 {
237     % title format
238     \zthmtitleformat*
239     {
240         \noindent\sffamily\bfseries\textcolor{\thm@tmp@color}{
241             \_ztex_thm_icon_use:o {\thm@tmp@name}
242             \_ \zthmname{\,:\,}\zthmnumber
243         }
244     }
245     % icon
246     \_ztex_thm_icon_set:n
247     {
248         axiom      = \ding{111},
249         definition = \ding{118},
250         theorem    = \ding{169},
251         lemma      = \ding{170},
252         corollary  = \ding{168},
253         proposition = \ding{125},
254         remark     = \ding{46},
255         proof      = ,
256         exercise   = \ding{45},
257         example    = ,
258         solution   = \ding{45},
259         problem    = ,
260     }
261 },
262 option =
263 {
264     \_ztex_thm_title_inline:n {F}
265     \_ztex_thm_tcolorbox_warning:
266 }
267 },
268 % lapsis theme from: book 'Foundation Mathematics for the Physical Sciences'
269 lapsis = {
270     begin =
271     {
272         \begin{tcolorbox}
273         [
274             enhanced, breakable,
275             top=1.5pt, bottom=1.5pt,
276             left=2pt, leftlower=-3pt,
277             right=3pt, arc=0pt, frame~hidden,
278             bicolor, colback=\thm@tmp@color!60,
279             opacitybacklower=0,
280             overlay~first = {
281                 \fill[color=\thm@tmp@color!50, path~fading=east]
282                 (frame.north~west)++(-\linewidth/2+width/2, 0pt)

```

```

rectangle
    ($ (frame.south~east)+(\linewidth/2-width/2, 0pt)$);
\draw[color=\thm@tmp@color, thick]
    (frame.north~west)++(-\linewidth/2+width/2, 0pt)
    --
    ($ (frame.north~east)+(\linewidth/2-width/2, 0pt)$);
},
overlay~last={
    \draw[color=\thm@tmp@color, thick]
        (frame.south~west)++(-\linewidth/2+width/2, 0pt)
        --
        ($ (frame.south~east)+(\linewidth/2-width/2, 0pt)$);
    \fill[color=\thm@tmp@color!50, path~fading=east]
        (frame.north~west)++(-\linewidth/2+width/2, 0pt)
        rectangle
        ($ (frame.south~east)+(\linewidth/2-width/2, 0pt)$);
    \node[anchor=south~east, outer~sep=0pt, text=\thm@tmp@color]
        at (\linewidth-width, 0) { \_ztex_thm_icon_use:o {\thm@tmp@name} };
},
overlay~unbroken={
    \fill[color=\thm@tmp@color!50, path~fading=east]
        (frame.north~west)++(-\linewidth/2+width/2, 0pt)
        rectangle
        ($ (frame.south~east)+(\linewidth/2-width/2, 0pt)$);
    \draw[color=\thm@tmp@color, thick]
        (frame.north~west)++(-\linewidth/2+width/2, 0pt)
        --
        ($ (frame.north~east)+(\linewidth/2-width/2, 0pt)$);
    \draw[color=\thm@tmp@color, thick]
        (frame.south~west)++(-\linewidth/2+width/2, 0pt)
        --
        ($ (frame.south~east)+(\linewidth/2-width/2, 0pt)$);
    \node[anchor=south~east, outer~sep=0pt, text=\thm@tmp@color]
        at (\linewidth-width, 1.5pt) { \_ztex_thm_icon_use:o {\thm@tmp@name} };
},
]\ztex@llapnote{\zthmtitle*}
},
end = { \end{tcolorbox} },
option =
{
    \_ztex_thm_title_inline:n {F}
    \_ztex_thm_tcolorbox_warning:
},
preamble =
{
    % title foramt
    \DeclareMathSymbol{\blacktriangleleft}{\mathrel}{AMSA}{"4A}
    \zthmtitleformat*{\bfseries

```

331	<code>\zthmname\zthmnumber</code>	331
332	<code>\zthmnotemptyTF{}{\}</code>	332
333	<code>\zthmnote{}{}</code>	333
334	<code>}</code>	334
335	<code>\newcommand{\ztex@llapnote}[1]{</code>	335
336	<code>\mbox{}\llap{</code>	336
337	<code>\adjustbox{set~height=0pt, set~depth=0pt}{</code>	337
338	<code>\parbox[t]{2.85cm}{\raggedleft #1}}\hspace*{.75em}}</code>	338
339	<code>}</code>	339
340	<code>% icon</code>	340
341	<code>__ztex_thm_icon_set:n</code>	341
342	<code>{</code>	342
343	<code>axiom = \ding{111},</code>	343
344	<code>definition = \ding{118},</code>	344
345	<code>theorem = \ding{169},</code>	345
346	<code>lemma = \ding{170},</code>	346
347	<code>corollary = \ding{168},</code>	347
348	<code>proposition = \ding{125},</code>	348
349	<code>remark = \ding{46},</code>	349
350	<code>proof = ,</code>	350
351	<code>exercise = \ding{45},</code>	351
352	<code>example = ,</code>	352
353	<code>solution = \ding{45},</code>	353
354	<code>problem = ,</code>	354
355	<code>}</code>	355
356	<code>}</code>	356
357	<code>},</code>	357
358	<code>}</code>	358

10.4.6 primitive

```
1 \ProvidesExplFile{ztex.library.primitive.tex} 1
2 {2025/09/20}{\ztex@version} 2
3 {primitive-library~for~ztex} 3
4 4
5 5
6 %% collect primitives from different engines 6
7 % avoid pdf compression 7
8 % 1. https://tex.stackexchange.com/a/483040/294585 8
9 % 2. https://tex.stackexchange.com/a/454235/294585 9
10 \NewDocumentCommand{\uncompressPDF}{} 10
11 { 11
12 % >> from expl3: 12
13 \pdf_uncompress: 13
14 14
15 % >> manually set: 15
16 % \sys_if_engine_pdftex:T 16
17 % { 17
18 % \tex_pdfcompresslevel:D = 0 18
19 % \tex_pdfobjcompresslevel:D = 0 19
20 % } 20
21 % \sys_if_engine_xetex:T 21
22 % { 22
23 % \special{dvipdfmx:config~z~0} 23
24 % \special{dvipdfmx:config~C~0x40} 24
25 % } 25
26 % \sys_if_engine luatex:T 26
27 % { 27
28 % \tex_pdfvariable:D~compresslevel = 0 28
29 % \tex_pdfvariable:D~objcompresslevel = 0 29
30 % } 30
31 } 31
32 \@onlypreamble\uncompressPDF 32
```

10.4.7 pratt

```
1 \ProvidesExplFile{ztex.library.pratt.tex} 1
2 {2025/10/14}{\ztex@version} 2
3 {pratt~library~for~ztex} 3
4 4
5 % TODO: 5
6 % 1. implement this library. 6
7 % 2. use it for (partial)derivative-related commands in 'alias' module. 7
8 % 3. use it to build a layer which is similiar to tikz based on 'l3draw'. 8
9 9
10 \cs_new_protected:Npn \__pratt_lexer:n #1 10
11 { 11
12 12
13 } 13
14 14
15 \cs_new_protected:Npn \__pratt_parser_expr:n #1 15
16 { 16
17 17
18 } 18
19 19
20 \cs_new_protected:Npn \__pratt_expr:n #1 20
21 { 21
22 22
23 } 23
24 24
25 \cs_new_protected:Npn \__pratt_opbd_power:nn #1#2 25
26 { 26
27 27
28 }
```


斜体数字表示对应条目被解释说明的页面, 带下划线的数字指向该条目的定义, 其余数字表示该条目的使用位置.

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<code>\da</code>	170	<code>format.page+</code>	96
<code>\dd</code>	173, 213	<code>format.title</code>	88, 96
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zTool 接口文档

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由于本人时间有限, 目前此宏包的开发暂停.

September 23, 2026

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1 基本介绍

$\text{\LaTeX}$ 宏集已独立实现了一个 `ztool` 宏包, 此宏包中包含原来已被废弃的 `l3sys-shell` 中的所有命令. 除此之外, `ztool` 提供了 `box` 操作, 文件 IO 以及基本图形绘制相关的函数. 在 `ztool` 的协助下, $\text{\LaTeX}$ 能够避免或减少命令行 `-shell-escape` 参数或其它相关宏包的调用 (如 `robust-externalize` 宏包).

本宏包在 Github 上的地址如下:

https://github.com/zongpingding/zTeX_bundle

该仓库中包含本宏集的源码与用户手册; 当前宏集的稳定版本于 2025 年 09 月发布, 最新的开发版请切换到 “dev” 分支; 本手册适用于当前最新的开发版.

2 宏包选项

ztool 分为了“shell-escape, file-io, box, zdraw”四个库, 每一个库之间互不影响, 均可单独加载. 默认情况下, ztool 不会加载任何一个库.

ztool/shell-escape	shell-escape = $\langle \text{false} \text{true} \rangle$初始值: false
ztool/file-io	file-io = $\langle \text{false} \text{true} \rangle$初始值: false
ztool/box	box = $\langle \text{false} \text{true} \rangle$初始值: false
ztool/zdraw	zdraw = $\langle \text{false} \text{true} \rangle$初始值: false

New: 2025-05-22

这四个选项为 ztool 宏包的选项, 可以在加载 ztool 宏包时使用, 一个基本的使用样例如下, 该示例加载了 ztool 的 shell-escape 库和 box 库:

```
\usepackage[shell-escape, box=true]{ztool}
```

例 1

\ztoolloadlib \ztoolloadlib $\langle \text{library} \rangle$

New: 2025-05-22

此命令用于加载 ztool 库, $\langle \text{library} \rangle$ 为库的名称, 可选值有: “shell-escape, file-io, box, zdraw”.

一个基本的使用样例如下, 该示例加载了 ztool 的 shell-escape 库和 box 库:

```
\ztoolloadlib{shell-escape, box}
```

例 2

3 l3sys-shell

本部分主要介绍 `ztool` 中实现的原始 `l3sys-shell` 宏包中的命令。所以使用本部分的命令时需编译 `LATEX` 文档时启用 `-shell-escape` 参数, 否则此系列命令将不会执行任何操作。

WARNING: 请谨慎使用此部分的命令, 部分不当操作可能会导致一些无法挽救的后果.

```
\ztool_shell_escape:n \ztool_shell_escape:n {<command>}
```

```
\ztool_shell_escape:e
```

当 `-shell-escape` 参数启用时, 此命令会在 `shell` 中执行 `<command>`, 如果 `-shell-escape` 参数未启用, 此命令将不会执行任何操作。

Updated: 2024-12-05

```
\ztool_shell_mkdir:n \ztool_shell_mkdir:n {<dir>}
```

```
\ztool_shell_mkdir:e
```

当 `-shell-escape` 参数启用时, 此命令会创建一个目录 `<dir>`, 如果 `-shell-escape` 参数未启用, 此命令将不会执行任何操作。

Updated: 2024-12-05

```
\ztool_shell_cp:nn \ztool_shell_cp:nn {<source>}{<target>}
```

```
\ztool_shell_cp:(ee|ne|en)
```

当 `-shell-escape` 参数启用时, 此命令将把文件 `<source>` 复制为文件 `<target>`, 如果 `-shell-escape` 参数未启用, 此命令将不会执行任何操作。

Updated: 2024-12-05

```
\ztool_shell_mv:nn \ztool_shell_mv:nn {<source>}{<target>}
```

```
\ztool_shell_mv:(ee|ne|en)
```

当 `-shell-escape` 参数启用时, 此命令将把文件 `<source>` 移动到目录 `<target>`, 如果 `-shell-escape` 参数未启用, 此命令将不会执行任何操作。

Updated: 2024-12-05

```
\ztool_shell_rm:n \ztool_shell_rm:n {<file>}
```

```
\ztool_shell_rm:e
```

当 `-shell-escape` 参数启用时, 此命令将删除文件 `<file>`, 如果 `-shell-escape` 参数未启用, 此命令将不会执行任何操作。

Updated: 2024-12-05

```
\ztool_shell_rmdir:n \ztool_shell_rmdir:n {<dir>}
```

```
\ztool_shell_rmdir:e
```

当 `-shell-escape` 参数启用时, 此命令将删除目录 `<dir>`, 如果 `-shell-escape` 参数未启用, 此命令将不会执行任何操作。

Updated: 2024-12-05

```
\ztool_get_shell_pwd:N \ztool_get_shell_pwd:N <tl>
```

```
\ztool_get_shell_pwd:c
```

当 `-shell-escape` 参数启用时, 此命令将返回当前的工作目录, 并将其存放在 `<tl>` 中, 如果 `-shell-escape` 参数未启用, 此命令将不会执行任何操作。

Updated: 2024-12-05

`\ztool_shell_split_ls:nN`

Updated: 2024-12-05

`\ztool_shell_split_ls:nN {<dir>}<t1>`

当 `-shell-escape` 参数启用时, 此命令将返回目录 `<dir>` 下的所有文件名, 并将其存放在 `<t1>` 中, 如果 `-shell-escape` 参数未启用, 此命令将不会执行任何操作.

4 File IO

本部分主要介绍 `ztool` 中实现的文件 IO 操作, 包括: 读取文件, 写入文件, 追加文件等操作. 本部分的系列命令均不需要启用 `-shell-escape` 参数.

```
\ztool_file_new:nn \ztool_file_new:nn {<bool>}{<file>}
```

Updated: 2024-12-05

此命令用于创建一个名为 `<file>` 的新文件, 如果 `<file>` 不存在, 则会创建一个名为 `<file>` 的新文件. 若文件已存在, 那么当 `<bool>` 为 `\c_true_bool` 时, **会覆盖原文件**, 否则不会进行任何操作.

```
\ztool_read_file_keep_spaces:nn \ztool_read_file_keep_spaces:nn {<file>}{<code>}
```

New: 2025-09-12

此命令与 `l3file` 中的 `\ior_str_map_inline:Nn` 命令类似, 但 `\ztool_read_file_keep_spaces:nn` 会将内容的 `catcode` 改为 `\c_document_cctab`. 在 `<code>` 中, 变量 `\l_ztool_strmap_read_tl` 表示每一行的内容. **备注:** `<file>` 需要包含文件后缀; 如果 `<file>` 不存在, 则此命令不会进行任何的操作.

```
\ztool_read_file_as_seq:nnN \ztool_read_file_as_seq:nnN {<bool>}{<file>}{<seq>}
\ztool_read_file_as_seq:(neN|nnc|nec)
```

Updated: 2024-12-05

此命令用于读取文件 `<file>` 的内容, 并将其存放在 `<seq>` 中, 如果 `<file>` 不存在, 则 `<seq>` 会被置为空. `<bool>` 用于控制是否保留行尾的空格, 可选值有: `\c_true_bool`, `\c_false_bool`; 如果 `<bool>` 为 `\c_true_bool`, 则保留行尾的空格, 否则不保留.

NOTE:

1. `<seq>` 的定义是局部的;
2. 由于命令 `\ior_map_inline:Nn` 的限制, 该命令无法获取行首的“空格”或“Tab”;
3. `<seq>` 中内容的 `catcode` 为当前的 `catcode`.

```
\ztool_read_file_as_seq_keep_spaces:nnN      \ztool_read_file_as_seq_keep_spaces:nnN
\ztool_read_file_as_seq_keep_spaces:(neN|nnc|nec)  {<bool>}{<file>}{<seq>}
```

New: 2025-09-01

此命令用于读取文件 $\langle file \rangle$ 的内容, 会保留内部空格, 并将其存放在 $\langle seq \rangle$ 中, 如果 $\langle file \rangle$ 不存在, 则 $\langle seq \rangle$ 会被置为空. $\langle bool \rangle$ 用于控制是否保留行首的空格, 可选值有: $\backslash c_true_bool$, $\backslash c_false_bool$; 如果 $\langle bool \rangle$ 为 $\backslash c_true_bool$, 则保留行首的空格, 否则不保留.

NOTE:

1. $\langle seq \rangle$ 的定义是局部的;
2. 由于命令 $\backslash ior_str_map_inline:Nn$ 的限制, 该命令无法获取行末的“空格”或“Tab”;
3. $\langle seq \rangle$ 中内容的 catcode 被修改为 $\backslash c_document_cctab$.

```
\ztool_gread_file_as_seq:nnN      \ztool_read_file_as_seq:nnN {<bool>}{<file>}{<seq>}
\ztool_gread_file_as_seq:(neN|nnc|nec)
```

Updated: 2025-01-05

此命令用于读取文件 $\langle file \rangle$ 的内容, 并将其存放在 $\langle seq \rangle$ 中, 如果 $\langle file \rangle$ 不存在, 则 $\langle seq \rangle$ 会被置为空. $\langle bool \rangle$ 用于控制是否保留行尾的空格, 可选值有: $\backslash c_true_bool$, $\backslash c_false_bool$; 如果 $\langle bool \rangle$ 为 $\backslash c_true_bool$, 则保留行尾的空格, 否则不保留.

NOTE:

1. $\langle seq \rangle$ 的定义是全局的;
2. 由于命令 $\backslash ior_map_inline:Nn$ 的限制, 该命令无法获取行首的“空格”或“Tab”;
3. $\langle seq \rangle$ 中内容的 catcode 为当前的 catcode.

```
\ztool_gread_file_as_seq_keep_spaces:nnN      \ztool_gread_file_as_seq_keep_spaces:nnN
\ztool_gread_file_as_seq_keep_spaces:(neN|nnc|nec)  {<bool>}{<file>}{<seq>}
```

New: 2025-09-01

此命令用于读取文件 $\langle file \rangle$ 的内容, 会保留内部空格, 并将其存放在 $\langle seq \rangle$ 中, 如果 $\langle file \rangle$ 不存在, 则 $\langle seq \rangle$ 会被置为空. $\langle bool \rangle$ 用于控制是否保留行首的空格, 可选值有: $\backslash c_true_bool$, $\backslash c_false_bool$; 如果 $\langle bool \rangle$ 为 $\backslash c_true_bool$, 则保留行首的空格, 否则不保留.

NOTE:

1. $\langle seq \rangle$ 的定义是全局的;
2. 由于命令 $\backslash ior_str_map_inline:Nn$ 的限制, 该命令无法获取行末的“空格”或“Tab”;
3. $\langle seq \rangle$ 中内容的 catcode 被修改为 $\backslash c_document_cctab$.

```
\ztool_write_seq_to_file:nNn      \ztool_write_seq_to_file:nNn {<bool>}{<seq>}{<file>}
\ztool_write_seq_to_file:(nNe|nNV|nce|ncV)
```

New: 2025-05-27

此命令用于将 $\langle seq \rangle$ 按行写入到文件 $\langle file \rangle$ 中, 如果 $\langle file \rangle$ 不存在, 则会创建一个名为 $\langle file \rangle$ 的新文件; 若 $\langle file \rangle$ 已经存在, 则可以使用 $\langle bool \rangle$ 控制当前的写入模式: $\langle bool \rangle$ 为 $\backslash c_true_bool$ 时, 覆盖写入; $\langle bool \rangle$ 为 $\backslash c_false_bool$ 时, 追加写入; 如果 $\langle seq \rangle$ 为空, 则不会进行任何操作.

```
\ztool_append_to_file:nn      \ztool_append_to_file:nn {<file>}{<content>}
\ztool_append_to_file:(no|nf|ee)
```

Updated: 2025-01-05

此命令用于将 $\langle content \rangle$ 追加到文件 $\langle file \rangle$ 中, 如果 $\langle file \rangle$ 不存在, 则会创建一个名为 $\langle file \rangle$ 的新文件, 并将 $\langle content \rangle$ 写入其中.

```
\ztool_replace_file_line:nnn      \ztool_replace_file_line:nnn {<file>}{<line>}{<content>}
\ztool_replace_file_line:(enn|ene|eee)
```

Updated: 2025-01-05

此命令用于将文件 $\langle file \rangle$ 中的第 $\langle line \rangle$ 行替换为 $\langle content \rangle$, 如果 $\langle file \rangle$ 不存在, 则不会进行任何操作.

```
\ztool_insert_to_file:nnn      \ztool_insert_to_file:nnn {\langle file \rangle}{\langle line \rangle}{\langle content \rangle}
\ztool_insert_to_file:(nen|nfn|een)
```

Updated: 2025-01-05

此命令用于将 $\langle content \rangle$ 插入到文件 $\langle file \rangle$ 的第 $\langle line \rangle$ 行之前, 如果 $\langle file \rangle$ 不存在, 则不会进行任何操作.

下面一个示例展示了如何使用 ztool 中的几个文件 IO 操作命令:

例 3

```
% \usepackage{verbatim}
\ExplSyntaxOn
\ztool_file_new:nn {\c_true_bool}{testIO.txt}
\seq_new:N \l_ztool_tmp_seq \seq_clear:N \l_ztool_tmp_seq
\ztool_append_to_file:nn {testIO.txt} {|APPEND-CONTENT|}
\ztool_insert_to_file:nnn {testIO.txt} {1} {|INSERT--CONTENT|}
\ztool_append_to_file:nn {testIO.txt} {|APPEND-CONTENT-II|}
\ztool_replace_file_line:nnn {testIO.txt} {3} {|REPLACE-CONTENT|}
\ztool_gread_file_as_seq:nnN {\c_false_bool} {testIO.txt}
\l_ztool_tmp_seq
\seq_use:Nn \l_ztool_tmp_seq {\par}
\ExplSyntaxOff
\verbatiminput*{testIO.txt}
```

```
[INSERT-CONTENT|
|APPEND-CONTENT|
|REPLACE-CONTENT|

|INSERT-_-CONTENT|
|APPEND-CONTENT|
|REPLACE-CONTENT|
```

下面这个示例展示了 ztool 中 file to seq 这一系列命令对空格的处理方式. 文件 testSpaces.txt 中的内容如下:

```
AAAA
BB_ _BB
_ _CC_ _CC_ _
DDDD
```

例 4

```
\ExplSyntaxOn
\def\TTTa#1{
  \seq_clear:N \l_ztool_tmp_seq
```

```

\ztool_read_file_as_seq:nnN {\c_true_bool} {#1} \l_ztool_tmp_seq
\seq_show:N \l_ztool_tmp_seq
}
\def\TTTb#1{
\seq_clear:N \l_ztool_tmp_seq
\ztool_read_file_as_seq:nnN {\c_false_bool} {#1} \l_ztool_tmp_seq
\seq_show:N \l_ztool_tmp_seq
}
\def\TTTc#1{
\seq_clear:N \l_ztool_tmp_seq
\ztool_read_file_as_seq_keep_spaces:nnN {\c_true_bool} {#1} \l_ztool_tmp_seq
\seq_show:N \l_ztool_tmp_seq
}
\def\TTTd#1{
\seq_clear:N \l_ztool_tmp_seq
\ztool_read_file_as_seq_keep_spaces:nnN {\c_false_bool} {#1} \l_ztool_tmp_seq
\seq_show:N \l_ztool_tmp_seq
}
\TTTa{testSpaces.txt}
\TTTb{testSpaces.txt}
\TTTc{testSpaces.txt}
\TTTd{testSpaces.txt}
\ExplSyntaxOff
\TTTa{testSpaces.txt}
\TTTb{testSpaces.txt}
\TTTc{testSpaces.txt}
\TTTd{testSpaces.txt}

```

各种情况在命令行下显示结果 (做了一定程度的简化):

\TTTa: outside expl3, true

{AAAA_}

{BB_BB}

{CC_CC}

{DDDD}.

\TTTa: inside expl3, true

{AAAA}

{BBBB}

{CCCC}

{DDDD}.

.....

\TTTc: outside expl3, true

{AAAA}

{BB_BB}

{_CC_CC}

{DDDD}.

\TTTc: inside expl3, true

{AAAA}

{BB_BB}

{_CC_CC}

{DDDD}.

\TTTb: outside expl3, false

{AAAA}

{BB_BB}

{CC_CC}

{DDDD}.

\TTTb: inside expl3, false

{AAAA}

{BBBB}

{CCCC}

{DDDD}.

\TTTd: outside expl3, false

{AAAA}

{BB_BB}

{CC_CC}

{DDDD}.

\TTTd: inside expl3, false

{AAAA}

{BB_BB}

{CC_CC}

{DDDD}.

5 盒子操作

本部分介绍 `ztool` 中实现的 `Box` 操作, 包括 `box` 的测量以及 `box` 的简单变换.

\ztool_get_ht:Nn \ztool_get_ht:(Ne ce) Updated: 2024-12-05	\ztool_get_ht:Nn <dim>{\<content>} 此命令用于将 <content> 的高度保存在 <dim> 这一寄存器中.
\ztool_get_ht_plus_dp:Nn \ztool_get_ht_plus_dp:(Ne ce) Updated: 2024-12-05	\ztool_get_ht:Nn <dim>{\<content>} 此命令用于将 <content> 的高度和深度的和保存在 <dim> 这一寄存器中.
\ztool_get_wd:Nn \ztool_get_wd:(Ne ce) Updated: 2024-12-05	\ztool_get_wd:Nn <dim>{\<content>} 此命令用于将 <content> 的宽度保存在 <dim> 这一寄存器中.
\ztool_get_dp:Nn \ztool_get_dp:(Ne ce) Updated: 2024-12-05	\ztool_get_dp:Nn <dim>{\<content>} 此命令用于将 <content> 的深度保存在 <dim> 这一寄存器中.
\ztool_gget_ht:Nn \ztool_gget_ht:(Ne ce) Updated: 2024-12-05	\ztool_gget_ht:Nn <dim>{\<content>} 此命令用于将 <content> 的高度保存在 <dim> 这一寄存器中, 并且此操作是全局的.
\ztool_gget_wd:Nn \ztool_gget_wd:(Ne ce) Updated: 2024-12-05	\ztool_gget_wd:Nn <dim>{\<content>} 此命令用于将 <content> 的宽度保存在 <dim> 这一寄存器中, 并且此操作是全局的.
\ztool_gget_dp:Nn \ztool_gget_dp:(Ne ce) Updated: 2024-12-05	\ztool_gget_dp:nn <dim>{\<content>} 此命令用于将 <content> 的深度保存在 <dim> 这一寄存器中, 并且此操作是全局的.
\ztool_set_to_wd:nn \ztool_set_to_wd:(en ne) Updated: 2024-12-05	\ztool_set_to_wd:nn {\<dim>}{\<content>} 此命令用于将 <content> 的宽度调整为 <dim>, 其高度会进行等比例缩放, 然后将 <content> 排版出来. 备注: 此命令与 <code>\hbox_to_wd:nn</code> 等命令不同.

<code>\ztool_set_to_ht:nn</code>	<code>\ztool_set_to_ht:nn {<dim>}{<content>}</code>
<code>\ztool_set_to_ht:(en ne)</code>	此命令用于将 <code><content></code> 的高度调整为 <code><dim></code> , 其宽度会进行等比例缩放, 然后将 <code><content></code> 排版出来. 备注: 此命令与 <code>\vbox_to_ht:nn</code> 等命令不同.
Updated: 2024-12-05	

<code>\ztool_autoset_to_wd_and_ht:nnn</code>	<code>\ztool_autoset_to_wd_and_ht:nn</code>
<code>\ztool_autoset_to_wd_and_ht:(nne een eee)</code>	<code>{<width>}{<height>}{<content>}</code>
Updated: 2025-04-29	
此命令用于将 <code><content></code> 的宽度调整为 <code>min(<width>, <height>)</code> , 然后排版出来, 具体原理请参见 <code>\box_autosize_to_wd_and_ht:Nnn</code> 命令的说明.	

<code>\ztool_rotate:nn</code>	<code>\ztool_rotate:nn {<angle>}{<content>}</code>
<code>\ztool_rotate:(en ne ee)</code>	此命令用于将 <code><content></code> 旋转 <code><angle></code> 度, 然后排版出来.
New: 2025-04-29	

<code>\ztool_scale_to_wd:nn</code>	<code>\ztool_scale_to_wd:nn {<dim>}{<content>}</code>
<code>\ztool_scale_to_wd:(en ne ee)</code>	
New: 2025-04-29	
此命令用于将 <code><content></code> 的宽度调整为 <code><dim></code> , 但是不对盒子的高度做任何的调整, 然后排版出来.	

<code>\ztool_scale_to_ht:nn</code>	<code>\ztool_scale_to_ht:nn {<dim>}{<content>}</code>
<code>\ztool_scale_to_ht:(en ne ee)</code>	
New: 2025-04-29	
此命令用于将 <code><content></code> 的高度 + 深度整体调整为 <code><dim></code> , 但是不对盒子的宽度做任何的调整, 然后排版出来.	

<code>\ztool_scale_to_wd_and_ht:nnn</code>	<code>\ztool_scale_to_wd_and_ht:nnn {<width>}{<height>}{<content>}</code>
<code>\ztool_scale_to_wd_and_ht:(nno nne eee)</code>	
New: 2025-04-29	
此命令用于将 <code><content></code> 的宽度调整为 <code><width></code> , 高度 + 深度整体调整为 <code><height></code> , 然后排版出来.	

<code>\ztool_box_item_align:Nnnn</code>	<code>\ztool_box_item_align:Nnnn</code>
<code>\ztool_box_item_align:(cnnn Nnnn cnno Nnen Nnee)</code>	<code>\cmd{\langle width \rangle}{\langle content \rangle}{\langle align \rangle}</code>

Updated: 2025-05-13

此命令用于将 $\langle content \rangle$ 的宽度调整为 $\langle width \rangle$, 然后排版出来, $\langle align \rangle$ 用于控制对齐方式, 可选值有: `left`, `center`, `right`, `scatter`. $\langle cmd \rangle$ 为一个命令, 其接受一个参数, 它将应用到 $\langle content \rangle$ 的每一个 Token 上. **注意:** $\langle content \rangle$ 中的空格会被忽略, 如果需要空格, 请使用 “_” 或 “~” 替代.

<code>\ztool_fp_to_rad:n</code>	<code>\ztool_fp_to_rad:n {\langle angle \rangle}</code>
---------------------------------	---

New: 2025-05-12

此命令用于将 $\langle angle \rangle$ 从弧度制转换为角度制.

<code>\ztoolboxaffine</code>	<code>\ztoolboxaffine[\langle key-value \rangle]{\langle content \rangle}{\langle matrix \rangle}</code>
------------------------------	--

New: 2025-05-12

上述 $\langle content \rangle$ 表示仿射变换作用的对象; $\langle matrix \rangle$ 为一个 2×2 的矩阵, 表示对应的仿射变换矩阵. 若 $\langle matrix \rangle = \{a, b, c, d\}$, 则其对应的仿射变换矩阵 Λ 如下:

$$\Lambda = \begin{bmatrix} a & c \\ b & d \end{bmatrix}.$$

若 $\det \Lambda = 0$, 则此变换无意义, `ztool` 会在终端输出一条警告, 最后将 $\langle content \rangle$ 中的内容原样输出到 PDF; $\langle keyval \rangle$ 设置请参见下述 “`ztool/affine`” 说明. **备注:** 此命令封装自下述的 `\ztool_affine_transformation:Nnnnnn` 命令.

<code>ztool/affine/debug</code>	<code>debug = \langle true false \rangle</code> 初始值: <code>false</code>
<code>ztool/affine/pole-1</code>	<code>pole-1 = \langle coffin's pole \rangle</code> 初始值: <code>1</code>
<code>ztool/affine/pole-2</code>	<code>pole-2 = \langle coffin's pole \rangle</code> 初始值: <code>b</code>
<code>ztool/affine/xoffset</code>	<code>xoffset = \langle number \rangle</code> 初始值: <code>Opt</code>
<code>ztool/affine/yoffset</code>	<code>yoffset = \langle number \rangle</code> 初始值: <code>Opt</code>

$\langle debug \rangle$ 用于调试, 如果设置为 `true`, 则会在 PDF 中输出一些中间变量信息, 用于调试; 其中 $\langle xoffset \rangle$, $\langle yoffset \rangle$ 为水平和垂直方向的偏移量, 默认值均为 `Opt`; $\langle pole-1 \rangle$, $\langle pole-2 \rangle$ 用于设置打印 coffin 时的参考点, 二者必须相交. 关于后四个 $\langle kye \rangle$ 的详细使用方法可以参见 `l3coffins` 的说明.

<code>\ztool_affine_transformation:Nnnnn</code>	<code>\ztool_affine_transformation:Nnnnn</code>
<code>\ztool_affine_transformation:(Neeee cnnnn ceeee)</code>	<code>\ztool_affine_transformation:\langle coffin \rangle \{ \langle a \rangle \} \{ \langle b \rangle \} \{ \langle c \rangle \} \{ \langle d \rangle \}</code>

New: 2025-05-12

此命令用于对 $\langle coffin \rangle$ 进行任意的仿射变换 (线性变换), 具体的使用方法可以参见前述的 `\ztoolboxaffine` 命令; 上述参数对应的仿射变换矩阵 Λ 为

$$\Lambda = \begin{bmatrix} a & c \\ b & d \end{bmatrix}.$$

关于上述函数 `\ztool_affine_transformation:Nnnnn` 的一些技术细节:

给定任意一个仿射变换 Λ , 不妨设

$$\Lambda = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}.$$

我们可以做如下的分解 (与 SVD 分解类似), 令 $m = 2x$, 则有:

$$\begin{aligned} \Lambda &= \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 1 & m \\ 0 & 1 \end{bmatrix} \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} \cos \omega & -\sin \omega \\ \sin \omega & \cos \omega \end{bmatrix} \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}. \end{aligned} \quad (5.1)$$

我们给出如下的记号:

- $\mathbf{T}_1(\theta)$: 旋转矩阵, 绕原点逆时针旋转 θ 角;
- $\mathbf{T}_2(x)$: 缩放矩阵, 把 x 轴方向的所有向量变为原来的 x 倍;
- $\mathbf{T}_3(y)$: 缩放矩阵, 把 y 轴方向的所有向量变为原来的 y 倍;

那么我们可以认为 $\{\mathbf{T}_1(\theta), \mathbf{T}_2(x), \mathbf{T}_3(y)\}$ 就是 $A_{2 \times 2}$ 的基. 所以我们可以把上面的 方程 (5.1) 写成如下表达式:

$$\Lambda = \mathbf{T}_1(\theta) \cdot \mathbf{T}_1(\phi) \cdot \mathbf{T}_2(S_x) \cdot \mathbf{T}_3(S_y) \cdot \mathbf{T}_1(\omega) \cdot \mathbf{T}_2(s_x) \cdot \mathbf{T}_3(s_y). \quad (5.2)$$

根据矩阵乘法的结果, 我们可以知道上述的 m, s_x, S_x, ϕ 等参数如下:

$$s_x = \sqrt{A_{11}^2 + A_{21}^2}, \quad \theta = \arctan\left(\frac{A_{21}}{A_{11}}\right).$$

s_y 和 m 的求解结果如下:

$$ms_y = A_{12} \cos \theta + A_{22} \sin \theta, \quad s_y = \begin{cases} \frac{ms_y \cos \theta - A_{12}}{\sin \theta} & \text{如果 } \sin \theta \neq 0, \\ \frac{A_{22} - ms_y \sin \theta}{\cos \theta} & \text{如果 } \sin \theta = 0; \end{cases}$$

那么此时很容易知道 $m = ms_y/s_y$. 对 shear matrix 的分解结果如下:

$$\begin{aligned} S_x &= \sqrt{\frac{m^2}{4} + 1} - \frac{m}{2}, & S_y &= \sqrt{\frac{m^2}{4} + 1} + \frac{m}{2}, \\ \phi &= -\frac{\pi}{4} - \frac{1}{2} \arctan\left(\frac{m}{2}\right), & \omega &= \frac{\pi}{4} - \frac{1}{2} \arctan\left(\frac{m}{2}\right). \end{aligned}$$

最后我们只需要从右到左将这一系列的变换应用到 `\box` 上即可. 从上面也可以看出, 命令 `\ztool_affine_transformation:Nnnnn` 仅依赖于 $\text{\LaTeX}$ 中的 `\coffin_scale:Nnn` 和 `\coffin_rotate:Nn` 两个函数.

命令 `\ztool_affine_transformation:Nnnnn` 实现过程中相关的参考链接如下:

- <https://math.stackexchange.com/a/3521141/1235323>;
- <https://math.stackexchange.com/a/281087/1235323>.

如果原 $\text{\TeX}$ 引擎提供了 shear transformation 相关的 primitive, 那么上述对 shear matrix 的分解就是不必要的. 部分的引擎中原始提供了仿射变换矩阵这一 primitive, 比如 $\text{pdf}\text{\TeX}$ 中的 `\pdfsetmatrix` 命令.

下面的示例展示了如何使用这一章节中的几个 Box 操作命令:

例 5

```
\ExplSyntaxOn
\setlength{\fboxsep}{0pt}
% get dim of content
\dotfill\par
\dim_new:N \l_ztool_tmp_H_dim
\dim_new:N \l_ztool_tmp_W_dim
\ztool_get_ht:Nn \l_ztool_tmp_H_dim {Hello,~world!}
\ztool_get_wd:Nn \l_ztool_tmp_W_dim {Hello,~world!}
\dim_use:N \l_ztool_tmp_H_dim \quad \dim_use:N \l_ztool_tmp_W_dim\par

% set content to dim
\dotfill\par
Hello,~world|
\ztool_set_to_ht:nn {.5cm} {Hello,~world}|
\ztool_set_to_wd:nn {40pt} {Hello,~world}\par

% scale one dimension
\dotfill\par
\ztool_scale_to_wd:nn {2em}{\fbox{AA}}\par
\ztool_scale_to_wd:nn {2em}{\fbox{AAA}}\par
\ztool_scale_to_wd:nn {2em}{\fbox{AAAAA}}\par
\ztool_scale_to_ht:nn {2.5em}{\fbox{\vbox{\hbox{A}}}}\quad
\ztool_scale_to_ht:nn {2.5em}{\fbox{\vbox{\hbox{A}\hbox{A}}}}\quad
\ztool_scale_to_ht:nn {2.5em}{\fbox{\vbox{\hbox{A}\hbox{A}\hbox{A}}}}\quad
\ztool_scale_to_ht:nn {2.5em}{\fbox{\vbox{\hbox{A}\hbox{A}\hbox{A}\hbox{A}}}}\quad
\par

% box item align
\dotfill\par
\def\boxItemCmd#1{\textcolor{blue}{|#1|}}
\underline{
  \ztool_box_item_align:Nnnn \boxItemCmd{15em}{{Tom}{Amy}{Jennery}}{scatter}
```

```

}\par
\underline{
  \ztool_box_item_align:Nnnn \boxItemCmd{15em}{\{Tom\} \{Amy\} \{Jennery\}}
}\par

% affine transform
\dotfill\par
\hcoffin_set:Nn \l_tmpa_coffin {\rule{2em}{2em}}
\coffin_typeset:Nnnnn \l_tmpa_coffin {1}{b}{0pt}{0pt}
\ztool_affine_transformation:Nnnnn \l_tmpa_coffin {1}{0}{.5}{1}
\coffin_typeset:Nnnnn \l_tmpa_coffin {1}{b}{0pt}{0pt}
\ExplSyntaxOff

```

.....

7.8402pt 60.87103pt

.....

Hello, world|Hello, world|Hello, world

.....

AA

AAA

AAAAA

	A	A	A
A	A	A	A

.....

|Tom| |Amy| |Jennery|

|Tom||Amy||Jennery|

.....



6 zdraw

这部分主要包含一些图像绘制命令, 这系列的命令并不依赖于 tikz 宏包, 它们的主要依赖项如下:

- L^AT_EX 2_ε 内置 picture 环境;
- pict2e : L^AT_EX 2_ε 内置 picture 环境的增强版, 提供了更好的绘图功能;
- bxeepic: 可以用于提供 dash line 支持, 目前未引入该宏包.

zpic	<code>\begin{zpic}[(key-value)] <draw commands> \end{zpic}</code>
New: 2025-05-13	此环境基于 L ^A T _E X 2 _ε 内置 picture 环境定义,

ztool/draw/picture/unit	unit	= <长度>.....	初始值: 1cm
ztool/draw/picture/width	width	= <浮点数>.....	初始值: 0
ztool/draw/picture/height	height	= <浮点数>.....	初始值: 0
ztool/draw/picture/depth	depth	= <浮点数>.....	初始值: 0
ztool/draw/picture/xoffset	xoffset	= <浮点数>.....	初始值: 0
ztool/draw/picture/yoffset	yoffset	= <浮点数>.....	初始值: 0

上述的 <width> 如果采用默认值, 那么当前图片将会与周围的排版内容相重叠.

\put	<code>\put (<x, y>) {<content>}</code>
New: 2025-05-13	此命令与 L ^A T _E X 2 _ε 内置 picture 环境中的 \put 命令相同. 注意: 此命令需要在 picture 或 zpic 环境中使用.

\zline	<code>\zline [(key-value)](<coor-1>)(<coor-2>)</code>
New: 2025-05-13	此命令用于绘制一条从 <coor-1> 到 <coor-2> 的线段, <key-value> 用于设置线条的属性, 可用选项请参见后续的 <parent=ztool/draw/picture/line>.

ztool/./line/draw	draw	= <颜色>.....	初始值: black
ztool/./line/width	width	= <长度>.....	初始值: .4pt
ztool/./line/dash	dash	= <true false>.....	初始值: false

上述 <width> 用于设置线条的宽度, <draw> 用于设置线条的颜色, <dash> 用于设置线条是否为虚线. 注意: 目前 <dash> 选项还未适配, 处于不可用的状态.

\zvector	<code>\zvector [(key-value)](<coor-1>)(<coor-2>)</code>
New: 2025-05-13	此命令用于绘制向量, 该向量的起点为 <coor-1>, 终点为 <coor-2>; <key-value> 用于设置该向量的外观属性, 其继承自 <parent=ztool/draw/picture/line>, 其余的可用选项请参见后续 <parent=ztool/draw/picture/line/vector>.

ztool/../../vector/	<code>> = <latex pst>.....</code> 初始值: <code>latex</code> 此选项用于控制箭头的样式, 默认为 L^AT_EX 样式, 即 <code>\ltxarrows</code>; <code><pst></code>, 即 PsTricks, 对应于 <code>\pstarrows</code> 命令.
<code>\zdraw</code> New: 2025-05-13	<code>\zdraw [<key-value>](<coor-1>)...(<coor-n>);</code> 此命令将绘制一条从点 <code><coor-1></code> 到点 <code><coor-n></code> 的折线段, <code><key-value></code> 继承自 <code><parent=ztool/draw/picture/line></code>, 可以用于设置线条的属性, 额外可用的选项请参见后续的 <code><parent=ztool/draw/picture/zdraw></code>. 注意: 此命令末尾的 “;” 是不能省略的, 否则会报错.
ztool/../../zdraw/vector ztool/../../zdraw/cycle ztool/../../zdraw/fill ztool/../../zdraw/shift	<code>vector = <false true>.....</code> 初始值: <code>false</code> <code>cycle = <false true>.....</code> 初始值: <code>false</code> <code>fill = <false true 颜色>.....</code> 初始值: <code>false</code> <code>shift = {<浮点数, 浮点数>}.....</code> 初始值: <code>{0, 0}</code> 当 <code><fill></code> 设置为 <code>true</code> 时, <code><cycle></code> 会自动设置为 <code>true</code>; <code><vector></code> 用于设置是否将每一个子线段替换为向量. <code><shift></code> 分别表示 x 和 y 方向的偏移量. 注意: <code><shift></code> 选项中的 <code>{ }</code> 不能省略.
<code>\zarc</code> New: 2025-05-13	<code>\zarc [<key-value>](<浮点数, 浮点数>)</code> 此命令用于绘制一个圆弧, <code>(<浮点数, 浮点数>)</code> 为其圆心, 默认绘制 $\frac{1}{4}$ 圆弧; <code><key-value></code> 继承自 <code><parent=ztool/draw/picture/line></code>, 可以用于设置线条的属性, 额外可用的选项请参见后续的 <code><parent=ztool/draw/picture/zarc></code>.
ztool/../../zarc/radius ztool/../../zarc/start ztool/../../zarc/end ztool/../../zarc/fill	<code>radius = <浮点数>.....</code> 初始值: <code>.5</code> <code>start = <浮点数>.....</code> 初始值: <code>0</code> <code>end = <浮点数>.....</code> 初始值: <code>90</code> <code>fill = <false true 颜色>.....</code> 初始值: <code>false</code> <code><start></code> 按照逆时针旋转到角度 <code><end></code> 结束; <code><radius></code> 为圆弧的半径; <code><fill></code> 用于设置圆弧的填充颜色.
<code>\zcircle</code> New: 2025-05-13	<code>\zcircle [<key-value>](<浮点数, 浮点数>)</code> 此命令基于上述的 <code>\zarc</code> 命令, 默认情况下将以 <code>(<浮点数, 浮点数>)</code> 为圆心绘制一个完整的圆; <code><key-value></code> 和上述的 <code>\zrac</code> 命令中的 <code><key-value></code> 选项相同,
<code>\zrectangle</code> New: 2025-05-13	<code>\zrectangle [<key-value>](<coor-1>)(<coor-2>)</code> 此命令用于绘制矩形, <code>(<coor-1>)</code> 和 <code>(<coor-2>)</code> 为矩形对角线的两个端点坐标; <code><key-value></code> 继承自 <code><parent=ztool/draw/picture/line></code>, 其余的 <code><key-value></code> 请参见后续 <code><parent=ztool/draw/picture/zrectangle></code>.

ztool/./zrectangle/arc	arc = <浮点数> 初始值: 0
ztool/./zrectangle/fill	fill = <false true 颜色> 初始值: false
	<fill> 用于设置矩形的填充颜色, <arc> 用于设置矩形圆角对应的半径.

\zpin	\zpin[<foreground background>]{<code>}
New: 2025-07-09	此命令用于给当前页面添加标注, 参考点为当前页面的右下角, 并且取向右上为正方向.

```
\zpin{%  
  \begin{zpic}%  
    \zline[width=5pt] (0, 0) (1, 1)  
  \end{zpic}  
}
```

例 6

为便于理解上述绘图命令的基本用法，现提供若干绘图示例.

案例 1: 基础的线段绘制命令.

例 7

```

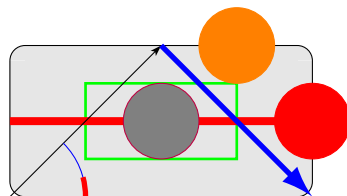
\mbox{} \vskip 2em
\begin{zpic}[unit=2em]
  \zdraw[fill, cycle] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[cycle, shift={2, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[fill, shift={4, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[draw=red, width=1pt, shift={6, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[vector, shift={8, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[vector, cycle, shift={10, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[vector, fill, shift={12, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[vector, cycle, fill, shift={14, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
\end{zpic}

```

案例 2: 基本的几何元素绘制命令.

例 8

```
\mbox{} \vskip 5em
\begin{zpic}[unit=2cm, xoffset=2]
% 1. rectangle
\zrectangle[arc=.1, fill=gray!20](0, 0)(2, 1)
\zrectangle[draw=green, width=1pt](.5, .25)(1.5, .75)
% 2. line / vector
\zline[width=3pt, draw=red](0, .5)(2, .5)
\zvector[>=pst](0, 0)(1, 1)
\zvector[draw=blue, width=2pt](1, 1)(2, 0)
% 3. arc / circle
\zarc[draw=blue, end=45](0, 0) % fill=<empty>
\zarc[draw=blue, width=2pt, end=15, fill=, draw=red](0, 0)
\zcircle[radius=.25, fill, draw=purple](1, .5)
\zcircle[radius=.25, fill=orange, draw=none](1.5, 1)
\zcircle[radius=.25, fill=red, draw=](2, .5)
\end{zpic}
```



7 TODO

ztool 在将来也许会有改动, 这里列出部分将来可能会完善的功能 (☐ – 未完成; ☒ – 已完成; ☐ – 不考虑该功能):

- ☐ 重新实现 xsimverb 宏包中的 `\xsim_file_write_start:nn` 和 `\xsim_file_write_stop:` 命令, 使其和 ztool 宏包适配.
- ☒ 2025-05-22-已完成:修复 `\ztool_append_to_file:nn` 文件首行空行的问题.
- ☒ 2025-09-01-已完成:针对命令 `\ztool_read_file_as_seq:nnN`, 有些情况下需要保留源文件中的所有空格, 可以参考命令 `\seq_set_split_keep_spaces:Nnn`.
- ☐ 使用的已实现的 `\ztex_tl_replace_all:nnn` 或 `\ztex_tl_replace_once:nnn` 命令实现 `\ztool_replace_file_line_text:nnnn {<file>}{<line>}{<pattern>}{<text>}`, 并且在 `<pattern>` 中实现简单的正则表达式功能, 需要确保该命令是可展的.
- ☐ 使用 l3tl-analysis 中的 `\tl_analysis_map_inline:nn` 命令 (该命令可以捕捉 '{', '}', '\$' 等特殊字符) 实现一个简易的 token 调试命令.
- ☐ 使用 l3draw 封装一个类似 tikz 的前端, 需要其原生支持 3D 绘图, 自动调整遮挡关系.
- ☒ 2025-07-09-已完成:`\zline` 绘制垂直或水平线段时报错或结果不符合预期
- ☒ 2026-09-23-已完成:`\zline` 和 `\zdraw` 二者的效果不一致, 在同一个坐标系绘制同一条线段, 二者无法重合 (目前来看 `\zline` 命令才是正确的).

8 zTool 源码

本节列出 ztool 宏包的源码, 供有需要的用户参考. 另, 本节所有打印源码的对应行号已经过特殊处理, 用户可直接从 PDF 中复制源码样例, 加以修改使用.

8.1 ztool.sty

1	%%%	1
2	%% ztool.sty	2
3	%% Copyright 2024, 2025 Zongping Ding.	3
4	%	4
5	% This work may be distributed and/or modified under the conditions of the	5
6	% LaTeX Project Public License, either version 1.3 of this license or any	6
7	% later version.	7
8	% The latest version of this license is in	8
9	% http://www.latex-project.org/lppl.txt	9
10	% and version 1.3 or later is part of all distributions of LaTeX	10
11	% version 2005/12/01 or later.	11
12	%	12
13	% This work has the LPPL maintenance status `maintained'.	13
14	%	14
15	% The Current Maintainer of this work is Zongping Ding.	15
16	%	16
17	% ztool.sty consists of the parts:	17
18	% shell-escape,	18
19	% file-io,	19
20	% box,	20
21	% zdraw.	21
22	%%%	22
23	\NeedsTeXFormat{LaTeX2e}	23
24	\ProvidesExplPackage{ztool}{2025/05/20}{1.0.1}{A~pre-release~tool~package~for~LaTeX}	24
25		25
26		26
27	%%% 13keys intial patch begin %%%	27
28	% 1. https://github.com/latex3/latex3/issues/1738	28
29	% 2. https://tex.stackexchange.com/q/742604/294585	29
30	\cs_set_protected:Npn __keys_initialise:n #1	30
31	{	31
32	\exp_after:wN __keys_find_key_module:wNN	32
33	\l_keys_path_str \s_keys_stop	33
34	\l_keys_key_tl \l_keys_key_str	34
35	\tl_set_eq:NN \l_keys_key_tl \l_keys_key_str	35
36	\tl_set:Nn \l_keys_value_tl {#1}	36
37	\cs_if_exist:cTF { \c__keys_code_root_str \l_keys_path_str }	37
38	{	38
39	\str_clear:N \l__keys_inherit_str	39
40	__keys_execute:nn \l_keys_path_str {#1}	40
41	}	41
42	{	42
43	\cs_if_exist:cT	43
44	{ \c__keys_inherit_root_str __keys_parent:o \l_keys_path_str }	44
45	{ __keys_execute_inherit: }	45
46	}	46

47	}	47
48	%%%% 13keys intial patch end %%%%	48
49		49
50		50
51	\clist_new:N \g__ztool_library_loaded_clist	51
52	\clist_gclear:N \g__ztool_library_loaded_clist	52
53	\bool_new:N \g__ztool_lib_user_load_dupuplicate_bool	53
54	\bool_gset_false:N \g__ztool_lib_user_load_dupuplicate_bool	54
55	\cs_new_nopar:Npn __ztool_load_library:n #1	55
56	{	56
57	\clist_map_inline:nn {#1} {	57
58	\clist_if_in:NnTF \g__ztool_library_loaded_clist {##1} {	58
59	\msg_set:nnn {ztool} {library-loaded}	59
60	{	60
61	ztool~library~"##1"~already~loaded,ignored~loading.	61
62	\msg_line_context:	62
63	}	63
64	\bool_if:NT \g__ztool_lib_user_load_dupuplicate_bool	64
65	{	65
66	\msg_warning:nnn {ztool} {library-loaded} {##1}	66
67	}	67
68	}{	68
69	\file_if_exist:nTF {library/ztool.library.##1.tex}{	69
70	\clist_gput_right:Nn \g__ztool_library_loaded_clist {##1}	70
71	\makeatletter\file_input:n {library/ztool.library.##1.tex}	71
72	}{	72
73	\msg_set:nnn {ztool} {library-not-found} {ztool~library~`##1'~not~found.}	73
74	\msg_error:nnn {ztool} {library-not-found} {##1}	74
75	}	75
76	}	76
77	}	77
78	}	78
79	\NewDocumentCommand\ztoolloadlib{m}	79
80	{	80
81	__ztool_load_library:n {#1}	81
82	\bool_gset_true:N \g__ztool_lib_user_load_dupuplicate_bool	82
83	\ExplSyntaxOff	83
84	}	84
85	\keys_define:nn { ztool }	85
86	{	86
87	shell-escape .code:n = { __ztool_load_library:n {shell-escape} },	87
88	file-io .code:n = { __ztool_load_library:n {file-io} },	88
89	box .code:n = { __ztool_load_library:n {box} },	89
90	zdraw .code:n = { __ztool_load_library:n {zdraw} },	90
91	}	91
92	\ProcessKeyOptions [ztool]	92

8.2 shell-escape

```
1 %% 1
2 %% This is file `ztool.library.shell-escape.tex'. 2
3 %% This file is based on the original source code with modifications. 3
4 %% The original disclaimer reads as follows: 4
5 %% 5
6 %% 6
7 %% This is file `l3sys-shell.sty', 7
8 %% generated with the docstrip utility. 8
9 %% 9
10 %% The original source files were: 10
11 %% 11
12 %% l3sys-shell.dtx (with options: `package') 12
13 %% 13
14 %% Copyright (C) 2018,2019 The LaTeX3 Project 14
15 %% 15
16 %% It may be distributed and/or modified under the conditions of 16
17 %% the LaTeX Project Public License (LPPL), either version 1.3c of 17
18 %% this license or (at your option) any later version. The latest 18
19 %% version of this license is in the file: 19
20 %% 20
21 %% http://www.latex-project.org/lppl.txt 21
22 %% 22
23 %% This file is part of the "l3experimental bundle" (The Work in LPPL) 23
24 %% and all files in that bundle must be distributed together. 24
25 %% 25
26 %% File: l3sys-shell.dtx 26
27 \ProvidesExplFile{ztool.library.shell-escape.tex} 27
28 {2025/09/06}{1.0.1} 28
29 {shell-escape-library~for~ztool} 29
30 30
31 31
32 % ==> l3sys-shell tool 32
33 % windows path handle 33
34 \cs_new:Npn \ztool_sys_path_to_win:N #1 34
35 { 35
36 \quark_if_nil:NF #1 { 36
37 \token_if_eq_meaning:NNTF #1 / 37
38 { \c_backslash_str } 38
39 {#1} 39
40 \ztool_sys_path_to_win:N 40
41 } 41
42 } 42
43 \cs_new:Npn \ztool_sys_path_to_win:w #1 ~ #2 \q_stop 43
44 { 44
45 \ztool_sys_path_to_win:N #1 \q_nil 45
46 \tl_if_empty:nF {#2} 46
```

```
47     {
48         \c_space_tl
49         \__sys_path_to_win:w #2 \q_stop
50     }
51 }
52 \cs_new:Npn \ztool_sys_path_to_win:n #1
53 {
54     \exp_after:wN \ztool_sys_path_to_win:w
55     \tl_to_str:n {#1} ~ \q_stop
56 }
57 % respective commands
58 \cs_new_protected:Npn \ztool_shell_escape:n #1
59 {
60     \sys_if_shell_unrestricted:T
61     { \sys_shell_now:n {#1} }
62 }
63 \cs_generate_variant:Nn \ztool_shell_escape:n {e}
64 \cs_new_protected:Npe \ztool_shell_mkdir:n #1
65 {
66     \ztool_shell_escape:e {
67         \sys_if_platform_unix:T
68         {mkdir~\p~\exp_not:N \tl_to_str:n {#1}}
69         \sys_if_platform_windows:T
70         {mkdir~ \exp_not:N \ztool_sys_path_to_win:n {#1}}
71     }
72 }
73 \cs_new_protected:Npe \ztool_shell_cp:nn #1#2
74 {
75     \ztool_shell_escape:e {
76         \sys_if_platform_unix:T
77         {
78             cp~-f~ \exp_not:N \tl_to_str:n {#1} ~
79             \exp_not:N \tl_to_str:n {#2}
80         }
81         \sys_if_platform_windows:T
82         {% can NOT use wildcards in CMD
83             copy~/y~ \exp_not:N \ztool_sys_path_to_win:n {#1} ~
84             \exp_not:N \ztool_sys_path_to_win:n {#2}
85         }
86     }
87 }
88 \cs_new_protected:Npe \ztool_shell_mv:nn #1#2
89 {
90     \ztool_shell_escape:e {
91         \sys_if_platform_unix:T
92         {
93             mv~ \exp_not:N \tl_to_str:n {#1} ~
94             \exp_not:N \tl_to_str:n {#2}
```

```
95     }
96     \sys_if_platform_windows:T
97     {
98         copy~/y~ \exp_not:N \ztool_sys_path_to_win:n {#1} ~
99         \exp_not:N \ztool_sys_path_to_win:n {#2}
100         \token_to_str:N & \token_to_str:N &
101         del~/f~/q~\exp_not:N \ztool_sys_path_to_win:n {#1}
102     }
103 }
104 }
105 \cs_new_protected:Npe \ztool_shell_rm:n #1
106 {
107     \ztool_shell_escape:e {
108         \sys_if_platform_unix:T
109         { rm~-f~ \exp_not:N \tl_to_str:n {#1} }
110         \sys_if_platform_windows:T
111         { del~/f~/q~ \exp_not:N \ztool_sys_path_to_win:n {#1} }
112     }
113 }
114 \cs_new_protected:Npe \ztool_shell_rmdir:n #1
115 {
116     \ztool_shell_mkdir:n {#1}
117     \ztool_shell_escape:e {
118         \sys_if_platform_unix:T
119         { rm~-rf~ \exp_not:N \tl_to_str:n {#1} }
120         \sys_if_platform_windows:T
121         { rmdir~/s~/q~ \exp_not:N \ztool_sys_path_to_win:n {#1} }
122     }
123 }
124 \tl_new:N \l__ztool_shell_tmp_tl
125 \cs_new_protected:Npe \ztool_get_shell_pwd:N #1
126 {
127     \exp_not:N \sys_get_shell:nnN
128     {
129         \sys_if_platform_unix:T { pwd }
130         \sys_if_platform_windows:T { cd }
131     }{
132         \char_set_catcode_other:N \exp_not:N \
133         \char_set_catcode_other:N \exp_not:N \#
134         \char_set_catcode_other:N \exp_not:N \~
135         \char_set_catcode_other:N \exp_not:N \%
136         \char_set_catcode_space:N \exp_not:N \_
137         \tex_endlinechar:D -1 \scan_stop:
138     }
139     \exp_not:N \l__ztool_shell_tmp_tl
140     \str_set:NV #1 \exp_not:N \l__ztool_shell_tmp_tl
141 }
142 \cs_new_protected:Npe \ztool_shell_split_ls:nN #1#2
```

143	{	143
144	\exp_not:N \sys_get_shell:nn	144
145	{	145
146	\sys_if_platform_unix:T { ls~-1~ #1 }	146
147	\sys_if_platform_windows:T { dir~/b~ #1 }	147
148	}{	148
149	\ExplSyntaxOff	149
150	\char_set_catcode_other:N \exp_not:N \	150
151	\char_set_catcode_other:N \exp_not:N \#	151
152	\char_set_catcode_other:N \exp_not:N \~	152
153	\char_set_catcode_other:N \exp_not:N \%	153
154	\char_set_catcode_other:n { 13 }	154
155	}	155
156	\exp_not:N \l__ztool_shell_tmp_tl	156
157	\str_set:NV \exp_not:N \l__sys_tmp_tl \exp_not:N \l__sys_tmp_tl	157
158	\seq_set_split:NnV #2	158
159	{ \char_generate:nn { `^^M } { 12 } }	159
160	\exp_not:N \l__ztool_shell_tmp_tl	160
161	\seq_pop_right:NN #2 \exp_not:N \l__sys_tmp_tl	161
162	}	162
163	\cs_generate_variant:Nn \ztool_shell_mkdir:n {e}	163
164	\cs_generate_variant:Nn \ztool_shell_cp:nn { ee, ne, en }	164
165	\cs_generate_variant:Nn \ztool_shell_mv:nn { ee, ne, en }	165
166	\cs_generate_variant:Nn \ztool_shell_rm:n { e, f, o }	166
167	\cs_generate_variant:Nn \ztool_shell_rmdir:nn { e, f, o }	167
168	\cs_generate_variant:Nn \ztool_get_shell_pwd:N {c}	168
169	\cs_generate_variant:Nn \ztool_shell_split_ls:nN {nc}	169

8.3 file-io

```
1 \ProvidesExplFile{ztool.library.file-io.tex}
2 {2025/09/12}{1.0.1}
3 {file-io~library~for~ztool}
4
5
6 % ==> file IO operations
7 % 1. create a new file
8 % 2. append to a file
9 % 3. read from file / write to file
10 \ior_new:N \g_ztool_file_read_ior
11 \ior_new:N \g_ztool_file_append_ior
12 \iow_new:N \g_ztool_file_append_iow
13 \tl_new:N \l_ztool_current_line
14 \str_clear:N \l_ztool_file_ori_content_str
15 \seq_new:N \l_ztool_file_seq
16 \seq_new:N \l_ztool_tmp_seq
17 \cs_generate_variant:Nn \seq_use:Nn { Ne }
18
19 % read file as seq(not keep internal spaces):
20 \cs_new_protected:Npn \ztool_read_file_as_seq:nnN #1#2#3
21 {% #1: bool(True to keep spaces, False to trim); #2: file name; #3: seq
22   \seq_clear:N #3
23   \file_if_exist:nT {#2}
24   {
25     \ior_open:Nn \g_ztool_file_read_ior {#2}
26     \ior_map_inline:Nn \g_ztool_file_read_ior
27     {
28       \tl_if_empty:nF {##1}
29       {
30         \bool_if:nTF {#1}
31         { \seq_put_right:Nn #3 {##1} }
32         {
33           \seq_put_right:Ne #3
34           { \tl_trim_spaces:n {##1} }
35         }
36       }
37     }
38     \ior_close:N \g_ztool_file_read_ior
39   }
40 }
41 \cs_new_protected:Npn \ztool_gread_file_as_seq:nnN #1#2#3
42 {% #1: bool(True to keep spaces, False to trim); #2: file name; #3: seq
43   \seq_gclear:N #3
44   \file_if_exist:nT {#2}
45   {
46     \ior_open:Nn \g_ztool_file_read_ior {#2}
```

47	\ior_map_inline:Nn \g_ztool_file_read_ior	47
48	{	48
49	\tl_if_empty:nF {##1}	49
50	{	50
51	\bool_if:nTF {#1}	51
52	{ \seq_gput_right:Nn #3 {##1} }	52
53	{	53
54	\seq_gput_right:Ne #3	54
55	{ \tl_trim_spaces:n {##1} }	55
56	}	56
57	}	57
58	}	58
59	\ior_close:N \g_ztool_file_read_ior	59
60	}	60
61	}	61
62	\cs_generate_variant:Nn \ztool_read_file_as_seq:nnN	62
63	{ ne, nnc, nec }	63
64	\cs_generate_variant:Nn \ztool_gread_file_as_seq:nnN	64
65	{ ne, nnc, nec }	65
66		66
67	% read file as seq(keep these internal spaces):	67
68	\tl_new:N \l__ztool_strmap_read_tl	68
69	\cs_new_protected:Npn \ztool_read_file_keep_spaces:nn #1#2	69
70	{	70
71	\file_if_exist:nT {#1}	71
72	{	72
73	\ior_open:Nn \g_ztool_file_read_ior {#1}	73
74	\ior_str_map_inline:Nn \g_ztool_file_read_ior	74
75	{	75
76	\exp_args:Nee \str_if_in:nnF { \tl_head:n {##1} }	76
77	{ \char_generate:nn {37}{12} }	77
78	{	78
79	\tl_set_rescan:Nnn \l__ztool_strmap_read_tl	79
80	{	80
81	\cctab_select:N \c_document_cctab	81
82	\char_set_catcode_space:n { 9 } % tab	82
83	\char_set_catcode_space:n { 32 } % space	83
84	}{ ##1 }	84
85	#2	85
86	}	86
87	}	87
88	\ior_close:N \g_ztool_file_read_ior	88
89	}	89
90	}	90
91	\cs_new_protected:Npn \ztool_read_file_as_seq_keep_spaces:nnN #1#2#3	91
92	{% #1: bool(True to keep trim spaces, False to trim); #2: file name; #3: seq	92
93	\seq_gclear:N #3	93
94	\ztool_read_file_keep_spaces:nn {#2}	94

9595
9696
9797
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9999
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109109
110110
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112112
113113
114114
115115
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118118
119119
120120
121121
122122
123123
124124
125125
126126
127127
128128
129129
130130
131131
132132
133133
134134
135135
136136
137137
138138
139139
140140
141141
142142

{
 \bool_if:nTF {#1}
 {
 \exp_args:NNo \seq_put_right:Nn #3
 { \l__ztool_strmap_read_tl }
 }{
 \seq_gput_right:Ne #3
 {
 \exp_args:No \tl_trim_spaces:n
 { \l__ztool_strmap_read_tl }
 }
 }
}

\cs_new_protected:Npn \ztool_gread_file_as_seq_keep_spaces:nnN #1#2#3
 {% #1: bool(True to keep trim spaces, False to trim); #2: file name; #3: seq
 \seq_gclear:N #3
 \ztool_read_file_keep_spaces:nn {#2}
 {
 \bool_if:nTF {#1}
 {
 \exp_args:NNo \seq_gput_right:Nn #3
 { \l__ztool_strmap_read_tl }
 }{
 \seq_gput_right:Ne #3
 {
 \exp_args:No \tl_trim_spaces:n
 { \l__ztool_strmap_read_tl }
 }
 }
 }
 }

\cs_generate_variant:Nn \ztool_read_file_as_seq_keep_spaces:nnN
 { ne, nnc, nec }

\cs_generate_variant:Nn \ztool_gread_file_as_seq_keep_spaces:nnN
 { ne, nnc, nec }

% create file / append to file / write to file
\cs_new_protected:Npn \ztool_file_new:nn #1#2
 {% #1: \c_true_bool to allow overwrite; #2: file name
 \bool_if:nT {#1}
 {
 \iow_open:Nn \g_ztool_file_append_iow {#2}
 \iow_close:N \g_ztool_file_append_iow
 }
 }

\cs_new_protected:Npn \ztool_append_to_file:nn #1#2
 {% #1: file name; #2: content

37

```

143 \seq_clear:N \l_ztool_file_seq
144 \file_if_exist:nF {#1}
145 { \ztool_file_new:nn {\c_true_bool}{#1} }
146 \ior_open:Nn \g_ztool_file_append_ior {#1}
147 \ior_str_map_inline:Nn \g_ztool_file_append_ior
148 {
149     \seq_put_right:Nn \l_ztool_file_seq
150     { ##1 }
151 }
152 \iow_open:Nn \g_ztool_file_append_iow {#1}
153 \seq_if_empty:NF \l_ztool_file_seq
154 {
155     \iow_now:Ne \g_ztool_file_append_iow
156     {
157         \seq_use:Ne \l_ztool_file_seq
158         { \iow_newline: }
159     }
160 }
161 \iow_now:Ne \g_ztool_file_append_iow {#2}
162 \iow_close:N \g_ztool_file_append_iow
163 }
164 \cs_generate_variant:Nn \ztool_append_to_file:nn
165 { no, nf, ne, ee }
166
167 \cs_new_protected:Npn \ztool_write_seq_to_file:nNn #1#2#3
168 {% #1:bool; #2:seq; #3:file name
169     \seq_clear:N \l__ztool_tmp_seq
170     \bool_if:nTF { #1 }
171     {
172         \seq_set_eq:NN \l_ztool_file_seq #2
173     }{
174         \ztool_read_file_as_seq:nnN
175         { \c_true_bool }{ #3 }
176         \l__ztool_tmp_seq
177         \seq_concat:NNN \l_ztool_file_seq
178         \l__ztool_tmp_seq #2
179     }
180     \file_if_exist:nF {#3}
181     { \ztool_file_new:nn {\c_true_bool}{#3} }
182     \iow_open:Nn \g_tmpa_iow { #3 }
183     \seq_if_empty:NF \l_ztool_file_seq
184     {
185         \iow_now:Ne \g_tmpa_iow
186         {
187             \seq_use:Ne \l_ztool_file_seq
188             { \iow_newline: }
189         }
190     }

```

191	\iow_close:N \g_tmpa_iow	191
192	}	192
193	\cs_generate_variant:Nn \ztool_write_seq_to_file:nNn	193
194	{ nNe, nNV, nce, ncV }	194
195		195
196	\cs_new_protected:Npn \ztool_replace_file_line:nnn #1#2#3	196
197	{% #1:file name; #2:line index; #3:replacement	197
198	\seq_clear:N \l_ztool_file_seq	198
199	\file_if_exist:nT {#1}{	199
200	\ior_open:Nn \g_ztool_file_read_ior {#1}	200
201	\ior_str_map_inline:Nn \g_ztool_file_read_ior	201
202	{	202
203	\seq_put_right:Nn \l_ztool_file_seq {##1}	203
204	}	204
205	\ior_close:N \g_ztool_file_read_ior	205
206	\seq_set_item:Nnn \l_ztool_file_seq {#2}	206
207	{ #3 }	207
208	\iow_open:Nn \g_ztool_file_append_iow {#1}	208
209	\seq_if_empty:NF \l_ztool_file_seq	209
210	{	210
211	\iow_now:Ne \g_ztool_file_append_iow	211
212	{	212
213	\seq_use:Ne \l_ztool_file_seq	213
214	{ \iow_newline: }	214
215	}	215
216	}	216
217	\iow_close:N \g_ztool_file_append_iow	217
218	}	218
219	}	219
220	\cs_generate_variant:Nn \seq_set_item:Nnn { Nne }	220
221	\cs_generate_variant:Nn \ztool_replace_file_line:nnn	221
222	{ e, ene, eee }	222
223	\cs_new_protected:Npn \ztool_insert_to_file:nnn #1#2#3	223
224	{% #1:file name; #2:line index; #3:content	224
225	\seq_clear:N \l_ztool_file_seq	225
226	\file_if_exist:nT {#1}{	226
227	\ior_open:Nn \g_ztool_file_read_ior {#1}	227
228	\ior_str_map_inline:Nn \g_ztool_file_read_ior	228
229	{	229
230	\seq_put_right:Nn \l_ztool_file_seq {##1}	230
231	}	231
232	\ior_close:N \g_ztool_file_read_ior	232
233	\tl_set:No \l_ztool_current_line	233
234	{ \seq_item:Nn \l_ztool_file_seq {#2} }	234
235	\seq_set_item:Nne \l_ztool_file_seq {#2}	235
236	{ #3\iow_newline:\l_ztool_current_line }	236
237	\iow_open:Nn \g_ztool_file_append_iow {#1}	237
238	\iow_now:Ne \g_ztool_file_append_iow	238

239	{	239
240	\seq_use:Ne \l_ztool_file_seq	240
241	{ \iow_newline: }	241
242	}	242
243	\iow_close:N \g_ztool_file_append_iow	243
244	}	244
245	}	245
246	\cs_generate_variant:Nn \ztool_insert_to_file:nn	246
247	{ ne, nf, ee }	247

8.4 box

```
1 \ProvidesExplFile{ztool.library.box.tex}
2 {2026/02/13}{1.0.1}
3 {box~library~for~ztool}
4
5
6 % ==> box manipulation tool
7 \cs_set:Nn \__ztool_leave_vmode:
8 { \ifvmode \leavevmode \fi }
9
10 \box_new:N \l_ztool_measure_box
11 \box_new:N \l_ztool_tmpa_box
12
13 % catch box dimension
14 \cs_new:Npn \ztool_box_set_to:NNn #1#2#3
15 {
16   \hbox_set:Nn \l_ztool_measure_box {#3}
17   \dim_set:Nn #2 {#1 \l_ztool_measure_box}
18   \box_set_eq:NN \l_ztool_measure_box \c_empty_box
19 }
20 \cs_new:Npn \ztool_box_gset_to:NNn #1#2#3
21 {
22   \hbox_set:Nn \l_ztool_measure_box {#3}
23   \dim_gset:Nn #2 {#1 \l_ztool_measure_box}
24   \box_set_eq:NN \l_ztool_measure_box \c_empty_box
25 }
26 \cs_new:Npn \ztool_get_ht:Nn
27 { \ztool_box_set_to:NNn \box_ht:N }
28 \cs_new:Npn \ztool_get_ht_plus_dp:Nn
29 { \ztool_box_set_to:NNn \box_ht_plus_dp:N }
30 \cs_new:Npn \ztool_get_wd:Nn
31 { \ztool_box_set_to:NNn \box_wd:N }
32 \cs_new:Npn \ztool_get_dp:Nn
33 { \ztool_box_set_to:NNn \box_dp:N }
34 \cs_new:Npn \ztool_gget_ht:Nn
35 { \ztool_box_gset_to:NNn \box_ht:N }
36 \cs_new:Npn \ztool_gget_wd:Nn
37 { \ztool_box_gset_to:NNn \box_wd:N }
38 \cs_new:Npn \ztool_gget_dp:Nn
39 { \ztool_box_gset_to:NNn \box_dp:N }
40 \cs_generate_variant:Nn \ztool_get_ht:Nn
41 { Ne, ce }
42 \cs_generate_variant:Nn \ztool_get_ht_plus_dp:Nn
43 { Ne, ce }
44 \cs_generate_variant:Nn \ztool_get_wd:Nn
45 { Ne, ce }
46 \cs_generate_variant:Nn \ztool_gget_ht:Nn
```

47	{ Ne, ce }	47
48	\cs_generate_variant:Nn \ztool_gget_wd:Nn	48
49	{ Ne, ce }	49
50		50
51		51
52	%% modify box content	52
53	% 1. auto scale and rotate (smaller of two)	53
54	\cs_new_protected:Npn \ztool_autoset_to_wd_and_ht:nnn #1#2#3	54
55	{% #1:width; #2:height; #3:object	55
56	\hbox_set:Nn \l_ztool_tmpa_box {#3}	56
57	\box_autosize_to_wd_and_ht:Nnn \l_ztool_tmpa_box {#1}{#2}	57
58	__ztool_leave_vmode:	58
59	\box_use:N \l_ztool_tmpa_box	59
60	}	60
61	\cs_new_protected:Npn \ztool_rotate:nn #1#2	61
62	{% #1:angle; #2:object	62
63	\hbox_set:Nn \l_ztool_tmpa_box {#2}	63
64	\box_rotate:Nn \l_ztool_tmpa_box {#1}	64
65	__ztool_leave_vmode:	65
66	\box_use:N \l_ztool_tmpa_box	66
67	}	67
68	\cs_generate_variant:Nn \ztool_rotate:nn	68
69	{ e, ne, ee }	69
70	\cs_generate_variant:Nn \ztool_autoset_to_wd_and_ht:nnn	70
71	{ nne, een, eee }	71
72		72
73	% 2. width/height scale to same time	73
74	% TODO: if '\dim(content) < dim', spread it to 'dim'.	74
75	\cs_new_protected:Npn \ztool_set_to_wd:nn #1#2	75
76	{% #1:width; #2:object	76
77	\hbox_set:Nn \l_ztool_tmpa_box {#2}	77
78	\box_resize_to_wd:Nn \l_ztool_tmpa_box {#1}	78
79	__ztool_leave_vmode:	79
80	\box_use:N \l_ztool_tmpa_box	80
81	}	81
82	\cs_new_protected:Npn \ztool_set_to_ht:nn #1#2	82
83	{% #1:height; #2:object	83
84	\hbox_set:Nn \l_ztool_tmpa_box {#2}	84
85	\box_resize_to_ht:Nn \l_ztool_tmpa_box {#1}	85
86	__ztool_leave_vmode:	86
87	\box_use:N \l_ztool_tmpa_box	87
88	}	88
89	\cs_generate_variant:Nn \ztool_set_to_wd:nn { e, ne, ee }	89
90	\cs_generate_variant:Nn \ztool_set_to_ht:nn { e, ne, ee }	90
91		91
92	% 3. only scale one dimension	92
93	% NOTE: if boxwd{content} <= given dim, no manipulation	93
94	\cs_new_protected:Npn \ztool_scale_to_wd:nn #1#2	94

95	{	95
96	\hbox_set:Nn \l_ztool_tmpa_box {#2}	96
97	\dim_set:Nn \l_tmpa_dim { \box_wd:N \l_ztool_tmpa_box }	97
98	\fp_set:Nn \l_tmpa_fp	98
99	{	99
100	\fp_eval:n { min(1, \dim_ratio:nn {#1}{\l_tmpa_dim}) }	100
101	}	101
102	\box_scale:Nnn \l_ztool_tmpa_box {\l_tmpa_fp}{1}	102
103	__ztool_leave_vmode:	103
104	\box_use:N \l_ztool_tmpa_box	104
105	}	105
106	\cs_new_protected:Npn \ztool_scale_to_ht:nn #1#2	106
107	{% take depth into consideration	107
108	\hbox_set:Nn \l_ztool_tmpa_box {#2}	108
109	\dim_set:Nn \l_tmpa_dim { \box_ht_plus_dp:N \l_ztool_tmpa_box }	109
110	\fp_set:Nn \l_tmpa_fp	110
111	{	111
112	\fp_eval:n { min(1, \dim_ratio:nn {#1}{\l_tmpa_dim}) }	112
113	}	113
114	\box_scale:Nnn \l_ztool_tmpa_box {1}{\l_tmpa_fp}	114
115	__ztool_leave_vmode:	115
116	\box_use:N \l_ztool_tmpa_box	116
117	}	117
118	\cs_new_protected:Npn \ztool_scale_to_wd_and_ht:nnn #1#2#3	118
119	{% take depth into consideration	119
120	\hbox_set:Nn \l_ztool_tmpa_box {#3}	120
121	\dim_set:Nn \l_tmpa_dim { \box_wd:N \l_ztool_tmpa_box }	121
122	\dim_set:Nn \l_tmpb_dim { \box_ht_plus_dp:N \l_ztool_tmpa_box }	122
123	\fp_set:Nn \l_tmpa_fp	123
124	{	124
125	\fp_eval:n { min(1, \dim_ratio:nn {#1}{\l_tmpa_dim}) }	125
126	}	126
127	\fp_set:Nn \l_tmpb_fp	127
128	{	128
129	\fp_eval:n { min(1, \dim_ratio:nn {#2}{\l_tmpb_dim}) }	129
130	}	130
131	\box_scale:Nnn \l_ztool_tmpa_box {\l_tmpa_fp}{\l_tmpb_fp}	131
132	__ztool_leave_vmode:	132
133	\box_use:N \l_ztool_tmpa_box	133
134	}	134
135	\cs_generate_variant:Nn \ztool_scale_to_wd:nn	135
136	{ e, ne, ee }	136
137	\cs_generate_variant:Nn \ztool_scale_to_ht:nn	137
138	{ e, ne, ee }	138
139	\cs_generate_variant:Nn \ztool_scale_to_wd_and_ht:nnn	139
140	{ nne, nno, eee }	140
141		141
142		142

```

143 %% box content align
144 \seq_new:N \l__ztool_boxitem_seq
145 \cs_set_protected:Npn \ztool_box_item_align:Nnnn #1#2#3#4
146 { % #1:cmd, #2:width, #3:object, #4:align format(left, right, scatter, center)
147   \hb@xt@#2{
148     \tl_map_inline:nn {#3}
149     {
150       \seq_put_right:No \l__ztool_boxitem_seq
151       { \exp_not:N #1{##1} }
152     }
153     \str_case:nnF { #4 }
154     {
155       { left }{ \seq_use:Nn \l__ztool_boxitem_seq {\hfill} }
156       { right }{ \hfill\seq_use:Nn \l__ztool_boxitem_seq {} }
157       { scatter }{ \seq_use:Nn \l__ztool_boxitem_seq {\hfill} }
158       { center }{ \hfill\seq_use:Nn \l__ztool_boxitem_seq {\hfill} }
159       { tower }
160       {
161         \edef\seq@count{\seq_count:N \l__ztool_boxitem_seq}
162         \seq_map_indexed_inline:Nn \l__ztool_boxitem_seq
163         { % ##1: index, ##2: content
164           %% Method II: plain
165           \edef\item@width{\dim_eval:n {#2/(\seq@count+1)}}
166           \hskip\item@width\clap{##2}
167           }\hskip\item@width\hss
168         }
169       { custom }
170       {
171         \def\total@width{#2}
172         \def\align@cmd{#1}
173         \def\align@object{#3}
174         \def\align@format{#4}
175         \tl_use:N \l__ztex_boxitem_align_custom_tl
176       }
177     }{\relax}
178   }
179   \seq_clear:N \l__ztool_boxitem_seq
180 }
181 \cs_generate_variant:Nn \ztool_box_item_align:Nnnn
182 { c, Nnno, cnno, Nne, Nnee }
183
184
185 %% affine transformation
186 % REF:
187 % 1. https://math.stackexchange.com/a/3521141/1235323
188 % 2. https://math.stackexchange.com/a/281087/1235323
189 \cs_new:Npn \ztool_fp_to_rad:n #1
190 { \fp_eval:n {#1/pi*180} }

```

```

191 \cs_new:Npn \ztool_matrix_det:nnnn #1#2#3#4 191
192 { 192
193     \fp_eval:n { #1*#4 - #2*#3 } 193
194 } 194
195 % (translation) + x-scale + y-scale + rotate 195
196 \fp_new:N \g_affine_precision_fp 196
197 \fp_set:Nn \g_affine_precision_fp {0.0001} 197
198 \fp_new:N \l__affine_@@_a_fp 198
199 \fp_new:N \l__affine_@@_b_fp 199
200 \fp_new:N \l__affine_@@_c_fp 200
201 \fp_new:N \l__affine_@@_d_fp 201
202 \msg_set:nnn { ztool }{affine-det-zero} 202
203 { 203
204     current~determination~of~the~affine~transformation~ 204
205     matrix~equals~to~zero,~give~up~this~transformation 205
206 } 206
207 207
208 \coffin_new:N \l__affine_trans_coffin 208
209 \cs_generate_variant:Nn \coffin_typeset:Nnnnn { Nxxxx } 209
210 \cs_new:Npn \ztool_affine_transformation:Nnnnn #1#2#3#4#5 210
211 { % #1:box; #2:$a_{11}$; #3:$a_{21}$; #4:$a_{12}$; #5:$a_{22}$. 211
212     \fp_compare:nNnT 212
213         { abs(\ztool_matrix_det:nnnn {#2}{#3}{#4}{#5}) } 213
214         < { \g_affine_precision_fp } 214
215         { \prg_map_break:Nn \l__affine_matrix_det_zero 215
216             { \msg_warning:nn { ztool }{affine-det-zero} }} 216
217     \fp_set:Nn \l__affine_@@_a_fp {#2} 217
218     \fp_set:Nn \l__affine_@@_b_fp {#3} 218
219     \fp_set:Nn \l__affine_@@_c_fp {#4} 219
220     \fp_set:Nn \l__affine_@@_d_fp {#5} 220
221     \__box_affine_transform:N #1 221
222     \prg_break_point:Nn \l__affine_matrix_det_zero { } 222
223     \coffin_typeset:Nxxxx \l__affine_trans_coffin 223
224         { \l__ztool_affine_pole_a_tl } 224
225         { \l__ztool_affine_pole_b_tl } 225
226         { \l__ztool_affine_xoffset_dim } 226
227         { \l__ztool_affine_yoffset_dim } 227
228 } 228
229 \cs_generate_variant:Nn \ztool_affine_transformation:Nnnnn 229
230 { Neeee, cnnnn, ceeee } 230
231 \cs_new:Npn \__box_affine_transform:N #1 231
232 { 232
233     % transform debug 233
234     \bool_if:NT \g_ztool_affine_debug_bool 234
235     { 235
236         \noindent\dotfill\[\begin{bmatrix} 236
237             \fp_use:N \l__affine_@@_a_fp & \fp_use:N \l__affine_@@_c_fp\\ 237
238             \fp_use:N \l__affine_@@_b_fp & \fp_use:N \l__affine_@@_d_fp 238

```

```

239 \end{bmatrix}\]
240 }
241 % get affine parameters
242 \__affine_trans_get_sx:
243 \__affine_trans_get_theta:
244 \__affine_trans_get_sy:
245 \__affine_trans_get_Sx:
246 \__affine_trans_get_Sy:
247 \__affine_trans_get_phi:
248 \__affine_trans_get_omega:
249 % start transform box/coffin
250 \coffin_scale:Nnn #1
251 { \l__box_affine_sx_fp }
252 { \l__box_affine_sy_fp }
253 \coffin_rotate:Nn #1
254 { \ztool_fp_to_rad:n {\l__box_affine_omega_fp} }
255 \coffin_scale:Nnn #1
256 { \l__box_affine_Sx_fp }
257 { \l__box_affine_Sy_fp }
258 \coffin_rotate:Nn #1
259 { \ztool_fp_to_rad:n {\l__box_affine_phi_fp} }
260 \coffin_rotate:Nn #1
261 { \ztool_fp_to_rad:n {\l__box_affine_theta_fp} }
262 }
263 \keys_define:nn { ztool / affine }
264 {
265     debug .bool_gset:N = \g_ztool_affine_debug_bool,
266     debug .initial:n = false,
267     debug .default:n = true,
268     pole-1 .tl_set:N = \l__ztool_affine_pole_a_tl,
269     pole-2 .tl_set:N = \l__ztool_affine_pole_b_tl,
270     pole-1 .initial:n = { l },
271     pole-2 .initial:n = { b },
272     xoffset .dim_set:N = \l__ztool_affine_xoffset_dim,
273     yoffset .dim_set:N = \l__ztool_affine_yoffset_dim,
274     xoffset .initial:n = { 0pt },
275     yoffset .initial:n = { 0pt },
276 }
277 \NewDocumentCommand{\ztoolboxaffine}{0{}m>{\SplitList{,}}m}
278 {% #1:key-value; #2:content; #3:matrix.
279     \group_begin:
280         \keys_set:nn { ztool / affine } {#1}
281         \hcoffin_set:Nn \l__affine_trans_coffin {#2}
282         \ztool_affine_transformation:Nnnnn \l__affine_trans_coffin #3
283     \group_end:
284 }
285 % internal affine transform functions
286 \cs_new:Nn \__ztool_affine_debug_fp:N

```

```

287 {
288     \bool_if:NTF \g_ztool_affine_debug_bool
289     { \string #1 % \show #1
290       ~~~\fp_use:N #1\\
291       }{ \relax }
292 }
293 \fp_new:N \l__box_affine_sx_fp
294 \cs_new:Nn \__affine_trans_get_sx:
295 {
296     \fp_set:Nn \l__box_affine_sx_fp
297     { \fp_eval:n {sqrt(\l__affine_@@_a_fp^2 + \l__affine_@@_b_fp^2)} }
298     \__ztool_affine_debug_fp:N \l__box_affine_sx_fp
299 }
300 \fp_new:N \l__box_affine_theta_fp
301 \cs_new:Nn \__affine_trans_get_theta:
302 {
303     \fp_set:Nn \l__box_affine_theta_fp
304     { \fp_eval:n {atan(\l__affine_@@_b_fp/\l__affine_@@_a_fp)} }
305     \__ztool_affine_debug_fp:N \l__box_affine_theta_fp
306 }
307 \fp_new:N \l__box_affine_msy_fp
308 \cs_new:Nn \__affine_trans_get_msy:
309 {
310     \fp_set:Nn \l__box_affine_msy_fp
311     { \fp_eval:n {
312         \l__affine_@@_c_fp*cos(\l__box_affine_theta_fp)
313         +
314         \l__affine_@@_d_fp*sin(\l__box_affine_theta_fp)
315     } }
316     \__ztool_affine_debug_fp:N \l__box_affine_msy_fp
317 }
318 \fp_new:N \l__box_affine_sy_fp
319 \cs_new:Nn \__affine_trans_get_sy:
320 {
321     \__affine_trans_get_msy:
322     \bool_if:nTF
323     {
324         \fp_compare_p:nNn { abs(sin(\l__box_affine_theta_fp)) }
325         < { \c_zero_fp + \g_affine_precision_fp }
326     }{
327         \fp_set:Nn \l__box_affine_sy_fp
328         {
329             ( \l__affine_@@_d_fp - \l__box_affine_msy_fp*sin(\l__box_affine_theta_fp) )
330             / cos(\l__box_affine_theta_fp)
331         }
332     }{
333         \fp_set:Nn \l__box_affine_sy_fp
334         {

```

```

335         ( \l__box_affine_msy_fp*cos(\l__box_affine_theta_fp) - \l__affine_@@_c_fp ) 335
336         / sin(\l__box_affine_theta_fp) 336
337     } 337
338 } 338
339     \__ztool_affine_debug_fp:N \l__box_affine_sy_fp 339
340 } 340
341 \fp_new:N \l__box_affine_m_fp 341
342 \cs_new:Nn \__affine_trans_get_m: 342
343 { 343
344     \fp_set:Nn \l__box_affine_m_fp 344
345     { \l__box_affine_msy_fp / \l__box_affine_sy_fp } 345
346     \__ztool_affine_debug_fp:N \l__box_affine_m_fp 346
347 } 347
348 \fp_new:N \l__box_affine_Sx_fp 348
349 \fp_new:N \l__box_affine_Sy_fp 349
350 \cs_new:Nn \__affine_trans_get_Sx: 350
351 { 351
352     \__affine_trans_get_m: 352
353     \fp_set:Nn \l__box_affine_Sx_fp 353
354     { sqrt(\l__box_affine_m_fp^2/4 + 1) - \l__box_affine_m_fp/2 } 354
355     \__ztool_affine_debug_fp:N \l__box_affine_Sx_fp 355
356 } 356
357 \cs_new:Nn \__affine_trans_get_Sy: 357
358 { 358
359     \fp_set:Nn \l__box_affine_Sy_fp 359
360     { sqrt(\l__box_affine_m_fp^2/4 + 1) + \l__box_affine_m_fp/2 } 360
361     \__ztool_affine_debug_fp:N \l__box_affine_Sy_fp 361
362 } 362
363 \fp_new:N \l__box_affine_phi_fp 363
364 \fp_new:N \l__box_affine_omega_fp 364
365 \cs_new:Nn \__affine_trans_get_phi: 365
366 { 366
367     \fp_set:Nn \l__box_affine_phi_fp 367
368     { -pi/4 - 1/2*atan(\l__box_affine_m_fp/2) } 368
369     \__ztool_affine_debug_fp:N \l__box_affine_phi_fp 369
370 } 370
371 \cs_new:Nn \__affine_trans_get_omega: 371
372 { 372
373     \fp_set:Nn \l__box_affine_omega_fp 373
374     { pi/4 - 1/2*atan(\l__box_affine_m_fp/2) } 374
375     \__ztool_affine_debug_fp:N \l__box_affine_omega_fp 375
376 } 376

```

8.5 zdraw

```
1 \ProvidesExplFile{ztool.library.zdraw.tex}
2 {2026/09/23}{1.1.0}
3 {zdraw~library~for~ztool}
4
5
6 % ==> ztool draw (based on package 'pict2e' and 'picture' env)
7 \RequirePackage{pict2e}
8
9 % operations precise
10 \fp_const:Nn \c__pic_precise_fp { 0.001 }
11
12 % command/env wrapper
13 \cs_new_protected:Npn \_@@_begin_picture:nnnn #1#2#3#4
14 { \begin{picture}
15     (\fp_eval:n {#1}, \fp_eval:n {#2})
16     (\fp_eval:n {-#3}, \fp_eval:n {-#4}) }
17 \cs_new_protected:Npn \_@@_end_picture:
18 { \end{picture} }
19 \cs_new_protected:Npn \_@@_pic_put:nnn #1#2#3
20 { \put(\fp_eval:n {#1}, \fp_eval:n {#2}){ #3 } }
21 \cs_generate_variant:Nn \_@@_begin_picture:nnnn
22 { VVVV, eeee }
23 \cs_generate_variant:Nn \_@@_pic_put:nnn
24 { VVV, een }
25
26 % picture environment alias
27 % width/height/depth are measured in units of `unit` and describe the
28 % dimensions reported to TeX. In particular, `height` is the extent above
29 % the baseline and `depth` is the extent below it. The picture environment
30 % itself always returns zero depth, so zpic collects its result and restores
31 % the requested box dimensions explicitly.
32 \box_new:N \l__pic_output_box
33 \cs_new:Npn \__pic_to_dim:n #1
34 {
35     \fp_to_dim:n
36     { (#1) * \dim_to_fp:n { \l__pic_unit_dim } }
37 }
38 \keys_define:nn { ztool / draw / picture }
39 {
40     unit .dim_set:N = \l__pic_unit_dim,
41     unit .initial:n = { 1cm },
42     width .fp_set:N = \l__pic_width_fp,
43     width .initial:n = 0,
44     height .fp_set:N = \l__pic_height_fp,
45     height .initial:n = 0,
46     depth .fp_set:N = \l__pic_depth_fp,
```

```

47     depth      .initial:n = 0,
48     xoffset .fp_set:N = \l__pic_xoffset_fp,
49     xoffset .initial:n = 0,
50     yoffset .fp_set:N = \l__pic_yoffset_fp,
51     yoffset .initial:n = 0,
52     % opacity-color .tl_set:N = \l__pic_opacity_color_tl,
53     % opacity-color .initial:n = { white },
54 }
55 \NewDocumentEnvironment{zpic}{0{}}
56 {
57     \group_begin:
58     \keys_set:nn { ztool / draw / picture } {#1}
59     \setlength\unitlength{ \l__pic_unit_dim }
60     \hbox_set:Nw \l__pic_output_box
61     \_@@_begin_picture:VVVV
62     \l__pic_width_fp \l__pic_height_fp
63     \l__pic_xoffset_fp\l__pic_yoffset_fp
64 }{
65     \_@@_end_picture:
66     \hbox_set_end:
67     \box_set_wd:Nn \l__pic_output_box
68     { \__pic_to_dim:n { \l__pic_width_fp } }
69     \box_set_ht:Nn \l__pic_output_box
70     { \__pic_to_dim:n { \l__pic_height_fp } }
71     \box_set_dp:Nn \l__pic_output_box
72     { \__pic_to_dim:n { \l__pic_depth_fp } }
73     \box_use_drop:N \l__pic_output_box
74     \group_end:
75 }
76
77
78 % picture commands alias
79 \cs_new:Npn \__coord_st:n #1
80 { \clist_item:nn {#1}{1} }
81 \cs_new:Npn \__coord_nd:n #1
82 { \clist_item:nn {#1}{2} }
83 \cs_new:Npn \__coord_rd:n #1#2
84 { \clist_item:nn {#1}{3} }
85 \cs_new:Npn \__coord_st_nd:n #1
86 {
87     {\clist_item:nn {#1}{1}}
88     {\clist_item:nn {#1}{2}}
89 }
90 \cs_new:Npn \__coord_st_nd_rd:n #1
91 {
92     {\clist_item:nn {#1}{1}}
93     {\clist_item:nn {#1}{2}}
94     {\clist_item:nn {#1}{3}}

```



```

95 }
96 \cs_generate_variant:Nn \__coord_st:n { V, e }
97 \cs_generate_variant:Nn \__coord_nd:n { V, e }
98 \cs_generate_variant:Nn \__coord_rd:n { V, e }
99 \cs_generate_variant:Nn \__coord_st_nd:n { V, e }
100 \cs_generate_variant:Nn \__coord_st_nd_rd:n { V, e }
101
102 \bool_new:N \l__ztool_invalid_color_bool
103 \cs_new:Npn \__color_safe_use:n #1
104 {
105     \__color_if_valid:nT {#1}
106     { \color{#1} }
107 }
108 \prg_new_conditional:Npnn \__color_if_valid:n #1 {p, T, F, TF}
109 {
110     \def\ztool@targer@color{#1}
111     \def\ztool@color@none{none}
112     \bool_if:eTF
113     {
114         \tl_if_empty_p:e {#1} ||
115         \tl_if_eq_p:NN \ztool@color@none \ztool@targer@color
116     }{ \prg_return_false: }
117     { \prg_return_true: }
118 }
119 \prg_generate_conditional_variant:Nnn \__color_if_valid:n
120 { V, e }{ p, T, F, TF }
121 \cs_generate_variant:Nn \__color_safe_use:n
122 { V, e }
123
124
125 % --> line/vector
126 \fp_new:N \l__draw_vector_slope_fp
127 \fp_new:N \l__draw_vector_normal_fp
128 \fp_new:N \l__draw_vector_xysep_fp
129 \cs_new:Npn \__@@_pic_line:nnn #1#2#3
130 { % #1:x; #2:y; #3:x-distance NOT the length
131     \line(\fp_eval:n {#1}, \fp_eval:n {#2})
132     { \fp_eval:n {#3} }
133 }
134 \cs_new:Npn \__@@_pic_vector:nnn #1#2#3
135 { % #1:x; #2:y; #3:x-distance NOT the length
136     \vector(\fp_eval:n {#1}, \fp_eval:n {#2})
137     { \fp_eval:n {#3} }
138 }
139 \keys_define:nn { ztool / draw / picture / line }
140 {
141     draw .tl_set:N = \l__pic_line_draw_color_tl,
142     draw .initial:n = { black },

```

```

143 % color .meta:n = { draw = #1 }, % alias for 'draw'
144 width .dim_set:N = \l__pic_line_width_dim,
145 width .initial:n = { .4pt },
146 dash .bool_set:N = \l__pic_line_dash_bool,
147 dash .initial:n = { false },
148 }
149 \cs_new_protected:Nn \__pic_set_line_width:
150 {
151     \linethickness{ \l__pic_line_width_dim }
152 }
153 \cs_new_protected:Nn \__pic_set_line_color:
154 {
155     \__color_safe_use:V \l__pic_line_draw_color_tl
156 }
157 \cs_new_protected:Nn \__pic_set_fill_color:
158 {
159     \__color_safe_use:V \l__pic_region_fill_color_tl
160 }
161 \def\z@pic@vector@style{\ltxarrows}
162 \keys_define:nn { ztool / draw / picture }
163 {
164     vector .inherit:n = { ztool/draw/picture/line },
165 }
166 \keys_define:nn { ztool / draw / picture / vector }
167 {
168     > .choice:,
169     > / latex .code:n = {\def\z@pic@vector@style{\ltxarrows}},
170     > / pst .code:n = {\def\z@pic@vector@style{\pstarrows}},
171     > / unknown .code:n =
172     {
173         \msg_set:nnn { ztool }{unknown-arrow-style}
174         { Unknown~arrow~style,~use~'latex'~or~'pst'. }
175         \msg_error:nn { ztool }{unknown-arrow-style}
176     }
177 }
178 \tl_new:N \l__draw_line_type % 'horizontal', 'vertical', 'normal'
179 \cs_new_protected:Npn \ztool_pic_line_vector:nnnn #1#2#3#4
180 {% #1:line/vector; #2:key-value; #3:start coor; #4:end coor;
181     \group_begin:
182     \keys_set:nn { ztool / draw / picture / #1 }{#2}
183     \tl_set:Nn \l__draw_line_type { normal }
184     \fp_compare:nNnTF { \__coor_st:n {#4} - \__coor_st:n {#3} } > { \c__pic_precise_fp }
185     {
186         \fp_set:Nn \l__draw_vector_slope_fp
187         { (\__coor_nd:n {#4} - \__coor_nd:n {#3})
188         / (\__coor_st:n {#4} - \__coor_st:n {#3}) }
189         \fp_set:Nn \l__draw_vector_xysep_fp
190         { abs(\__coor_st:n {#4} - \__coor_st:n {#3}) }

```

```

191     }{
192         % NOTE: we do NOT set slope infinte, just set it to '0'
193         \fp_set:Nn \l__draw_vector_slope_fp {0}
194         \fp_set:Nn \l__draw_vector_xysep_fp
195             { abs(\__coord:n {#4} - \__coord:n {#3}) }
196         \tl_set:Nn \l__draw_line_type { vertical }
197     }
198     \fp_compare:nNnT { abs(\__coord:n {#4} - \__coord:n {#3}) } < {0.001}
199         { \tl_set:Nn \l__draw_line_type { horizontal } }
200     \z@pic@vector@style
201     \__pic_set_line_width:
202     \exp_last_unbraced:Ne \__@@_pic_put:nnn {\__coord_st_nd:n {#3}}
203     {
204         \__pic_set_line_color:
205         \str_case:VnF \l__draw_line_type
206             {
207                 {vertical}{
208                     \cs:w __@@_pic_#1:nnn\cs_end:
209                     { \l__draw_vector_slope_fp }
210                     { 1 }
211                     { \l__draw_vector_xysep_fp }
212                 }
213                 {horizontal}{
214                     \cs:w __@@_pic_#1:nnn\cs_end:
215                     { 1 }
216                     { \l__draw_vector_slope_fp }
217                     { \l__draw_vector_xysep_fp }
218                 }
219                 {normal}{
220                     \cs:w __@@_pic_#1:nnn\cs_end: {1}
221                     { \l__draw_vector_slope_fp }
222                     { \l__draw_vector_xysep_fp }
223                 }
224             }{ \relax }
225     }
226     \group_end:
227 }
228 \cs_generate_variant:Nn \ztool_pic_line_vector:nnnn {neee, nooo}
229 \NewDocumentCommand{\zline}{0{}d()d()}
230 {
231     \ztool_pic_line_vector:neee {line}{#1}{#2}{#3}
232 }
233 \NewDocumentCommand{\zvector}{0{}d()d()}
234 {
235     \ztool_pic_line_vector:neee {vector}{#1}{#2}{#3}
236 }

```

```

239 % --> \zdraw -- similar to \tikz command in tikz
240 % NOTE: these line/vector commands are identical to
241 % 1. \Line (x_1, y_1)(x_2, y_2), \Vector (x_1,y_1)(x_2,y_2)
242 % 2. \polyline(x_1, y_1) ... (x_n, y_n), \polyvector(x_1, y_1) ... (x_n, y_n)
243 % 3. \polygon (x_1, y_1) ... (x_n, y_n), when set 'cycle',
244 % \polygon*(x_1, y_1) ... (x_n, y_n), when set 'fill' (auto cycle).
245 % 4. Trim leading space after '\polygon' or '*' to avoid error !!
246 \cs_new:Npn \__@@_pic_Line:nnnn #1#2#3#4
247 { \Line (#1, #2)(#3, #4) }
248 \cs_new:Npn \__@@_pic_Vector:nnnn #1#2#3#4
249 { \Vector (#1, #2)(#3, #4) }
250 \cs_new:Npn \__@@_pic_polyline:n #1
251 {
252   \tl_set:Nc \l_tmpa_tl {\tl_trim_spaces:e {#1}}
253   \exp_last_unbraced:NV \polyline \l_tmpa_tl
254 }
255 \cs_new:Npn \__@@_pic_polyvector:n #1
256 {
257   \tl_set:Nc \l_tmpa_tl {\tl_trim_spaces:e {#1}}
258   \exp_last_unbraced:NV \polyvector \l_tmpa_tl
259 }
260 \cs_new:Npn \__@@_pic_polygon:nn #1#2
261 {
262   \tl_set:Nc \l_tmpa_tl {\tl_trim_spaces:e {#1}}
263   \tl_set:Nc \l_tmpb_tl {\tl_trim_spaces:e {#2}}
264   \tl_set:Nc \l_tmpa_tl { \l_tmpa_tl\l_tmpb_tl }
265   \exp_last_unbraced:NV \polygon \l_tmpa_tl
266 }
267 \cs_generate_variant:Nn \__@@_pic_polygon:nn { nV, ne }
268 \tl_new:N \l__pic_region_fill_color_tl
269 \bool_new:N \l__pic_region_fill_bool
270 \keys_define:nn { ztool / draw / picture / region }
271 {
272   fill .choices:nn = { true, false }{
273     \use:c { bool_set_ \l_keys_choice_tl :N }
274     \l__pic_region_fill_bool
275   },
276   fill .initial:n = { false },
277   fill .default:n = { true },
278   fill / unknown .code:n = {
279     \tl_if_empty:eF \l_keys_value_tl
280     { \bool_set_true:N \l__pic_region_fill_bool }
281     \tl_set:Nc \l__pic_region_fill_color_tl { \l_keys_value_tl }
282   },
283 }
284 \keys_define:nn { ztool / draw / picture }
285 {
286   zdraw .inherit:n = {

```

```

287         ztool/draw/picture/line,
288         ztool/draw/picture/vector,
289         ztool/draw/picture/region,
290     },
291 }
292 \keys_define:nn { ztool / draw / picture / zdraw }
293 {
294     vector .bool_set:N = \l__pic_draw_vector_bool,
295     vector .initial:n = { false },
296     cycle .bool_set:N = \l__pic_draw_cycle_bool,
297     cycle .initial:n = { false },
298     shift .tl_set:N = \l__pic_draw_shift_tl,
299     shift .initial:n = { 0, 0 },
300 }
301 \cs_new:Npn \__region_fill_color_miss:n #1
302 {
303     \bool_if:eT {
304         \l__pic_region_fill_bool &&
305         \tl_if_empty_p:N \l__pic_region_fill_color_tl
306     } { \tl_set:Nn \l__pic_region_fill_color_tl {#1} }
307 }
308 \cs_new_protected:Npn \ztool_pic_draw:nw #1#2;
309 {% #1:key-value; #2:coors list (use ';' to end scan just like tikz)
310     \group_begin:
311     \keys_set:nn { ztool / draw / picture / zdraw } {#1}
312     \__region_fill_color_miss:n { gray }
313     \edef\coors@first
314     {
315         \exp_last_unbraced:Ne
316         \__coors_list_first:w {\tl_trim_spaces:e {#2}}
317         \scan_stop:
318     }
319     \edef\draw@flag
320     {
321         \tl_map_function:nN {
322             \l__pic_draw_vector_bool
323             \l__pic_draw_cycle_bool
324             \l__pic_region_fill_bool
325         } \int_eval:n
326     }
327     \__@@_pic_put:nnn
328     { \__coor_st:V \coors@first + \__coor_st:V \l__pic_draw_shift_tl }
329     { \__coor_nd:V \coors@first + \__coor_nd:V \l__pic_draw_shift_tl }
330     {
331         \__pic_set_line_width:
332         \__pic_set_line_color:
333         \exp_after:wN \int_case:nnF \exp_after:wN {
334             \exp_after:wN \int_from_bin:n \exp_after:wN

```

```

335         { \draw@flag }
336     }{
337         {0}{ \__@@_pic_polyline:n {#2} }
338         {1}{ \__@@_pic_polygon:nn {*}{#2} }
339         {2}{ \__@@_pic_polygon:ne { }{#2} }
340         {3}{ \__@@_pic_polygon:nn {*}{#2} }
341         {4}{ \__@@_pic_polyvector:n {#2} }
342         {5}{
343             \__pic_set_fill_color:
344             \__@@_pic_polygon:nn {*}{#2}
345             \__pic_set_line_color:
346             \exp_args:Ne \__@@_pic_polyvector:n {#2(\coors@first)}
347         }
348         {6}{ \exp_args:Ne \__@@_pic_polyvector:n {#2(\coors@first)} }
349         {7}{
350             \__pic_set_fill_color:
351             \__@@_pic_polygon:nn {*}{#2}
352             \__pic_set_line_color:
353             \exp_args:Ne \__@@_pic_polyvector:n {#2(\coors@first)}
354         }
355     }{\relax}
356 }
357 \group_end:
358 }
359 \cs_new:Npn \__coors_list_first:w (#1)#2\scan_stop:
360 { #1 }
361 \NewDocumentCommand{\zdraw}{0{}}
362 { \ztool_pic_draw:nw {#1} }
363
364
365 % --> arc / circle
366 \cs_new:Npn \__@@_pic_arc:nnnn #1#2#3#4
367 { % #1:fill bool; #2:start angle; #3:end angle; #4:radius
368     \arc #1[\fp_eval:n {#2}, \fp_eval:n {#3}]
369     { \fp_eval:n {#4} }
370 }
371 \cs_new:Npn \__@@_pic_circel:nn #1#2
372 { % #1:fill bool; #2:radius
373     \__@@_pic_arc:nnnn {#1}{0}{360}{#2}
374 }
375
376
377 % --> arc rule: similar to arc by it's a rule
378 \cs_new:Npn \__@@_pic_arcrule:nnn #1#2#3
379 {
380     % To be implemented ...
381 }
382

```

```

383                                     383
384 % --> circle                                     384
385 \keys_define:nn { ztool / draw / picture }       385
386 {                                               386
387     arc .inherit:n = {                         387
388         ztool/draw/picture/line,               388
389         ztool/draw/picture/region,              389
390     },                                           390
391 }                                               391
392 \keys_define:nn { ztool / draw / picture / arc } 392
393 {                                               393
394     radius .fp_set:N = \l__pic_arc_radius_fp,    394
395     radius .initial:n = .5,                     395
396     start .fp_set:N = \l__pic_arc_start_fp,       396
397     start .initial:n = 0,                         397
398     end .fp_set:N = \l__pic_arc_end_fp,           398
399     end .initial:n = 90,                          399
400 }                                               400
401 \prg_generate_conditional_variant:Nnn            401
402     \bool_if:n { e } { p, T, F, TF }             402
403 \cs_new_protected:Npn \ztool_pic_arc:nn #1#2     403
404     {% #1:key-value; #2:coord                    404
405     \group_begin:                                405
406     \keys_set:nn { ztool / draw / picture / arc }{#1} 406
407     \__region_fill_color_miss:n { gray }          407
408     \__color_if_valid:VF \l__pic_region_fill_color_tl 408
409     { \bool_set_false:N \l__pic_region_fill_bool } 409
410     \exp_last_unbraced:Ne \__@@_pic_put:nnn      410
411     { \__coord_st_nd:n {#2} }                   411
412     {                                             412
413         \__pic_set_line_width:                   413
414         \bool_if:eT \l__pic_region_fill_bool     414
415         {                                         415
416             \__pic_set_fill_color:               416
417             \exp_args:Ne \__@@_pic_arc:nnnn {*}  417
418             { \fp_use:N \l__pic_arc_start_fp }   418
419             { \fp_use:N \l__pic_arc_end_fp }     419
420             { \fp_use:N \l__pic_arc_radius_fp }  420
421         }                                         421
422         % NOTE: border must draw after the fill  422
423         \__pic_set_line_color:                   423
424         \exp_args:Ne \__@@_pic_arc:nnnn {}       424
425         { \fp_use:N \l__pic_arc_start_fp }       425
426         { \fp_use:N \l__pic_arc_end_fp }         426
427         { \fp_use:N \l__pic_arc_radius_fp }      427
428     }                                             428
429     \group_end:                                 429
430 }                                               430

```

```

431 \NewDocumentCommand{\zarc}{0{}d{}}
432   {% #1:key-value; #2:coord
433     \ztool_pic_arc:nn {#1}{#2}
434   }
435 \NewDocumentCommand{\zcircle}{0{}d{}}
436   {
437     \ztool_pic_arc:nn {start=0, end=360, #1}{#2}
438   }
439
440
441 % --> oval / rectangle
442 % \oval[arc](full-x-width, full-y-width)[part]
443 % part: (l, r) x (t, b)
444 \cs_new:Npn \__@@_pic_oval:nnnn #1#2#3#4
445   {% #1:arc; #2:part; #3:x-width; #4:y-width;
446     \oval
447       [\fp_eval:n {#1}]
448       (\fp_eval:n {#3}, \fp_eval:n {#4})
449       [ #2 ]
450   }
451 \keys_define:nn { ztool / draw / picture }
452   {
453     rectangle .inherit:n = {
454       ztool/draw/picture/line,
455       ztool/draw/picture/region,
456     },
457   }
458 \keys_define:nn { ztool / draw / picture / rectangle }
459   {
460     arc .fp_set:N = \l__pic_rec_arc_fp,
461     arc .initial:n = 0,
462   }
463
464 \int_new:N \l__pic_rec_quadrant_index_int
465 \clist_const:Nn \l__pic_rec_quadrant_shift_prop
466   { {(0, 0)}, {(-\rec@arc, 0)}, {(-\rec@arc, -\rec@arc)}, {(0, -\rec@arc)} }
467 \cs_new_protected:Npn \ztool_pic_rectangle:nnn #1#2#3
468   {% #1:key-value; #2:start coord; #3:end coord;
469     \group_begin:
470     \keys_set:nn { ztool / draw / picture / rectangle }{ fill=false }
471     \keys_set:nn { ztool / draw / picture / rectangle }{ #1 }
472     \edef\rec@width { \fp_eval:n {\__coord_st:n {#3} - \__coord_st:n {#2}} }
473     \edef\rec@height{ \fp_eval:n {\__coord_nd:n {#3} - \__coord_nd:n {#2}} }
474     % check if arc radius too large.
475     \edef\rec@arc
476       {
477         \fp_eval:n { min(min(\rec@height/2, \rec@width/2), \l__pic_rec_arc_fp) }
478       }

```



```

479 \__region_fill_color_miss:n { gray }
480 \__color_if_valid:VF \l__pic_region_fill_color_tl
481 {
482     \bool_set_false:N \l__pic_region_fill_bool
483     \prg_map_break:Nn \l__ztool_pic_rec_fill {}
484 }
485 %% begin fill rounded rectangle
486 \__pic_rectangle_rec_comp_fill:n { #2 }
487 \__pic_rectangle_arc_comp_fill:n { #2 }
488 %% end fill rounded rectangle
489 \prg_break_point:Nn \l__ztool_pic_rec_fill { }
490 \__@@_pic_put:nnn
491 { \__coord_st:n {#2}+\rec@width/2 }
492 { \__coord_nd:n {#2}+\rec@height/2 }
493 {
494     \__pic_set_line_color:
495     \__pic_set_line_width:
496     \__@@_pic_oval:nnnn
497     { \rec@arc }{ }
498     { \rec@width }
499     { \rec@height }
500 }
501 \group_end:
502 }
503
504 % aux function for rectangle area filling in rectangle
505 \cs_new:Npn \__pic_rectangle_rec_comp_fill:n #1
506 {
507     \__@@_pic_put:nnn {\__coord_st:n {#1}+\rec@arc}{\__coord_nd:n {#1}}
508     {
509         \__pic_set_fill_color:
510         \rule
511             {\fp_eval:n {(\rec@width-2*\rec@arc)*\dim_to_decimal:n {\l__pic_unit_dim}}pt}
512             {\fp_eval:n {\rec@height*\dim_to_decimal:n {\l__pic_unit_dim}}pt}
513     }
514     \fp_compare:nNnT { \rec@height-2*\rec@arc } > { \c__pic_precise_fp }
515     {
516         \__@@_pic_put:nnn {\__coord_st:n {#1}}{\__coord_nd:n {#1}+\rec@arc}
517         {
518             \__pic_set_fill_color:
519             \rule
520                 {\fp_eval:n {\rec@arc*\dim_to_decimal:n {\l__pic_unit_dim}}pt}
521                 {\fp_eval:n {(\rec@height-2*\rec@arc)*\dim_to_decimal:n {\l__pic_unit_dim}}pt}
522         }
523         \__@@_pic_put:nnn {\__coord_st:n {#1}+\rec@width-\rec@arc}{\__coord_nd:n {#1}+\rec@arc}
524         {
525             \__pic_set_fill_color:
526             \rule

```

```

527         {\fp_eval:n {\rec@arc*\dim_to_decimal:n {\l__pic_unit_dim}}pt} 527
528         {\fp_eval:n {(\rec@height-2*\rec@arc)*\dim_to_decimal:n {\l__pic_unit_dim}}pt} 528
529     } 529
530 } 530
531 } 531
532 % aux function for arc area filling in rectangle 532
533 \cs_new:Npn \__pic_rectangle_arc_comp_fill:n #1 533
534 { 534
535     \int_set:Nn \l__pic_rec_quadrant_index_int { 0 } 535
536     \tl_map_inline:nn 536
537     { 537
538         {\__coord_st:n {#1}+\rec@width-\rec@arc, \__coord_nd:n {#1}+\rec@height-\rec@arc} 538
539         {\__coord_st:n {#1}+\rec@arc, \__coord_nd:n {#1}+\rec@height-\rec@arc} 539
540         {\__coord_st:n {#1}+\rec@arc, \__coord_nd:n {#1}+\rec@arc} 540
541         {\__coord_st:n {#1}+\rec@width-\rec@arc, \__coord_nd:n {#1}+\rec@arc} 541
542     }{ 542
543     \int_incr:N \l__pic_rec_quadrant_index_int 543
544     \edef\qu@drant@index{\int_use:N \l__pic_rec_quadrant_index_int} 544
545     \exp_last_unbraced:Ne \__@@_pic_put:nnn 545
546     { \__coord_st_nd:n {##1} } 546
547     { 547
548         \__pic_set_fill_color: 548
549         \__@@_pic_arc:nnnn {*} 549
550         { (\qu@drant@index-1)*90 } 550
551         { \qu@drant@index*90 } 551
552         { \rec@arc } 552
553     } 553
554 } 554
555 } 555
556 556
557 \NewDocumentCommand{\zrectangle}{0{}d()d()} 557
558 { 558
559     \ztool_pic_rectangle:nnn { #1 }{#2}{#3} 559
560 } 560
561 561
562 562
563 % ==> absolute page coordinate (left, bottom) = (0, 0) 563
564 \NewDocumentCommand{\zpin}{0{background}m} 564
565 { 565
566     \hook_gput_next_code:nn {shipout/#1} 566
567     { 567
568         \put(0pt, -\paperheight) 568
569         { \makebox(0, 0)[bl]{#2} } 569
570     } 570
571 } 571

```

9 索引

斜体数字表示对应条目被解释说明的页面, 带下划线的数字指向该条目的定义, 其余数字表示该条目的使用位置.

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zTool Interface

Eureka zongpingding5(at)outlook(dot)com

Due to limited time, development of this package is currently suspended.

September 23, 2026

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1 Basic Introduction

The $\text{\texttt{zT\TeX}}$ macro collection has independently implemented a `ztool` package, which contains all the commands from the now-obsolete `l3sys-shell` package. In addition, `ztool` provides functions related to box operations, file IO, and basic graphics drawing. With the help of `ztool`, $\text{\texttt{zT\TeX}}$ can avoid or reduce the use of the `-shell-escape` command-line parameter or calls to other related packages (such as the `robust-externalize` package).

The GitHub address of this package is:

https://github.com/zongpingding/zTeX_bundle

This repository contains the source code and user manual of this macro collection. The stable version of the current macro collection was released in September 2025. For the latest development version, please switch to the “dev” branch. This manual applies to the current latest development version.

2 Package Options

ztool is divided into four libraries: “shell-escape, file-io, box, zdraw”. Each library is independent and can be loaded separately. By default, ztool does not load any library.

ztool/shell-escape	shell-escape = <code><false true></code> Initial value: <code>false</code>
ztool/file-io	file-io = <code><false true></code> Initial value: <code>false</code>
ztool/box	box = <code><false true></code> Initial value: <code>false</code>
ztool/zdraw	zdraw = <code><false true></code> Initial value: <code>false</code>

New: 2025-05-22

These four options are options for the ztool package and can be used when loading the ztool package. A basic usage example is as follows, which loads the shell-escape and box libraries of ztool:

<code>\usepackage[shell-escape, box=true]{ztool}</code>	Ex. 1
---	--------------

<code>\ztoolloadlib</code>	<code>\ztoolloadlib {<library>}</code>
----------------------------	--

New: 2025-05-22

This command is used to load a ztool library. `<library>` is the name of the library, and the optional values are: “shell-escape, file-io, box, zdraw”.

A basic usage example is as follows, which loads the shell-escape and box libraries of ztool:

<code>\ztoolloadlib{shell-escape, box}</code>	Ex. 2
---	--------------

3 l3sys-shell

This section mainly introduces the commands from the original l3sys-shell package implemented in ztool. Therefore, when using the commands in this section, you need to enable the `-shell-escape` parameter when compiling the L^AT_EX document, otherwise this series of commands will not perform any operations.

WARNING: Please use the commands in this section with caution. Some improper operations may lead to irreparable consequences.

```
\ztool_shell_escape:n
\ztool_shell_escape:e
```

Updated: 2024-12-05

```
\ztool_shell_escape:n {⟨command⟩}
```

When the `-shell-escape` parameter is enabled, this command will execute `⟨command⟩` in the shell. If the `-shell-escape` parameter is not enabled, this command will not perform any operations.

```
\ztool_shell_mkdir:n
\ztool_shell_mkdir:e
```

Updated: 2024-12-05

```
\ztool_shell_mkdir:n {⟨dir⟩}
```

When the `-shell-escape` parameter is enabled, this command will create a directory `⟨dir⟩`. If the `-shell-escape` parameter is not enabled, this command will not perform any operations.

```
\ztool_shell_cp:nn
\ztool_shell_cp:(ee|ne|en)
```

Updated: 2024-12-05

```
\ztool_shell_cp:nn {⟨source⟩}{⟨target⟩}
```

When the `-shell-escape` parameter is enabled, this command will copy the file `⟨source⟩` to the file `⟨target⟩`. If the `-shell-escape` parameter is not enabled, this command will not perform any operations.

```
\ztool_shell_mv:nn
\ztool_shell_mv:(ee|ne|en)
```

Updated: 2024-12-05

```
\ztool_shell_mv:nn {⟨source⟩}{⟨target⟩}
```

When the `-shell-escape` parameter is enabled, this command will move the file `⟨source⟩` to the directory `⟨target⟩`. If the `-shell-escape` parameter is not enabled, this command will not perform any operations.

```
\ztool_shell_rm:n
\ztool_shell_rm:e
```

Updated: 2024-12-05

```
\ztool_shell_rm:n {⟨file⟩}
```

When the `-shell-escape` parameter is enabled, this command will delete the file `⟨file⟩`. If the `-shell-escape` parameter is not enabled, this command will not perform any operations.

```
\ztool_shell_rmdir:n
\ztool_shell_rmdir:e
```

Updated: 2024-12-05

```
\ztool_shell_rmdir:n {⟨dir⟩}
```

When the `-shell-escape` parameter is enabled, this command will delete the directory `⟨dir⟩`. If the `-shell-escape` parameter is not enabled, this command will not perform any operations.

```
\ztool_get_shell_pwd:N
\ztool_get_shell_pwd:c
```

Updated: 2024-12-05

```
\ztool_get_shell_pwd:N ⟨tl⟩
```

When the `-shell-escape` parameter is enabled, this command will return the current working directory and store it in `⟨tl⟩`. If the `-shell-escape` parameter is not enabled, this command will not perform any operations.

`\ztool_shell_split_ls:nN`

`Updated: 2024-12-05`

`\ztool_shell_split_ls:nN {<dir>}<t1>`

When the `-shell-escape` parameter is enabled, this command will return all filenames under the directory `<dir>` and store them in `<t1>`. If the `-shell-escape` parameter is not enabled, this command will not perform any operations.

4 File IO

This section mainly introduces the file IO operations implemented in `ztool`, including: reading files, writing files, appending to files, etc. This series of commands does not require enabling the `-shell-escape` parameter.

<code>\ztool_file_new:nn</code>	<code>\ztool_file_new:nn {<bool>}{<file>}</code>
---------------------------------	--

Updated: 2024-12-05

This command is used to create a new file named `<file>`. If `<file>` does not exist, a new file named `<file>` will be created. If the file already exists, then when `<bool>` is `\c_true_bool`, **it will overwrite the original file**, otherwise no operation will be performed.

<code>\ztool_read_file_keep_spaces:nn</code>	<code>\ztool_read_file_keep_spaces:nn {<file>}{<code>}</code>
--	---

New: 2025-09-12

This command is similar to the `\ior_str_map_inline:Nn` command in `l3file`, but `\ztool_read_file_keep_spaces:nn` will change the catcode of the content to `\c_document_cctab`. In `<code>`, the variable `\l__ztool_strmap_read_tl` represents the content of each line. **Note:** `<file>` needs to include the file extension; if `<file>` does not exist, this command will not perform any operation.

<code>\ztool_read_file_as_seq:nnN</code>	<code>\ztool_read_file_as_seq:nnN {<bool>}{<file>}{<seq>}</code>
<code>\ztool_read_file_as_seq:(neN nnc nec)</code>	

Updated: 2024-12-05

This command is used to read the content of the file `<file>` and store it in `<seq>`. If `<file>` does not exist, `<seq>` will be set to empty. `<bool>` is used to control whether to keep the spaces at the **end of the line**. Optional values are: `\c_true_bool`, `\c_false_bool`; if `<bool>` is `\c_true_bool`, then the spaces at the **end of the line** are kept, otherwise they are not.

NOTE:

1. The definition of `<seq>` is local;
2. Due to the limitation of the command `\ior_map_inline:Nn`, this command cannot obtain the “space” or “Tab” at the beginning of the line;
3. The catcode of the content in `<seq>` is the current catcode.

<code>\ztool_read_file_as_seq_keep_spaces:nnN</code> <code>\ztool_read_file_as_seq_keep_spaces:(neN nnc nec)</code>	<code>\ztool_read_file_as_seq_keep_spaces:nnN</code> <code>{\bool}\{file}\seq</code>
--	---

New: 2025-09-01

This command is used to read the content of the file $\langle file \rangle$, **preserving internal spaces**, and store it in $\langle seq \rangle$. If $\langle file \rangle$ does not exist, $\langle seq \rangle$ will be set to empty. $\langle bool \rangle$ is used to control whether to keep the spaces at the **beginning of the line**. Optional values are: `\c_true_bool`, `\c_false_bool`; if $\langle bool \rangle$ is `\c_true_bool`, then the spaces at the **beginning of the line** are kept, otherwise they are not.

NOTE:

1. The definition of $\langle seq \rangle$ is local;
2. Due to the limitation of the command `\ior_str_map_inline:Nn`, this command cannot obtain the “space” or “Tab” at the end of the line;
3. The catcode of the content in $\langle seq \rangle$ is modified to `\c_document_cctab`.

<code>\ztool_gread_file_as_seq:nnN</code> <code>\ztool_gread_file_as_seq:(neN nnc nec)</code>	<code>\ztool_read_file_as_seq:nnN {\bool}\{file}\seq</code>
--	---

Updated: 2025-01-05

This command is used to read the content of the file $\langle file \rangle$ and store it in $\langle seq \rangle$. If $\langle file \rangle$ does not exist, $\langle seq \rangle$ will be set to empty. $\langle bool \rangle$ is used to control whether to keep the spaces at the **end of the line**. Optional values are: `\c_true_bool`, `\c_false_bool`; if $\langle bool \rangle$ is `\c_true_bool`, then the spaces at the **end of the line** are kept, otherwise they are not.

NOTE:

1. The definition of $\langle seq \rangle$ is global;
2. Due to the limitation of the command `\ior_map_inline:Nn`, this command cannot obtain the “space” or “Tab” at the beginning of the line;
3. The catcode of the content in $\langle seq \rangle$ is the current catcode.

<code>\ztool_gread_file_as_seq_keep_spaces:nnN</code>	<code>\ztool_gread_file_as_seq_keep_spaces:nnN</code>
<code>\ztool_gread_file_as_seq_keep_spaces:(neN nnc nec)</code>	<code>{⟨bool⟩}{⟨file⟩}{⟨seq⟩}</code>

New: 2025-09-01

This command is used to read the content of the file `⟨file⟩`, **preserving internal spaces**, and store it in `⟨seq⟩`. If `⟨file⟩` does not exist, `⟨seq⟩` will be set to empty. `⟨bool⟩` is used to control whether to keep the spaces at the **beginning of the line**. Optional values are: `\c_true_bool`, `\c_false_bool`; if `⟨bool⟩` is `\c_true_bool`, then the spaces at the **beginning of the line** are kept, otherwise they are not.

NOTE:

1. The definition of `⟨seq⟩` is global;
2. Due to the limitation of the command `\ior_str_map_inline:Nn`, this command cannot obtain the “space” or “Tab” at the end of the line;
3. The catcode of the content in `⟨seq⟩` is modified to `\c_document_cctab`.

<code>\ztool_write_seq_to_file:nNn</code>	<code>\ztool_write_seq_to_file:nNn {⟨bool⟩}{⟨seq⟩}{⟨file⟩}</code>
<code>\ztool_write_seq_to_file:(nNe nNV nce ncV)</code>	

New: 2025-05-27

This command is used to write `⟨seq⟩` line by line to the file `⟨file⟩`. If `⟨file⟩` does not exist, a new file named `⟨file⟩` will be created; if `⟨file⟩` already exists, you can use `⟨bool⟩` to control the current writing mode: when `⟨bool⟩` is `\c_true_bool`, **overwrite**; when `⟨bool⟩` is `\c_false_bool`, **append**; if `⟨seq⟩` is empty, no operation will be performed.

<code>\ztool_append_to_file:nn</code>	<code>\ztool_append_to_file:nn {⟨file⟩}{⟨content⟩}</code>
<code>\ztool_append_to_file:(no nf ee)</code>	

Updated: 2025-01-05

This command is used to append `⟨content⟩` to the file `⟨file⟩`. If `⟨file⟩` does not exist, a new file named `⟨file⟩` will be created and `⟨content⟩` will be written into it.

<code>\ztool_replace_file_line:nnn</code>	<code>\ztool_replace_file_line:nnn {⟨file⟩}{⟨line⟩}{⟨content⟩}</code>
<code>\ztool_replace_file_line:(enn ene eee)</code>	

Updated: 2025-01-05

This command is used to replace the `⟨line⟩`-th line in the file `⟨file⟩` with `⟨content⟩`. If `⟨file⟩` does not exist, no operation will be performed.

<code>\ztool_insert_to_file:nnn</code>	<code>\ztool_insert_to_file:nnn {⟨file⟩}{⟨line⟩}{⟨content⟩}</code>
<code>\ztool_insert_to_file:(nen nfn een)</code>	

Updated: 2025-01-05

This command is used to insert `⟨content⟩` before the `⟨line⟩`-th line of the file `⟨file⟩`. If `⟨file⟩` does not exist, no operation will be performed.

The following example shows how to use several file IO operation commands in `ztool`:

Ex. 3

```
% \usepackage{verbatim}
\ExplSyntaxOn
\ztool_file_new:nn {\c_true_bool}{testIO.txt}
\seq_new:N \l_ztool_tmp_seq \seq_clear:N \l_ztool_tmp_seq
\ztool_append_to_file:nn {testIO.txt} {|APPEND-CONTENT|}
\ztool_insert_to_file:nnn {testIO.txt} {1} {|INSERT-~-CONTENT|}
\ztool_append_to_file:nn {testIO.txt} {|APPEND-CONTENT-II|}
\ztool_replace_file_line:nnn {testIO.txt} {3} {|REPLACE-CONTENT|}
\ztool_gread_file_as_seq:nnN {\c_false_bool} {testIO.txt}
\l_ztool_tmp_seq
\seq_use:Nn \l_ztool_tmp_seq {\par}
\ExplSyntaxOff
\verbatiminput*{testIO.txt}
```

```
|INSERT-CONTENT|
|APPEND-CONTENT|
|REPLACE-CONTENT|

|INSERT-~-CONTENT|
|APPEND-CONTENT|
|REPLACE-CONTENT|
```

The following example demonstrates how the file to seq series of commands in `ztool` handle spaces. The content in the file `testSpaces.txt` is as follows:

```
AAAA
BB_ _BB
_ _CC_ _CC_ _
DDDD
```

Ex. 4

```
\ExplSyntaxOn
\def\TTTa#1{
  \seq_clear:N \l_ztool_tmp_seq
  \ztool_read_file_as_seq:nnN {\c_true_bool} {#1} \l_ztool_tmp_seq
  \seq_show:N \l_ztool_tmp_seq
}
```

```

\def\TTTb#1{
  \seq_clear:N \l_ztool_tmp_seq
  \ztool_read_file_as_seq:nnN {\c_false_bool} {#1} \l_ztool_tmp_seq
  \seq_show:N \l_ztool_tmp_seq
}

\def\TTTc#1{
  \seq_clear:N \l_ztool_tmp_seq
  \ztool_read_file_as_seq_keep_spaces:nnN {\c_true_bool} {#1}
  \l_ztool_tmp_seq
  \seq_show:N \l_ztool_tmp_seq
}

\def\TTTd#1{
  \seq_clear:N \l_ztool_tmp_seq
  \ztool_read_file_as_seq_keep_spaces:nnN {\c_false_bool} {#1}
  \l_ztool_tmp_seq
  \seq_show:N \l_ztool_tmp_seq
}

\TTTa{testSpaces.txt}
\TTTb{testSpaces.txt}
\TTTc{testSpaces.txt}
\TTTd{testSpaces.txt}
\ExplSyntaxOff
\TTTa{testSpaces.txt}
\TTTb{testSpaces.txt}
\TTTc{testSpaces.txt}
\TTTd{testSpaces.txt}

```

Results displayed in the command line in various cases (somewhat simplified):

\TTTa: outside expl3, true

```

{AAAA_}
{BB_BB_}
{CC_CC_}
{DDDD_}.

```

\TTTb: outside expl3, false

```

{AAAA}
{BB_BB}
{CC_CC}
{DDDD}.

```

\TTTa: inside expl3, true

{AAAA}
{BBBB}
{CCCC}
{DDDD}.

\TTTb: inside expl3, false

{AAAA}
{BBBB}
{CCCC}
{DDDD}.

.....

\TTTc: outside expl3, true

{AAAA}
{BB_BB}
{_CC_CC}
{DDDD}.

\TTTd: outside expl3, false

{AAAA}
{BB_BB}
{CC_CC}
{DDDD}.

\TTTc: inside expl3, true

{AAAA}
{BB_BB}
{_CC_CC}
{DDDD}.

\TTTd: inside expl3, false

{AAAA}
{BB_BB}
{CC_CC}
{DDDD}.

5 Box Operations

This section introduces the box operations implemented in `ztool`, including box measurement and simple box transformations.

```
\ztool_get_ht:Nn
\ztool_get_ht:(Ne|ce)
```

Updated: 2024-12-05

```
\ztool_get_ht:Nn <dim>{\<content>}
```

This command is used to save the height of `<content>` in the register `<dim>`.

```
\ztool_get_ht_plus_dp:Nn
\ztool_get_ht_plus_dp:(Ne|ce)
```

Updated: 2024-12-05

```
\ztool_get_ht:Nn <dim>{\<content>}
```

This command is used to save the sum of the height and depth of `<content>` in the register `<dim>`.

```
\ztool_get_wd:Nn
\ztool_get_wd:(Ne|ce)
```

Updated: 2024-12-05

```
\ztool_get_wd:Nn <dim>{\<content>}
```

This command is used to save the width of `<content>` in the register `<dim>`.

```
\ztool_get_dp:Nn
\ztool_get_dp:(Ne|ce)
```

Updated: 2024-12-05

```
\ztool_get_dp:Nn <dim>{\<content>}
```

This command is used to save the depth of `<content>` in the register `<dim>`.

```
\ztool_gget_ht:Nn
\ztool_gget_ht:(Ne|ce)
```

Updated: 2024-12-05

```
\ztool_gget_ht:Nn <dim>{\<content>}
```

This command is used to save the height of `<content>` in the register `<dim>`, and this operation is global.

```
\ztool_gget_wd:Nn
\ztool_gget_wd:(Ne|ce)
```

Updated: 2024-12-05

```
\ztool_gget_wd:Nn <dim>{\<content>}
```

This command is used to save the width of `<content>` in the register `<dim>`, and this operation is global.

```
\ztool_gget_dp:Nn
\ztool_gget_dp:(Ne|ce)
```

Updated: 2024-12-05

```
\ztool_gget_dp:nn <dim>{\<content>}
```

This command is used to save the depth of `<content>` in the register `<dim>`, and this operation is global.

```
\ztool_set_to_wd:nn
\ztool_set_to_wd:(en|ne)
```

Updated: 2024-12-05

```
\ztool_set_to_wd:nn {\<dim>}{\<content>}
```

This command is used to adjust the width of `<content>` to `<dim>`. Its height will be scaled proportionally, and then `<content>` will be typeset. **Note:** This command is different from commands like `\hbox_to_wd:nn`.

```
\ztool_set_to_ht:nn
\ztool_set_to_ht:(en|ne)
```

Updated: 2024-12-05

```
\ztool_set_to_ht:nn {\<dim>}{\<content>}
```

This command is used to adjust the height of `<content>` to `<dim>`. Its width will be scaled proportionally, and then `<content>` will be typeset. **Note:** This command is different from commands like `\vbox_to_ht:nn`.

<code>\ztool_autoset_to_wd_and_ht:nnn</code>	<code>\ztool_autoset_to_wd_and_ht:nn</code>
<code>\ztool_autoset_to_wd_and_ht:(nne een eee)</code>	<code>{\langle width \rangle}{\langle height \rangle}{\langle content \rangle}</code>

Updated: 2025-04-29

This command is used to adjust the width of $\langle content \rangle$ to $\min(\langle width \rangle, \langle height \rangle)$, and then typeset it. See the description of the `\box_autosize_to_wd_and_ht:Nnn` command for details.

<code>\ztool_rotate:nn</code>	<code>\ztool_rotate:nn {\langle angle \rangle}{\langle content \rangle}</code>
<code>\ztool_rotate:(en ne ee)</code>	

New: 2025-04-29

This command is used to rotate $\langle content \rangle$ by $\langle angle \rangle$ degrees and then typeset it.

<code>\ztool_scale_to_wd:nn</code>	<code>\ztool_scale_to_wd:nn {\langle dim \rangle}{\langle content \rangle}</code>
<code>\ztool_scale_to_wd:(en ne ee)</code>	

New: 2025-04-29

This command is used to adjust the width of $\langle content \rangle$ to $\langle dim \rangle$, but without making any adjustments to the height of the box, and then typeset it.

<code>\ztool_scale_to_ht:nn</code>	<code>\ztool_scale_to_ht:nn {\langle dim \rangle}{\langle content \rangle}</code>
<code>\ztool_scale_to_ht:(en ne ee)</code>	

New: 2025-04-29

This command is used to adjust the total height + depth of $\langle content \rangle$ to $\langle dim \rangle$, but without making any adjustments to the width of the box, and then typeset it.

<code>\ztool_scale_to_wd_and_ht:nnn</code>	<code>\ztool_scale_to_wd_and_ht:nnn {\langle width \rangle}{\langle height \rangle}{\langle content \rangle}</code>
<code>\ztool_scale_to_wd_and_ht:(nno nne eee)</code>	

New: 2025-04-29

This command is used to adjust the width of $\langle content \rangle$ to $\langle width \rangle$, and the total height + depth to $\langle height \rangle$, then typeset it.

<code>\ztool_box_item_align:Nnnn</code>	<code>\ztool_box_item_align:Nnnn</code>
<code>\ztool_box_item_align:(cnno Nnno cnno Nnen Nnee)</code>	<code>\langle cmd \rangle{\langle width \rangle}{\langle content \rangle}{\langle align \rangle}</code>

Updated: 2025-05-13

This command is used to adjust the width of $\langle content \rangle$ to $\langle width \rangle$ and then typeset it. $\langle align \rangle$ is used to control the alignment. Optional values are: **left**, **center**, **right**, **scatter**. $\langle cmd \rangle$ is a command that accepts one argument, which will be applied to each token of $\langle content \rangle$. **Note:** Spaces in $\langle content \rangle$ will be ignored. If spaces are needed, please use “`\_`” or “`\~`” instead.

<code>\ztool_fp_to_rad:n</code>	<code>\ztool_fp_to_rad:n {\langle angle \rangle}</code>
---------------------------------	---

New: 2025-05-12

This command is used to convert $\langle angle \rangle$ from radians to degrees.

<code>\ztoolboxaffine</code>	<code>\ztoolboxaffine[⟨key-value⟩]{⟨content⟩}{⟨matrix⟩}</code>
------------------------------	--

New: 2025-05-12

The above $\langle content \rangle$ represents the object on which the affine transformation acts; $\langle matrix \rangle$ is a 2×2 matrix, representing the corresponding affine transformation matrix. If $\langle matrix \rangle = \{a, b, c, d\}$, its corresponding affine transformation matrix Λ is as follows:

$$\Lambda = \begin{bmatrix} a & c \\ b & d \end{bmatrix}.$$

If $\det \Lambda = 0$, this transformation is meaningless, `ztool` will output a warning in the terminal, and finally output the content in $\langle content \rangle$ as it is to the PDF; for $\langle keyval \rangle$ settings, please refer to the following “`ztool/affine`” description. **Note:** This command is encapsulated from the `\ztool_affine_transformation:Nnnnn` command below.

<code>ztool/affine/debug</code>	<code>debug</code>	<code>=</code>	$\langle true false \rangle$		Initial value: <code>false</code>
<code>ztool/affine/pole-1</code>	<code>pole-1</code>	<code>=</code>	$\langle coffin's pole \rangle$		Initial value: <code>l</code>
<code>ztool/affine/pole-2</code>	<code>pole-2</code>	<code>=</code>	$\langle coffin's pole \rangle$		Initial value: <code>b</code>
<code>ztool/affine/xoffset</code>	<code>xoffset</code>	<code>=</code>	$\langle number \rangle$		Initial value: <code>Opt</code>
<code>ztool/affine/yoffset</code>	<code>yoffset</code>	<code>=</code>	$\langle number \rangle$		Initial value: <code>Opt</code>

$\langle debug \rangle$ is used for debugging. If set to `true`, some intermediate variable information will be output in the PDF for debugging; $\langle xoffset \rangle$ and $\langle yoffset \rangle$ are the horizontal and vertical offsets, with default values of `Opt`; $\langle pole-1 \rangle$ and $\langle pole-2 \rangle$ are used to set the reference point when printing the coffin, and the two must intersect. For detailed usage of the latter four keys, please refer to the description of `l3coffins`.

<code>\ztool_affine_transformation:Nnnnn</code>	<code>\ztool_affine_transformation:Nnnnn</code>
<code>\ztool_affine_transformation:(Neeee cnnnn ceeee)</code>	$\langle coffin \rangle \{ \langle a \rangle \} \{ \langle b \rangle \} \{ \langle c \rangle \} \{ \langle d \rangle \}$

New: 2025-05-12

This command is used to perform arbitrary affine transformation (linear transformation) on $\langle coffin \rangle$. For the specific usage method, please refer to the aforementioned `\ztoolboxaffine` command. The affine transformation matrix Λ corresponding to the above parameters is

$$\Lambda = \begin{bmatrix} a & c \\ b & d \end{bmatrix}.$$

Some technical details about the aforementioned function `\zttool_affine_transformation:Nnnnn`: Given any affine transformation Λ , let

$$\Lambda = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}.$$

We can do the following decomposition (similar to SVD decomposition), let $m = 2x$, then we have:

$$\begin{aligned} \Lambda &= \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 1 & m \\ 0 & 1 \end{bmatrix} \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} \cos \omega & -\sin \omega \\ \sin \omega & \cos \omega \end{bmatrix} \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}. \end{aligned} \quad (5.1)$$

We give the following notation:

- $\mathbf{T}_1(\theta)$: rotation matrix, rotating θ angle counterclockwise around the origin;
- $\mathbf{T}_2(x)$: scaling matrix, changing all vectors in the x -axis direction to x times the original;
- $\mathbf{T}_3(y)$: scaling matrix, changing all vectors in the y -axis direction to y times the original;

Then we can consider $\{\mathbf{T}_1(\theta), \mathbf{T}_2(x), \mathbf{T}_3(y)\}$ to be the basis of $A_{2 \times 2}$. So we can write the above equation (5.1) as the following expression:

$$\Lambda = \mathbf{T}_1(\theta) \cdot \mathbf{T}_1(\phi) \cdot \mathbf{T}_2(S_x) \cdot \mathbf{T}_3(S_y) \cdot \mathbf{T}_1(\omega) \cdot \mathbf{T}_2(s_x) \cdot \mathbf{T}_3(s_y). \quad (5.2)$$

Based on the result of matrix multiplication, we can know the parameters m, s_x, S_x, ϕ , etc. as follows:

$$s_x = \sqrt{A_{11}^2 + A_{21}^2}, \quad \theta = \arctan\left(\frac{A_{21}}{A_{11}}\right).$$

The results for s_y and m are as follows:

$$ms_y = A_{12} \cos \theta + A_{22} \sin \theta, \quad s_y = \begin{cases} \frac{ms_y \cos \theta - A_{12}}{\sin \theta} & \text{if } \sin \theta \neq 0, \\ \frac{A_{22} - ms_y \sin \theta}{\cos \theta} & \text{if } \sin \theta = 0; \end{cases}$$

Then it is easy to know $m = ms_y/s_y$. The decomposition result for the shear matrix is as follows:

$$\begin{aligned} S_x &= \sqrt{\frac{m^2}{4} + 1} - \frac{m}{2}, & S_y &= \sqrt{\frac{m^2}{4} + 1} + \frac{m}{2}, \\ \phi &= -\frac{\pi}{4} - \frac{1}{2} \arctan\left(\frac{m}{2}\right), & \omega &= \frac{\pi}{4} - \frac{1}{2} \arctan\left(\frac{m}{2}\right). \end{aligned}$$

Finally, we only need to apply this series of transformations to the `\box` from right to left. As can be seen from the above, the command `\zttool_affine_transformation:Nnnnn` only depends on the two functions `\coffin_scale:Nnn` and `\coffin_rotate:Nn` in `LATEX3`.

The relevant reference links during the implementation of the command `\ztool_affine_transformation:Nnnnn` are as follows:

- <https://math.stackexchange.com/a/3521141/1235323>;
- <https://math.stackexchange.com/a/281087/1235323>.

If the original $\text{\TeX}$ engine provided primitives related to shear transformation, then the above decomposition of the shear matrix would be unnecessary. Some engines provide the affine transformation matrix primitive natively, such as the `\pdfsetmatrix` command in $\text{pdf\TeX}$.

The following example shows how to use several box operation commands in this chapter:

Ex. 5

```

\ExplSyntaxOn
\setlength{\fboxsep}{0pt}
% get dim of content
\dotfill\par
\dim_new:N \l_ztool_tmp_H_dim
\dim_new:N \l_ztool_tmp_W_dim
\ztool_get_ht:Nn \l_ztool_tmp_H_dim {Hello,~world!}
\ztool_get_wd:Nn \l_ztool_tmp_W_dim {Hello,~world!}
\dim_use:N \l_ztool_tmp_H_dim \quad \dim_use:N \l_ztool_tmp_W_dim\par

% set content to dim
\dotfill\par
Hello,~world!
\ztool_set_to_ht:nn {.5cm} {Hello,~world!}
\ztool_set_to_wd:nn {40pt} {Hello,~world!}\par

% scale one dimension
\dotfill\par
\ztool_scale_to_wd:nn {2em}{\fbox{AA}}\par
\ztool_scale_to_wd:nn {2em}{\fbox{AAA}}\par
\ztool_scale_to_wd:nn {2em}{\fbox{AAAAA}}\par
\ztool_scale_to_ht:nn {2.5em}{\fbox{\vbox{\hbox{A}}}}\quad
\ztool_scale_to_ht:nn {2.5em}{\fbox{\vbox{\hbox{A}}\hbox{A}}}\quad
\ztool_scale_to_ht:nn {2.5em}{\fbox{\vbox{\hbox{A}}\hbox{A}}\hbox{A}}}\quad
\ztool_scale_to_ht:nn {2.5em}{\fbox{\vbox{\hbox{A}}\hbox{A}}\hbox{A}}\hbox{A}}}\quad
\par

% box item align
\dotfill\par
\def\boxItemCmd#1{\textcolor{blue}{|#1|}}
\underline{
  \ztool_box_item_align:Nnnn \boxItemCmd{15em}{Tom}{Amy}{Jennery}{scatter}

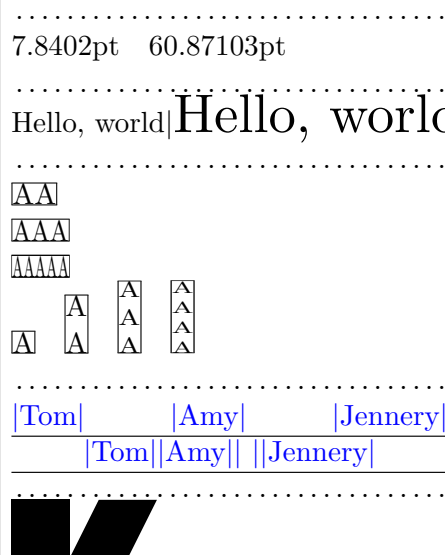
```

```

}\par
\underline{
  \ztool_box_item_align:Nnnn \boxItemCmd{15em}{\{Tom\} \{Amy\} }
\_{{Jennery}}{center}
}\par

% affine transform
\dotfill\par
\hcoffin_set:Nn \l_tmpa_coffin {\rule{2em}{2em}}
\coffin_typeset:Nnnnn \l_tmpa_coffin {1}{b}{0pt}{0pt}
\ztool_affine_transformation:Nnnnn \l_tmpa_coffin {1}{0}{.5}{1}
\coffin_typeset:Nnnnn \l_tmpa_coffin {1}{b}{0pt}{0pt}
\ExplSyntaxOff
.....
7.8402pt 60.87103pt
.....
Hello, world|Hello, world|Hello, world
.....
AA
AAA
AAAAA
A  A  A  A
A  A  A  A
.....
|Tom|      |Amy|      |Jennery|
|Tom|Amy||Jennery|
.....

```



6 zdraw

This section mainly contains some image drawing commands. This series of commands does not depend on the `tikz` package. Their main dependencies are as follows:

- $\text{\LaTeX} 2_{\epsilon}$ built-in `picture` environment;
- `pict2e`: an enhanced version of the $\text{\LaTeX} 2_{\epsilon}$ built-in `picture` environment, providing better drawing functions;
- `bxeepic`: can be used to provide dash line support, currently **not** included.

<code>\zpic</code>	<code>\begin{zpic}[\langle key-value \rangle] \langle draw commands \rangle \end{zpic}</code>
New: 2025-05-13	This environment is defined based on the $\text{\LaTeX} 2_{\epsilon}$ built-in <code>picture</code> environment.

<code>ztool/draw/picture/unit</code>	<code>unit</code>	<code>=</code>	<code>\langle length \rangle</code>		Initial value:	<code>1cm</code>
<code>ztool/draw/picture/width</code>	<code>width</code>	<code>=</code>	<code>\langle float \rangle</code>		Initial value:	<code>0</code>
<code>ztool/draw/picture/height</code>	<code>height</code>	<code>=</code>	<code>\langle float \rangle</code>		Initial value:	<code>0</code>
<code>ztool/draw/picture/depth</code>	<code>depth</code>	<code>=</code>	<code>\langle float \rangle</code>		Initial value:	<code>0</code>
<code>ztool/draw/picture/xoffset</code>	<code>xoffset</code>	<code>=</code>	<code>\langle float \rangle</code>		Initial value:	<code>0</code>
<code>ztool/draw/picture/yoffset</code>	<code>yoffset</code>	<code>=</code>	<code>\langle float \rangle</code>		Initial value:	<code>0</code>

If `\langle width \rangle` adopts the default value, the current image will overlap with the surrounding typeset content.

<code>\put</code>	<code>\put (\langle x, y \rangle) {\langle content \rangle}</code>
New: 2025-05-13	This command is the same as the <code>\put</code> command in the $\text{\LaTeX} 2_{\epsilon}$ built-in <code>picture</code> environment. Note: This command needs to be used in the <code>picture</code> or <code>zpic</code> environment.

<code>\zline</code>	<code>\zline [\langle key-value \rangle](\langle coor-1 \rangle)(\langle coor-2 \rangle)</code>
New: 2025-05-13	This command is used to draw a line segment from <code>\langle coor-1 \rangle</code> to <code>\langle coor-2 \rangle</code> . <code>\langle key-value \rangle</code> is used to set the properties of the line. For available options, please refer to the following <code>\parent=ztool/draw/picture/line</code> .

<code>ztool/./line/draw</code>	<code>draw</code>	<code>=</code>	<code>\langle color \rangle</code>		Initial value:	<code>black</code>
<code>ztool/./line/width</code>	<code>width</code>	<code>=</code>	<code>\langle length \rangle</code>		Initial value:	<code>.4pt</code>
<code>ztool/./line/dash</code>	<code>dash</code>	<code>=</code>	<code>\langle true false \rangle</code>		Initial value:	<code>false</code>

The above `\langle width \rangle` is used to set the width of the line, `\langle draw \rangle` is used to set the color of the line, and `\langle dash \rangle` is used to set whether the line is a dashed line. **Note:** Currently the `\langle dash \rangle` option has not been adapted and is **in an unavailable state**.

<code>\zvector</code>	<code>\zvector [\langle key-value \rangle](\langle coor-1 \rangle)(\langle coor-2 \rangle)</code>
New: 2025-05-13	This command is used to draw a vector. The starting point of the vector is <code>\langle coor-1 \rangle</code> and the end point is <code>\langle coor-2 \rangle</code> . <code>\langle key-value \rangle</code> is used to set the appearance attributes of the vector, which is inherited from <code>\parent=ztool/draw/picture/line</code> . For other available options, please refer to the following <code>\parent=ztool/draw/picture/line/ve</code>

ztool/../../vector/>	<code>> = $\langle \text{latex pst} \rangle$ Initial value: latex</code> This option is used to control the style of the arrow. The default is $\text{\LaTeX}$ style, i.e., $\text{\ltxarrows}$; $\langle \text{pst} \rangle$, i.e., PsTricks, corresponds to the $\text{\pstarrows}$ command.
$\backslash\text{zdraw}$ New: 2025-05-13	<code>$\backslash\text{zdraw} [\langle \text{key-value} \rangle] (\langle \text{coor-1} \rangle) \dots (\langle \text{coor-n} \rangle);$</code> This command will draw a polyline segment from point $\langle \text{coor-1} \rangle$ to point $\langle \text{coor-n} \rangle$. $\langle \text{key-value} \rangle$ is inherited from $\langle \text{parent=ztool/draw/picture/line} \rangle$ and can be used to set the properties of the line. For additional available options, please refer to the following $\langle \text{parent=ztool/draw/picture/zdraw} \rangle$. Note: The “;” at the end of this command cannot be omitted, otherwise an error will occur.
ztool/../../zdraw/vector ztool/../../zdraw/cycle ztool/../../zdraw/fill ztool/../../zdraw/shift	<code>vector = $\langle \text{false true} \rangle$ Initial value: false</code> <code>cycle = $\langle \text{false true} \rangle$ Initial value: false</code> <code>fill = $\langle \text{false true color} \rangle$ Initial value: false</code> <code>shift = $\{ \langle \text{float}, \text{float} \rangle \}$ Initial value: $\{0, 0\}$</code> When $\langle \text{fill} \rangle$ is set to true, $\langle \text{cycle} \rangle$ will be automatically set to true; $\langle \text{vector} \rangle$ is used to set whether to replace each sub-segment with a vector. $\langle \text{shift} \rangle$ represents the offset in the x and y directions respectively. Note: The $\{ \}$ in the $\langle \text{shift} \rangle$ option cannot be omitted.
$\backslash\text{zarc}$ New: 2025-05-13	<code>$\backslash\text{zarc} [\langle \text{key-value} \rangle] (\langle \text{float}, \text{float} \rangle)$</code> This command is used to draw an arc. $(\langle \text{float}, \text{float} \rangle)$ is its center. By default, a $\frac{1}{4}$ arc is drawn. $\langle \text{key-value} \rangle$ is inherited from $\langle \text{parent=ztool/draw/picture/line} \rangle$ and can be used to set the properties of the line. For additional available options, please refer to the following $\langle \text{parent=ztool/draw/picture/zarc} \rangle$.
ztool/../../zarc/radius ztool/../../zarc/start ztool/../../zarc/end ztool/../../zarc/fill	<code>radius = $\langle \text{float} \rangle$ Initial value: $.5$</code> <code>start = $\langle \text{float} \rangle$ Initial value: 0</code> <code>end = $\langle \text{float} \rangle$ Initial value: 90</code> <code>fill = $\langle \text{false true color} \rangle$ Initial value: false</code> $\langle \text{start} \rangle$ rotates counterclockwise until it ends at angle $\langle \text{end} \rangle$; $\langle \text{radius} \rangle$ is the radius of the arc; $\langle \text{fill} \rangle$ is used to set the filling color of the arc.
$\backslash\text{zcircle}$ New: 2025-05-13	<code>$\backslash\text{zcircle} [\langle \text{key-value} \rangle] (\langle \text{float}, \text{float} \rangle)$</code> This command is based on the aforementioned $\backslash\text{zarc}$ command. By default, it will draw a complete circle with $(\langle \text{float}, \text{float} \rangle)$ as the center; the $\langle \text{key-value} \rangle$ is the same as the $\langle \text{key-value} \rangle$ option in the aforementioned $\backslash\text{zarc}$ command.
$\backslash\text{zrectangle}$ New: 2025-05-13	<code>$\backslash\text{zrectangle} [\langle \text{key-value} \rangle] (\langle \text{coor-1} \rangle) (\langle \text{coor-2} \rangle)$</code> This command is used to draw a rectangle. $(\langle \text{coor-1} \rangle)$ and $(\langle \text{coor-2} \rangle)$ are the coordinates of the two endpoints of the rectangle’s diagonal; $\langle \text{key-value} \rangle$ is inherited from $\langle \text{parent=ztool/draw/picture/line} \rangle$. For other $\langle \text{key-value} \rangle$ options, please refer to the following $\langle \text{parent=ztool/draw/picture/zrectangle} \rangle$.

```
ztool/./zrectangle/arc
ztool/./zrectangle/fill
```

```
arc = <float>.....Initial value: 0
fill = <false|true|color>.....Initial value: false
```

`<fill>` is used to set the filling color of the rectangle, and `<arc>` is used to set the radius corresponding to the rounded corners of the rectangle.

```
\zpin
```

```
New: 2025-07-09
```

```
\zpin[<foreground|background>]{<code>}
```

This command is used to add annotations to the current page. The reference point is the bottom right corner of the current page, and the positive direction is to the right and upward.

```
\zpin{%
  \begin{zpic}%
    \zline[width=5pt] (0, 0) (1, 1)
  \end{zpic}
}
```

Ex. 6

To facilitate understanding of the basic usage of the aforementioned drawing commands, several drawing examples are provided here.

Example 1: Basic line segment drawing commands.

```
\mbox{\vskip2em
\begin{zpic}[unit=2em]
  \zdraw[fill, cycle] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[cycle, shift={2, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[fill, shift={4, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[draw=red, width=1pt, shift={6, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[vector, shift={8, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[vector, cycle, shift={10, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[vector, fill, shift={12, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
  \zdraw[vector, cycle, fill, shift={14, 0}] (0, 0)(1, 0)(1, 1)(0, 1);
\end{zpic}
```

Ex. 7





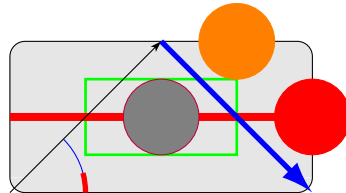





Example 2: Basic geometric element drawing commands.

```
\mbox{\hspace{5em}}
\begin{zpic}[unit=2cm, xoffset=2]
% 1. rectangle
\zrectangle[arc=.1, fill=gray!20](0, 0)(2, 1)
\zrectangle[draw=green, width=1pt](.5, .25)(1.5, .75)
% 2. line / vector
\zline[width=3pt, draw=red](0, .5)(2, .5)
\zvector[>=pst](0, 0)(1, 1)
\zvector[draw=blue, width=2pt](1, 1)(2, 0)
% 3. arc / circle
\zarc[draw=blue, end=45](0, 0) % fill=<empty>
\zarc[draw=blue, width=2pt, end=15, fill=, draw=red](0, 0)
\zcircle[radius=.25, fill, draw=purple](1, .5)
\zcircle[radius=.25, fill=orange, draw=none](1.5, 1)
\zcircle[radius=.25, fill=red, draw=](2, .5)
\end{zpic}
```

Ex. 8



7 TODO

ztool may have changes in the future. Some functions that may be improved in the future are listed here (☐ – unfinished; ☒ – finished; ☒ – functionality not considered):

- ☐ Re-implement the `\xsim_file_write_start:nn` and `\xsim_file_write_stop:` commands in the `xsimverb` package to make them compatible with the `ztool` package.
- ☒ 2025-05-22-Finished: Fixed the issue of the empty first line in the `\ztool_append_to_file:nn` file.
- ☒ 2025-09-01-Finished: For the command `\ztool_read_file_as_seq:nnN`, in some cases all spaces in the source file need to be preserved, reference command `\seq_set_split_keep_spaces:Nnn`.
- ☐ Use the implemented `\ztex_tl_replace_all:nnn` or `\ztex_tl_replace_once:nnn` command to implement `\ztool_replace_file_line_text:nnnn` $\{\langle file \rangle\}\{\langle line \rangle\}$ $\{\langle pattern \rangle\}\{\langle text \rangle\}$, and implement simple regular expression functionality in $\langle pattern \rangle$, ensuring that this command is expandable.
- ☐ Use the `\tl_analysis_map_inline:nn` command from `l3tl-analysis` (which can capture special characters like '{', '}', '\$') to implement a simple token debugging command.
- ☐ Use `l3draw` to wrap a `tikz`-like front end, which needs to natively support 3D drawing and automatic occlusion relationship adjustment.
- ☒ 2025-07-09-Finished: Error or unexpected result when `\zline` draws vertical or horizontal line segments.
- ☒ 2026-09-23-Finished: The effects of `\zline` and `\zdraw` are inconsistent. When drawing the same line segment in the same coordinate system, the two cannot coincide (currently it seems the `\zline` command is correct).

8 zTool Source Code

This section lists the source code of the `ztool` package for reference. The line numbers have been specially handled so you can copy the source directly from the PDF.

8.1 ztool.sty

```
1 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% 1
2 %% ztool.sty % 2
3 %% Copyright 2024, 2025 Zongping Ding. % 3
4 % % 4
5 % This work may be distributed and/or modified under the conditions of the % 5
6 % LaTeX Project Public License, either version 1.3 of this license or any % 6
7 % later version. % 7
8 % The latest version of this license is in % 8
9 % http://www.latex-project.org/lppl.txt % 9
10 % and version 1.3 or later is part of all distributions of LaTeX % 10
11 % version 2005/12/01 or later. % 11
12 % % 12
13 % This work has the LPPL maintenance status `maintained'. % 13
14 % % 14
15 % The Current Maintainer of this work is Zongping Ding. % 15
16 % % 16
17 % ztool.sty consists of the parts: % 17
18 % shell-escape, % 18
19 % file-io, % 19
20 % box, % 20
21 % zdraw. % 21
22 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% 22
23 \NeedsTeXFormat{LaTeX2e} 23
24 \ProvidesExplPackage{ztool}{2025/05/20}{1.0.1}{A~pre-release~tool~package~for~LaTeX} 24
25 25
26 26
27 %%%% 13keys intial patch begin %%%% 27
28 % 1. https://github.com/latex3/latex3/issues/1738 28
29 % 2. https://tex.stackexchange.com/q/742604/294585 29
30 \cs_set_protected:Npn \__keys_initialise:n #1 30
31 { 31
32 \exp_after:wN \__keys_find_key_module:wNN 32
33 \l_keys_path_str \s_keys_stop 33
34 \l_keys_key_tl \l_keys_key_str 34
35 \tl_set_eq:NN \l_keys_key_tl \l_keys_key_str 35
36 \tl_set:Nn \l_keys_value_tl {#1} 36
37 \cs_if_exist:cTF { \c__keys_code_root_str \l_keys_path_str } 37
38 { 38
39 \str_clear:N \l__keys_inherit_str 39
40 \__keys_execute:nn \l_keys_path_str {#1} 40
41 } 41
42 { 42
43 \cs_if_exist:cT 43
44 { \c__keys_inherit_root_str \__keys_parent:o \l_keys_path_str } 44
45 { \__keys_execute_inherit: } 45
46 } 46
```

47	}	47
48	%%%% 13keys intial patch end %%%%	48
49		49
50		50
51	\clist_new:N \g__ztool_library_loaded_clist	51
52	\clist_gclear:N \g__ztool_library_loaded_clist	52
53	\bool_new:N \g__ztool_lib_user_load_dupuplicate_bool	53
54	\bool_gset_false:N \g__ztool_lib_user_load_dupuplicate_bool	54
55	\cs_new_nopar:Npn __ztool_load_library:n #1	55
56	{	56
57	\clist_map_inline:nn {#1} {	57
58	\clist_if_in:NnTF \g__ztool_library_loaded_clist {##1} {	58
59	\msg_set:nnn {ztool} {library-loaded}	59
60	{	60
61	ztool~library~"##1"~already~loaded,ignored~loading.	61
62	\msg_line_context:	62
63	}	63
64	\bool_if:NT \g__ztool_lib_user_load_dupuplicate_bool	64
65	{	65
66	\msg_warning:nnn {ztool} {library-loaded} {##1}	66
67	}	67
68	}{	68
69	\file_if_exist:nTF {library/ztool.library.##1.tex}{	69
70	\clist_gput_right:Nn \g__ztool_library_loaded_clist {##1}	70
71	\makeatletter\file_input:n {library/ztool.library.##1.tex}	71
72	}{	72
73	\msg_set:nnn {ztool} {library-not-found} {ztool~library~`##1'~not~found.}	73
74	\msg_error:nnn {ztool} {library-not-found} {##1}	74
75	}	75
76	}	76
77	}	77
78	}	78
79	\NewDocumentCommand\ztoolloadlib{m}	79
80	{	80
81	__ztool_load_library:n {#1}	81
82	\bool_gset_true:N \g__ztool_lib_user_load_dupuplicate_bool	82
83	\ExplSyntaxOff	83
84	}	84
85	\keys_define:nn { ztool }	85
86	{	86
87	shell-escape .code:n = { __ztool_load_library:n {shell-escape} },	87
88	file-io .code:n = { __ztool_load_library:n {file-io} },	88
89	box .code:n = { __ztool_load_library:n {box} },	89
90	zdraw .code:n = { __ztool_load_library:n {zdraw} },	90
91	}	91
92	\ProcessKeyOptions [ztool]	92

8.2 shell-escape

```
1 %% 1
2 %% This is file `ztool.library.shell-escape.tex'. 2
3 %% This file is based on the original source code with modifications. 3
4 %% The original disclaimer reads as follows: 4
5 %% 5
6 %% 6
7 %% This is file `l3sys-shell.sty', 7
8 %% generated with the docstrip utility. 8
9 %% 9
10 %% The original source files were: 10
11 %% 11
12 %% l3sys-shell.dtx (with options: `package') 12
13 %% 13
14 %% Copyright (C) 2018,2019 The LaTeX3 Project 14
15 %% 15
16 %% It may be distributed and/or modified under the conditions of 16
17 %% the LaTeX Project Public License (LPPL), either version 1.3c of 17
18 %% this license or (at your option) any later version. The latest 18
19 %% version of this license is in the file: 19
20 %% 20
21 %% http://www.latex-project.org/lppl.txt 21
22 %% 22
23 %% This file is part of the "l3experimental bundle" (The Work in LPPL) 23
24 %% and all files in that bundle must be distributed together. 24
25 %% 25
26 %% File: l3sys-shell.dtx 26
27 \ProvidesExplFile{ztool.library.shell-escape.tex} 27
28 {2025/09/06}{1.0.1} 28
29 {shell-escape-library~for~ztool} 29
30 30
31 31
32 % ==> l3sys-shell tool 32
33 % windows path handle 33
34 \cs_new:Npn \ztool_sys_path_to_win:N #1 34
35 { 35
36 \quark_if_nil:NF #1 { 36
37 \token_if_eq_meaning:NNTF #1 / 37
38 { \c_backslash_str } 38
39 {#1} 39
40 \ztool_sys_path_to_win:N 40
41 } 41
42 } 42
43 \cs_new:Npn \ztool_sys_path_to_win:w #1 ~ #2 \q_stop 43
44 { 44
45 \ztool_sys_path_to_win:N #1 \q_nil 45
46 \tl_if_empty:nF {#2} 46
```

```

47     {
48         \c_space_tl
49         \__sys_path_to_win:w #2 \q_stop
50     }
51 }
52 \cs_new:Npn \ztool_sys_path_to_win:n #1
53 {
54     \exp_after:wN \ztool_sys_path_to_win:w
55     \tl_to_str:n {#1} ~ \q_stop
56 }
57 % respective commands
58 \cs_new_protected:Npn \ztool_shell_escape:n #1
59 {
60     \sys_if_shell_unrestricted:T
61     { \sys_shell_now:n {#1} }
62 }
63 \cs_generate_variant:Nn \ztool_shell_escape:n {e}
64 \cs_new_protected:Npe \ztool_shell_mkdir:n #1
65 {
66     \ztool_shell_escape:e {
67         \sys_if_platform_unix:T
68         {mkdir~\exp_not:N \tl_to_str:n {#1}}
69         \sys_if_platform_windows:T
70         {mkdir~ \exp_not:N \ztool_sys_path_to_win:n {#1}}
71     }
72 }
73 \cs_new_protected:Npe \ztool_shell_cp:nn #1#2
74 {
75     \ztool_shell_escape:e {
76         \sys_if_platform_unix:T
77         {
78             cp~-f~ \exp_not:N \tl_to_str:n {#1} ~
79             \exp_not:N \tl_to_str:n {#2}
80         }
81         \sys_if_platform_windows:T
82         {% can NOT use wildcards in CMD
83             copy~/y~ \exp_not:N \ztool_sys_path_to_win:n {#1} ~
84             \exp_not:N \ztool_sys_path_to_win:n {#2}
85         }
86     }
87 }
88 \cs_new_protected:Npe \ztool_shell_mv:nn #1#2
89 {
90     \ztool_shell_escape:e {
91         \sys_if_platform_unix:T
92         {
93             mv~ \exp_not:N \tl_to_str:n {#1} ~
94             \exp_not:N \tl_to_str:n {#2}

```

95 } 95

96 \sys_if_platform_windows:T 96

97 { 97

98 copy~/y~ \exp_not:N \ztool_sys_path_to_win:n {#1} ~ 98

99 \exp_not:N \ztool_sys_path_to_win:n {#2} 99

100 \token_to_str:N & \token_to_str:N & 100

101 del~/f~/q~\exp_not:N \ztool_sys_path_to_win:n {#1} 101

102 } 102

103 } 103

104 } 104

105 \cs_new_protected:Npe \ztool_shell_rm:n #1 105

106 { 106

107 \ztool_shell_escape:e { 107

108 \sys_if_platform_unix:T 108

109 { rm~-f~ \exp_not:N \tl_to_str:n {#1} } 109

110 \sys_if_platform_windows:T 110

111 { del~/f~/q~ \exp_not:N \ztool_sys_path_to_win:n {#1} } 111

112 } 112

113 } 113

114 \cs_new_protected:Npe \ztool_shell_rmdir:n #1 114

115 { 115

116 \ztool_shell_mkdir:n {#1} 116

117 \ztool_shell_escape:e { 117

118 \sys_if_platform_unix:T 118

119 { rm~-rf~ \exp_not:N \tl_to_str:n {#1} } 119

120 \sys_if_platform_windows:T 120

121 { rmdir~/s~/q~ \exp_not:N \ztool_sys_path_to_win:n {#1} } 121

122 } 122

123 } 123

124 \tl_new:N \l__ztool_shell_tmp_tl 124

125 \cs_new_protected:Npe \ztool_get_shell_pwd:N #1 125

126 { 126

127 \exp_not:N \sys_get_shell:nnN 127

128 { 128

129 \sys_if_platform_unix:T { pwd } 129

130 \sys_if_platform_windows:T { cd } 130

131 }{ 131

132 \char_set_catcode_other:N \exp_not:N \ 132

133 \char_set_catcode_other:N \exp_not:N \# 133

134 \char_set_catcode_other:N \exp_not:N \~ 134

135 \char_set_catcode_other:N \exp_not:N \% 135

136 \char_set_catcode_space:N \exp_not:N _ 136

137 \tex_endlinechar:D -1 \scan_stop: 137

138 } 138

139 \exp_not:N \l__ztool_shell_tmp_tl 139

140 \str_set:NV #1 \exp_not:N \l__ztool_shell_tmp_tl 140

141 } 141

142 \cs_new_protected:Npe \ztool_shell_split_ls:nN #1#2 142

143	{	143
144	\exp_not:N \sys_get_shell:nn	144
145	{	145
146	\sys_if_platform_unix:T { ls~-1~ #1 }	146
147	\sys_if_platform_windows:T { dir~/b~ #1 }	147
148	}{	148
149	\ExplSyntaxOff	149
150	\char_set_catcode_other:N \exp_not:N \	150
151	\char_set_catcode_other:N \exp_not:N \#	151
152	\char_set_catcode_other:N \exp_not:N \~	152
153	\char_set_catcode_other:N \exp_not:N \%	153
154	\char_set_catcode_other:n { 13 }	154
155	}	155
156	\exp_not:N \l__ztool_shell_tmp_tl	156
157	\str_set:NV \exp_not:N \l__sys_tmp_tl \exp_not:N \l__sys_tmp_tl	157
158	\seq_set_split:NnV #2	158
159	{ \char_generate:nn { ``~M } { 12 } }	159
160	\exp_not:N \l__ztool_shell_tmp_tl	160
161	\seq_pop_right:NN #2 \exp_not:N \l__sys_tmp_tl	161
162	}	162
163	\cs_generate_variant:Nn \ztool_shell_mkdir:n {e}	163
164	\cs_generate_variant:Nn \ztool_shell_cp:nn { ee, ne, en }	164
165	\cs_generate_variant:Nn \ztool_shell_mv:nn { ee, ne, en }	165
166	\cs_generate_variant:Nn \ztool_shell_rm:n { e, f, o }	166
167	\cs_generate_variant:Nn \ztool_shell_rmdir:nn { e, f, o }	167
168	\cs_generate_variant:Nn \ztool_get_shell_pwd:N {c}	168
169	\cs_generate_variant:Nn \ztool_shell_split_ls:nN {nc}	169

8.3 file-io

```
1 \ProvidesExplFile{ztool.library.file-io.tex}
2 {2025/09/12}{1.0.1}
3 {file-io~library~for~ztool}
4
5
6 % ==> file IO operations
7 % 1. create a new file
8 % 2. append to a file
9 % 3. read from file / write to file
10 \ior_new:N \g_ztool_file_read_ior
11 \ior_new:N \g_ztool_file_append_ior
12 \iow_new:N \g_ztool_file_append_iow
13 \tl_new:N \l_ztool_current_line
14 \str_clear:N \l_ztool_file_ori_content_str
15 \seq_new:N \l_ztool_file_seq
16 \seq_new:N \l_ztool_tmp_seq
17 \cs_generate_variant:Nn \seq_use:Nn { Ne }
18
19 % read file as seq(not keep internal spaces):
20 \cs_new_protected:Npn \ztool_read_file_as_seq:nnN #1#2#3
21 {% #1: bool(True to keep spaces, False to trim); #2: file name; #3: seq
22   \seq_clear:N #3
23   \file_if_exist:nT {#2}
24   {
25     \ior_open:Nn \g_ztool_file_read_ior {#2}
26     \ior_map_inline:Nn \g_ztool_file_read_ior
27     {
28       \tl_if_empty:nF {##1}
29       {
30         \bool_if:nTF {#1}
31         { \seq_put_right:Nn #3 {##1} }
32         {
33           \seq_put_right:Ne #3
34           { \tl_trim_spaces:n {##1} }
35         }
36       }
37     }
38     \ior_close:N \g_ztool_file_read_ior
39   }
40 }
41 \cs_new_protected:Npn \ztool_gread_file_as_seq:nnN #1#2#3
42 {% #1: bool(True to keep spaces, False to trim); #2: file name; #3: seq
43   \seq_gclear:N #3
44   \file_if_exist:nT {#2}
45   {
46     \ior_open:Nn \g_ztool_file_read_ior {#2}
```

47	\ior_map_inline:Nn \g_ztool_file_read_ior	47
48	{	48
49	\tl_if_empty:nF {##1}	49
50	{	50
51	\bool_if:nTF {#1}	51
52	{ \seq_gput_right:Nn #3 {##1} }	52
53	{	53
54	\seq_gput_right:Ne #3	54
55	{ \tl_trim_spaces:n {##1} }	55
56	}	56
57	}	57
58	}	58
59	\ior_close:N \g_ztool_file_read_ior	59
60	}	60
61	}	61
62	\cs_generate_variant:Nn \ztool_read_file_as_seq:nnN	62
63	{ ne, nnc, nec }	63
64	\cs_generate_variant:Nn \ztool_gread_file_as_seq:nnN	64
65	{ ne, nnc, nec }	65
66		66
67	% read file as seq(keep these internal spaces):	67
68	\tl_new:N \l__ztool_strmap_read_tl	68
69	\cs_new_protected:Npn \ztool_read_file_keep_spaces:nn #1#2	69
70	{	70
71	\file_if_exist:nT {#1}	71
72	{	72
73	\ior_open:Nn \g_ztool_file_read_ior {#1}	73
74	\ior_str_map_inline:Nn \g_ztool_file_read_ior	74
75	{	75
76	\exp_args:Nee \str_if_in:nnF { \tl_head:n {##1} }	76
77	{ \char_generate:nn {37}{12} }	77
78	{	78
79	\tl_set_rescan:Nnn \l__ztool_strmap_read_tl	79
80	{	80
81	\cctab_select:N \c_document_cctab	81
82	\char_set_catcode_space:n { 9 } % tab	82
83	\char_set_catcode_space:n { 32 } % space	83
84	}{ ##1 }	84
85	#2	85
86	}	86
87	}	87
88	\ior_close:N \g_ztool_file_read_ior	88
89	}	89
90	}	90
91	\cs_new_protected:Npn \ztool_read_file_as_seq_keep_spaces:nnN #1#2#3	91
92	{% #1: bool(True to keep trim spaces, False to trim); #2: file name; #3: seq	92
93	\seq_gclear:N #3	93
94	\ztool_read_file_keep_spaces:nn {#2}	94

95100105110115120125130135140145

{\bool_if:nTF {#1}{\exp_args:NNo \seq_put_right:Nn #3{ \l__ztool_strmap_read_tl }}{\seq_gput_right:Ne #3{\exp_args:No \tl_trim_spaces:n{ \l__ztool_strmap_read_tl }}}}

\cs_new_protected:Npn \ztool_gread_file_as_seq_keep_spaces:nnN #1#2#3{% #1: bool(True to keep trim spaces, False to trim); #2: file name; #3: seq\seq_gclear:N #3\ztool_read_file_keep_spaces:nn {#2}{\bool_if:nTF {#1}{\exp_args:NNo \seq_gput_right:Nn #3{ \l__ztool_strmap_read_tl }}{\seq_gput_right:Ne #3{\exp_args:No \tl_trim_spaces:n{ \l__ztool_strmap_read_tl }}}}}

\cs_generate_variant:Nn \ztool_read_file_as_seq_keep_spaces:nnN{ ne, nnc, nec }\cs_generate_variant:Nn \ztool_gread_file_as_seq_keep_spaces:nnN{ ne, nnc, nec }

% create file / append to file / write to file\cs_new_protected:Npn \ztool_file_new:nn #1#2{% #1: \c_true_bool to allow overwrite; #2: file name\bool_if:nT {#1}{\iow_open:Nn \g_ztool_file_append_iow {#2}\iow_close:N \g_ztool_file_append_iow}}

\cs_new_protected:Npn \ztool_append_to_file:nn #1#2{% #1: file name; #2: content

9596979899100101102103104105106107108109110111112113114115116117118119120121122123124125126127128129130131132133134135136137138139140141142

```

143 \seq_clear:N \l_ztool_file_seq
144 \file_if_exist:nF {#1}
145 { \ztool_file_new:nn {\c_true_bool}{#1} }
146 \ior_open:Nn \g_ztool_file_append_ior {#1}
147 \ior_str_map_inline:Nn \g_ztool_file_append_ior
148 {
149     \seq_put_right:Nn \l_ztool_file_seq
150     { ##1 }
151 }
152 \iow_open:Nn \g_ztool_file_append_iow {#1}
153 \seq_if_empty:NF \l_ztool_file_seq
154 {
155     \iow_now:Ne \g_ztool_file_append_iow
156     {
157         \seq_use:Ne \l_ztool_file_seq
158         { \iow_newline: }
159     }
160 }
161 \iow_now:Ne \g_ztool_file_append_iow {#2}
162 \iow_close:N \g_ztool_file_append_iow
163 }
164 \cs_generate_variant:Nn \ztool_append_to_file:nn
165 { no, nf, ne, ee }
166
167 \cs_new_protected:Npn \ztool_write_seq_to_file:nNn #1#2#3
168 {% #1:bool; #2:seq; #3:file name
169     \seq_clear:N \l__ztool_tmp_seq
170     \bool_if:nTF { #1 }
171     {
172         \seq_set_eq:NN \l_ztool_file_seq #2
173     }{
174         \ztool_read_file_as_seq:nnN
175         { \c_true_bool }{ #3 }
176         \l__ztool_tmp_seq
177         \seq_concat:NNN \l_ztool_file_seq
178         \l__ztool_tmp_seq #2
179     }
180     \file_if_exist:nF {#3}
181     { \ztool_file_new:nn {\c_true_bool}{#3} }
182     \iow_open:Nn \g_tmpa_iow { #3 }
183     \seq_if_empty:NF \l_ztool_file_seq
184     {
185         \iow_now:Ne \g_tmpa_iow
186         {
187             \seq_use:Ne \l_ztool_file_seq
188             { \iow_newline: }
189         }
190     }

```



```
191 \iow_close:N \g_tmpa_iow
192 }
193 \cs_generate_variant:Nn \ztool_write_seq_to_file:nNn
194 { nNe, nNV, nce, ncV }
195
196 \cs_new_protected:Npn \ztool_replace_file_line:nnn #1#2#3
197 {% #1:file name; #2:line index; #3:replacement
198 \seq_clear:N \l_ztool_file_seq
199 \file_if_exist:nT {#1}{
200 \iow_open:Nn \g_ztool_file_read_iow {#1}
201 \iow_str_map_inline:Nn \g_ztool_file_read_iow
202 {
203 \seq_put_right:Nn \l_ztool_file_seq {##1}
204 }
205 \iow_close:N \g_ztool_file_read_iow
206 \seq_set_item:Nnn \l_ztool_file_seq {#2}
207 { #3 }
208 \iow_open:Nn \g_ztool_file_append_iow {#1}
209 \seq_if_empty:NF \l_ztool_file_seq
210 {
211 \iow_now:Ne \g_ztool_file_append_iow
212 {
213 \seq_use:Ne \l_ztool_file_seq
214 { \iow_newline: }
215 }
216 }
217 \iow_close:N \g_ztool_file_append_iow
218 }
219 }
220 \cs_generate_variant:Nn \seq_set_item:Nnn { Nne }
221 \cs_generate_variant:Nn \ztool_replace_file_line:nnn
222 { e, ene, eee }
223 \cs_new_protected:Npn \ztool_insert_to_file:nnn #1#2#3
224 {% #1:file name; #2:line index; #3:content
225 \seq_clear:N \l_ztool_file_seq
226 \file_if_exist:nT {#1}{
227 \iow_open:Nn \g_ztool_file_read_iow {#1}
228 \iow_str_map_inline:Nn \g_ztool_file_read_iow
229 {
230 \seq_put_right:Nn \l_ztool_file_seq {##1}
231 }
232 \iow_close:N \g_ztool_file_read_iow
233 \tl_set:No \l_ztool_current_line
234 { \seq_item:Nn \l_ztool_file_seq {#2} }
235 \seq_set_item:Nne \l_ztool_file_seq {#2}
236 { #3\iow_newline:\l_ztool_current_line }
237 \iow_open:Nn \g_ztool_file_append_iow {#1}
238 \iow_now:Ne \g_ztool_file_append_iow
```

239	{	239
240	\seq_use:Ne \l_ztool_file_seq	240
241	{ \iow_newline: }	241
242	}	242
243	\iow_close:N \g_ztool_file_append_iow	243
244	}	244
245	}	245
246	\cs_generate_variant:Nn \ztool_insert_to_file:nn	246
247	{ ne, nf, ee }	247

8.4 box

```
1 \ProvidesExplFile{ztool.library.box.tex} 1
2 {2026/02/13}{1.0.1} 2
3 {box~library~for~ztool} 3
4 4
5 5
6 % ==> box manipulation tool 6
7 \cs_set:Nn \__ztool_leave_vmode: 7
8 { \ifvmode \leavevmode \fi } 8
9 9
10 \box_new:N \l_ztool_measure_box 10
11 \box_new:N \l_ztool_tmpa_box 11
12 12
13 % catch box dimension 13
14 \cs_new:Npn \ztool_box_set_to:NNn #1#2#3 14
15 { 15
16 \hbox_set:Nn \l_ztool_measure_box {#3} 16
17 \dim_set:Nn #2 {#1 \l_ztool_measure_box} 17
18 \box_set_eq:NN \l_ztool_measure_box \c_empty_box 18
19 } 19
20 \cs_new:Npn \ztool_box_gset_to:NNn #1#2#3 20
21 { 21
22 \hbox_set:Nn \l_ztool_measure_box {#3} 22
23 \dim_gset:Nn #2 {#1 \l_ztool_measure_box} 23
24 \box_set_eq:NN \l_ztool_measure_box \c_empty_box 24
25 } 25
26 \cs_new:Npn \ztool_get_ht:Nn 26
27 { \ztool_box_set_to:NNn \box_ht:N } 27
28 \cs_new:Npn \ztool_get_ht_plus_dp:Nn 28
29 { \ztool_box_set_to:NNn \box_ht_plus_dp:N } 29
30 \cs_new:Npn \ztool_get_wd:Nn 30
31 { \ztool_box_set_to:NNn \box_wd:N } 31
32 \cs_new:Npn \ztool_get_dp:Nn 32
33 { \ztool_box_set_to:NNn \box_dp:N } 33
34 \cs_new:Npn \ztool_gget_ht:Nn 34
35 { \ztool_box_gset_to:NNn \box_ht:N } 35
36 \cs_new:Npn \ztool_gget_wd:Nn 36
37 { \ztool_box_gset_to:NNn \box_wd:N } 37
38 \cs_new:Npn \ztool_gget_dp:Nn 38
39 { \ztool_box_gset_to:NNn \box_dp:N } 39
40 \cs_generate_variant:Nn \ztool_get_ht:Nn 40
41 { Ne, ce } 41
42 \cs_generate_variant:Nn \ztool_get_ht_plus_dp:Nn 42
43 { Ne, ce } 43
44 \cs_generate_variant:Nn \ztool_get_wd:Nn 44
45 { Ne, ce } 45
46 \cs_generate_variant:Nn \ztool_gget_ht:Nn 46
```

47	{ Ne, ce }	47
48	\cs_generate_variant:Nn \ztool_gget_wd:Nn	48
49	{ Ne, ce }	49
50		50
51		51
52	%% modify box content	52
53	% 1. auto scale and rotate (smaller of two)	53
54	\cs_new_protected:Npn \ztool_autoset_to_wd_and_ht:nnn #1#2#3	54
55	{% #1:width; #2:height; #3:object	55
56	\hbox_set:Nn \l_ztool_tmpa_box {#3}	56
57	\box_autosize_to_wd_and_ht:Nnn \l_ztool_tmpa_box {#1}{#2}	57
58	__ztool_leave_vmode:	58
59	\box_use:N \l_ztool_tmpa_box	59
60	}	60
61	\cs_new_protected:Npn \ztool_rotate:nn #1#2	61
62	{% #1:angle; #2:object	62
63	\hbox_set:Nn \l_ztool_tmpa_box {#2}	63
64	\box_rotate:Nn \l_ztool_tmpa_box {#1}	64
65	__ztool_leave_vmode:	65
66	\box_use:N \l_ztool_tmpa_box	66
67	}	67
68	\cs_generate_variant:Nn \ztool_rotate:nn	68
69	{ e, ne, ee }	69
70	\cs_generate_variant:Nn \ztool_autoset_to_wd_and_ht:nnn	70
71	{ nne, een, eee }	71
72		72
73	% 2. width/height scale to same time	73
74	% TODO: if '\dim(content) < dim', spread it to 'dim'.	74
75	\cs_new_protected:Npn \ztool_set_to_wd:nn #1#2	75
76	{% #1:width; #2:object	76
77	\hbox_set:Nn \l_ztool_tmpa_box {#2}	77
78	\box_resize_to_wd:Nn \l_ztool_tmpa_box {#1}	78
79	__ztool_leave_vmode:	79
80	\box_use:N \l_ztool_tmpa_box	80
81	}	81
82	\cs_new_protected:Npn \ztool_set_to_ht:nn #1#2	82
83	{% #1:height; #2:object	83
84	\hbox_set:Nn \l_ztool_tmpa_box {#2}	84
85	\box_resize_to_ht:Nn \l_ztool_tmpa_box {#1}	85
86	__ztool_leave_vmode:	86
87	\box_use:N \l_ztool_tmpa_box	87
88	}	88
89	\cs_generate_variant:Nn \ztool_set_to_wd:nn { e, ne, ee }	89
90	\cs_generate_variant:Nn \ztool_set_to_ht:nn { e, ne, ee }	90
91		91
92	% 3. only scale one dimension	92
93	% NOTE: if boxwd{content} <= given dim, no manipulation	93
94	\cs_new_protected:Npn \ztool_scale_to_wd:nn #1#2	94

95	{	95
96	\hbox_set:Nn \l_ztool_tmpa_box {#2}	96
97	\dim_set:Nn \l_tmpa_dim { \box_wd:N \l_ztool_tmpa_box }	97
98	\fp_set:Nn \l_tmpa_fp	98
99	{	99
100	\fp_eval:n { min(1, \dim_ratio:nn {#1}{\l_tmpa_dim}) }	100
101	}	101
102	\box_scale:Nnn \l_ztool_tmpa_box {\l_tmpa_fp}{1}	102
103	__ztool_leave_vmode:	103
104	\box_use:N \l_ztool_tmpa_box	104
105	}	105
106	\cs_new_protected:Npn \ztool_scale_to_ht:nn #1#2	106
107	{% take depth into consideration	107
108	\hbox_set:Nn \l_ztool_tmpa_box {#2}	108
109	\dim_set:Nn \l_tmpa_dim { \box_ht_plus_dp:N \l_ztool_tmpa_box }	109
110	\fp_set:Nn \l_tmpa_fp	110
111	{	111
112	\fp_eval:n { min(1, \dim_ratio:nn {#1}{\l_tmpa_dim}) }	112
113	}	113
114	\box_scale:Nnn \l_ztool_tmpa_box {1}{\l_tmpa_fp}	114
115	__ztool_leave_vmode:	115
116	\box_use:N \l_ztool_tmpa_box	116
117	}	117
118	\cs_new_protected:Npn \ztool_scale_to_wd_and_ht:nnn #1#2#3	118
119	{% take depth into consideration	119
120	\hbox_set:Nn \l_ztool_tmpa_box {#3}	120
121	\dim_set:Nn \l_tmpa_dim { \box_wd:N \l_ztool_tmpa_box }	121
122	\dim_set:Nn \l_tmpb_dim { \box_ht_plus_dp:N \l_ztool_tmpa_box }	122
123	\fp_set:Nn \l_tmpa_fp	123
124	{	124
125	\fp_eval:n { min(1, \dim_ratio:nn {#1}{\l_tmpa_dim}) }	125
126	}	126
127	\fp_set:Nn \l_tmpb_fp	127
128	{	128
129	\fp_eval:n { min(1, \dim_ratio:nn {#2}{\l_tmpb_dim}) }	129
130	}	130
131	\box_scale:Nnn \l_ztool_tmpa_box {\l_tmpa_fp}{\l_tmpb_fp}	131
132	__ztool_leave_vmode:	132
133	\box_use:N \l_ztool_tmpa_box	133
134	}	134
135	\cs_generate_variant:Nn \ztool_scale_to_wd:nn	135
136	{ e, ne, ee }	136
137	\cs_generate_variant:Nn \ztool_scale_to_ht:nn	137
138	{ e, ne, ee }	138
139	\cs_generate_variant:Nn \ztool_scale_to_wd_and_ht:nnn	139
140	{ nne, nno, eee }	140
141		141
142		142

```

143 %% box content align
144 \seq_new:N \l__ztool_boxitem_seq
145 \cs_set_protected:Npn \ztool_box_item_align:Nnnn #1#2#3#4
146 { % #1:cmd, #2:width, #3:object, #4:align format(left, right, scatter, center)
147   \hb@xt@#2{
148     \tl_map_inline:nn {#3}
149     {
150       \seq_put_right:No \l__ztool_boxitem_seq
151       { \exp_not:N #1{##1} }
152     }
153     \str_case:nnF { #4 }
154     {
155       { left }{ \seq_use:Nn \l__ztool_boxitem_seq {\hfill} }
156       { right }{ \hfill\seq_use:Nn \l__ztool_boxitem_seq {} }
157       { scatter }{ \seq_use:Nn \l__ztool_boxitem_seq {\hfill} }
158       { center }{ \hfill\seq_use:Nn \l__ztool_boxitem_seq {\hfill} }
159       { tower }
160       {
161         \edef\seq@count{\seq_count:N \l__ztool_boxitem_seq}
162         \seq_map_indexed_inline:Nn \l__ztool_boxitem_seq
163         { % ##1: index, ##2: content
164           %% Method II: plain
165           \edef\item@width{\dim_eval:n {#2/(\seq@count+1)}}
166           \hskip\item@width\clap{##2}
167           }\hskip\item@width\hss
168         }
169       { custom }
170       {
171         \def\total@width{#2}
172         \def\align@cmd{#1}
173         \def\align@object{#3}
174         \def\align@format{#4}
175         \tl_use:N \l__ztex_boxitem_align_custom_tl
176       }
177       {\relax}
178     }
179     \seq_clear:N \l__ztool_boxitem_seq
180   }
181 \cs_generate_variant:Nn \ztool_box_item_align:Nnnn
182 { c, Nnno, cnno, Nne, Nnee }
183
184
185 %% affine transformation
186 % REF:
187 % 1. https://math.stackexchange.com/a/3521141/1235323
188 % 2. https://math.stackexchange.com/a/281087/1235323
189 \cs_new:Npn \ztool_fp_to_rad:n #1
190 { \fp_eval:n {#1/pi*180} }

```

```

191 \cs_new:Npn \ztool_matrix_det:nnnn #1#2#3#4 191
192 { 192
193     \fp_eval:n { #1*#4 - #2*#3 } 193
194 } 194
195 % (translation) + x-scale + y-scale + rotate 195
196 \fp_new:N \g_affine_precision_fp 196
197 \fp_set:Nn \g_affine_precision_fp {0.0001} 197
198 \fp_new:N \l__affine_@@_a_fp 198
199 \fp_new:N \l__affine_@@_b_fp 199
200 \fp_new:N \l__affine_@@_c_fp 200
201 \fp_new:N \l__affine_@@_d_fp 201
202 \msg_set:nnn { ztool }{affine-det-zero} 202
203 { 203
204     current~determination~of~the~affine~transformation~ 204
205     matrix~equals~to~zero,~give~up~this~transformation 205
206 } 206
207 207
208 \coffin_new:N \l__affine_trans_coffin 208
209 \cs_generate_variant:Nn \coffin_typeset:Nnnnn { Nxxxx } 209
210 \cs_new:Npn \ztool_affine_transformation:Nnnnn #1#2#3#4#5 210
211 {% #1:box; #2:$a_{11}$; #3:$a_{21}$; #4:$a_{12}$; #5:$a_{22}$. 211
212     \fp_compare:nNnT 212
213         { abs(\ztool_matrix_det:nnnn {#2}{#3}{#4}{#5}) } 213
214         < { \g_affine_precision_fp } 214
215         { \prg_map_break:Nn \l__affine_matrix_det_zero 215
216             { \msg_warning:nn { ztool }{affine-det-zero} }} 216
217     \fp_set:Nn \l__affine_@@_a_fp {#2} 217
218     \fp_set:Nn \l__affine_@@_b_fp {#3} 218
219     \fp_set:Nn \l__affine_@@_c_fp {#4} 219
220     \fp_set:Nn \l__affine_@@_d_fp {#5} 220
221     \__box_affine_transform:N #1 221
222     \prg_break_point:Nn \l__affine_matrix_det_zero { } 222
223     \coffin_typeset:Nxxxx \l__affine_trans_coffin 223
224         { \l__ztool_affine_pole_a_tl } 224
225         { \l__ztool_affine_pole_b_tl } 225
226         { \l__ztool_affine_xoffset_dim } 226
227         { \l__ztool_affine_yoffset_dim } 227
228     } 228
229 \cs_generate_variant:Nn \ztool_affine_transformation:Nnnnn 229
230 { Neeee, cnnnn, ceeee } 230
231 \cs_new:Npn \__box_affine_transform:N #1 231
232 { 232
233     % transform debug 233
234     \bool_if:NT \g_ztool_affine_debug_bool 234
235     { 235
236         \noindent\dotfill\[\begin{bmatrix} 236
237             \fp_use:N \l__affine_@@_a_fp & \fp_use:N \l__affine_@@_c_fp\\ 237
238             \fp_use:N \l__affine_@@_b_fp & \fp_use:N \l__affine_@@_d_fp 238

```

```

239 \end{bmatrix}\]
240 }
241 % get affine parameters
242 \__affine_trans_get_sx:
243 \__affine_trans_get_theta:
244 \__affine_trans_get_sy:
245 \__affine_trans_get_Sx:
246 \__affine_trans_get_Sy:
247 \__affine_trans_get_phi:
248 \__affine_trans_get_omega:
249 % start transform box/coffin
250 \coffin_scale:Nnn #1
251 { \l__box_affine_sx_fp }
252 { \l__box_affine_sy_fp }
253 \coffin_rotate:Nn #1
254 { \ztool_fp_to_rad:n {\l__box_affine_omega_fp} }
255 \coffin_scale:Nnn #1
256 { \l__box_affine_Sx_fp }
257 { \l__box_affine_Sy_fp }
258 \coffin_rotate:Nn #1
259 { \ztool_fp_to_rad:n {\l__box_affine_phi_fp} }
260 \coffin_rotate:Nn #1
261 { \ztool_fp_to_rad:n {\l__box_affine_theta_fp} }
262 }
263 \keys_define:nn { ztool / affine }
264 {
265     debug .bool_gset:N = \g_ztool_affine_debug_bool,
266     debug .initial:n = false,
267     debug .default:n = true,
268     pole-1 .tl_set:N = \l__ztool_affine_pole_a_tl,
269     pole-2 .tl_set:N = \l__ztool_affine_pole_b_tl,
270     pole-1 .initial:n = { l },
271     pole-2 .initial:n = { b },
272     xoffset .dim_set:N = \l__ztool_affine_xoffset_dim,
273     yoffset .dim_set:N = \l__ztool_affine_yoffset_dim,
274     xoffset .initial:n = { 0pt },
275     yoffset .initial:n = { 0pt },
276 }
277 \NewDocumentCommand{\ztoolboxaffine}{0{}m>{\SplitList{,}}m}
278 {% #1:key-value; #2:content; #3:matrix.
279     \group_begin:
280         \keys_set:nn { ztool / affine } {#1}
281         \hcoffin_set:Nn \l__affine_trans_coffin {#2}
282         \ztool_affine_transformation:Nnnnn \l__affine_trans_coffin #3
283     \group_end:
284 }
285 % internal affine transform functions
286 \cs_new:Nn \__ztool_affine_debug_fp:N

```



```

287 {
288     \bool_if:NTF \g_ztool_affine_debug_bool
289     { \string #1 % \show #1
290       ~~~\fp_use:N #1\\
291       }{ \relax }
292 }
293 \fp_new:N \l__box_affine_sx_fp
294 \cs_new:Nn \__affine_trans_get_sx:
295 {
296     \fp_set:Nn \l__box_affine_sx_fp
297     { \fp_eval:n {sqrt(\l__affine_@@_a_fp^2 + \l__affine_@@_b_fp^2)} }
298     \__ztool_affine_debug_fp:N \l__box_affine_sx_fp
299 }
300 \fp_new:N \l__box_affine_theta_fp
301 \cs_new:Nn \__affine_trans_get_theta:
302 {
303     \fp_set:Nn \l__box_affine_theta_fp
304     { \fp_eval:n {atan(\l__affine_@@_b_fp/\l__affine_@@_a_fp)} }
305     \__ztool_affine_debug_fp:N \l__box_affine_theta_fp
306 }
307 \fp_new:N \l__box_affine_msy_fp
308 \cs_new:Nn \__affine_trans_get_msy:
309 {
310     \fp_set:Nn \l__box_affine_msy_fp
311     { \fp_eval:n {
312         \l__affine_@@_c_fp*cos(\l__box_affine_theta_fp)
313         +
314         \l__affine_@@_d_fp*sin(\l__box_affine_theta_fp)
315     } }
316     \__ztool_affine_debug_fp:N \l__box_affine_msy_fp
317 }
318 \fp_new:N \l__box_affine_sy_fp
319 \cs_new:Nn \__affine_trans_get_sy:
320 {
321     \__affine_trans_get_msy:
322     \bool_if:nTF
323     {
324         \fp_compare_p:nNn { abs(sin(\l__box_affine_theta_fp)) }
325         < { \c_zero_fp + \g_affine_precision_fp }
326     }{
327         \fp_set:Nn \l__box_affine_sy_fp
328         {
329             ( \l__affine_@@_d_fp - \l__box_affine_msy_fp*sin(\l__box_affine_theta_fp) )
330             / cos(\l__box_affine_theta_fp)
331         }
332     }{
333         \fp_set:Nn \l__box_affine_sy_fp
334         {

```

```

335         ( \l__box_affine_msy_fp*cos(\l__box_affine_theta_fp) - \l__affine_@@_c_fp ) 335
336         / sin(\l__box_affine_theta_fp) 336
337     } 337
338 } 338
339     \__ztool_affine_debug_fp:N \l__box_affine_sy_fp 339
340 } 340
341 \fp_new:N \l__box_affine_m_fp 341
342 \cs_new:Nn \__affine_trans_get_m: 342
343 { 343
344     \fp_set:Nn \l__box_affine_m_fp 344
345     { \l__box_affine_msy_fp / \l__box_affine_sy_fp } 345
346     \__ztool_affine_debug_fp:N \l__box_affine_m_fp 346
347 } 347
348 \fp_new:N \l__box_affine_Sx_fp 348
349 \fp_new:N \l__box_affine_Sy_fp 349
350 \cs_new:Nn \__affine_trans_get_Sx: 350
351 { 351
352     \__affine_trans_get_m: 352
353     \fp_set:Nn \l__box_affine_Sx_fp 353
354     { sqrt(\l__box_affine_m_fp^2/4 + 1) - \l__box_affine_m_fp/2 } 354
355     \__ztool_affine_debug_fp:N \l__box_affine_Sx_fp 355
356 } 356
357 \cs_new:Nn \__affine_trans_get_Sy: 357
358 { 358
359     \fp_set:Nn \l__box_affine_Sy_fp 359
360     { sqrt(\l__box_affine_m_fp^2/4 + 1) + \l__box_affine_m_fp/2 } 360
361     \__ztool_affine_debug_fp:N \l__box_affine_Sy_fp 361
362 } 362
363 \fp_new:N \l__box_affine_phi_fp 363
364 \fp_new:N \l__box_affine_omega_fp 364
365 \cs_new:Nn \__affine_trans_get_phi: 365
366 { 366
367     \fp_set:Nn \l__box_affine_phi_fp 367
368     { -pi/4 - 1/2*atan(\l__box_affine_m_fp/2) } 368
369     \__ztool_affine_debug_fp:N \l__box_affine_phi_fp 369
370 } 370
371 \cs_new:Nn \__affine_trans_get_omega: 371
372 { 372
373     \fp_set:Nn \l__box_affine_omega_fp 373
374     { pi/4 - 1/2*atan(\l__box_affine_m_fp/2) } 374
375     \__ztool_affine_debug_fp:N \l__box_affine_omega_fp 375
376 } 376

```

8.5 zdraw

```
1 \ProvidesExplFile{ztool.library.zdraw.tex}
2 {2026/09/23}{1.1.0}
3 {zdraw~library~for~ztool}
4
5
6 % ==> ztool draw (based on package 'pict2e' and 'picture' env)
7 \RequirePackage{pict2e}
8
9 % operations precise
10 \fp_const:Nn \c__pic_precise_fp { 0.001 }
11
12 % command/env wrapper
13 \cs_new_protected:Npn \_@@_begin_picture:nnnn #1#2#3#4
14 { \begin{picture}
15     (\fp_eval:n {#1}, \fp_eval:n {#2})
16     (\fp_eval:n {-#3}, \fp_eval:n {-#4}) }
17 \cs_new_protected:Npn \_@@_end_picture:
18 { \end{picture} }
19 \cs_new_protected:Npn \_@@_pic_put:nnn #1#2#3
20 { \put(\fp_eval:n {#1}, \fp_eval:n {#2}){ #3 } }
21 \cs_generate_variant:Nn \_@@_begin_picture:nnnn
22 { VVVV, eeee }
23 \cs_generate_variant:Nn \_@@_pic_put:nnn
24 { VVV, een }
25
26 % picture environment alias
27 % width/height/depth are measured in units of `unit` and describe the
28 % dimensions reported to TeX. In particular, `height` is the extent above
29 % the baseline and `depth` is the extent below it. The picture environment
30 % itself always returns zero depth, so zpic collects its result and restores
31 % the requested box dimensions explicitly.
32 \box_new:N \l__pic_output_box
33 \cs_new:Npn \__pic_to_dim:n #1
34 {
35     \fp_to_dim:n
36     { (#1) * \dim_to_fp:n { \l__pic_unit_dim } }
37 }
38 \keys_define:nn { ztool / draw / picture }
39 {
40     unit .dim_set:N = \l__pic_unit_dim,
41     unit .initial:n = { 1cm },
42     width .fp_set:N = \l__pic_width_fp,
43     width .initial:n = 0,
44     height .fp_set:N = \l__pic_height_fp,
45     height .initial:n = 0,
46     depth .fp_set:N = \l__pic_depth_fp,
```

47	depth .initial:n = 0,	47
48	xoffset .fp_set:N = \l__pic_xoffset_fp,	48
49	xoffset .initial:n = 0,	49
50	yoffset .fp_set:N = \l__pic_yoffset_fp,	50
51	yoffset .initial:n = 0,	51
52	% opacity-color .tl_set:N = \l__pic_opacity_color_tl,	52
53	% opacity-color .initial:n = { white },	53
54	}	54
55	\NewDocumentEnvironment{zpic}{0{}}	55
56	{	56
57	\group_begin:	57
58	\keys_set:nn { ztool / draw / picture } {#1}	58
59	\setlength\unitlength{ \l__pic_unit_dim }	59
60	\hbox_set:Nw \l__pic_output_box	60
61	_@@_begin_picture:VVVV	61
62	\l__pic_width_fp \l__pic_height_fp	62
63	\l__pic_xoffset_fp\l__pic_yoffset_fp	63
64	}{	64
65	_@@_end_picture:	65
66	\hbox_set_end:	66
67	\box_set_wd:Nn \l__pic_output_box	67
68	{ __pic_to_dim:n { \l__pic_width_fp } }	68
69	\box_set_ht:Nn \l__pic_output_box	69
70	{ __pic_to_dim:n { \l__pic_height_fp } }	70
71	\box_set_dp:Nn \l__pic_output_box	71
72	{ __pic_to_dim:n { \l__pic_depth_fp } }	72
73	\box_use_drop:N \l__pic_output_box	73
74	\group_end:	74
75	}	75
76		76
77		77
78	% picture commands alias	78
79	\cs_new:Npn __coord_st:n #1	79
80	{ \clist_item:nn {#1}{1} }	80
81	\cs_new:Npn __coord_nd:n #1	81
82	{ \clist_item:nn {#1}{2} }	82
83	\cs_new:Npn __coord_rd:n #1#2	83
84	{ \clist_item:nn {#1}{3} }	84
85	\cs_new:Npn __coord_st_nd:n #1	85
86	{	86
87	{\clist_item:nn {#1}{1}}	87
88	{\clist_item:nn {#1}{2}}	88
89	}	89
90	\cs_new:Npn __coord_st_nd_rd:n #1	90
91	{	91
92	{\clist_item:nn {#1}{1}}	92
93	{\clist_item:nn {#1}{2}}	93
94	{\clist_item:nn {#1}{3}}	94

```

95 }
96 \cs_generate_variant:Nn \__coord_st:n { V, e }
97 \cs_generate_variant:Nn \__coord_nd:n { V, e }
98 \cs_generate_variant:Nn \__coord_rd:n { V, e }
99 \cs_generate_variant:Nn \__coord_st_nd:n { V, e }
100 \cs_generate_variant:Nn \__coord_st_nd_rd:n { V, e }
101
102 \bool_new:N \l__ztool_invalid_color_bool
103 \cs_new:Npn \__color_safe_use:n #1
104 {
105     \__color_if_valid:nT {#1}
106     { \color{#1} }
107 }
108 \prg_new_conditional:Npnn \__color_if_valid:n #1 {p, T, F, TF}
109 {
110     \def\ztool@target@color{#1}
111     \def\ztool@color@none{none}
112     \bool_if:eTF
113     {
114         \tl_if_empty_p:e {#1} ||
115         \tl_if_eq_p:NN \ztool@color@none \ztool@target@color
116     }{ \prg_return_false: }
117     { \prg_return_true: }
118 }
119 \prg_generate_conditional_variant:Nnn \__color_if_valid:n
120 { V, e }{ p, T, F, TF }
121 \cs_generate_variant:Nn \__color_safe_use:n
122 { V, e }
123
124
125 % --> line/vector
126 \fp_new:N \l__draw_vector_slope_fp
127 \fp_new:N \l__draw_vector_normal_fp
128 \fp_new:N \l__draw_vector_xysep_fp
129 \cs_new:Npn \__@@_pic_line:nnn #1#2#3
130 { % #1:x; #2:y; #3:x-distance NOT the length
131     \line(\fp_eval:n {#1}, \fp_eval:n {#2})
132     { \fp_eval:n {#3} }
133 }
134 \cs_new:Npn \__@@_pic_vector:nnn #1#2#3
135 { % #1:x; #2:y; #3:x-distance NOT the length
136     \vector(\fp_eval:n {#1}, \fp_eval:n {#2})
137     { \fp_eval:n {#3} }
138 }
139 \keys_define:nn { ztool / draw / picture / line }
140 {
141     draw .tl_set:N = \l__pic_line_draw_color_tl,
142     draw .initial:n = { black },

```

```

143 % color .meta:n = { draw = #1 }, % alias for 'draw'
144 width .dim_set:N = \l__pic_line_width_dim,
145 width .initial:n = { .4pt },
146 dash .bool_set:N = \l__pic_line_dash_bool,
147 dash .initial:n = { false },
148 }
149 \cs_new_protected:Nn \__pic_set_line_width:
150 {
151     \linethickness{ \l__pic_line_width_dim }
152 }
153 \cs_new_protected:Nn \__pic_set_line_color:
154 {
155     \__color_safe_use:V \l__pic_line_draw_color_tl
156 }
157 \cs_new_protected:Nn \__pic_set_fill_color:
158 {
159     \__color_safe_use:V \l__pic_region_fill_color_tl
160 }
161 \def\z@pic@vector@style{\ltxarrows}
162 \keys_define:nn { ztool / draw / picture }
163 {
164     vector .inherit:n = { ztool/draw/picture/line },
165 }
166 \keys_define:nn { ztool / draw / picture / vector }
167 {
168     > .choice:,
169     > / latex .code:n = {\def\z@pic@vector@style{\ltxarrows}},
170     > / pst .code:n = {\def\z@pic@vector@style{\pstarrows}},
171     > / unknown .code:n =
172     {
173         \msg_set:nnn { ztool }{unknown-arrow-style}
174         { Unknown~arrow~style,~use~'latex'~or~'pst'. }
175         \msg_error:nn { ztool }{unknown-arrow-style}
176     }
177 }
178 \tl_new:N \l__draw_line_type % 'horizontal', 'vertical', 'normal'
179 \cs_new_protected:Npn \ztool_pic_line_vector:nnnn #1#2#3#4
180 {% #1:line/vector; #2:key-value; #3:start coor; #4:end coor;
181     \group_begin:
182     \keys_set:nn { ztool / draw / picture / #1 }{#2}
183     \tl_set:Nn \l__draw_line_type { normal }
184     \fp_compare:nNnTF { \__coor_st:n {#4} - \__coor_st:n {#3} } > { \c__pic_precise_fp }
185     {
186         \fp_set:Nn \l__draw_vector_slope_fp
187         { (\__coor_nd:n {#4} - \__coor_nd:n {#3})
188         / (\__coor_st:n {#4} - \__coor_st:n {#3}) }
189         \fp_set:Nn \l__draw_vector_xysep_fp
190         { abs(\__coor_st:n {#4} - \__coor_st:n {#3}) }

```

```

191     }{
192         % NOTE: we do NOT set slope infinte, just set it to '0'
193         \fp_set:Nn \l__draw_vector_slope_fp {0}
194         \fp_set:Nn \l__draw_vector_xysep_fp
195             { abs(\__coord:n {#4} - \__coord:n {#3}) }
196         \tl_set:Nn \l__draw_line_type { vertical }
197     }
198     \fp_compare:nNnT { abs(\__coord:n {#4} - \__coord:n {#3}) } < {0.001}
199         { \tl_set:Nn \l__draw_line_type { horizontal } }
200     \z@pic@vector@style
201     \__pic_set_line_width:
202     \exp_last_unbraced:Ne \__@@_pic_put:nnn {\__coord_st_nd:n {#3}}
203     {
204         \__pic_set_line_color:
205         \str_case:VnF \l__draw_line_type
206             {
207                 {vertical}{
208                     \cs:w __@@_pic_#1:nnn\cs_end:
209                     { \l__draw_vector_slope_fp }
210                     { 1 }
211                     { \l__draw_vector_xysep_fp }
212                 }
213                 {horizontal}{
214                     \cs:w __@@_pic_#1:nnn\cs_end:
215                     { 1 }
216                     { \l__draw_vector_slope_fp }
217                     { \l__draw_vector_xysep_fp }
218                 }
219                 {normal}{
220                     \cs:w __@@_pic_#1:nnn\cs_end: {1}
221                     { \l__draw_vector_slope_fp }
222                     { \l__draw_vector_xysep_fp }
223                 }
224             }{ \relax }
225     }
226     \group_end:
227 }
228 \cs_generate_variant:Nn \ztool_pic_line_vector:nnnn {neee, nooo}
229 \NewDocumentCommand{\zline}{0{}d()d()}
230 {
231     \ztool_pic_line_vector:neee {line}{#1}{#2}{#3}
232 }
233 \NewDocumentCommand{\zvector}{0{}d()d()}
234 {
235     \ztool_pic_line_vector:neee {vector}{#1}{#2}{#3}
236 }

```

```

239 % --> \zdraw -- similar to \tikz command in tikz
240 % NOTE: these line/vector commands are identical to
241 % 1. \Line (x_1, y_1)(x_2, y_2), \Vector (x_1,y_1)(x_2,y_2)
242 % 2. \polyline(x_1, y_1) ... (x_n, y_n), \polyvector(x_1, y_1) ... (x_n, y_n)
243 % 3. \polygon (x_1, y_1) ... (x_n, y_n), when set 'cycle',
244 % \polygon*(x_1, y_1) ... (x_n, y_n), when set 'fill' (auto cycle).
245 % 4. Trim leading space after '\polygon' or '*' to avoid error !!
246 \cs_new:Npn \__@@_pic_Line:nnnn #1#2#3#4
247 { \Line (#1, #2)(#3, #4) }
248 \cs_new:Npn \__@@_pic_Vector:nnnn #1#2#3#4
249 { \Vector (#1, #2)(#3, #4) }
250 \cs_new:Npn \__@@_pic_polyline:n #1
251 {
252   \tl_set:Nc \l_tmpa_tl {\tl_trim_spaces:e {#1}}
253   \exp_last_unbraced:NV \polyline \l_tmpa_tl
254 }
255 \cs_new:Npn \__@@_pic_polyvector:n #1
256 {
257   \tl_set:Nc \l_tmpa_tl {\tl_trim_spaces:e {#1}}
258   \exp_last_unbraced:NV \polyvector \l_tmpa_tl
259 }
260 \cs_new:Npn \__@@_pic_polygon:nn #1#2
261 {
262   \tl_set:Nc \l_tmpa_tl {\tl_trim_spaces:e {#1}}
263   \tl_set:Nc \l_tmpb_tl {\tl_trim_spaces:e {#2}}
264   \tl_set:Nc \l_tmpa_tl { \l_tmpa_tl\l_tmpb_tl }
265   \exp_last_unbraced:NV \polygon \l_tmpa_tl
266 }
267 \cs_generate_variant:Nn \__@@_pic_polygon:nn { nV, ne }
268 \tl_new:N \l__pic_region_fill_color_tl
269 \bool_new:N \l__pic_region_fill_bool
270 \keys_define:nn { ztool / draw / picture / region }
271 {
272   fill .choices:nn = { true, false }{
273     \use:c { bool_set_ \l_keys_choice_tl :N }
274     \l__pic_region_fill_bool
275   },
276   fill .initial:n = { false },
277   fill .default:n = { true },
278   fill / unknown .code:n = {
279     \tl_if_empty:eF \l_keys_value_tl
280     { \bool_set_true:N \l__pic_region_fill_bool }
281     \tl_set:Nc \l__pic_region_fill_color_tl { \l_keys_value_tl }
282   },
283 }
284 \keys_define:nn { ztool / draw / picture }
285 {
286   zdraw .inherit:n = {

```



```

287         ztool/draw/picture/line,
288         ztool/draw/picture/vector,
289         ztool/draw/picture/region,
290     },
291 }
292 \keys_define:nn { ztool / draw / picture / zdraw }
293 {
294     vector .bool_set:N = \l__pic_draw_vector_bool,
295     vector .initial:n = { false },
296     cycle .bool_set:N = \l__pic_draw_cycle_bool,
297     cycle .initial:n = { false },
298     shift .tl_set:N = \l__pic_draw_shift_tl,
299     shift .initial:n = { 0, 0 },
300 }
301 \cs_new:Npn \__region_fill_color_miss:n #1
302 {
303     \bool_if:eT {
304         \l__pic_region_fill_bool &&
305         \tl_if_empty_p:N \l__pic_region_fill_color_tl
306     } { \tl_set:Nn \l__pic_region_fill_color_tl {#1} }
307 }
308 \cs_new_protected:Npn \ztool_pic_draw:nw #1#2;
309 {% #1:key-value; #2:coors list (use ';' to end scan just like tikz)
310     \group_begin:
311     \keys_set:nn { ztool / draw / picture / zdraw } {#1}
312     \__region_fill_color_miss:n { gray }
313     \edef\coors@first
314     {
315         \exp_last_unbraced:Ne
316         \__coors_list_first:w {\tl_trim_spaces:e {#2}}
317         \scan_stop:
318     }
319     \edef\draw@flag
320     {
321         \tl_map_function:nN {
322             \l__pic_draw_vector_bool
323             \l__pic_draw_cycle_bool
324             \l__pic_region_fill_bool
325         } \int_eval:n
326     }
327     \__@@_pic_put:nnn
328     { \__coor_st:V \coors@first + \__coor_st:V \l__pic_draw_shift_tl }
329     { \__coor_nd:V \coors@first + \__coor_nd:V \l__pic_draw_shift_tl }
330     {
331         \__pic_set_line_width:
332         \__pic_set_line_color:
333         \exp_after:wN \int_case:nnF \exp_after:wN {
334             \exp_after:wN \int_from_bin:n \exp_after:wN

```

```

335         { \draw@flag }
336     }{
337         {0}{ \__@@_pic_polyline:n {#2} }
338         {1}{ \__@@_pic_polygon:nn {*}{#2} }
339         {2}{ \__@@_pic_polygon:ne { }{#2} }
340         {3}{ \__@@_pic_polygon:nn {*}{#2} }
341         {4}{ \__@@_pic_polyvector:n {#2} }
342         {5}{
343             \__pic_set_fill_color:
344             \__@@_pic_polygon:nn {*}{#2}
345             \__pic_set_line_color:
346             \exp_args:Ne \__@@_pic_polyvector:n {#2(\coors@first)}
347         }
348         {6}{ \exp_args:Ne \__@@_pic_polyvector:n {#2(\coors@first)} }
349         {7}{
350             \__pic_set_fill_color:
351             \__@@_pic_polygon:nn {*}{#2}
352             \__pic_set_line_color:
353             \exp_args:Ne \__@@_pic_polyvector:n {#2(\coors@first)}
354         }
355     }{\relax}
356 }
357 \group_end:
358 }
359 \cs_new:Npn \__coors_list_first:w (#1)#2\scan_stop:
360 { #1 }
361 \NewDocumentCommand{\zdraw}{0{}}
362 { \ztool_pic_draw:nw {#1} }
363
364
365 % --> arc / circle
366 \cs_new:Npn \__@@_pic_arc:nnnn #1#2#3#4
367 { % #1:fill bool; #2:start angle; #3:end angle; #4:radius
368     \arc #1[\fp_eval:n {#2}, \fp_eval:n {#3}]
369     { \fp_eval:n {#4} }
370 }
371 \cs_new:Npn \__@@_pic_circel:nn #1#2
372 { % #1:fill bool; #2:radius
373     \__@@_pic_arc:nnnn {#1}{0}{360}{#2}
374 }
375
376
377 % --> arc rule: similar to arc by it's a rule
378 \cs_new:Npn \__@@_pic_arcrule:nnn #1#2#3
379 {
380     % To be implemented ...
381 }
382

```

```

383                                     383
384 % --> circle                                     384
385 \keys_define:nn { ztool / draw / picture }                                     385
386 {                                     386
387     arc .inherit:n = {                                     387
388         ztool/draw/picture/line,                                     388
389         ztool/draw/picture/region,                                     389
390     },                                     390
391 }                                     391
392 \keys_define:nn { ztool / draw / picture / arc }                                     392
393 {                                     393
394     radius .fp_set:N = \l__pic_arc_radius_fp,                                     394
395     radius .initial:n = .5,                                     395
396     start .fp_set:N = \l__pic_arc_start_fp,                                     396
397     start .initial:n = 0,                                     397
398     end .fp_set:N = \l__pic_arc_end_fp,                                     398
399     end .initial:n = 90,                                     399
400 }                                     400
401 \prg_generate_conditional_variant:Nnn                                     401
402     \bool_if:n { e } { p, T, F, TF }                                     402
403 \cs_new_protected:Npn \ztool_pic_arc:nn #1#2                                     403
404     {% #1:key-value; #2:coord                                     404
405     \group_begin:                                     405
406     \keys_set:nn { ztool / draw / picture / arc }{#1}                                     406
407     \__region_fill_color_miss:n { gray }                                     407
408     \__color_if_valid:VF \l__pic_region_fill_color_tl                                     408
409     { \bool_set_false:N \l__pic_region_fill_bool }                                     409
410     \exp_last_unbraced:Ne \__@@_pic_put:nnn                                     410
411     { \__coord_st_nd:n {#2} }                                     411
412     {                                     412
413         \__pic_set_line_width:                                     413
414         \bool_if:eT \l__pic_region_fill_bool                                     414
415         {                                     415
416             \__pic_set_fill_color:                                     416
417             \exp_args:Ne \__@@_pic_arc:nnnn {*}                                     417
418             { \fp_use:N \l__pic_arc_start_fp }                                     418
419             { \fp_use:N \l__pic_arc_end_fp }                                     419
420             { \fp_use:N \l__pic_arc_radius_fp }                                     420
421         }                                     421
422         % NOTE: border must draw after the fill                                     422
423         \__pic_set_line_color:                                     423
424         \exp_args:Ne \__@@_pic_arc:nnnn {}                                     424
425         { \fp_use:N \l__pic_arc_start_fp }                                     425
426         { \fp_use:N \l__pic_arc_end_fp }                                     426
427         { \fp_use:N \l__pic_arc_radius_fp }                                     427
428     }                                     428
429     \group_end:                                     429
430 }                                     430

```

```

431 \NewDocumentCommand{\zarc}{0{}d{}}
432   {% #1:key-value; #2:coord
433     \ztool_pic_arc:nn {#1}{#2}
434   }
435 \NewDocumentCommand{\zcircle}{0{}d{}}
436   {
437     \ztool_pic_arc:nn {start=0, end=360, #1}{#2}
438   }
439
440
441 % --> oval / rectangle
442 % \oval[arc](full-x-width, full-y-width)[part]
443 % part: (l, r) x (t, b)
444 \cs_new:Npn \__@@_pic_oval:nnnn #1#2#3#4
445   {% #1:arc; #2:part; #3:x-width; #4:y-width;
446     \oval
447       [\fp_eval:n {#1}]
448       (\fp_eval:n {#3}, \fp_eval:n {#4})
449       [ #2 ]
450   }
451 \keys_define:nn { ztool / draw / picture }
452   {
453     rectangle .inherit:n = {
454       ztool/draw/picture/line,
455       ztool/draw/picture/region,
456     },
457   }
458 \keys_define:nn { ztool / draw / picture / rectangle }
459   {
460     arc .fp_set:N = \l__pic_rec_arc_fp,
461     arc .initial:n = 0,
462   }
463
464 \int_new:N \l__pic_rec_quadrant_index_int
465 \clist_const:Nn \l__pic_rec_quadrant_shift_prop
466   { {(0, 0)}, {(-\rec@arc, 0)}, {(-\rec@arc, -\rec@arc)}, {(0, -\rec@arc)} }
467 \cs_new_protected:Npn \ztool_pic_rectangle:nnn #1#2#3
468   {% #1:key-value; #2:start coord; #3:end coord;
469     \group_begin:
470     \keys_set:nn { ztool / draw / picture / rectangle }{ fill=false }
471     \keys_set:nn { ztool / draw / picture / rectangle }{ #1 }
472     \edef\rec@width { \fp_eval:n {\__coord_st:n {#3} - \__coord_st:n {#2}} }
473     \edef\rec@height{ \fp_eval:n {\__coord_nd:n {#3} - \__coord_nd:n {#2}} }
474     % check if arc radius too large.
475     \edef\rec@arc
476       {
477         \fp_eval:n { min(min(\rec@height/2, \rec@width/2), \l__pic_rec_arc_fp) }
478       }

```

```

479 \__region_fill_color_miss:n { gray }
480 \__color_if_valid:VF \l__pic_region_fill_color_tl
481 {
482     \bool_set_false:N \l__pic_region_fill_bool
483     \prg_map_break:Nn \l__ztool_pic_rec_fill {}
484 }
485 %% begin fill rounded rectangle
486 \__pic_rectangle_rec_comp_fill:n { #2 }
487 \__pic_rectangle_arc_comp_fill:n { #2 }
488 %% end fill rounded rectangle
489 \prg_break_point:Nn \l__ztool_pic_rec_fill { }
490 \__@@_pic_put:nnn
491 { \__coord_st:n {#2}+\rec@width/2 }
492 { \__coord_nd:n {#2}+\rec@height/2 }
493 {
494     \__pic_set_line_color:
495     \__pic_set_line_width:
496     \__@@_pic_oval:nnnn
497     { \rec@arc }{ }
498     { \rec@width }
499     { \rec@height }
500 }
501 \group_end:
502 }
503
504 % aux function for rectangle area filling in rectangle
505 \cs_new:Npn \__pic_rectangle_rec_comp_fill:n #1
506 {
507     \__@@_pic_put:nnn {\__coord_st:n {#1}+\rec@arc}{\__coord_nd:n {#1}}
508     {
509         \__pic_set_fill_color:
510         \rule
511             {\fp_eval:n {(\rec@width-2*\rec@arc)*\dim_to_decimal:n {\l__pic_unit_dim}}pt}
512             {\fp_eval:n {\rec@height*\dim_to_decimal:n {\l__pic_unit_dim}}pt}
513     }
514     \fp_compare:nNnT { \rec@height-2*\rec@arc } > { \c__pic_precise_fp }
515     {
516         \__@@_pic_put:nnn {\__coord_st:n {#1}}{\__coord_nd:n {#1}+\rec@arc}
517         {
518             \__pic_set_fill_color:
519             \rule
520                 {\fp_eval:n {\rec@arc*\dim_to_decimal:n {\l__pic_unit_dim}}pt}
521                 {\fp_eval:n {(\rec@height-2*\rec@arc)*\dim_to_decimal:n {\l__pic_unit_dim}}pt}
522         }
523         \__@@_pic_put:nnn {\__coord_st:n {#1}+\rec@width-\rec@arc}{\__coord_nd:n {#1}+\rec@arc}
524         {
525             \__pic_set_fill_color:
526             \rule

```

```

527         {\fp_eval:n {\rec@arc*\dim_to_decimal:n {\l__pic_unit_dim}}pt} 527
528         {\fp_eval:n {(\rec@height-2*\rec@arc)*\dim_to_decimal:n {\l__pic_unit_dim}}pt} 528
529     } 529
530 } 530
531 } 531
532 % aux function for arc area filling in rectangle 532
533 \cs_new:Npn \__pic_rectangle_arc_comp_fill:n #1 533
534 { 534
535     \int_set:Nn \l__pic_rec_quadrant_index_int { 0 } 535
536     \tl_map_inline:nn 536
537     { 537
538         {\__coord_st:n {#1}+\rec@width-\rec@arc, \__coord_nd:n {#1}+\rec@height-\rec@arc} 538
539         {\__coord_st:n {#1}+\rec@arc, \__coord_nd:n {#1}+\rec@height-\rec@arc} 539
540         {\__coord_st:n {#1}+\rec@arc, \__coord_nd:n {#1}+\rec@arc} 540
541         {\__coord_st:n {#1}+\rec@width-\rec@arc, \__coord_nd:n {#1}+\rec@arc} 541
542     }{ 542
543     \int_incr:N \l__pic_rec_quadrant_index_int 543
544     \edef\qu@drant@index{\int_use:N \l__pic_rec_quadrant_index_int} 544
545     \exp_last_unbraced:Ne \__@@_pic_put:nnn 545
546     { \__coord_st_nd:n {##1} } 546
547     { 547
548         \__pic_set_fill_color: 548
549         \__@@_pic_arc:nnnn {*} 549
550         { (\qu@drant@index-1)*90 } 550
551         { \qu@drant@index*90 } 551
552         { \rec@arc } 552
553     } 553
554 } 554
555 } 555
556 556
557 \NewDocumentCommand{\zrectangle}{0{}d()d()} 557
558 { 558
559     \ztool_pic_rectangle:nnn { #1 }{#2}{#3} 559
560 } 560
561 561
562 562
563 % ==> absolute page coordinate (left, bottom) = (0, 0) 563
564 \NewDocumentCommand{\zpin}{0{background}m} 564
565 { 565
566     \hook_gput_next_code:nn {shipout/#1} 566
567     { 567
568         \put(0pt, -\paperheight) 568
569         { \makebox(0, 0)[bl]{#2} } 569
570     } 570
571 } 571

```

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zTiZ 接口文档

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由于本人时间有限, 目前此宏包的开发暂停.

May 8, 2026

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1 基本介绍

直到今天, 其实已经有很多基于 `tikz` 开发的绘图宏包了, 它们有着不同的用途, 在不同的领域中你都能看到 `Ti $\kern 0.05em k$ Z` 的痕迹. 部分宏包已经提供了和 `ztikz` 功能差不多接口, 这系列的宏包包括:

- `Ti $\kern 0.05em k$ Z` 的常见命令封装: `tzplot`;
- 用于 3D 绘图的 `Ti $\kern 0.05em k$ Z` 宏包: `tikz-3dplot`;
- 新出的通用绘图宏包 `luadraw`, 原生支持 3D 绘图, 目前仅支持 `Lua $\TeX$` ;
- 基于 `PSTricks` 的 (特殊) 函数绘制宏包: `pst-func`;
- 用于缓存编译结果的宏包: `robust-externalize`;
- ...

如果你觉得 `ztikz` 宏包并不符合你的需求, 不妨试试上面的几个宏包, 或者是直接使用原始的 `tikz` 宏包提供的命令与库进行绘图. 在网络上也有着丰富的 `Ti $\kern 0.05em k$ Z` 资源; 比如 `Ti $\kern 0.05em k$ Z` 绘图的网站 – `Ti $\kern 0.05em k$ Z` `Example`, 这个网站中有着丰富的绘制样例并且提供了对应的绘图代码.

但是上述的系列宏包提供的接口并不是那么的统一, 自己用着不习惯, 所以我才决定开发 `ztikz` 宏包. `zTi $\kern 0.05em k$ Z` 的命令格式基本遵守了类似 `Mathematica` 中函数的命名规范.

1.1 项目地址

本宏包在 Github 上的地址如下:

https://github.com/zongpingding/zTeX_bundle

该仓库中包含本宏集的源码, 用户手册以及一些测试用例; 当前宏集的稳定版本于 2025 年 09 月发布, 最新版本请切换到 “dev” 分支; 本手册适用于当前最新的 “dev” 分支.

1.2 功能概述

`zTi $\kern 0.05em k$ Z` 宏包主要用于绘图与计算, 支持调用外部程序, 比如 `Python`, `Mathematica`, `gnuplot`; 同时也提供了调用缓存机制; 虽然 `zTi $\kern 0.05em k$ Z` 提供了这些软件的调用接口, 但这并不意味着你需要安装以上的所有软件; `zTi $\kern 0.05em k$ Z` 中每一个软件的调用接口都是互相独立的, 用户仅需在操作系统上安装自己需要功能对应的软件即可. `zTi $\kern 0.05em k$ Z` 的功能概述如下:

- **绘图:** 二维绘图, 三维绘图;
- **计算:** 浮点数计算, 符号计算.

绘图部分基于: $\text{Ti}\text{k}\text{Z}$ 的 2d 绘图部分,¹ Python 的 `matplotlib` 库, `WolframScript` 的绘图功能; 计算部分基于: $\text{L}\text{A}\text{T}\text{E}\text{X}$ 3 的 `xfp` 模块, Python 的 `numpy`, `sympy` 和 `scipy` 库, 以及 `WolframScript` 的计算功能.

虽然这个宏集名字中仅有 “ $\text{Ti}\text{k}\text{Z}$ ” 字样, 但是 $\text{zTi}\text{k}\text{Z}$ 能够完成 (或想要完成) 的功能是不止于此的.

1.3 坐标对齐

$\text{zTi}\text{k}\text{Z}$ 提供的所有绘图命令可以和 $\text{Ti}\text{k}\text{Z}$ 中的命令配合使用, 即 – 它们可以在同一个 `tikzpicture` 环境中使用. $\text{zTi}\text{k}\text{Z}$ 对函数绘制时的坐标进行了“对齐”: $\text{zTi}\text{k}\text{Z}$ 命令中的坐标, 和 $\text{Ti}\text{k}\text{Z}$ 命令中的坐标, 亦或者是 `Geogebra` 中的坐标是一致的.

为何要在 $\text{zTi}\text{k}\text{Z}$ 中把坐标“对齐”? 试想这么一个情景: 你在 `Geogebra` 中找到了两个函数图像的交点为 $P(1, 2)$, 首先使用 $\text{Ti}\text{k}\text{Z}$ 自带的 `\filldraw` 命令把 P 点绘制出来了; 然后使用 $\text{zTi}\text{k}\text{Z}$ 中的 `\ShowPoint` 命令再次绘制这个 P 点. 然而结果就是: 这两个 P 点没有重合, 尽管我们指定的坐标都是 $(1, 2)$.

所以当你不方便使用 $\text{zTi}\text{k}\text{Z}$ 求解某些特殊的点时, 你可以先在诸如 `Geogebra` 这样的软件中把对应的 P 点求解出来, 然后直接在 $\text{zTi}\text{k}\text{Z}$ 中使用 `\ShowPoint` 命令绘制此点.

1.4 缓存机制

$\text{zTi}\text{k}\text{Z}$ 除了提供和外部程序交互的接口外, 还内置了一套 `cache` 系统, $\text{zTi}\text{k}\text{Z}$ 会自动把 TEX 和外部程序交互产生的结果缓存下来, 并且记录下 $\text{L}\text{A}\text{T}\text{E}\text{X}$ 文档中调用部分源代码的 `Hash` 值.

如果 $\text{L}\text{A}\text{T}\text{E}\text{X}$ 文档中的源代码对应的 `Hash` 值发生了改变, 那么 $\text{zTi}\text{k}\text{Z}$ 就会重新和外部程序交互, 重新产生结果, 然后缓存新的 `Hash` 值. 如果文档中的源代码的 `Hash` 值没有改变, 那么 $\text{zTi}\text{k}\text{Z}$ 就会直接调用上一次的缓存结果. `cache` 系统的优势: 我们不必反复的编译没有变化的内容, 直接引用之前的缓存, 减少文档的编译时间. 在实际测试中, 结果缓存后, 再次编译源文档的时间和直接插入对应数量的图片的时间几乎一致.

$\text{zTi}\text{k}\text{Z}$ 中的 `basic`, `python`, `wolfram`, `gnuplot` 库均已实现缓存机制. `tikzpicture` 环境或者是 `\tikz` 命令生成图片的 `cache` 机制是依靠 $\text{Ti}\text{k}\text{Z}$ 的 `external` 库实现的; (它的实现是出了名的复杂, 用户如果感兴趣, 也可以去看看.)

因为 $\text{zTi}\text{k}\text{Z}$ 还没有进行完整的测试, 所以可能存在没有发现的 `bug`; 例如, 用户可能会遇到类似下面的问题:

- 过时的缓存 `Hash` 值: 如果一个环境最开始的 `Hash` 值为 “A”, 在你修改了这个环境的内容后, 使得此环境中代码的 `Hash` 值变为 “B”. 但是如果你现在再次修改会 `Hash` 值为 “A” 时对应的源代码, 此刻的 `Hash` 值已经缓存在了文件 `ztikz.hash` 中, 所

¹由于 3d 绘图涉及的几个变换矩阵接口我还没想好怎么在 $\text{zTi}\text{k}\text{Z}$ 中声明, 所以目前 $\text{zTi}\text{k}\text{Z}$ 不提供 3d 绘图功能

以再次编译时此环境对应的绘制结果并不会改变. 调用的缓存结果仍然是 Hash 值为 “B” 对应的那个缓存结果.

- 和 `indextool` 宏包冲突: 有可能你在启用缓存库后, 发现编译报错 `missing \begin{document}....`. 这个问题和宏包 `indextool` 的索引功能有关. 可以先注释 `\makeindex`, `\printindex` 命令, 随后在图片缓存结束后, 取消注释, 最后再生成索引.

1.5 局限

`tikz` 未来也许会提供 3d 绘图相关的接口, 但是如果你的图像需要复杂的计算或布局, 那么还请使用其余的宏包或使用对应的专业绘图软件. `asymptote` 宏包就是一个比较好的选择.

2 安装使用

2.1 兼容情况

目前 `ztikz` 宏包兼容 Windows/Linux/MacOS 三个平台. 各个平台中不同 $\text{T}_\text{E}\text{X}$ Live 版本的兼容性如下:

Windows : $\text{T}_\text{E}\text{X}$ Live 最低版本 2023

Linux : $\text{T}_\text{E}\text{X}$ Live 最低版本 2022

MacOS : $\text{MacT}_\text{E}\text{X}$ 最低版本 2024

`zTiZ` 在 Windows 下的表现可能没有在 Linux/MacOS 下的那么好, 建议用户在 Linux/MacOS 下使用本宏包.

2.2 环境配置

如果用户需要使用 `zTiZ` 提供的调用外部程序的库, 用户不仅需要配置文档的导言区, 还需在系统中安装对应的应用程序; 应用程序安装后需要将其添加到环境变量, 使得该应用可以在命令行被调用. 最后在编译文档时加上 `--shell-esape` 参数, 就像下面这样:

```
pdflatex --shell-escape main.tex
```

在 Windows 下推荐用户使用 `scoop` 这一包管理器安装一系列的软件, 这样可以免去配置环境变量这一烦恼. 以下是不同程序在配置过程中需要注意的事项:

2.2.1 gnuplot

在 Windows 下, 用户使用 GUI 界面安装 gnuplot 时请一定勾选 “Add gnuplot to PATH” 这一选项.

2.2.2 Python

若用户需要使用 `python` 库提供的功能, 用户需要同时安装 Python 以及 `matplotlib`, `sympy` 与 `scipy` 库; 前者用于绘图, 后者用于计算.

在 Windows 平台, 由于 $\text{T}_\text{E}\text{X}$ Live 的编译配置, 需确保系统环境变量 `PATHEXT` 中已经删除 “.PY” 后缀.

2.2.3 WSTP

如果用户无法安装 Mathematica, `zTiZ` 宏包还提供了另一种选择 – WSTP(Wolfram Symbolic Transfer Protocol), 此时用户仅需安装免费的 Wolframscript(Wolfram Engine), 环境配置方法请参见我的仓库: [WSTP Example](#).

此方法的优势十分明显: 免费, 速度很快 (大约是直接调用 wolframscript 的十倍). 用户即使不加载 cache 库, wolfram 库中所有命令的执行速度也与直接调用缓存相当.

目前此方法的缺陷也比较明显: 需要用户自行编译 wolfram 引擎的客户端, 需要用户提前启动服务程序, 该客户端仅在 Linux 下进行了测试 (其它平台的支持计划, 欢迎 PR).

2.2.4 Wolfram

若用户需要使用 wolfram 库对应的功能, 那么用户需要安装 WolframScript 或 Mathematica 软件. 执行命令时可以选择在云端执行, 这样就避免调用本地 Mathematica 计算内核. 用户需首先在命令行完成 wolfram 账号绑定, 绑定方法如下 (当用户第一次在命令行调用 Wolfram Cloud 上执行时):

```
> wolframscript -cloud -code 2+2
Wolfram ID: <Account>
Password: <Password>
```

上述命令会提示用户输入 Wolfram ID 和密码, 输入对应的 <Account> 和 <Password> 后即可使用. 但云端执行速度可能比较慢, 用户需自行决定是否采用此方案. 下面介绍在本地安装 wolfram 引擎的方法:

在 Linux 下, 除 wolfram 以外的软件都是很好安装的, 直接使用 Linux 发行版自带的包管理器即可. 这里我提供一个在 WSL 中使用 Windows 下 Mathematica 的方法 (用户也可以不按照此方法配置 WolframScript): 其实就是创建一个从 Linux 到 Windows 的软连接, 命令中 WolframScript 在 Windows 下的路径请根据自己的实际情况更改, 命令如下:

```
sudo ln -sf "/mnt/c/Program Files/Wolfram
Research/WolframScript/wolframscript.exe" /usr/bin/wolframscript
```

请务必确保 WolframScript 在命令行中能被正常调用. 可以使用如下代码测试 WolframScript 是否成功配置:

```
plotFunction[fun_, xlimits_, ylimits_] := ContourPlot[
  fun, xlimits, ylimits,
  ContourStyle->{
    RGBColor["#00C0A3"],
    Thickness[0.004]
  },
  AspectRatio->((xlimits[[2]]//Abs) +
(xlimits[[3]]//Abs))/((ylimits[[2]]//Abs) + (ylimits[[3]]//Abs)),
  AxesOrigin->{0,0},
  Axes->True,
```

```

Frame->False,
AxesStyle->Arrowheads[{0, 0.03}],
AxesLabel->{"x", "y"},
PlotRange -> Full
]

xlimits = {x, -3, 6};
ylimits = {y, -4, 5};
fp1 = plotFunction[y==Sin[x], xlimits, ylimits];
fp2 = plotFunction[x^2/4 + y^2/3 == 5, {x, -5, 5}, {y, -5, 5}];

figure = Show[fp2, fp1];
(* 保存的图片格式为: '*.wls.pdf' *)
Export["works_well.pdf", figure];

```

把上述的源码保存为 `test.wls`, 然后在命令行运行如下命令:

```
wolframscript -script test.wls
```

如果配置成功, 那么在当前工作目录下会产生一个名为 `works_well.pdf` 的 PDF 文件; 反之, 则说明你的 WolframScript 没有配置成功, 也就不能够使用本库.

2.2.5 Mathics

用户除了选择 WolframScript 作为计算引擎外, 还可以选择 Mathics 作为计算引擎. Mathics 是什么? An open-source Mathematica Kernel. MathsciScript 为 Mathics 的一个前端, 具有自动命令/变量补全, 语法高亮等功能.

在本地安装 Mathics 的方法请参见: [Installing Mathics3](#). 若用户在 Windows 下已经安装好 Mathics, 不想要在 WSL 中重新安装一次, 那么在 WSL 下创建软连接的方法和上述 WolframScript 的配置方法同理. 如果用户通过命令 “pip install Mathics-omnibus” 安装了 Mathics, 那么创建软连接的命令如下:

```

sudo ln -sf
"/mnt/c/Users/<name>/AppData/Local/Programs/Python/Python312/Scripts/mathics.exe"
/usr/bin/mathics

```

上述命令中的 `<name>` 需要替换为你自己的用户名, 同时也需要注意 Python 的版本号.

NOTE: 部分 Mathematica 中的函数 Mathics 也许并没有支持, 还请参考 Mathics 文档.

3 宏包选项

 ztikz/library

 New: 2025-05-18

quiet = $\langle \text{true}|\text{false} \rangle$ 初始值: false
 library = $\langle \text{basic}|\text{gnuplot}|\text{cache}|\text{python}|\text{wolfram}|\text{l3draw} \rangle$ 初始值: 空
 选项 $\langle \text{library} \rangle$ 与命令 `\ztikzloadlib` 等价, 用于指定 $\text{\textit{zTikZ}}$ 加载的库名列
 表, 在加载 ztikz 宏包时使用; $\langle \text{quiet} \rangle$ 选项控制是否在终端显示调试信息, 设置
 $\langle \text{quiet} \rangle = \text{true}$ 或单独加载 $\langle \text{quiet} \rangle$ 选项以禁用调试信息. 一个简单的配置样例
 如下:

```
\usepackage[quiet, library={basic, gnuplot}]{ztikz}
```

例 1

 ztikz/wolfram/engine

 ztikz/wolfram/cloud

 New: 2026-04-29

engine = $\langle \text{wolfram}|\text{wstp}|\text{mathics} \rangle$ 初始值: wolfram
 cloud = $\langle \text{true}|\text{false} \rangle$ 初始值: false
 $\langle \text{engine} \rangle$ 用于指定 Wolfram 代码的计算引擎, 目前支持 Wolfram 和 Mathics 两
 种引擎, 前者为商业闭源软件, 后者为开源软件; 若用户对速度要求较高, 可以自
 行配置 $\langle \text{wstp} \rangle$ 环境. $\langle \text{cloud} \rangle$ 用于指定是否使用 Wolfram Cloud 进行计算, 目前
 仅对 $\langle \text{wolfram} \rangle$ 选项有效; **注意:** Mathics 目前不支持云计算. 一个简单的配置
 样例如下:

```
\usepackage[
  library = { wolfram },
  wolfram = { engine=wolfram, cloud=true }
]{ztikz}
```

例 2

5 $\text{\texttt{tikz}}$ 库

$\text{\texttt{tikz}}$ 提供了多种功能的库, 这些库可以通过 `\ztikzloadlib` 命令加载. 用户需要使用 `\ztexloadlib{<library name>}` 加载对应的库, $\text{\texttt{tikz}}$ 中可用的 `<library name>` 列表如下:

- basic
- cache
- gnuplot
- python
- wolfram
- l3draw

上述的所有库均不自动加载, 需用户手动加载. `basic` 库中仅包含了用于绘制点, 直线, 坐标轴和基本多边形等系列命令. 在导言区使用如下命令加载 `ztikz` 的库方法如下, 比如加载 `cache` 库和 `gnuplot` 库:

```
\ztikzloadlib{cache, gnuplot}
```

例 4

注意: 只有当用户加载对应的库后, 该库的脚本文件才会被写入项目文件夹下.

5.1 basic 库

basic 库主要包含一些和坐标系相关的部分命令: 包括点, 线, 面和规则多边形的绘制以及交点的求解与绘制. 其中的所有的绘制命令均继承自 TikZ 中内建的命令, 比如后续的 \BarPlot 命令其实就是如下内建命令的封装:

```
\draw[⟨key-value⟩] plot [ycomb, ⟨other style⟩] file {⟨data⟩};
```

NOTE: 为后续行文方便, 我们约定 $\langle\text{draw-keyval}\rangle$ 表示 $\backslash\text{draw}[\langle\text{keyval}\rangle]$ 中的 $\langle\text{keyval}\rangle$ 选项. 使用 $\langle\text{node-keyval}\rangle$ 表示 $\backslash\text{node}[\langle\text{keyval}\rangle]$ 中的 $\langle\text{keyval}\rangle$ 选项. 具体来说: 针对 $\langle\text{draw}\rangle$ 命令, 其可用的选项有 $\langle\text{line width}\rangle$, $\langle\text{color}\rangle$ 等, 详细信息请参见 TikZ 的用户手册.

\ShowPoint

New: 2025-05-15

\ShowPoint[⟨key-value⟩]{⟨point-1⟩; ...; ⟨point-n⟩}
[⟨label-1⟩; ...; ⟨label-n⟩][⟨node-keyval⟩]

此命令用于绘制点, $\langle\text{point-1}\rangle$ 到 $\langle\text{point-n}\rangle$ 为点的坐标, 使用 “;” 进行分割, 坐标的格式为 (x,y) . $\langle\text{key-value}\rangle$ 用于设置点的样式; $\langle\text{label}\rangle$ 的数量和 $\langle\text{point}\rangle$ 的数量不必一致, $\langle\text{label}\rangle$ 从第一个开始一次应用于每一个点.

ztikz/point/type	type	=	⟨字符串⟩.....	初始值:	无
ztikz/point/radius	radius	=	⟨长度⟩.....	初始值:	1pt
ztikz/point/color	color	=	⟨颜色⟩.....	初始值:	black
ztikz/point/opacity	opacity	=	⟨浮点数⟩.....	初始值:	1
ztikz/point/rotate	rotate	=	⟨角度⟩.....	初始值:	0

$\langle\text{type}\rangle$ 用于设置 maker 的样式, $\langle\text{radius}\rangle$ 用于设置 maker 的半径, $\langle\text{color}\rangle$ 用于设置 maker 的颜色, $\langle\text{opacity}\rangle$ 用于设置 maker 的透明度, $\langle\text{rotate}\rangle$ 用于设置 maker 的旋转角度.

```
\begin{tikzpicture}
\draw[gray] (-2, -1) grid (2, 1);
\ShowPoint
  [color=teal, radius=2pt, type=pentagon*, opacity=.8,
rotate=60]
  {(-1.5, 0); (2, .5)}[$0=(0, 0)$; $\pi, 0$]
  [above right=3pt and 0em, font=\small]
\end{tikzpicture}
```

例 5

ztikz/axis/tickStart	tickStart	= <浮点数>.....初始值: -5
ztikz/axis/tickEnd	tickEnd	= <浮点数>.....初始值: 5
ztikz/axis/axisRotate	axisRotate	= <浮点数>.....初始值: 0
ztikz/axis/mainStep	mainStep	= <浮点数>.....初始值: 1
ztikz/axis/subStep	subStep	= <浮点数>.....初始值: 0.1
ztikz/axis/tickLabelShift	tickLabelShift	= <长度>.....初始值: 0pt
ztikz/axis/mainTickLength	mainTickLength	= <长度>.....初始值: 4pt
ztikz/axis/subTickLength	subTickLength	= <长度>.....初始值: 2pt
ztikz/axis/axisColor	axisColor	= <颜色>.....初始值: black
ztikz/axis/mainTickColor	mainTickColor	= <颜色>.....初始值: black
ztikz/axis/subTickColor	subTickColor	= <颜色>.....初始值: black
ztikz/axis/tickStyle	tickStyle	= <below above cross>.....初始值: 无
ztikz/axis/mainTickLabel	mainTickLabel	= <字符串>.....初始值: \CurrentFp
ztikz/axis/mainTickLabelColor	mainTickLabelColor	= <颜色>.....初始值: black
ztikz/axis/mainTickLabelPosition	mainTickLabelPosition	= <below above cross>.....初始值: below

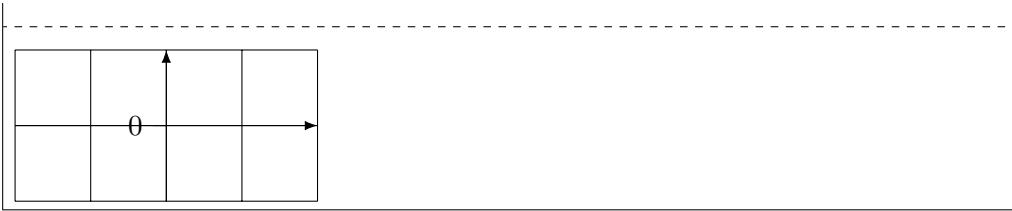
<mainTickLabel> 主要用于自定义坐标标签的样式, \CurrentFp 表示当前刻度处的浮点数值. <tickStyle> 会受到 tikzpicture 环境可选参数中的 <rotate> 选项的影响.

注意: 在使用 \ShowAxis 时若没有指定键 <tickStyle> 的值, 那么此时并不会绘制任何的刻度.

\CurrentFp	此命令表示当前刻度处的浮点数值, 其值在不同刻度处会自动更新.
New: 2025-05-31	
\xAxis	\xAxis[<start>][<end>]
New: 2025-05-15	此命令来自 \ShowAxis, 用于绘制 x 轴; <start> 和 <end> 均为浮点数, 分别表示坐标轴的起始点和结束点.
\yAxis	\yAxis[<start>][<end>]
New: 2025-05-15	此命令来自 \ShowAxis, 用于绘制 y 轴; <start> 和 <end> 均为浮点数, 分别表示坐标轴的起始点和结束点.

```
\begin{tikzpicture}[>=Latex]
\yAxis[-1][1]
\ShowAxis{(-2, 0); (2, 0)}
\draw (-2, -1) grid (2, 1);
\end{tikzpicture}
```

例 7



<code>\ShowGrid</code>	<code>\ShowGrid[⟨draw-keyval⟩]{⟨start⟩; ⟨end⟩}</code>
New: 2025-05-15	此命令用于绘制网格线, <code>⟨start⟩</code> 和 <code>⟨end⟩</code> 分别表示网格线的左下角和和右上角的坐标, 使用 “;” 进行分割, 坐标的格式为 (x,y) . <code>⟨key-value⟩</code> 为可选参数, 用于设置网格线的样式;

<code>\Polygon</code>	<code>\Polygon[⟨key-value⟩]{⟨number⟩}</code>
New: 2025-05-15	此命令用于绘制正多边形, <code>⟨number⟩</code> 表示多边形的边数, 其值必须为大于等于 3 的整数. <code>⟨key-value⟩</code> 为可选参数, 用于设置多边形的样式;

<code>ztikz/polygon/radius</code>	<code>radius</code>	=	⟨浮点数⟩		初始值:	1
<code>ztikz/polygon/edgeColor</code>	<code>edgeColor</code>	=	⟨颜色⟩		初始值:	black
<code>ztikz/polygon/fillColor</code>	<code>fillColor</code>	=	⟨颜色⟩		初始值:	无
<code>ztikz/polygon/fillOpacity</code>	<code>fillOpacity</code>	=	⟨浮点数⟩		初始值:	0
<code>ztikz/polygon/rotate</code>	<code>rotate</code>	=	⟨浮点数⟩		初始值:	0
<code>ztikz/polygon/shift</code>	<code>shift</code>	=	⟨坐标⟩		初始值:	(0, 0)
<code>ztikz/polygon/marker</code>	<code>marker</code>	=	⟨key-value⟩		初始值:	无
⟨radius⟩ 表示此正多边形外接圆的半径, 而非 ⟨marker⟩ 的半径; ⟨shift⟩ 外围的 “()” 不能省略. ⟨marker⟩ 对应 <code>ztikz/point</code> . ⟨marker⟩ 的设置请参见 图 (3).						

```
\begin{tikzpicture}
\ShowGrid[gray, thin]{(-2, -1); (2, 1)}
\Polygon[
  edgeColor=blue, shift={(1, 0)},
  marker={type=ball, color=green}
]{3}
\end{tikzpicture}
```

例 8

`\StairsPlot`

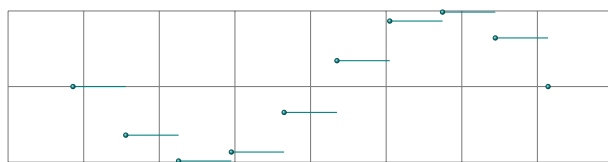
New: 2025-05-15

`\StairsPlot[⟨plot option⟩; ⟨jump option⟩][⟨draw-keyval⟩]
[⟨key-value⟩]{⟨file⟩}`

此命令用于绘制阶梯图, 绘图数据由 $\langle file \rangle$ 指定; $\langle plot\ option \rangle$ 用于设置阶梯图的绘制样式, 可选值有: `plot left`, `plot right`, `plot mid`; $\langle jump\ option \rangle$ 用于设置阶梯图的跳跃样式, 可选值有: `jump left`, `jump right`, `jump mid`; $\langle key-value \rangle$ 对应 $\langle ztikz/point \rangle$;

```
\begin{tikzpicture}
\ShowGrid[step=1, color=gray]{(-4, -1); (4, 1)}
\StairsPlot[;jump-left][teal][type=ball,
color=teal]{./sine.data}
\end{tikzpicture}
```

例 9



`\StemPlot`

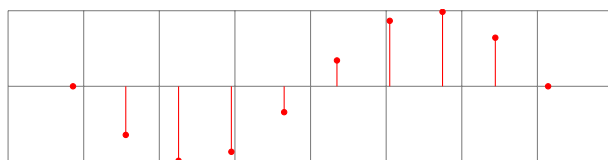
New: 2025-05-15

`\StemPlot[⟨direction⟩][⟨draw-keyval⟩]
[⟨key-value⟩]{⟨file⟩}`

此命令用于绘制火柴棍图, 绘图数据由 $\langle file \rangle$ 指定; $\langle direction \rangle$ 用于指定系列线段的方向, 可选值有: `x`, `y`, `o`, 分别表示垂直 x 轴, 垂直 y 轴, 以及指向坐标原点; $\langle key-value \rangle$ 对应 $\langle ztikz/point \rangle$.

```
\begin{tikzpicture}
\ShowGrid[step=1, color=gray]{(-4, -1); (4, 1)}
\StemPlot[x][red][type=*, color=red]{./sine.data}
\end{tikzpicture}
```

例 10



`\BarPlot`

New: 2025-05-15

`\BarPlot[⟨position⟩][⟨draw-keyval⟩]
[⟨key-value⟩]{⟨file⟩}`

此命令用于绘制条形图, 绘图数据由 $\langle file \rangle$ 指定; $\langle position \rangle$ 用于指定每个小矩形的位置以及宽度, 可选值有: `x`, `y`, `xc`, `yc`; $\langle key-value \rangle$ 对应 $\langle ztikz/point \rangle$.

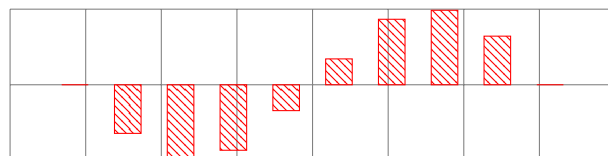

```
\begin{tikzpicture}
```

例 11

```
\ShowGrid[step=1, color=gray]{(-4, -1); (4, 1)}
```

```
\BarPlot[x][red, pattern=north west lines, pattern  
color=red]{./sine.data}
```

```
\end{tikzpicture}
```



5.2 gnuplot 库

需要说明的是: $\text{\textit{Ti}kZ}$ 宏包内部已经提供了直接调用 gnuplot 程序的命令 (需启用 `-shell-escape` 参数), 其调用格式如下:

```
\draw[⟨key-value⟩] plot[⟨id⟩] function{⟨function⟩};
```

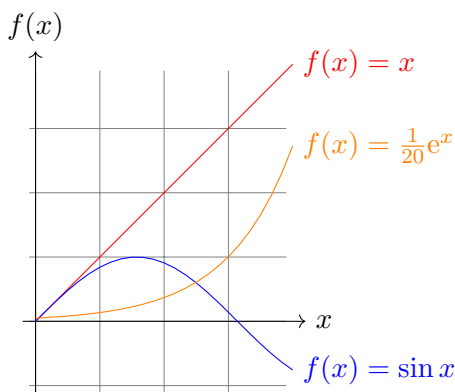
上述命令中 $\langle id \rangle$ 用于区分不同的数据文件, 在 $\langle file \rangle.tex$ 文件 (不妨设文件名为 $\langle file \rangle$) 的根路径下会产生两个文件: 一个是 gnuplot 用于绘图的样式文件 $\langle file \rangle.\langle id \rangle.gnuplot$; 第二个是 gnuplot 产生的数据文件 $\langle file \rangle.\langle id \rangle.table$. 命令中的 $\langle function \rangle$ 可用值请参见: 表 (1).

$\text{\textit{Ti}kZ}$ 的内置命令也支持另外两种格式: “parametric”, “raw gnuplot”: 第一个参数表示绘制参数方程, 第二个参数表示直接在文档中使用 gnuplot 的原始绘图命令 (比如 “set samples 25; plot sin(x)”). 两者的调用格式如下:

```
\draw[⟨key-value⟩] plot [parametric, ⟨id⟩]{⟨function⟩};
\draw[⟨key-value⟩] plot [raw gnuplot, ⟨id⟩]{⟨gnuplot code⟩};
```

```
\begin{tikzpicture}[domain=0:4, scale=.85]
\draw[very thin,color=gray] (-0.1,-1.1) grid (3.9,3.9);
\draw[->] (-0.2,0) -- (4.2,0) node[right] {$x$};
\draw[->] (0,-1.2) -- (0,4.2) node[above] {$f(x)$};
\draw[color=red] plot[id=x] function{x} node[right]
{$f(x)=x$};
\draw[color=blue] plot[id=sin]
function{sin(x)}node[right]{$f(x)=\sin x$};
\draw[color=orange] plot[id=exp] function{0.05*exp(x)}
node[right] {$f(x)=\frac{1}{20} \mathrm{e}^x$};
\end{tikzpicture}
```

例 12



关于 $\text{\textit{Ti}kZ}$ 中这部分原生绘图命令更加详细使用方法请参见 $\text{\textit{Ti}kZ}$ 官方文档

中 Section 22: Plots of Functions.

但是为了 gnuplot 这一系列绘图命令的统一, ztikZ 并没有采用上面的方式, 而是借用 ztool 宏包, 然后配合预定义的绘图脚本去完成绘图任务. ztikZ 中 gnuplot 库的绘图逻辑大致如下:

- 首先通过 ztool 的 `\ztool_replace_file_line:nnn` 函数修改预定义的脚本;
- 然后通过命令行的 `-shell-escape` 参数去调用 gnuplot 运行修改后的脚本;
- 最后使用命令 `\draw[⟨key-value⟩] plot file [⟨data⟩]`; 调用上一步生成的数据文件完成绘图.

不熟悉 gnuplot 的用户可阅读这份 7 页的快速入门指南: [gnuplot card](#).

NOTE: 调用此库后, 需在编译时启用 “-shell-escape” 参数.

ztikz/2dplot/domain	domain = ⟨浮点数: 浮点数; 浮点数: 浮点数⟩..... 初始值: (不定)
ztikz/2dplot/style	style = ⟨draw-keyval⟩..... 初始值: black
ztikz/2dplot/marker	marker = ⟨key-value⟩..... 初始值: 空

⟨maker⟩ 中的 ⟨key-value⟩ 对应 ⟨ztikz/point⟩. ⟨domain⟩ 二者之间使用 “;” 进行分割, 在不同的函数中 ⟨domain⟩ 的意义不同: 在 `\Plot` 中用于设置自变量 x 的范围; 在 `\ParamPlot` 和 `\PolarPlot` 中, 用于设置参数 t 或极坐标系中角度 θ 的范围; 在 `\ContourPlot` 中, “;” 前后两个 ⟨domain⟩ 分别表示 x 和 y 的范围.

<code>\PlotPrecise</code>	<code>\PlotPrecise{⟨type⟩}{⟨number⟩}</code>
New: 2025-05-15	<code>\PlotPrecise*{⟨type⟩}{⟨number⟩}</code>

此命令用于设置 gnuplot 中一系列二维绘图函数对应的精度, ⟨type⟩ 可选值有: “plot, param, polar, contour”, 分别对应命令 `\Plot`, `\ParamPlot`, `\PolarPlot` 和 `\ContourPlot` 的绘制精度. 含有 “*” 的命令会应用于对应绘图命令之后的所有实例, 没有 “*” 的命令仅会应用于之后的第一个绘图命令.

<code>\Plot</code>	<code>\Plot[⟨key-value⟩]{⟨function⟩}</code>
New: 2025-05-15	

此命令用于绘制函数 $y = y(x)$, ⟨function⟩ 为 gnuplot 中的函数表达式, 自变量为 “x”; ⟨key-value⟩ 用于设置绘图样式, 对应 ⟨ztikz/2dplot⟩. ⟨domain⟩ 默认为 -5:5. **注记:** 只需将 ⟨opacity⟩ 置为 0, 即可实现散点图绘制.

`\ContourPlot` `\ContourPlot[⟨key-value⟩]{⟨equation⟩}`

New: 2025-05-15

此命令用于绘制方程 $f(x, y) = c$, $\langle \text{equation} \rangle$ 为 gnuplot 中的方程表达式, 变量为 “x, y”, 且表达式中不需要书写 “=” 符号; $\langle \text{key-value} \rangle$ 用于设置绘图样式, 对应 $\langle \text{ztikz/2dplot} \rangle$. $\langle \text{domain} \rangle$ 默认为 “-5:5;*:*” (即自变量 y 的范围自适应).

注意: 绘制 $x = c$ 这种垂直线段时, 可以使用此函数.

`\ParamPlot` `\ParamPlot[⟨key-value⟩]{⟨equation⟩}`

New: 2025-05-15

此命令用于绘制参数方程 $x = x(t), y = y(t)$, $\langle \text{equation} \rangle$ 为 gnuplot 中的方程表达式, 参数为 “t”; $\langle \text{key-value} \rangle$ 用于设置绘图样式, 对应 $\langle \text{ztikz/2dplot} \rangle$. $\langle \text{domain} \rangle$ 默认为 $0:2\pi$.

`\PolarPlot` `\PolarPlot[⟨key-value⟩]{⟨equation⟩}`

New: 2025-05-15

此命令用于绘制极坐标方程 $\rho = \rho(t)$, $\langle \text{equation} \rangle$ 为 gnuplot 中的方程表达式, 参数为 “t”; $\langle \text{key-value} \rangle$ 用于设置绘图样式, 对应 $\langle \text{ztikz/2dplot} \rangle$. $\langle \text{domain} \rangle$ 默认为 $0:2\pi$.

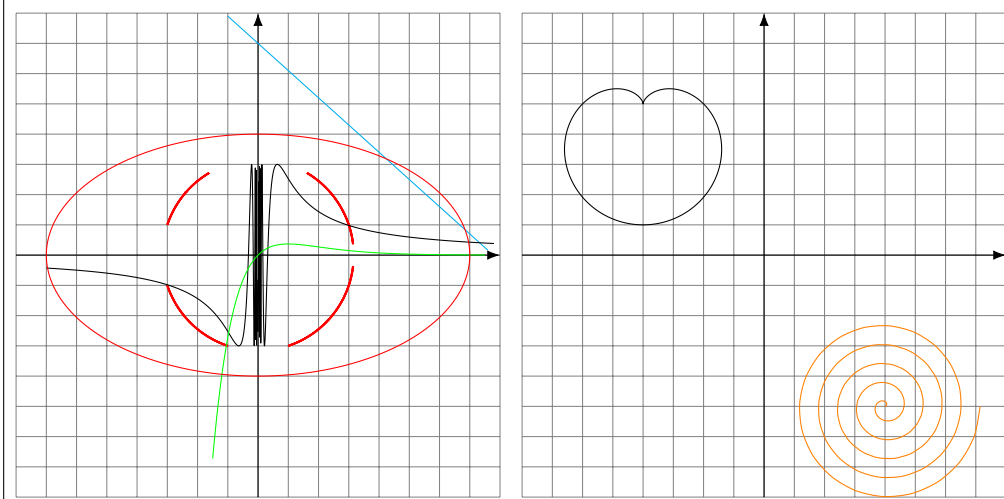
```
\begin{tikzpicture}[>=Latex, scale=.4]
\ShowGrid{(-8, -8); (8, 8)}\ShowAxis{(0, -8); (0, 8)} ✓
\ShowAxis{(-8, 0); (8, 0)}
% draw functions/curves
\Plot[domain=-1:7.6, style=cyan] {-.9*x+7}
\ContourPlot[
  domain={-3:pi; -3:exp(1)}, style={red, thick}
]{x**2 + y**2 - 10}
% change plot precise
\PlotPrecise{plot}{1500}
\Plot[domain=-7:7.8]{3*sin(1/x)}
\Plot[domain=-1.5:7.5, style=green] {x*exp(-x)}
\ParamPlot[domain=0:2*pi, style=red]{7*sin(t), 4*cos(t)}
\end{tikzpicture}
\hskip.5em
\begin{tikzpicture}[>=Latex, scale=.4]
\ShowGrid{(-8, -8); (8, 8)}\ShowAxis{(0, -8); (0, 8)} ✓
\ShowAxis{(-8, 0); (8, 0)}
% draw functions/curves
\begin{scope}[xshift=4cm, yshift=-5cm]
```

例 13

```

\Plot[domain=0:10*pi, style=orange]{0.1*t}
\end{scope}
\begin{scope}[xshift=-4cm, yshift=5cm]
\Plot{2*(1-sin(t))}
\end{scope}
\end{tikzpicture}

```



回顾上面给出的这个简单案例: 这个案例中我们使用了 `\Plot`, `\ParamPlot`, `\PolarPlot` 和 `\ContourPlot` 四个命令; 同时也应用了 `\PlotPrecise` 命令, 它更改了 `\Plot` 命令的绘制精度.

<code>ztikz/3dplot/domain</code>	<code>domain</code> = $\langle \text{浮点数: 浮点数; 浮点数: 浮点数} \rangle$ 初始值: <code>-5.5; -5.5</code>
<code>ztikz/3dplot/pm3d</code>	<code>pm3d</code> = $\langle \text{true false} \rangle$ 初始值: <code>true</code>
<code>ztikz/3dplot/width</code>	<code>width</code> = $\langle \text{长度} \rangle$ 初始值: <code>0.75\linewidth</code>
<code>ztikz/3dplot/palette</code>	<code>palette</code> = $\langle \text{字符串} \rangle$ 初始值: <code>rgbformulae 22,13,-31</code>

$\langle \text{domain} \rangle$ 用于设置自变量 x 和 y 的取值范围, 二者之间使用 “;” 进行分割;
 $\langle \text{pm3d} \rangle$ 用于控制是否启用曲面染色, 若 $\langle \text{pm3d} \rangle = \text{false}$ 则此时进行绘制曲面的一系列曲线;
 $\langle \text{width} \rangle$ 用于设置该图片的宽度.

<code>\Plotz</code>	<code>\Plotz[\langle \text{key-value} \rangle]{\langle \text{function} \rangle}</code>
---------------------	--

New: 2025-05-15

此命令用户绘制普通的二维显式函数, $\langle \text{function} \rangle$ 为 gnuplot 中的函数表达式; $\langle \text{key-value} \rangle$ 用于设置绘图样式, 对应 $\langle \text{ztikz/3dplot} \rangle$. **注意:** 该命令不能在 `\tikzpicture` 环境中使用.

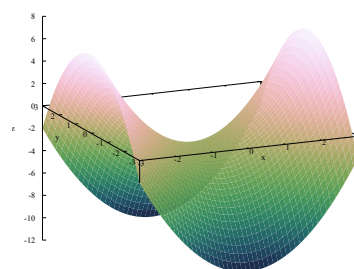
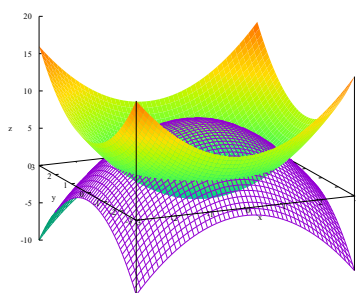
下面这个案例展示了 `\Plotz` 命令的基本使用方法, 其中第一个案例内的 “`x**2+y**2-2 with pm3d`” 为 gnuplot 所特有的语法, 详细信息请参见 gnuplot 手册.

```

\Plotz[
  pm3d = false,
  width = .45\linewidth,
  domain = {-3:3; -3:3}
]{x**2+y**2-2 with pm3d, -x**2-y**2+8 with lines}
\hskip5em
\Plotz[
  pm3d,
  width = .45\linewidth,
  domain = {-3:3; -3:3},
  palette = {cubehelix start 0 cycles -1. saturation 1}
]{x**2-y**2-2}

```

例 14

 $\backslash\text{currentTikzIndex}$

New: 2025-05-15

该命令表示当前 tikzpicture 环境的索引, 返回值为整数, 从 1 开始.

 $\backslash\text{gnudata}$ ★ $\backslash\text{gnudata}\{\langle index \rangle\}$

New: 2025-05-22

该命令会用引用当前 tikzpicture 环境中产生的绘图数据, 返回一个 (数据) 文件名, 从 1 开始. $\langle index \rangle$ 接受一个整数, 表示当前环境中绘图数据的编号. 每一个已经绘制的函数都会在对应的文件夹下生成一个对应的数据文件, 用户可以使用此数据文件进行后续的绘图操作.

注记 ($\backslash\text{gnudata}$ 用法补充, 为后面区域填充做铺垫): 比如命令 $\backslash\text{gnudata}\{2\}$, 参数中的 “2” 表示此数据是在当前 tikzpicture 环境中的第二个函数绘图数据; 所以在第一个 tikzpicture 环境中它的返回值可能为 “./ztikz-output/gnuplot_data/gnu_data_1_2.table”.

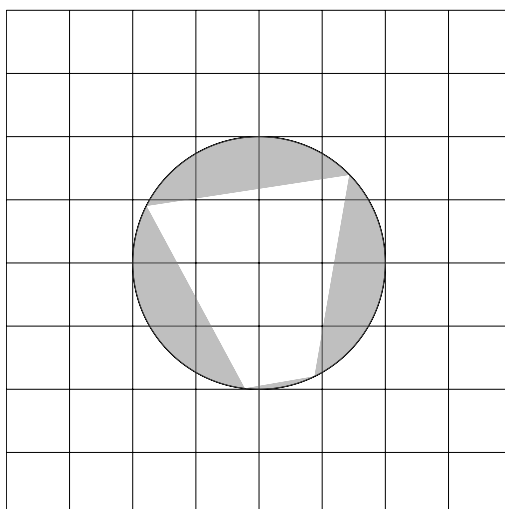


Figure 1: $\backslash\text{ContourPlot}$ Fill Issue

注意: 由于技术原因, $\backslash\text{ContourPlot}$ 命令生成的数据暂时不可用于后续填充操作. 可考虑先将隐函数转化为参数方程形式或极坐标形式, 再导出对应的数据. 如果你强行使用此类型数据, 那么用户可能会得到类似 图 (1) 这样的不良输出.

5.3 cache 库

当用户加载 cache 库后, 随后在命令行中编译文档, 不妨设其名称为 $\langle file \rangle$; 那么用户会看到如下的日志输出:

```
\write18 enabled.
entering extended mode
```

编译结束后, 在你的项目文件夹下会生成一个名为 `ztikz_output` 的文件夹, 这个文件夹在你第一次调用 `ztikz` 宏包时便会产生; 这个文件夹用于存放 $\LaTeX$ 的缓存文件: 包括 $\TeX$ external 库的缓存结果, Python 脚本的缓存结果, WolframScript 脚本的缓存结果, 以及 gnuplot 的一系列缓存结果.

现在来说说这个文件夹的构成: 比如, 若用户运行了 `\Plot` 命令, 此时会在 `ztikz_output/tikz_data/` 目录下生成了如图 (2) 中所示的 4 个文件:

```
ztikz_output
├── gnuplot_data.....gnuplot 缓存文件夹
│   └── gnu_data_1_1.table
├── mma_data.....WolframScript 缓存文件夹
├── python_data.....Python 缓存文件夹
├── scripts.....gnuplot 绘图脚本
│   ├── 3d_plot.gp
│   ├── contour_plot.gp
│   ├── param_plot.gp
│   ├── plot_plot.gp
│   └── polar_plot.gp
├── tikz_data.....TikZ 缓存文件夹
│   ├──  $\langle file \rangle$ -figure0.dpth
│   ├──  $\langle file \rangle$ -figure0.log
│   ├──  $\langle file \rangle$ -figure0.md5
│   ├──  $\langle file \rangle$ -figure0.pdf
│   └──  $\langle file \rangle$ -figure0.run.xml
└── ztikz.hash.....Hash 值记录
```

Figure 2: $\LaTeX$ 缓存目录结构示意图

`tikz_data` 中的 $\langle file \rangle$ -figure0.pdf 为 `tikzpicture` 环境缓存的 PDF 文件; 此时在对应的 $\langle file \rangle$.md5 文件中可以看到如下内容:

```
\def \tikzexternallastkey {AE7F2539E81C96848ADCCEE3994993D1}%
```

上述命令保存了此 `tikzpicture` 环境中代码的 Hash 值, 当我们改变 `tikzpicture` 环境中的代码时, 这个 Hash 值就会改变, 从而 $\TeX$ 就会再次运行此环境, 重新生成图片. 这便是 $\TeX$ 的 external 库所提供的缓存功能的大致描述. $\LaTeX$ 中的 Cache 机制和此原理是十分类似的.

5.3.1 用户接口

<code>\ztikzHashClean</code>	此命令不接受任何参数, 用于清除之前缓存的所有 Hash 值.
New: 2025-05-15	
<code>\ztikzHashCurrent</code>	<code>\ztikzHashCurrent*</code>
New: 2025-05-15	<code>\ztikzHashCurrent[⟨separator⟩]</code>
	此命令主要用于调试缓存相关的代码, 它常常与命令 <code>\ztikzForceToSkip</code> 配合使用; <code>\ztikzHashCurrent*</code> 将输出最近的一次 Hash 值计算结果; <code>\ztikzHashCurrent[⟨separator⟩]</code> 用于输出截至目前位置所有缓存的 Hash 值, 以 <code>⟨separator⟩</code> 分隔输出到 PDF; <code>⟨separator⟩</code> 默认为 “,”.
<code>\ztikzCachedHash *</code>	<code>\ztikzCachedHash[⟨keyval⟩]</code>
New: 2025-05-29	此命令用于输出当前已缓存的 Hash 值, 应用场景较之 <code>\ztikzHashCurrent</code> 命令更加的广泛.
<code>ztikz/cache/hash/index</code>	<code>index = ⟨整数⟩.....初始值: -1</code>
<code>ztikz/cache/hash/file</code>	<code>file = ⟨文件名⟩.....初始值: ztikz_output/ztikz.hash</code>
<code>ztikz/cache/hash/label</code>	<code>label = ⟨字符串⟩.....初始值: \g__zcache_latest_cache_label_tl</code>
	<code>⟨label⟩</code> 默认情况下为当前最新的缓存标签; <code>⟨file⟩</code> 为 Hash 值对应的缓存文件; <code>⟨index⟩</code> 用于指定该 <code>⟨label⟩</code> 所缓存的 Hash 值的索引, 默认为 -1, 即最新的 Hash 值;
<code>\ztikzForceToSkip</code>	此命令会强制跳过 (重新) 运行它之后的第一个具有 cache 机制的环境或命令, 即使该环境或命令对应的 Hash 已经改变; 后续的 <code>\wolframResult</code> 或 <code>\wolframOutputFile</code> 命令对应的引用结果都将受到此命令的影响.
New: 2025-05-15	注意: 当应用此命令后, 新产生的 Hash 值并不会被缓存; 该命令目前仅对 python 和 wolfram 库中的命令和环境有效; 该命令会删除后续与 新 Hash 相关的脚本与结果.
<code>\ztikzForceToRun</code>	此命令会强制运行它之后的第一个具有 cache 机制的环境或命令, 即使该环境或命令对应的 Hash 并没有改变.
New: 2025-05-21	注意: 该命令目前仅对 python 和 wolfram 库中的命令和环境有效; 即使是该命令后续的命令或环境对应的 Hash 值改变, 这个新的 Hash 值也不会被缓存.

5.3.2 编程接口

ztikz 的 cache 库提供了一系列的编程接口, 用户可以利用这一系列的接口来编写外部程序调用相关的命令或环境. 这系列的新建命令或环境将支持缓存机制, 目前 cache 库中提供的编程接口和部分其它相关命令如下:

`\g_ztikz_file_hash_seq`

New: 2025-05-30

该序列 (变量) 中保存了当前所有已缓存 Hash 值, 该序列中的项 (元素) 为某个具有缓存机制的命令或环境对应的 Hash 值.

注记: 该命令由后续的 `\_zcach hash_extract_all:nN` 命令设置得到.

`\g_ztikz_hash_nochg_run_bool`

`\g_ztikz_hashchg_norun_bool`

New: 2025-05-30

这两个 bool 值用于控制 `\ztikz_if_run_again:nnnTF` 命令的行为, 前者为 “true” 时: `\ztikz_if_run_again:nnnTF` 命令的 Hash 校验会被强制跳过, 从而直接运行 `<true code>`; 后者为 “true” 时: `\ztikz_if_run_again:nnnTF` 命令的 Hash 校验会被强制跳过, 从而直接运行 `<false code>`.

注意: 这两个 bool 值默认均为 “false”; **二者同时为 “true” 会报错.**

`\xsim_file_write_start:nn`

`\xsim_file_write_start:ne`

New: 2025-05-30

`\xsim_file_write_start:nn {<bool>}{<file>}`

此命令来自 xsimverb 宏包, 用于将环境内容抄录到 `<file>` 文件中; `<bool>` 为布尔值, 如果该抄录环境需要接受参数, 那么请将 `<bool>` 置为 “`\c_true_bool`”, 否则请置为 “`\c_false_bool`”.

注意: 该命令需配合 `\xsim_file_write_stop:` 命令使用, 否则会报错.

`\xsim_file_write_stop:`

New: 2025-05-30

该命令用于结束环境内容抄录, 需配合 `\xsim_file_write_start:nn` 命令使用.

`\ztikz_if_run_again:nnnTF`

`\ztikz_if_run_again:nenTF`

New: 2025-05-30

`\ztikz_if_run_again:nnnTF`
`{<bool>}{<file/hash>}{<label>}`
`{<true code>}{<false code>}`

`<bool>` 用于控制第二个参数 `<file/hash>` 的类型, `<bool>` 为 “`\c_true_bool`” 时, `<file/hash>` 需传入文件名, 否则应传入一个 Hash 值; `<label>` 为该 Hash 值或文件 Hash 值对应的缓存标签; 当该环境的 Hash 值不存在, Hash 值改变抑或该环境被置于命令 `\ztikzForceToRun` 后时, 将会运行 `<true code>`; 当该环境的 Hash 值已存在或该环境被置于命令 `\ztikzForceToSkip` 后时, 将会运行 `<false code>`.

`\ztikz_term_info:n` `\ztikz_term_info:n {⟨message⟩}`

`\ztikz_term_info:e` 此命令与 $\text{\TeX}$ 中的 `\typeout` 命令类似，用于向终端输出信息 $\langle message \rangle$ ，当“`quiet = true`”时，该命令不会执行任何操作。

New: 2025-05-31

5.3.3 私有接口

本小节将介绍 `cache` 库中的私有命令和变量, 这部分命令或变量在用户定义一些命令或环境时可能是必须的; 因这部分命令暂时未整理好, 所以后续可能会有较大改动, 用户应谨慎使用这部分命令或变量.

<code>\l_ztikz_current_hash_tl</code>	该变量将返回当前最新的 Hash 值.
New: 2025-05-30	
<code>_zcache_hash_add:nn</code>	<code>_zcache_hash_add:nn {<label>}{<hash>}</code>
New: 2025-05-30	此命令用于将 <code><hash></code> 添加到 <code><label></code> 对应的缓存标签中.
<code>_zcache_hash_extract_all:nN</code>	<code>_zcache_hash_extract_all:nN {<file>}<seq></code>
New: 2025-05-30	该命令用于将文件 <code><file></code> 中的所有 Hash 值提取后保存到到变量 <code><seq></code> 中, <code><seq></code> 中的每一项都对应一个 Hash 值.
<code>_zcache_hash_extract_by_label:nnn</code>	<code>_zcache_hash_extract_by_label:nnn {<file>}{<label>}{<index>}</code>
New: 2025-05-30	该命令会从文件 <code><file></code> 中提取标签为 <code><label></code> , 索引为 <code><index></code> 的 Hash 值. 该命令无返回值, 它会将 <code>\l_ztikz_current_hash_tl</code> 置为其提取的 Hash 值.

5.3.4 编程接口案例

下面我们给出上述命令的一个使用样例, 该样例制作了一个支持缓存机制的 Mermaid 绘图环境 (用户需要安装 “mermaid-cli” 这一工具):

例 15

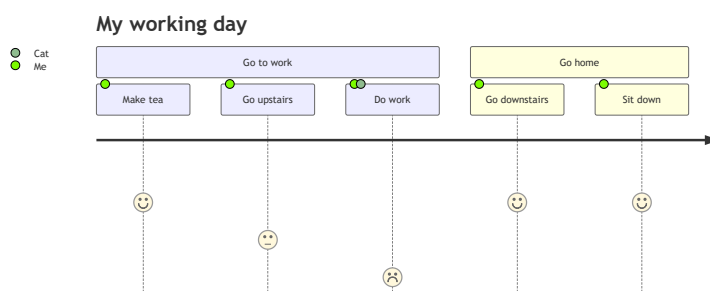
```

\ExplSyntaxOn\makeatletter
% 1. environment declaration
\tl_new:N \g__ztikz_mmd_path_tl
\tl_gset:Nn \g__ztikz_mmd_path_tl {ztikz_output/mmd_data}
\ztool_shell_mkdir:e { \g__ztikz_mmd_path_tl }
\NewDocumentEnvironment{mermaid}{m }
{
  \exp_args:Ne \xsim_file_write_start:nn {\c_true_bool}{
    \g__ztikz_mmd_path_tl/t@mp.mmd}
  }{
    \xsim_file_write_stop:
    \ztikz_if_run_again:nenTF {\c_true_bool}{t@mp.mmd}{#1}
    {
      \ztool_shell_escape:e
      {
        mmdc
        \space-i\space \g__ztikz_mmd_path_tl/t@mp.mmd
        \space-o\space \g__ztikz_mmd_path_tl/t@mp.pdf
        \space-f
      }
      \ztikz_term_info:e { Mermaid~running~on~file:'
\l__ztikz_current_hash_tl.mmd'~... }
    }{
      \ztikz_term_info:e { Use~the~cached~Mermaid~result:'
\l__ztikz_current_hash_tl.pdf'~... }
    }
    \edef\t@mp@file{\g__ztikz_mmd_path_tl/\l__ztikz_current_hash_tl}
    \ztool_shell_mv:ee
    {\g__ztikz_mmd_path_tl/t@mp.mmd}
    {\t@mp@file.mmd}
    \ztool_shell_mv:ee
  
```

```

    {\g__ztikz_mmd_path_tl/t@mp.pdf}
    {\t@mp@file.pdf}
    \xdef\mmdOutputFile{\t@mp@file.pdf}
  }
\makeatother\ExplSyntaxOff
% 2. environment usage
\begin{mermaid}{mmd-I}
journey
  title My working day
  section Go to work
    Make tea: 5: Me
    Go upstairs: 3: Me
    Do work: 1: Me, Cat
  section Go home
    Go downstairs: 5: Me
    Sit down: 5: Me
\end{mermaid}
\begin{center}
  \includegraphics[width=.7\linewidth]{\mmdOutputFile}
\end{center}

```



5.4 python 库

python 库主要用于和 Python 交互, 其使用方法和 gnuplot 库类似. python 库中主要提供了图片绘制与计算接口, 其中计算接口包含数值计算与符号计算.

除去 xTikZ 提供的 Python 绘图功能外, 我们需要着重说明 xTikZ 提供的浮点数计算功能: xTikZ 在调用此库时默认导入 Python 的 numpy, sympy, scipy 三个包; 此外, 用户在使用 numpy 中的函数时不用再加以前缀, 比如求解 $\sin(2.345)$ 时, 直接使用 `\py{sin(2.345)}` 即可, 不必写为 `\py{np.sin(2.345)}` 之类的格式了. 对于其它 Python 库中的函数, 使用方法同理.

NOTE: 调用此库后, 需在编译时启用 “-shell-escape” 参数.

\py

New: 2025-05-15

\py[<raw|str>]{<code>}

此命令会调用 Python 进行浮点数运算, <code> 为合法的 Python 表达式; 这部分的结果并不会被缓存, 也就是说每次编译此文档时, Python 都会重新计算此部分的结果. 用户可以把 \py 命令嵌套到自己定义的宏命令中.

注意: <raw> 会将返回的结果按照 TeX 原始的 catcode 进行 tokenize; <str> 则是将返回的结果处理为 string.

```
\newcommand{\pypow}[1]{\py{#1}}
\newcommand{\pyreverse}[1]{\py{'#1'[::-1]}}
\newcommand{\pyuppercase}[1]{\py{'#1'.upper()}}
\begin{itemize}
  \item Power Calculation:  $2^{10} = \pypow{2**10}$ 
  \item Reverse a string using Python: \pyreverse{Hello-Latex}
  \item Uppercase a string: \pyuppercase{hello-latex}
  \item Modulus:  $102 = \py{mod(102, 8)} \bmod 8$ 
  \item Return string Options: \py[str]{'$$'+str(2**10)+'$$'}
\end{itemize}
```

例 16

- Power Calculation: $2^{10} = 1024$
- Reverse a string using Python: XeTaL-olleH
- Uppercase a string: HELLO-LATEX
- Modulus: $102 = 6 \bmod 8$
- Return string Options: \$\$1024\$\$

`\sympy`

New: 2025-05-29

`\sympy{<label>}{<expression>}`

此命令主要用于调用 Python 的 sympy 库进行符号计算, $\LaTeX$ 对此命令提供了 cache 机制; `<label>` 中不能包含 “:”, 其用于指定该命令的缓存标签, 该 `<label>` 在当前文档中必须是唯一的; `<expression>` 为符号表达式. python 库中预定义了一系列的符号变量, 包括: `x`, `y`, `z`, `u`, `v`, `t`, 这些预定义变量无需用户再次声明.

注意: 默认的情况下, 此命令的返回结果中可能包含: “ $\wedge$ ”, “ $_$ ” 等数学环境中才能使用的字符, 故用户应尽量将此命令置于数学环境中.

例 17

```

\begin{math}
\int x^8 + \cos(7x) + 6t \, dx = \text{sympy}{\text{integral}}{\text{integrate}(x**8 + \cos(7*x) + 6*t, x)}
\end{math}
\begin{math}
\text{eig}(\begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}) = \text{sympy}{\text{matrix}}{\text{Matrix}([[1, 2], [2, 2]]).eigenvals()}
\end{math}

```

$$\int x^8 + \cos(7x) + 6t \, dx = 6tx + \frac{x^9}{9} + \frac{\sin(7x)}{7}$$

$$\text{eig}\left(\begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}\right) = \left\{ \frac{3}{2} - \frac{\sqrt{17}}{2} : 1, \frac{3}{2} + \frac{\sqrt{17}}{2} : 1 \right\}$$

`pyfig`

Updated: 2025-05-29

`\begin{pyfig}{<label>}{<output file>}``<plot code>``\end{pyfig}`

此环境用于调用 Python 进行绘图, 不会返回任何结果, 该环境具有缓存机制; `<label>` 中不能包含 “:”, 其用于指定该环境的缓存标签, 该 `<label>` 在当前文档中必须是唯一的; `<output file>` 用于指定代码 `<plot code>` 的输出文件名, `<output file>` 中无需给出输出文件路径, 但需指定输出文件的拓展名;

注意: 针对不同的 pyfig 环境建议使用不同的 `<output file>` 值; 用户不需要在代码末尾添加 `plt.savefig()` 命令, $\LaTeX$ 会自动处理此问题. 代码在抄录过程中会保留用户的缩进格式, 从行首开始抄录, 所以请不要添加多余的行首缩进; 请确保 `<output file>` 与后续环境代码中的输出文件名保持一致, 否则会报错.

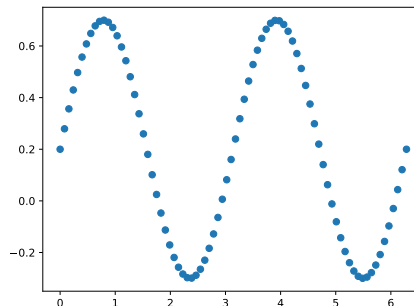
 $\backslash$ pyfigOutputFile $\star$

New: 2025-04-21

此命令将返回 `pyfig` 环境运行输出的文件名，用户可以使用 `\input` 或 `\includegraphics` 之类的命令导入该文件。

```
\begin{pyfig}{sinGraph}{sin_graph.pdf}
import matplotlib
matplotlib.use('Agg')
from matplotlib import pyplot as plt
import numpy as np
x = np.linspace(0, 2*np.pi, num = 80)
y = np.sin(x)*np.cos(x)+.2
plt.plot(x, y, 'o')
\end{pyfig}
\begin{center}
\includegraphics[width=.5\linewidth]{\pyfigOutputFile}
\end{center}
```

例 18



`pycode`

New: 2025-05-21

 $\backslash$ begin{pycode}{ $\langle$ label $\rangle$ }{ $\langle$ output file $\rangle$ }

 $\langle$ any python code $\rangle$
 $\backslash$ end{pycode}

此环境用于调用 Python 执行环境中的 $\langle$ any python code $\rangle$ ，不会返回任何结果，该环境具有缓存机制； $\langle$ label $\rangle$ 中不能包含 “:”，其用于指定该环境的缓存标签，该 $\langle$ label $\rangle$ 在当前文档中必须是唯一的； $\langle$ output file $\rangle$ 用于指定该环境代码的输出文件名， $\langle$ output file $\rangle$ 中无需给出输出文件路径，但需指定输出文件的扩展名；随后 $\text{\LaTeX}$ 会自动调用 Python 执行该文件，该环境的运行结果保存于文件 $\backslash$ pycodeOutputFile 中，用户后续仅需导入该文件即可；

注意：针对不同的 `pycode` 环境建议使用不同的 $\langle$ output file $\rangle$ 值；代码在抄录过程中会保留用户的缩进格式，从行首开始抄录，所以不要过度使用缩进；请确保 $\langle$ output file $\rangle$ 与后续环境代码中的输出文件名保持一致，否则会报错。

`\pycodeOutputFile`

★

New: 2025-04-21

此命令将返回 `pycode` 环境运行输出的文件名，用户可以使用 `\input` 或 `\includegraphics` 之类的命令导入该文件。

下面是一个关于 `pycode` 环境的简单使用示例，`table.py.txt` 对应的文件内容请参见 节 (6.3)。

`\input{./table.py.txt}`
`\begin{center}`
`\input{\pycodeOutputFile}`
`\end{center}`

例 19

number/function	sin	cos	tan
1	0.8415	0.5403	1.5574
2	0.9093	−0.4161	−2.185
3	0.1411	−0.99	−0.1425
4	−0.7568	−0.6536	1.1578
5	−0.9589	0.2837	−3.3805
6	−0.2794	0.9602	−0.291
7	0.657	0.7539	0.8714
8	0.9894	−0.1455	−6.7997
9	0.4121	−0.9111	−0.4523
10	−0.544	−0.8391	0.6484
11	−1.0	0.0044	−225.9508
12	−0.5366	0.8439	−0.6359
13	0.4202	0.9074	0.463
14	0.9906	0.1367	7.2446
15	0.6503	−0.7597	−0.856

5.5 wolfram 库

$\LaTeX$ 的 wolfram 库可看作是原始宏包 latexalpha2 的一个新实现，可以弥补 latexalpha2 宏包的一系列不足。目前 wolfram 库已经实现 latexalpha2 中除 `\wolframanimation` 命令外的所有命令，并且在兼容性，易用性和可拓展性上相较于原始的 latexalpha2 宏包都有了极大的提升²。例如， $\LaTeX$ 的 wolfram 库可以在 Windwos/Linux/MacOs 三大平台上使用；wolfram 库的环境源码中支持直接键入 “\, #, \$, _, ^, &” 等特殊字符。下面是使用 wolfram 库时的一些注意事项：

- 用户需注意 WolframScript 脚本中注释的写法，不是 “(* something*)”，而是 “(* something *)”，即注释内容不能够紧挨 “*”，否则可能会造成 WolframScript 的解析错误。
- 由于 WolframScript 的限制，脚本的后缀只能为：“.wls”，否则 WolframScript 会无法识别此脚本（也就不会去执行此脚本了）。
- 本节后续的所有支持缓存的命令或环境，其缓存标签 `\label` 中只能使用“数字，字母，短横线 (-)”。

NOTE: 调用此库后，需在编译时启用 “-shell-escape” 参数。

<code>\wolframResult</code>	<code>\wolframResult[<separator>]</code>
New: 2025-05-15 Updated: 2026-05-07	<code>\wolframResult* [<index>]</code>
	此命令用于引用前一次 WolframScript 的计算结果， <code>\wolframResult[<separator>]</code> 表示使用 <code><separator></code> 进行分隔，然后引用全部计算结果； <code>\wolframResult* [<index>]</code> 仅引用部分计算结果， <code><index></code> 为整数或整数表达式，默认为 1。

NOTE: 因为 `\wolframResult` 会根据 “@₁₂” 去划分结果，所以请确保 “@” 的 catcode 为 12，否则可能会导致结果解析错误。

<code>\wolframOutputPath</code> ★	此命令保存了 WolframScript 运行结果（文件）所在的文件夹；此命令常与 <code>wolframCode</code> 环境搭配使用，用于引用一些图片或文本结果。此命令比后续的 <code>\wolframResult</code> ， <code>\wolframOutputFile</code> 命令更加通用。
New: 2026-05-08	

<code>\wolframOutputFile</code> ★	此命令会返回 WolframScript 上次运行结果对应的文件名；此命令在引用一些图片结果时是十分方便的。此命令比之 <code>\wolframResult</code> 更加的灵活，前者调用上一次的文本文件，后者仅返回上次 WolframScript 调用产生的文件名。
New: 2025-05-15 Updated: 2026-04-30	

²2026-04-29: 新增 wstp 引擎后，速度也有了巨大的提升。

 $\backslash\text{wolfram}$ $\backslash\text{wolfram}\{\langle\text{label}\rangle\}\{\langle\text{code}\rangle\}$

New: 2025-05-29

 $\backslash\text{wolfram*}\{\langle\text{label}\rangle\}\{\langle\text{code}\rangle\}$

此命令用于调用 WolframScript 中的进行计算, 具有缓存机制; $\langle\text{label}\rangle$ 中不能包含 “:”, 其用于指定该命令的缓存标签, 该 $\langle\text{label}\rangle$ 在当前文档中必须是唯一的; $\langle\text{code}\rangle$ 为合法的 WolframScript 代码; 默认计算结果为 $\LaTeX$ 代码, 含有 “*” 的命令计算结果为普通的字符串 (catcode 并没有改变).

```
\wolfram{wolframLaplace}{LaplaceTransform[t^4 Sin[3*t], t, s]}
\[
  \mathcal{L}(t^4 \sin(3t)) = \text{wolframResult}
\]
```

$$\mathcal{L}(t^4 \sin(3t)) = \frac{72(5s^4 - 90s^2 + 81)}{(s^2 + 9)^5}$$

例 20

 $\backslash\text{wolframTex}$ $\backslash\text{wolframTex}\{\langle\text{label}\rangle\}\{\langle\text{Tex code}\rangle\}$

New: 2025-05-29

此命令和上述的 $\backslash\text{wolfram}$ 命令类似, 不同的是, 此命令会将 $\langle\text{Tex code}\rangle$ 中的所有内容转化为对应的 Mathematica/Mathics 代码, 返回的结果为 $\LaTeX$ 代码. $\langle\text{label}\rangle$ 中不能包含 “:”, 其用于指定该命令的缓存标签, 该 $\langle\text{label}\rangle$ 在当前文档中必须是唯一的;

NOTE: 由于此命令的实现原理较为复杂与特殊, 所以 $\langle\text{Tex code}\rangle$ 和 $\langle\text{label}\rangle$ 中均不能包含 “\$” 符号, 否则会出现解析错误; 用户不应过于依赖此命令, 部分情况下该命令会消耗相当长时间, 更甚者会导致编译报错.

```
\wolframTex{wolframTexInt}{\int_a^b \sin(x) \, dx}
\[
  \int_a^b \sin(x) \, dx = \text{wolframResult}
\]
```

$$\int_a^b \sin(x) dx = \cos(a) - \cos(b)$$

例 21

`\wolframTable` `\wolframTable{<label>}{<code>}`

New: 2025-05-29

`\wolframTable*{<label>}[<key-value>]{<code>}`

此命令用于调用 Wolfram 引擎生成表格, 具有缓存机制; `<label>` 中不能包含 “:”, 其用于指定该命令的缓存标签, 该 `<label>` 在当前文档中必须是唯一的; 前者 (不带有 “*” 的命令) 不会在 PDF 中输出对应的表格, 此命令设置了 `\wolframTablePData`, `\wolframTableFData` 两个临时变量, 其中保存了表格数据; 后者 (带有 “*” 的命令) 会在 PDF 中输出对应的表格, 表格的样式可以通过 `<key-value>` 进行指定; `<code>` 为合法的 WolframScript/Mathics 代码;

ztikz/wolfram/table/format	format	= <列格式>.....	初始值: <code>*{t2}{1}</code>
ztikz/wolfram/table/header	header	= <表头>.....	初始值: 空
ztikz/wolfram/table/hdbt-rule	hdbt-rule	= <code><false true></code>	初始值: <code>false</code>
ztikz/wolfram/table/cell-cmd	cell-cmd	= <单元格命令>.....	初始值: <code>#1</code>

`<format>` 用于设置表格的列格式; `<header>` 用于设置表格的表头, 该参数会在表格的第一行输出; `<hdbt-rule>` (header bottom rule) 用于设置是否显示表头与后续表格内容之间的横线, 默认不显示; 该选项仅对 “`\wolframTable*`” 命令有效, 即仅在需要排版表格时有效; `<cell-cmd>` 接受一个参数, 其将应用于每一个 `\wolframTablePData` 中的单元格 (不包括表头), 使用 “`#1`” 表示当前单元格内容;

`\wolframTablePData *`
`\wolframTableFData *`

New: 2025-05-18

这两个命令表示最近一次 `\wolframTable` 命令的运算结果, 不含有 `\hline`, `\midrule` 等命令, 即二者仅含有纯表格数据; `\wolframTablePData` (Part Data) 返回的数据不包括表格的表头, `\wolframTableFData` (Full Data) 返回的数据包括表格的表头; 此二命令可以传入 `tabularray` 等表格排版宏包的数据输入. 注意: 命令 `\wolframTable` 中的 `<cell-cmd>` 不建议与 `tabularray` 中的 `<cells/cmd>` 混用.

```

\wolframTable*{wolframTable}[
  format=cccc, hdbt-rule,
  header={x & x^2 & x^3 & x^4},
  cell-cmd={\textcolor{red}{( #1 )}}
]{Table[{i, i^2, i^3, i^4}, {i, 6}]}
\SetTblrOuter{expand=\wolframTableFData}
\hskip6em
\begin{tblr}
{
  colspec = {cccc},

```

例 22


```
\wolframSolve{wolframSolve-II}
[var={x, y}, domain=Integers]
{x^2 + 2 y^3 == 3681 && x > 0 && y > 0}
\begin{align}
&\text{\wolframResult}
\end{align}
```

$$x = \frac{9}{a+b}, y = -\frac{a-8b}{a+b} \quad (5.1)$$

$$x = \frac{9}{a+b} || y = -\frac{a-8b}{a+b} \quad (5.2)$$

$$x = \frac{9}{a+b} \quad (5.3)$$

$$y = -\frac{a-8b}{a+b} \quad (5.4)$$

$$x = 15, y = 12, x = 41, y = 10, x = 57, y = 6 \quad (5.5)$$

```
\wolframDSolve \wolframDSolve{<label>}[<key-value>]{<equation>}
```

New: 2025-05-29

```
\wolframDSolve*{<label>}{<full code>}
```

此命令用于调用 WolframScript 中的进行微分方程的求解，具有缓存机制；
 <label> 中不能包含 “:”，其用于指定该命令的缓存标签，该 <label> 在当前文档中必须是唯一的；<equation> 表示方程的表达式；<key-value> 用于设置求解的自变量与定义域；<full code> 为完整的微分方程表达式，包含自变量，因变量；

```
ztikz/wolfram/dsolve/depend depend = <因变量>.....初始值: y[x]
ztikz/wolfram/dsolve/independ independ = <自变量>.....初始值: x
```

<depend> 用于指定该微分方程的因变量，比如 <depend>=y[x] 表示 y 是 x 的函数；<independ> 用于指定该微分方程的自变量，比如 <independ>=x 表示 x 是自变量；

```
\wolframDSolve{wolframDSolve-I}{y'[x] + y[x] == a*Sin[x], y[0] == 1}
\begin{align}
&\&\text{\wolframResult}
\end{align}
\wolframDSolve{wolframDSolve-II}
[depend={y[x], z[x]}]
```

```

{y'[x] == Exp[z[x]] + 1, z'[x] == y[x] - x}
\begin{align}\left\{\begin{aligned}
&\&\texttt{wolframResult}[\&\&]
\end{aligned}\right.\end{align}

```

$$y(x) = -\frac{1}{2}e^{-x}(-ae^x \sin(x) + ae^x \cos(x) - a - 2) \quad (5.6)$$

$$\begin{cases} z(x) = \log\left(c_1 \tan^2\left(\frac{1}{2}\left(\sqrt{2}\sqrt{c_1}x + 2\sqrt{2}\sqrt{c_1}c_2\right)\right) + c_1\right) \\ y(x) = x + \sqrt{2}\sqrt{c_1} \tan\left(\frac{1}{2}\left(\sqrt{2}\sqrt{c_1}x + 2\sqrt{2}\sqrt{c_1}c_2\right)\right) \end{cases} \quad (5.7)$$

wolframGraphics `\begin{wolframGraphics}{⟨label⟩}[⟨spec⟩]`

New: 2025-05-29

`⟨plot code⟩`

`\end{wolframGraphics}`

此环境用于调用 WolframScript 进行绘图，具有缓存机制；⟨label⟩ 中不能包含 “:”，其用于指定该环境的缓存标签，该 ⟨label⟩ 在当前文档中必须是唯一的；⟨spec⟩ 用于设置图片的排版参数，默认为空，此时该环境不会返回任何的结果，可以通过 `\wolframOutputFile` 调用其产生的文件；⟨spec⟩ 可以设置值，对应图片的排版参数，比如 `width=10em`；若 ⟨spec⟩ 非空，则该环境的返回值为：`\includegraphics[⟨spec⟩]{⟨path⟩/⟨HASH⟩.pdf}`，其中 ⟨HASH⟩ 为当前 wolframGraphics 环境中代码的 Hash 值，⟨path⟩ 为 WolframScript 缓存文件夹对应的目录。

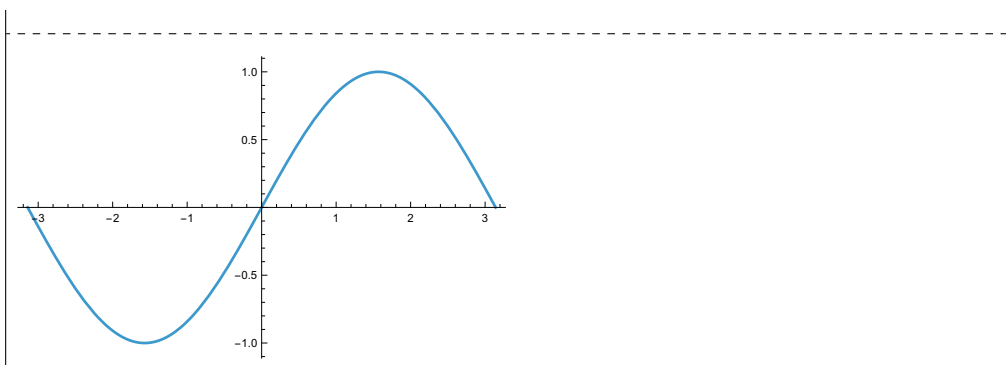
NOTE: ⟨plot code⟩ 中最后得到的图片名称必须为 “FIGURE”，否则会报错。

```

\begin{wolframGraphics}{wolframSinGraph}
    FIGURE=Plot[Sin[x], {x, -Pi, Pi}]
\end{wolframGraphics}
% include graphics
\includegraphics[width=.5\linewidth]{\wolframOutputFile}

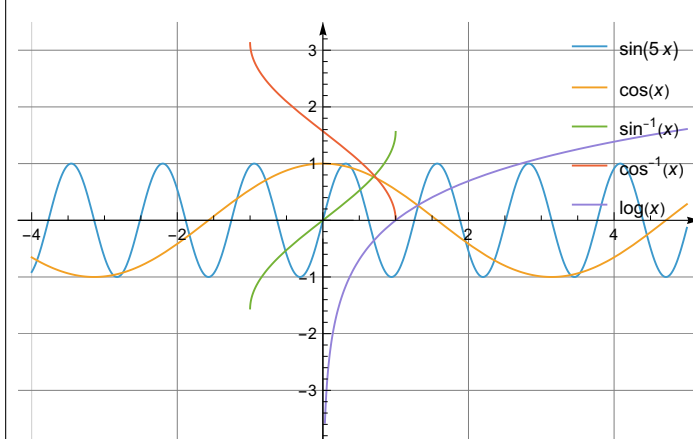
```

例 25



```
\begin{wolframGraphics}{fig-1}[width=.7\linewidth]
fp = Plot[{
  Sin[5 x],
  Cos[x],
  ArcSin[x],
  ArcCos[x],
  Log[x]
},
{x, -4, 5},
PlotLegends -> Placed["Expressions", {Right, Top}],
PlotStyle -> AbsoluteThickness[1],
AxesStyle -> Arrowheads[0.02],
GridLines -> Automatic
];
FIGURE = Show[fp];
\end{wolframGraphics}
```

例 26



```
wolframCode \begin{wolframCode}{\label}{
              \langle wolfram code \rangle
\end{wolframCode}
```

New: 2026-05-08

此环境可以运行任意的 wolfram 代码, 具有缓存机制; 若用户需要导出计算结果, 可以使用本 wolfram 库额外提供的 “scriptPath” 变量, 其表示 wolfram 代码的缓存文件夹; $\langle label \rangle$ 中不能包含 “:”, 其用于指定该命令的缓存标签, 该 $\langle label \rangle$ 在当前文档中必须是唯一的;

NOTE: 该环境的缓存机制存在缺陷, 具体表现为: 当 $\langle label \rangle$ 已经缓存 “X, Y” 两个 Hash 后, 用户再次更改环境内代码, 使得更改后的 Hash 恰好为 “X”, 此时编译只会得到 “Y” 对应的结果; 用户可以在该环境前添加 `\ztikzForceToRun` 命令刷新缓存.

```
\begin{wolframCode}{wolfram-code-1}
num = 2^110;
Export[scriptPath <> "wolfram-code_text.txt", num]
\end{wolframCode}
% load result
\[ 2^{\{110\}} = \input{\wolframOutputPath/wolfram-code_text.txt} \]
```

$$2^{110} = 1298074214633706907132624082305024$$

例 27

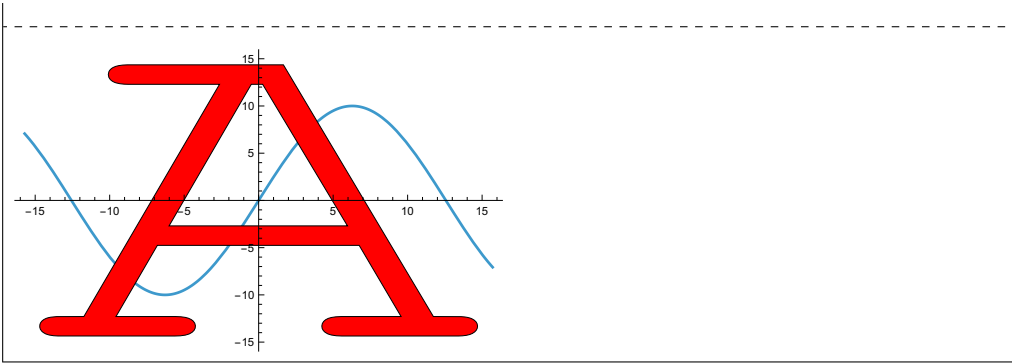
```
\begin{wolframCode}{wolfram-code-2}
fp1 = Plot[10 Sin[.25 x], {x, -5 Pi, 5 Pi}];

myChar = Style["A", FontFamily -> "Courier New", FontSize ->
50];
boundaryMesh = BoundaryDiscretizeGraphics[Text[myChar], _Text];
fp2 = Region[boundaryMesh, BaseStyle -> {FaceForm[Red],
EdgeForm[Black]}]

fig = Show[fp1, fp2, PlotRange->All];
Export[scriptPath <> "wolfram-code_fig.pdf", fig];
\end{wolframCode}

\includegraphics[width=.5\linewidth]{
\wolframOutputPath/wolfram-code_fig.pdf}
```

例 28



```
\g_ztikz_wolfram_res_seq *
```

New: 2026-05-07

该变量保存了前一次 WolframScript 的计算结果，普通用户请使用后续的 `\wolframResult` 命令.

5.6 l3draw 库

zTikZ 基于 l3draw 宏封装了一个 l3draw 库, 此库主要用于完成一些比较简单的绘图需求. 在普通用户层面: l3draw 库提供了 \zrule 和 \zplot 两个命令, 前者用于绘制渐变矩形, 后者用于绘制函数, 同样也支持渐变; zTikZ 也对 l3draw 提供的绘图环境与命令进行了简单的封装, 目前不是很完善, 且不稳定, 不推荐普通用户使用.

\zdrawSetUnit	\zdrawSetUnit[⟨unit⟩]
New: 2025-05-15	此命令用于设置当前绘图的单位, 例如 ⟨unit⟩ 可以取值为 “cm”.
\zdrawSetPathWidth	\zdrawSetPathWidth[⟨width⟩]
New: 2025-05-15	此命令用于设置当前绘图的线宽, 例如 ⟨width⟩ 可以取值为 “0.5pt”; l3draw 中默认的线径为 0.4pt.
\zrule	\zrule[⟨key-value⟩]
New: 2025-05-15	此命令用于绘制渐变矩形, ⟨key-value⟩ 用于设置渐变矩形的属性.

ztikz/zdraw/zrule/width	width	=	⟨浮点数⟩.....	初始值:	1
ztikz/zdraw/zrule/height	height	=	⟨浮点数⟩.....	初始值:	1
ztikz/zdraw/zrule/startColor	startColor	=	⟨颜色⟩.....	初始值:	red
ztikz/zdraw/zrule/endColor	endColor	=	⟨颜色⟩.....	初始值:	blue
ztikz/zdraw/zrule/step	step	=	⟨浮点数⟩.....	初始值:	0.25

⟨width⟩ 和 ⟨height⟩ 用于设置渐变矩形的宽度和高度; ⟨startColor⟩ 和 ⟨endColor⟩ 用于设置渐变矩形的起始颜色和结束颜色; ⟨step⟩ 用于控制渐变精度.

```
\zrule[width=10, startColor=red, step=1]
```



例 29

\zplot	\zplot[⟨key-value⟩]{⟨function⟩}
New: 2025-05-15	此命令用于绘制函数, 水平方向和垂直方向的渐变, ⟨key-value⟩ 用于设置函数的属性; ⟨function⟩ 为合法的函数表达式.

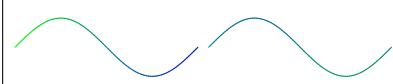
NOTE: 目前 \zplot 命令不太稳定, 在部分情况下可能会报错, 用户应该谨慎使用该命令.

ztikz/zdraw/zplot/action	action = $\langle \text{draw stroke fill clip shade} \rangle$初始值: draw
ztikz/zdraw/zplot/domain	domain = $\langle \text{浮点数, 浮点数, 浮点数} \rangle$初始值: -5, 0.1, 5
ztikz/zdraw/zplot/range	range = $\langle \text{浮点数, 浮点数} \rangle$初始值: -5, 5
ztikz/zdraw/zplot/startColor	startColor = $\langle \text{颜色} \rangle$初始值: black
ztikz/zdraw/zplot/endColor	endColor = $\langle \text{颜色} \rangle$初始值: white
ztikz/zdraw/zplot/axis	axis = $\langle \text{x y} \rangle$初始值: y

$\langle \text{action} \rangle$ 用于控制绘制的行为; $\langle \text{domain} \rangle$ 用于设置函数的自变量范围, 其中第一个浮点数为起始值, 第二个浮点数为步长, 第三个浮点数为结束值; $\langle \text{range} \rangle$ 用于设置 y 轴范围, 在 $\langle \text{action} \rangle = \text{shade}$ 时比较有用; $\langle \text{startColor} \rangle$ 和 $\langle \text{endColor} \rangle$ 用于设置函数的起始颜色和结束颜色; $\langle \text{axis} \rangle$ 用于设置渐变方式, ‘x’ 对应水平渐变, ‘y’ 对应垂直渐变.

```
\def\PI{3.1415926}
\zplot[
  domain={0, 0.02*\PI, 2*\PI},
  action=shade, startColor=blue,
  endColor=green, axis=x]{sin(x)}
\zplot[
  domain={0, 0.02*\PI, 2*\PI},
  action=shade, startColor=blue,
  endColor=green, axis=y]{sin(x)}
```

例 30



Zdraw	$\backslash \text{begin}\{zdraw\} \langle l3draw \text{ code} \rangle \backslash \text{end}\{zdraw\}$
New: 2025-05-15	此环境为 $\backslash \text{draw_begin}$: 和 $\backslash \text{draw_end}$: 的封装.
Zgroup	$\backslash \text{begin}\{zgroup\} \langle l3draw \text{ code} \rangle \backslash \text{end}\{zgroup\}$
New: 2025-05-15	此环境为 $\backslash \text{draw_path_scope_begin}$: 和 $\backslash \text{draw_path_scope_end}$: 的封装.
$\backslash \text{zmoveto}$	$\backslash \text{zmoveto}\{\langle \text{coordinate} \rangle\}$
$\backslash \text{zlineto}$	$\backslash \text{zlineto}\{\langle \text{coordinate} \rangle\}$
New: 2025-05-15	这两个命令用于移动当前画笔的坐标, $\langle \text{coordinate} \rangle$ 为 l3draw 中合法的坐标表达式. 比如 “1mm, 2cm+3em”.

$\text{\texttt{\backslash zscolor}}$	$\text{\texttt{\backslash zscolor}}\{\langle l3color \rangle\}$
$\text{\texttt{\backslash zfcolor}}$	$\text{\texttt{\backslash zfcolor}}\{\langle l3color \rangle\}$
New: 2025-05-15	$\langle l3color \rangle$ 为 $\text{\texttt{l3draw}}$ 中合法的颜色表达式; $\text{\texttt{zTiZ}}$ 对常见的颜色预定义了其对应的“ $\text{\texttt{l3color}}$ ”变量, 用户可以直接使用这部分颜色.
$\text{\texttt{\backslash zfevenodd}}$	命令 $\text{\texttt{\backslash zfevenodd}}$ 用于指定区域内外分割规则为 – “奇偶规则”; 命令 $\text{\texttt{\backslash zfnozero}}$
$\text{\texttt{\backslash zfnozero}}$	用于指定区域内外分割规则为 – “非零规则”,
New: 2025-05-31	
$\text{\texttt{\backslash zxvec}}$	$\text{\texttt{\backslash zxvec}}\{\langle coordinate \rangle\}$
$\text{\texttt{\backslash zyvec}}$	$\text{\texttt{\backslash zyvec}}\{\langle coordinate \rangle\}$
New: 2025-05-15	这两个命令用于设置当前坐标系的 x 轴和 y 轴的单位向量, $\langle coordinate \rangle$ 为合法的坐标表达式; 比如 “ 1mm , $2\text{cm}+3\text{em}$ ”.
$\text{\texttt{\backslash zpolar}}$	$\text{\texttt{\backslash zpolar}}\{\langle radius \rangle\}\{\langle angle \rangle\}$
$\text{\texttt{\backslash zcoor}}$	$\text{\texttt{\backslash zcoor}}\{\langle x-scale \rangle\}\{\langle y-scale \rangle\}$
New: 2025-05-15	$\text{\texttt{\backslash zpolar}}$ 命令按照极坐标的方式获取点的坐标: $\langle radius \rangle$ 为合法的长度, 如 “ 2em ”; $\langle angle \rangle$ 为浮点数; $\text{\texttt{\backslash zcoor}}$ 命令按照直角坐标的方式获取点的坐标: $\langle x-scale \rangle$ 为浮点数, $\langle y-scale \rangle$ 为浮点数; 此命令获取的最终坐标还取决于 x 和 y 方向两个基向量的影响, $(\langle x-scale \rangle, \langle y-scale \rangle)$ 也就是所谓的在基 $\{\text{\texttt{\backslash svec}}, \text{\texttt{\backslash yvec}}\}$ 下的坐标.
$\text{\texttt{\backslash zrect}}$	$\text{\texttt{\backslash zrect}}\{\langle coordinate \rangle\}\{\langle coordinate \rangle\}$
$\text{\texttt{\backslash zcirc}}$	$\text{\texttt{\backslash zcirc}}\{\langle center \rangle\}\{\langle radius \rangle\}$
New: 2025-05-15	前者用于绘制矩形, 两个坐标点分别为矩形的左下角和右上角; 后者用于绘制圆形, $\langle center \rangle$ 为圆心坐标, $\langle radius \rangle$ 为半径; $\langle coordinate \rangle$ 和 $\langle center \rangle$ 均为合法的坐标表达式, 比如 “ 1mm , $2\text{cm}+3\text{em}$ ”.
$\text{\texttt{\backslash znewtext}}$	$\text{\texttt{\backslash znewtext}}\langle coffin \rangle$
$\text{\texttt{\backslash zsethtext}}$	$\text{\texttt{\backslash zsethtext}}\langle coffin \rangle\{\langle content \rangle\}$
$\text{\texttt{\backslash zsetvtext}}$	$\text{\texttt{\backslash zsetvtext}}\langle coffin \rangle\{\langle width \rangle\}\{\langle content \rangle\}$
$\text{\texttt{\backslash zscaletext}}$	$\text{\texttt{\backslash zscaletext}}\langle coffin \rangle\{\langle x-scale \rangle\}\{\langle y-scale \rangle\}$
$\text{\texttt{\backslash zputtext}}$	$\text{\texttt{\backslash zputtext}}\langle coffin \rangle\{\langle hpole \rangle\}\{\langle vpole \rangle\}\{\langle point \rangle\}$
New: 2025-05-15	这系列命令用于在 $\text{\texttt{l3draw}}$ 中创建, 变换与放置文本.
$\text{\texttt{\backslash zbg}}$	这两个命令为 $\text{\texttt{\backslash draw_path_scope_begin}}$: 和 $\text{\texttt{\backslash draw_path_scope_end}}$: 的封装.
$\text{\texttt{\backslash zeg}}$	
New: 2025-05-15	

<code>\zcapbutt</code>	这系列命令用于设置线段之间的连接方式.
<code>\zcaproun</code>	
<code>\zcaprect</code>	
<code>\zclosepath</code>	

New: 2025-05-15

<code>\zshift</code>	<code>\zshift{<vector>}</code>
<code>\zxscale</code>	<code>\zxscale{<x-scale>}</code>
<code>\zyscale</code>	<code>\zyscale{<y-scale>}</code>
<code>\ztrans</code>	<code>\ztrans{<a>}{}{<c>}{<d>}</code>

New: 2025-05-15

这一系列的命令用于对坐标轴进行仿射变换, `\ztrans{a}{b}{c}{d}` 对应的仿射变换矩阵为:

$$\begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

<code>\zusepath</code>	<code>\zusepath[<style>]</code>
------------------------	---------------------------------------

New: 2025-05-15

此命令用于显示最终的路径, `<style>` 默认为 “draw”, 其余的可选值有: “stroke, fill, clip”.

6 附录

6.1 gnuplot 内置函数

我们在这里补充说明 gnuplot 中内建的函数: Arguments to math functions in gnuplot can be integer, real, or complex unless otherwise noted. Functions that accept or return angles (e.g. $\sin(x)$) treat angle values as radians, but this may be changed to degrees using the command `set angles`. (摘录自: [gnuplot support functions](#))

Table 1: gnuplot math library functions

Function	Arguments	Returns
$\text{abs}(x)$	any	$ x $, absolute value of x ; same type
$\text{abs}(x)$	complex	length of x , $\sqrt{\text{Re}(x)^2 + \text{Im}(x)^2}$
$\text{acos}(x)$	any	$\cos^{-1} x$ (inverse cosine)
$\text{acosh}(x)$	any	$\cosh^{-1} x$ (inverse hyperbolic cosine) in radians
$\text{airy}(x)$	any	Airy function $\text{Ai}(x)$
$\text{arg}(x)$	complex	the phase of x
$\text{asin}(x)$	any	$\sin^{-1} x$ (inverse sine)
$\text{asinh}(x)$	any	$\sinh^{-1} x$ (inverse hyperbolic sine) in radians
$\text{atan}(x)$	any	$\tan^{-1} x$ (inverse tangent)
$\text{atan2}(y, x)$	int or real	$\tan^{-1}(y/x)$ (inverse tangent)
$\text{atanh}(x)$	any	$\tanh^{-1} x$ (inverse hyperbolic tangent) in radians
$\text{EllipticK}(k)$	real k in $(-1 : 1)$	$K(k)$ complete elliptic integral of the first kind
$\text{EllipticE}(k)$	real k in $[-1 : 1]$	$E(k)$ complete elliptic integral of the second kind
$\text{EllipticPi}(n, k)$	real $n, k < 1$	$\Pi(n, k)$ complete elliptic integral of the third kind
$\text{besj0}(x)$	int or real	J_0 Bessel function of x , in radians
$\text{besj1}(x)$	int or real	J_1 Bessel function of x , in radians
$\text{besy0}(x)$	int or real	Y_0 Bessel function of x , in radians
$\text{besy1}(x)$	int or real	Y_1 Bessel function of x , in radians
$\text{ceil}(x)$	any	$\lceil x \rceil$, smallest integer not less than x (real part)
$\text{cos}(x)$	radians	$\cos x$, cosine of x
$\text{cosh}(x)$	any	$\cosh x$, hyperbolic cosine of x in radians
$\text{erf}(x)$	any	$\text{erf}(\text{Re}(x))$, error function of $\text{Re}(x)$
$\text{erfc}(x)$	any	$\text{erfc}(\text{Re}(x))$, 1.0– error function of $\text{Re}(x)$
$\text{exp}(x)$	any	e^x , exponential function of x

$\text{expint}(n, x)$	any	$E_n(x)$, exponential integral function of x
$\text{floor}(x)$	any	$\lfloor x \rfloor$, largest integer not greater than x (real part)
$\text{gamma}(x)$	any	$\Gamma(\text{Re}(x))$, gamma function of $\text{Re}(x)$
$\text{ibeta}(p, q, x)$	any	$\text{ibeta}(\text{Re}(p, q, x))$, ibeta function of $\text{Re}(p, q, x)$
$\text{inverf}(x)$	any	inverse error function $\text{Re}(x)$
$\text{igamma}(a, x)$	any	$\text{igamma}(\text{Re}(a, x))$, igamma function of $\text{Re}(a, x)$
$\text{imag}(x)$	complex	$\text{Im}(x)$, imaginary part of x as a real number
$\text{invnorm}(x)$	any	inverse normal distribution function $\text{Re}(x)$
$\text{int}(x)$	real	integer part of x , truncated toward zero
$\text{lambertw}(x)$	real	Lambert W function
$\text{lgamma}(x)$	any	$\text{lgamma}(\text{Re}(x))$, lgamma function of $\text{Re}(x)$
$\text{log}(x)$	any	$\ln x$, natural logarithm (base e) of x
$\text{log10}(x)$	any	$\log_{10} x$, logarithm (base 10) of x
$\text{norm}(x)$	any	$\text{norm}(x)$, normal distribution function of $\text{Re}(x)$
$\text{rand}(x)$	int	pseudo random number in the interval $(0 : 1)$
$\text{real}(x)$	any	$\text{Re}(x)$, real part of x
$\text{sgn}(x)$	any	1 if $x > 0$, -1 if $x < 0$, 0 if $x = 0$. $\Im(x)$ ignored
$\text{sin}(x)$	any	$\sin x$, sine of x
$\text{sinh}(x)$	any	$\sinh x$, hyperbolic sine of x in radians
$\text{sqrt}(x)$	any	$\sqrt{x}$, square root of x
$\text{tan}(x)$	any	$\tan x$, tangent of x
$\text{tanh}(x)$	any	$\tanh x$, hyperbolic tangent of x in radians
$\text{voigt}(x, y)$	real	convolution of Gaussian and Lorentzian
$\text{cerf}(z)$	complex	complex error function
$\text{cdawson}(z)$	complex	complex Dawson's integral
$\text{faddeeva}(z)$	complex	$w(z) = \exp(-z^2) \times \text{erfc}(-iz)$
$\text{erfi}(x)$	real	imaginary error function $\text{erfi}(x) = -i \times \text{erf}(ix)$
$\text{VP}(x, \sigma, \gamma)$	real	Voigt profile

备注: $\text{faddeeva}(z)$: rescaled complex error function

6.2 marker 样式

Ti $\text{\textcolor{brown}{k}}$ Z 中的可以使用的 Marker 样式表如下:

<code>\pgfuseplotmark{-}</code>	
<code>\pgfuseplotmark{ }</code>	
<code>\pgfuseplotmark{o}</code>	
<code>\pgfuseplotmark{asterisk}</code>	
<code>\pgfuseplotmark{star}</code>	
<code>\pgfuseplotmark{10-pointed star}</code>	
<code>\pgfuseplotmark{oplus}</code>	
<code>\pgfuseplotmark{oplus*}</code>	
<code>\pgfuseplotmark{otimes}</code>	
<code>\pgfuseplotmark{otimes*}</code>	
<code>\pgfuseplotmark{square}</code>	
<code>\pgfuseplotmark{square*}</code>	
<code>\pgfuseplotmark{triangle}</code>	
<code>\pgfuseplotmark{triangle*}</code>	
<code>\pgfuseplotmark{diamond}</code>	
<code>\pgfuseplotmark{diamond*}</code>	
<code>\pgfuseplotmark{halfdiamond*}</code>	
<code>\pgfuseplotmark{halfsquare*}</code>	
<code>\pgfuseplotmark{halfsquare right*}</code>	
<code>\pgfuseplotmark{halfsquare left*}</code>	
<code>\pgfuseplotmark{pentagon}</code>	
<code>\pgfuseplotmark{pentagon*}</code>	
<code>\pgfuseplotmark{Mercedes star}</code>	
<code>\pgfuseplotmark{Mercedes star flipped}</code>	
<code>\pgfuseplotmark{halfcircle}</code>	
<code>\pgfuseplotmark{halfcircle*}</code>	
<code>\pgfuseplotmark{heart}</code>	
<code>\pgfuseplotmark{text}</code>	

Figure 3: Ti $\text{\textcolor{brown}{k}}$ Z Marker Style

6.3 测试数据/代码

```
# Curve 0 of 1, 10 points
# Curve title: "f(x)"
# x y type
-3.14159 -0.00000 i
-2.44346 -0.64279 i
-1.74533 -0.98481 i
-1.04720 -0.86603 i
-0.34907 -0.34202 i
0.34907 0.34202 i
1.04720 0.86603 i
1.74533 0.98481 i
2.44346 0.64279 i
3.14159 0.00000 i
```

sine.data

```
\begin{pycode}{pycode_table.txt}
import numpy as np

# write file
with open ('pycode_table.txt', 'w') as file:
    file.write("\\begin{tabular}{p{3cm}ccc}\\n")
    file.write("\\hline\\n")
    file.write("number/function & \\sin$ & \\cos$ & \\tan$\\n")
    file.write("\\hline\\n")
    for i in range(1, 16):
        file.write(
            f"${i}$ & ${np.around(np.sin(i), decimals=4)}$ &
            ${np.around(np.cos(i), decimals=4)}$ & ${np.around(np.tan(i),
            decimals=4)}$\\n"
        )

    file.write("\\hline\\n")
    file.write("\\end{tabular}\\n")
\\end{pycode}
```

table.py.txt

7 TODO

$\text{\texttt{zTikZ}}$ 的开发暂且告一段落了, 这里列出部分将来可能会增加的功能 (☐ – 未完成; ☒ – 已完成; ☐ – 不考虑该功能):

- ☒ (已经有了 `luadraw` 宏包)实现类似 `tikz-3dplot` 的接口, 使用 $\text{\texttt{L\TeX 3}}$ 对其进行重写.
- ☒ (参考后续 “`wolframAny`” 环境的处理方法)增加 Matlab 脚本的调用接口, 或者直接使用其开源替代 `GNU Octave` ?
- ☒ (参考 `cache` 库一节中 “`mermaid`” 环境的实现)实现 `wolframAny` 环境, 该环境实现的功能类似 `pycode`.
- ☒ 2025-05-29-已完成:重写缓存机制对应的函数 `\ztikz_hash_if_change:nn`, 目前不够灵活 (或许直接使用 `robust-externalize` 宏包).
- ☒ 2025-05-29-已完成:针对 `cache` 库, 需要清除多余的 Hash 值: 例如某个环境/命令产生的原 Hash 值为 “A”, 对应环境/命令中的参数改变后, 其 Hash 值变为了 “B”, 那么此时需要清除原始的 “A”.
- ☒ 2025-05-29-已完成:实现 `\ztikzForceToSkip` 命令: 如何处理 `\xsim_file_write_start:nn` 环境? 如何处理多个 `\ztikzForceToRun` ? 也许使用 `\str_mdfive_hash:n` 会有帮助?
- ☐ 集成有关 `wstp` 的更多功能: 接管 $\text{\texttt{T\TeX}}$ 底层分页与盒子计算 (全局最优?), 索引 (Index) 与自定义排序生成, 复杂正则表达式与宏代码生成 (Metaprogramming),

8.1 ztikz.sty

```
1 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% 1
2 %% ztikz.sty % 2
3 %% Copyright 2024, 2025, 2026 Zongping Ding. % 3
4 % % 4
5 % This work may be distributed and/or modified under the conditions of the % 5
6 % LaTeX Project Public License, either version 1.3 of this license or any % 6
7 % later version. % 7
8 % The latest version of this license is in % 8
9 % http://www.latex-project.org/lppl.txt % 9
10 % and version 1.3 or later is part of all distributions of LaTeX % 10
11 % version 2005/12/01 or later. % 11
12 % % 12
13 % This work has the LPPL maintenance status `maintained'. % 13
14 % % 14
15 % The Current Maintainer of this work is Zongping Ding. % 15
16 % % 16
17 % This work consists of the files ztikz.sty, % 17
18 % the libraries: ztikz.library.gnuplot.tex, % 18
19 % ztikz.library.cache.tex, % 19
20 % ztikz.library.python.tex, % 20
21 % ztikz.library.wolfram.tex, % 21
22 % ztikz.library.l3draw.tex, % 22
23 % and the script files: ztikz.library.gnuscrypt.tex, % 23
24 % ztikz.library.pyscript.tex. % 24
25 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% 25
26 \NeedsTeXFormat{LaTeX2e} 26
27 \ProvidesExplPackage{ztikz}{2026/05/08} 27
28 {1.1.0}{A~pre-release~diagram~and~CAS~package} 28
29 29
30 30
31 % ----- 31
32 % basic tools 32
33 % ----- 33
34 \RequirePackage{ztool} 34
35 \RequirePackage{graphicx} 35
36 \__ztool_load_library:n {shell-escape, file-io} 36
37 \cs_new_protected:Npn \ztikz_hook_preamble_last:n #1 37
38 { \AddToHook{env/document/before}{#1} } 38
39 \cs_new_protected:Npn \ztikz_label_hook_preamble_last:nn #1#2 39
40 { \AddToHook{env/document/before}[#1]{#2} } 40
41 \cs_new_protected:Npn \ztikz_hook_doc_begin:n #1 41
42 { \AddToHook{begindocument}{#1} } 42
43 \cs_new_protected:Npn \ztikz_hook_doc_end:n #1 43
44 { \AddToHook{enddocument}{#1} } 44
45 45
```

```

46 % -----
47 %
48 % -----
49 % load module interface
50 % -----
51 \clist_new:N \g__ztikz_library_loaded_clist
52 \clist_gclear:N \g__ztikz_library_loaded_clist
53 \cs_new_nopar:Npn \__ztikz_load_library:n #1
54 {
55     \clist_map_inline:nn {#1} {
56         \clist_if_in:NnTF \g__ztikz_library_loaded_clist {##1} {
57             \msg_set:nnn {ztikz} {library-loaded}
58             {
59                 ztikz~library~"##1"~already~loaded,ignored~loading.
60                 \msg_line_context:
61             }
62             \msg_warning:nnn {ztikz} {library-loaded} {##1}
63         }{
64             \file_if_exist:nTF {library/ztikz.library.##1.tex}{
65                 \clist_gput_right:Nn \g__ztikz_library_loaded_clist {##1}
66                 \makeatletter\file_input:n {library/ztikz.library.##1.tex}
67             }{
68                 \msg_set:nnn {ztikz} {library-not-found} {ztikz~library~`##1'~not~found.}
69                 \msg_error:nnn {ztikz} {library-not-found} {##1}
70             }
71         }
72     }
73 }
74 \NewDocumentCommand\ztikzloadlib{m}
75 {
76     \__ztikz_load_library:n {#1}
77     \ExplSyntaxOff
78 }
79
80
81
82 % -----
83 % Init ztikz's Environment
84 % -----
85 \RequirePackage{xsimverb}
86 \cs_generate_variant:Nn \xsim_file_write_start:nn {ne}
87 \ztool_shell_mkdir:n {ztikz_output/}
88 \tl_const:Nn \g__ztikz_scripts_path_tl {ztikz_output/scripts}
89 \NewDocumentCommand\ztikzMkdir{m}{ \ztool_shell_mkdir:n {#1} }
90
91
92
93 % -----

```


142	{	142
143	\msg_set:nnn {ztikz} {wolfram-engine-not-found}	143
144	{ztikz's~'wolfram'~engine~option~`#1'~invalid,	144
145	~valid~engine~is~'wolframscript'~or~'mathics'.}	145
146	\msg_error:nn {ztikz} {wolfram-engine-not-found}	146
147	},	147
148	engine .initial:n = { wolfram },	148
149	cloud .choices:nn = { true, false }	149
150	{	150
151	\use:c { bool_gset_ \l_keys_choice_tl :N }	151
152	\g__ztikz_wolfram_cloud_bool	152
153	},	153
154	cloud .initial:n = { false },	154
155	cloud .default:n = { true },	155
156	}	156
157	\ProcessKeyOptions [ztikz]	157
158		158
159	% only 'wolfram' support 'cloud'	159
160	\bool_if:NT \g__ztikz_wolfram_cloud_bool	160
161	{	161
162	\tl_if_eq:VnT \g__ztikz_wolfram_engine_tl { mathics }	162
163	{	163
164	\msg_set:nnn {ztikz}{mathics-cloud}	164
165	{mathics~'cloud'~is~not~supported.}	165
166	\msg_error:nn {ztikz}{mathics-cloud}	166
167	}	167
168	}	168
169		169
170	% whether to show these debug info	170
171	\bool_if:NTF \g_ztikz_quiet_bool	171
172	{	172
173	\cs_new_protected:Npn \ztikz_term_info:n #1	173
174	{	174
175	}	175
176	\cs_new_protected:Npn \ztikz_term_info:e #1	176
177	{	177
178	}	178
179	}{	179
180	\cs_new_protected:Npn \ztikz_term_info:n #1	180
181	{	181
182	\iow_now:Nn \c_term_iow {#1}	182
183	}	183
184	\cs_new_protected:Npn \ztikz_term_info:e #1	184
185	{	185
186	\iow_now:Ne \c_term_iow {#1}	186
187	}	187
188	}	188
189		189

190		190
191		191
192	% -----	192
193	% cache, sed file and ztikz logo	193
194	% -----	194
195	\cs_generate_variant:Nn \ior_open:Nn { Ne }	195
196	\cs_generate_variant:Nn \iow_open:Nn { Ne }	196
197	\cs_generate_variant:Nn \tl_map_function:nN { eN }	197
198	\ztikz_hook_preamble_last:n	198
199	{	199
200	\clist_if_in:NnF \g__ztikz_library_loaded_clist {cache}	200
201	{	201
202	\tl_new:N \l__ztikz_current_hash_tl	202
203	\prg_new_conditional:Npnn \ztikz_if_run_again:nnn #1#2#3 {p, T, F, TF}	203
204	{	204
205	\bool_gset_true:N \g__hash_change_bool	205
206	\prg_return_true:	206
207	}	207
208	\prg_generate_conditional_variant:Nnn \ztikz_if_run_again:nnn	208
209	{ nen } { p, T, F, TF }	209
210	}	210
211	}	211
212	% sed script	212
213	\cs_new:Npn __ztikz_addto_script:nn #1#2	213
214	{% #1: file; #2: line; #3:new	214
215	\ztool_append_to_file:nn	215
216	{#1}{#2}	216
217	}	217
218	\cs_generate_variant:Nn __ztikz_addto_script:nn { ne, en, ee }	218
219	\ztikz_keys_define:nn { sed-script }	219
220	{	220
221	file .tl_set:N = \l__ztikz_sed_script_file_tl,	221
222	line .int_set:N = \l__ztikz_sed_script_line_int,	222
223	ori .tl_set:N = \l__ztikz_sed_script_ori_tl,	223
224	new .tl_set:N = \l__ztikz_sed_script_new_tl,	224
225	}	225
226	\cs_new:Npn __ztikz_sed_script:nnn #1#2#3	226
227	{% #1: file; #2: line; #3:new	227
228	\ztool_replace_file_line:nnn	228
229	{\g__ztikz_scripts_path_tl/#1}	229
230	{#2}{#3}	230
231	}	231
232	\cs_generate_variant:Nn __ztikz_sed_script:nnn {eee}	232
233	\cs_generate_variant:Nn __ztikz_sed_script:nnn {nne}	233
234	% ztik logo	234
235	\NewDocumentCommand\TikZ{}{Ti\textcolor{orange}{\textit{k}}Z}	235
236	\NewDocumentCommand\zTikZ{}{	236
237	{	237

238	<code>\ztool_scale_to_wd_and_ht:nnn {.9ex}{1.3ex}{</code>	238
239	<code>\ztool_rotate:nn {89}{\(\aleph\)}</code>	239
240	<code>}\kern-0.3423ex\hbox{\TikZ</code>	240
241	<code>}</code>	241
242	<code>\let\ztikz\zTikZ</code>	242
243	<code>\protected\def\HoLogo@zTikZ#1{\zTikZ</code>	243
244	<code>\protected\def\HoLogo@ztikz#1{\zTikZ</code>	244

8.2 Library

8.2.1 basic

```
1 \ProvidesExplFile{ztikz.library.basic.tex}{2025/09/22}
2 {1.0.0}{basic~library~for~ztikz}
3
4
5
6 % -----
7 %               basic packages
8 % -----
9 \RequirePackage{tikz}
10 \RequirePackage{etoolbox}
11 \patchcmd{\pgfutil@InputIfFileExists}{\input #1}{%
12   \@pushfilename
13   \xdef\@currname{#1}
14   \input #1
15   \@popfilename
16 }{}{}
17 \usetikzlibrary{arrows.meta}
18 \usetikzlibrary{intersections}
19 \usetikzlibrary{patterns}
20 \usetikzlibrary{plotmarks}
21 \usetikzlibrary{positioning}
22 \usetikzlibrary{shapes.geometric}
23 \usetikzlibrary{decorations.markings}
24 \usetikzlibrary{fadings}
25
26
27
28 % ==> coordinate basic components
29 \ztikz_keys_define:nn { point }
30 {
31   type      .str_set:N = \l__point_type_str,
32   type      .initial:n = { * },
33   radius    .dim_set:N = \l__point_radius_dim,
34   radius    .initial:n = { 1pt },
35   color     .tl_set:N = \l__point_color_tl,
36   color     .initial:n = { black },
37   opacity   .tl_set:N = \l__point_opacity_tl,
38   opacity   .initial:n = { 1 },
39   rotate    .fp_set:N = \l__point_rotate_angle,
40   rotate    .initial:n = { 0 },
41 }
42 \NewDocumentCommand\ShowPoint{ 0{}m0{}0{} }
43 {
44   \group_begin:
45   \exp_args:Nne \ztikz_keys_set:nn { point } { #1 }
```

```

46 \seq_set_split:Nnn \l__point_list_seq { ; }{#2}
47 \seq_set_split:Nnn \l__point_label_seq { ; }{#3}
48 \int_step_inline:nnnn {1}{1}
49 { \seq_count:N \l__point_list_seq }
50 {
51 \draw plot [
52 only~ marks,
53 mark = \str_use:N \l__point_type_str,
54 mark~ size = \dim_use:N \l__point_radius_dim,
55 mark~ options = {
56 rotate = \fp_use:N \l__point_rotate_angle,
57 opacity = \tl_use:N \l__point_opacity_tl,
58 color = \tl_use:N \l__point_color_tl,
59 ball~ color = \tl_use:N \l__point_color_tl,
60 }
61 ] coordinates{\seq_item:Nn \l__point_list_seq{##1}}
62 node[#4] {\seq_item:Nn \l__point_label_seq{##1}};
63 }
64 \group_end:
65 }
66 \NewDocumentCommand\ShowGrid{ 0{color=gray, very~ thin, step=1}m }
67 {
68 \seq_set_split:Nnn \l__grid_param_ii_seq { ; }{#2}
69 \draw[#1] \seq_item:Nn \l__grid_param_ii_seq{1}
70 grid \seq_item:Nn \l__grid_param_ii_seq{2};
71 }
72 % intersection
73 \NewDocumentCommand\ShowIntersection{ omm }
74 {
75 \seq_set_split:Nnn \l__intersection_num_seq { ; }{#2}
76 \path[name~ intersections={
77 of=\seq_item:Nn \l__intersection_num_seq{1}~
78 and~ \seq_item:Nn \l__intersection_num_seq{2}
79 }];
80 \int_step_inline:nnnn {1}{1}{#3}{
81 \ShowPoint[#1]{(intersection-##1)}
82 }
83 }
84 % polygon plot
85 \ztikz_keys_define:nn { polygon }
86 {
87 radius .fp_set:N = \l__polygon_radius_fp,
88 radius .initial:n = { 1 },
89 edgeColor .tl_set:N = \l__polygon_edge_color_tl,
90 edgeColor .initial:n = { black },
91 fillColor .tl_set:N = \l__polygon_fill_color_tl,
92 fillColor .initial:n = { },
93 fillOpacity .fp_set:N = \l__polygon_fill_opacity_fp,

```

```

94 fillOpacity .initial:n = { 0 },
95 rotate .fp_set:N = \l__polygon_rotate_angle,
96 rotate .initial:n = { 0 },
97 shift .tl_set:N = \l__polygon_shift_tl,
98 shift .initial:n = { (0,0) },
99 marker .tl_set:N = \l__polygon_marker_option_tl,
100 marker .initial:n = { },
101 }
102 \tl_new:N \l__ztikz_basic_poly_path_tl
103 \NewDocumentCommand\Polygon{ 0{}m }
104 {
105 \group_begin:
106 \ztikz_keys_set:nn { polygon } { #1 }
107 % strip '(' and ')'
108 \tl_replace_once:Nnn \l__polygon_shift_tl{({}}
109 \tl_replace_once:Nnn \l__polygon_shift_tl{)}}{}
110 \coordinate (mv) at (\tl_use:N \l__polygon_shift_tl);
111 % create polygon
112 \begin{scope}[shift=(mv), rotate=\fp_use:N \l__polygon_rotate_angle]
113 % arg require: #2 $\ge$ 3
114 \int_step_inline:nnn {1}{#2}{
115 % draw edges
116 \fp_set:Nn \l_angle_fp {360/#2*##1*\c_one_degree_fp}
117 \fp_set:Nn \l_angle_next_fp {360/#2*(##1+1)*\c_one_degree_fp}
118 \draw [\tl_use:N \l__polygon_edge_color_tl]
119 ( \fp_eval:n {\l__polygon_radius_fp*cos(\l_angle_fp)},
120 \fp_eval:n {\l__polygon_radius_fp*sin(\l_angle_fp)}
121 ) -- (
122 \fp_eval:n {\l__polygon_radius_fp*cos(\l_angle_next_fp)},
123 \fp_eval:n {\l__polygon_radius_fp*sin(\l_angle_next_fp)}
124 );
125 % fill polygon path
126 \int_compare:nNnTF {##1}<{#2}
127 {
128 \tl_put_right:Nn \l__ztikz_basic_poly_path_tl
129 { (p##1)-- }
130 }{
131 \tl_put_right:Nn \l__ztikz_basic_poly_path_tl
132 { (p##1)--cycle }
133 }
134 % mark coordinates
135 \coordinate (p##1) at (
136 \fp_eval:n {\l__polygon_radius_fp*cos(\l_angle_fp)},
137 \fp_eval:n {\l__polygon_radius_fp*sin(\l_angle_fp)}
138 );
139 }
140 % fill polygon (none-color -> opacity=1; or opacity=.75)
141 \tl_if_empty:NTF \l__polygon_fill_color_tl {

```

```

142 \fp_set:Nn \l__polygon_fill_opacity_fp {0}
143 }{
144 \fp_set:Nn \l__polygon_fill_opacity_fp {.75}
145 }
146 \fill [
147 \tl_use:N \l__polygon_fill_color_tl,
148 fill~opacity=\fp_use:N \l__polygon_fill_opacity_fp
149 ] \l__ztikz_basic_poly_path_tl;
150 % show markers
151 \int_step_inline:nnn {1}{#2}
152 {
153 \ShowPoint[\l__polygon_marker_option_tl]
154 { (p##1) }
155 }
156 \end{scope}
157 \group_end:
158 }
159
160
161 % ==> axis
162 \ztikz_keys_define:nn { axis }
163 {
164 % basic tick args
165 tickStart .fp_set:N = \l__start_fp,
166 tickStart .initial:n = { -5 },
167 tickEnd .fp_set:N = \l__end_fp,
168 tickEnd .initial:n = { 5 },
169 axisRotate .fp_set:N = \l__axis_rotate_angle,
170 axisRotate .initial:n = { 0 },
171 % tick dimension spec
172 mainStep .fp_set:N = \l__main_step_fp,
173 mainStep .initial:n = { 1.0 },
174 subStep .fp_set:N = \l__sub_step_fp,
175 subStep .initial:n = { 0.1 },
176 mainTickLabel .tl_set:N = \l__main_tick_label_tl,
177 mainTickLabel .initial:n = { \fp_use:N {\CurrentFp} },
178 tickLabelShift .dim_set:N = \l__tick_label_shift_dim,
179 tickLabelShift .initial:n = { 0pt },
180 mainTickLength .dim_set:N = \l__main_tick_length_dim,
181 mainTickLength .initial:n = { 4pt },
182 subTickLength .dim_set:N = \l__sub_tick_length_dim,
183 subTickLength .initial:n = { 2pt },
184 mainTickLabelPosition .tl_set:N = \l__main_tick_label_position_tl,
185 mainTickLabelPosition .initial:n = { below },
186 % color spec
187 axisColor .tl_set:N = \l__axis_color_tl,
188 axisColor .initial:n = { black },
189 mainTickColor .tl_set:N = \l__main_tick_color_tl,

```

```

190     mainTickColor      .initial:n = { black },
191     subTickColor       .tl_set:N = \l__sub_tick_color_tl,
192     subTickColor       .initial:n = { black },
193     mainTickLabelColor .tl_set:N = \l__main_tick_label_color_tl,
194     mainTickLabelColor .initial:n = { black },
195     % tick cross type spec
196     tickStyle          .choice:,
197     tickStyle/cross    .code:n = \tl_set:Nn \l__tick_spec_tl { cross },
198     tickStyle/above    .code:n = \tl_set:Nn \l__tick_spec_tl { above },
199     tickStyle/below    .code:n = \tl_set:Nn \l__tick_spec_tl { below },
200 }
201 % ticks style
202 \tl_new:N \l__tick_type_tl % `main' or `sub'
203 \tl_new:N \l__tick_spec_tl % `cross', `above' or `below'
204 \tl_new:N \l__tick_color_tl
205 \dim_new:N \l__tick_length_dim
206 \tl_new:N \l__node_text_tl
207 % draw ticks (main or sub)
208 \cs_new_protected:Npn \ztikz_draw_axis_ticks_cs:n #1
209 {
210     \str_case:NnT \l__tick_type_tl {
211         {main}{
212             \dim_set_eq:NN \l__tick_length_dim
213             \l__main_tick_length_dim
214             \tl_set:NV \l__tick_color_tl
215             \l__main_tick_color_tl
216             \tl_set:Nn \l__node_text_tl
217             { \tl_use:N \l__main_tick_label_tl }
218         }
219         {sub}{
220             \dim_set_eq:NN \l__tick_length_dim \l__sub_tick_length_dim
221             \tl_set:NV \l__tick_color_tl \l__sub_tick_color_tl
222             \tl_set:Nn \l__node_text_tl {}
223         }
224     }{}
225     \str_case:VnT \l__tick_spec_tl {
226         {cross}{
227             \draw[\tl_use:N \l__tick_color_tl]
228             (#1, 0)++(0, \dim_eval:n {\l__tick_length_dim/2})
229             -- ++(0, \dim_eval:n {-\l__tick_length_dim})
230             node[\tl_use:N \l__main_tick_label_position_tl]
231             {
232                 \textcolor{\tl_use:N \l__main_tick_label_color_tl}
233                 {\tl_use:N \l__node_text_tl}
234             };
235         }
236         {above}{
237             \draw[\tl_use:N \l__tick_color_tl] (#1, 0)

```

```

238      -- ++(0, \dim_eval:n {\l__tick_length_dim/2})
239      node[\tl_use:N \l__main_tick_label_position_tl]
240      {
241          \textcolor{\tl_use:N \l__main_tick_label_color_tl}
242          {\tl_use:N \l__node_text_tl}
243      };
244  }
245  {below}{
246      \draw[\tl_use:N \l__tick_color_tl] (#1, 0)
247      -- ++(0, \dim_eval:n {-\l__tick_length_dim/2})
248      node[\tl_use:N \l__main_tick_label_position_tl
249          =\dim_use:N \l__tick_label_shift_dim]
250      {
251          \textcolor{\tl_use:N \l__main_tick_label_color_tl}
252          {\tl_use:N \l__node_text_tl}
253      };
254  }
255  }{}
256 }
257 % draw axis
258 \fp_new:N \CurrentFp
259 \int_new:N \l__substep_num_int
260 \NewDocumentCommand\ShowAxis{O{}m}
261 {
262     \group_begin:
263     \ztikz_keys_set:nn { axis } { #1 }
264     \seq_set_split:Nnn \l__points_seq { ; }{#2}
265     \begin{scope}[rotate=\fp_use:N \l__axis_rotate_angle]
266     \draw[->, \tl_use:N \l__axis_color_tl]
267         \seq_item:Nn \l__points_seq{1}
268         -- \seq_item:Nn \l__points_seq{2};
269 % draw ticks
270 \fp_step_inline:nnnn
271     {\fp_eval:n {\l__start_fp}}
272     {\fp_use:N \l__main_step_fp}
273     {\fp_use:N \l__end_fp}
274     {
275         % main ticks
276         \tl_set:Nn \l__tick_type_tl {main}
277         \fp_gset:Nn \CurrentFp {##1}
278         \ztikz_draw_axis_ticks_cs:n {##1}
279         % sub ticks
280         \tl_set:Nn \l__tick_type_tl {sub}
281         \int_set:Nn \l__substep_num_int
282             {\fp_eval:n {floor(\l__main_step_fp/\l__sub_step_fp)}}
283         \fp_compare:nNnTF {##1}<{\fp_eval:n {floor(\l__end_fp)}}{
284             \fp_step_function:nnnN
285             {\fp_eval:n {##1+\l__sub_step_fp}}

```



```

286         {\fp_use:N \l__sub_step_fp}
287         {\fp_eval:n {##1+\l__substep_num_int*\l__sub_step_fp}}
288         \ztikz_draw_axis_ticks_cs:n
289     }{}
290 }
291 \end{scope}
292 \group_end:
293 }
294 \NewDocumentCommand{\xAxis}{0{-2}0{8}}
295 {
296     \ShowAxis[
297         tickStart=\fp_eval:n {#1+1},
298         tickEnd=\fp_eval:n {#2-0.75},
299         mainTickLabelPosition=below,
300         mainStep=1,          subStep=.25,
301         axisRotate=0,        axisColor=black,
302         mainTickColor=black, subTickColor=black,
303         mainTickLength=10pt, subTickLength=5pt,
304         tickLabelShift=0pt,  tickStyle=below,
305     ]{(#1, 0); (#2, 0)}
306 }
307 \NewDocumentCommand{\yAxis}{0{-2}0{8}}
308 {
309     \ShowAxis[
310         tickStart=\fp_eval:n {#1+1},
311         tickEnd=\fp_eval:n {#2-0.75},
312         mainStep=1,          subStep=.25,
313         axisRotate=90,       axisColor=black,
314         mainTickColor=black, subTickColor=black,
315         mainTickLength=10pt, subTickLength=5pt,
316         tickLabelShift=0pt,  tickStyle=above,
317         mainTickLabelPosition=left
318     ]{(#1, 0); (#2, 0)}
319 }
320
321
322 % ==> statistic plot function
323 \cs_new_protected:Npn \ztikz_statistic_plot_cs:nnnn #1#2#3#4
324 {% #1:starts option; #2:draw-keyval; #3:point-keyval; #4:filename
325     \tl_if_empty:nTF {#3}{\draw[#2] plot[#1] file {#4};}
326     {
327         \group_begin:
328         \keys_set:nn { ztikz / point } { #3 }
329         \draw[#2] plot [
330             % stairs options
331             #1,
332             % marker options
333             mark = \str_use:N \l__point_type_str,

```

```

334 mark~ size = \dim_use:N \l__point_radius_dim,
335 mark~ options = {
336     rotate = \fp_use:N \l__point_rotate_angle,
337     opacity = \tl_use:N \l__point_opacity_tl,
338     color = \tl_use:N \l__point_color_tl,
339     ball~ color = \tl_use:N \l__point_color_tl,
340 }
341 ] file {#4};
342 \group_end:
343 }
344 }
345 \cs_generate_variant:Nn \ztikz_statistic_plot_cs:nnnn {ennn}
346
347 % stairs plot
348 \seq_new:N \l__statistic_option_tl
349 \NewDocumentCommand\StairsPlot
350 { 0{plot-left;jump-left}0{color=black}0{}m }
351 {
352     \seq_set_split:Nnn \l__statistic_option_tl { ; }{#1}
353     \str_case:enF {\seq_item:Nn \l__statistic_option_tl{1}}{
354         {plot-left}{\tl_set:Nn \l__tmpa_tl {const~plot~mark~left}}
355         {plot-right}{\tl_set:Nn \l__tmpa_tl {const~plot~mark~right}}
356         {plot-mid}{\tl_set:Nn \l__tmpa_tl {const~plot~mark~mid}}
357         {}{\tl_set:Nn \l__tmpa_tl {}}
358     }{
359         \msg_new:nnn {ztikz}{ztikz-stairs-plot}
360         {current~stairs~plot~type~is:~'#1'~,~ invalide}
361         \msg_error:nn {ztikz}{ztikz-stairs-plot}
362     }
363     \str_case:enF {\seq_item:Nn \l__statistic_option_tl{2}}{
364         {jump-left}{\tl_set:Nn \l__tmpb_tl {jump~mark~left}}
365         {jump-right}{\tl_set:Nn \l__tmpb_tl {jump~mark~right}}
366         {jump-mid}{\tl_set:Nn \l__tmpb_tl {jump~mark~mid}}
367         {}{\tl_set:Nn \l__tmpb_tl {}}
368     }{
369         \msg_new:nnn {ztikz}{ztikz-stairs-plot}
370         {current~stairs~jump~type~is:~'#1'~,~ invalide}
371         \msg_error:nn {ztikz}{ztikz-stairs-plot}
372     }
373     \ztikz_statistic_plot_cs:ennn {\l__tmpa_tl,\l__tmpb_tl}
374     {#2}{#3}{#4}
375 }
376 % stem plot
377 \NewDocumentCommand\StemPlot{ 0{x}0{color=black}0{}m }
378 {
379     \str_case:enF {#1}{
380         {x}{\tl_set:Nn \l__tmpa_tl {ycomb}}
381         {y}{\tl_set:Nn \l__tmpa_tl {xcomb}}

```

382	<code>{o}{\tl_set:Nn \l__tmpa_tl {polar~ comb}}</code>	382
383	<code>{\tl_set:Nn \l__tmpa_tl {ycomb}}</code>	383
384	<code>{</code>	384
385	<code>\msg_new:nnn {ztikz}{ztikz-stem-plot}</code>	385
386	<code>{current~stem~plot~type~is::~'#1'~,~ invalide}</code>	386
387	<code>\msg_error:nn {ztikz}{ztikz-stem-plot}</code>	387
388	<code>}</code>	388
389	<code>\ztikz_statistic_plot_cs:ennn {\l__tmpa_tl}{#2}{#3}{#4}</code>	389
390	<code>}</code>	390
391	<code>% bar plot</code>	391
392	<code>\NewDocumentCommand\BarPlot{ 0{ybar}0{color=black}0{m }}</code>	392
393	<code>{</code>	393
394	<code>\str_case:enF {#1}{</code>	394
395	<code>{x}{\tl_set:Nn \l__tmpa_tl {ybar}}</code>	395
396	<code>{y}{\tl_set:Nn \l__tmpa_tl {xbar}}</code>	396
397	<code>{xc}{\tl_set:Nn \l__tmpa_tl {ybar~ interval}}</code>	397
398	<code>{yc}{\tl_set:Nn \l__tmpa_tl {xbar~ interval}}</code>	398
399	<code>{\tl_set:Nn \l__tmpa_tl {ybar}}</code>	399
400	<code>}</code>	400
401	<code>\msg_new:nnn {ztikz}{ztikz-bar-plot}</code>	401
402	<code>{current~bar~plot~type~is::~'#1'~,~ invalide}</code>	402
403	<code>\msg_error:nn {ztikz}{ztikz-bar-plot}</code>	403
404	<code>}</code>	404
405	<code>\ztikz_statistic_plot_cs:ennn {\l__tmpa_tl}{#2}{#3}{#4}</code>	405
406	<code>}</code>	406

8.2.2 gnuplot

```
1 \ProvidesExplFile{ztikz.library.gnuplot.tex}{2025/12/02}
2 {1.0.0}{gnuplot~library~for~ztikz}
3
4
5
6 % ==> init variables
7 \ztool_shell_mkdir:n {ztikz_output/gnuplot_data/}
8 \__ztikz_load_library:n {gnuscript}
9 \int_new:N \g__tikz_env_index_int
10 \int_new:N \g__gnu_data_index_int
11 \int_new:N \g__gnu_plotz_index_int
12 \int_gadd:Nn \g__gnu_plotz_index_int {1}
13 \tl_const:Nn \g__ztikz_gnu_path_tl {ztikz_output/gnuplot_data}
14 \pretocmd{\tikzpicture}{
15   \int_gincr:N \g__tikz_env_index_int
16   \int_gset:Nn \g__gnu_data_index_int {0}
17 }{}{}
18
19 % plot args
20 \tl_new:N \l__pairs_x_domain_tl
21 \tl_new:N \l__pairs_y_domain_tl
22 \ztikz_keys_define:nn { 2dplot }
23 {
24   domain .tl_set:N = \l__ztikz_plot_domain_tl,
25   domain .initial:n = { -5:5; },
26   style .tl_set:N = \l__ztikz_plot_style_tl,
27   marker .tl_set:N = \l__ztikz_plot_marker_tl,
28 }
29 \ztikz_keys_define:nn { 3dplot }
30 {
31   domain .tl_set:N = \l__ztikz_plotz_domain_tl,
32   domain .initial:n = {-5:5; -5:5},
33   palette .tl_set:N = \l__ztikz_plotz_palette_tl,
34   palette .initial:n = {rgbformulae~ 22,13,-31},
35   pm3d .bool_set:N = \l__ztikz_plotz_pm_bool,
36   pm3d .initial:n = {true},
37   pm3d .default:n = {true},
38   width .dim_set:N = \l__ztikz_plotz_width_dim,
39   width .initial:n = {0.75\linewidth},
40 }
41 \cs_new:Npn \__ztikz_pairs_domain_parse:w #1; #2\q_stop
42 {
43   \tl_set:Nn \l__pairs_x_domain_tl {#1}
44   \tl_if_empty:nTF {#2}
45     { \tl_set:Nn \l__pairs_y_domain_tl {*: *} }
46     { \tl_set:Nn \l__pairs_y_domain_tl {#2} }
```

```

47 }
48
49
50 % ==> plot precise
51 \bool_new:N \g__plot_precise_restore_bool
52 \bool_new:N \g__contour_precise_restore_bool
53 \bool_new:N \g__param_precise_restore_bool
54 \bool_new:N \g__polar_precise_restore_bool
55 \NewDocumentCommand\PlotPrecise{ smm }
56 {
57     \tl_if_in:nnF {plot, param, polar, contour}{#2}
58     {
59         \msg_new:nnn {ztikz}{ztikz-plot-type}
60         {Valid~plot~type~are:~'plot',~'param',~'polar'~and~'contour'}
61         \msg_error:nn {ztikz}{ztikz-plot-type}
62     }
63     \IfBooleanF{#1}{ \bool_gset_true:c {g__#2_precise_restore_bool} }
64     \tl_if_eq:nnTF {contour}{#2}{
65         \__ztikz_sed_script:nnn {contour_plot.gp}{2}{set~samples~#3}
66         \__ztikz_sed_script:nnn {contour_plot.gp}{3}{set~isosamples~#3,#3}
67     }{
68         \__ztikz_sed_script:nnn {#2_plot.gp}{3}{set~samples~#3}
69     }
70 }
71 % data plot, precise, style
72 \cs_new_protected:Npn \__ztikz_gnu_data_plot:nnn #1#2#3
73 {
74     \ztool_shell_escape:e {gnuplot~ \g__ztikz_scripts_path_tl/#1_plot.gp}
75     \int_gadd:Nn \g__gnu_data_index_int {1}
76     \tl_set:Ne \l__gnu_data_new_name_tl
77     {
78         gnu_data_\int_use:N \g__tikz_env_index_int
79         _\int_use:N \g__gnu_data_index_int.table
80     }
81     \tl_set:Ne \l__gnu_data_full_path_tl
82     { \g__ztikz_gnu_path_tl/\l__gnu_data_new_name_tl }
83     \ztool_shell_mv:ee
84     {\g__ztikz_gnu_path_tl/gnu_data.table}
85     {\l__gnu_data_full_path_tl}
86     \tl_if_empty:nTF {#3}
87     {
88         \draw[#2] plot[smooth] file {\l__gnu_data_full_path_tl};
89     }{
90         \group_begin:
91         \ztikz_keys_set:nn { point } { #3 }
92         \draw[#2] plot [
93             mark = \str_use:N \l__point_type_str,
94             mark~ size = \dim_use:N \l__point_radius_dim,

```

```
95         mark~ options = {
96             rotate = \fp_use:N \l__point_rotate_angle,
97             opacity = \tl_use:N \l__point_opacity_tl,
98             color = \tl_use:N \l__point_color_tl,
99             ball~ color = \tl_use:N \l__point_color_tl,
100         }
101     ] file {\l__gnu_data_full_path_tl};
102     \group_end:
103 }
104 \bool_if:cT {g__#1_precise_restore_bool}
105     { \PlotPrecise{#1}{100} }
106 }
107 \cs_generate_variant:Nn \__ztikz_gnu_data_plot:nnn {nee}
108
109 % ==> simple 2d function plot
110 \NewDocumentCommand\Plot{ 0{}m }
111 {
112     \group_begin:
113     \ztikz_keys_set:nn { 2dplot } { domain=-5:5, style={color=black}, marker= }
114     \ztikz_keys_set:nn { 2dplot } { #1 }
115     \__ztikz_sed_script:nne {plot_plot.gp}{8}{set~xr~[\l__ztikz_plot_domain_tl]}
116     \__ztikz_sed_script:nne {plot_plot.gp}{7}{f(x)~::~#2}
117     \__ztikz_gnu_data_plot:nee {plot}{\l__ztikz_plot_style_tl}{\l__ztikz_plot_marker_tl}
118     \group_end:
119 }
120
121 \NewDocumentCommand\ContourPlot{ 0{}m }
122 {
123     \group_begin:
124     \ztikz_keys_set:nn { 2dplot } { domain={-5:5}, style={color=black}, marker= }
125     \ztikz_keys_set:nn { 2dplot } { #1 }
126     \exp_last_unbraced:Nf \__ztikz_pairs_domain_parse:w \l__ztikz_plot_domain_tl\q_stop
127     \__ztikz_sed_script:nne {contour_plot.gp}{11}{set~xr~[\l__pairs_x_domain_tl]}
128     \__ztikz_sed_script:nne {contour_plot.gp}{12}{set~yr~[\l__pairs_y_domain_tl]}
129     \__ztikz_sed_script:nne {contour_plot.gp}{14}{f(x,~y)~::~#2}
130     \__ztikz_gnu_data_plot:nee {contour}{\l__ztikz_plot_style_tl}{\l__ztikz_plot_marker_tl}
131     \group_end:
132 }
133 \NewDocumentCommand\ParamPlot{ 0{domain=0:2*pi, style=black, marker=}m }
134 {
135     \group_begin:
136     \ztikz_keys_set:nn { 2dplot } { domain=0:2*pi, style=black, marker= }
137     \ztikz_keys_set:nn { 2dplot } { #1 }
138     \__ztikz_sed_script:nne {param_plot.gp}{8}{set~trange~[\l__ztikz_plot_domain_tl]}
139     \__ztikz_sed_script:nne {param_plot.gp}{9}{plot~#2}
140     \__ztikz_gnu_data_plot:nee {param}{\l__ztikz_plot_style_tl}{\l__ztikz_plot_marker_tl}
141     \group_end:
142 }
```

```

143 \NewDocumentCommand\PolarPlot{ 0{}m }
144 {
145     \group_begin:
146     \ztikz_keys_set:nn { 2dplot } { domain=0:2*pi, style=black, marker= }
147     \ztikz_keys_set:nn { 2dplot } { #1 }
148     \__ztikz_sed_script:nne {polar_plot.gp}{8}{set~trange~[\l__ztikz_plot_domain_tl]}
149     \__ztikz_sed_script:nne {polar_plot.gp}{9}{plot~#2}
150     \__ztikz_gnu_data_plot:nee {polar}{\l__ztikz_plot_style_tl}{\l__ztikz_plot_marker_tl}
151     \group_end:
152 }
153 \NewDocumentCommand\Plotz{ 0{}m }
154 {
155     \group_begin:
156     \ztikz_keys_set:nn { 3dplot } { #1 }
157     \bool_if:NTF \l__ztikz_plotz_pm_bool {\def\plotz@pm{with~pm3d}}{\def\plotz@pm{}}
158     \exp_last_unbraced:Nf \__ztikz_pairs_domain_parse:w \l__ztikz_plotz_domain_tl\q_stop
159     \__ztikz_sed_script:nne {3d_plot.gp}{18}{set~palette~\l__ztikz_plotz_palette_tl}
160     \__ztikz_sed_script:nne {3d_plot.gp}{23}{set~xr~[\l__pairs_x_domain_tl]}
161     \__ztikz_sed_script:nne {3d_plot.gp}{24}{set~yr~[\l__pairs_y_domain_tl]}
162     \__ztikz_sed_script:nne {3d_plot.gp}{25}{splot~#2~\plotz@pm}
163     \ztool_shell_escape:e {gnuplot~ ./ztikz_output/scripts/3d_plot.gp}
164     \tl_set:N \l__tmpa_tl {./ztikz_output/gnuplot_data/plot_3d_\int_use:N
\g__gnu_plotz_index_int.pdf}
165     \ztool_shell_mv:ne {./ztikz_output/gnuplot_data/plot_3d.pdf}{\l__tmpa_tl}
166     \includegraphics[width=\dim_use:N \l__ztikz_plotz_width_dim]{\l__tmpa_tl}
167     \int_gadd:Nn \g__gnu_plotz_index_int {1}
168     \group_end:
169 }
170
171
172 % ==> users' interface
173 \NewDocumentCommand{\currentTikzIndex}{}
174 {
175     \int_use:N \g__tikz_env_index_int
176 }
177 \def\gnudata#1
178 {
179     \tl_use:N \g__ztikz_gnu_path_tl/gnu_data_
180     \int_use:N \g__tikz_env_index_int _#1.table
181 }

```

8.2.3 cache

```
1 \ProvidesExplFile{ztikz.library.cache.tex}{2025/05/31}
2 {1.0.0}{cache~library~for~ztikz}
3
4
5
6 % ==> init cache
7 \clist_if_in:NnT \g__ztikz_library_loaded_clist {basic}
8 {
9   \ztool_shell_mkdir:n {ztikz_output/tikz_data/}
10   \usetikzlibrary{external}
11   \tikzexternalize[prefix=ztikz_output/tikz_data/]
12 }
13 \ztool_file_new:nn {\c_false_bool}{ztikz_output/ztikz.hash}
14
15
16 % ==> variables declaration
17 \ior_new:N \g__ztikz_file_ior
18 \tl_new:N \l__ztikz_current_hash_tl
19 \seq_new:N \g_ztikz_file_hash_seq
20 \seq_new:N \l__zcache_hash_label_seq
21 \seq_new:N \g__zcache_tmp_hash_seq
22 \tl_new:N \l__zcache_hash_label_tl
23 \tl_new:N \l__zcache_hash_hash_tl
24 \tl_new:N \g__zcache_latest_cache_label_tl
25 \bool_new:N \l__zcache_hash_label_miss_bool
26 \bool_new:N \g_ztikz_hash_nochg_run_bool
27 \bool_new:N \g_ztikz_hashchg_norun_bool
28 \bool_gset_false:N \g_ztikz_hashchg_norun_bool
29 \bool_gset_false:N \g_ztikz_hash_nochg_run_bool
30 \cs_generate_variant:Nn \ztikz_file_read_lines:n { e }
31
32
33 % ==> cache function
34 \prg_set_conditional:Npnn \ztikz_if_run_again:nnn #1#2#3 { p, T, F, TF }
35 {% #1:true/false; #2:True-->file, False-->str; #3:label
36   \tl_gset:Nn \g__zcache_latest_cache_label_tl { #3 }
37   \__zcache_hash_get:nn {#1} {#2}
38   \edef\zcache@flag
39   {
40     \tl_map_function:nN {
41       \g_ztikz_hashchg_norun_bool
42       \g_ztikz_hash_nochg_run_bool
43     } \int_eval:n
44   }
45   \exp_args:Ne \int_case:nnF
46   { \exp_not:N \int_from_bin:n {\zcache@flag} }
```



```

47 {
48 {0}{
49 \seq_if_in:NVTF \g_ztikz_file_hash_seq \l__ztikz_current_hash_tl
50 {
51 \ztikz_term_info:n {CURRENT~HASH~ALREADY~EXISTS}
52 \prg_return_false:
53 }{
54 \ztikz_term_info:n {CURRENT~HASH~IS~UNIQUE:RECORDING...}
55 \__zcache_hash_add:nn { #3 }{ \l__ztikz_current_hash_tl }
56 \prg_return_true:
57 }
58 }
59 {1}{
60 \ztikz_term_info:n {FORCE~TO~RUN~AGAIN~...}
61 \bool_gset_false:N \g_ztikz_hash_nochg_run_bool
62 \prg_return_true:
63 }
64 {2}{
65 \ztikz_term_info:n {FORCE~TO~SKIP~...}
66 \bool_gset_false:N \g_ztikz_hashchg_norun_bool
67 \__zcache_hash_extract_by_label:nnn
68 { ztikz_output/ztikz.hash }{ #3 }{-1}
69 \prg_return_false:
70 }
71 }{ \relax }
72 }
73 \cs_new_protected:Npn \__zcache_hash_get:nn #1#2
74 {% #1:true/false; #2:True-->file, False-->str;
75 \bool_if:nTF {#1}
76 { \file_get_md5five_hash:nN {#2} \l__ztikz_current_hash_tl }
77 { \tl_set:Nn \l__ztikz_current_hash_tl {#2} }
78 \tl_set_rescan:Nne \l__ztikz_current_hash_tl
79 { \cctab_select:N \c_initex_cctab }
80 { \l__ztikz_current_hash_tl }
81 \__zcache_hash_extract_all:nN
82 { ztikz_output/ztikz.hash }
83 \g_ztikz_file_hash_seq
84 \seq_gremove_duplicates:N \g_ztikz_file_hash_seq
85 \ztikz_term_info:e
86 {
87 \iow_newline:
88 CURRENT~FILE's~HASH:\l__ztikz_current_hash_tl
89 }
90 }
91 \cs_new_protected:Npn \__zcache_hash_extract_by_label:nnn #1#2#3
92 {% #1:file; #2:label; #3:index
93 \ztool_read_file_as_seq:nnN
94 { \c_false_bool }{ #1 }

```

95	\l_tmpa_seq	95
96	\bool_set_true:N \l__zcache_hash_label_miss_bool	96
97	\seq_map_inline:Nn \l_tmpa_seq	97
98	{	98
99	\zcache_hash_label_extract:nnN { label }{ ##1 }	99
100	\l__zcache_hash_label_tl	100
101	\zcache_hash_label_extract:nnN { hash }{ ##1 }	101
102	\l__zcache_hash_hash_tl	102
103	\tl_if_eq:NnT \l__zcache_hash_label_tl { #2 }	103
104	{	104
105	\bool_set_false:N \l__zcache_hash_label_miss_bool	105
106	\tl_set:Ne \l__ztikz_current_hash_tl	106
107	{	107
108	\clist_item:en {\l__zcache_hash_hash_tl}{#3}	108
109	}	109
110	}	110
111	}	111
112	\bool_if:NT \l__zcache_hash_label_miss_bool	112
113	{	113
114	\msg_set:nnn {ztikz}{hash-label-missing}	114
115	{	115
116	current~hash~label~'#2'~not~found~in~'ztikz.hash',~	116
117	do~NOT~change~the~hash~label~in~any~respect~after~cache.	117
118	}	118
119	\msg_error:nn {ztikz}{hash-label-missing}	119
120	}	120
121	}	121
122	\prg_generate_conditional_variant:Nnn \ztikz_if_run_again:nnn	122
123	{ nen } { T, F, TF }	123
124		124
125	% extract hash or label	125
126	\cs_new_protected:Npn __zcache_hash_add:nn #1#2	126
127	{% #1:label, #2:hash	127
128	\seq_clear:N \l_tmpa_seq	128
129	\seq_clear:N \l_tmpb_seq	129
130	\seq_clear:N \l__zcache_hash_label_seq	130
131	\ztool_read_file_as_seq:nnN	131
132	{ \c_false_bool }	132
133	{ ztikz_output/ztikz.hash }	133
134	\l_tmpa_seq	134
135	\seq_map_inline:Nn \l_tmpa_seq	135
136	{	136
137	\zcache_hash_label_extract:nnN { label }{ ##1 } \l_tmpa_tl	137
138	\seq_put_right:NV \l__zcache_hash_label_seq \l_tmpa_tl	138
139	\tl_if_eq:NnTF \l_tmpa_tl { #1 }	139
140	{ \seq_put_right:Ne \l_tmpb_seq {##1, #2} }	140
141	{ \seq_put_right:Ne \l_tmpb_seq {##1} }	141
142	}	142

```

143 \seq_if_in:NnF \l__zcache_hash_label_seq { #1 }
144 { \seq_put_right:Ne \l_tmpb_seq { #1:#2 } }
145 \ztool_write_seq_to_file:nNn { \c_true_bool }
146 \l_tmpb_seq { ztikz_output/ztikz.hash }
147 }
148 \cs_new_protected:Npn \__zcache_hash_extract_all:nN #1#2
149 {% #1:file; #2:seq
150 \seq_clear:N \l_tmpa_seq
151 \clist_clear:N \l_tmpa_clist
152 \ztool_read_file_as_seq:nnN
153 { \c_false_bool }{ #1 }
154 \l_tmpa_seq
155 \seq_map_inline:Nn \l_tmpa_seq
156 {
157 \zcache_hash_label_extract:nnN { hash }{ ##1 } \l_tmpa_tl
158 \seq_gset_from_clist:NN \g__zcache_tmp_hash_seq \l_tmpa_tl
159 \clist_put_right:NV \l_tmpa_clist \l_tmpa_tl
160 }
161 \seq_set_from_clist:NN \l_tmpb_seq \l_tmpa_clist
162 \seq_remove_duplicates:N \l_tmpb_seq
163 \seq_gset_eq:NN #2 \l_tmpb_seq
164 }
165 \cs_set:Npn \zcache_cache_hash_last:nnn #1#2#3
166 {% #1:file; #2:label; #3:index
167 \seq_clear:N \l_tmpa_seq
168 \clist_clear:N \l_tmpa_clist
169 \ztool_read_file_as_seq:nnN
170 { \c_false_bool }{ #1 }
171 \l_tmpa_seq
172 \seq_map_inline:Nn \l_tmpa_seq
173 {
174 \zcache_hash_label_extract:nnN { label }{ ##1 }
175 \l__zcache_hash_label_tl
176 \zcache_hash_label_extract:nnN { hash }{ ##1 }
177 \l__zcache_hash_hash_tl
178 \tl_if_eq:NnT \l__zcache_hash_label_tl { #2 }
179 {
180 \clist_item:en {\l__zcache_hash_hash_tl}{ #3 }
181 \seq_map_break:
182 }
183 }
184 }
185 \cs_generate_variant:Nn \zcache_cache_hash_last:nnn { nen, eee }
186 \ztikz_keys_define:nn { cache/hash }
187 {
188 label .tl_set:N = \l__cache_hash_label_user_tl,
189 label .initial:e = { \g__zcache_latest_cache_label_tl },
190 file .tl_set:N = \l__cache_hash_file_user_tl,

```

```

191 file .initial:e = { ztikz_output/ztikz.hash },
192 index .int_set:N = \l__cache_hash_index_user_int,
193 index .initial:n = { -1 },
194 }
195 \newcommand{\ztikzCachedHash}[1] []
196 {
197   \group_begin:
198     \ztikz_keys_set:nn { cache/hash } { #1 }
199     \zcache_cache_hash_last:eee
200     { \l__cache_hash_file_user_tl }
201     { \l__cache_hash_label_user_tl }
202     { \int_use:N \l__cache_hash_index_user_int }
203   \group_end:
204 }
205 \cs_set_protected:Npn \zcache_hash_label_extract:nnN #1#2#3
206 {
207   \tl_clear:N \l_tmpa_tl
208   \tl_set_rescan:Nnn \l_tmpa_tl
209   { \cctab_select:N \c_code_cctab }
210   { #2 }
211   \exp_args:NNe \tl_set:Ne #3
212   {
213     \exp_not:c {__hash_#1_extract:w}
214     \l_tmpa_tl \exp_not:N \q_stop
215   }
216 }
217 \cs_set:Npn \__hash_label_extract:w #1:#2\q_stop
218 { #1 }
219 \cs_set:Npn \__hash_hash_extract:w #1:#2\q_stop
220 { #2 }
221
222
223 % ==> clear cache hash
224 \cs_new_protected:Npn \ztikz_clear_hash:
225 {
226   \iow_open:Nn \g__ztikz_file_ior {ztikz_output/ztikz.hash}
227   \ior_close:N \g__ztikz_file_ior
228 }
229 \NewDocumentCommand{\ztikzHashClean}{s}{
230   \ztikz_clear_hash:
231   \ztikz_term_info:n {CLEAN~ALL~CACHED~HASH~SUCCESSFULLY...}
232 }
233 \NewDocumentCommand{\ztikzHashCurrent}{s+0{,}}{
234   \IfBooleanTF{#1}
235   { \tl_use:N \l__ztikz_current_hash_tl }
236   { \seq_use:Nn \g_ztikz_file_hash_seq {#2} }
237 }
238

```

239		239
240	% ==> override the cache mechanism	240
241	<code>\NewDocumentCommand{\ztikzForceToSkip}{}{</code>	241
242	<code>{</code>	242
243	<code>\bool_gset_false:N \g_ztikz_hash_nochg_run_bool</code>	243
244	<code>\bool_gset_true:N \g_ztikz_hashchg_norun_bool</code>	244
245	<code>}</code>	245
246	<code>\NewDocumentCommand{\ztikzForceToRun}{}{</code>	246
247	<code>{</code>	247
248	<code>\bool_gset_false:N \g_ztikz_hashchg_norun_bool</code>	248
249	<code>\bool_gset_true:N \g_ztikz_hash_nochg_run_bool</code>	249
250	<code>}</code>	250

8.2.4 python

```
1 \ProvidesExplFile{ztikz.library.python.tex}{2025/08/10}
2 {1.0.0}{python~library~for~ztikz}
3
4
5
6 % ==> writing scripts
7 \__ztikz_load_library:n {pyscript}
8 \ztool_shell_mkdir:n {ztikz_output/python_data/}
9 \tl_const:Nn \g__ztikz_python_path_tl {ztikz_output/python_data}
10 \ior_new:N \g__file_read_ior
11 \tl_new:N \g__file_content_tl
12
13
14 % ==> core functions
15 \cs_new_protected:Npn \zlatex_Readlines_cs:nn #1#2
16 {
17   \ior_open:Nn \g__file_read_ior {#2}
18   \str_case:nnF {#1}{
19     {raw}{
20       \ior_get:NN \g__file_read_ior \g__file_content_tl
21     }
22     {str}{
23       \ior_str_get:NN \g__file_read_ior \g__file_content_tl
24     }
25   }{}
26   \tl_use:N \g__file_content_tl
27 }
28 \cs_generate_variant:Nn \zlatex_Readlines_cs:nn {ee}
29
30
31 % ==> users' interface
32 % python-matplotlib
33 \NewDocumentEnvironment{pyfig}{ mm }
34 {% #1:label; #2:file name
35   \xsim_file_write_start:ne {\c_true_bool}{\g__ztikz_python_path_tl/t@mp.py}
36 }{
37   \xsim_file_write_stop:
38   \ztikz_if_run_again:nenTF {\c_true_bool}{\g__ztikz_python_path_tl/t@mp.py}{#1}
39   {
40     \__ztikz_addto_script:nn {\g__ztikz_python_path_tl/t@mp.py}
41     { plt.savefig('#2') }
42     \ztool_shell_escape:e {python~\g__ztikz_python_path_tl/t@mp.py}
43     \ztool_shell_mv:ee
44     { \g__ztikz_python_path_tl/t@mp.py }
45     { \g__ztikz_python_path_tl/pyfig~\l__ztikz_current_hash_tl.py }
46     \ztool_shell_mv:ee
```

```

47      { #2 }
48      { \g__ztikz_python_path_tl/\l__ztikz_current_hash_tl _#2 }
49      \ztikz_term_info:e
50      {
51          writing~source~to~file:'\g__ztikz_python_path_tl/
52              pyfig_\l__ztikz_current_hash_tl.py'
53      }
54  }{
55      \ztikz_term_info:e
56      {
57          skip~recompile~of~python,~use~the~
58          cache~picture:'\l__ztikz_current_hash_tl _#2'
59      }
60  }
61  \xdef\pyfigOutputFile{ \g__ztikz_python_path_tl/\l__ztikz_current_hash_tl _#2 }
62  }
63
64  % inline python command
65  \NewDocumentCommand\py{0{raw}m}
66  {
67      \__ztikz_sed_script:nne {python_script.py}{6}{Float_res~~~\tl_to_str:n {#2}}
68      \ztikz_term_info:e {using~python~float~module~calculating...}
69      \ztool_shell_escape:e {python~ \g__ztikz_scripts_path_tl/python_script.py}
70      \zlatex_Readlines_cs:ee {#1}{\g__ztikz_python_path_tl/PyFloat.out}
71      % ---> cause bug that can't write ToC to file
72      % \cs{iow_close:N} \cs{g__file_read_ior} leads to bug ??
73  }
74  % python-sympy
75  \NewDocumentCommand\sympy{ mm }
76  {
77      \__ztikz_sed_script:nne {sympy_script.py}{8}{F_res~~~\tl_to_str:n {#2}}
78      \ztikz_if_run_again:nenTF {\c_true_bool}{\g__ztikz_scripts_path_tl/sympy_script.py}{#1}
79      {
80          \ztool_shell_escape:e {python~ \g__ztikz_scripts_path_tl/sympy_script.py}
81          \ztool_shell_mv:ee
82              {\g__ztikz_python_path_tl/sympy.out}
83              {\g__ztikz_python_path_tl/sympy_\l__ztikz_current_hash_tl.out}
84          \ztikz_term_info:e {using~python~sympy~calculating~question~...}
85          \exp_args:Ne \input{\g__ztikz_python_path_tl/sympy_\l__ztikz_current_hash_tl.out}
86      }{
87          \exp_args:Ne \input{\g__ztikz_python_path_tl/sympy_\l__ztikz_current_hash_tl.out}
88          \ztikz_term_info:e {skip~recompile,~using~the~cache~sympy~result:~ ✓
89      \l__ztikz_current_hash_tl}
90      }
91  }
92  % python-code-env
93  \NewDocumentEnvironment{pycode}{ mm }
94  {% #1:label; #2:output file name (with ext)

```

94	\xsim_file_write_start:ne {\c_true_bool}{\g__ztikz_python_path_tl/t@mp.py}	94
95	}{	95
96	\xsim_file_write_stop:	96
97	\ztikz_if_run_again:nenTF {\c_true_bool}{\g__ztikz_python_path_tl/t@mp.py}{#1}	97
98	{	98
99	\ztool_shell_escape:e {python~\g__ztikz_python_path_tl/t@mp.py}	99
100	\ztool_shell_mv:ee	100
101	{ \g__ztikz_python_path_tl/t@mp.py }	101
102	{ \g__ztikz_python_path_tl/pycode_\l__ztikz_current_hash_tl.py }	102
103	\ztool_shell_mv:ee	103
104	{ #2 }	104
105	{ \g__ztikz_python_path_tl/\l__ztikz_current_hash_tl _#2 }	105
106	\ztikz_term_info:e	106
107	{	107
108	writing~source~to~file:'\g__ztikz_python_path_tl/	108
109	pycode_\l__ztikz_current_hash_tl.py'	109
110	}	110
111	}{	111
112	\ztikz_term_info:e	112
113	{	113
114	skip~recompile~of~python,~use~the~cache~result:	114
115	'\l__ztikz_current_hash_tl _#2'	115
116	}	116
117	}	117
118	\xdef\pycodeOutputFile{\g__ztikz_python_path_tl/\l__ztikz_current_hash_tl _#2}	118
119	}	119

8.2.5 wolfram

```
1 \ProvidesExplFile{ztikz.library.wolfram.tex}{2026/05/08}
2 {1.2.0}{wolfram~library~for~ztikz}
3
4
5
6 % ==> init variables
7 \ztool_shell_mkdir:n {ztikz_output/mma_data/}
8 \tl_const:Nn \g__ztikz_wolfram_path_tl {ztikz_output/mma_data}
9 \newcommand\wolframOutputPath{\g__ztikz_wolfram_path_tl}
10
11 \tl_new:N \l_part_table_data_tl
12 \tl_new:N \l_full_table_data_tl
13 \tl_new:N \l__wolfram_current_hash_tl
14 \tl_new:N \l__ztikz_wolfram_tmp_arg_tl
15 \tl_new:N \l__ztikz_wolfram_tmp_res_tl
16 \seq_new:N \g_ztikz_wolfram_res_seq
17 \ior_new:N \g__ztikz_wolfram_ior
18 \iow_new:N \g__ztikz_wolfram_iow
19
20
21 % ==> core function
22 \cs_new_protected:Npn \__ztikz_wolfram_execute:nnnn #1#2#3#4
23 {% #1:contents(empty->not add); #2:extension; #3:output object; #4:label
24   \__ztikz_wolfram_execute_prehook:nnn { #1 }{ #2 }{ #3 }
25   \__ztikz_wolfram_execute_cli:nn { #2 }{ #4 }
26   \__ztikz_wolfram_execute_posthook:n { #2 }
27 }
28 \cs_generate_variant:Nn \__ztikz_wolfram_execute:nnnn { e, o }
29
30 % wolfram execute aux functions
31 \cs_new:Npn \__ztikz_wolfram_execute_prehook:nnn #1#2#3
32 {
33   \edef\@wolfram@tmp@file{\g__ztikz_wolfram_path_tl/t@mp}
34
35   % add additional wolfram code - export process
36   \tl_if_empty:nF {#1}{
37     \__ztikz_addto_script:en { \@wolfram@tmp@file.wls }{ #1 }
38   }
39
40   % add additional wolfram code - script path
41   \tl_if_empty:nT { #3 }
42   {
43     \exp_args:Neee \ztool_insert_to_file:nnn
44       { \@wolfram@tmp@file.wls }{ 1 }
45       { scriptPath~=""\g__ztikz_wolfram_path_tl/" }
46   }
```

```

47 % get script hash
48 \file_get_md5five_hash:nN { \@wolfram@tmp@file.wls }\l__ztikz_current_hash_tl
49 \file_get_md5five_hash:nN { \@wolfram@tmp@file.wls }\l__wolfram_current_hash_tl
50 \tl_set_rescan:Nne \l__wolfram_current_hash_tl
51 { \cctab_select:N \c_initex_cctab }
52 { \l__wolfram_current_hash_tl }
53 \xdef\wolfram@tmp@file{\g__ztikz_wolfram_path_tl/\l__ztikz_current_hash_tl}
54 \ztool_shell_mv:ee {\@wolfram@tmp@file.wls}{\wolfram@tmp@file.wls}
55
56
57 % add additional wolfram code - general export
58 \tl_if_empty:nF { #3 }
59 {
60     \__ztikz_addto_script:ee { \wolfram@tmp@file.wls }
61     { Export["\wolfram@tmp@file.#2", #3] }
62     \xdef\wolframOutputFile {
63         \g__ztikz_wolfram_path_tl / \l__ztikz_current_hash_tl.#2
64     }
65 }
66 }
67
68 \cs_new:Npn \__ztikz_wolfram_execute_posthook:n #1
69 {
70     \ior_open:Ne \g__ztikz_wolfram_ior { \wolframOutputFile }
71     \ior_get:NN \g__ztikz_wolfram_ior \l__ztikz_wolfram_tmp_res_tl
72     \tl_set_rescan:Nno \l__ztikz_wolfram_tmp_res_tl
73     { \cctab_select:N \c_document_cctab }
74     { \l__ztikz_wolfram_tmp_res_tl }
75     \exp_args:NNe \seq_gset_split:NnV \g__ztikz_wolfram_res_seq
76     { \c_atsign_str } \l__ztikz_wolfram_tmp_res_tl
77     \ior_close:N \g__ztikz_wolfram_ior
78 }
79
80 \cs_new:Nn \__ztikz_wolfram_case_run:
81 {
82     \tl_if_eq:VnTF \g__ztikz_wolfram_engine_tl { wstp }
83     {
84         % NOTE: 'wstp' is much faster than the other engines
85         \edef\wolfram@script
86         {
87             wstp~ '<<< "\wolfram@tmp@file.wls"'
88         }
89     }
90     {
91         \edef\wolfram@script
92         {
93             \g__ztikz_wolfram_engine_tl\space
94             \bool_if:NT \g__ztikz_wolfram_cloud_bool {-cloud\space}

```

95	-script\space	95
96	\wolfram@tmp@file.wls	96
97	}	97
98	}	98
99	\ztikz_term_info:e	99
100	{	100
101	\g__ztikz_wolfram_engine_tl\space~is~running~on:	101
102	'\wolfram@tmp@file.wls'...\iow_newline:	102
103	}	103
104	\ztool_shell_escape:e { \wolfram@script }	104
105	}	105
106	\cs_new:Npn __ztikz_wolfram_case_skip:n #1	106
107	{% #1: ext	107
108	% remove the skip wolframscript file and result	108
109	\clist_if_in:NnT \g__ztikz_library_loaded_clist {cache}	109
110	{	110
111	\tl_if_eq:enT {\zcache@flag}{10}	111
112	{	112
113	\seq_if_in:NVF \g_ztikz_file_hash_seq \l__wolfram_current_hash_tl	113
114	{	114
115	\ztikz_term_info:e {Removing~result~of~hash:\l__wolfram_current_hash_tl}	115
116	\ztool_shell_rm:e {\wolfram@tmp@file.wls}	116
117	\ztool_shell_rm:e {\wolfram@tmp@file.#1}	117
118	}	118
119	}	119
120	}	120
121	\ztikz_term_info:e	121
122	{	122
123	Use~cache~result:'\g__ztikz_wolfram_path_tl/	123
124	\l__ztikz_current_hash_tl.#1'\iow_newline:	124
125	}	125
126	}	126
127	\cs_new:Npn __ztikz_wolfram_execute_cli:nn #1#2	127
128	{% #1: ext; #2: hash label	128
129	\ztikz_if_run_again:nenTF { \c_false_bool }{ \l__ztikz_current_hash_tl }{ #2 }	129
130	{	130
131	__ztikz_wolfram_case_run:	131
132	}{	132
133	__ztikz_wolfram_case_skip:n { #1 }	133
134	}	134
135	}	135
136		136
137		137
138	% ==> user interface	138
139	\msg_set:nnn {ztikz}{wolfram-arg_empty}	139
140	{ wolfram~library~error:~argument~of~command~'\c_backslash_str #1'~can~NOT~be~empty. }	140
141	\cs_new:Npn __ztikz_wolfram_arg_empty_assert:nn #1#2	141
142	{	142

```

143 \exp_args:No \tl_if_empty:nT { #2 }
144 {
145     \msg_error:nnn {ztikz}{wolfram-arg_empty}{#1}
146 }
147 }
148
149 % --> load result
150 \NewDocumentCommand\wolframResult{so}
151 {
152     \IfBooleanTF{#1}
153     {% must be integer expression, or it will raise bug.
154     \seq_item:Ne \g_ztikz_wolfram_res_seq
155     {\IfValueTF {#2}{\int_eval:n {#2}}{1}}
156 }{
157     \seq_use:Nn \g_ztikz_wolfram_res_seq
158     { \IfValueTF {#2}{#2}{,} }
159 }
160 }
161
162 % --> wolfram code
163 \NewDocumentEnvironment{wolframCode}{m}
164 {
165     \xsim_file_write_start:ne {\c_false_bool}{\g__ztikz_wolfram_path_tl/t@mp.wls}
166 }{
167     \xsim_file_write_stop:
168     \__ztikz_wolfram_execute:nnnn
169     {}{}{}{#1}
170 }
171
172 % --> wolfram graphicx
173 \NewDocumentEnvironment{wolframGraphics}{m0{}}
174 {
175     \gdef\zgraphics@spec{#2}
176     \xsim_file_write_start:ne {\c_false_bool}{\g__ztikz_wolfram_path_tl/t@mp.wls}
177 }{
178     \xsim_file_write_stop:
179     \__ztikz_wolfram_execute:nnnn {}{pdf}{FIGURE}{#1}
180     \tl_if_empty:eF {\zgraphics@spec}
181     {
182         \exp_after:wN \includegraphics \exp_after:wN
183         [\zgraphics@spec]{\wolfram@tmp@file.pdf}
184     }
185 }
186
187 % --> wolfram simple code
188 \NewDocumentCommand\wolfram{smm}
189 {
190     \__ztikz_wolfram_arg_empty_assert:nn {\wolfram}{#3}

```

```

191 \_ztikz_wolfram_execute:ennn
192 {
193 \IfBooleanTF{#1}
194 { TeXResult = ToString[#3]; }
195 { TeXResult = ToString[TeXForm[#3]]; }
196 }{txt}{TeXResult}{#2}
197 }
198
199 % --> wolfram tex code(expandable token replace)
200 \group_begin:
201 \char_set_catcode_escape:n { 36 }
202 \char_set_catcode_letter:n { 92 }
203 $cs_gset:Nn $__double_backslash:n
204 { $tl_if_eq:NNTF #1\_{\}\{#1} }
205 $gdef$wolframTex{
206 \char_set_catcode_letter:n { 92 }
207 $wolframTex@getarg
208 }
209 $gdef$wolframTex@getarg#1#2{
210 $__ztikz_wolfram_arg_empty_assert:nn {wolframTex}{#2}
211 $tl_set:Ne $l_tmpa_tl
212 {
213 \tl_map_function:nN {#2}
214 $__double_backslash:n
215 }
216 $__ztikz_wolfram_execute:onn
217 {TeXResult = TeXForm[ToExpression["$l_tmpa_tl", TeXForm]]}
218 {txt}{TeXResult}{#1}
219 \char_set_catcode_escape:n { 92 }
220 }
221 \char_set_catcode_escape:n { 92 }
222 \char_set_catcode_letter:n { 36 }
223 \group_end:
224
225 % --> wolfram table (extended the interface of 'latexalpha2')
226 \cs_set:Npn \__table_item_handle:n #1
227 {% the inner '\exp_not:N' prevent expansion from 'tabularray'.
228 \exp_not:n {
229 \exp_not:N \__wolfram_table_cell_cmd:n {#1}
230 },
231 }
232 \cs_set:Npn \__table_row_handle:n #1 % #1='{1, 2, 3}'
233 {
234 \clist_use:en
235 {
236 \clist_map_function:oN #1
237 \__table_item_handle:n
238 }{ & } \

```

239	}	239
240	\cs_generate_variant:Nn \clist_use:nn { en }	240
241	\cs_generate_variant:Nn \clist_map_function:nN { oN }	241
242	\cs_new:Npn __part_table_from_file:nN #1#2	242
243	{% #1:file; #2:data var	243
244	\ztool_gread_file_as_seq:neN {\c_true_bool}	244
245	{ #1 } \l_tmpa_seq	245
246	\tl_set:Ne #2	246
247	{	247
248	\seq_map_function:NN \l_tmpa_seq	248
249	__table_row_handle:n	249
250	}	250
251	}	251
252	\cs_set:Npn __full_table_from_file:nn #1#2	252
253	{% #1:file; #2:table header	253
254	__part_table_from_file:nN	254
255	{ #1 } \l_part_table_data_tl	255
256	\tl_set:Ne \l_full_table_data_tl	256
257	{	257
258	\tl_if_empty:eF {#2}{#2 \\}	258
259	\l_part_table_data_tl	259
260	}	260
261	\tl_set:Ne \l_part_table_data_tl	261
262	{ \l_part_table_data_tl }	262
263	}	263
264	\cs_generate_variant:Nn __full_table_from_file:nn { VV }	264
265	\cs_set:Npn __typeset_table:nnn #1#2#3	265
266	{% #1:table format; #2:table header; #3:table part data	266
267	\begin{tabular}{#1}	267
268	\hline	268
269	\bool_if:NT \l_wolfram_table_hdbt_rule_bool	269
270	{ #2\\ \hline }	270
271	#3	271
272	\hline	272
273	\end{tabular}	273
274	}	274
275	\cs_generate_variant:Nn __typeset_table:nnn { VVV }	275
276	\ztikz_keys_define:nn { wolfram / table }	276
277	{	277
278	format .tl_set:N = \l__ztikz_wolfram_table_format_tl,	278
279	format .initial:n = { *{12}{1} },	279
280	header .tl_set:N = \l__ztikz_wolfram_table_header_tl,	280
281	header .initial:n = { },	281
282	hdbt-rule .bool_set:N = \l_wolfram_table_hdbt_rule_bool,	282
283	hdbt-rule .initial:n = { false },	283
284	hdbt-rule .default:n = { true },	284
285	cell-cmd .cs_gset:Np = __wolfram_table_cell_cmd:n #1,	285
286	cell-cmd .initial:n = { #1 },	286

```

287 }
288 \NewDocumentCommand{\wolframTable}{sm0{m}}
289 {% #1:if typeset; #2:key-value; #3:code
290   \__ztikz_wolfram_arg_empty_assert:nn { wolframTable }{ #4 }
291   \group_begin:
292     \ztikz_keys_set:nn { wolfram/table }{ #3 }
293     \__ztikz_wolfram_execute:ennn
294       { TeXResult = #4; }{ txt }
295       { TeXResult }{ #2 }
296     \__full_table_from_file:VV \wolframOutputFile
297     \l__ztikz_wolfram_table_header_tl
298     \IfBooleanT{#1}
299     {
300       \__typeset_table:VVV
301       \l__ztikz_wolfram_table_format_tl
302       \l__ztikz_wolfram_table_header_tl
303       \l_part_table_data_tl
304     }
305     \exp_args:NNo \gdef\wolframTablePData{ \l_part_table_data_tl }
306     \exp_args:NNo \gdef\wolframTableFData{ \l_full_table_data_tl }
307   \group_end:
308 }
309
310 % --> equation solve
311 \ztikz_keys_define:nn { wolfram/solve }
312 {
313   var .tl_set:N = \l__ztikz_wolfram_var_tl,
314   var .initial:n = {},
315   domain .tl_set:N = \l__ztikz_wolfram_domain_tl,
316   domain .initial:n = {},
317 }
318 \NewDocumentCommand\wolframSolve{smom}
319 {
320   \__ztikz_wolfram_arg_empty_assert:nn { wolframSolve }{ #4 }
321   \group_begin:
322   \IfValueT {#3} { \ztikz_keys_set:nn { wolfram/solve } {#3} }
323   \tl_if_empty:VF \l__ztikz_wolfram_domain_tl
324     { \tl_set:N \l__ztikz_wolfram_tmp_arg_tl {,\l__ztikz_wolfram_domain_tl} }
325   \__ztikz_wolfram_execute:ennn
326   {
327     \IfBooleanTF {#1}{
328       TeXResult = Row[Solve[#4]//Flatten, "@"]
329       /.{Rule -> Equal}//TeXForm//ToString;
330     }{
331       TeXResult = Row[
332         Solve[#4, {\l__ztikz_wolfram_var_tl} \l__ztikz_wolfram_tmp_arg_tl]//Flatten,
333         "@"
334       ]/.{Rule -> Equal}//TeXForm//ToString;

```

335	}	335
336	}{txt}{TeXResult}{#2}	336
337	\group_end:	337
338	}	338
339		339
340	% --> differential equation solve	340
341	\ztikz_keys_define:nn { wolfram/dsolve }	341
342	{	342
343	depend .tl_set:N = \l__ztikz_wolfram_de_var_tl,	343
344	depend .initial:n = { y[x] },	344
345	independ .tl_set:N = \l__ztikz_wolfram_in_var_tl,	345
346	independ .initial:n = { x },	346
347	}	347
348	\NewDocumentCommand\wolframDSolve{smom}	348
349	{	349
350	__ztikz_wolfram_arg_empty_assert:nn { wolframDSolve }{ #4 }	350
351	\group_begin:	351
352	\IfValueT {#3} { \ztikz_keys_set:nn { wolfram/dsolve } {#3} }	352
353	\tl_if_empty:VF \l__ztikz_wolfram_in_var_tl	353
354	{ \tl_set:Ne \l__ztikz_wolfram_in_var_tl {,\l__ztikz_wolfram_in_var_tl} }	354
355	__ztikz_wolfram_execute:ennn	355
356	{	356
357	\IfBooleanTF {#1}{	357
358	TeXResult = Row[DSolve[#4]//Flatten, ","]	358
359	/.{Rule -> Equal}//TeXForm//ToString;	359
360	}{	360
361	TeXResult = Row[	361
362	DSolve[{#4}, {\l__ztikz_wolfram_de_var_tl}\l__ztikz_wolfram_in_var_tl]//Flatten,	362
363	"@"	363
364	]/.{Rule -> Equal}//TeXForm//ToString;	364
365	}	365
366	}{txt}{TeXResult}{#2}	366
367	\group_end:	367
368	}	368

9 索引

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Due to limited time, development of this package is currently suspended.

May 8, 2026

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1 Basic Introduction

To this day, there are many drawing packages based on `tikz`, each with different purposes and found across various fields. Some packages already provide interfaces similar to `ztikz`, including:

- Common `TikZ` command wrappers: `tzplot`;
- `TikZ` package for 3D drawing: `tikz-3dplot`;
- New general-purpose drawing package `luadraw`, which natively supports 3D drawing, currently supporting only `LuaTeX`;
- PSTricks-based (special) function drawing package: `pst-func`;
- Package for caching compilation results: `robust-externalize`;
- ...

If you find that the `ztikz` package does not meet your needs, you might try the packages mentioned above, or directly use the commands and libraries provided by the original `tikz` package. There are also rich `TikZ` resources online, such as the `TikZ` drawing website – `TikZ Example`, which contains many drawing examples and their corresponding code.

However, the interfaces provided by the above packages are not very uniform and I am not used to them, which is why I decided to develop the `ztikz` package. The command format of `zTikZ` basically follows the naming conventions of functions in Mathematica.

1.1 Project Repository

The address of this package on GitHub is as follows:

https://github.com/zongpingding/zTeX_bundle

This repository contains the source code of this macro set, the user manual, and some test cases. The current stable version of the macro set was released in September 2025. For the latest version, please switch to the “dev” branch. This manual applies to the current latest “dev” branch.

1.2 Functional Overview

The `zTikZ` package is mainly used for drawing and calculation, supporting calls to external programs such as Python, Mathematica, and gnuplot. It also provides a calling cache mechanism. Although `zTikZ` provides interfaces for these software programs, it does not mean you need to install all of them. Each software interface in `zTikZ` is independent; users only need to install the software corresponding to the functions they need. The functional overview of `zTikZ` is as follows:

- **Drawing:** 2D drawing, 3D drawing;
- **Calculation:** Floating-point calculation, symbolic calculation.

The drawing part is based on: `TikZ`'s 2D drawing part,¹ Python's `matplotlib` library, and WolframScript's drawing functions. The calculation part is based on: `LATEX3`'s `xfp` module, Python's `numpy`, `sympy`, and `scipy` libraries, and WolframScript's calculation functions.

Although the name of this macro set only contains “`TikZ`”, the functions that `ztikZ` can (or aims to) achieve go beyond this.

1.3 Coordinate Alignment

All drawing commands provided by `ztikZ` can be used in conjunction with `TikZ` commands; that is, they can be used within the same `tikzpicture` environment. `ztikZ` performs “alignment” of coordinates when drawing functions: coordinates in `ztikZ` commands are consistent with those in `TikZ` commands or in Geogebra.

Why align coordinates in `ztikZ`? Imagine a scenario: you find the intersection of two function graphs in Geogebra at $P(1, 2)$. First, you draw point P using `TikZ`'s built-in `\filldraw` command; then you draw the same point P using the `\ShowPoint` command in `ztikZ`. The result is that the two P points do not overlap, even though the specified coordinates are both $(1, 2)$.

Therefore, when it is inconvenient to use `ztikZ` to solve for certain special points, you can first find the corresponding point P in software like Geogebra and then directly use the `\ShowPoint` command in `ztikZ` to draw it.

1.4 Cache Mechanism

In addition to providing interfaces for interaction with external programs, `ztikZ` also has a built-in cache system. `ztikZ` automatically caches the results of interactions between `TEX` and external programs and records the Hash value of the calling source code in the `LATEX` document.

If the Hash value of the source code in the `LATEX` document changes, `ztikZ` will re-interact with the external program, generate new results, and cache the new Hash value. If the Hash value does not change, `ztikZ` will directly call the previous cached result. The advantages of the cache system are: we do not have to repeatedly compile unchanged content, directly referencing previous caches, which reduces document compilation time. In actual tests, after results are cached, the time to re-compile the source document is almost the same as the time to directly insert the corresponding number of images.

The `basic`, `python`, `wolfram`, and `gnuplot` libraries in `ztikZ` have all implemented the cache mechanism. The cache mechanism for images generated by the `tikzpicture` environment or the `\tikz` command relies on `TikZ`'s external library (its implementation is notoriously complex; interested users can look it up).

Since `ztikZ` has not yet undergone complete testing, there may be undiscovered bugs. For example, users might encounter problems like the following:

- Outdated Cache Hash value: If the initial Hash value of an environment is “A”, and after you modify the content, the Hash value becomes “B”. However, if you modify the source code back so the Hash value is “A” again, the Hash value is already cached in the `ztikz.hash` file, so the drawing result for this environment will not change

¹Since I haven't decided how to declare the transformation matrix interfaces involved in 3D drawing in `ztikZ`, 3D drawing is currently not provided.

when re-compiled. The cached result called is still the one corresponding to Hash value “B”.

- Conflict with the `indextool` package: You might find a compilation error like `missing \begin{document}...` after enabling the cache library. This issue is related to the indexing function of the `indextool` package. You can comment out the `\makeindex` and `\printindex` commands first, and then uncomment them and generate the index after the image caching is finished.

1.5 Limitations

`ǻTiǻZ` may provide 3D drawing related interfaces in the future, but if your images require complex calculations or layouts, please use other macro packages or professional drawing software. The `asymptote` package is a good choice.

2 Installation and Usage

2.1 Compatibility

Currently, the `ztikz` package is compatible with Windows, Linux, and macOS. The compatibility with different `TeXLive` versions on each platform is as follows:

Windows : Minimum `TeXLive` version 2023

Linux : Minimum `TeXLive` version 2022

macOS : Minimum `MacTeX` version 2024

The performance of `ztikz` on Windows may not be as good as on Linux/macOS. It is recommended that users use this package on Linux/macOS.

2.2 Environment Configuration

If users need to use the libraries provided by `ztikz` that call external programs, they not only need to configure the document preamble but also need to install the corresponding applications on the system. After the applications are installed, they need to be added to the environment variables so that they can be called from the command line. Finally, when compiling the document, add the `--shell-escape` parameter, like this:

```
pdflatex --shell-escape main.tex
```

On Windows, it is recommended to use the `scoop` package manager to install these software programs, which avoids the hassle of configuring environment variables. The following are points to note during the configuration process for different programs:

2.2.1 gnuplot

On Windows, when using the GUI to install gnuplot, please make sure to check the “Add gnuplot to PATH” option.

2.2.2 Python

If users need to use the functions provided by the `python` library, they need to install Python as well as the `matplotlib`, `sympy`, and `scipy` libraries; the former is used for drawing, and the latter for calculation.

On Windows, due to `TeXLive`’s compilation configuration, ensure that the “.PY” suffix has been removed from the system environment variable `PATHEXT`.

2.2.3 WSTP

If users cannot install Mathematica, the `ztikz` package offers another option – WSTP (Wolfram Symbolic Transfer Protocol). In this case, users only need to install the free WolframScript (Wolfram Engine). For environment configuration methods, please refer to my repository: [WSTP Example](#).

The advantages of this method are obvious: it is free and very fast (about **ten** times faster than calling WolframScript directly). Even if users do not load the `cache` library, the execution speed of all commands in the `wolfram` library is comparable to directly calling the cache.

Currently, the limitations of this method are also obvious: users need to compile the Wolfram engine client themselves and start the service program in advance. This client has only been tested on Linux (contributions for other platforms are welcome).

2.2.4 Wolfram

If users need to use the functions corresponding to the `wolfram` library, they need to install WolframScript or Mathematica. Commands can be chosen to be executed in the cloud, thus avoiding calls to the local Mathematica calculation kernel. Users must first complete the Wolfram account binding on the command line (when calling on the Wolfram Cloud for the first time):

```
> wolframscript -cloud -code 2+2
Wolfram ID: <Account>
Password: <Password>
```

The above command will prompt the user to enter their Wolfram ID and password. After entering the corresponding `<Account>` and `<Password>`, it can be used. However, cloud execution may be slow, and users should decide whether to adopt this solution. The following describes how to install the Wolfram engine locally:

On Linux, software other than Wolfram is easy to install, using the package manager provided by the Linux distribution. Here I provide a method for using Mathematica under Windows in WSL (users can also configure WolframScript without following this method): it's essentially creating a symbolic link from Linux to Windows. Please change the path of WolframScript under Windows in the command according to your actual situation:

```
sudo ln -sf "/mnt/c/Program Files/Wolfram Research/WolframScript/wolframscript.exe" /usr/bin/wolframscript
```

Please ensure that WolframScript can be called normally from the command line. You can use the following code to test if WolframScript is configured successfully:

```
plotFunction[fun_, xlimits_, ylimits_] := ContourPlot[
  fun, xlimits, ylimits,
  ContourStyle->{
    RGBColor["#00C0A3"],
    Thickness[0.004]
  },
  AspectRatio->((xlimits[[2]]//Abs) + (xlimits[[3]]//Abs))/((ylimits[[2]]//Abs) + (ylimits[[3]]//Abs)),
  AxesOrigin->{0,0},
  Axes->True,
  Frame->False,
  AxesStyle->Arrowheads[{0, 0.03}],
  AxesLabel->{"x", "y"},
```

```

PlotRange -> Full
]

xlimits = {x, -3, 6};
ylimits = {y, -4, 5};
fp1 = plotFunction[y==Sin[x], xlimits, ylimits];
fp2 = plotFunction[x^2/4 + y^2/3 == 5, {x, -5, 5}, {y, -5, 5}];

figure = Show[fp2, fp1];
(* the saved image format is: '*.wls.pdf' *)
Export["works_well.pdf", figure];

```

Save the above source code as `test.wls`, and then run the following command in the command line:

```
wolframscript -script test.wls
```

If the configuration is successful, a PDF file named `works_well.pdf` will be generated in the current working directory. Otherwise, it indicates that your WolframScript configuration was not successful, and this library cannot be used.

2.2.5 Mathics

Besides choosing WolframScript as the calculation engine, users can also choose Mathics. What is Mathics? An open-source Mathematica Kernel. MathicsScript is a front-end for Mathics with features like automatic command/variable completion and syntax highlighting.

For local installation of Mathics, please refer to: [Installing Mathics3](#). If users have already installed Mathics on Windows and do not want to re-install it in WSL, the method for creating a symbolic link in WSL is the same as for WolframScript. If users installed Mathics via the command “pip install Mathics-omnibus”, the command to create the symbolic link is:

```

sudo ln -sf
"/mnt/c/Users/<name>/AppData/Local/Programs/Python/Python312/Scripts/mathics.exe"
/usr/bin/mathics

```

The `<name>` in the above command needs to be replaced with your own username, and pay attention to the Python version number.

NOTE: Some functions in Mathematica may not be supported by Mathics; please refer to the Mathics documentation.

3 Package Options

 ztikz/library

 New: 2025-05-18

quiet = `<true|false>`.....Initial value: `false`
 library = `<basic|gnuplot|cache|python|wolfram|l3draw>` Initial value: `empty`

The `<library>` option is equivalent to the command `\ztikzloadlib` and is used to specify the list of libraries to be loaded by `zTikZ` when loading the `ztikz` package. The `<quiet>` option controls whether debugging information is displayed in the terminal; setting `<quiet>=true` or loading `<quiet>` alone will suppress most output. A simple configuration example is as follows:

```
\usepackage[library={basic, gnuplot}]{ztikz}
```

Ex. 1

 ztikz/wolfram/engine
 ztikz/wolfram/cloud

 New: 2026-04-29

engine = `<wolfram|wstp|mathics>`.....Initial value: `wolfram`
 cloud = `<true|false>`.....Initial value: `false`

`<engine>` is used to specify the calculation engine for Wolfram code. Currently, Wolfram and Mathics engines are supported; the former is commercial closed-source software, and the latter is open-source. If users have high speed requirements, they can configure the `<wstp>` environment themselves. `<cloud>` is used to specify whether to use Wolfram Cloud for calculation, currently only valid for the `<wolfram>` option; **Note:** Mathics currently does not support cloud computing. A simple configuration example is:

```
\usepackage[
  library = { wolfram },
  wolfram = { engine=wolfram, cloud=true }
]{ztikz}
```

Ex. 2

4 Miscellaneous

`\zTikZ`
`\ztikz`

New: 2025-09-22

These are the logos of the `ztikz` package. Additionally, `ztikz` provides macros related to `hologo`.

```
% \usepackage{hologo}
\zTikZ, \ztikz;\par
\hologo{zTikZ}, \hologo{ztikz}.
```

Ex. 3

```
zTikZ, zTikZ;
zTikZ, zTikZ.
```

`\ztikzMkdir`

New: 2025-05-15

`\ztikzMkdir{⟨path⟩}`

This command is used to create a directory. `⟨path⟩` can be any valid path name, such as `./A/B`.

5 $\text{\textit{zTikZ}}$ Libraries

$\text{\textit{zTikZ}}$ provides various functional libraries that can be loaded using the `\ztikzloadlib` command. Users need to use `\ztexloadlib{<library name>}` to load the corresponding library. The available `<library name>` list in $\text{\textit{zTikZ}}$ is as follows:

- basic
- cache
- gnuplot
- python
- wolfram
- l3draw

All the above libraries are not loaded automatically; users must load them manually. The **basic** library only contains commands for drawing points, lines, axes, and basic polygons. The method to load $\text{\textit{zTikZ}}$ libraries in the preamble is as follows, for example, loading the **cache** and **gnuplot** libraries:

```
\ztikzloadlib{cache, gnuplot}
```

Ex. 4

Note: The script files for a library will only be written to the project folder after the user loads the corresponding library.

5.1 basic Library

The basic library mainly includes commands related to the coordinate system: drawing points, lines, planes, and regular polygons, as well as solving for and drawing intersections. All drawing commands in it are inherited from built-in commands in TikZ; for example, the subsequent `\BarPlot` command is actually a wrapper for the following built-in command:

```
\draw[⟨key-value⟩] plot [ycomb, ⟨other style⟩] file {⟨data⟩};
```

NOTE: For convenience, we use `⟨draw-keyval⟩` to represent the `⟨keyval⟩` options in `\draw[⟨keyval⟩]`, and `⟨node-keyval⟩` to represent the `⟨keyval⟩` options in `\node[⟨keyval⟩]`. Specifically, for the `⟨draw⟩` command, available options include `⟨line width⟩`, `⟨color⟩`, etc.; for details, please refer to the TikZ user manual.

`\ShowPoint`

New: 2025-05-15

```
\ShowPoint[⟨key-value⟩]{⟨point-1⟩; ...; ⟨point-n⟩}
                [⟨label-1⟩; ...; ⟨label-n⟩][⟨node-keyval⟩]
```

This command is used to draw points. `⟨point-1⟩` to `⟨point-n⟩` are the coordinates of the points, separated by “;”; the coordinate format is (x, y) . `⟨key-value⟩` is used to set the style of the points. The number of `⟨label⟩`s does not need to match the number of `⟨point⟩`s; `⟨label⟩`s are applied sequentially to each point starting from the first.

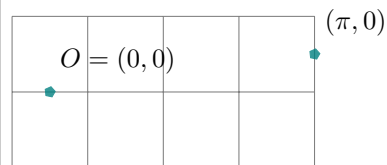
ztikz/point/type
ztikz/point/radius
ztikz/point/color
ztikz/point/opacity
ztikz/point/rotate

```
type      = ⟨string⟩.....Initial value: none
radius    = ⟨length⟩.....Initial value: 1pt
color     = ⟨color⟩.....Initial value: black
opacity   = ⟨float⟩.....Initial value: 1
rotate    = ⟨angle⟩.....Initial value: 0
```

`⟨type⟩` is used to set the style of the marker, `⟨radius⟩` for the radius, `⟨color⟩` for the color, `⟨opacity⟩` for the transparency, and `⟨rotate⟩` for the rotation angle.

```
\begin{tikzpicture}
\draw[gray] (-2, -1) grid (2, 1);
\ShowPoint
  [color=teal, radius=2pt, type=pentagon*, opacity=.8,
rotate=60]
  {(-1.5, 0); (2, .5)}[$O=(0, 0)$; $(\pi, 0)$]
  [above right=3pt and 0em, font=\small]
\end{tikzpicture}
```

Ex. 5



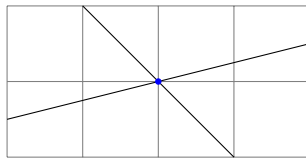
\ShowIntersection

New: 2025-05-15

\ShowIntersection[*<key-val>*]{*<path-1>*; *<path-2>*}{*<number>*}

This command is used to find the intersections of *<path-1>* and *<path-2>*, separated by “;”, and then draws the first *<number>* intersections. *<key-value>* corresponds to the options in the **\ShowPoint** command, i.e., *<ztikz/point>*.

```
\begin{tikzpicture}
\draw[gray] (-2, -1) grid (2, 1);
\draw[name path=line1] (-2, -.5) -- (2, .5);
\draw[name path=line2] (-1, 1) -- (1, -1);
\ShowIntersection[color=blue]{line1; line2}{1}
\end{tikzpicture}
```

Ex. 6**\ShowAxis**

New: 2025-05-15

\ShowAxis[*<key-value>*]{*<start>*; *<end>*}

This command is used to draw an axis. *<start>* and *<end>* represent the starting and ending points of the axis, separated by “;”; the coordinate format is (x, y) . *<key-value>* is an optional parameter for setting the axis style.

ztikz/axis/tickStart	tickStart	= <i><float></i>Initial value: -5
ztikz/axis/tickEnd	tickEnd	= <i><float></i>Initial value: 5
ztikz/axis/axisRotate	axisRotate	= <i><float></i>Initial value: 0
ztikz/axis/mainStep	mainStep	= <i><float></i>Initial value: 1
ztikz/axis/subStep	subStep	= <i><float></i>Initial value: 0.1
ztikz/axis/tickLabelShift	tickLabelShift	= <i><length></i>Initial value: 0pt
ztikz/axis/mainTickLength	mainTickLength	= <i><length></i>Initial value: 4pt
ztikz/axis/subTickLength	subTickLength	= <i><length></i>Initial value: 2pt
ztikz/axis/axisColor	axisColor	= <i><color></i>Initial value: black
ztikz/axis/mainTickColor	mainTickColor	= <i><color></i>Initial value: black
ztikz/axis/subTickColor	subTickColor	= <i><color></i>Initial value: black
ztikz/axis/tickStyle	tickStyle	= <i><below above cross></i>Initial value: none
ztikz/axis/mainTickLabel	mainTickLabel	= <i><string></i>Initial value: \CurrentFp
ztikz/axis/mainTickLabelColor	mainTickLabelColor	= <i><color></i>Initial value: black
ztikz/axis/mainTickLabelPosition	mainTickLabelPosition	= <i><below above cross></i> ... Initial value: below

<mainTickLabel> is mainly used to customize the style of coordinate labels; **\CurrentFp** represents the floating-point value at the current tick. *<tickStyle>* is affected by the *<rotate>* option in the optional parameters of the **tikzpicture** environment.

Note: If the value of the key *<tickStyle>* is not specified when using **\ShowAxis**, no ticks will be drawn.

\CurrentFp

New: 2025-05-31

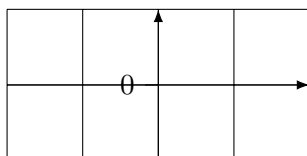
This command represents the floating-point value at the current tick, which automatically updates at different ticks.

<code>\xAxis</code>	<code>\xAxis[$\langle start \rangle$][$\langle end \rangle$]</code>
New: 2025-05-15	This command comes from <code>\ShowAxis</code> and is used to draw the x -axis; $\langle start \rangle$ and $\langle end \rangle$ are floating-point numbers representing the starting and ending points of the axis.

<code>\yAxis</code>	<code>\yAxis[$\langle start \rangle$][$\langle end \rangle$]</code>
New: 2025-05-15	This command comes from <code>\ShowAxis</code> and is used to draw the y -axis; $\langle start \rangle$ and $\langle end \rangle$ are floating-point numbers representing the starting and ending points of the axis.

```
\begin{tikzpicture}[>=Latex]
\yAxis[-1][1]
\ShowAxis{(-2, 0); (2, 0)}
\draw (-2, -1) grid (2, 1);
\end{tikzpicture}
```

Ex. 7



<code>\ShowGrid</code>	<code>\ShowGrid[$\langle draw-keyval \rangle$]{$\langle start \rangle$; $\langle end \rangle$}</code>
New: 2025-05-15	This command is used to draw grid lines. $\langle start \rangle$ and $\langle end \rangle$ represent the coordinates of the lower-left and upper-right corners of the grid, separated by “;”, in format (x, y) . $\langle key-value \rangle$ is an optional parameter for setting the grid line style.

<code>\Polygon</code>	<code>\Polygon[$\langle key-value \rangle$]{$\langle number \rangle$}</code>
New: 2025-05-15	This command is used to draw regular polygons; $\langle number \rangle$ represents the number of sides, which must be an integer greater than or equal to 3. $\langle key-value \rangle$ is an optional parameter for setting the polygon style.

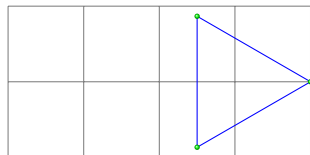
<code>ztikz/polygon/radius</code>	<code>radius</code>	<code>= $\langle float \rangle$</code>	Initial value: 1
<code>ztikz/polygon/edgeColor</code>	<code>edgeColor</code>	<code>= $\langle color \rangle$</code>	Initial value: black
<code>ztikz/polygon/fillColor</code>	<code>fillColor</code>	<code>= $\langle color \rangle$</code>	Initial value: none
<code>ztikz/polygon/fillOpacity</code>	<code>fillOpacity</code>	<code>= $\langle float \rangle$</code>	Initial value: 0
<code>ztikz/polygon/rotate</code>	<code>rotate</code>	<code>= $\langle float \rangle$</code>	Initial value: 0
<code>ztikz/polygon/shift</code>	<code>shift</code>	<code>= $\langle coordinate \rangle$</code>	Initial value: (0, 0)
<code>ztikz/polygon/marker</code>	<code>marker</code>	<code>= $\langle key-value \rangle$</code>	Initial value: none

$\langle radius \rangle$ represents the radius of the circumscribed circle of this regular polygon, not the radius of the $\langle marker \rangle$. The “()” surrounding $\langle shift \rangle$ cannot be omitted. $\langle marker \rangle$ corresponds to `\ztikz/point`; please see figure (3) for marker settings.

```
\begin{tikzpicture}
\ShowGrid[gray, thin]{(-2, -1); (2, 1)}
```

Ex. 8

```
\Polygon[
  edgeColor=blue, shift={(1, 0)},
  marker={type=ball, color=green}
]{3}
\end{tikzpicture}
```

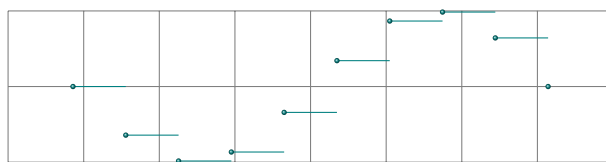


\StairsPlot
New: 2025-05-15

```
\StairsPlot[⟨plot option⟩; ⟨jump option⟩][⟨draw-keyval⟩]
[⟨key-value⟩]{⟨file⟩}
```

This command is used to draw stairs plots, with drawing data specified by *⟨file⟩*. *⟨plot option⟩* is used to set the stairs plot style, with possible values: *plot left*, *plot right*, *plot mid*. *⟨jump option⟩* sets the jump style, with possible values: *jump left*, *jump right*, *jump mid*. *⟨key-value⟩* corresponds to *⟨ztikz/point⟩*.

```
\begin{tikzpicture}
\ShowGrid[step=1, color=gray]{(-4, -1); (4, 1)}
\StairsPlot[; jump-left][teal][type=ball,
color=teal]{./sine.data}
\end{tikzpicture}
```

Ex. 9

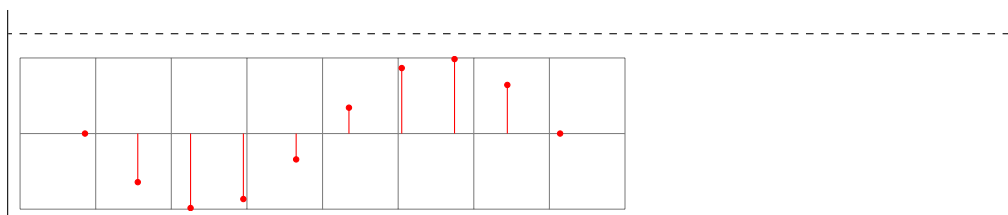
\StemPlot
New: 2025-05-15

```
\StemPlot[⟨direction⟩][⟨draw-keyval⟩]
[⟨key-value⟩]{⟨file⟩}
```

This command is used to draw stem plots, with drawing data specified by *⟨file⟩*. *⟨direction⟩* specifies the direction of the line segments, with possible values: *x*, *y*, *o*, representing perpendicular to the *x*-axis, perpendicular to the *y*-axis, and pointing to the origin, respectively. *⟨key-value⟩* corresponds to *⟨ztikz/point⟩*.

```
\begin{tikzpicture}
\ShowGrid[step=1, color=gray]{(-4, -1); (4, 1)}
\StemPlot[x][red][type=*, color=red]{./sine.data}
\end{tikzpicture}
```

Ex. 10



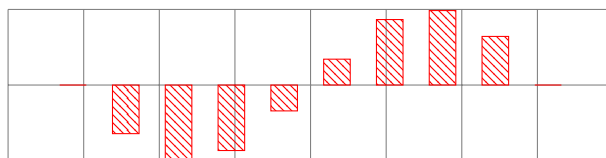
`\BarPlot`

New: 2025-05-15

```
\BarPlot[⟨position⟩][⟨draw-keyval⟩]
          [⟨key-value⟩]{⟨file⟩}
```

This command is used to draw bar plots, with drawing data specified by `⟨file⟩`. `⟨position⟩` specifies the position and width of each small rectangle, with possible values: `x`, `y`, `xc`, `yc`. `⟨key-value⟩` corresponds to `⟨ztikz/point⟩`.

```
\begin{tikzpicture}
\ShowGrid[step=1, color=gray]{(-4, -1); (4, 1)}
\BarPlot[x][red, pattern=north west lines, pattern
color=red]{./sine.data}
\end{tikzpicture}
```

Ex. 11

5.2 gnuplot Library

It should be noted that the $\text{\textit{tikz}}$ package internally provides commands to directly call the gnuplot program (requires the `-shell-escape` parameter enabled), with the calling format as follows:

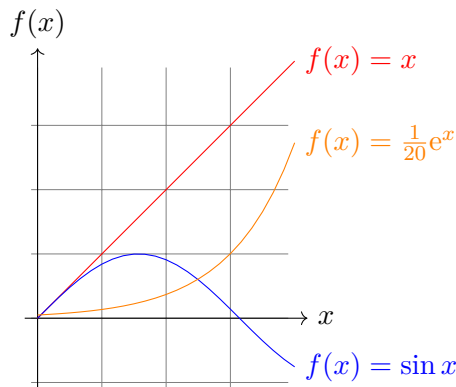
```
\draw[⟨key-value⟩] plot[⟨id⟩] function{⟨function⟩};
```

In the above command, $\langle id \rangle$ is used to distinguish different data files. In the root path of the $\langle file \rangle .\text{tex}$ file (let the file name be $\langle file \rangle$), two files will be generated: one is the style file used by gnuplot for drawing, $\langle file \rangle .\langle id \rangle .\text{gnuplot}$, and the second is the data file generated by gnuplot, $\langle file \rangle .\langle id \rangle .\text{table}$. Possible values for $\langle function \rangle$ can be found in [table \(1\)](#).

$\text{\textit{tikz}}$ built-in commands also support two other formats: “parametric” and “raw gnuplot”. The first parameter draws parametric equations, and the second parameter uses raw gnuplot drawing commands directly in the document (e.g., “set samples 25; plot sin(x)”). The calling formats are:

```
\draw[⟨key-value⟩] plot [parametric, ⟨id⟩]{⟨function⟩};
\draw[⟨key-value⟩] plot [raw gnuplot, ⟨id⟩]{⟨gnuplot code⟩};
```

```
\begin{tikzpicture}[domain=0:4, scale=.85] Ex. 12
  \draw[very thin,color=gray] (-0.1,-1.1) grid (3.9,3.9);
  \draw[->] (-0.2,0) -- (4.2,0) node[right] {$x$};
  \draw[->] (0,-1.2) -- (0,4.2) node[above] {$f(x)$};
  \draw[color=red] plot[id=x] function{x} node[right] ✓
  {$f(x)=x$};
  \draw[color=blue] plot[id=sin] ✓
  function{sin(x)}node[right]{$f(x)=\sin x$};
  \draw[color=orange] plot[id=exp] function{0.05*exp(x)} ✓
  node[right] {$f(x)=\frac{1}{20}\mathrm{e}^x$};
\end{tikzpicture}
```



For more detailed usage of these native drawing commands in $\text{\textit{tikz}}$, please

refer to Section 22: Plots of Functions in the official $\text{\textit{TiKZ}}$ documentation.

However, for the uniformity of the `gnuplot` series of drawing commands, $\text{\textit{ztikz}}$ does not adopt the above method. Instead, it leverages the `ztool` package and works with predefined drawing scripts to complete the drawing tasks. The drawing logic of the `gnuplot` library in $\text{\textit{ztikz}}$ is roughly as follows:

- First, modify the predefined script via the `\ztool_replace_file_line:nnn` function of `ztool`.
- Then call `gnuplot` to run the modified script using the `-shell-escape` parameter from the command line.
- Finally, use the command `\draw[⟨key-value⟩] plot file [⟨data⟩];` to call the generated data file from the previous step to complete the drawing.

Users unfamiliar with `gnuplot` can read this 7-page quick start guide: [gnuplot card](#).

NOTE: After calling this library, the “`-shell-escape`” parameter must be enabled during compilation.

<code>ztikz/2dplot/domain</code>	<code>domain</code>	<code>= ⟨float:float; float:float⟩</code>	Initial value: <code>⟨variable⟩</code>
<code>ztikz/2dplot/style</code>	<code>style</code>	<code>= ⟨draw-keyval⟩</code>	Initial value: <code>black</code>
<code>ztikz/2dplot/marker</code>	<code>marker</code>	<code>= ⟨key-value⟩</code>	Initial value: <code>empty</code>

`⟨marker⟩` with `⟨key-value⟩` corresponds to `⟨ztikz/point⟩`. `⟨domain⟩` values are separated by “;”. The meaning of `⟨domain⟩` differs across functions: in `\Plot`, it sets the range of the independent variable x ; in `\ParamPlot` and `\PolarPlot`, it sets the range for parameter t or angle θ in polar coordinates; in `\ContourPlot`, the `⟨domain⟩` before and after “;” represent the ranges for x and y , respectively.

<code>\PlotPrecise</code>	<code>\PlotPrecise{⟨type⟩}{⟨number⟩}</code>
New: 2025-05-15	<code>\PlotPrecise*{⟨type⟩}{⟨number⟩}</code>

This command is used to set the precision for a series of **2D** drawing functions in `gnuplot`. Possible values for `⟨type⟩` are: “`plot`, `param`, `polar`, `contour`”, corresponding to the drawing precision of `\Plot`, `\ParamPlot`, `\PolarPlot`, and `\ContourPlot`. Commands with “`*`” apply to all instances following the drawing command, while those without “`*`” apply only to the first drawing command that follows.

<code>\Plot</code>	<code>\Plot[⟨key-value⟩]{⟨function⟩}</code>
New: 2025-05-15	

This command is used to draw the function $y = y(x)$, where `⟨function⟩` is a `gnuplot` function expression with independent variable “`x`”. `⟨key-value⟩` sets the drawing style, corresponding to `⟨ztikz/2dplot⟩`. `⟨domain⟩` defaults to `-5:5`. **Note:** Scatter plots can be achieved simply by setting `⟨opacity⟩` to 0.

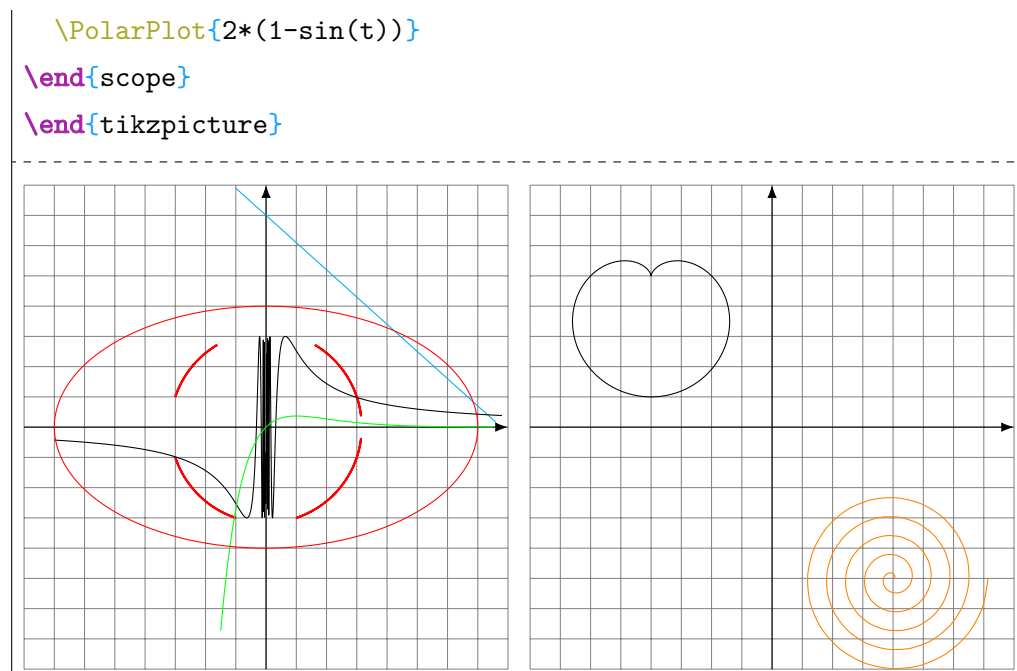
$\backslash\text{ContourPlot}$	$\backslash\text{ContourPlot}[\langle\text{key-value}\rangle]\{\langle\text{equation}\rangle\}$
New: 2025-05-15	This command is used to draw the equation $f(x, y) = c$, where $\langle\text{equation}\rangle$ is a gnuplot equation expression with variables “x”, “y”, and the “$=$” sign is not required in the expression. $\langle\text{key-value}\rangle$ sets the drawing style, corresponding to $\langle\text{ztikz}/2\text{dplot}\rangle$. $\langle\text{domain}\rangle$ defaults to “$-5:5;*:*$” (i.e., the range for y is self-adapting). Note: This function can be used to draw vertical segments like $x = c$.
$\backslash\text{ParamPlot}$	$\backslash\text{ParamPlot}[\langle\text{key-value}\rangle]\{\langle\text{equation}\rangle\}$
New: 2025-05-15	This command draws parametric equations $x = x(t)$, $y = y(t)$, where $\langle\text{equation}\rangle$ is a gnuplot equation expression with parameter “t”. $\langle\text{key-value}\rangle$ sets the style, corresponding to $\langle\text{ztikz}/2\text{dplot}\rangle$. $\langle\text{domain}\rangle$ defaults to $0:2\pi$.
$\backslash\text{PolarPlot}$	$\backslash\text{PolarPlot}[\langle\text{key-value}\rangle]\{\langle\text{equation}\rangle\}$
New: 2025-05-15	This command draws polar equations $\rho = \rho(t)$, where $\langle\text{equation}\rangle$ is a gnuplot equation expression with parameter “t”. $\langle\text{key-value}\rangle$ sets the style, corresponding to $\langle\text{ztikz}/2\text{dplot}\rangle$. $\langle\text{domain}\rangle$ defaults to $0:2\pi$.

```

\begin{tikzpicture}[>=Latex, scale=.4]
\ShowGrid{(-8, -8); (8, 8)}\ShowAxis{(0, -8); (0, 8)} ✓
\ShowAxis{(-8, 0); (8, 0)}
% draw functions/curves
\Plot[domain=-1:7.6, style=cyan] {-.9*x+7}
\ContourPlot[
  domain={-3:pi; -3:exp(1)}, style={red, thick}
]{x**2 + y**2 - 10}
% change plot precise
\PlotPrecise{plot}{1500}
\Plot[domain=-7:7.8] {3*sin(1/x)}
\Plot[domain=-1.5:7.5, style=green] {x*exp(-x)}
\ParamPlot[domain=0:2*pi, style=red] {7*sin(t), 4*cos(t)}
\end{tikzpicture}
\hskip.5em
\begin{tikzpicture}[>=Latex, scale=.4]
\ShowGrid{(-8, -8); (8, 8)}\ShowAxis{(0, -8); (0, 8)} ✓
\ShowAxis{(-8, 0); (8, 0)}
% draw functions/curves
\begin{scope}[xshift=4cm, yshift=-5cm]
  \PolarPlot[domain=0:10*pi, style=orange] {0.1*t}
\end{scope}
\begin{scope}[xshift=-4cm, yshift=5cm]

```

Ex. 13



Reviewing the simple case above: we used the four commands `\Plot`, `\ParamPlot`, `\PolarPlot`, and `\ContourPlot`, and also applied the `\PlotPrecise` command to change the drawing precision of the `\Plot` command.

<code>ztikz/3dplot/domain</code>	<code>domain = <float:float; float:float>.....Initial value: -5:5; -5:5</code>
<code>ztikz/3dplot/pm3d</code>	<code>pm3d = <true false>.....Initial value: true</code>
<code>ztikz/3dplot/width</code>	<code>width = <length>.....Initial value: 0.75\linewidth</code>
<code>ztikz/3dplot/palette</code>	<code>palette = <string>.....Initial value: rgbformulae 22,13,-31</code>

`<domain>` is used to set the value ranges for independent variables x and y , separated by “;”. `<pm3d>` controls whether to enable surface coloring; if `<pm3d>=false`, only a series of curves for the surface will be drawn. `<width>` sets the width of the image.

<code>\Plotz</code>	<code>\Plotz[<key-value>]{<function>}</code>
New: 2025-05-15	This command draws ordinary 2D explicit functions, where <code><function></code> is a gnuplot function expression. <code><key-value></code> sets the style, corresponding to <code><ztikz/3dplot></code> . Note: This command cannot be used within the <code>\tikzpicture</code> environment.

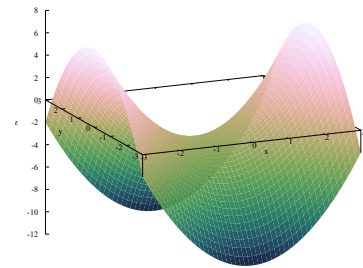
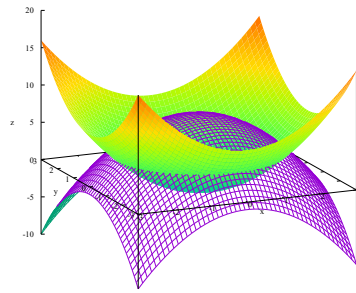
The following case demonstrates the basic usage of the `\Plotz` command, where “ x^2+y^2-2 with `pm3d`” in the first case is a syntax unique to gnuplot; see the gnuplot manual for details.

<pre> \Plotz[pm3d = false, width = .45\linewidth, domain = {-3:3; -3:3}]{x**2+y**2-2 with pm3d, -x**2-y**2+8 with lines} \hskip5em </pre>	Ex. 14
---	---------------

```

\Plotz[
  pm3d,
  width = .45\linewidth,
  domain = {-3:3; -3:3},
  palette = {cubehelix start 0 cycles -1. saturation 1}
]{x**2-y**2-2}

```



`\currentTikzIndex`

New: 2025-05-15

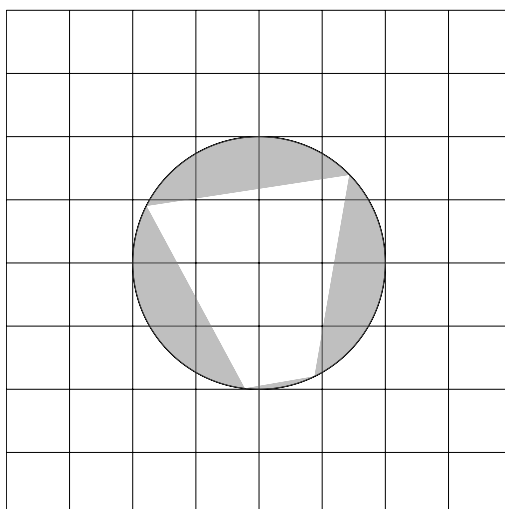
This command represents the index of the current `tikzpicture` environment, returning an integer starting from 1.

`\gnudata` ★ `\gnudata{⟨index⟩}`

New: 2025-05-22

This command references the drawing data generated in the current `tikzpicture` environment, returning a (data) file name, starting from 1. `⟨index⟩` accepts an integer representing the number of the drawing data in the current environment. Each drawn function will generate a corresponding data file in the corresponding folder, which users can use for subsequent drawing operations.

Note (Supplement to `\gnudata` usage for subsequent area filling): For example, the command `\gnudata{2}` where the parameter “2” indicates this data is the second function drawing data in the current `tikzpicture` environment. Thus, in the first `tikzpicture` environment, its return value might be “./ztikz_output/gnuplot_data/gnu_data_1_2.table”.

Figure 1: `\ContourPlot` Fill Issue

Note: For technical reasons, data generated by the `\ContourPlot` command currently cannot be used for subsequent filling operations. Consider first transforming the implicit function into parametric or polar form, and then exporting the corresponding data. If you force the use of this type of data, you may get an undesirable output like [figure \(1\)](#).

5.3 cache Library

When users load the `cache` library and then compile the document (let its name be $\langle file \rangle$) in the command line, they will see log output like the following:

```
\write18 enabled.
entering extended mode
```

After compilation, a folder named `ztikz_output` will be generated in your project folder, which is created when you first call the `ztikz` macro package. This folder is used to store $\text{\textit{zTikZ}}$'s cache files, including the cached results of $\text{\textit{TikZ}}$'s external library, Python scripts, WolframScript scripts, and gnuplot results.

Now let's talk about the composition of this folder: for example, if a user runs the `\Plot` command, four files as shown in figure (2) will be generated in the `ztikz_output/tikz_data/` directory:

```
ztikz_output
├── gnuplot_data ..... gnuplot cache dir
│   └── gnu_data_1_1.table
├── mma_data ..... WolframScript cache dir
├── python_data ..... Python cache dir
├── scripts ..... gnuplot scripts
│   ├── 3d_plot.gp
│   ├── contour_plot.gp
│   ├── param_plot.gp
│   ├── plot_plot.gp
│   └── polar_plot.gp
├── tikz_data ..... TikZ cache dir
│   ├──  $\langle file \rangle$ -figure0.dpth
│   ├──  $\langle file \rangle$ -figure0.log
│   ├──  $\langle file \rangle$ -figure0.md5
│   ├──  $\langle file \rangle$ -figure0.pdf
│   └──  $\langle file \rangle$ -figure0.run.xml
└── ztikz.hash ..... Hash records
```

Figure 2: Schematic of the $\text{\textit{zTikZ}}$ Cache Directory Structure

$\langle file \rangle$ -figure0.pdf in `tikz_data` is the cached PDF file of the `tikzpicture` environment. At this time, the following content can be seen in the corresponding $\langle file \rangle$.md5 file:

```
\def \tikzexternallastkey {AE7F2539E81C96848ADCCEE3994993D1}%
```

The above command saves the Hash value of the code in this `tikzpicture` environment. When we change the code in the environment, this Hash value will change, causing $\text{\textit{TikZ}}$ to run the environment again and re-generate the image. This is a rough description of the cache function provided by $\text{\textit{TikZ}}$'s external library. The Cache mechanism in $\text{\textit{zTikZ}}$ is very similar to this principle.

5.3.1 User Interface

`\ztikzHashClean`

New: 2025-05-15

This command takes no parameters and is used to clear all previously cached Hash values.

`\ztikzHashCurrent`

New: 2025-05-15

`\ztikzHashCurrent*`
`\ztikzHashCurrent[⟨separator⟩]`

This command is mainly used for debugging cache-related code, often in conjunction with the `\ztikzForceToSkip` command. `\ztikzHashCurrent*` outputs the most recent Hash calculation result. `\ztikzHashCurrent[⟨separator⟩]` outputs all cached Hash values up to now, separated by `⟨separator⟩` in the PDF; `⟨separator⟩` defaults to “,”.

`\ztikzCachedHash *`

New: 2025-05-29

`\ztikzCachedHash[⟨keyval⟩]`

This command is used to output the currently cached Hash value, with a wider range of applications than the `\ztikzHashCurrent` command.

`ztikz/cache/hash/index`

`ztikz/cache/hash/file`

`ztikz/cache/hash/label`

`index = ⟨integer⟩.....Initial value: -1`
`file = ⟨filename⟩.....Initial value: ztikz_output/ztikz.hash`
`label = ⟨string⟩.....Initial value: \g_zcache_latest_cache_label_tl`

By default, `⟨label⟩` is the latest cache label; `⟨file⟩` is the cache file for Hash values; `⟨index⟩` specifies the index of the Hash value cached by that `⟨label⟩`, defaulting to -1 (the latest Hash value).

`\ztikzForceToSkip`

New: 2025-05-15

This command forces skipping the (re-)running of the first environment or command with a cache mechanism that follows it, even if the Hash for that environment or command has changed. Subsequent referenced results from `\wolframResult` or `\wolframOutputFile` commands will be affected by this command.

Note: When this command is applied, new Hash values will not be cached. Currently, this command only works for commands and environments in the `python` and `wolfram` libraries; it will delete scripts and results related to the **new** Hash.

`\ztikzForceToRun`

New: 2025-05-21

This command forces the running of the first environment or command with a cache mechanism that follows it, even if its Hash has not changed.

Note: Currently, this command only works for commands and environments in the `python` and `wolfram` libraries. Even if the Hash value of subsequent commands or environments changes, this new Hash value will not be cached.

5.3.2 Programming Interface

The `cache` library of `ztikz` provides a series of programming interfaces that users can use to write commands or environments related to calling external programs. These new commands or environments will support the cache mechanism. The programming interfaces and some other related commands currently provided in the `cache` library are:

`\g_ztikz_file_hash_seq`

New: 2025-05-30

This sequence (variable) stores all currently cached Hash values, where each item is the Hash value corresponding to a command or environment with a cache mechanism.

Note: This command is set by the subsequent `\__zcache_hash_extract_all:nN` command.

`\g_ztikz_hash_nochg_run_bool`

`\g_ztikz_hashchg_norun_bool`

New: 2025-05-30

These two boolean values control the behavior of the `\ztikz_if_run_again:nnnTF` command. When the former is “true”, Hash verification for the `\ztikz_if_run_again:nnnTF` command will be forced to skip and the `<true code>` will be run directly. When the latter is “true”, Hash verification will be forced to skip and the `<false code>` will be run directly.

Note: Both boolean values default to “false”; it is an error for both to be “true” at the same time.

`\xsim_file_write_start:nn`

`\xsim_file_write_start:ne`

New: 2025-05-30

`\xsim_file_write_start:nn {<bool>}{<file>}`

This command from the `xsimverb` package is used to transcribe environment content into the `<file>` file. `<bool>` is a boolean; if the environment needs to accept parameters, set `<bool>` to “`\c_true_bool`”, otherwise to “`\c_false_bool`”.

Note: This command must be used in conjunction with the `\xsim_file_write_stop:` command, otherwise an error will occur.

`\xsim_file_write_stop:`

New: 2025-05-30

This command ends the transcription of environment content and must be used with the `\xsim_file_write_start:nn` command.

`\ztikz_if_run_again:nnnTF`

`\ztikz_if_run_again:nenTF`

New: 2025-05-30

`\ztikz_if_run_again:nnnTF`
`{<bool>}{<file/hash>}{<label>}`
`{<true code>}{<false code>}`

`<bool>` controls the type of the second parameter `<file/hash>`. If `<bool>` is “`\c_true_bool`”, `<file/hash>` should be a file name; otherwise, it should be a Hash value. `<label>` is the cache label corresponding to that Hash or file Hash. When the Hash value for the environment does not exist, changes, or follows the `\ztikzForceToRun` command, `<true code>` will be executed. When the Hash value exists or follows the `\ztikzForceToSkip` command, `<false code>` will be executed.

```
\ztikz_term_info:n \ztikz_term_info:n {message}
```

```
\ztikz_term_info:e
```

This command is similar to the `\typeout` command in T_EX, used to output information *message* to the terminal.

New: 2025-05-31

5.3.3 Private Interface

This section describes the private commands and variables in the `cache` library, which may be necessary when users define some commands or environments. As these commands have not yet been fully organized, they may undergo significant changes in the future, and users should use them with caution.

<code>\l_ztikz_current_hash_tl</code>	
New: 2025-05-30	
	This variable returns the latest Hash value.
<code>_zcache_hash_add:nn</code>	<code>_zcache_hash_add:nn {<label>}{<hash>}</code>
New: 2025-05-30	
	This command adds <code><hash></code> to the cache label corresponding to <code><label></code> .
<code>_zcache_hash_extract_all:nN</code>	<code>_zcache_hash_extract_all:nN {<file>}<seq></code>
New: 2025-05-30	
	This command extracts all Hash values from the file <code><file></code> and saves them into the variable <code><seq></code> , where each item corresponds to a Hash value.
<code>_zcache_hash_extract_by_label:nnn</code>	<code>_zcache_hash_extract_by_label:nnn {<file>}{<label>}{<index>}</code>
New: 2025-05-30	
	This command extracts the Hash value with label <code><label></code> and index <code><index></code> from file <code><file></code> . It has no return value but sets <code>\l_ztikz_current_hash_tl</code> to the extracted Hash value.

5.3.4 Programming Interface Case Study

Below is a usage example of the above commands, creating a Mermaid drawing environment that supports the cache mechanism (users need to install the “mermaid-cli” tool):

Ex. 15

```

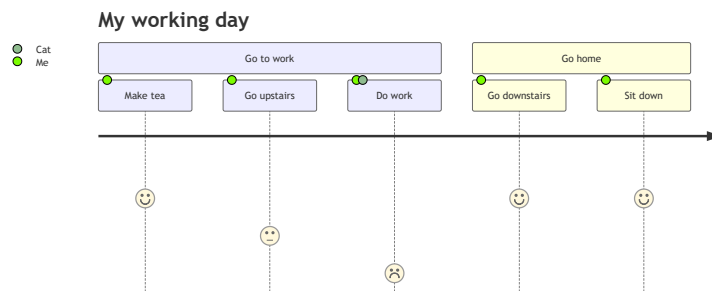
\ExplSyntaxOn\makeatletter
% 1. environment declaration
\tl_new:N \g__ztikz_mmd_path_tl
\tl_gset:Nn \g__ztikz_mmd_path_tl {ztikz_output/mmd_data}
\ztool_shell_mkdir:e { \g__ztikz_mmd_path_tl }
\NewDocumentEnvironment{mermaid}{m }
{
  \exp_args:Ne \xsim_file_write_start:nn {\c_true_bool}{
    \g__ztikz_mmd_path_tl/t@mp.mmd}
  }{
    \xsim_file_write_stop:
    \ztikz_if_run_again:nenTF {\c_true_bool}{t@mp.mmd}{#1}
    {
      \ztool_shell_escape:e
      {
        mmdc
        \space-i\space \g__ztikz_mmd_path_tl/t@mp.mmd
        \space-o\space \g__ztikz_mmd_path_tl/t@mp.pdf
        \space-f
      }
      \ztikz_term_info:e { Mermaid~running~on~file:'
\l__ztikz_current_hash_tl.mmd'~... }
    }{
      \ztikz_term_info:e { Use~the~cached~Mermaid~result:'
\l__ztikz_current_hash_tl.pdf'~... }
    }
    \edef\t@mp@file{\g__ztikz_mmd_path_tl/\l__ztikz_current_hash_tl}
    \ztool_shell_mv:ee
    {\g__ztikz_mmd_path_tl/t@mp.mmd}
    {\t@mp@file.mmd}
    \ztool_shell_mv:ee
    {\g__ztikz_mmd_path_tl/t@mp.pdf}
  }
}

```

```

    {\t@mp@file.pdf}
\edef\mmdOutputFile{\t@mp@file.pdf}
}
\makeatother\ExplSyntaxOff
% 2. environment usage
\begin{mermaid}{mmd-I}
journey
  title My working day
  section Go to work
    Make tea: 5: Me
    Go upstairs: 3: Me
    Do work: 1: Me, Cat
  section Go home
    Go downstairs: 5: Me
    Sit down: 5: Me
\end{mermaid}
\begin{center}
  \includegraphics[width=.7\linewidth]{\mmdOutputFile}
\end{center}

```



5.4 python Library

The `python` library is mainly used for interacting with Python, with usage similar to the `gnuplot` library. It primarily provides image drawing and calculation interfaces, where calculation interfaces include numerical and symbolic calculations.

Apart from the Python drawing functions provided by $\text{\texttt{zTikZ}}$, we need to highlight the floating-point calculation functions: when calling this library, $\text{\texttt{zTikZ}}$ imports the Python packages `numpy`, `sympy`, and `scipy` by default. Furthermore, when using functions in `numpy`, users no longer need to add prefixes; for example, when solving for `sin(2.345)`, use `\py{sin(2.345)}` directly instead of `\py{np.sin(2.345)}`. The same applies to functions in other Python libraries.

NOTE: After calling this library, the “`-shell-escape`” parameter must be enabled during compilation.

 $\text{\texttt{\py}}$

New: 2025-05-15

`\py[ $\langle\text{raw}\rangle$ | $\langle\text{str}\rangle$ ]{ $\langle\text{code}\rangle$ }`

This command calls Python for floating-point calculation, with $\langle\text{code}\rangle$ being valid Python code. These results are not cached, meaning Python re-calculates them each time the document is compiled. Users can nest the `\py` command into their own defined macros.

Note: $\langle\text{raw}\rangle$ tokenizes the returned results according to original $\text{\texttt{T\TeX}}$ cat-codes, while $\langle\text{str}\rangle$ treats the results as strings.

```
\newcommand{\pypow}[1]{\py{#1}}
\newcommand{\pyreverse}[1]{\py{'#1'[::-1]}}
\newcommand{\pyuppercase}[1]{\py{'#1'.upper()}}
\begin{itemize}
  \item Power Calculation:  $2^{10} = \text{\texttt{\pypow{2**10}}}$ 
  \item Reverse a string using Python:  $\text{\texttt{\pyreverse{Hello-LaTeX}}}$ 
  \item Uppercase a string:  $\text{\texttt{\pyuppercase{hello-latex}}}$ 
  \item Modulus:  $102 = \text{\texttt{\py{mod(102, 8)}}} \bmod 8$ 
  \item Return string Options:  $\text{\texttt{\py[str]{'\$'+str(2**10)+'\$'}}}$ 
\end{itemize}
```

Ex. 16

-
- Power Calculation: $2^{10} = 1024$
 - Reverse a string using Python: XeTaL-olleH
 - Uppercase a string: HELLO-LATEX
 - Modulus: $102 = 6 \bmod 8$
 - Return string Options: \$\$1024\$\$

 $\backslash\text{sympy}$

New: 2025-05-29

 $\backslash\text{sympy}\{\langle\text{label}\rangle\}\{\langle\text{expression}\rangle\}$

This command is used to call Python’s `sympy` library for symbolic calculations, and $\text{\textit{zTikZ}}$ provides a cache mechanism for it. $\langle\text{label}\rangle$ cannot contain “:” and is used to specify the cache label, which must be unique in the current document. $\langle\text{expression}\rangle$ is the symbolic expression. A series of symbolic variables are predefined in the `python` library, including `x`, `y`, `z`, `u`, `v`, `t`, which do not need re-declaration.

Note: By default, results may include characters like “ $\wedge$ ”, “ $_$ ” that can only be used in math mode, so users should place this command within a mathematical environment.

Ex. 17

```

\[\int x^8 + \cos(7x) + 6t\,,\mathrm{d}x
= \text{sympy}\{\text{integral}\}\{\text{integrate}(x**8 + \cos(7*x) + 6*t, x )\}
\]
\[\mathrm{eig}(\begin{bmatrix}1 & 2\\2 & 2\end{bmatrix})
= \text{sympy}\{\text{matrix}\}\{\text{Matrix}([[1, 2], [2, 2]]).\text{eigenvals}()\}
\]
```

$$\int x^8 + \cos(7x) + 6t \, dx = 6tx + \frac{x^9}{9} + \frac{\sin(7x)}{7}$$

$$\text{eig}\left(\begin{bmatrix}1 & 2 \\ 2 & 2\end{bmatrix}\right) = \left\{\frac{3}{2} - \frac{\sqrt{17}}{2} : 1, \frac{3}{2} + \frac{\sqrt{17}}{2} : 1\right\}$$

 pyfig

Updated: 2025-05-29

 $\backslash\text{begin}\{\text{pyfig}\}\{\langle\text{label}\rangle\}\{\langle\text{output file}\rangle\}$
 $\quad\langle\text{plot code}\rangle$
 $\backslash\text{end}\{\text{pyfig}\}$

This environment calls Python for drawing and returns no results. It has a cache mechanism. $\langle\text{label}\rangle$ cannot contain “:” and must be unique. $\langle\text{output file}\rangle$ specifies the name of the output file for the $\langle\text{plot code}\rangle$, without the path but with the extension.

Note: It is recommended to use different $\langle\text{output file}\rangle$ values for different `pyfig` environments. Users do not need to add `plt.savefig()`; $\text{\textit{zTikZ}}$ handles this. Indentation is preserved starting from the beginning of the line, so avoid extra leading spaces. Ensure $\langle\text{output file}\rangle$ matches the file name in the environment code.

 $\backslash\text{pyfigOutputFile} \star$

New: 2025-04-21

This command returns the name of the file output by the `pyfig` environment, which can be imported using commands like $\backslash\text{input}$ or $\backslash\text{includegraphics}$.

Ex. 18

```

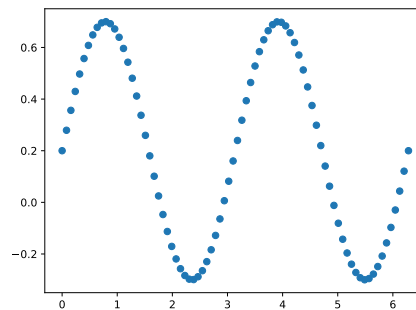
\begin{pyfig}\sinGraph\sin_graph.pdf
import matplotlib

```

```

matplotlib.use('Agg')
from matplotlib import pyplot as plt
import numpy as np
x = np.linspace(0, 2*np.pi, num = 80)
y = np.sin(x)*np.cos(x)+.2
plt.plot(x, y, 'o')
\end{pyfig}
\begin{center}
  \includegraphics[width=.5\linewidth]{\pyfigOutputFile}
\end{center}

```



`pycode`

New: 2025-05-21

```

\begin{pycode}{\label}{\langle output file \rangle}
  \langle any python code \rangle
\end{pycode}

```

This environment calls Python to execute $\langle \text{any python code} \rangle$ and returns no results. It has a cache mechanism. $\langle \text{label} \rangle$ cannot contain “:” and must be unique. $\langle \text{output file} \rangle$ specifies the name for the output file without the path but with the extension. $\text{\textit{zTikZ}}$ automatically calls Python to execute the file, and results are saved in `\pycodeOutputFile`.

Note: It is recommended to use different $\langle \text{output file} \rangle$ values for different `pycode` environments. Indentation is preserved, so avoid excessive indentation. Ensure $\langle \text{output file} \rangle$ matches the file name in the environment code.

`\pycodeOutputFile` ★

New: 2025-04-21

Returns the name of the file output by the `pycode` environment, which can be imported using `\input` or `\includegraphics`.

Below is a simple usage example for the `pycode` environment; for the content of `table.py.txt`, see [section \(6.3\)](#).

```

\input{./table.py.txt}
\begin{center}
  \input{\pycodeOutputFile}

```

Ex. 19

```
\end{center}
```

number/function	sin	cos	tan
1	0.8415	0.5403	1.5574
2	0.9093	-0.4161	-2.185
3	0.1411	-0.99	-0.1425
4	-0.7568	-0.6536	1.1578
5	-0.9589	0.2837	-3.3805
6	-0.2794	0.9602	-0.291
7	0.657	0.7539	0.8714
8	0.9894	-0.1455	-6.7997
9	0.4121	-0.9111	-0.4523
10	-0.544	-0.8391	0.6484
11	-1.0	0.0044	-225.9508
12	-0.5366	0.8439	-0.6359
13	0.4202	0.9074	0.463
14	0.9906	0.1367	7.2446
15	0.6503	-0.7597	-0.856

5.5 wolfram Library

The `wolfram` library in `zTikZ` can be seen as a new implementation of the original `latexalpha2` package, addressing its various limitations. Currently, the `wolfram` library has implemented all commands from `latexalpha2` except for `\wolframanimation`, with significant improvements in compatibility, ease of use, and extensibility². For example, it can be used on Windows, Linux, and macOS, and environment source code supports special characters like “\, #, \$, _, ^, &”. Here are some points to note:

- Note the comment style in WolframScript: use “`(* something *)`” rather than “`(* something*)`”; comments should not be adjacent to “`*`” to avoid parsing errors.
- Due to WolframScript limitations, scripts must have the “`.wls`” extension to be recognized and executed.
- For all subsequent commands or environments supporting caching in this section, the cache label `\label` can only use “**numbers, letters, hyphens (-)**”.

NOTE: After calling this library, the “`-shell-escape`” parameter must be enabled during compilation.

<code>\wolframResult</code>
New: 2025-05-15
Updated: 2026-05-07

`\wolframResult[separator]`
`\wolframResult*[index]`

This command references results from the previous WolframScript calculation. `\wolframResult[separator]` references all results separated by `separator`. `\wolframResult*[index]` references only a specific result, where `index` is an integer, defaulting to 1.

NOTE: Because `\wolframResult` partitions results based on “`@12`”, ensure the catcode of “`@`” is 12, otherwise parsing errors may occur.

<code>\wolframOutputPath</code> ★
New: 2026-05-08

This command stores the folder where WolframScript running results (files) are located; it is often used with the `wolframCode` environment to reference image or text results. This command is more general than the subsequent `\wolframResult` and `\wolframOutputFile` commands.

<code>\wolframOutputFile</code> ★
New: 2025-05-15
Updated: 2026-04-30

Returns the filename corresponding to the previous WolframScript run; very convenient for referencing image results. It is more flexible than `\wolframResult` as the former calls the text file while the latter returns only the filename produced by the last WolframScript call.

²2026-04-29: Speed has also significantly increased with the new `wstp` engine.

`\wolfram`

New: 2025-05-29

`\wolfram{<label>}{<code>}`
`\wolfram*{<label>}{<code>}`

Calls WolframScript for calculation with a cache mechanism. `<label>` cannot contain “:” and must be unique. `<code>` is valid WolframScript code. Results are $\text{\LaTeX}$ code by default; commands with “*” return ordinary strings.

```
\wolfram{wolframLaplace}{LaplaceTransform[t^4 Sin[3*t], t, s]}
\[\mathcal{L}(t^4\sin(3t)) = \wolframResult
\]
```

$$\mathcal{L}(t^4 \sin(3t)) = \frac{72(5s^4 - 90s^2 + 81)}{(s^2 + 9)^5}$$

`\wolframTex`

New: 2025-05-29

`\wolframTex{<label>}{<Tex code>}`

Similar to `\wolfram`, but it converts all content in `<Tex code>` into Mathematica/Mathics code, returning results as $\text{\LaTeX}$ code. `<label>` cannot contain “:” and must be unique.

NOTE: Due to the complex and special implementation, neither `<Tex code>` nor `<label>` can contain the “\$” symbol, otherwise parsing errors will occur; users should not rely too much on this command, as it may consume significant time or even cause compilation errors in some cases.

```
\wolframTex{wolframTexInt}{\int_a^b \sin(x) \, dx}
\[\int_a^b \sin(x) \, dx = \wolframResult
\]
```

$$\int_a^b \sin(x) \, dx = \cos(a) - \cos(b)$$

`\wolframTable`

New: 2025-05-29

`\wolframTable{<label>}{<code>}`
`\wolframTable*{<label>}{<key-value>}{<code>}`

Calls the Wolfram engine to generate a table with a cache mechanism. `<label>` cannot contain “:” and must be unique. Commands without “*” do not output the table in the PDF but set `\wolframTablePData` and `\wolframTableFData` variables. Commands with “*” output the table with styles specified by `<key-value>`. `<code>` is valid WolframScript/Mathics code.

```
\wolframTable*{wolframTable}[
  format=cccc, hdbt-rule,
  header={x$ & x^2$ & x^3$ & x^4$},
  cell-cmd={\textcolor{red}{(1)}}]
\]
```

```
] {Table[{i, i^2, i^3, i^4}, {i, 6}]}
\SetTblrOuter{expand=\wolframTableFData}
\hskip6em
\begin{tblr}
{
  colspec = {cccc},
  rowspec = {
    | [2pt,green7]Q| [2pt, teal7]Q| [green7]Q| [green6]
    Q| [green5]Q| [green4]Q| [green3]Q| [3pt,teal7]
  }
} \wolframTableFData
\end{tblr}
```

x	x^2	x^3	x^4
(1)	(1)	(1)	(1)
(2)	(4)	(8)	(16)
(3)	(9)	(27)	(81)
(4)	(16)	(64)	(256)
(5)	(25)	(125)	(625)
(6)	(36)	(216)	(1296)

x	x^2	x^3	x^4
(1)	(1)	(1)	(1)
(2)	(4)	(8)	(16)
(3)	(9)	(27)	(81)
(4)	(16)	(64)	(256)
(5)	(25)	(125)	(625)
(6)	(36)	(216)	(1296)

ztikz/wolfram/table/format	format	= $\langle$ column format $\rangle$	Initial value: $\ast\{12\}\{1\}$
ztikz/wolfram/table/header	header	= $\langle$ header $\rangle$	Initial value: empty
ztikz/wolfram/table/hdbt-rule	hdbt-rule	= $\langle$ false true $\rangle$	Initial value: false
ztikz/wolfram/table/cell-cmd	cell-cmd	= $\langle$ cell command $\rangle$	Initial value: #1

$\langle$ format $\rangle$ sets column formatting; $\langle$ header $\rangle$ sets the table header for the first row; $\langle$ hdbt-rule $\rangle$ (header bottom rule) specifies whether to display a horizontal line below the header (only for `\wolframTable*`); $\langle$ cell-cmd $\rangle$ accepts one parameter and is applied to each cell in `\wolframTablePData` (excluding the header), using “#1” for the content.

<code>\wolframTablePData</code> $\star$	These commands represent the results of the most recent <code>\wolframTable</code> calculation without commands like <code>\hline</code> or <code>\midrule</code> . <code>\wolframTablePData</code> (Part Data) excludes the header, while <code>\wolframTableFData</code> (Full Data) includes it. They can be passed to table formatting packages like <code>tabularray</code> . Note: $\langle cell-cmd \rangle$ in <code>\wolframTable</code> should not be mixed with $\langle cells/cmd \rangle$ in <code>tabularray</code> .
<code>\wolframTableFData</code> $\star$	
New: 2025-05-18	

`\wolframSolve` `\wolframSolve{⟨label⟩}[⟨key-value⟩]{⟨equation⟩}`
`\wolframSolve*{⟨label⟩}{⟨full code⟩}`

New: 2025-05-29

Calls WolframScript to solve equations with a cache mechanism. $\langle label \rangle$ cannot contain “:” and must be unique. $\langle equation \rangle$ is the equation expression. $\langle key-value \rangle$ sets independent variables and domains. $\langle full code \rangle$ is the complete equation expression including variables and domains.

`ztikz/wolfram/solve/domain` `domain = ⟨domain⟩.....Initial value: empty`
`ztikz/wolfram/solve/var` `var = ⟨variable⟩.....Initial value: empty`

$\langle domain \rangle$ sets the range for solving (e.g., $\langle domain \rangle = \text{Integers}$); $\langle var \rangle$ sets the independent variables (e.g., $\langle var \rangle = x$).

```
\wolframSolve{wolframSolve-I}[var={x, y}]{a x + y == 8 && bEx.-23
y == 1}
```

```
\begin{align}
& \text{\textcolor{brown}{\textit{& \wolframResult \}}}
& \text{\textcolor{brown}{\textit{& \wolframResult[|]| \}}}
& \text{\textcolor{brown}{\textit{& \wolframResult* \}}}
& \text{\textcolor{brown}{\textit{& \wolframResult*[3-1]}}}
```

```
\end{align}
```

```
\wolframSolve{wolframSolve-II}
```

```
[var={x, y}, domain=Integers]
```

```
{x^2 + 2 y^3 == 3681 && x > 0 && y > 0}
```

```
\begin{align}
```

```
\textcolor{brown}{\textit{\wolframResult}}
```

```
\end{align}
```

$$x = \frac{9}{a+b}, y = -\frac{a-8b}{a+b} \quad (5.1)$$

$$x = \frac{9}{a+b} || y = -\frac{a-8b}{a+b} \quad (5.2)$$

$$x = \frac{9}{a+b} \quad (5.3)$$

$$y = -\frac{a-8b}{a+b} \quad (5.4)$$

$$x = 15, y = 12, x = 41, y = 10, x = 57, y = 6 \quad (5.5)$$

`\wolframDSolve` `\wolframDSolve{<label>}[<key-value>]{<equation>}`
 `\wolframDSolve*{<label>}{<full code>}`

New: 2025-05-29

Calls WolframScript to solve differential equations with a cache mechanism. `<label>` cannot contain “:” and must be unique. `<equation>` is the equation expression. `<key-value>` sets variables; `<full code>` is the complete expression.

ztikz/wolfram/dsolve/depend `depend` = `<dependent variable>.....Initial value: y[x]`
 ztikz/wolfram/dsolve/independ `independ` = `<independent variable>.....Initial value: x`

`<depend>` specifies the dependent variable (e.g., `<depend>=y[x]`); `<independ>` specifies the independent variable (e.g., `<independ>=x`).

Ex. 24

```

\wolframDSolve{wolframDSolve-I}
{y'[x] + y[x] == a*Sin[x], y[0] == 1}
\begin{align}
&\&\wolframResult
\end{align}
\wolframDSolve{wolframDSolve-II}
[depend={y[x], z[x]}]
{y'[x] == Exp[z[x]] + 1, z'[x] == y[x] - x}
\begin{align}\left\{\begin{aligned}
&\&\wolframResult[\&]
\end{aligned}\right.\end{align}
\end{aligned}\right.\end{align}

```

$$y(x) = -\frac{1}{2}e^{-x}(-ae^x \sin(x) + ae^x \cos(x) - a - 2) \quad (5.6)$$

$$\begin{cases} z(x) = \log\left(c_1 \tan^2\left(\frac{1}{2}\left(\sqrt{2}\sqrt{c_1}x + 2\sqrt{2}\sqrt{c_1}c_2\right)\right) + c_1\right) \\ y(x) = x + \sqrt{2}\sqrt{c_1} \tan\left(\frac{1}{2}\left(\sqrt{2}\sqrt{c_1}x + 2\sqrt{2}\sqrt{c_1}c_2\right)\right) \end{cases} \quad (5.7)$$

`wolframGraphics` `\begin{wolframGraphics}{<label>}[<spec>]`
 `<plot code>`

`\end{wolframGraphics}`

New: 2025-05-29

Environment for drawing with WolframScript, with a cache mechanism. `<label>` cannot contain “:” and must be unique. `<spec>` sets layout parameters; if empty, no result is returned and the file can be called via `\wolframOutputFile`. If `<spec>` is not empty, the return value is `\includegraphics[<spec>]{<path>/<HASH>.pdf}`.

NOTE: The final image name in `<plot code>` must be “FIGURE”, otherwise an error will occur.

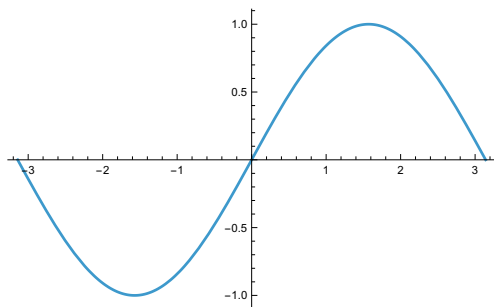
`\begin{wolframGraphics}{wolframSinGraph}` Ex. 25

FIGURE=Plot[Sin[x], {x, -Pi, Pi}]

`\end{wolframGraphics}`

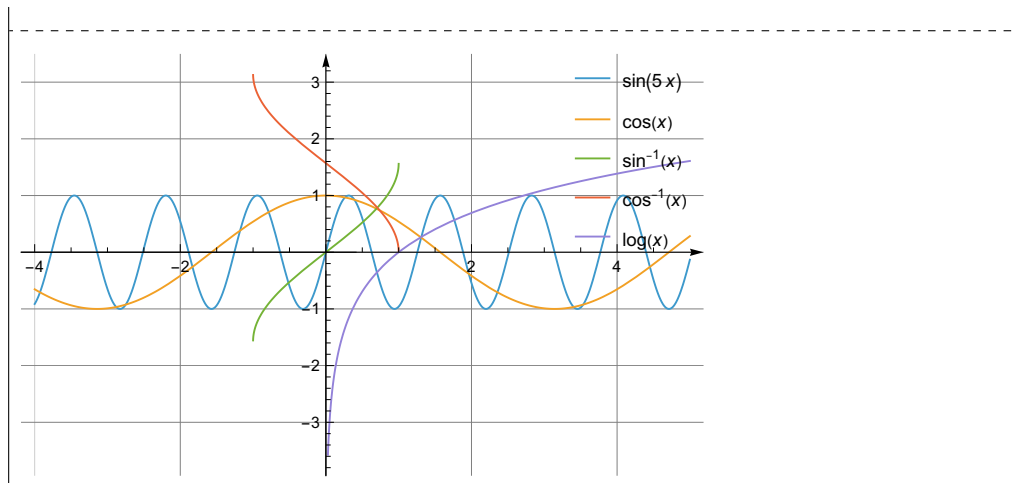
% include graphics

`\includegraphics[width=.5\linewidth]{\wolframOutputFile}`



`\begin{wolframGraphics}{fig-1}[width=.7\linewidth]` Ex. 26

```
fp = Plot[{
  Sin[5 x],
  Cos[x],
  ArcSin[x],
  ArcCos[x],
  Log[x]
},
{x, -4, 5},
PlotLegends -> Placed["Expressions", {Right, Top}],
PlotStyle    -> AbsoluteThickness[1],
AxesStyle    -> Arrowheads[0.02],
GridLines    -> Automatic
];
FIGURE = Show[fp];
\end{wolframGraphics}
```



```
wolframCode \begin{wolframCode}{\label}
              \langle wolfram code \rangle
New: 2026-05-08 \end{wolframCode}
```

This environment can run arbitrary Wolfram code and has a cache mechanism. If users need to export calculation results, they can use the “`scriptPath`” variable provided by this `wolfram` library, which represents the cache folder for the Wolfram code. $\langle label \rangle$ cannot contain “.” and must be unique within the current document.

NOTE: The cache mechanism of this environment has a flaw: if the $\langle label \rangle$ has already cached results for two hashes “X, Y”, and the user changes the code such that the new Hash happens to be “X”, the compilation will only return the result corresponding to “Y”. Users can add the `\ztikzForceToRun` command before this environment to refresh the cache.

```
\begin{wolframCode}{wolfram-code-1} Ex. 27
num = 2110;
Export[scriptPath <> "wolfram-code_text.txt", num]
\end{wolframCode}
% load result
\[ 2{110} = \input{\wolframOutputPath/wolfram-code_text.txt} \]
```

$$2^{110} = 1298074214633706907132624082305024$$

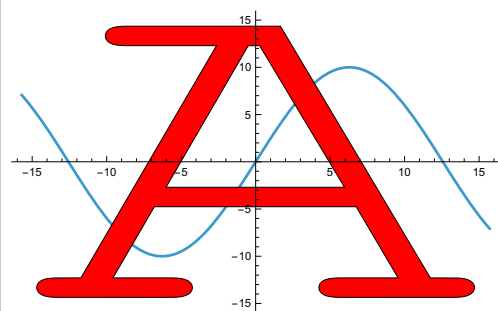
```
\begin{wolframCode}{wolfram-code-2} Ex. 28
fp1 = Plot[10 Sin[.25 x], {x, -5 Pi, 5 Pi}];

myChar = Style["A", FontFamily -> "Courier New", FontSize -> 50];
boundaryMesh = BoundaryDiscretizeGraphics[Text[myChar], _Text];
fp2 = Region[boundaryMesh, BaseStyle -> {FaceForm[Red],
```

```
EdgeForm[Black]]]

fig = Show[fp1, fp2, PlotRange->All];
Export[scriptPath <> "wolfram-code_fig.pdf", fig];
\end{wolframCode}

\includegraphics[width=.5\linewidth]{
\wolframOutputPath/wolfram-code_fig.pdf}
```



`\g_ztikz_wolfram_res_seq` *

New: 2026-05-07

This variable stores the previous WolframScript calculation result. Ordinary users should use the subsequent `\wolframResult` command.

5.6 l3draw Library

zTikZ provides an `l3draw` library based on the `l3draw` package, mainly for simple drawing needs. For general users, it offers `\zrule` and `\zplot` commands for gradient rectangles and functions. Wrappers for `l3draw` environments and commands are also provided but are currently incomplete and not recommended for general use.

<code>\zdrawSetUnit</code>	<code>\zdrawSetUnit[<i><unit></i>]</code>
New: 2025-05-15	Sets the drawing unit, e.g., <i><unit></i> as “cm”.

<code>\zdrawSetPathWidth</code>	<code>\zdrawSetPathWidth[<i><width></i>]</code>
New: 2025-05-15	Sets the line width, e.g., <i><width></i> as “0.5pt”; the default is 0.4pt in <code>l3draw</code> .

<code>\zrule</code>	<code>\zrule[<i><key-value></i>]</code>
New: 2025-05-15	Draws a gradient rectangle with attributes set by <i><key-value></i> .

<code>ztikz/zdraw/zrule/width</code>	<code>width</code>	<code>=</code>	<code><float></code>		Initial value:	<code>1</code>
<code>ztikz/zdraw/zrule/height</code>	<code>height</code>	<code>=</code>	<code><float></code>		Initial value:	<code>1</code>
<code>ztikz/zdraw/zrule/startColor</code>	<code>startColor</code>	<code>=</code>	<code><color></code>		Initial value:	<code>red</code>
<code>ztikz/zdraw/zrule/endColor</code>	<code>endColor</code>	<code>=</code>	<code><color></code>		Initial value:	<code>blue</code>
<code>ztikz/zdraw/zrule/step</code>	<code>step</code>	<code>=</code>	<code><float></code>		Initial value:	<code>0.25</code>

<width> and *<height>* set the dimensions; *<startColor>* and *<endColor>* set the colors; *<step>* controls the gradient precision.

`\zrule[width=10, startColor=red, step=1]` **Ex. 29**



<code>\zplot</code>	<code>\zplot[<i><key-value></i>]{<i><function></i>}</code>
New: 2025-05-15	Draws functions with horizontal or vertical gradients. <i><key-value></i> sets attributes; <i><function></i> is a valid expression.

NOTE: `\zplot` is currently unstable and may error; use with caution.

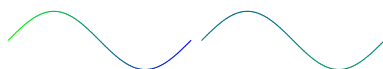
<code>ztikz/zdraw/zplot/action</code>	<code>action</code>	<code>=</code>	<code><draw stroke fill clip shade></code>		Initial value:	<code>draw</code>
<code>ztikz/zdraw/zplot/domain</code>	<code>domain</code>	<code>=</code>	<code><float, float, float></code>		Initial value:	<code>-5, 0.1, 5</code>
<code>ztikz/zdraw/zplot/range</code>	<code>range</code>	<code>=</code>	<code><float, float></code>		Initial value:	<code>-5, 5</code>
<code>ztikz/zdraw/zplot/startColor</code>	<code>startColor</code>	<code>=</code>	<code><color></code>		Initial value:	<code>black</code>
<code>ztikz/zdraw/zplot/endColor</code>	<code>endColor</code>	<code>=</code>	<code><color></code>		Initial value:	<code>white</code>
<code>ztikz/zdraw/zplot/axis</code>	<code>axis</code>	<code>=</code>	<code><x y></code>		Initial value:	<code>y</code>

<action> controls drawing behavior; *<domain>* sets the independent variable range (start, step, end); *<range>* sets the *y*-axis range, useful for *<action>*=`shade`; *<startColor>* and *<endColor>* set colors; *<axis>* sets the gradient direction (`‘x’` for horizontal, `‘y’` for vertical).

```
\def\PI{3.1415926}
```

Ex. 30

```
\zplot[
  domain={0, 0.02*\PI, 2*\PI},
  action=shade, startColor=blue,
  endColor=green, axis=x]{sin(x)}
\zplot[
  domain={0, 0.02*\PI, 2*\PI},
  action=shade, startColor=blue,
  endColor=green, axis=y]{sin(x)}
```



Zdraw `\begin{zdraw} <l3draw code> \end{zdraw}`

New: 2025-05-15 Wrapper for `\draw_begin:` and `\draw_end:`.

Zgroup `\begin{zgroup} <l3draw code> \end{zgroup}`

New: 2025-05-15 Wrapper for `\draw_path_scope_begin:` and `\draw_path_scope_end:`.

`\zmoveto` `\zmoveto{<coordinate>}`

`\zlineto` `\zlineto{<coordinate>}`

New: 2025-05-15 Moves the pen to `<coordinate>`, a valid `l3draw` coordinate expression like “1mm, 2cm+3em”.

`\zscolor` `\zscolor{<l3color>}`

`\zfcolor` `\zfcolor{<l3color>}`

New: 2025-05-15 `<l3color>` is a valid `l3draw` color expression; *z*TikZ predefines common colors as “`l3color`” variables.

`\zfevenodd` `\zfevenodd` specifies the even-odd rule for fill division, while `\zfnozero` specifies the non-zero rule.

New: 2025-05-31

`\zxvec` `\zxvec{<coordinate>}`

`\zyvec` `\zyvec{<coordinate>}`

New: 2025-05-15 Sets the unit vectors for x and y axes.

`\zpolar` `\zpolar{<radius>}{<angle>}`

`\zcoor` `\zcoor{<x-scale>}{<y-scale>}`

New: 2025-05-15 `\zpolar` gets coordinates via polar method; `\zcoor` via rectangular method relative to the base vectors.

<code>\zrect</code>	<code>\zrect{<coordinate>}{<coordinate>}</code>
<code>\zcirc</code>	<code>\zcirc{<center>}{<radius>}</code>
New: 2025-05-15	Draws rectangles (lower-left to upper-right) or circles (center and radius).
<code>\znewtext</code>	<code>\znewtext<coffin></code>
<code>\zsethtext</code>	<code>\zsethtext<coffin>{<content>}</code>
<code>\zsetvtext</code>	<code>\zsetvtext<coffin>{<width>}{<content>}</code>
<code>\zscaletext</code>	<code>\zscaletext<coffin>{<x-scale>}{<y-scale>}</code>
<code>\zputtext</code>	<code>\zputtext<coffin>{<hpole>}{<vpole>}{<point>}</code>
New: 2025-05-15	Macros for creating, transforming, and placing text in <code>l3draw</code> .
<code>\zbg</code>	Wrappers for <code>\draw_path_scope_begin:</code> and <code>\draw_path_scope_end:</code> .
<code>\zeg</code>	
New: 2025-05-15	
<code>\zcapbutt</code>	Set line connection styles.
<code>\zcaproun</code>	
<code>\zcaprect</code>	
<code>\zclosepath</code>	
New: 2025-05-15	
<code>\zshift</code>	<code>\zshift{<vector>}</code>
<code>\zxscale</code>	<code>\zxscale{<x-scale>}</code>
<code>\zyscale</code>	<code>\zyscale{<y-scale>}</code>
<code>\ztrans</code>	<code>\ztrans{<a>}{}{<c>}{<d>}</code>
New: 2025-05-15	Macros for affine transformations of axes.
<code>\zusepath</code>	<code>\zusepath[<style>]</code>
New: 2025-05-15	Shows the final path; <code><style></code> defaults to “draw”, other options: “stroke, fill, clip”.

6 Appendix

6.1 gnuplot Built-in Functions

Here we supplement the built-in functions in gnuplot: Arguments to math functions in gnuplot can be integer, real, or complex unless otherwise noted. Functions that accept or return angles (e.g. $\sin(x)$) treat angle values as radians, but this may be changed to degrees using the command `set angles`. (Excerpt from: [gnuplot support functions](#))

Table 1: **gnuplot math library functions**

Function	Arguments	Returns
$\text{abs}(x)$	any	$ x $, absolute value of x ; same type
$\text{abs}(x)$	complex	length of x , $\sqrt{\text{Re}(x)^2 + \text{Im}(x)^2}$
$\text{acos}(x)$	any	$\cos^{-1} x$ (inverse cosine)
$\text{acosh}(x)$	any	$\cosh^{-1} x$ (inverse hyperbolic cosine) in radians
$\text{airy}(x)$	any	Airy function $\text{Ai}(x)$
$\text{arg}(x)$	complex	the phase of x
$\text{asin}(x)$	any	$\sin^{-1} x$ (inverse sine)
$\text{asinh}(x)$	any	$\sinh^{-1} x$ (inverse hyperbolic sine) in radians
$\text{atan}(x)$	any	$\tan^{-1} x$ (inverse tangent)
$\text{atan2}(y, x)$	int or real	$\tan^{-1}(y/x)$ (inverse tangent)
$\text{atanh}(x)$	any	$\tanh^{-1} x$ (inverse hyperbolic tangent) in radians
$\text{EllipticK}(k)$	real k in $(-1 : 1)$	$K(k)$ complete elliptic integral of the first kind
$\text{EllipticE}(k)$	real k in $[-1 : 1]$	$E(k)$ complete elliptic integral of the second kind
$\text{EllipticPi}(n, k)$	real $n, k < 1$	$\Pi(n, k)$ complete elliptic integral of the third kind
$\text{besj0}(x)$	int or real	J_0 Bessel function of x , in radians
$\text{besj1}(x)$	int or real	J_1 Bessel function of x , in radians
$\text{besy0}(x)$	int or real	Y_0 Bessel function of x , in radians
$\text{besy1}(x)$	int or real	Y_1 Bessel function of x , in radians
$\text{ceil}(x)$	any	$\lceil x \rceil$, smallest integer not less than x (real part)
$\text{cos}(x)$	radians	$\cos x$, cosine of x
$\text{cosh}(x)$	any	$\cosh x$, hyperbolic cosine of x in radians
$\text{erf}(x)$	any	$\text{erf}(\text{Re}(x))$, error function of $\text{Re}(x)$
$\text{erfc}(x)$	any	$\text{erfc}(\text{Re}(x))$, 1.0– error function of $\text{Re}(x)$
$\text{exp}(x)$	any	e^x , exponential function of x
$\text{expint}(n, x)$	any	$E_n(x)$, exponential integral function of x
$\text{floor}(x)$	any	$\lfloor x \rfloor$, largest integer not greater than x (real part)
$\text{gamma}(x)$	any	$\Gamma(\text{Re}(x))$, gamma function of $\text{Re}(x)$
$\text{ibeta}(p, q, x)$	any	$\text{ibeta}(\text{Re}(p, q, x))$, ibeta function of $\text{Re}(p, q, x)$
$\text{inverf}(x)$	any	inverse error function $\text{Re}(x)$
$\text{igamma}(a, x)$	any	$\text{igamma}(\text{Re}(a, x))$, igamma function of $\text{Re}(a, x)$
$\text{imag}(x)$	complex	$\text{Im}(x)$, imaginary part of x as a real number
$\text{invnorm}(x)$	any	inverse normal distribution function $\text{Re}(x)$
$\text{int}(x)$	real	integer part of x , truncated toward zero
$\text{lambertw}(x)$	real	Lambert W function

$\text{lgamma}(x)$	any	$\text{lgamma}(\text{Re}(x))$, lgamma function of $\text{Re}(x)$
$\log(x)$	any	$\ln x$, natural logarithm (base e) of x
$\log_{10}(x)$	any	$\log_{10} x$, logarithm (base 10) of x
$\text{norm}(x)$	any	$\text{norm}(x)$, normal distribution function of $\text{Re}(x)$
$\text{rand}(x)$	int	pseudo random number in the interval $(0 : 1)$
$\text{real}(x)$	any	$\text{Re}(x)$, real part of x
$\text{sgn}(x)$	any	1 if $x > 0$, -1 if $x < 0$, 0 if $x = 0$. $\Im(x)$ ignored
$\sin(x)$	any	$\sin x$, sine of x
$\sinh(x)$	any	$\sinh x$, hyperbolic sine of x in radians
$\text{sqrt}(x)$	any	$\sqrt{x}$, square root of x
$\tan(x)$	any	$\tan x$, tangent of x
$\tanh(x)$	any	$\tanh x$, hyperbolic tangent of x in radians
$\text{voigt}(x, y)$	real	convolution of Gaussian and Lorentzian
$\text{cerf}(z)$	complex	complex error function
$\text{cdawson}(z)$	complex	complex Dawson's integral
$\text{faddeeva}(z)$	complex	$w(z) = \exp(-z^2) \times \text{erfc}(-iz)$
$\text{erfi}(x)$	real	imaginary error function $\text{erfi}(x) = -i \times \text{erf}(ix)$
$\text{VP}(x, \sigma, \gamma)$	real	Voigt profile

Note: $\text{faddeeva}(z)$: rescaled complex error function

6.2 marker Styles

The marker styles available in `TikZ` are as follows:

<code>\pgfuseplotmark{-}</code>	
<code>\pgfuseplotmark{ }</code>	
<code>\pgfuseplotmark{o}</code>	
<code>\pgfuseplotmark{asterisk}</code>	
<code>\pgfuseplotmark{star}</code>	
<code>\pgfuseplotmark{10-pointed star}</code>	
<code>\pgfuseplotmark{oplus}</code>	
<code>\pgfuseplotmark{oplus*}</code>	
<code>\pgfuseplotmark{otimes}</code>	
<code>\pgfuseplotmark{otimes*}</code>	
<code>\pgfuseplotmark{square}</code>	
<code>\pgfuseplotmark{square*}</code>	
<code>\pgfuseplotmark{triangle}</code>	
<code>\pgfuseplotmark{triangle*}</code>	
<code>\pgfuseplotmark{diamond}</code>	
<code>\pgfuseplotmark{diamond*}</code>	
<code>\pgfuseplotmark{halfdiamond*}</code>	
<code>\pgfuseplotmark{halfsquare*}</code>	
<code>\pgfuseplotmark{halfsquare right*}</code>	
<code>\pgfuseplotmark{halfsquare left*}</code>	
<code>\pgfuseplotmark{pentagon}</code>	
<code>\pgfuseplotmark{pentagon*}</code>	
<code>\pgfuseplotmark{Mercedes star}</code>	
<code>\pgfuseplotmark{Mercedes star flipped}</code>	
<code>\pgfuseplotmark{halfcircle}</code>	
<code>\pgfuseplotmark{halfcircle*}</code>	
<code>\pgfuseplotmark{heart}</code>	
<code>\pgfuseplotmark{text}</code>	

Figure 3: `TikZ` Marker Styles

6.3 Test Data/Code

```
# Curve 0 of 1, 10 points
# Curve title: "f(x)"
# x y type
-3.14159 -0.00000 i
-2.44346 -0.64279 i
-1.74533 -0.98481 i
-1.04720 -0.86603 i
-0.34907 -0.34202 i
0.34907 0.34202 i
1.04720 0.86603 i
1.74533 0.98481 i
2.44346 0.64279 i
3.14159 0.00000 i
```

sine.data

```
\begin{pycode}{pycode_table.txt}
import numpy as np

# write file
with open ('pycode_table.txt', 'w') as file:
    file.write("\begin{tabular}{p{3cm}ccc}\n")
    file.write("\hline\n")
    file.write("number/function & $\sin$ & $\cos$ & $\tan$\\n")
\n")
    file.write("\hline\n")
    for i in range(1, 16):
        file.write(
            f"${i}$ & ${np.around(np.sin(i), decimals=4)}$ &
${np.around(np.cos(i), decimals=4)}$ & ${np.around(np.tan(i),
decimals=4)}$\\n"
        )

    file.write("\hline\n")
    file.write("\end{tabular}\n")
\end{pycode}
```

table.py.txt

7 TODO

Development of $\text{\texttt{tikz}}$ has come to a pause for now. Here are some functions that may be added in the future (☐ – undone; ☒ – done; ☒ – will not fix):

- ☒ (luadraw package already exists) Implement an interface similar to `tikz-3dplot`, rewriting it with $\text{\texttt{L\^A T\^E X 3}}$.
- ☒ (Refer to processing method of the “`wolframAny`” environment) Add calling interfaces for Matlab scripts, or directly use its open-source alternative `GNU Octave`?
- ☒ (Refer to implementation of “`mermaid`” environment in cache library) Implement the `wolframAny` environment, providing functionality similar to `pycode`.
- ☒ 2025-05-29-Done: Rewrite the function `\ztikz_hash_if_change:nn` for the cache mechanism, as it is currently not flexible enough (maybe directly use the `robust-externalize` package).
- ☒ 2025-05-29-Done: For the cache library, redundant Hash values need to be cleared: for example, if an environment/command generates an original Hash value of “A”, and after parameters change, the Hash becomes “B”, then the original “A” needs to be cleared.
- ☒ 2025-05-29-Done: Implement the `\ztikzForceToSkip` command: how to handle the `\xsim_file_write_start:nn` environment? How to handle multiple `\ztikzForceToRun`? Maybe using `\str_mdfive_hash:n` would be helpful?
- ☐ Integrate more features related to `wstp`: Take over $\text{\texttt{T\^E X}}$ low-level pagination and box calculations (global optimum?), index and custom sort generation, complex regular expressions, and metaprogramming.

8.1 ztikz.sty

```

1 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% 1
2 % ztikz.sty % 2
3 % Copyright 2024, 2025, 2026 Zongping Ding. % 3
4 % % 4
5 % This work may be distributed and/or modified under the conditions of the % 5
6 % LaTeX Project Public License, either version 1.3 of this license or any % 6
7 % later version. % 7
8 % The latest version of this license is in % 8
9 % http://www.latex-project.org/lppl.txt % 9
10 % and version 1.3 or later is part of all distributions of LaTeX % 10
11 % version 2005/12/01 or later. % 11
12 % % 12
13 % This work has the LPPL maintenance status `maintained'. % 13
14 % % 14
15 % The Current Maintainer of this work is Zongping Ding. % 15
16 % % 16
17 % This work consists of the files ztikz.sty, % 17
18 % the libraries: ztikz.library.gnuplot.tex, % 18
19 % ztikz.library.cache.tex, % 19
20 % ztikz.library.python.tex, % 20
21 % ztikz.library.wolfram.tex, % 21
22 % ztikz.library.l3draw.tex, % 22
23 % and the script files: ztikz.library.gnuscrypt.tex, % 23
24 % ztikz.library.pyscript.tex. % 24
25 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% 25
26 \NeedsTeXFormat{LaTeX2e} 26
27 \ProvidesExplPackage{ztikz}{2026/05/08} 27
28 {1.1.0}{A~pre-release~diagram~and~CAS~package} 28
29 29
30 30
31 % ----- 31
32 % basic tools 32
33 % ----- 33
34 \RequirePackage{ztool} 34
35 \RequirePackage{graphicx} 35
36 \__ztool_load_library:n {shell-escape, file-io} 36
37 \cs_new_protected:Npn \ztikz_hook_preamble_last:n #1 37
38 { \AddToHook{env/document/before}{#1} } 38
39 \cs_new_protected:Npn \ztikz_label_hook_preamble_last:nn #1#2 39
40 { \AddToHook{env/document/before}[#1]{#2} } 40
41 \cs_new_protected:Npn \ztikz_hook_doc_begin:n #1 41
42 { \AddToHook{begindocument}{#1} } 42
43 \cs_new_protected:Npn \ztikz_hook_doc_end:n #1 43
44 { \AddToHook{endddocument}{#1} } 44
45 45

```

46		46
47		47
48	% -----	48
49	%	49
50	% -----	50
51	\clist_new:N \g__ztikz_library_loaded_clist	51
52	\clist_gclear:N \g__ztikz_library_loaded_clist	52
53	\cs_new_nopar:Npn __ztikz_load_library:n #1	53
54	{	54
55	\clist_map_inline:nn {#1} {	55
56	\clist_if_in:NnTF \g__ztikz_library_loaded_clist {##1} {	56
57	\msg_set:nnn {ztikz} {library-loaded}	57
58	{	58
59	ztikz~library~"##1"~already~loaded,ignored~loading.	59
60	\msg_line_context:	60
61	}	61
62	\msg_warning:nnn {ztikz} {library-loaded} {##1}	62
63	}{	63
64	\file_if_exist:nTF {library/ztikz.library.##1.tex}{	64
65	\clist_gput_right:Nn \g__ztikz_library_loaded_clist {##1}	65
66	\makeatletter\file_input:n {library/ztikz.library.##1.tex}	66
67	}{	67
68	\msg_set:nnn {ztikz} {library-not-found} {ztikz~library~`##1'~not~found.}	68
69	\msg_error:nnn {ztikz} {library-not-found} {##1}	69
70	}	70
71	}	71
72	}	72
73		73
74	\NewDocumentCommand\ztikzloadlib{m}	74
75	{	75
76	__ztikz_load_library:n {#1}	76
77	\ExplSyntaxOff	77
78	}	78
79		79
80		80
81		81
82	% -----	82
83	%	83
84	% -----	84
85	\RequirePackage{xsimverb}	85
86	\cs_generate_variant:Nn \xsim_file_write_start:nn {ne}	86
87	\ztool_shell_mkdir:n {ztikz_output/}	87
88	\tl_const:Nn \g__ztikz_scripts_path_tl {ztikz_output/scripts}	88
89	\NewDocumentCommand\ztikzMkdir{m}{ \ztool_shell_mkdir:n {#1} }	89
90		90
91		91
92		92
93	% -----	93

```
94 %                                ztikz's options                                94
95 % -----                                95
96 \cs_new_protected:Npn \ztikz_keys_define:n #1                                96
97   { \keys_define:nn { ztikz }{ #1 } }                                97
98 \cs_new_protected:Npn \ztikz_keys_define:nn #1#2                                98
99   { \keys_define:nn { ztikz / #1 }{ #2 } }                                99
100 \cs_new_protected:Npn \ztikz_keys_set:nn #1#2                                100
101   { \keys_set:nn { ztikz / #1 }{ #2 } }                                101
102 \ztikz_keys_define:n                                102
103   {                                103
104     quiet                .bool_gset:N = \g_ztikz_quiet_bool,                                104
105     quiet                .initial:n   = { false },                                105
106     quiet                .default:n   = { true },                                106
107     library              .multichoice:,                                107
108     library / basic      .code:n      = { \__ztikz_load_library:n {basic} },                                108
109     library / gnuplot     .code:n      = { \__ztikz_load_library:n {gnuplot} },                                109
110     library / cache      .code:n      = { \__ztikz_load_library:n {cache} },                                110
111     library / python      .code:n      = { \__ztikz_load_library:n {python} },                                111
112     library / wolfram     .code:n      = { \__ztikz_load_library:n {wolfram} },                                112
113     library / l3draw      .code:n      = { \__ztikz_load_library:n {l3draw} },                                113
114     library / unknown     .code:n      = {                                114
115       \msg_set:nnn {ztikz} {library-not-found}                                115
116       {ztikz~library~`#1'~not~found,~valid~libraries~are~'basic',                                116
117       ~'gnuplot',~'cache',~'python',~'wolfram'~and~'l3draw'.}                                117
118       \msg_error:nn {ztikz} {library-not-found}                                118
119     },                                119
120     wolfram              .meta:nn      = { ztikz/wolfram }{ #1 },                                120
121   }                                121
122                                122
123 % wolfram meta-key                                123
124 \tl_new:N \g__ztikz_wolfram_engine_tl                                124
125 \bool_new:N \g__ztikz_wolfram_cloud_bool                                125
126 \ztikz_keys_define:nn { wolfram }                                126
127   {                                127
128     engine .choice:,                                128
129     engine / wolfram .code:n =                                129
130       {                                130
131         \tl_gset:Nn \g__ztikz_wolfram_engine_tl { wolframscript }                                131
132       },                                132
133     engine / mathics .code:n =                                133
134       {                                134
135         \tl_gset:Nn \g__ztikz_wolfram_engine_tl { mathics }                                135
136       },                                136
137     engine / wstp .code:n =                                137
138       {                                138
139         \tl_gset:Nn \g__ztikz_wolfram_engine_tl { wstp }                                139
140       },                                140
141     engine / unknown .code:n =                                141
```

```

142 {
143     \msg_set:nnn {ztikz} {wolfram-engine-not-found}
144     {ztik's~'wolfram'~engine~option~`#1'~invalid,
145      ~valid~engine~is~'wolframscript'~or~'mathics'.}
146     \msg_error:nn {ztikz} {wolfram-engine-not-found}
147 },
148 engine .initial:n      = { wolfram },
149 cloud  .choices:nn     = { true, false }
150 {
151     \use:c { bool_gset_ \l_keys_choice_tl :N }
152     \g__ztikz_wolfram_cloud_bool
153 },
154 cloud .initial:n      = { false },
155 cloud .default:n      = { true },
156 }
157 \ProcessKeyOptions [ ztikz ]
158
159 % only 'wolfram' support 'cloud'
160 \bool_if:NT \g__ztikz_wolfram_cloud_bool
161 {
162     \tl_if_eq:VnT \g__ztikz_wolfram_engine_tl { mathics }
163     {
164         \msg_set:nnn {ztikz}{mathics-cloud}
165         {mathics~'cloud'~is~not~supported.}
166         \msg_error:nn {ztikz}{mathics-cloud}
167     }
168 }
169
170 % whether to show these debug info
171 \bool_if:NTF \g_ztikz_quiet_bool
172 {
173     \cs_new_protected:Npn \ztikz_term_info:n #1
174     {
175     }
176     \cs_new_protected:Npn \ztikz_term_info:e #1
177     {
178     }
179 }{
180     \cs_new_protected:Npn \ztikz_term_info:n #1
181     {
182         \iow_now:Nn \c_term_iow {#1}
183     }
184     \cs_new_protected:Npn \ztikz_term_info:e #1
185     {
186         \iow_now:Ne \c_term_iow {#1}
187     }
188 }
189

```


190		190
191		191
192	% -----	192
193	%	193
194	% -----	194
195	\cs_generate_variant:Nn \ior_open:Nn { Ne }	195
196	\cs_generate_variant:Nn \iow_open:Nn { Ne }	196
197	\cs_generate_variant:Nn \tl_map_function:nN { eN }	197
198	\ztikz_hook_preamble_last:n	198
199	{	199
200	\clist_if_in:NnF \g__ztikz_library_loaded_clist {cache}	200
201	{	201
202	\tl_new:N \l__ztikz_current_hash_tl	202
203	\prg_new_conditional:Npnn \ztikz_if_run_again:nnn #1#2#3 {p, T, F, TF}	203
204	{	204
205	\bool_gset_true:N \g__hash_change_bool	205
206	\prg_return_true:	206
207	}	207
208	\prg_generate_conditional_variant:Nnn \ztikz_if_run_again:nnn	208
209	{ nen } { p, T, F, TF }	209
210	}	210
211	}	211
212	% sed script	212
213	\cs_new:Npn __ztikz_addto_script:nn #1#2	213
214	{% #1: file; #2: line; #3:new	214
215	\ztool_append_to_file:nn	215
216	{#1}{#2}	216
217	}	217
218	\cs_generate_variant:Nn __ztikz_addto_script:nn { ne, en, ee }	218
219	\ztikz_keys_define:nn { sed-script }	219
220	{	220
221	file .tl_set:N = \l__ztikz_sed_script_file_tl,	221
222	line .int_set:N = \l__ztikz_sed_script_line_int,	222
223	ori .tl_set:N = \l__ztikz_sed_script_ori_tl,	223
224	new .tl_set:N = \l__ztikz_sed_script_new_tl,	224
225	}	225
226	\cs_new:Npn __ztikz_sed_script:nnn #1#2#3	226
227	{% #1: file; #2: line; #3:new	227
228	\ztool_replace_file_line:nnn	228
229	{\g__ztikz_scripts_path_tl/#1}	229
230	{#2}{#3}	230
231	}	231
232	\cs_generate_variant:Nn __ztikz_sed_script:nnn {eee}	232
233	\cs_generate_variant:Nn __ztikz_sed_script:nnn {nne}	233
234	% ztik logo	234
235	\NewDocumentCommand\TikZ{}{Ti\textcolor{orange}{\textit{k}}Z}	235
236	\NewDocumentCommand\zTikZ{}{	236
237	{	237

238	<code>\ztool_scale_to_wd_and_ht:nnn {.9ex}{1.3ex}{</code>	238
239	<code>\ztool_rotate:nn {89}{\(\aleph\)}</code>	239
240	<code>}\kern-0.3423ex\hbox{\TikZ</code>	240
241	<code>}</code>	241
242	<code>\let\ztikz\zTikZ</code>	242
243	<code>\protected\def\HoLogo@zTikZ#1{\zTikZ</code>	243
244	<code>\protected\def\HoLogo@ztikz#1{\zTikZ</code>	244

8.2 Library

8.2.1 basic

```
1 \ProvidesExplFile{ztikz.library.basic.tex}{2025/09/22}
2 {1.0.0}{basic~library~for~ztikz}
3
4
5
6 % -----
7 %                               basic packages
8 % -----
9 \RequirePackage{tikz}
10 \RequirePackage{etoolbox}
11 \patchcmd{\pgfutil@InputIfFileExists}{\input #1}{%
12   \@pushfilename
13   \xdef\@currname{#1}
14   \input #1
15   \@popfilename
16 }{}{}
17 \usetikzlibrary{arrows.meta}
18 \usetikzlibrary{intersections}
19 \usetikzlibrary{patterns}
20 \usetikzlibrary{plotmarks}
21 \usetikzlibrary{positioning}
22 \usetikzlibrary{shapes.geometric}
23 \usetikzlibrary{decorations.markings}
24 \usetikzlibrary{fadings}
25
26
27
28 % ==> coordinate basic components
29 \ztikz_keys_define:nn { point }
30 {
31   type      .str_set:N = \l__point_type_str,
32   type      .initial:n = { * },
33   radius    .dim_set:N = \l__point_radius_dim,
34   radius    .initial:n = { 1pt },
35   color     .tl_set:N = \l__point_color_tl,
36   color     .initial:n = { black },
37   opacity   .tl_set:N = \l__point_opacity_tl,
38   opacity   .initial:n = { 1 },
39   rotate    .fp_set:N = \l__point_rotate_angle,
40   rotate    .initial:n = { 0 },
41 }
42 \NewDocumentCommand\ShowPoint{ 0{}m0{}0{} }
43 {
44   \group_begin:
45   \exp_args:Nne \ztikz_keys_set:nn { point } { #1 }
```

```

46 \seq_set_split:Nnn \l__point_list_seq { ; }{#2}
47 \seq_set_split:Nnn \l__point_label_seq { ; }{#3}
48 \int_step_inline:nnnn {1}{1}
49 { \seq_count:N \l__point_list_seq }
50 {
51 \draw plot [
52 only~ marks,
53 mark = \str_use:N \l__point_type_str,
54 mark~ size = \dim_use:N \l__point_radius_dim,
55 mark~ options = {
56 rotate = \fp_use:N \l__point_rotate_angle,
57 opacity = \tl_use:N \l__point_opacity_tl,
58 color = \tl_use:N \l__point_color_tl,
59 ball~ color = \tl_use:N \l__point_color_tl,
60 }
61 ] coordinates{\seq_item:Nn \l__point_list_seq{##1}}
62 node[#4] {\seq_item:Nn \l__point_label_seq{##1}};
63 }
64 \group_end:
65 }
66 \NewDocumentCommand\ShowGrid{ 0{color=gray, very~ thin, step=1}m }
67 {
68 \seq_set_split:Nnn \l__grid_param_ii_seq { ; }{#2}
69 \draw[#1] \seq_item:Nn \l__grid_param_ii_seq{1}
70 grid \seq_item:Nn \l__grid_param_ii_seq{2};
71 }
72 % intersection
73 \NewDocumentCommand\ShowIntersection{ omm }
74 {
75 \seq_set_split:Nnn \l__intersection_num_seq { ; }{#2}
76 \path[name~ intersections={
77 of=\seq_item:Nn \l__intersection_num_seq{1}~
78 and~ \seq_item:Nn \l__intersection_num_seq{2}
79 }];
80 \int_step_inline:nnnn {1}{1}{#3}{
81 \ShowPoint[#1]{(intersection-##1)}
82 }
83 }
84 % polygon plot
85 \ztikz_keys_define:nn { polygon }
86 {
87 radius .fp_set:N = \l__polygon_radius_fp,
88 radius .initial:n = { 1 },
89 edgeColor .tl_set:N = \l__polygon_edge_color_tl,
90 edgeColor .initial:n = { black },
91 fillColor .tl_set:N = \l__polygon_fill_color_tl,
92 fillColor .initial:n = { },
93 fillOpacity .fp_set:N = \l__polygon_fill_opacity_fp,

```

```

94 fillOpacity .initial:n = { 0 },
95 rotate .fp_set:N = \l__polygon_rotate_angle,
96 rotate .initial:n = { 0 },
97 shift .tl_set:N = \l__polygon_shift_tl,
98 shift .initial:n = { (0,0) },
99 marker .tl_set:N = \l__polygon_marker_option_tl,
100 marker .initial:n = { },
101 }
102 \tl_new:N \l__ztikz_basic_poly_path_tl
103 \NewDocumentCommand\Polygon{ 0{}m }
104 {
105 \group_begin:
106 \ztikz_keys_set:nn { polygon } { #1 }
107 % strip '(' and ')'
108 \tl_replace_once:Nnn \l__polygon_shift_tl{({}}
109 \tl_replace_once:Nnn \l__polygon_shift_tl{)}}{}
110 \coordinate (mv) at (\tl_use:N \l__polygon_shift_tl);
111 % create polygon
112 \begin{scope}[shift=(mv), rotate=\fp_use:N \l__polygon_rotate_angle]
113 % arg require: #2 $\ge$ 3
114 \int_step_inline:nnn {1}{#2}{
115 % draw edges
116 \fp_set:Nn \l_angle_fp {360/#2*##1*\c_one_degree_fp}
117 \fp_set:Nn \l_angle_next_fp {360/#2*(##1+1)*\c_one_degree_fp}
118 \draw [\tl_use:N \l__polygon_edge_color_tl]
119 ( \fp_eval:n {\l__polygon_radius_fp*cos(\l_angle_fp)},
120 \fp_eval:n {\l__polygon_radius_fp*sin(\l_angle_fp)}
121 ) -- (
122 \fp_eval:n {\l__polygon_radius_fp*cos(\l_angle_next_fp)},
123 \fp_eval:n {\l__polygon_radius_fp*sin(\l_angle_next_fp)}
124 );
125 % fill polygon path
126 \int_compare:nNnTF {##1}<{#2}
127 {
128 \tl_put_right:Nn \l__ztikz_basic_poly_path_tl
129 { (p##1)-- }
130 }{
131 \tl_put_right:Nn \l__ztikz_basic_poly_path_tl
132 { (p##1)--cycle }
133 }
134 % mark coordinates
135 \coordinate (p##1) at (
136 \fp_eval:n {\l__polygon_radius_fp*cos(\l_angle_fp)},
137 \fp_eval:n {\l__polygon_radius_fp*sin(\l_angle_fp)}
138 );
139 }
140 % fill polygon (none-color -> opacity=1; or opacity=.75)
141 \tl_if_empty:NTF \l__polygon_fill_color_tl {

```

```

142 \fp_set:Nn \l__polygon_fill_opacity_fp {0}
143 }{
144 \fp_set:Nn \l__polygon_fill_opacity_fp {.75}
145 }
146 \fill [
147 \tl_use:N \l__polygon_fill_color_tl,
148 fill~opacity=\fp_use:N \l__polygon_fill_opacity_fp
149 ] \l__ztikz_basic_poly_path_tl;
150 % show markers
151 \int_step_inline:nnn {1}{#2}
152 {
153 \ShowPoint[\l__polygon_marker_option_tl]
154 { (p##1) }
155 }
156 \end{scope}
157 \group_end:
158 }
159
160
161 % ==> axis
162 \ztikz_keys_define:nn { axis }
163 {
164 % basic tick args
165 tickStart .fp_set:N = \l__start_fp,
166 tickStart .initial:n = { -5 },
167 tickEnd .fp_set:N = \l__end_fp,
168 tickEnd .initial:n = { 5 },
169 axisRotate .fp_set:N = \l__axis_rotate_angle,
170 axisRotate .initial:n = { 0 },
171 % tick dimension spec
172 mainStep .fp_set:N = \l__main_step_fp,
173 mainStep .initial:n = { 1.0 },
174 subStep .fp_set:N = \l__sub_step_fp,
175 subStep .initial:n = { 0.1 },
176 mainTickLabel .tl_set:N = \l__main_tick_label_tl,
177 mainTickLabel .initial:n = { \fp_use:N {\CurrentFp} },
178 tickLabelShift .dim_set:N = \l__tick_label_shift_dim,
179 tickLabelShift .initial:n = { 0pt },
180 mainTickLength .dim_set:N = \l__main_tick_length_dim,
181 mainTickLength .initial:n = { 4pt },
182 subTickLength .dim_set:N = \l__sub_tick_length_dim,
183 subTickLength .initial:n = { 2pt },
184 mainTickLabelPosition .tl_set:N = \l__main_tick_label_position_tl,
185 mainTickLabelPosition .initial:n = { below },
186 % color spec
187 axisColor .tl_set:N = \l__axis_color_tl,
188 axisColor .initial:n = { black },
189 mainTickColor .tl_set:N = \l__main_tick_color_tl,

```

```

190     mainTickColor      .initial:n    = { black },
191     subTickColor       .tl_set:N     = \l__sub_tick_color_tl,
192     subTickColor       .initial:n    = { black },
193     mainTickLabelColor .tl_set:N     = \l__main_tick_label_color_tl,
194     mainTickLabelColor .initial:n    = { black },
195     % tick cross type spec
196     tickStyle          .choice:,
197     tickStyle/cross    .code:n       = \tl_set:Nn \l__tick_spec_tl { cross },
198     tickStyle/above    .code:n       = \tl_set:Nn \l__tick_spec_tl { above },
199     tickStyle/below    .code:n       = \tl_set:Nn \l__tick_spec_tl { below },
200 }
201 % ticks style
202 \tl_new:N \l__tick_type_tl    % `main' or `sub'
203 \tl_new:N \l__tick_spec_tl    % `cross', `above' or `below'
204 \tl_new:N \l__tick_color_tl
205 \dim_new:N \l__tick_length_dim
206 \tl_new:N \l__node_text_tl
207 % draw ticks (main or sub)
208 \cs_new_protected:Npn \ztikz_draw_axis_ticks_cs:n #1
209 {
210     \str_case:NnT \l__tick_type_tl {
211         {main}{
212             \dim_set_eq:NN \l__tick_length_dim
213                 \l__main_tick_length_dim
214             \tl_set:NV \l__tick_color_tl
215                 \l__main_tick_color_tl
216             \tl_set:Nn \l__node_text_tl
217                 { \tl_use:N \l__main_tick_label_tl }
218         }
219         {sub}{
220             \dim_set_eq:NN \l__tick_length_dim \l__sub_tick_length_dim
221             \tl_set:NV \l__tick_color_tl \l__sub_tick_color_tl
222             \tl_set:Nn \l__node_text_tl {}
223         }
224     }{}
225     \str_case:VnT \l__tick_spec_tl {
226         {cross}{
227             \draw[\tl_use:N \l__tick_color_tl]
228                 (#1, 0)++(0, \dim_eval:n {\l__tick_length_dim/2})
229                 -- ++(0, \dim_eval:n {-\l__tick_length_dim})
230                 node[\tl_use:N \l__main_tick_label_position_tl]
231                 {
232                     \textcolor{\tl_use:N \l__main_tick_label_color_tl}
233                     {\tl_use:N \l__node_text_tl}
234                 };
235         }
236         {above}{
237             \draw[\tl_use:N \l__tick_color_tl] (#1, 0)

```

```

238      -- ++(0, \dim_eval:n {\l__tick_length_dim/2})
239      node[\tl_use:N \l__main_tick_label_position_tl]
240      {
241          \textcolor{\tl_use:N \l__main_tick_label_color_tl}
242          {\tl_use:N \l__node_text_tl}
243      };
244  }
245  {below}{
246      \draw[\tl_use:N \l__tick_color_tl] (#1, 0)
247      -- ++(0, \dim_eval:n {-\l__tick_length_dim/2})
248      node[\tl_use:N \l__main_tick_label_position_tl
249      =\dim_use:N \l__tick_label_shift_dim]
250      {
251          \textcolor{\tl_use:N \l__main_tick_label_color_tl}
252          {\tl_use:N \l__node_text_tl}
253      };
254  }
255  }{}
256  }
257  % draw axis
258  \fp_new:N \CurrentFp
259  \int_new:N \l__substep_num_int
260  \NewDocumentCommand\ShowAxis{O{}m}
261  {
262      \group_begin:
263      \ztikz_keys_set:nn { axis } { #1 }
264      \seq_set_split:Nnn \l__points_seq { ; }{#2}
265      \begin{scope}[rotate=\fp_use:N \l__axis_rotate_angle]
266      \draw[->, \tl_use:N \l__axis_color_tl]
267          \seq_item:Nn \l__points_seq{1}
268          -- \seq_item:Nn \l__points_seq{2};
269      % draw ticks
270      \fp_step_inline:nnnn
271          {\fp_eval:n {\l__start_fp}}
272          {\fp_use:N \l__main_step_fp}
273          {\fp_use:N \l__end_fp}
274      {
275          % main ticks
276          \tl_set:Nn \l__tick_type_tl {main}
277          \fp_gset:Nn \CurrentFp {##1}
278          \ztikz_draw_axis_ticks_cs:n {##1}
279          % sub ticks
280          \tl_set:Nn \l__tick_type_tl {sub}
281          \int_set:Nn \l__substep_num_int
282              {\fp_eval:n {floor(\l__main_step_fp/\l__sub_step_fp)}}
283          \fp_compare:nNnTF {##1}<{\fp_eval:n {floor(\l__end_fp)}}{
284              \fp_step_function:nnnN
285              {\fp_eval:n {##1+\l__sub_step_fp}}

```



```

286         {\fp_use:N \l__sub_step_fp}
287         {\fp_eval:n {##1+\l__substep_num_int*\l__sub_step_fp}}
288         \ztikz_draw_axis_ticks_cs:n
289     }{}
290 }
291 \end{scope}
292 \group_end:
293 }
294 \NewDocumentCommand{\xAxis}{0{-2}0{8}}
295 {
296     \ShowAxis[
297         tickStart=\fp_eval:n {#1+1},
298         tickEnd=\fp_eval:n {#2-0.75},
299         mainTickLabelPosition=below,
300         mainStep=1,          subStep=.25,
301         axisRotate=0,        axisColor=black,
302         mainTickColor=black, subTickColor=black,
303         mainTickLength=10pt, subTickLength=5pt,
304         tickLabelShift=0pt,  tickStyle=below,
305     ]{(#1, 0); (#2, 0)}
306 }
307 \NewDocumentCommand{\yAxis}{0{-2}0{8}}
308 {
309     \ShowAxis[
310         tickStart=\fp_eval:n {#1+1},
311         tickEnd=\fp_eval:n {#2-0.75},
312         mainStep=1,          subStep=.25,
313         axisRotate=90,       axisColor=black,
314         mainTickColor=black, subTickColor=black,
315         mainTickLength=10pt, subTickLength=5pt,
316         tickLabelShift=0pt,  tickStyle=above,
317         mainTickLabelPosition=left
318     ]{(#1, 0); (#2, 0)}
319 }
320
321
322 % ==> statistic plot function
323 \cs_new_protected:Npn \ztikz_statistic_plot_cs:nnnn #1#2#3#4
324 {% #1:starts option; #2:draw-keyval; #3:point-keyval; #4:filename
325     \tl_if_empty:nTF {#3}{\draw[#2] plot[#1] file {#4};}
326     {
327         \group_begin:
328         \keys_set:nn { ztikz / point } { #3 }
329         \draw[#2] plot [
330             % stairs options
331             #1,
332             % marker options
333             mark = \str_use:N \l__point_type_str,

```

```

334 mark~ size = \dim_use:N \l__point_radius_dim,
335 mark~ options = {
336     rotate = \fp_use:N \l__point_rotate_angle,
337     opacity = \tl_use:N \l__point_opacity_tl,
338     color = \tl_use:N \l__point_color_tl,
339     ball~ color = \tl_use:N \l__point_color_tl,
340 }
341 ] file {#4};
342 \group_end:
343 }
344 }
345 \cs_generate_variant:Nn \ztikz_statistic_plot_cs:nnnn {ennn}
346
347 % stairs plot
348 \seq_new:N \l__statistic_option_tl
349 \NewDocumentCommand\StairsPlot
350 { 0{plot-left;jump-left}0{color=black}0{}m }
351 {
352     \seq_set_split:Nnn \l__statistic_option_tl { ; }{#1}
353     \str_case:enF {\seq_item:Nn \l__statistic_option_tl{1}}{
354         {plot-left}{\tl_set:Nn \l__tmpa_tl {const~plot~mark~left}}
355         {plot-right}{\tl_set:Nn \l__tmpa_tl {const~plot~mark~right}}
356         {plot-mid}{\tl_set:Nn \l__tmpa_tl {const~plot~mark~mid}}
357         {}{\tl_set:Nn \l__tmpa_tl {}}
358     }{
359         \msg_new:nnn {ztikz}{ztikz-stairs-plot}
360         {current~stairs~plot~type~is:~'#1'~,~ invalide}
361         \msg_error:nn {ztikz}{ztikz-stairs-plot}
362     }
363     \str_case:enF {\seq_item:Nn \l__statistic_option_tl{2}}{
364         {jump-left}{\tl_set:Nn \l__tmpb_tl {jump~mark~left}}
365         {jump-right}{\tl_set:Nn \l__tmpb_tl {jump~mark~right}}
366         {jump-mid}{\tl_set:Nn \l__tmpb_tl {jump~mark~mid}}
367         {}{\tl_set:Nn \l__tmpb_tl {}}
368     }{
369         \msg_new:nnn {ztikz}{ztikz-stairs-plot}
370         {current~stairs~jump~type~is:~'#1'~,~ invalide}
371         \msg_error:nn {ztikz}{ztikz-stairs-plot}
372     }
373     \ztikz_statistic_plot_cs:ennn {\l__tmpa_tl,\l__tmpb_tl}
374     {#2}{#3}{#4}
375 }
376 % stem plot
377 \NewDocumentCommand\StemPlot{ 0{x}0{color=black}0{}m }
378 {
379     \str_case:enF {#1}{
380         {x}{\tl_set:Nn \l__tmpa_tl {ycomb}}
381         {y}{\tl_set:Nn \l__tmpa_tl {xcomb}}

```

```

382 {o}{\tl_set:Nn \l__tmpa_tl {polar~ comb}} 382
383 {}{\tl_set:Nn \l__tmpa_tl {ycomb}} 383
384 }{ 384
385 \msg_new:nnn {ztikz}{ztikz-stem-plot} 385
386 {current~stem~plot~type~is::~'#1'~,~ invalide} 386
387 \msg_error:nn {ztikz}{ztikz-stem-plot} 387
388 } 388
389 \ztikz_statistic_plot_cs:ennn {\l__tmpa_tl}{#2}{#3}{#4} 389
390 } 390
391 % bar plot 391
392 \NewDocumentCommand\BarPlot{ 0{ybar}0{color=black}0{m } 392
393 { 393
394 \str_case:enF {#1}{ 394
395 {x}{\tl_set:Nn \l__tmpa_tl {ybar}} 395
396 {y}{\tl_set:Nn \l__tmpa_tl {xbar}} 396
397 {xc}{\tl_set:Nn \l__tmpa_tl {ybar~ interval}} 397
398 {yc}{\tl_set:Nn \l__tmpa_tl {xbar~ interval}} 398
399 {}{\tl_set:Nn \l__tmpa_tl {ybar}} 399
400 }{ 400
401 \msg_new:nnn {ztikz}{ztikz-bar-plot} 401
402 {current~bar~plot~type~is::~'#1'~,~ invalide} 402
403 \msg_error:nn {ztikz}{ztikz-bar-plot} 403
404 } 404
405 \ztikz_statistic_plot_cs:ennn {\l__tmpa_tl}{#2}{#3}{#4} 405
406 } 406

```

8.2.2 gnuplot

```
1 \ProvidesExplFile{ztikz.library.gnuplot.tex}{2025/12/02}
2 {1.0.0}{gnuplot~library~for~ztikz}
3
4
5
6 % ==> init variables
7 \ztool_shell_mkdir:n {ztikz_output/gnuplot_data/}
8 \__ztikz_load_library:n {gnuscript}
9 \int_new:N \g__tikz_env_index_int
10 \int_new:N \g__gnu_data_index_int
11 \int_new:N \g__gnu_plotz_index_int
12 \int_gadd:Nn \g__gnu_plotz_index_int {1}
13 \tl_const:Nn \g__ztikz_gnu_path_tl {ztikz_output/gnuplot_data}
14 \pretocmd{\tikzpicture}{
15   \int_gincr:N \g__tikz_env_index_int
16   \int_gset:Nn \g__gnu_data_index_int {0}
17 }{}{}
18
19 % plot args
20 \tl_new:N \l__pairs_x_domain_tl
21 \tl_new:N \l__pairs_y_domain_tl
22 \ztikz_keys_define:nn { 2dplot }
23 {
24   domain .tl_set:N = \l__ztikz_plot_domain_tl,
25   domain .initial:n = { -5:5; },
26   style .tl_set:N = \l__ztikz_plot_style_tl,
27   marker .tl_set:N = \l__ztikz_plot_marker_tl,
28 }
29 \ztikz_keys_define:nn { 3dplot }
30 {
31   domain .tl_set:N = \l__ztikz_plotz_domain_tl,
32   domain .initial:n = {-5:5; -5:5},
33   palette .tl_set:N = \l__ztikz_plotz_palette_tl,
34   palette .initial:n = {rgbformulae~ 22,13,-31},
35   pm3d .bool_set:N = \l__ztikz_plotz_pm_bool,
36   pm3d .initial:n = {true},
37   pm3d .default:n = {true},
38   width .dim_set:N = \l__ztikz_plotz_width_dim,
39   width .initial:n = {0.75\linewidth},
40 }
41 \cs_new:Npn \__ztikz_pairs_domain_parse:w #1; #2\q_stop
42 {
43   \tl_set:Nn \l__pairs_x_domain_tl {#1}
44   \tl_if_empty:nTF {#2}
45     { \tl_set:Nn \l__pairs_y_domain_tl {*:~} }
46     { \tl_set:Nn \l__pairs_y_domain_tl {#2} }
```

```

47 }
48
49
50 % ==> plot precise
51 \bool_new:N \g__plot_precise_restore_bool
52 \bool_new:N \g__contour_precise_restore_bool
53 \bool_new:N \g__param_precise_restore_bool
54 \bool_new:N \g__polar_precise_restore_bool
55 \NewDocumentCommand\PlotPrecise{ smm }
56 {
57     \tl_if_in:nnF {plot, param, polar, contour}{#2}
58     {
59         \msg_new:nnn {ztikz}{ztikz-plot-type}
60         {Valid~plot~type~are:~'plot',~'param',~'polar'~and~'contour'}
61         \msg_error:nn {ztikz}{ztikz-plot-type}
62     }
63     \IfBooleanF{#1}{ \bool_gset_true:c {g__#2_precise_restore_bool} }
64     \tl_if_eq:nnTF {contour}{#2}{
65         \__ztikz_sed_script:nnn {contour_plot.gp}{2}{set~samples~#3}
66         \__ztikz_sed_script:nnn {contour_plot.gp}{3}{set~isosamples~#3,#3}
67     }{
68         \__ztikz_sed_script:nnn {#2_plot.gp}{3}{set~samples~#3}
69     }
70 }
71 % data plot, precise, style
72 \cs_new_protected:Npn \__ztikz_gnu_data_plot:nnn #1#2#3
73 {
74     \ztool_shell_escape:e {gnuplot~ \g__ztikz_scripts_path_tl/#1_plot.gp}
75     \int_gadd:Nn \g__gnu_data_index_int {1}
76     \tl_set:Ne \l__gnu_data_new_name_tl
77     {
78         gnu_data_\int_use:N \g__tikz_env_index_int
79         _\int_use:N \g__gnu_data_index_int.table
80     }
81     \tl_set:Ne \l__gnu_data_full_path_tl
82     { \g__ztikz_gnu_path_tl/\l__gnu_data_new_name_tl }
83     \ztool_shell_mv:ee
84     { \g__ztikz_gnu_path_tl/gnu_data.table }
85     { \l__gnu_data_full_path_tl }
86     \tl_if_empty:nTF {#3}
87     {
88         \draw[#2] plot[smooth] file {\l__gnu_data_full_path_tl};
89     }{
90         \group_begin:
91         \ztikz_keys_set:nn { point } { #3 }
92         \draw[#2] plot [
93             mark = \str_use:N \l__point_type_str,
94             mark~ size = \dim_use:N \l__point_radius_dim,

```

```

95         mark~ options = {
96             rotate = \fp_use:N \l__point_rotate_angle,
97             opacity = \tl_use:N \l__point_opacity_tl,
98             color = \tl_use:N \l__point_color_tl,
99             ball~ color = \tl_use:N \l__point_color_tl,
100         }
101     ] file {\l__gnu_data_full_path_tl};
102     \group_end:
103 }
104 \bool_if:cT {g__#1_precise_restore_bool}
105 { \PlotPrecise{#1}{100} }
106 }
107 \cs_generate_variant:Nn \__ztikz_gnu_data_plot:nnn {nee}
108
109 % ==> simple 2d function plot
110 \NewDocumentCommand\Plot{ 0{}m }
111 {
112     \group_begin:
113     \ztikz_keys_set:nn { 2dplot } { domain=-5:5, style={color=black}, marker= }
114     \ztikz_keys_set:nn { 2dplot } { #1 }
115     \__ztikz_sed_script:nne {plot_plot.gp}{8}{set~xr~[\l__ztikz_plot_domain_tl]}
116     \__ztikz_sed_script:nne {plot_plot.gp}{7}{f(x)~::~#2}
117     \__ztikz_gnu_data_plot:nee {plot}{\l__ztikz_plot_style_tl}{\l__ztikz_plot_marker_tl}
118     \group_end:
119 }
120
121 \NewDocumentCommand\ContourPlot{ 0{}m }
122 {
123     \group_begin:
124     \ztikz_keys_set:nn { 2dplot } { domain={-5:5}, style={color=black}, marker= }
125     \ztikz_keys_set:nn { 2dplot } { #1 }
126     \exp_last_unbraced:Nf \__ztikz_pairs_domain_parse:w \l__ztikz_plot_domain_tl\q_stop
127     \__ztikz_sed_script:nne {contour_plot.gp}{11}{set~xr~[\l__pairs_x_domain_tl]}
128     \__ztikz_sed_script:nne {contour_plot.gp}{12}{set~yr~[\l__pairs_y_domain_tl]}
129     \__ztikz_sed_script:nne {contour_plot.gp}{14}{f(x,~y)~::~#2}
130     \__ztikz_gnu_data_plot:nee {contour}{\l__ztikz_plot_style_tl}{\l__ztikz_plot_marker_tl}
131     \group_end:
132 }
133 \NewDocumentCommand\ParamPlot{ 0{domain=0:2*pi, style=black, marker=}m }
134 {
135     \group_begin:
136     \ztikz_keys_set:nn { 2dplot } { domain=0:2*pi, style=black, marker= }
137     \ztikz_keys_set:nn { 2dplot } { #1 }
138     \__ztikz_sed_script:nne {param_plot.gp}{8}{set~trange~[\l__ztikz_plot_domain_tl]}
139     \__ztikz_sed_script:nne {param_plot.gp}{9}{plot~#2}
140     \__ztikz_gnu_data_plot:nee {param}{\l__ztikz_plot_style_tl}{\l__ztikz_plot_marker_tl}
141     \group_end:
142 }

```

```

143 \NewDocumentCommand\PolarPlot{ 0{}m }
144 {
145     \group_begin:
146     \ztikz_keys_set:nn { 2dplot } { domain=0:2*pi, style=black, marker= }
147     \ztikz_keys_set:nn { 2dplot } { #1 }
148     \__ztikz_sed_script:nne {polar_plot.gp}{8}{set~trange~[\l__ztikz_plot_domain_tl]}
149     \__ztikz_sed_script:nne {polar_plot.gp}{9}{plot~#2}
150     \__ztikz_gnu_data_plot:nee {polar}{\l__ztikz_plot_style_tl}{\l__ztikz_plot_marker_tl}
151     \group_end:
152 }
153 \NewDocumentCommand\Plotz{ 0{}m }
154 {
155     \group_begin:
156     \ztikz_keys_set:nn { 3dplot } { #1 }
157     \bool_if:NTF \l__ztikz_plotz_pm_bool {\def\plotz@pm{with~pm3d}}{\def\plotz@pm{}}
158     \exp_last_unbraced:Nf \__ztikz_pairs_domain_parse:w \l__ztikz_plotz_domain_tl\q_stop
159     \__ztikz_sed_script:nne {3d_plot.gp}{18}{set~palette~\l__ztikz_plotz_palette_tl}
160     \__ztikz_sed_script:nne {3d_plot.gp}{23}{set~xr~[\l__pairs_x_domain_tl]}
161     \__ztikz_sed_script:nne {3d_plot.gp}{24}{set~yr~[\l__pairs_y_domain_tl]}
162     \__ztikz_sed_script:nne {3d_plot.gp}{25}{splot~#2~\plotz@pm}
163     \ztool_shell_escape:e {gnuplot~ ./ztikz_output/scripts/3d_plot.gp}
164     \tl_set:N \l__tmpa_tl {./ztikz_output/gnuplot_data/plot_3d_\int_use:N
\g__gnu_plotz_index_int.pdf}
165     \ztool_shell_mv:ne {./ztikz_output/gnuplot_data/plot_3d.pdf}{\l__tmpa_tl}
166     \includegraphics[width=\dim_use:N \l__ztikz_plotz_width_dim]{\l__tmpa_tl}
167     \int_gadd:Nn \g__gnu_plotz_index_int {1}
168     \group_end:
169 }
170
171
172 % ==> users' interface
173 \NewDocumentCommand{\currentTikzIndex}{}
174 {
175     \int_use:N \g__tikz_env_index_int
176 }
177 \def\gnudata#1
178 {
179     \tl_use:N \g__ztikz_gnu_path_tl/gnu_data_
180     \int_use:N \g__tikz_env_index_int _#1.table
181 }

```

8.2.3 cache

```
1 \ProvidesExplFile{ztikz.library.cache.tex}{2025/05/31}
2 {1.0.0}{cache~library~for~ztikz}
3
4
5
6 % ==> init cache
7 \clist_if_in:NnT \g__ztikz_library_loaded_clist {basic}
8 {
9   \ztool_shell_mkdir:n {ztikz_output/tikz_data/}
10   \usetikzlibrary{external}
11   \tikzexternalize[prefix=ztikz_output/tikz_data/]
12 }
13 \ztool_file_new:nn {\c_false_bool}{ztikz_output/ztikz.hash}
14
15
16 % ==> variables declaration
17 \ior_new:N \g__ztikz_file_ior
18 \tl_new:N \l__ztikz_current_hash_tl
19 \seq_new:N \g_ztikz_file_hash_seq
20 \seq_new:N \l__zcache_hash_label_seq
21 \seq_new:N \g__zcache_tmp_hash_seq
22 \tl_new:N \l__zcache_hash_label_tl
23 \tl_new:N \l__zcache_hash_hash_tl
24 \tl_new:N \g__zcache_latest_cache_label_tl
25 \bool_new:N \l__zcache_hash_label_miss_bool
26 \bool_new:N \g_ztikz_hash_nochg_run_bool
27 \bool_new:N \g_ztikz_hashchg_norun_bool
28 \bool_gset_false:N \g_ztikz_hashchg_norun_bool
29 \bool_gset_false:N \g_ztikz_hash_nochg_run_bool
30 \cs_generate_variant:Nn \ztikz_file_read_lines:n { e }
31
32
33 % ==> cache function
34 \prg_set_conditional:Npnn \ztikz_if_run_again:nnn #1#2#3 { p, T, F, TF }
35 {% #1:true/false; #2:True-->file, False-->str; #3:label
36   \tl_gset:Nn \g__zcache_latest_cache_label_tl { #3 }
37   \__zcache_hash_get:nn {#1} {#2}
38   \edef\zcache@flag
39   {
40     \tl_map_function:nN {
41       \g_ztikz_hashchg_norun_bool
42       \g_ztikz_hash_nochg_run_bool
43     } \int_eval:n
44   }
45   \exp_args:Ne \int_case:nnF
46   { \exp_not:N \int_from_bin:n {\zcache@flag} }
```



```

47 {
48 {0}{
49 \seq_if_in:NVTF \g_ztikz_file_hash_seq \l__ztikz_current_hash_tl
50 {
51 \ztikz_term_info:n {CURRENT~HASH~ALREADY~EXISTS}
52 \prg_return_false:
53 }{
54 \ztikz_term_info:n {CURRENT~HASH~IS~UNIQUE:RECORDING...}
55 \__zcache_hash_add:nn { #3 }{ \l__ztikz_current_hash_tl }
56 \prg_return_true:
57 }
58 }
59 {1}{
60 \ztikz_term_info:n {FORCE~TO~RUN~AGAIN~...}
61 \bool_gset_false:N \g_ztikz_hash_nochg_run_bool
62 \prg_return_true:
63 }
64 {2}{
65 \ztikz_term_info:n {FORCE~TO~SKIP~...}
66 \bool_gset_false:N \g_ztikz_hashchg_norun_bool
67 \__zcache_hash_extract_by_label:nnn
68 { ztikz_output/ztikz.hash }{ #3 }{-1}
69 \prg_return_false:
70 }
71 }{ \relax }
72 }
73 \cs_new_protected:Npn \__zcache_hash_get:nn #1#2
74 {% #1:true/false; #2:True-->file, False-->str;
75 \bool_if:nTF {#1}
76 { \file_get_md5five_hash:nN {#2} \l__ztikz_current_hash_tl }
77 { \tl_set:Nn \l__ztikz_current_hash_tl {#2} }
78 \tl_set_rescan:Nne \l__ztikz_current_hash_tl
79 { \cctab_select:N \c_initex_cctab }
80 { \l__ztikz_current_hash_tl }
81 \__zcache_hash_extract_all:nN
82 { ztikz_output/ztikz.hash }
83 \g_ztikz_file_hash_seq
84 \seq_gremove_duplicates:N \g_ztikz_file_hash_seq
85 \ztikz_term_info:e
86 {
87 \iow_newline:
88 CURRENT~FILE's~HASH:\l__ztikz_current_hash_tl
89 }
90 }
91 \cs_new_protected:Npn \__zcache_hash_extract_by_label:nnn #1#2#3
92 {% #1:file; #2:label; #3:index
93 \ztool_read_file_as_seq:nnN
94 { \c_false_bool }{ #1 }

```

```

95 \l_tmpa_seq
96 \bool_set_true:N \l__zcache_hash_label_miss_bool
97 \seq_map_inline:Nn \l_tmpa_seq
98 {
99   \zcache_hash_label_extract:nnN { label }{ ##1 }
100   \l__zcache_hash_label_tl
101   \zcache_hash_label_extract:nnN { hash }{ ##1 }
102   \l__zcache_hash_hash_tl
103   \tl_if_eq:NnT \l__zcache_hash_label_tl { #2 }
104   {
105     \bool_set_false:N \l__zcache_hash_label_miss_bool
106     \tl_set:Ne \l__ztikz_current_hash_tl
107     {
108       \clist_item:en {\l__zcache_hash_hash_tl}{#3}
109     }
110   }
111 }
112 \bool_if:NT \l__zcache_hash_label_miss_bool
113 {
114   \msg_set:nnn {ztikz}{hash-label-missing}
115   {
116     current~hash~label~'#2'~not~found~in~'ztikz.hash',~
117     do~NOT~change~the~hash~label~in~any~respect~after~cache.
118   }
119   \msg_error:nn {ztikz}{hash-label-missing}
120 }
121 }
122 \prg_generate_conditional_variant:Nnn \ztikz_if_run_again:nnn
123 { nen } { T, F, TF }
124
125 % extract hash or label
126 \cs_new_protected:Npn \__zcache_hash_add:nn #1#2
127 {% #1:label, #2:hash
128   \seq_clear:N \l_tmpa_seq
129   \seq_clear:N \l_tmpb_seq
130   \seq_clear:N \l__zcache_hash_label_seq
131   \ztool_read_file_as_seq:nnN
132   { \c_false_bool }
133   { ztikz_output/ztikz.hash }
134   \l_tmpa_seq
135   \seq_map_inline:Nn \l_tmpa_seq
136   {
137     \zcache_hash_label_extract:nnN { label }{ ##1 } \l_tmpa_tl
138     \seq_put_right:NV \l__zcache_hash_label_seq \l_tmpa_tl
139     \tl_if_eq:NnTF \l_tmpa_tl { #1 }
140     { \seq_put_right:Ne \l_tmpb_seq {##1, #2} }
141     { \seq_put_right:Ne \l_tmpb_seq {##1} }
142   }

```

```

143 \seq_if_in:NnF \l__zcache_hash_label_seq { #1 }
144 { \seq_put_right:Ne \l_tmpb_seq { #1:#2 } }
145 \ztool_write_seq_to_file:nNn { \c_true_bool }
146 \l_tmpb_seq { ztikz_output/ztikz.hash }
147 }
148 \cs_new_protected:Npn \__zcache_hash_extract_all:nN #1#2
149 {% #1:file; #2:seq
150 \seq_clear:N \l_tmpa_seq
151 \clist_clear:N \l_tmpa_clist
152 \ztool_read_file_as_seq:nnN
153 { \c_false_bool }{ #1 }
154 \l_tmpa_seq
155 \seq_map_inline:Nn \l_tmpa_seq
156 {
157 \zcache_hash_label_extract:nnN { hash }{ ##1 } \l_tmpa_tl
158 \seq_gset_from_clist:NN \g__zcache_tmp_hash_seq \l_tmpa_tl
159 \clist_put_right:NV \l_tmpa_clist \l_tmpa_tl
160 }
161 \seq_set_from_clist:NN \l_tmpb_seq \l_tmpa_clist
162 \seq_remove_duplicates:N \l_tmpb_seq
163 \seq_gset_eq:NN #2 \l_tmpb_seq
164 }
165 \cs_set:Npn \zcache_cache_hash_last:nnn #1#2#3
166 {% #1:file; #2:label; #3:index
167 \seq_clear:N \l_tmpa_seq
168 \clist_clear:N \l_tmpa_clist
169 \ztool_read_file_as_seq:nnN
170 { \c_false_bool }{ #1 }
171 \l_tmpa_seq
172 \seq_map_inline:Nn \l_tmpa_seq
173 {
174 \zcache_hash_label_extract:nnN { label }{ ##1 }
175 \l__zcache_hash_label_tl
176 \zcache_hash_label_extract:nnN { hash }{ ##1 }
177 \l__zcache_hash_hash_tl
178 \tl_if_eq:NnT \l__zcache_hash_label_tl { #2 }
179 {
180 \clist_item:en {\l__zcache_hash_hash_tl}{ #3 }
181 \seq_map_break:
182 }
183 }
184 }
185 \cs_generate_variant:Nn \zcache_cache_hash_last:nnn { nen, eee }
186 \ztikz_keys_define:nn { cache/hash }
187 {
188 label .tl_set:N = \l__cache_hash_label_user_tl,
189 label .initial:e = { \g__zcache_latest_cache_label_tl },
190 file .tl_set:N = \l__cache_hash_file_user_tl,

```

```

191 file .initial:e = { ztikz_output/ztikz.hash },
192 index .int_set:N = \l__cache_hash_index_user_int,
193 index .initial:n = { -1 },
194 }
195 \newcommand{\ztikzCachedHash}[1] []
196 {
197   \group_begin:
198     \ztikz_keys_set:nn { cache/hash } { #1 }
199     \zcache_cache_hash_last:eee
200     { \l__cache_hash_file_user_tl }
201     { \l__cache_hash_label_user_tl }
202     { \int_use:N \l__cache_hash_index_user_int }
203   \group_end:
204 }
205 \cs_set_protected:Npn \zcache_hash_label_extract:nnN #1#2#3
206 {
207   \tl_clear:N \l_tmpa_tl
208   \tl_set_rescan:Nnn \l_tmpa_tl
209   { \cctab_select:N \c_code_cctab }
210   { #2 }
211   \exp_args:NNe \tl_set:Ne #3
212   {
213     \exp_not:c {__hash_#1_extract:w}
214     \l_tmpa_tl \exp_not:N \q_stop
215   }
216 }
217 \cs_set:Npn \__hash_label_extract:w #1:#2\q_stop
218 { #1 }
219 \cs_set:Npn \__hash_hash_extract:w #1:#2\q_stop
220 { #2 }
221
222
223 % ==> clear cache hash
224 \cs_new_protected:Npn \ztikz_clear_hash:
225 {
226   \iow_open:Nn \g__ztikz_file_ior {ztikz_output/ztikz.hash}
227   \ior_close:N \g__ztikz_file_ior
228 }
229 \NewDocumentCommand{\ztikzHashClean}{s}{
230   \ztikz_clear_hash:
231   \ztikz_term_info:n {CLEAN~ALL~CACHED~HASH~SUCCESSFULLY...}
232 }
233 \NewDocumentCommand{\ztikzHashCurrent}{s+0{,}}{
234   \IfBooleanTF{#1}
235   { \tl_use:N \l__ztikz_current_hash_tl }
236   { \seq_use:Nn \g_ztikz_file_hash_seq {#2} }
237 }
238

```

239		239
240	% ==> override the cache mechanism	240
241	\NewDocumentCommand{\ztikzForceToSkip}{}{}	241
242	{	242
243	\bool_gset_false:N \g_ztikz_hash_nochg_run_bool	243
244	\bool_gset_true:N \g_ztikz_hashchg_norun_bool	244
245	}	245
246	\NewDocumentCommand{\ztikzForceToRun}{}{}	246
247	{	247
248	\bool_gset_false:N \g_ztikz_hashchg_norun_bool	248
249	\bool_gset_true:N \g_ztikz_hash_nochg_run_bool	249
250	}	250

8.2.4 python

```
1 \ProvidesExplFile{ztikz.library.python.tex}{2025/08/10}
2 {1.0.0}{python~library~for~ztikz}
3
4
5
6 % ==> writing scripts
7 \__ztikz_load_library:n {pyscript}
8 \ztool_shell_mkdir:n {ztikz_output/python_data/}
9 \tl_const:Nn \g__ztikz_python_path_tl {ztikz_output/python_data}
10 \ior_new:N \g__file_read_ior
11 \tl_new:N \g__file_content_tl
12
13
14 % ==> core functions
15 \cs_new_protected:Npn \zlatex_Readlines_cs:nn #1#2
16 {
17   \ior_open:Nn \g__file_read_ior {#2}
18   \str_case:nnF {#1}{
19     {raw}{
20       \ior_get:NN \g__file_read_ior \g__file_content_tl
21     }
22     {str}{
23       \ior_str_get:NN \g__file_read_ior \g__file_content_tl
24     }
25   }{}
26   \tl_use:N \g__file_content_tl
27 }
28 \cs_generate_variant:Nn \zlatex_Readlines_cs:nn {ee}
29
30
31 % ==> users' interface
32 % python-matplotlib
33 \NewDocumentEnvironment{pyfig}{ mm }
34 {% #1:label; #2:file name
35   \xsim_file_write_start:ne {\c_true_bool}{\g__ztikz_python_path_tl/t@mp.py}
36 }{
37   \xsim_file_write_stop:
38   \ztikz_if_run_again:nenTF {\c_true_bool}{\g__ztikz_python_path_tl/t@mp.py}{#1}
39   {
40     \__ztikz_addto_script:nn {\g__ztikz_python_path_tl/t@mp.py}
41     { plt.savefig('#2') }
42     \ztool_shell_escape:e {python~\g__ztikz_python_path_tl/t@mp.py}
43     \ztool_shell_mv:ee
44     { \g__ztikz_python_path_tl/t@mp.py }
45     { \g__ztikz_python_path_tl/pyfig_\l__ztikz_current_hash_tl.py }
46     \ztool_shell_mv:ee
```

```

47         { #2 }
48         { \g__ztikz_python_path_tl/\l__ztikz_current_hash_tl _#2 }
49         \ztikz_term_info:e
50         {
51             writing~source~to~file:'\g__ztikz_python_path_tl/
52             pyfig_\l__ztikz_current_hash_tl.py'
53         }
54     }{
55         \ztikz_term_info:e
56         {
57             skip~recompile~of~python,~use~the~
58             cache~picture:'\l__ztikz_current_hash_tl _#2'
59         }
60     }
61     \xdef\pyfigOutputFile{ \g__ztikz_python_path_tl/\l__ztikz_current_hash_tl _#2 }
62 }
63
64 % inline python command
65 \NewDocumentCommand\py{0{raw}m}
66 {
67     \__ztikz_sed_script:nne {python_script.py}{6}{Float_res~~~\tl_to_str:n {#2}}
68     \ztikz_term_info:e {using~python~float~module~calculating...}
69     \ztool_shell_escape:e {python~ \g__ztikz_scripts_path_tl/python_script.py}
70     \zlatex_Readlines_cs:ee {#1}{\g__ztikz_python_path_tl/PyFloat.out}
71     % ---> cause bug that can't write ToC to file
72     % \cs{iow_close:N} \cs{g__file_read_ior} leads to bug ??
73 }
74 % python-sympy
75 \NewDocumentCommand\sympy{ mm }
76 {
77     \__ztikz_sed_script:nne {sympy_script.py}{8}{F_res~~~\tl_to_str:n {#2}}
78     \ztikz_if_run_again:nenTF {\c_true_bool}{\g__ztikz_scripts_path_tl/sympy_script.py}{#1}
79     {
80         \ztool_shell_escape:e {python~ \g__ztikz_scripts_path_tl/sympy_script.py}
81         \ztool_shell_mv:ee
82         {\g__ztikz_python_path_tl/sympy.out}
83         {\g__ztikz_python_path_tl/sympy_\l__ztikz_current_hash_tl.out}
84         \ztikz_term_info:e {using~python~sympy~calculating~question~...}
85         \exp_args:Ne \input{\g__ztikz_python_path_tl/sympy_\l__ztikz_current_hash_tl.out}
86     }{
87         \exp_args:Ne \input{\g__ztikz_python_path_tl/sympy_\l__ztikz_current_hash_tl.out}
88         \ztikz_term_info:e {skip~recompile,~using~the~cache~sympy~result:~ ✓
89         \l__ztikz_current_hash_tl}
90     }
91 }
92 % python-code-env
93 \NewDocumentEnvironment{pycode}{ mm }
94 {
95     {% #1:label; #2:output file name (with ext)

```

94	\xsim_file_write_start:ne {\c_true_bool}{\g__ztikz_python_path_tl/t@mp.py}	94
95	}{	95
96	\xsim_file_write_stop:	96
97	\ztikz_if_run_again:nenTF {\c_true_bool}{\g__ztikz_python_path_tl/t@mp.py}{#1}	97
98	{	98
99	\ztool_shell_escape:e {python~\g__ztikz_python_path_tl/t@mp.py}	99
100	\ztool_shell_mv:ee	100
101	{ \g__ztikz_python_path_tl/t@mp.py }	101
102	{ \g__ztikz_python_path_tl/pycode_\l__ztikz_current_hash_tl.py }	102
103	\ztool_shell_mv:ee	103
104	{ #2 }	104
105	{ \g__ztikz_python_path_tl/\l__ztikz_current_hash_tl _#2 }	105
106	\ztikz_term_info:e	106
107	{	107
108	writing~source~to~file:'\g__ztikz_python_path_tl/	108
109	pycode_\l__ztikz_current_hash_tl.py'	109
110	}	110
111	}{	111
112	\ztikz_term_info:e	112
113	{	113
114	skip~recompile~of~python,~use~the~cache~result:	114
115	'\l__ztikz_current_hash_tl _#2'	115
116	}	116
117	}	117
118	\xdef\pycodeOutputFile{\g__ztikz_python_path_tl/\l__ztikz_current_hash_tl _#2}	118
119	}	119

8.2.5 wolfram

```
1 \ProvidesExplFile{ztikz.library.wolfram.tex}{2026/05/08}
2 {1.2.0}{wolfram~library~for~ztikz}
3
4
5
6 % ==> init variables
7 \ztool_shell_mkdir:n {ztikz_output/mma_data/}
8 \tl_const:Nn \g__ztikz_wolfram_path_tl {ztikz_output/mma_data}
9 \newcommand\wolframOutputPath{\g__ztikz_wolfram_path_tl}
10
11 \tl_new:N \l_part_table_data_tl
12 \tl_new:N \l_full_table_data_tl
13 \tl_new:N \l__wolfram_current_hash_tl
14 \tl_new:N \l__ztikz_wolfram_tmp_arg_tl
15 \tl_new:N \l__ztikz_wolfram_tmp_res_tl
16 \seq_new:N \g_ztikz_wolfram_res_seq
17 \ior_new:N \g__ztikz_wolfram_ior
18 \iow_new:N \g__ztikz_wolfram_iow
19
20
21 % ==> core function
22 \cs_new_protected:Npn \__ztikz_wolfram_execute:nnnn #1#2#3#4
23 {% #1:contents(empty->not add); #2:extension; #3:output object; #4:label
24   \__ztikz_wolfram_execute_prehook:nnn { #1 }{ #2 }{ #3 }
25   \__ztikz_wolfram_execute_cli:nn { #2 }{ #4 }
26   \__ztikz_wolfram_execute_posthook:n { #2 }
27 }
28 \cs_generate_variant:Nn \__ztikz_wolfram_execute:nnnn { e, o }
29
30 % wolfram execute aux functions
31 \cs_new:Npn \__ztikz_wolfram_execute_prehook:nnn #1#2#3
32 {
33   \edef\@wolfram@tmp@file{\g__ztikz_wolfram_path_tl/t@mp}
34
35   % add additional wolfram code - export process
36   \tl_if_empty:nF {#1}{
37     \__ztikz_addto_script:en { \@wolfram@tmp@file.wls }{ #1 }
38   }
39
40   % add additional wolfram code - script path
41   \tl_if_empty:nT { #3 }
42   {
43     \exp_args:Neee \ztool_insert_to_file:nnn
44       { \@wolfram@tmp@file.wls }{ 1 }
45       { scriptPath~=""\g__ztikz_wolfram_path_tl/" }
46   }
```

```

47 % get script hash
48 \file_get_md5five_hash:nN { \@wolfram@tmp@file.wls }\l__ztikz_current_hash_tl
49 \file_get_md5five_hash:nN { \@wolfram@tmp@file.wls }\l__wolfram_current_hash_tl
50 \tl_set_rescan:Nne \l__wolfram_current_hash_tl
51 { \cctab_select:N \c_initex_cctab }
52 { \l__wolfram_current_hash_tl }
53 \xdef\wolfram@tmp@file{\g__ztikz_wolfram_path_tl/\l__ztikz_current_hash_tl}
54 \ztool_shell_mv:ee {\@wolfram@tmp@file.wls}{\wolfram@tmp@file.wls}
55
56
57 % add additional wolfram code - general export
58 \tl_if_empty:nF { #3 }
59 {
60     \__ztikz_addto_script:ee { \wolfram@tmp@file.wls }
61     { Export["\wolfram@tmp@file.#2", #3] }
62     \xdef\wolframOutputFile {
63         \g__ztikz_wolfram_path_tl / \l__ztikz_current_hash_tl.#2
64     }
65 }
66 }
67
68 \cs_new:Npn \__ztikz_wolfram_execute_posthook:n #1
69 {
70     \ior_open:Ne \g__ztikz_wolfram_ior { \wolframOutputFile }
71     \ior_get:NN \g__ztikz_wolfram_ior \l__ztikz_wolfram_tmp_res_tl
72     \tl_set_rescan:Nno \l__ztikz_wolfram_tmp_res_tl
73     { \cctab_select:N \c_document_cctab }
74     { \l__ztikz_wolfram_tmp_res_tl }
75     \exp_args:NNe \seq_gset_split:NnV \g__ztikz_wolfram_res_seq
76     { \c_atsign_str } \l__ztikz_wolfram_tmp_res_tl
77     \ior_close:N \g__ztikz_wolfram_ior
78 }
79
80 \cs_new:Nn \__ztikz_wolfram_case_run:
81 {
82     \tl_if_eq:VnTF \g__ztikz_wolfram_engine_tl { wstp }
83     {
84         % NOTE: 'wstp' is much faster than the other engines
85         \edef\wolfram@script
86         {
87             wstp~ '<<< "\wolfram@tmp@file.wls"'
88         }
89     }
90     {
91         \edef\wolfram@script
92         {
93             \g__ztikz_wolfram_engine_tl\space
94             \bool_if:NT \g__ztikz_wolfram_cloud_bool {-cloud\space}

```

95	-script\space	95
96	\wolfram@tmp@file.wls	96
97	}	97
98	}	98
99	\ztikz_term_info:e	99
100	{	100
101	\g__ztikz_wolfram_engine_tl\space~is~running~on:	101
102	'\wolfram@tmp@file.wls'...\iow_newline:	102
103	}	103
104	\ztool_shell_escape:e { \wolfram@script }	104
105	}	105
106	\cs_new:Npn __ztikz_wolfram_case_skip:n #1	106
107	{% #1: ext	107
108	% remove the skip wolframscript file and result	108
109	\clist_if_in:NnT \g__ztikz_library_loaded_clist {cache}	109
110	{	110
111	\tl_if_eq:enT {\zcache@flag}{10}	111
112	{	112
113	\seq_if_in:NVF \g_ztikz_file_hash_seq \l__wolfram_current_hash_tl	113
114	{	114
115	\ztikz_term_info:e {Removing~result~of~hash:\l__wolfram_current_hash_tl}	115
116	\ztool_shell_rm:e {\wolfram@tmp@file.wls}	116
117	\ztool_shell_rm:e {\wolfram@tmp@file.#1}	117
118	}	118
119	}	119
120	}	120
121	\ztikz_term_info:e	121
122	{	122
123	Use~cache~result:'\g__ztikz_wolfram_path_tl/	123
124	\l__ztikz_current_hash_tl.#1'\iow_newline:	124
125	}	125
126	}	126
127	\cs_new:Npn __ztikz_wolfram_execute_cli:nn #1#2	127
128	{% #1: ext; #2: hash label	128
129	\ztikz_if_run_again:nenTF { \c_false_bool }{ \l__ztikz_current_hash_tl }{ #2 }	129
130	{	130
131	__ztikz_wolfram_case_run:	131
132	}{	132
133	__ztikz_wolfram_case_skip:n { #1 }	133
134	}	134
135	}	135
136		136
137		137
138	% ==> user interface	138
139	\msg_set:nnn {ztikz}{wolfram-arg_empty}	139
140	{ wolfram~library~error:~argument~of~command~'\c_backslash_str #1'~can~NOT~be~empty. }	140
141	\cs_new:Npn __ztikz_wolfram_arg_empty_assert:nn #1#2	141
142	{	142

```

143 \exp_args:No \tl_if_empty:nT { #2 }
144 {
145     \msg_error:nnn {ztikz}{wolfram-arg_empty}{#1}
146 }
147 }
148
149 % --> load result
150 \NewDocumentCommand\wolframResult{so}
151 {
152     \IfBooleanTF{#1}
153     {% must be integer expression, or it will raise bug.
154     \seq_item:Ne \g_ztikz_wolfram_res_seq
155     {\IfValueTF {#2}{\int_eval:n {#2}}{1}}
156 }{
157     \seq_use:Nn \g_ztikz_wolfram_res_seq
158     { \IfValueTF {#2}{#2}{,} }
159 }
160 }
161
162 % --> wolfram code
163 \NewDocumentEnvironment{wolframCode}{m}
164 {
165     \xsim_file_write_start:ne {\c_false_bool}{\g__ztikz_wolfram_path_tl/t@mp.wls}
166 }{
167     \xsim_file_write_stop:
168     \__ztikz_wolfram_execute:nnnn
169     {}{}{}{#1}
170 }
171
172 % --> wolfram graphicx
173 \NewDocumentEnvironment{wolframGraphics}{m0{}}
174 {
175     \gdef\zgraphics@spec{#2}
176     \xsim_file_write_start:ne {\c_false_bool}{\g__ztikz_wolfram_path_tl/t@mp.wls}
177 }{
178     \xsim_file_write_stop:
179     \__ztikz_wolfram_execute:nnnn {}{pdf}{FIGURE}{#1}
180     \tl_if_empty:eF {\zgraphics@spec}
181     {
182         \exp_after:wN \includegraphics \exp_after:wN
183         [\zgraphics@spec]{\wolfram@tmp@file.pdf}
184     }
185 }
186
187 % --> wolfram simple code
188 \NewDocumentCommand\wolfram{smm}
189 {
190     \__ztikz_wolfram_arg_empty_assert:nn {\wolfram}{#3}

```

```

191 \_ztikz_wolfram_execute:ennn
192 {
193 \IfBooleanTF{#1}
194 { TeXResult = ToString[#3]; }
195 { TeXResult = ToString[TeXForm[#3]]; }
196 }{txt}{TeXResult}{#2}
197 }
198
199 % --> wolfram tex code(expandable token replace)
200 \group_begin:
201 \char_set_catcode_escape:n { 36 }
202 \char_set_catcode_letter:n { 92 }
203 $cs_gset:Nn $__double_backslash:n
204 { $tl_if_eq:NNTF #1\_{{\}}{#1} }
205 $gdef$wolframTex{
206 \char_set_catcode_letter:n { 92 }
207 $wolframTex@getarg
208 }
209 $gdef$wolframTex@getarg#1#2{
210 $__ztikz_wolfram_arg_empty_assert:nn {wolframTex}{#2}
211 $tl_set:Ne $l_tmpa_tl
212 {
213 \tl_map_function:nN {#2}
214 $__double_backslash:n
215 }
216 $__ztikz_wolfram_execute:onn
217 {TeXResult = TeXForm[ToExpression["$l_tmpa_tl", TeXForm]]}
218 {txt}{TeXResult}{#1}
219 \char_set_catcode_escape:n { 92 }
220 }
221 \char_set_catcode_escape:n { 92 }
222 \char_set_catcode_letter:n { 36 }
223 \group_end:
224
225 % --> wolfram table (extended the interface of 'latexalpha2')
226 \cs_set:Npn \__table_item_handle:n #1
227 {% the inner '\exp_not:N' prevent expansion from 'tabularray'.
228 \exp_not:n {
229 \exp_not:N \__wolfram_table_cell_cmd:n {#1}
230 },
231 }
232 \cs_set:Npn \__table_row_handle:n #1 % #1='{1, 2, 3}'
233 {
234 \clist_use:en
235 {
236 \clist_map_function:oN #1
237 \__table_item_handle:n
238 }{ & } \

```

239	}	239
240	\cs_generate_variant:Nn \clist_use:nn { en }	240
241	\cs_generate_variant:Nn \clist_map_function:nN { oN }	241
242	\cs_new:Npn __part_table_from_file:nN #1#2	242
243	{% #1:file; #2:data var	243
244	\ztool_gread_file_as_seq:neN {\c_true_bool}	244
245	{ #1 } \l_tmpa_seq	245
246	\tl_set:Ne #2	246
247	{	247
248	\seq_map_function:NN \l_tmpa_seq	248
249	__table_row_handle:n	249
250	}	250
251	}	251
252	\cs_set:Npn __full_table_from_file:nn #1#2	252
253	{% #1:file; #2:table header	253
254	__part_table_from_file:nN	254
255	{ #1 } \l_part_table_data_tl	255
256	\tl_set:Ne \l_full_table_data_tl	256
257	{	257
258	\tl_if_empty:eF {#2}{#2 \\}	258
259	\l_part_table_data_tl	259
260	}	260
261	\tl_set:Ne \l_part_table_data_tl	261
262	{ \l_part_table_data_tl }	262
263	}	263
264	\cs_generate_variant:Nn __full_table_from_file:nn { VV }	264
265	\cs_set:Npn __typeset_table:nnn #1#2#3	265
266	{% #1:table format; #2:table header; #3:table part data	266
267	\begin{tabular}{#1}	267
268	\hline	268
269	\bool_if:NT \l_wolfram_table_hdbt_rule_bool	269
270	{ #2\\ \hline }	270
271	#3	271
272	\hline	272
273	\end{tabular}	273
274	}	274
275	\cs_generate_variant:Nn __typeset_table:nnn { VVV }	275
276	\ztikz_keys_define:nn { wolfram / table }	276
277	{	277
278	format .tl_set:N = \l_ztikz_wolfram_table_format_tl,	278
279	format .initial:n = { *{12}{1} },	279
280	header .tl_set:N = \l_ztikz_wolfram_table_header_tl,	280
281	header .initial:n = { },	281
282	hdbt-rule .bool_set:N = \l_wolfram_table_hdbt_rule_bool,	282
283	hdbt-rule .initial:n = { false },	283
284	hdbt-rule .default:n = { true },	284
285	cell-cmd .cs_gset:Np = __wolfram_table_cell_cmd:n #1,	285
286	cell-cmd .initial:n = { #1 },	286

```

287 }
288 \NewDocumentCommand{\wolframTable}{sm0{m}}
289 {
290   {% #1:if typeset; #2:key-value; #3:code
291     \__ztikz_wolfram_arg_empty_assert:nn { wolframTable }{ #4 }
292     \group_begin:
293       \ztikz_keys_set:nn { wolfram/table }{ #3 }
294       \__ztikz_wolfram_execute:ennn
295         { TeXResult = #4; }{ txt }
296         { TeXResult }{ #2 }
297       \__full_table_from_file:VV \wolframOutputFile
298       \l__ztikz_wolfram_table_header_tl
299       \IfBooleanT{#1}
300         {
301           \__typeset_table:VVV
302           \l__ztikz_wolfram_table_format_tl
303           \l__ztikz_wolfram_table_header_tl
304           \l_part_table_data_tl
305         }
306       \exp_args:NNo \gdef\wolframTablePData{ \l_part_table_data_tl }
307       \exp_args:NNo \gdef\wolframTableFData{ \l_full_table_data_tl }
308     \group_end:
309   }
310   % --> equation solve
311   \ztikz_keys_define:nn { wolfram/solve }
312   {
313     var .tl_set:N = \l__ztikz_wolfram_var_tl,
314     var .initial:n = {},
315     domain .tl_set:N = \l__ztikz_wolfram_domain_tl,
316     domain .initial:n = {},
317   }
318 \NewDocumentCommand\wolframSolve{smom}
319 {
320   \__ztikz_wolfram_arg_empty_assert:nn { wolframSolve }{ #4 }
321   \group_begin:
322   \IfValueT {#3} { \ztikz_keys_set:nn { wolfram/solve } {#3} }
323   \tl_if_empty:VF \l__ztikz_wolfram_domain_tl
324     { \tl_set:N \l__ztikz_wolfram_tmp_arg_tl {,\l__ztikz_wolfram_domain_tl} }
325   \__ztikz_wolfram_execute:ennn
326   {
327     \IfBooleanTF {#1}{
328       TeXResult = Row[Solve[#4]//Flatten, "@"]
329       /.{Rule -> Equal}//TeXForm//ToString;
330     }{
331       TeXResult = Row[
332         Solve[#4, {\l__ztikz_wolfram_var_tl} \l__ztikz_wolfram_tmp_arg_tl]//Flatten,
333         "@"
334       ]/.{Rule -> Equal}//TeXForm//ToString;

```

335	}	335
336	}{txt}{TeXResult}{#2}	336
337	\group_end:	337
338	}	338
339		339
340	% --> differential equation solve	340
341	\ztikz_keys_define:nn { wolfram/dsolve }	341
342	{	342
343	depend .tl_set:N = \l__ztikz_wolfram_de_var_tl,	343
344	depend .initial:n = { y[x] },	344
345	independ .tl_set:N = \l__ztikz_wolfram_in_var_tl,	345
346	independ .initial:n = { x },	346
347	}	347
348	\NewDocumentCommand\wolframDSolve{smom}	348
349	{	349
350	__ztikz_wolfram_arg_empty_assert:nn { wolframDSolve }{ #4 }	350
351	\group_begin:	351
352	\IfValueT {#3} { \ztikz_keys_set:nn { wolfram/dsolve } {#3} }	352
353	\tl_if_empty:VF \l__ztikz_wolfram_in_var_tl	353
354	{ \tl_set:Ne \l__ztikz_wolfram_in_var_tl {,\l__ztikz_wolfram_in_var_tl} }	354
355	__ztikz_wolfram_execute:ennn	355
356	{	356
357	\IfBooleanTF {#1}{	357
358	TeXResult = Row[DSolve[#4]//Flatten, ","]	358
359	/.{Rule -> Equal}//TeXForm//ToString;	359
360	}{	360
361	TeXResult = Row[	361
362	DSolve[{#4}, {\l__ztikz_wolfram_de_var_tl}\l__ztikz_wolfram_in_var_tl]//Flatten,	362
363	"@"	363
364	]/.{Rule -> Equal}//TeXForm//ToString;	364
365	}	365
366	}{txt}{TeXResult}{#2}	366
367	\group_end:	367
368	}	368

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